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## MAX9296B

## Dual GMSL2/GMSL1 to CSI-2 Deserializer

### General Description

The MAX9296B deserializer converts single or dual serial inputs to MIPI CSI-2 outputs. The device operates in either GMSL1 or GMSL2 mode. The MAX9296B also sends and receives side-channel data, enabling full-duplex transmission of forward path video and bidirectional control data over low-cost 50Ω coax or 100Ω STP cables to meet the GMSL2 or GMSL1 channel specification.

The GMSL2 links operate at a fixed rate of 3Gbps or 6Gbps in the forward direction and 187.5Mbps in the reverse direction. In GMSL1 mode, each serial link can be paired with 3.12Gbps or 1.5Gbps GMSL1 serializers or operate up to 4.5Gbps with GMSL2 serializers. The MAX9296B supports mixed GMSL2 and GMSL1 links. The serial inputs operate independently, allowing videos with different timings and resolutions to be received on each input.

Video data from both inputs can be aggregated for output on a single CSI-2 port or replicated on a second port for redundant processing.

Each four-lane CSI-2 port can be programmed to operate with any number of active data lanes. Each CSI-2 port supports 16 virtual channels and allows virtual channel and data type reassignment.

In GMSL1 and GMSL2 modes, the side-channel operates at 9.6kbps to 1Mbps in UART-to-UART mode, and 9.6kbps to 1Mbps in I<sup>2</sup>C-to-I<sup>2</sup>C mode with clock stretching. Using the side-channel, a μC residing on either end of the link can program serializer, deserializer, and peripheral device registers independent of video timing. In GMSL2 mode, two additional pass-through I<sup>2</sup>C or UART channels and a pass-through 50MHz SPI channel provides peripheral control.

The MAX9296B operates over the automotive temperature range of -40°C to +105°C and is AEC-Q100 qualified.

### Applications

- High-Definition (2MP+/4MP 60fps) Advanced Driver Assistance Systems (ADAS)
  - Surround View Systems (SVS)
  - Rear View Cameras (RVC)
  - Camera Monitoring Systems (CMS)
  - Driver Monitoring Systems (DMS)
- 8MP 40fps Forward Vision Cameras (FVC)
- Satellite Lidar and Radar Sensors
- Gesture Cameras
- ECU-to-ECU Links

**Ordering Information** appears at end of data sheet.

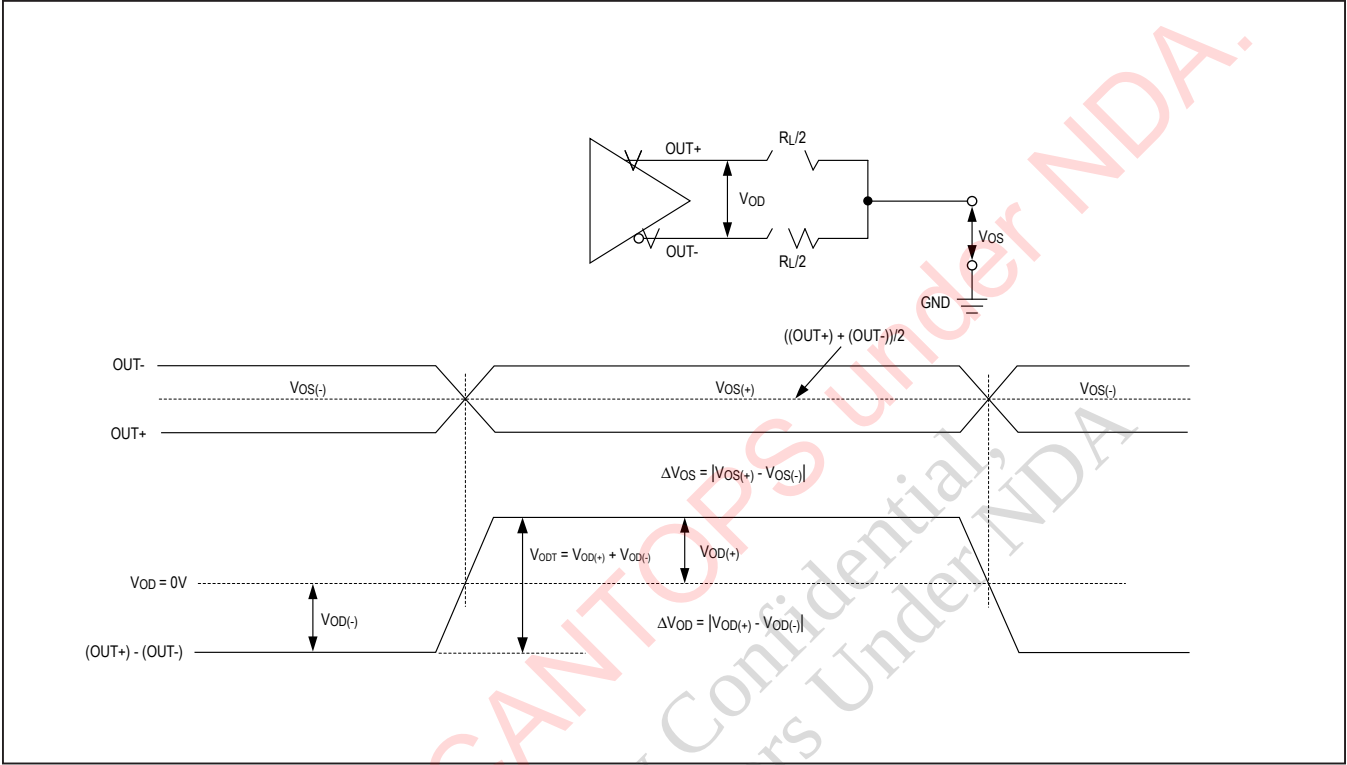
### Benefits and Features

- Dual Four-Lane, Four-Lane and Dual Two-Lane, or Quad Two-Lane MIPI CSI-2 v1.3 Output Ports
  - Aggregation and Replication Functions
  - 16-Channel Virtual Channel Support
  - Supports RAW8/10/12/14/16/20, RGB565/666/888, YUV422 8-/10-Bit, User-Defined, and Generic Long Packet Data Types
- Advanced MIPI D-PHY v1.2 Transmitters
  - 80Mbps to 2.0Gbps per Lane
  - Polarity Flip and Data-Lane Reassignment
- Flexible Input Configuration
  - Independent Serial Inputs Accept Video with Different Timing and Resolution
  - Support for Mixed GMSL2 and GMSL1 Inputs
- Multiple GMSL2 Data Rates for System and Power Flexibility
  - 3Gbps or 6Gbps Forward Link Rates
  - 187.5Mbps Reverse Link Rate
- Backward-Compatible GMSL1 Mode
- Full-Duplex Capability Over a Single Wire
- Ideal for High-Definition Camera Applications
  - Supports 2MP+/4MP at 60fps and 8MP at 40fps
- ASIL-Relevant Functional Safety Features (GMSL2)
  - ASIL-B Compliant in GMSL2 Mode
  - 16-Bit CRC Protection of Side-Channel Data (I<sup>2</sup>C, UART, SPI, GPIO) with Retransmission Upon Error Detection
  - Optional 32-Bit Video-Line CRC
  - Video Watermark Insertion and Detection
- Forward and Reverse Channel PRBS for BER Testing of Serial Link
- Continuous Link Margin Monitoring and Optimization
  - Adaptive Equalization Allows 15m Coax Cable with Multiple In-Line Connectors
  - Eye-Opening Monitor for Continuous Link Margin Diagnosis
- Concurrent Side-Channel for Device Configuration and Communicating with Remote Peripherals
  - GMSL2: I<sup>2</sup>C/UART, Pass-Through I<sup>2</sup>C/UART, SPI, GPIO and Register-Programmable GPIO
  - GMSL1: I<sup>2</sup>C/UART, GPI-GPO Link and Register-Programmable GPIO
  - Four Hardware-Programmable Device Addresses
- Compact 7mm x 7mm TQFN Package with Exposed Pad

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Simplified Block Diagram



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## Absolute Maximum Ratings

(All voltages with respect to ground.)

|                                      |                                                 |
|--------------------------------------|-------------------------------------------------|
| V <sub>DDIO</sub> .....              | -0.3V to +3.9V                                  |
| V <sub>DD18</sub> .....              | -0.3V to +2.0V                                  |
| V <sub>DD</sub> .....                | -0.3V to +2.0V                                  |
| V <sub>TERM</sub> .....              | -0.3V to +1.32V                                 |
| CAP_VDD .....                        | -0.3V to +1.2V                                  |
| SIO_ (Active State) (Note A) .....   | (V <sub>DD18</sub> - 1.1V) to V <sub>DD18</sub> |
| SIO_ (Inactive state) (Note A) ..... | -0.3V to +1.1V                                  |

|                                                                                                                          |                                     |
|--------------------------------------------------------------------------------------------------------------------------|-------------------------------------|
| XRES, X2 .....                                                                                                           | -0.3V to (V <sub>DD18</sub> + 0.3V) |
| All Other Pins (Note B) .....                                                                                            | -0.3V to (V <sub>DDIO</sub> + 0.3V) |
| DA/B_P/N, CKA/BP/N (Note C) .....                                                                                        | -0.3V to (V <sub>TERM</sub> + 0.1V) |
| Continuous Power Dissipation (T <sub>A</sub> = +70°C)<br>TQFN, TQFN-SW (Multilayer,<br>derate 40mW/°C above +70°C) ..... | 2200mW                              |
| Storage Temperature Range .....                                                                                          | -40°C to +150°C                     |
| Soldering Temperature (reflow) .....                                                                                     | +260°C                              |

**Note A:** Active state means the device is powered up and not in sleep or power-down modes. Inactive means the device is not powered up or powered up in sleep or power-down mode.

**Note B:** Specified maximum voltage or 3.9V, whichever is lower.

**Note C:** Specified maximum voltage or 1.36V, whichever is lower.

Stresses beyond those listed under "Absolute Maximum Ratings" may cause permanent damage to the device. These are stress ratings only, and functional operation of the device at these or any other conditions beyond those indicated in the operational sections of the specifications is not implied. Exposure to absolute maximum rating conditions for extended periods may affect device reliability.

## Package Information

## 48 TQFN

|                                              |                         |
|----------------------------------------------|-------------------------|
| Package Code                                 | T4877+11                |
| Outline Number                               | <a href="#">21-0144</a> |
| Land Pattern Number                          | <a href="#">90-0130</a> |
| <b>THERMAL RESISTANCE, FOUR-LAYER BOARD:</b> |                         |
| Junction to Ambient (θ <sub>JA</sub> )       | 25°C/W                  |
| Junction to Case (θ <sub>JC</sub> )          | 1°C/W                   |

## 48 TQFN-SW (Side-Wettable)

|                                              |                           |
|----------------------------------------------|---------------------------|
| Package Code                                 | T4877Y+11                 |
| Outline Number                               | <a href="#">21-100045</a> |
| Land Pattern Number                          | <a href="#">90-100016</a> |
| <b>THERMAL RESISTANCE, FOUR-LAYER BOARD:</b> |                           |
| Junction to Ambient (θ <sub>JA</sub> )       | 25°C/W                    |
| Junction to Case (θ <sub>JC</sub> )          | 1°C/W                     |

For the latest package outline information and land patterns (footprints), go to [www.maximintegrated.com/packages](http://www.maximintegrated.com/packages). Note that a "+", "#", or "-" in the package code indicates RoHS status only. Package drawings may show a different suffix character, but the drawing pertains to the package regardless of RoHS status.

Package thermal resistances were obtained using the method described in JEDEC specification JESD51-7, using a four-layer board. For detailed information on package thermal considerations, refer to [www.maximintegrated.com/thermal-tutorial](http://www.maximintegrated.com/thermal-tutorial).



## MAX9296B

## Dual GMSL2/GMSL1 to CSI-2 Deserializer

## DC Electrical Characteristics

( $V_{TERM} = 1.14V$  to  $1.26V$ ,  $V_{DD18} = 1.7V$  to  $1.9V$ ,  $V_{DD} = 0.95V$  to  $1.05V$  or  $V_{DD} = 1.14V$  to  $1.26V$ ,  $V_{DDIO} = 1.7V$  to  $3.6V$ ,  $T_A = -40^{\circ}C$  to  $+105^{\circ}C$ , EP connected to PCB ground, typical values are at  $V_{TERM} = 1.2V$ ,  $V_{DD18} = V_{DDIO} = 1.8V$ ,  $V_{DD} = 1.0V$ ,  $T_A = 25^{\circ}C$ , unless otherwise noted.) (Note 1)

| PARAMETER                                                            | SYMBOL          | CONDITIONS                                                      | MIN               | TYP              | MAX               | UNITS    |
|----------------------------------------------------------------------|-----------------|-----------------------------------------------------------------|-------------------|------------------|-------------------|----------|
| <b>GMSL2 REVERSE CHANNEL SERIAL OUTPUTS (See Figure 1)</b>           |                 |                                                                 |                   |                  |                   |          |
| Output Voltage Swing (Single Ended)                                  | $V_O$           | $R_L = 100\Omega \pm 1\%$                                       | 190               | 250              | 310               | mV       |
| Output Voltage Swing (Differential)                                  | $V_{ODT}$       | $R_L = 100\Omega \pm 1\%$ , peak-to-peak differential voltage   | 380               | 500              | 620               | mV       |
| Change in $V_{OD}$ Between Complementary Output States               | $\Delta V_{OD}$ | $R_L = 100\Omega \pm 1\%$ , $ V_{OD(H)} - V_{OD(L)} $           |                   |                  | 25                | mV       |
| Differential Output Offset Voltage                                   | $V_{OS}$        | $R_L = 100\Omega \pm 1\%$ , offset voltage in each output state | $V_{DD18} - 0.45$ | $V_{DD18} - 0.3$ | $V_{DD18} - 0.15$ | V        |
| Change in $V_{OS}$ Between Complementary Output States               | $\Delta V_{OS}$ | $R_L = 100\Omega \pm 1\%$ , $ V_{OS(H)} - V_{OS(L)} $           |                   |                  | 25                | mV       |
| Termination Resistance (Internal)                                    | $R_T$           | Any pin to $V_{DD18}$                                           | 50                | 55               | 60                | $\Omega$ |
| <b>GMSL1 REVERSE CHANNEL SERIAL OUTPUTS</b>                          |                 |                                                                 |                   |                  |                   |          |
| Differential High Output Peak Voltage $V_{(SIO\_P)} - V_{(SIO\_N)}$  | $V_{RODH}$      | Forward channel disabled (Figure 2)                             | HIM disabled      | 30               | 70                | mV       |
|                                                                      |                 |                                                                 | HIM enabled       | 50               | 110               |          |
| Differential Low Output Peak Voltage $V_{(SIO\_P)} - V_{(SIO\_N)}$   | $V_{RODL}$      | Forward channel disabled (Figure 2)                             | HIM disabled      | -70              | -30               | mV       |
|                                                                      |                 |                                                                 | HIM enabled       | -110             | -50               |          |
| Single-Ended High Output Peak Voltage                                | $V_{ROSH}$      | Forward channel disabled                                        | HIM disabled      | 30               | 70                | mV       |
|                                                                      |                 |                                                                 | HIM enabled       | 50               | 110               |          |
| Single-Ended Low Output Peak Voltage                                 | $V_{ROSL}$      | Forward channel disabled                                        | HIM disabled      | -70              | -30               | mV       |
|                                                                      |                 |                                                                 | HIM enabled       | -110             | -50               |          |
| Differential Output Offset Voltage $(V_{(SIO\_P)} + V_{(SIO\_N)})/2$ | $V_{DOS}$       |                                                                 | $V_{DD18} - 0.3$  |                  | $V_{DD18}$        | V        |
| Termination Resistance (Internal)                                    | $R_T$           | Any pin to $V_{DD18}$                                           | 50                | 55               | 60                | $\Omega$ |
| <b>D-PHY LP TRANSMITTER</b>                                          |                 |                                                                 |                   |                  |                   |          |
| Thevenin High-Level Output Voltage                                   | $V_{OH}$        |                                                                 | 0.95              | 1.2              | 1.3               | V        |
| Thevenin Low-Level Output Voltage                                    | $V_{OL}$        |                                                                 | -50               |                  | 50                | mV       |
| Output Impedance                                                     | $Z_{OLP}$       |                                                                 | 110               |                  |                   | $\Omega$ |



## MAX9296B

## Dual GMSL2/GMSL1 to CSI-2 Deserializer

## DC Electrical Characteristics (continued)

( $V_{\text{TERM}} = 1.14\text{V}$  to  $1.26\text{V}$ ,  $V_{\text{DD18}} = 1.7\text{V}$  to  $1.9\text{V}$ ,  $V_{\text{DD}} = 0.95\text{V}$  to  $1.05\text{V}$  or  $V_{\text{DD}} = 1.14\text{V}$  to  $1.26\text{V}$ ,  $V_{\text{DDIO}} = 1.7\text{V}$  to  $3.6\text{V}$ ,  $T_A = -40^\circ\text{C}$  to  $+105^\circ\text{C}$ , EP connected to PCB ground, typical values are at  $V_{\text{TERM}} = 1.2\text{V}$ ,  $V_{\text{DD18}} = V_{\text{DDIO}} = 1.8\text{V}$ ,  $V_{\text{DD}} = 1.0\text{V}$ ,  $T_A = 25^\circ\text{C}$ , unless otherwise noted.) (Note 1)

| PARAMETER                                                                  | SYMBOL                   | CONDITIONS                                                                                               | MIN                        | TYP | MAX                        | UNITS |
|----------------------------------------------------------------------------|--------------------------|----------------------------------------------------------------------------------------------------------|----------------------------|-----|----------------------------|-------|
| D-PHY HS TRANSMITTER                                                       |                          |                                                                                                          |                            |     |                            |       |
| HS Transmit Static Common-Mode Voltage                                     | V <sub>CMTX</sub>        | See <a href="#">Figure 3</a>                                                                             | 150                        | 200 | 250                        | mV    |
| V <sub>CMTX</sub> Mismatch When Output Is Differential-1 Or Differential-0 | ΔV <sub>CMTX</sub> (1,0) | ΔV <sub>CMTX</sub> (1,0) = (V <sub>CMTX</sub> (1) - V <sub>CMTX</sub> (0))/2, <a href="#">(Figure 4)</a> |                            |     | 5                          | mV    |
| HS Transmit Differential Voltage                                           | V <sub>OD</sub>          | See <a href="#">Figure 3</a>                                                                             | 140                        | 200 | 270                        | mV    |
| V <sub>OD</sub> Mismatch When Output Is Differential-1 Or Differential-0   | ΔV <sub>OD</sub>         | See <a href="#">Figure 4</a>                                                                             |                            |     | 14                         | mV    |
| HS Output High Voltage                                                     | V <sub>OHHS</sub>        |                                                                                                          |                            |     | 360                        | mV    |
| Single-Ended Output Impedance                                              | Z <sub>OS</sub>          |                                                                                                          | 40                         | 50  | 62.5                       | Ω     |
| Single-Ended Output Impedance Mismatch                                     | ΔZ <sub>OS</sub>         |                                                                                                          |                            |     | 10                         | %     |
| I/O PINS                                                                   |                          |                                                                                                          |                            |     |                            |       |
| High-Level Input Voltage                                                   | V <sub>IH</sub>          |                                                                                                          | 0.7<br>x V <sub>DDIO</sub> |     |                            | V     |
| Low-Level Input Voltage                                                    | V <sub>IL</sub>          |                                                                                                          |                            |     | 0.3<br>x V <sub>DDIO</sub> | V     |
| High-Level Output Voltage                                                  | V <sub>OH</sub>          | I <sub>OH</sub> = -4mA                                                                                   | V <sub>DDIO</sub> - 0.4    |     |                            | V     |
| Low-Level Output Voltage                                                   | V <sub>OL</sub>          | I <sub>OL</sub> = 4mA                                                                                    |                            |     | 0.4                        | V     |
| Input Current                                                              | I <sub>IN</sub>          | All pullup/pulldown devices disabled, V <sub>IN</sub> = 0V to V <sub>DDIO</sub>                          |                            |     | 1                          | μA    |
| Input Capacitance                                                          | C <sub>IN</sub>          |                                                                                                          |                            | 3   |                            | pF    |
| Internal Pullup/<br>Pulldown Resistance                                    | R <sub>IN</sub>          | 40kΩ enabled                                                                                             |                            | 40  |                            | kΩ    |
|                                                                            |                          | 1MΩ enabled                                                                                              |                            | 1   |                            | MΩ    |
| OPEN-DRAIN PINS                                                            |                          |                                                                                                          |                            |     |                            |       |
| High-Level Input Voltage                                                   | V <sub>IH</sub>          |                                                                                                          | 0.7<br>x V <sub>DDIO</sub> |     |                            | V     |
| Low-Level Input Voltage                                                    | V <sub>IL</sub>          |                                                                                                          |                            |     | 0.3<br>x V <sub>DDIO</sub> | V     |
| Low-Level Open-Drain Output Voltage                                        | V <sub>OL</sub>          | I <sub>OL</sub> = 4mA                                                                                    |                            |     | 0.4                        | V     |

## MAX9296B

## Dual GMSL2/GMSL1 to CSI-2 Deserializer

## DC Electrical Characteristics (continued)

( $V_{\text{TERM}} = 1.14\text{V}$  to  $1.26\text{V}$ ,  $V_{\text{DD18}} = 1.7\text{V}$  to  $1.9\text{V}$ ,  $V_{\text{DD}} = 0.95\text{V}$  to  $1.05\text{V}$  or  $V_{\text{DD}} = 1.14\text{V}$  to  $1.26\text{V}$ ,  $V_{\text{DDIO}} = 1.7\text{V}$  to  $3.6\text{V}$ ,  $T_A = -40^\circ\text{C}$  to  $+105^\circ\text{C}$ , EP connected to PCB ground, typical values are at  $V_{\text{TERM}} = 1.2\text{V}$ ,  $V_{\text{DD18}} = V_{\text{DDIO}} = 1.8\text{V}$ ,  $V_{\text{DD}} = 1.0\text{V}$ ,  $T_A = 25^\circ\text{C}$ , unless otherwise noted.) (Note 1)

| PARAMETER                                                           | SYMBOL             | CONDITIONS                                                                         | MIN                        | TYP | MAX                        | UNITS |
|---------------------------------------------------------------------|--------------------|------------------------------------------------------------------------------------|----------------------------|-----|----------------------------|-------|
| Input Current                                                       | I <sub>IN</sub>    | All pullup/pulldown devices disabled,<br>V <sub>IN</sub> = 0V to V <sub>DDIO</sub> |                            |     | 1                          | μA    |
| Input Capacitance                                                   | C <sub>IN</sub>    |                                                                                    |                            | 3   |                            | pF    |
| Internal Pull-up Resistor                                           | R <sub>PU</sub>    | 40kΩ enabled                                                                       |                            | 40  |                            | kΩ    |
|                                                                     |                    | 1MΩ enabled                                                                        |                            | 1   |                            | MΩ    |
| PWDNB INPUT                                                         |                    |                                                                                    |                            |     |                            |       |
| High-Level Input Voltage                                            | V <sub>IH</sub>    |                                                                                    | 0.7<br>x V <sub>DDIO</sub> |     |                            | V     |
| Low-Level Input Voltage                                             | V <sub>IL</sub>    |                                                                                    |                            |     | 0.3<br>x V <sub>DDIO</sub> | V     |
| Input Current                                                       | I <sub>IN</sub>    | V <sub>IN</sub> = 0V to V <sub>DDIO</sub>                                          |                            |     | 6                          | μA    |
| Internal Pulldown Resistance                                        | R <sub>PD</sub>    |                                                                                    |                            | 1   |                            | MΩ    |
| Input Capacitance                                                   | C <sub>IN</sub>    |                                                                                    |                            | 3   |                            | pF    |
| PUSH-PULL OUTPUTS                                                   |                    |                                                                                    |                            |     |                            |       |
| High-Level Output Voltage                                           | V <sub>OH</sub>    | I <sub>OH</sub> = -4mA                                                             | V <sub>DDIO</sub> -<br>0.4 |     |                            | V     |
| Low-Level Output Voltage                                            | V <sub>OL</sub>    | I <sub>OL</sub> = 4mA                                                              |                            |     | 0.4                        | V     |
| REFERENCE CLOCK INPUT REQUIREMENTS (CRYSTAL) (X1, X2)               |                    |                                                                                    |                            |     |                            |       |
| X1 Input Capacitance                                                | C <sub>IN_X1</sub> |                                                                                    |                            | 3   |                            | pF    |
| X2 Input Capacitance                                                | C <sub>IN_X2</sub> |                                                                                    |                            | 1   |                            | pF    |
| Internal X2 Limit Resistor                                          | R <sub>LIM</sub>   |                                                                                    |                            | 1.2 |                            | kΩ    |
| Internal Feedback Resistor                                          | R <sub>FB</sub>    |                                                                                    |                            | 10  |                            | kΩ    |
| Transconductance                                                    | g <sub>M</sub>     |                                                                                    |                            | 28  |                            | mA/V  |
| REFERENCE CLOCK REQUIREMENTS (EXTERNAL INPUT ON X1, X2 UNCONNECTED) |                    |                                                                                    |                            |     |                            |       |
| High-Level Input Voltage                                            | V <sub>IH</sub>    |                                                                                    | 0.9                        |     |                            | V     |
| Low-Level Input Voltage                                             | V <sub>IL</sub>    |                                                                                    |                            |     | 0.4                        | V     |
| Input Impedance                                                     | R <sub>IN</sub>    |                                                                                    |                            | 10  |                            | kΩ    |
| X1 Input Capacitance                                                | C <sub>IN_X1</sub> |                                                                                    |                            | 3   |                            | pF    |

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Dual GMSL2/GMSL1 to CSI-2 Deserializer

DC Electrical Characteristics (continued)

( $V_{TERM}$  = 1.14V to 1.26V,  $V_{DD18}$  = 1.7V to 1.9V,  $V_{DD}$  = 0.95V to 1.05V or  $V_{DD}$  = 1.14V to 1.26V,  $V_{DDIO}$  = 1.7V to 3.6V,  $T_A$  = -40°C to +105°C, EP connected to PCB ground, typical values are at  $V_{TERM}$  = 1.2V,  $V_{DD18}$  =  $V_{DDIO}$  = 1.8V,  $V_{DD}$  = 1.0V,  $T_A$  = 25°C, unless otherwise noted.) (Note 1)

| PARAMETER                                   | SYMBOL            | CONDITIONS                                                                                                                                                                                                                         |                           | MIN | TYP | MAX        | UNITS |
|---------------------------------------------|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|-----|-----|------------|-------|
| GMSL2 POWER SUPPLY CURRENT                  |                   |                                                                                                                                                                                                                                    |                           |     |     |            |       |
| Supply Current<br>(Note 4)                  | I <sub>DD</sub>   | Single 6Gbps link,<br>5.2Gbps video payload<br>input, RGB888, out-<br>put on port A, repli-<br>cated on port B, each<br>port four data lanes,<br>1.3Gbps per lane                                                                  | V <sub>TERM</sub> = 1.26V |     | 31  | 50         | mA    |
|                                             |                   |                                                                                                                                                                                                                                    | V <sub>DD18</sub> = 1.9V  |     | 92  | 120        |       |
|                                             |                   |                                                                                                                                                                                                                                    | V <sub>DD</sub> = 1.05V   |     | 150 | 550        |       |
|                                             |                   |                                                                                                                                                                                                                                    | V <sub>DD</sub> = 1.26V   |     | 150 | 550        |       |
|                                             |                   | Two 6Gbps links, 4Gbps<br>video payload input<br>per link, RGB888, link<br>A output on port C and<br>replicated on port D,<br>link B output on port E<br>and replicated on port F,<br>each port two data lanes,<br>2Gbps per lane. | V <sub>TERM</sub> = 1.26V |     | 41  | 75         |       |
|                                             |                   |                                                                                                                                                                                                                                    | V <sub>DD18</sub> = 1.9V  |     | 150 | 190        |       |
|                                             |                   |                                                                                                                                                                                                                                    | V <sub>DD</sub> = 1.05V   |     | 275 | 690        |       |
|                                             |                   |                                                                                                                                                                                                                                    | V <sub>DD</sub> = 1.26V   |     | 275 | 690        |       |
| Maximum V <sub>DDIO</sub><br>Supply Current | I <sub>DDIO</sub> | Per toggling GPIO,<br>C <sub>L</sub> = 20pF                                                                                                                                                                                        | V <sub>DDIO</sub> at 1.9V |     | 44  | μA/<br>MHz |       |
|                                             |                   |                                                                                                                                                                                                                                    | V <sub>DDIO</sub> at 3.6V |     | 81  |            |       |
| GMSL1 POWER SUPPLY CURRENT                  |                   |                                                                                                                                                                                                                                    |                           |     |     |            |       |
| Supply Current<br>(Note 4)                  | I <sub>DD</sub>   | Two 4.5Gbps links,<br>3.6Gbps video payload<br>input per link, HBM,<br>RGB888, link A out-<br>put on port A, link B<br>output on port B, each<br>port four data lanes,<br>900Mbps per lane,<br>60Hz GPI                            | V <sub>TERM</sub> = 1.26V |     | 33  | 50         | mA    |
|                                             |                   |                                                                                                                                                                                                                                    | V <sub>DD18</sub> = 1.9V  |     | 86  | 110        |       |
|                                             |                   |                                                                                                                                                                                                                                    | V <sub>DD</sub> = 1.05V   |     | 115 | 490        |       |
|                                             |                   |                                                                                                                                                                                                                                    | V <sub>DD</sub> = 1.26V   |     | 115 | 490        |       |
|                                             |                   |                                                                                                                                                                                                                                    | V <sub>DDIO</sub> = 3.6V  |     | 0   | 1          |       |
| MAXIMUM POWER-DOWN AND SLEEP CURRENT        |                   |                                                                                                                                                                                                                                    |                           |     |     |            |       |
| Maximum Power-Down<br>Current               | I <sub>DD</sub>   | V <sub>DDIO</sub> at 3.6V                                                                                                                                                                                                          | T <sub>A</sub> = +25°C    |     | 6   | μA         |       |
|                                             |                   |                                                                                                                                                                                                                                    | T <sub>A</sub> = +105°C   |     | 6   |            |       |
|                                             |                   | V <sub>TERM</sub> at 1.26V                                                                                                                                                                                                         | T <sub>A</sub> = +25°C    |     | 1   |            |       |
|                                             |                   |                                                                                                                                                                                                                                    | T <sub>A</sub> = +105°C   |     | 1   |            |       |
|                                             |                   | V <sub>DD18</sub> at 1.9V                                                                                                                                                                                                          | T <sub>A</sub> = +25°C    |     | 1   |            |       |
|                                             |                   |                                                                                                                                                                                                                                    | T <sub>A</sub> = +105°C   |     | 5   |            |       |
|                                             |                   | V <sub>DD</sub> at 1.26V                                                                                                                                                                                                           | T <sub>A</sub> = +25°C    |     | 1   |            |       |
|                                             |                   |                                                                                                                                                                                                                                    | T <sub>A</sub> = +105°C   |     | 1   |            |       |

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## Dual GMSL2/GMSL1 to CSI-2 Deserializer

## AC Electrical Characteristics

( $V_{\text{TERM}} = 1.14\text{V}$  to  $1.26\text{V}$ ,  $V_{\text{DD18}} = 1.7\text{V}$  to  $1.9\text{V}$ ,  $V_{\text{DD}} = 0.95\text{V}$  to  $1.05\text{V}$  or  $V_{\text{DD}} = 1.14\text{V}$  to  $1.26\text{V}$ ,  $V_{\text{DDIO}} = 1.7\text{V}$  to  $3.6\text{V}$ ,  $T_A = -40^\circ\text{C}$  to  $+105^\circ\text{C}$ , EP connected to PCB ground, typical values are at  $V_{\text{TERM}} = 1.2\text{V}$ ,  $V_{\text{DD18}} = V_{\text{DDIO}} = 1.8\text{V}$ ,  $V_{\text{DD}} = 1.0\text{V}$ ,  $T_A = 25^\circ\text{C}$ , unless otherwise noted.) (Note 1)

| PARAMETER                                           | SYMBOL                          | CONDITIONS                                                                                                                |                         | MIN | TYP                                       | MAX | UNITS |
|-----------------------------------------------------|---------------------------------|---------------------------------------------------------------------------------------------------------------------------|-------------------------|-----|-------------------------------------------|-----|-------|
| Maximum Sleep Current                               | I <sub>DD</sub>                 | V <sub>DDIO</sub> at 3.6V                                                                                                 | T <sub>A</sub> = +25°C  |     | 6                                         |     | μA    |
|                                                     |                                 |                                                                                                                           | T <sub>A</sub> = +105°C |     | 6                                         |     |       |
|                                                     |                                 | V <sub>TERM</sub> at 1.26V                                                                                                | T <sub>A</sub> = +25°C  |     | 1                                         |     |       |
|                                                     |                                 |                                                                                                                           | T <sub>A</sub> = +105°C |     | 1                                         |     |       |
|                                                     |                                 | V <sub>DD18</sub> at 1.9V                                                                                                 | T <sub>A</sub> = +25°C  |     | 20                                        |     |       |
|                                                     |                                 |                                                                                                                           | T <sub>A</sub> = +105°C |     | 25                                        |     |       |
|                                                     |                                 | V <sub>DD</sub> at 1.26V                                                                                                  | T <sub>A</sub> = +25°C  |     | 1                                         |     |       |
|                                                     |                                 |                                                                                                                           | T <sub>A</sub> = +105°C |     | 1                                         |     |       |
| GMSL2 FORWARD CHANNEL SWITCHING CHARACTERISTICS     |                                 |                                                                                                                           |                         |     |                                           |     |       |
| Lock Time                                           | t <sub>LOCK</sub>               | From power-up, one-shot reset, or rising edge of PWDNB to rising edge of LOCK (Figure 6)                                  |                         |     | 40                                        |     | ms    |
| Maximum Video Initialization Time                   | t <sub>VIDEOSTART</sub>         | Time from GMSL2 video packets at SIO_ to valid packets at the CSI-2 output (assumes link locked and registers configured) |                         |     | 0.1ms + (6600 x t <sub>PCLK</sub> )       |     | ms    |
| Maximum Video Latency                               | t <sub>VL</sub>                 | Time from the first pixel in a GMSL2 packet at SIO_ to the first pixel in the CSI-2 output packet (Figure 7)              |                         |     | 1 video line + (128 x t <sub>PCLK</sub> ) |     | s     |
| PWDNB Hold Time                                     | t <sub>HOLD_PWDNB</sub>         | The minimum duration PWDNB must be held LOW to reset the chip                                                             |                         |     | 1                                         |     | μs    |
| GMSL2 REVERSE CHANNEL SERIAL OUTPUTS (SIO_P, SIO_N) |                                 |                                                                                                                           |                         |     |                                           |     |       |
| GMSL Reverse Channel Transmitter Rise/Fall Time     | t <sub>R</sub> , t <sub>F</sub> | 20% to 80%, V <sub>O</sub> = 250mV, R <sub>L</sub> = 100Ω                                                                 |                         |     | 2300                                      |     | ps    |
| Total Serial Output p-p Jitter                      | t <sub>TSOJ</sub>               | PRBS7, single-ended or differential output                                                                                |                         |     | 0.15                                      |     | UI    |
| Deterministic Serial Output p-p Jitter              | t <sub>DSOJ</sub>               | PRBS7, single-ended or differential output                                                                                |                         |     | 0.1                                       |     | UI    |
| GMSL1 SWITCHING CHARACTERISTICS                     |                                 |                                                                                                                           |                         |     |                                           |     |       |
| Maximum Lock Time                                   | t <sub>LOCK</sub>               | See Figure 8                                                                                                              |                         |     | 4                                         |     | ms    |
| Maximum Power-Up Delay                              | t <sub>PU</sub>                 | See Figure 9                                                                                                              |                         |     | 8.5                                       |     | ms    |
| Maximum Video Latency                               | t <sub>VL</sub>                 | Time from the first pixel in a video line at SIO_ to the first pixel in the CSI-2 output packet (Figure 10)               |                         |     | 1 video line + (128 x t <sub>PCLK</sub> ) |     | s     |
| PWDNB Hold Time                                     | t <sub>HOLD_PWDNB</sub>         | The minimum duration PWDNB must be held LOW to reset the chip                                                             |                         |     | 1                                         |     | μs    |

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## Dual GMSL2/GMSL1 to CSI-2 Deserializer

## AC Electrical Characteristics (continued)

( $V_{\text{TERM}} = 1.14\text{V}$  to  $1.26\text{V}$ ,  $V_{\text{DD18}} = 1.7\text{V}$  to  $1.9\text{V}$ ,  $V_{\text{DD}} = 0.95\text{V}$  to  $1.05\text{V}$  or  $V_{\text{DD}} = 1.14\text{V}$  to  $1.26\text{V}$ ,  $V_{\text{DDIO}} = 1.7\text{V}$  to  $3.6\text{V}$ ,  $T_A = -40^\circ\text{C}$  to  $+105^\circ\text{C}$ , EP connected to PCB ground, typical values are at  $V_{\text{TERM}} = 1.2\text{V}$ ,  $V_{\text{DD18}} = V_{\text{DDIO}} = 1.8\text{V}$ ,  $V_{\text{DD}} = 1.0\text{V}$ ,  $T_A = 25^\circ\text{C}$ , unless otherwise noted.) (Note 1)

| PARAMETER                                | SYMBOL                          | CONDITIONS                                                                                            | MIN   | TYP           | MAX   | UNITS             |
|------------------------------------------|---------------------------------|-------------------------------------------------------------------------------------------------------|-------|---------------|-------|-------------------|
| Reverse Control-Channel Output Rise Time | $t_R$                           | No forward channel data transmission, see <a href="#">Figure 2</a> (Note 2)                           | 100   |               | 400   | ns                |
| Reverse Control-Channel Output Fall Time | $t_F$                           | No forward channel data transmission see <a href="#">Figure 2</a> (Note 2)                            | 100   |               | 400   | ns                |
| GPI-to-GPO Delay                         | $t_{\text{GPIO}}$               | Deserializer GPI to serializer GPO (cable delay not included), see <a href="#">Figure 11</a> (Note 2) |       |               | 350   | $\mu\text{s}$     |
| <b>D-PHY HS TRANSMITTER</b>              |                                 |                                                                                                       |       |               |       |                   |
| Common-Level Variations, HF              | $\Delta V_{\text{CMTX(HF)}}$    | $> 450\text{MHz}$ ( <a href="#">Figure 4</a> ) (Note 2)                                               |       |               | 15    | mV <sub>RMS</sub> |
| Common-Level Variations, LF              | $\Delta V_{\text{CMTX(LF)}}$    | $50\text{MHz} - 450\text{MHz}$ ( <a href="#">Figure 4</a> ) (Note 2)                                  |       |               | 25    | mV-PEAK           |
| 20% to 80% Rise Time and Fall Time       | $t_R$ and $t_F$                 | (Note 2)                                                                                              |       |               | 0.4   | UI                |
|                                          |                                 | (Note 2)                                                                                              |       |               | 50    | ps                |
| Tx Differential Return Loss              | $S_{\text{ddTX}}$               | $f_{\text{HMAX}} = 1.0\text{GHz}$                                                                     |       | -10           |       | dB                |
|                                          |                                 | $f_{\text{MAX}} = 1.5\text{GHz}$                                                                      |       | -8            |       |                   |
| Tx Common Mode Return Loss               | $S_{\text{ccTX}}$               | $f_{\text{MAX}} = 1.5\text{GHz}$                                                                      |       | -7            |       | dB                |
| Data Lane Bit Rate                       | $DL_{\text{BR}}$                |                                                                                                       | 80    |               | 2000  | Mbps              |
| Clock Lane Frequency                     | $CL_{\text{FREQ}}$              |                                                                                                       | 40    |               | 1000  | MHz               |
| CSI-2 Output Inter-packet Spacing        | $t_{\text{SPACE}}$              |                                                                                                       |       | 300ns + 370UI |       | ns                |
| <b>D-PHY LP TRANSMITTER (NOTE 2)</b>     |                                 |                                                                                                       |       |               |       |                   |
| 15% to 85% Rise Time and Fall Time       | $t_{\text{RLP}}/t_{\text{FLP}}$ | (Note 3)                                                                                              |       |               | 25    | ns                |
| 30% to 85% Rise Time and Fall Time       | $t_{\text{REOT}}$               | ( <a href="#">Figure 12</a> , <a href="#">Figure 14</a> ) (Note 5)                                    |       |               | 35    | ns                |
| Load Capacitance                         | $C_{\text{LOAD}}$               | (Note 3)                                                                                              | 0     |               | 70    | pF                |
| <b>D-PHY DATA-CLOCK TIMING (NOTE 2)</b>  |                                 |                                                                                                       |       |               |       |                   |
| UI Instantaneous                         | $U_{\text{INST}}$               |                                                                                                       | 0.5   |               | 12.5  | ns                |
| UI Variation                             | $\Delta UI$                     | $UI \geq 1\text{ns}$ within a single burst                                                            | -10%  |               | +10%  | UI                |
|                                          |                                 | $0.667\text{ns} \leq UI \leq 1\text{ns}$ within a single burst                                        | -5%   |               | +5%   |                   |
| Data to Clock Skew                       | $T_{\text{SKEW}}$               | $0.08$ to $1.0\text{Gbps}$ ( <a href="#">Figure 15</a> )                                              | -0.15 |               | +0.15 | $U_{\text{INST}}$ |
|                                          |                                 | $> 1.0$ to $1.5\text{Gbps}$ ( <a href="#">Figure 15</a> )                                             | -0.2  |               | +0.2  |                   |
| Static Data to Clock Skew (Tx)           | $T_{\text{SKEW Static}}$        | $> 1.5\text{Gbps}$ ( <a href="#">Figure 15</a> )                                                      | -0.2  |               | +0.2  | $U_{\text{INST}}$ |
| Dynamic Data to Clock Skew (Tx)          | $T_{\text{SKEW Dynamic}}$       | $> 1.5\text{Gbps}$ ( <a href="#">Figure 15</a> )                                                      | -0.15 |               | +0.15 | $U_{\text{INST}}$ |

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## Dual GMSL2/GMSL1 to CSI-2 Deserializer

## AC Electrical Characteristics (continued)

( $V_{\text{TERM}} = 1.14\text{V}$  to  $1.26\text{V}$ ,  $V_{\text{DD18}} = 1.7\text{V}$  to  $1.9\text{V}$ ,  $V_{\text{DD}} = 0.95\text{V}$  to  $1.05\text{V}$  or  $V_{\text{DD}} = 1.14\text{V}$  to  $1.26\text{V}$ ,  $V_{\text{DDIO}} = 1.7\text{V}$  to  $3.6\text{V}$ ,  $T_A = -40^\circ\text{C}$  to  $+105^\circ\text{C}$ , EP connected to PCB ground, typical values are at  $V_{\text{TERM}} = 1.2\text{V}$ ,  $V_{\text{DD18}} = V_{\text{DDIO}} = 1.8\text{V}$ ,  $V_{\text{DD}} = 1.0\text{V}$ ,  $T_A = 25^\circ\text{C}$ , unless otherwise noted.) (Note 1)

| PARAMETER                                                                                                                            | SYMBOL                                         | CONDITIONS                                                                            | MIN                         | TYP | MAX                         | UNITS |
|--------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|---------------------------------------------------------------------------------------|-----------------------------|-----|-----------------------------|-------|
| <b>D-PHY GLOBAL OPERATION TIMING (Note 2)</b>                                                                                        |                                                |                                                                                       |                             |     |                             |       |
| Time that the HS Clock Drives by the Transmitter Prior to Any Associated Data Lane Beginning the Transition from LP to HS Mode       | $T_{\text{CLK-PRE}}$                           | See <a href="#">Figure 13</a>                                                         | 8                           |     |                             | UI    |
| Time that the Transmitter Drives the Clock Lane LP-00 Line State Immediately Before the HS-0 Line State Starting the HS Transmission | $T_{\text{CLK\_PREPARE}}$                      | See <a href="#">Figure 13</a>                                                         | 38                          |     | 95                          | ns    |
| $T_{\text{CLK-PREPARE}} +$ Time that the Transmitter Drives the HS-0 State Prior to Starting the Clock                               | $T_{\text{CLK-PREPARE}} + T_{\text{CLK-ZERO}}$ | See <a href="#">Figure 13</a>                                                         | 300                         |     |                             | ns    |
| Transmitted Time Interval from the Start of THS-TRAIL to the Start of the LP-11 State Following an HS Burst                          | $T_{\text{EOT}}$                               | See <a href="#">Figure 12</a> , <a href="#">Figure 13</a> , <a href="#">Figure 14</a> |                             |     | $105 + 12 \times \text{UI}$ | ns    |
| Time that the Transmitter Drives LP-11 Following an HS Burst                                                                         | $T_{\text{HS-EXIT}}$                           | See <a href="#">Figure 12</a> , <a href="#">Figure 13</a> , <a href="#">Figure 14</a> | 100                         |     |                             | ns    |
| Time that the Transmitter Drives the Data Lane LP-00 Line State Immediately Before the HS-0 Line State Starting the HS Transmission  | $T_{\text{HS-PREPARE}}$                        | See <a href="#">Figure 12</a> , <a href="#">Figure 13</a> , <a href="#">Figure 14</a> | $40 + 4 \times \text{UI}$   |     | $85 + 6 \times \text{UI}$   | ns    |
| $T_{\text{HS-PREPARE}} +$ Time that the Transmitter Drives the HS-0 State Prior to Transmitting the Sync Sequence                    | $T_{\text{HS-PREPARE}} + T_{\text{HS-ZERO}}$   | See <a href="#">Figure 12</a> , <a href="#">Figure 14</a>                             | $145 + 10 \times \text{UI}$ |     |                             | ns    |

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## Dual GMSL2/GMSL1 to CSI-2 Deserializer

## AC Electrical Characteristics (continued)

( $V_{\text{TERM}} = 1.14\text{V}$  to  $1.26\text{V}$ ,  $V_{\text{DD18}} = 1.7\text{V}$  to  $1.9\text{V}$ ,  $V_{\text{DD}} = 0.95\text{V}$  to  $1.05\text{V}$  or  $V_{\text{DD}} = 1.14\text{V}$  to  $1.26\text{V}$ ,  $V_{\text{DDIO}} = 1.7\text{V}$  to  $3.6\text{V}$ ,  $T_A = -40^\circ\text{C}$  to  $+105^\circ\text{C}$ , EP connected to PCB ground, typical values are at  $V_{\text{TERM}} = 1.2\text{V}$ ,  $V_{\text{DD18}} = V_{\text{DDIO}} = 1.8\text{V}$ ,  $V_{\text{DD}} = 1.0\text{V}$ ,  $T_A = 25^\circ\text{C}$ , unless otherwise noted.) (Note 1)

| PARAMETER                                                                                                               | SYMBOL                     | CONDITIONS                                                                                                                                                                  | MIN                                     | TYP  | MAX  | UNITS         |
|-------------------------------------------------------------------------------------------------------------------------|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|------|------|---------------|
| Time that the Transmitter Drives the Flipped Differential State After Last Payload Data Bit of an HS Transmission Burst | $T_{\text{HS-TRAIL}}$      | See Figure 12, Figure 14                                                                                                                                                    | $60 + 4 \times UI$                      |      |      | ns            |
| Initialization Time                                                                                                     | $T_{\text{INIT}}$          |                                                                                                                                                                             | 100                                     |      |      | $\mu\text{s}$ |
| Transmitted Length of Any Low-Power State Period                                                                        | $T_{\text{LPX}}$           | See Figure 12, Figure 13, Figure 14                                                                                                                                         | 50                                      |      |      | ns            |
| Time that the Transmitter Drives the Skew-Calibration Sync Pattern, 0xFFFF                                              | $T_{\text{SKEWCAL\_SYNC}}$ | See Figure 14                                                                                                                                                               |                                         | 16   |      | UI            |
| Time that the Transmitter Drives the Skew-Calibration Pattern in the Initial Skew-Calibration Mode                      | $T_{\text{SKEWCAL}}$       | See Figure 14                                                                                                                                                               |                                         |      | 100  | $\mu\text{s}$ |
|                                                                                                                         |                            | See Figure 14                                                                                                                                                               | $2^{15}$                                |      |      | UI            |
| Time that the Transmitter Drives the Skew-Calibration Pattern in the Periodic Skew-Calibration Mode                     | $T_{\text{SKEWCAL}}$       | See Figure 14                                                                                                                                                               |                                         |      | 10   | $\mu\text{s}$ |
|                                                                                                                         |                            | See Figure 14                                                                                                                                                               | $2^{10}$                                |      |      | UI            |
| <b>I<sup>2</sup>C/UART PORT TIMING</b>                                                                                  |                            |                                                                                                                                                                             |                                         |      |      |               |
| Output Fall Time                                                                                                        | $t_F$                      | 70% to 30%, $C_L = 20\text{pF}$ to $100\text{pF}$ , $1\text{k}\Omega$ pullup to $V_{\text{DDIO}}$                                                                           | $20 \times V_{\text{DDIO}}/5.5\text{V}$ |      | 150  | ns            |
| I <sup>2</sup> C/UART Wake Time                                                                                         | $t_{\text{WAKEUP}}$        | From power-up, or rising edge of PWDNB to local register access; for remote register access, I <sup>2</sup> C/UART wake time is the same as lock time ( $t_{\text{LOCK}}$ ) |                                         | 2.25 |      | ms            |
| <b>I<sup>2</sup>C TIMING (SEE FIGURE 16)</b>                                                                            |                            |                                                                                                                                                                             |                                         |      |      |               |
| SCL Clock Frequency                                                                                                     | $f_{\text{SCL}}$           | Low $f_{\text{SCL}}$ range                                                                                                                                                  | 9.6                                     |      | 100  | kHz           |
|                                                                                                                         |                            | Mid $f_{\text{SCL}}$ range                                                                                                                                                  | 100                                     |      | 400  |               |
|                                                                                                                         |                            | High $f_{\text{SCL}}$ range                                                                                                                                                 | 400                                     |      | 1000 |               |



## MAX9296B

## Dual GMSL2/GMSL1 to CSI-2 Deserializer

## AC Electrical Characteristics (continued)

( $V_{\text{TERM}} = 1.14\text{V}$  to  $1.26\text{V}$ ,  $V_{\text{DD18}} = 1.7\text{V}$  to  $1.9\text{V}$ ,  $V_{\text{DD}} = 0.95\text{V}$  to  $1.05\text{V}$  or  $V_{\text{DD}} = 1.14\text{V}$  to  $1.26\text{V}$ ,  $V_{\text{DDIO}} = 1.7\text{V}$  to  $3.6\text{V}$ ,  $T_A = -40^\circ\text{C}$  to  $+105^\circ\text{C}$ , EP connected to PCB ground, typical values are at  $V_{\text{TERM}} = 1.2\text{V}$ ,  $V_{\text{DD18}} = V_{\text{DDIO}} = 1.8\text{V}$ ,  $V_{\text{DD}} = 1.0\text{V}$ ,  $T_A = 25^\circ\text{C}$ , unless otherwise noted.) (Note 1)

| PARAMETER                           | SYMBOL              | CONDITIONS                   | MIN  | TYP | MAX  | UNITS         |
|-------------------------------------|---------------------|------------------------------|------|-----|------|---------------|
| Start Condition Hold Time           | $t_{\text{HD:STA}}$ | f <sub>SCL</sub> range, low  | 4    |     |      | $\mu\text{s}$ |
|                                     |                     | f <sub>SCL</sub> range, mid  | 0.6  |     |      |               |
|                                     |                     | f <sub>SCL</sub> range, high | 0.26 |     |      |               |
| Low Period of SCL Clock             | $t_{\text{LOW}}$    | f <sub>SCL</sub> range, low  | 4.7  |     |      | $\mu\text{s}$ |
|                                     |                     | f <sub>SCL</sub> range, mid  | 1.3  |     |      |               |
|                                     |                     | f <sub>SCL</sub> range, high | 0.5  |     |      |               |
| High Period of SCL Clock            | $t_{\text{HIGH}}$   | f <sub>SCL</sub> range, low  | 4    |     |      | $\mu\text{s}$ |
|                                     |                     | f <sub>SCL</sub> range, mid  | 0.6  |     |      |               |
|                                     |                     | f <sub>SCL</sub> range, high | 0.26 |     |      |               |
| Repeated Start Condition Setup Time | $t_{\text{SU:STA}}$ | f <sub>SCL</sub> range, low  | 4.7  |     |      | $\mu\text{s}$ |
|                                     |                     | f <sub>SCL</sub> range, mid  | 0.6  |     |      |               |
|                                     |                     | f <sub>SCL</sub> range, high | 0.26 |     |      |               |
| Data Hold Time                      | $t_{\text{HD:DAT}}$ | f <sub>SCL</sub> range, low  | 0    |     |      | ns            |
|                                     |                     | f <sub>SCL</sub> range, mid  | 0    |     |      |               |
|                                     |                     | f <sub>SCL</sub> range, high | 0    |     |      |               |
| Data Setup Time                     | $t_{\text{SU:DAT}}$ | f <sub>SCL</sub> range, low  | 250  |     |      | ns            |
|                                     |                     | f <sub>SCL</sub> range, mid  | 100  |     |      |               |
|                                     |                     | f <sub>SCL</sub> range, high | 50   |     |      |               |
| Setup Time for Stop Condition       | $t_{\text{SU:STO}}$ | f <sub>SCL</sub> range, low  | 4    |     |      | $\mu\text{s}$ |
|                                     |                     | f <sub>SCL</sub> range, mid  | 0.6  |     |      |               |
|                                     |                     | f <sub>SCL</sub> range, high | 0.26 |     |      |               |
| Bus Free Time                       | $t_{\text{BUF}}$    | f <sub>SCL</sub> range, low  | 4.7  |     |      | $\mu\text{s}$ |
|                                     |                     | f <sub>SCL</sub> range, mid  | 1.3  |     |      |               |
|                                     |                     | f <sub>SCL</sub> range, high | 0.5  |     |      |               |
| Data Valid Time                     | $t_{\text{VD:DAT}}$ | f <sub>SCL</sub> range, low  |      |     | 3.45 | $\mu\text{s}$ |
|                                     |                     | f <sub>SCL</sub> range, mid  |      |     | 0.9  |               |
|                                     |                     | f <sub>SCL</sub> range, high |      |     | 0.45 |               |
| Data Valid Acknowledge Time         | $t_{\text{VD:ACK}}$ | f <sub>SCL</sub> range, low  |      |     | 3.45 | $\mu\text{s}$ |
|                                     |                     | f <sub>SCL</sub> range, mid  |      |     | 0.9  |               |
|                                     |                     | f <sub>SCL</sub> range, high |      |     | 0.45 |               |
| Pulse Width of Spikes Suppressed    | $t_{\text{SP}}$     | f <sub>SCL</sub> range, low  |      |     | 50   | ns            |
|                                     |                     | f <sub>SCL</sub> range, mid  |      |     | 50   |               |
|                                     |                     | f <sub>SCL</sub> range, high |      |     | 50   |               |
| Capacitive Load On Each Bus Line    | $C_B$               | (Note 2)                     |      |     | 100  | pF            |

## MAX9296B

## Dual GMSL2/GMSL1 to CSI-2 Deserializer

## AC Electrical Characteristics (continued)

( $V_{\text{TERM}} = 1.14\text{V}$  to  $1.26\text{V}$ ,  $V_{\text{DD18}} = 1.7\text{V}$  to  $1.9\text{V}$ ,  $V_{\text{DD}} = 0.95\text{V}$  to  $1.05\text{V}$  or  $V_{\text{DD}} = 1.14\text{V}$  to  $1.26\text{V}$ ,  $V_{\text{DDIO}} = 1.7\text{V}$  to  $3.6\text{V}$ ,  $T_A = -40^\circ\text{C}$  to  $+105^\circ\text{C}$ , EP connected to PCB ground, typical values are at  $V_{\text{TERM}} = 1.2\text{V}$ ,  $V_{\text{DD18}} = V_{\text{DDIO}} = 1.8\text{V}$ ,  $V_{\text{DD}} = 1.0\text{V}$ ,  $T_A = 25^\circ\text{C}$ , unless otherwise noted.) (Note 1)

| PARAMETER                                                                              | SYMBOL                           | CONDITIONS                                                                                            | MIN                    | TYP                | MAX       | UNITS  |
|----------------------------------------------------------------------------------------|----------------------------------|-------------------------------------------------------------------------------------------------------|------------------------|--------------------|-----------|--------|
| <b>GMSL2 SPI MASTER (See Figure 17) (Note 6)</b>                                       |                                  |                                                                                                       |                        |                    |           |        |
| Operating Frequency                                                                    | $f_{\text{MCK}}$                 | (Note 2)                                                                                              | 0.588                  |                    | 50        | MHz    |
| SCLK Period                                                                            | $t_{\text{MCK}}$                 |                                                                                                       |                        | $1/f_{\text{MCK}}$ |           | ns     |
| SCLK Output Pulse-Width High/Low                                                       | $t_{\text{MCH}}, t_{\text{MCL}}$ | (Note 2)                                                                                              | $t_{\text{MCK}}/2 - 3$ | $t_{\text{MCK}}/2$ |           | ns     |
| MOSI Data Output Delay                                                                 | $t_{\text{MOD}}$                 | After SCLK falling edge (Note 2)                                                                      | -6                     |                    | 3         | ns     |
| MISO Input Setup Time                                                                  | $t_{\text{MIS}}$                 | Before programmed sampling edge                                                                       | 13.5                   |                    |           | ns     |
| MISO Input Hold Time                                                                   | $t_{\text{MIH}}$                 | After programmed sampling edge                                                                        | -2                     |                    |           | ns     |
| <b>GMSL2 SPI SLAVE (See Figure 18) (Note 6)</b>                                        |                                  |                                                                                                       |                        |                    |           |        |
| Operating Frequency                                                                    | $f_{\text{SCK}}$                 | (Note 2)                                                                                              |                        |                    | 50        | MHz    |
| SCLK Period                                                                            | $t_{\text{SCK}}$                 |                                                                                                       |                        | $1/f_{\text{SCK}}$ |           | ns     |
| MISO Data Output Delay                                                                 | $t_{\text{SOD}}$                 | After SCLK falling edge (Note 2)                                                                      | 2                      |                    | 11.3      | ns     |
| MOSI Input Setup Time                                                                  | $t_{\text{SIS}}$                 | Before SCLK rising edge                                                                               | 5                      |                    |           | ns     |
| MOSI Input Hold Time                                                                   | $t_{\text{SIH}}$                 | After SCLK rising edge                                                                                | 3                      |                    |           | ns     |
| <b>REFERENCE CLOCK INPUT REQUIREMENTS (CRYSTAL) (X1, X2) (Note 2)</b>                  |                                  |                                                                                                       |                        |                    |           |        |
| Frequency                                                                              | $f_{\text{XTAL}}$                |                                                                                                       |                        | 25                 |           | MHz    |
| Frequency Stability + Frequency Tolerance                                              | $f_{\text{TN}}$                  |                                                                                                       |                        |                    | $\pm 200$ | ppm    |
| <b>REFERENCE CLOCK REQUIREMENTS (EXTERNAL CLOCK INPUT ON X1, X2 FLOATING) (Note 2)</b> |                                  |                                                                                                       |                        |                    |           |        |
| Frequency                                                                              | $f_{\text{REF}}$                 |                                                                                                       |                        | 25                 |           | MHz    |
| Frequency Stability + Frequency Tolerance                                              | $f_{\text{TN}}$                  |                                                                                                       |                        |                    | $\pm 200$ | ppm    |
| Input Jitter                                                                           |                                  | 6Gbps/187.5Mbps, sinusoidal jitter < 1MHz (falling edge), upstream serializer using crystal reference |                        |                    | 600       | ps p-p |
| Input Duty Cycle                                                                       | $T_{\text{DUTY}}$                |                                                                                                       | 40                     |                    | 60        | %      |
| Input Fall Time                                                                        | $t_{\text{F}}$                   | 80% to 20%                                                                                            |                        |                    | 4         | ns     |

## MAX9296B

## Dual GMSL2/GMSL1 to CSI-2 Deserializer

## AC Electrical Characteristics (continued)

( $V_{\text{TERM}} = 1.14\text{V}$  to  $1.26\text{V}$ ,  $V_{\text{DD18}} = 1.7\text{V}$  to  $1.9\text{V}$ ,  $V_{\text{DD}} = 0.95\text{V}$  to  $1.05\text{V}$  or  $V_{\text{DD}} = 1.14\text{V}$  to  $1.26\text{V}$ ,  $V_{\text{DDIO}} = 1.7\text{V}$  to  $3.6\text{V}$ ,  $T_A = -40^\circ\text{C}$  to  $+105^\circ\text{C}$ , EP connected to PCB ground, typical values are at  $V_{\text{TERM}} = 1.2\text{V}$ ,  $V_{\text{DD18}} = V_{\text{DDIO}} = 1.8\text{V}$ ,  $V_{\text{DD}} = 1.0\text{V}$ ,  $T_A = 25^\circ\text{C}$ , unless otherwise noted.) (Note 1)

| PARAMETER             | SYMBOL           | CONDITIONS                                                                                 | MIN | TYP     | MAX | UNITS |
|-----------------------|------------------|--------------------------------------------------------------------------------------------|-----|---------|-----|-------|
| <b>ESD PROTECTION</b> |                  |                                                                                            |     |         |     |       |
| SIO__                 | $V_{\text{ESD}}$ | Human Body Model, $R_D = 1.5\text{k}\Omega$ , $C_S = 100\text{pF}$                         |     | $\pm 8$ |     | kV    |
|                       |                  | ISO10605, $R_D = 330\Omega$ , $C_S = 150\text{pF}$ , Contact Discharge, Coax Configuration |     | $\pm 6$ |     |       |
|                       |                  | ISO10605, $R_D = 330\Omega$ , $C_S = 150\text{pF}$ , Contact Discharge, STP Configuration  |     | $\pm 4$ |     |       |
|                       |                  | ISO10605, $R_D = 330\Omega$ , $C_S = 150\text{pF}$ , Air Discharge                         |     | $\pm 6$ |     |       |
|                       |                  | AEC-Q100-011 Rev-C1, Charged Device Model (CDM)                                            |     | 750     |     | V     |
| All Other Pins        | $V_{\text{ESD}}$ | Human Body Model, $R_D = 1.5\text{k}\Omega$ , $C_S = 100\text{pF}$                         |     | $\pm 4$ |     | kV    |
|                       |                  | AEC-Q100-011 Rev-C1, Charged Device Model (CDM)                                            |     | 750     |     | V     |

**Note 1:** Limits are 100% tested at  $T_A = +105^\circ\text{C}$  unless otherwise noted. Limits within the operating temperature range and relevant supply voltage range are guaranteed by design and characterization.

**Note 2:** Not production tested. Guaranteed by design and characterization.

**Note 3:**  $C_{\text{LOAD}}$  includes the low-frequency equivalent transmission line capacitance. The capacitance of Tx and Rx are assumed to always be  $< 10\text{pF}$ . The distributed line capacitance can be up to  $50\text{pF}$  for a transmission line with  $2\text{ns}$  delay.

**Note 4:** Color bar pattern. Maximum supply currents are measured at indicated supply voltages. Typical supply currents are measured at typical supply voltages.

**Note 5:** Additional  $60\text{pF}$  at Rx termination center tap.

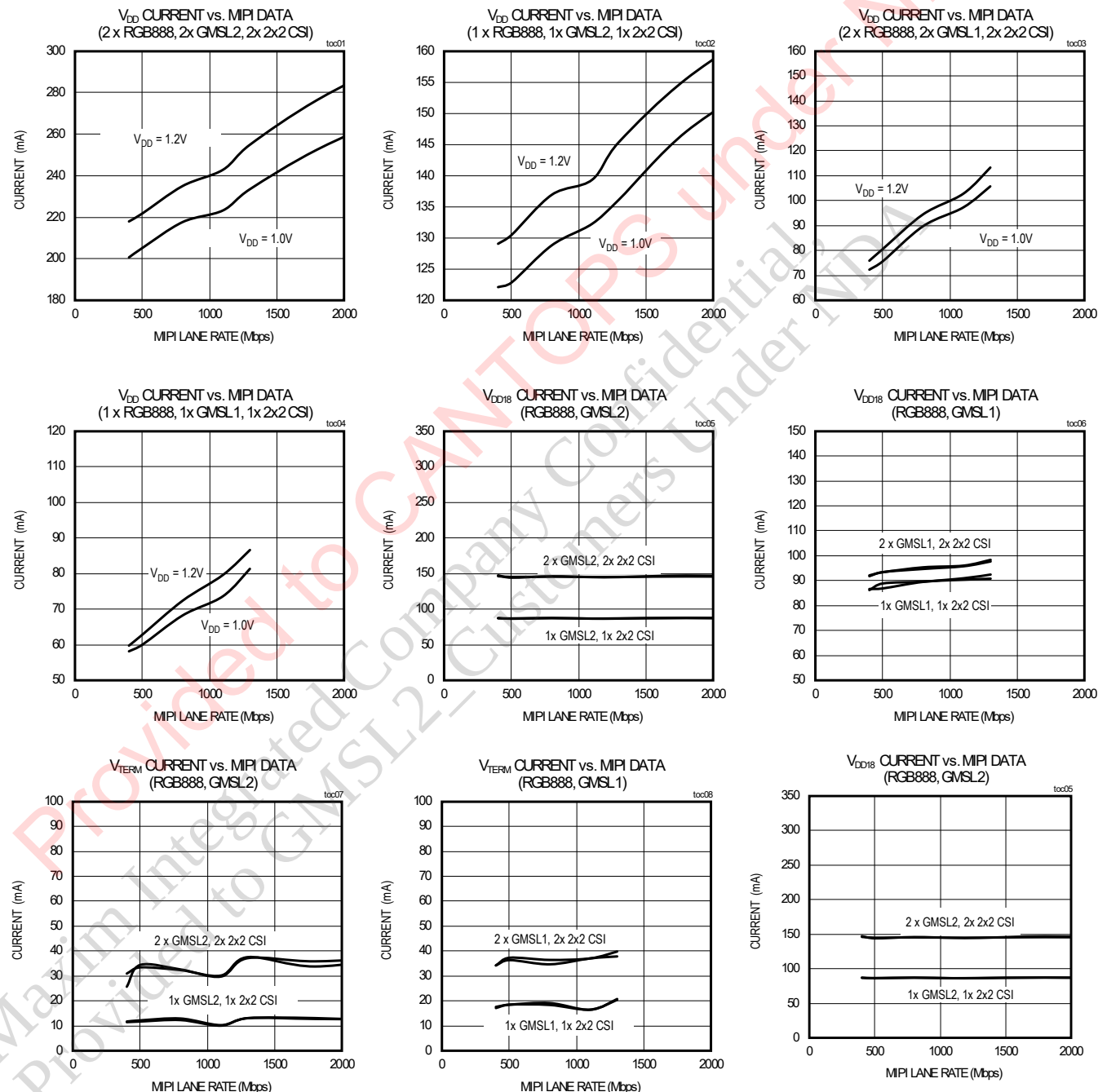
**Note 6:** Measured at  $50\text{MHz}$ . See [Table 5](#) for pin-programming recommendations.

# MAX9296B

## Dual GMSL2/GMSL1 to CSI-2 Deserializer

### Typical Operating Characteristics

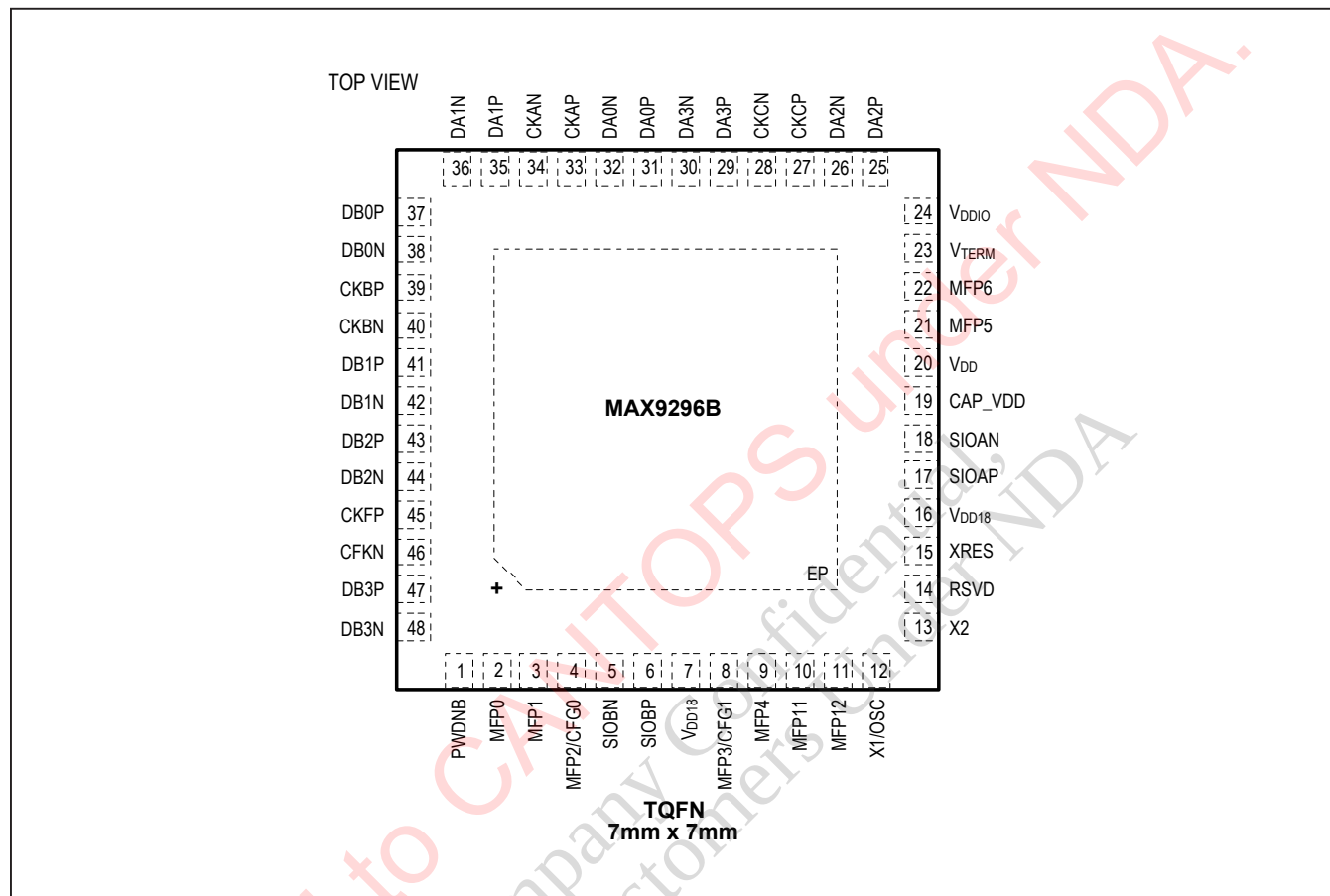
( $V_{TERM} = 1.2V$ ,  $V_{DD18} = V_{DDIO} = 1.8V$ ,  $V_{DD} = 1.0V$ ,  $T_A = +25^\circ C$ , unless otherwise noted. 6Gbps forward rate, 187.5Mbps reverse rate, PRBS24 data, single GMSL2 PHY configuration: SIOA or SIOB GMSL2 input enabled, data output on 4-lane Port A and replicated on 4-lane Port B, dual GMSL2 PHY configuration: SIOA and SIOB GMSL2 inputs enabled, SIOA data output on 4-lane Port A, SIOB data output on 4-lane Port B.)



## MAX9296B

## Dual GMSL2/GMSL1 to CSI-2 Deserializer

## Pin Configuration



## Pin Description

| PIN         | NAME  | FUNCTION MODES |            | DESCRIPTION                                                                                            |
|-------------|-------|----------------|------------|--------------------------------------------------------------------------------------------------------|
|             |       | GMSL2          | GMSL1      |                                                                                                        |
| SERIAL I/O  |       |                |            |                                                                                                        |
| 5           | SIOBN | SIOBN          | SIOBN      | Inverted Twisted-Pair Serial-Data Input/Output B                                                       |
| 6           | SIOBP | SIOBP          | SIOBP      | Noninverted Coax/Twisted-Pair Serial-Data Input/Output B                                               |
| 17          | SIOAP | SIOAP          | SIOAP      | Noninverted Coax/Twisted-Pair Serial-Data Input/Output A                                               |
| 18          | SIOAN | SIOAN          | SIOAN      | Inverted Twisted-Pair Serial-Data Input/Output A                                                       |
| CSI-2 PORTS |       |                |            |                                                                                                        |
| 25          | DA2P  | DA2P*/DC0P     | DA2P*/DC0P | DA2P: CSI-2 Port A Data Lane 2 Noninverted Output<br>DC0P: CSI-2 Port C Data Lane 0 Noninverted Output |
| 26          | DA2N  | DA2N*/DC0N     | DA2N*/DC0N | DA2N: CSI-2 Port A Data Lane 2 Inverted Output<br>DC0N: CSI-2 Port C Data Lane 0 Inverted Output       |

## MAX9296B

## Dual GMSL2/GMSL1 to CSI-2 Deserializer

## Pin Description (continued)

| PIN | NAME | FUNCTION MODES |            | DESCRIPTION                                                                                            |
|-----|------|----------------|------------|--------------------------------------------------------------------------------------------------------|
|     |      | GMSL2          | GMSL1      |                                                                                                        |
| 29  | DA3P | DA3P*/DC1P     | DA3P*/DC1P | DA3P: CSI-2 Port A Data Lane 3 Noninverted Output<br>DC1P: CSI-2 Port C Data Lane 1 Noninverted Output |
| 30  | DA3N | DA3N*/DC1N     | DA3N*/DC1N | DA3N: CSI-2 Port A Data Lane 3 Inverted Output<br>DC1N: CSI-2 Port C Data Lane 1 Inverted Output       |
| 31  | DA0P | DA0P*/DD0P     | DA0P*/DD0P | DA0P: CSI-2 Port A Data Lane 0 Noninverted Output<br>DD0P: CSI-2 Port D Data Lane 0 Noninverted Output |
| 32  | DA0N | DA0N*/DD0N     | DA0N*/DD0N | DA0N: CSI-2 Port A Data Lane 0 Inverted Data Output<br>DD0N: CSI-2 Port D Data Lane 0 Inverted Output  |
| 33  | CKAP | CKAP*/CKDP     | CKAP*/CKDP | CKAP: CSI-2 Port A Clock Lane Noninverted Output<br>CKDP: CSI-2 Port D Clock Lane Noninverted Output   |
| 34  | CKAN | CKAN*/CKDN     | CKAN*/CKDN | CKAN: CSI-2 Port A Clock Lane Inverted Output<br>CKDN: CSI-2 Port D Clock Lane Inverted Output         |
| 35  | DA1P | DA1P*/DD1P     | DA1P*/DD1P | DA1P: CSI-2 Port A Data Lane 1 Noninverted Output<br>DD1P: CSI-2 Port D Data Lane 1 Noninverted Output |
| 36  | DA1N | DA1N*/DD1N     | DA1N*/DD1N | DA1N: CSI-2 Port A Data Lane 1 Inverted Output<br>DD1N: CSI-2 Port D Data Lane 1 Inverted Output       |
| 37  | DB0P | DB0P*/DE0P     | DB0P*/DE0P | DB0P: CSI-2 Port B Data Lane 0 Noninverted Output<br>DE0P: CSI-2 Port E Data Lane 0 Noninverted Output |
| 38  | DB0N | DB0N*/DE0N     | DB0N*/DE0N | DB0N: CSI-2 Port B Data Lane 0 Inverted Output<br>DE0N: CSI-2 Port D Data Lane Inverted Output         |
| 39  | CKBP | CKBP*/CKEP     | CKBP*/CKEP | CKBP: CSI-2 Port B Clock Lane Noninverted Output<br>CKEP: CSI-2 Port E Clock Lane Noninverted Output   |
| 40  | CKBN | CKBN*/CKEN     | CKBN*/CKEN | CKBN: CSI-2 Port B Clock Lane Inverted Output<br>CKEN: CSI-2 Port E Clock Lane Inverted Output         |
| 41  | DB1P | DB1P*/DE1P     | DB1P*/DE1P | DB1P: CSI-2 Port B Data Lane 1 Noninverted Output<br>DE1P: CSI-2 Port E Data Lane 1 Noninverted Output |
| 42  | DB1N | DB1N*/DE1N     | DB1N*/DE1N | DB1N: CSI-2 Port B Data Lane 1 Inverted Output<br>DE1N: CSI-2 Port E Data Lane 1 Inverted Output       |
| 43  | DB2P | DB2P*/DF0P     | DB2P*/DF0P | DB2P: CSI-2 Port B Data Lane 2 Noninverted Output<br>DF0P: CSI-2 Port F Data Lane 0 Noninverted Output |
| 44  | DB2N | DB2N*/DF0N     | DB2N*/DF0N | DB2N: CSI-2 Port B Data Lane 2 Inverted Output<br>DF0N: CSI-2 Port F Data Lane 0 Inverted Output       |
| 47  | DB3P | DB3P*/DF1P     | DB3P*/DF1P | DB3P: CSI-2 Port B Data Lane 3 Noninverted Output<br>DF1P: CSI-2 Port F Data Lane 1 Noninverted Output |
| 48  | DB3N | DB3N*/DF1N     | DB3N*/DF1N | DB3N: CSI-2 Port B Data Lane 3 Inverted Output<br>DF1N: CSI-2 Port F Data Lane 1 Inverted Output       |
| 27  | CKCP | CKCP           | CKCP       | CSI-2 Port C Clock Lane Noninverted Output                                                             |
| 28  | CKCN | CKCN           | CKCN       | CSI-2 Port C Clock Lane Inverted Output                                                                |
| 45  | CKFP | CKFP           | CKFP       | CSI-2 Port F Clock Lane Noninverted Output                                                             |
| 46  | CKFN | CKFN           | CKFN       | CSI-2 Port F Clock Lane Inverted Output                                                                |

## MAX9296B

## Dual GMSL2/GMSL1 to CSI-2 Deserializer

## Pin Description (continued)

| PIN                                                                                                               | NAME          | FUNCTION MODES                                                                     |                                         | DESCRIPTION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-------------------------------------------------------------------------------------------------------------------|---------------|------------------------------------------------------------------------------------|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                   |               | GMSL2                                                                              | GMSL1                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| MULTIFUNCTION PINS (*DEFAULT STATE AFTER POWER-UP)<br>(**INPUT READ OR OUTPUT WRITE BY REGISTER PROGRAMMING ONLY) |               |                                                                                    |                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 2                                                                                                                 | MFP0          | SDA1<br>RX1<br>SDA2<br>RX2<br>FRSYNC_OUT<br>VS2<br>DE2/DV2<br>SCLK<br>MS<br>GPIO0* | FRSYNC_OUT<br>CNTL1<br>MS<br>GPIO0*(**) | SDA1: Pass-Through I <sup>2</sup> C Data Input/Open-Drain Output with an Internal 40kΩ Pullup to V <sub>DDIO</sub> .<br>RX1: Pass-Through UART Input with an Internal 40kΩ Pullup to V <sub>DDIO</sub> .<br>SDA2: Pass-Through I <sup>2</sup> C Data Input/Open-Drain Output with an Internal 40kΩ Pullup to V <sub>DDIO</sub> .<br>RX2: Pass-Through UART Input with an Internal 40kΩ Pullup to V <sub>DDIO</sub> .<br>FRSYNC_OUT: Frame Sync Push-Pull Output.<br>VS2: Video Stream Vertical Sync Monitor Push-Pull Output.<br>DE2/DV2: Video Stream Data Enable or Data Valid Monitor Push-Pull Output.<br>SCLK: SPI Clock. When configured as master, push-pull clock output. When configured as a slave, clock input with internal 1MΩ pulldown to ground.<br>MS: UART Mode Select with Internal 1MΩ Pulldown to Ground. Set MS low to select base mode. Set MS high to select bypass mode. MS state can also be temporarily overwritten by a register write.<br>GPIO0: Configurable General-Purpose Input or Output. Power-up default is GPIO general-purpose input with a 1MΩ pulldown to ground.<br>CNTL1: Control 1 Push-Pull Output. |
| 3                                                                                                                 | MFP1          | SCL1<br>TX1<br>SCL2<br>TX2<br>LOCK*<br>GPIO1                                       | FRSYNC_IN<br>LOCK*<br>GPIO1**           | SCL1: Pass-Through I <sup>2</sup> C Clock Input/Open-Drain Output with an Internal 40kΩ Pullup to V <sub>DDIO</sub> .<br>TX1: Pass-Through UART Open-Drain Output with an Internal 40kΩ Pullup to V <sub>DDIO</sub> .<br>SCL2: Pass-Through I <sup>2</sup> C Clock Input/Open-Drain Output with an Internal 40kΩ Pullup to V <sub>DDIO</sub> .<br>TX2: Pass-Through UART Open-Drain Output with an Internal 40kΩ Pullup to V <sub>DDIO</sub> .<br>LOCK: Open-Drain Lock Indication Output with an Internal 40kΩ Pullup to V <sub>DDIO</sub> .<br>GPIO1: Configurable General-Purpose Input or Output.<br>FRSYNC_IN: Frame Sync Input with an Internal 1MΩ Pulldown to Ground.                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 4                                                                                                                 | MFP2/<br>CFG0 | CFG0<br>BNE<br>SS1<br>GPO2*                                                        | CFG0<br>CNTL0<br>GPO2*(**)              | CFG0: Configuration Pin. Voltage at Pin Sets Device Mode, Which is Latched at Power-Up. Connect to a resistor-divider between V <sub>DDIO</sub> and ground. See Table 10.<br>BNE: When Configured as Slave, SPI Buffer Not Empty Push-Pull Output. When BNE is high, SPI data is available.<br>SS1: SPI Slave Select. When configured as the master, Slave 1 Select Push-Pull Output.<br>GPO2: General-Purpose Push-Pull Output. Power-up default is high impedance.<br>CNTL0: Control 0 Push-Pull Output.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |



MAX9296B

Dual GMSL2/GMSL1 to CSI-2 Deserializer

## Pin Description (continued)

| PIN | NAME          | FUNCTION MODES                               |                              | DESCRIPTION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-----|---------------|----------------------------------------------|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     |               | GMSL2                                        | GMSL1                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 8   | MFP3/<br>CFG1 | CFG1<br>VS1<br>DE1/DV1<br>SS2<br>GPO3*       | CFG1<br>CNTL3<br>GPO3*(**)   | CFG1: Configuration Pin. Voltage at the pin sets device mode, which is latched at power-up. Connect to a resistor-divider between $V_{DDIO}$ and ground. See <a href="#">Table 11</a> .<br>VS1: Video Stream Vertical Sync Monitor Push-Pull Output.<br>DE1/DV1: Video Stream Data Enable or Data Valid Monitor Push-Pull Output.<br>SS2: SPI Slave Select. When configured as the master, Slave 2 Select push-pull output.<br>GPO3: General-Purpose Push-Pull Output. Power-up default is high impedance.<br>CNTL3: Control 3 Push-Pull Output.                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 9   | MFP4          | RO(Alt)<br>ERRB*<br>GPIO4                    | ERRB*<br>GPIO4**             | RO(Alt): When Configured as a Slave, SPI Mode-Select Input with an Internal $1M\Omega$ Pulldown to Ground. When RO is high, enables a master read from MISO. When RO is low, enables a master write to MOSI.<br>ERRB: Open-Drain Error Indication Output with an Internal $40k\Omega$ Pullup to $V_{DDIO}$ . When ERRB is low, a data error, or interrupt is detected. ERRB is high when PWDNB is low.<br>GPIO4: Configurable General-Purpose Input or Output.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 21  | MFP5          | SDA2<br>RX2<br>SDA1<br>RX1*<br>MOSI<br>GPIO5 | CNTL2<br>GPI_1<br>GPIO5*(**) | SDA2: Pass-Through I <sup>2</sup> C Data Input/Open-Drain Output with an Internal $40k\Omega$ Pullup to $V_{DDIO}$ .<br>RX2: Pass-Through UART Input with an Internal $40k\Omega$ Pullup to $V_{DDIO}$ .<br>SDA1: Pass-Through I <sup>2</sup> C Data Input/Open-Drain Output with an Internal $40k\Omega$ Pullup to $V_{DDIO}$ .<br>RX1: Pass-Through UART Input with an Internal $40k\Omega$ Pullup to $V_{DDIO}$ (Default 1Mbps Rate).<br>MOSI: SPI Master Out Slave In. When configured as the master, the push-pull output drives data to the external slave. When configured as a slave, input with an internal $1M\Omega$ pulldown to ground that receives data from external master.<br>GPIO5: Configurable General-Purpose Input or Output. Power-up default with a receiver enabled and open-drain driver high with an internal $40k\Omega$ pullup to $V_{DDIO}$ .<br>CNTL2: Control 2 Push-Pull Output.<br>GPI_1: General-Purpose Input with an Internal $40k\Omega$ Pullup to $V_{DDIO}$ . |



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Dual GMSL2/GMSL1 to CSI-2 Deserializer

## Pin Description (continued)

| PIN | NAME  | FUNCTION MODES                               |                                           | DESCRIPTION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-----|-------|----------------------------------------------|-------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     |       | GMSL2                                        | GMSL1                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 22  | MFP6  | SCL2<br>TX2<br>SCL1<br>TX1*<br>MISO<br>GPIO6 | FRSYNC_IN<br>CNTL4<br>GPI_0<br>GPIO6**(*) | <p>SCL2: Pass-Through I<sup>2</sup>C Clock Input/Open-Drain Output with an Internal 40kΩ Pullup to V<sub>DDIO</sub>.</p> <p>TX2: Pass-Through UART Output with an Internal 40kΩ Pullup to V<sub>DDIO</sub>.</p> <p>SCL1: Pass-Through I<sup>2</sup>C Clock Input/Open-Drain Output with an Internal 40kΩ Pullup to V<sub>DDIO</sub>.</p> <p>TX1: Pass-Through UART Output with an Internal 40kΩ Pullup to V<sub>DDIO</sub> (Default 1Mbps Rate).</p> <p>MISO: SPI Master In Slave Out. When configured as a master, input with an internal 1MΩ pulldown to ground that receives data from the external slave. When configured as a slave, push-pull output that drives data to an external master.</p> <p>GPIO6: Configurable General-Purpose Input or Output. Power-up default with the receiver enabled and open-drain driver high with internal 40kΩ pullup to V<sub>DDIO</sub>.</p> <p>FRSYNC_IN: Frame Sync Input with an Internal 1MΩ Pulldown to Ground.</p> <p>CNTL4: Control 4 Push-Pull Output.</p> <p>GPI_0: General-Purpose Input with an Internal 40kΩ Pullup to V<sub>DDIO</sub>.</p> |
| 10  | MFP11 | SDA<br>Rx<br>GPI11_ODO11                     | SDA<br>RX<br>GPI11_ODO11**                | <p>SDA: I<sup>2</sup>C Data Input/Open-Drain Output with an Internal 40kΩ Pullup to V<sub>DDIO</sub> (SDA or RX Selected by CFG0 at Power-Up).</p> <p>Rx: UART Input with Internal 40kΩ Pullup to V<sub>DDIO</sub> (SDA or Rx Selected by CFG0 at Power-Up).</p> <p>GPI11_ODO11: General-Purpose Input and/or Open-Drain Output with an Internal 40kΩ Pullup to V<sub>DDIO</sub>.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 11  | MFP12 | SCL<br>Tx<br>GPI12_ODO12                     | SCL<br>TX<br>GPI12_ODO12**                | <p>SCL: I<sup>2</sup>C Clock Input/Open-Drain Output with an Internal 40kΩ Pullup to V<sub>DDIO</sub> (SCL or Tx Selected by CFG0 at Power-Up).</p> <p>Tx: UART Open-Drain Output with an Internal 40kΩ Pullup to V<sub>DDIO</sub> (SCL or Tx Selected by CFG0 at Power-Up).</p> <p>GPI12_ODO12: General-Purpose Input and/or Open-Drain Output with Internal 40kΩ Pullup to V<sub>DDIO</sub>.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

MAX9296B

Dual GMSL2/GMSL1 to CSI-2 Deserializer

## Pin Description (continued)

| PIN                        | NAME    | FUNCTION MODES |         | DESCRIPTION                                                                                                                                                                                                                                                  |
|----------------------------|---------|----------------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                            |         | GMSL2          | GMSL1   |                                                                                                                                                                                                                                                              |
| MISCELLANEOUS—SEE TABLE 2  |         |                |         |                                                                                                                                                                                                                                                              |
| 1                          | PWDNB   | PWDNB          | PWDNB   | Active-Low, Power-Down Input with an Internal 1MΩ Pulldown to Ground. Set PWDNB Low to Enter Power-Down Mode.                                                                                                                                                |
| 12                         | X1/OSC  | X1/OSC         | X1/OSC  | Crystal/Oscillator Input. Connect to either a 25MHz Crystal or 25MHz External Clock Source. When using a crystal, connect a load capacitor from X1/OSC to ground (load capacitor value depends on the crystal used). Required in both GMSL2 and GMSL1 Modes. |
| 13                         | X2      | X2             | X2      | Crystal Input. When using a crystal, connect to one terminal of a 25MHz crystal and connect a load capacitor from X2 to ground (load capacitor value depends on the crystal used).                                                                           |
| 14                         | RSVD    | RSVD           | RSVD    | Reserved. Make no electrical connection to this pin.                                                                                                                                                                                                         |
| 15                         | XRES    | XRES           | XRES    | Used to Calibrate SIO Output Driver Swings. Connect an external 402Ω resistor between XRES and ground.                                                                                                                                                       |
| 19                         | CAP_VDD | CAP_VDD        | CAP_VDD | Decoupling Capacitor for 1V Core Supply.                                                                                                                                                                                                                     |
| POWER SUPPLIES—SEE TABLE 2 |         |                |         |                                                                                                                                                                                                                                                              |
| 7                          | VDD18   | VDD18          | VDD18   | 1.8V Analog Supply.                                                                                                                                                                                                                                          |
| 16                         | VDD18   | VDD18          | VDD18   | 1.8V Analog Supply.                                                                                                                                                                                                                                          |
| 20                         | VDD     | VDD            | VDD     | 1.0V Core Supply. Connect a 0.95V to 1.05V supply or connect a 1.14V to 1.26V supply to use the internal 1.0V regulator. In order to use the internal 1.0V regulator, first write REG_ENABLE = 1, then write REG_MNL = 1.                                    |
| 23                         | VTERM   | VTERM          | VTERM   | 1.2V D-PHY Termination Supply.                                                                                                                                                                                                                               |
| 24                         | VDDIO   | VDDIO          | VDDIO   | 1.8V to 3.3V I/O Supply.                                                                                                                                                                                                                                     |
| —                          | EP      | EP             | EP      | Exposed Pad. EP is internally connected to device ground. EP must be connected to the PCB ground plane through an array of vias for proper thermal and electrical performance.                                                                               |

MAX9296B Dual GMSL2/GMSL1 to CSI-2 Deserializer

Functional Block Diagram

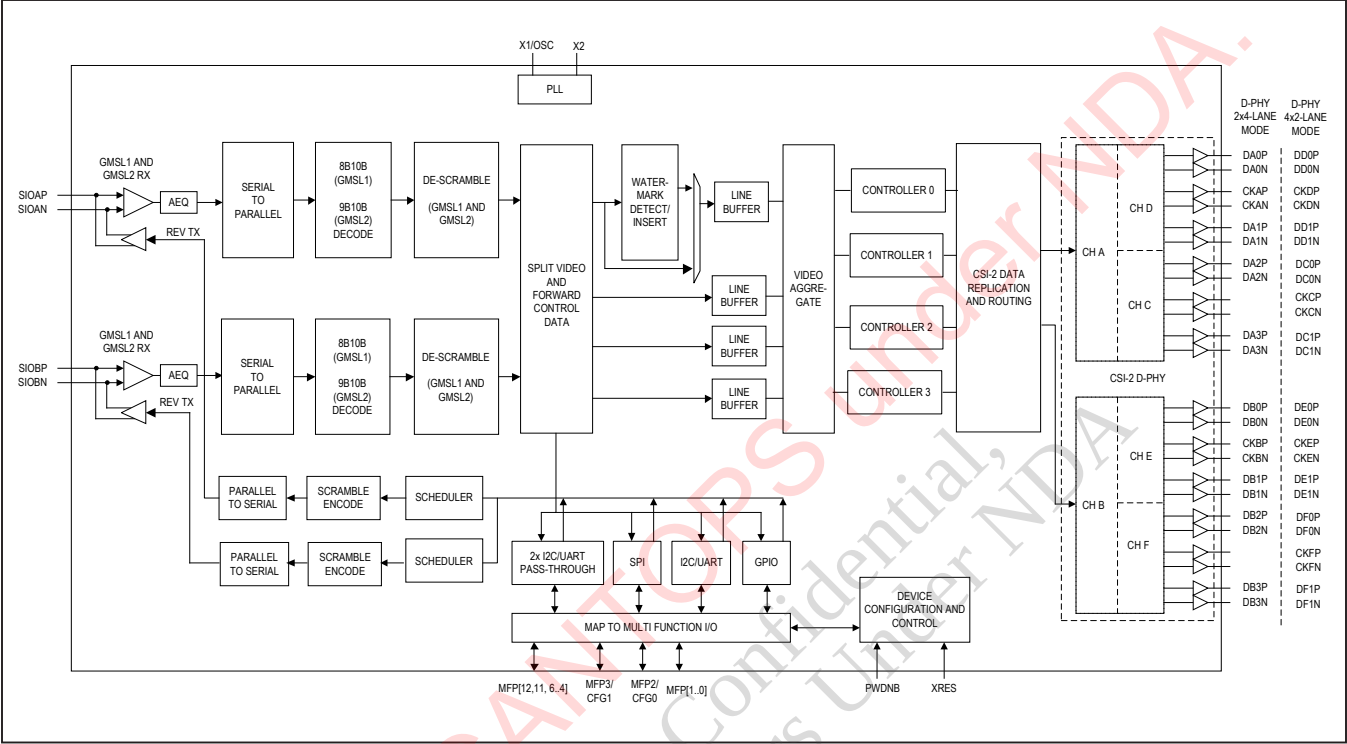


Table 1. Recommended Operating Conditions

| PARAMETER                                      | PIN NAME          | NOMINAL VOLTAGE (V) | MIN  | TYP | MAX  | UNIT              |
|------------------------------------------------|-------------------|---------------------|------|-----|------|-------------------|
| Supply Range                                   | V <sub>TERM</sub> | —                   | 1.14 | 1.2 | 1.26 | V                 |
|                                                | V <sub>DD18</sub> | —                   | 1.7  | 1.8 | 1.9  |                   |
|                                                | V <sub>DD</sub>   | 1.0                 | 0.95 | 1.0 | 1.05 |                   |
|                                                |                   | 1.2                 | 1.14 | 1.2 | 1.26 |                   |
|                                                |                   | 1.8                 | 1.7  | 1.8 | 1.9  |                   |
| Maximum Supply Noise*                          | V <sub>DDIO</sub> | 3.3                 | 3.0  | 3.3 | 3.6  | mV <sub>p-p</sub> |
|                                                | V <sub>TERM</sub> | —                   |      | 50  |      |                   |
|                                                | V <sub>DD18</sub> | —                   |      | 25  |      |                   |
|                                                | V <sub>DD</sub>   | 1.0                 |      | 50  |      |                   |
|                                                |                   | 1.2                 |      | 50  |      |                   |
|                                                | V <sub>DDIO</sub> | 1.8                 |      | 50  |      |                   |
|                                                |                   | 3.3                 |      | 100 |      |                   |
| Operating Junction Temperature, T <sub>J</sub> |                   |                     | -40  |     | +125 | °C                |

\*Supply noise < 1MHz

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## Dual GMSL2/GMSL1 to CSI-2 Deserializer

Table 2. External Component Requirements

| COMPONENT                                        | SYMBOL            | CONDITION                                                                                                                                                                                                                             |                          | VALUE                                                            | UNIT     |
|--------------------------------------------------|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|------------------------------------------------------------------|----------|
| External Reference Resistor                      | R <sub>XRES</sub> |                                                                                                                                                                                                                                       |                          | 402 $\pm$ 1%, use a single resistor                              | $\Omega$ |
| Link Isolation Capacitors                        | C <sub>LINK</sub> | Place close to the SIO <sub>_</sub> pins (5, 6, 17, 18) used in the application.                                                                                                                                                      | GMSL2 Mode               | 0.1                                                              | $\mu$ F  |
|                                                  |                   |                                                                                                                                                                                                                                       | GMSL1 Mode, HIM disabled | 0.22                                                             | $\mu$ F  |
|                                                  |                   |                                                                                                                                                                                                                                       | GMSL1 Mode, HIM enabled  | 0.047 to 0.22                                                    | $\mu$ F  |
| Crystal                                          |                   | Place as close as possible to pins 12 and 13.                                                                                                                                                                                         |                          | 25MHz $\pm$ 200ppm                                               |          |
| Crystal Load Capacitors                          |                   | Use crystal loading capacitor guidance from the crystal manufacturer. Select values that compensate for the X1 and X2 input and PCB node capacitances. Place the capacitors as close as possible to pins 12 (X1/OSC) and pin 13 (X2). |                          |                                                                  |          |
| V <sub>DDIO</sub> Decoupling Capacitor           |                   | Place a 0.01 $\mu$ F capacitor as close as possible to pin 24. Include a minimum of 10 $\mu$ F bulk decoupling on the PCB.                                                                                                            |                          |                                                                  |          |
| V <sub>DD18</sub> Decoupling Capacitors          |                   | Place a 0.01 $\mu$ F capacitors as close as possible to pins 7 and 16. Include a minimum of 10 $\mu$ F bulk decoupling on the PCB.                                                                                                    |                          |                                                                  |          |
| V <sub>TERM</sub> Decoupling Capacitor           |                   | Place a 0.01 $\mu$ F capacitor as close as possible to pin 23. Include a minimum of 10 $\mu$ F bulk decoupling on the PCB.                                                                                                            |                          |                                                                  |          |
| V <sub>DD</sub> Decoupling Capacitor             |                   | Place a 0.1 $\mu$ F capacitor as close as possible to pin 20. Include a minimum of 10 $\mu$ F bulk decoupling on the PCB.                                                                                                             |                          |                                                                  |          |
| CAP_VDD Decoupling Capacitor                     |                   | Place a 0.1 $\mu$ F capacitor as close as possible to pin 19. Include a minimum of 10 $\mu$ F bulk decoupling near the pin.                                                                                                           |                          |                                                                  |          |
| Open-Drain Pullup Resistors                      |                   | Application-specific. Quantity and values depend on multifunction GPIO pin configurations.                                                                                                                                            |                          |                                                                  |          |
| Resistors for Configuration Pin Resistor Divider | R1, R2            | Place resistor divider close to pin 4 (MFP2/CFG0).                                                                                                                                                                                    |                          | Use $\pm$ 1% tolerance resistors. See <a href="#">Table 10</a> . | $\Omega$ |
|                                                  | R1, R2            | Place resistor divider close to pin 8 (MFP3/CFG1).                                                                                                                                                                                    |                          | Use $\pm$ 1% tolerance resistors. See <a href="#">Table 11</a> . | $\Omega$ |

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Dual GMSL2/GMSL1 to CSI-2 Deserializer

Functional Diagrams

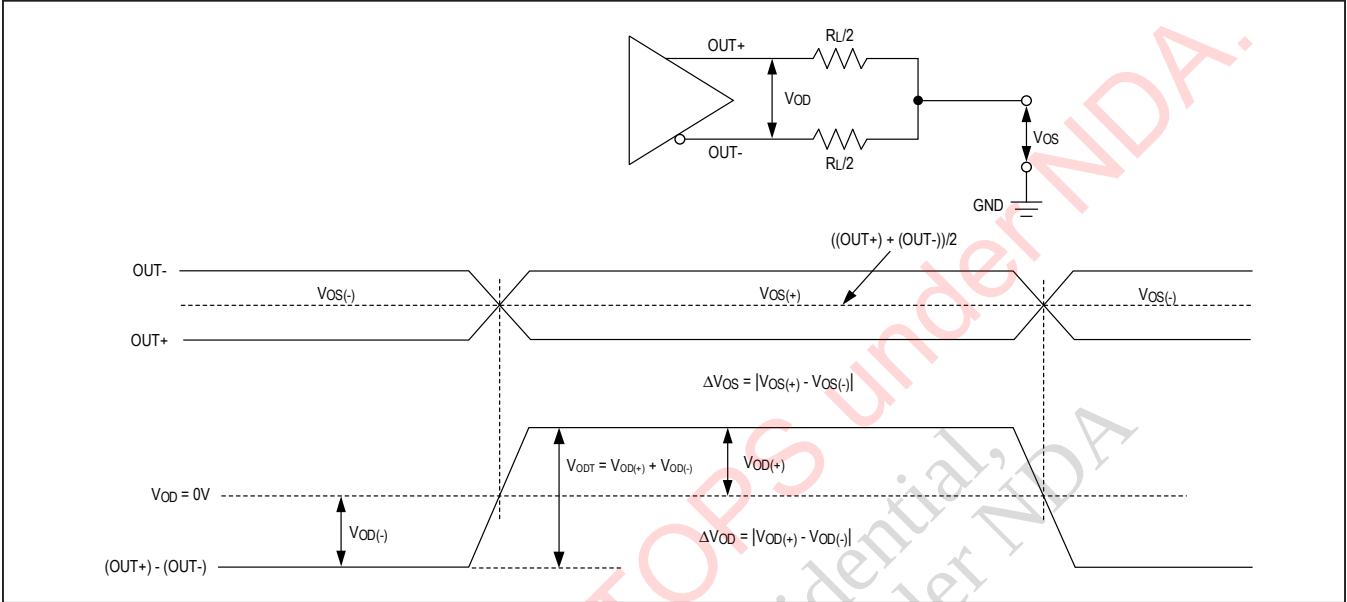


Figure 1. GMSL2 Reverse Channel Serial Outputs

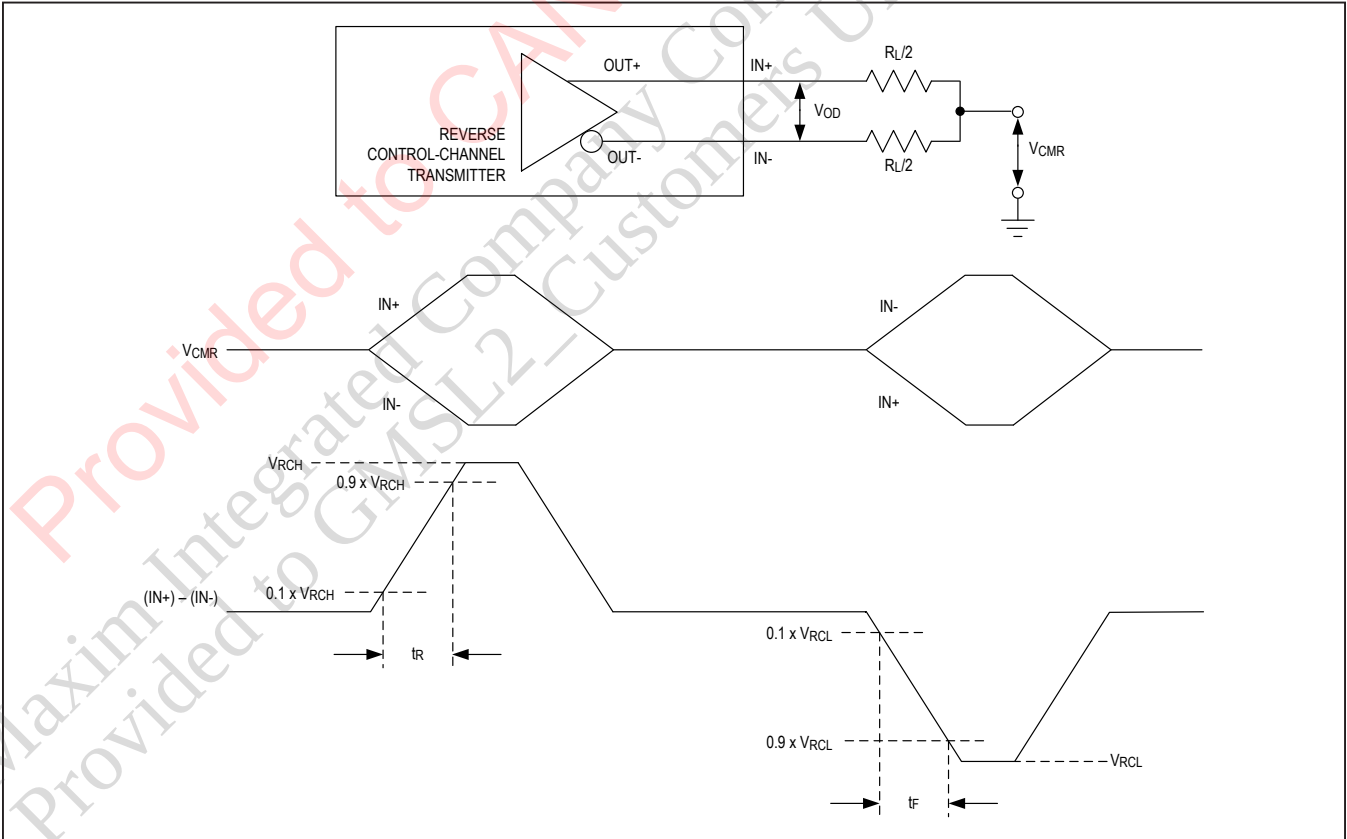


Figure 2. GMSL1 Reverse Channel Serial Outputs

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Dual GMSL2/GMSL1 to CSI-2 Deserializer

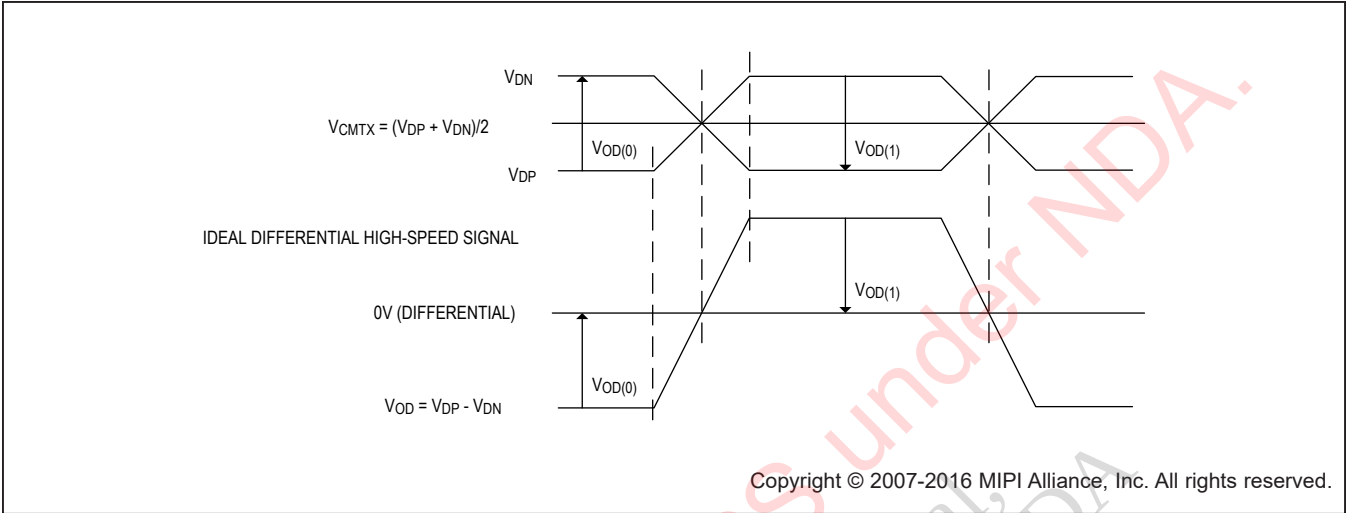


Figure 3. Ideal Single-ended and Resulting Differential HS Signals

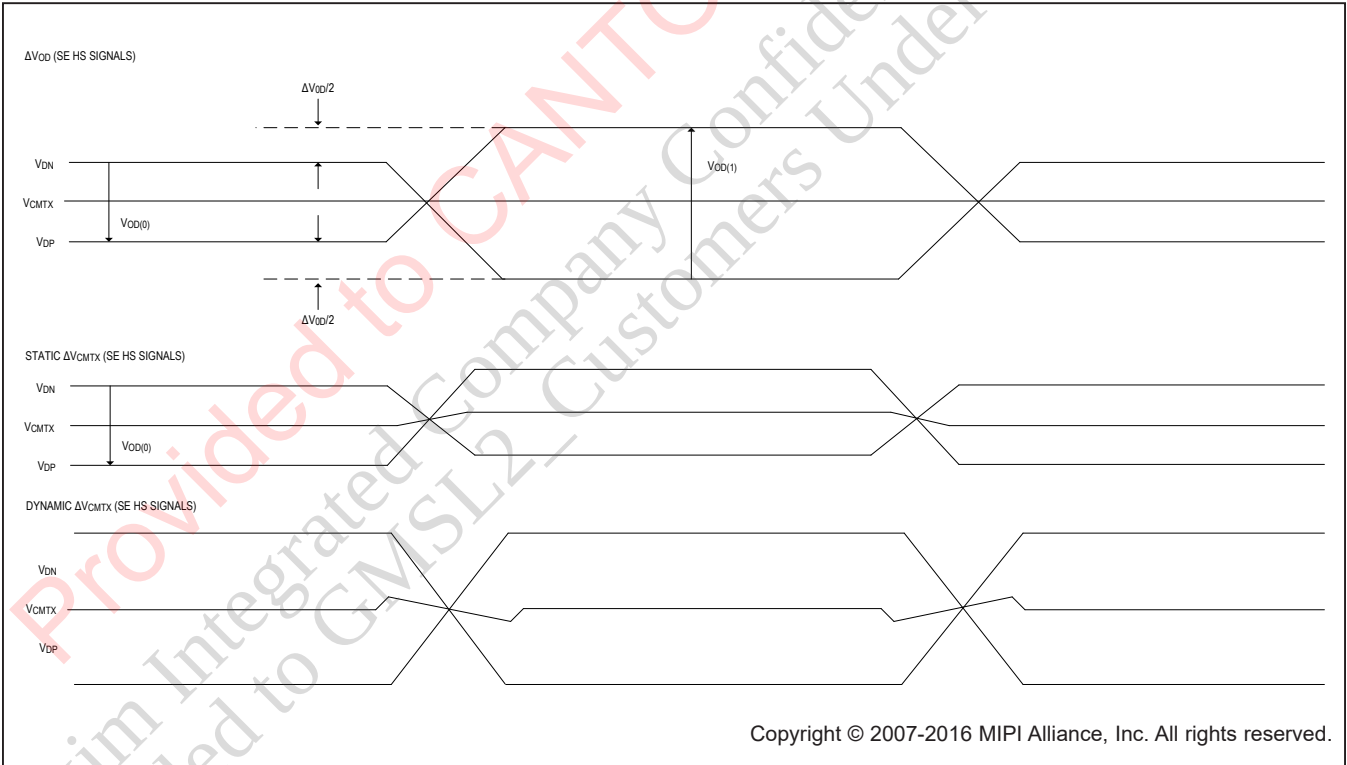


Figure 4. Possible Delta  $V_{CMTX}$  and Delta  $V_{OD}$  Distortions of Single-Ended HS Signals

MAX9296B Dual GMSL2/GMSL1 to CSI-2 Deserializer

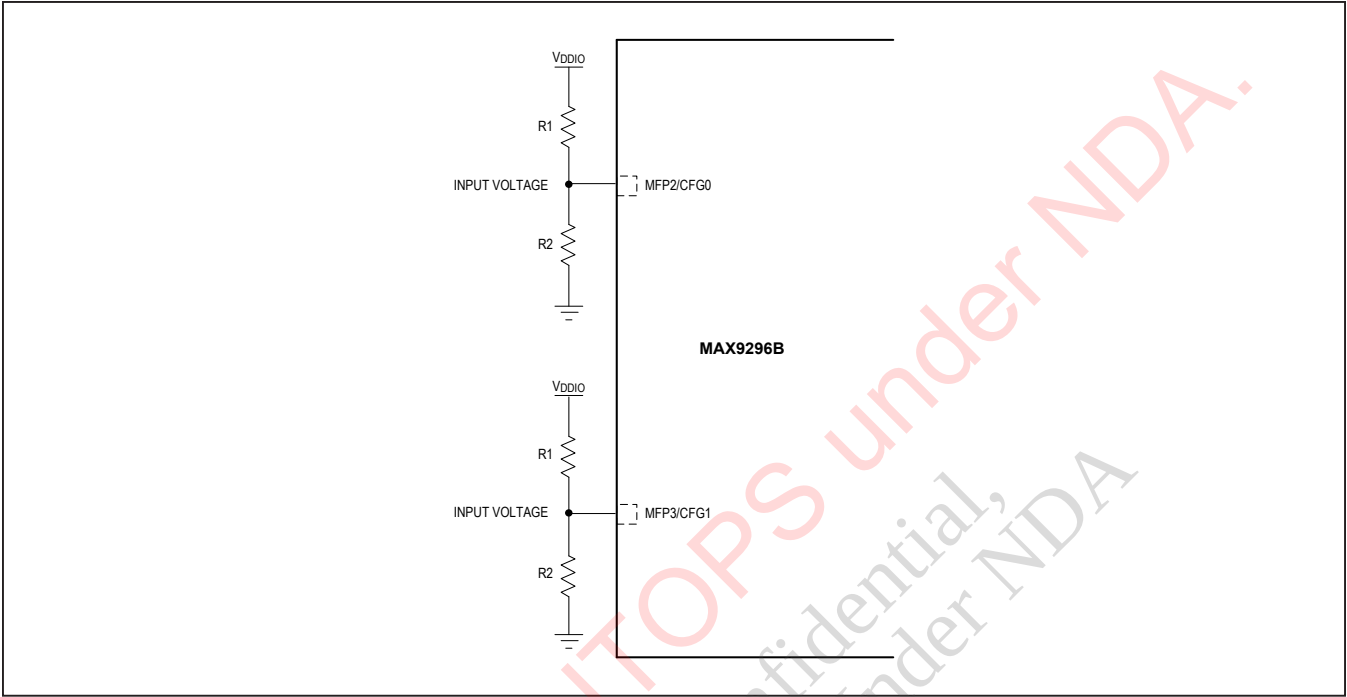


Figure 5. Configuration Pin Connection

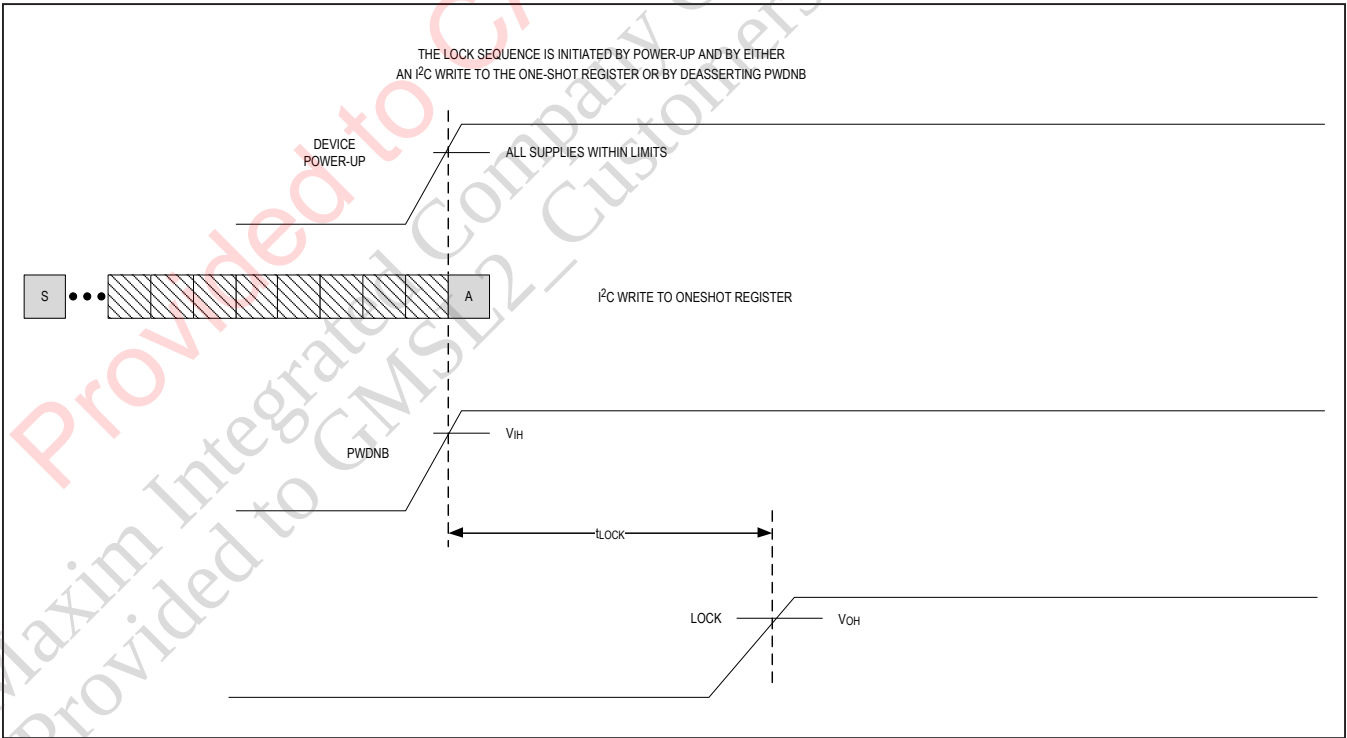


Figure 6. GMSL2 Lock Time



MAX9296B Dual GMSL2/GMSL1 to CSI-2 Deserializer

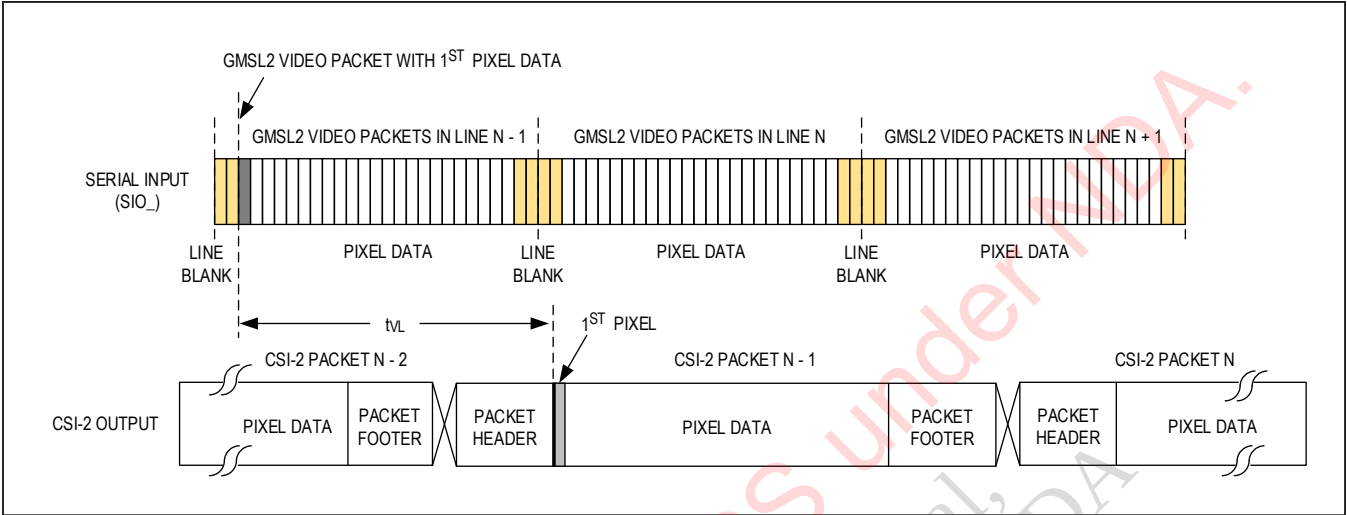


Figure 7. GMSL2 Video Latency

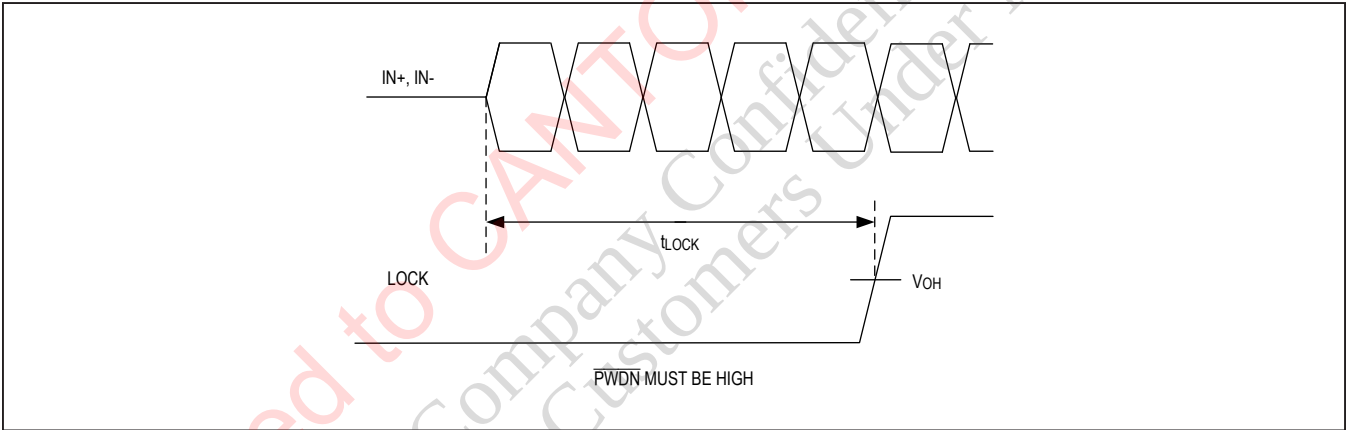


Figure 8. GMSL1 Lock Time

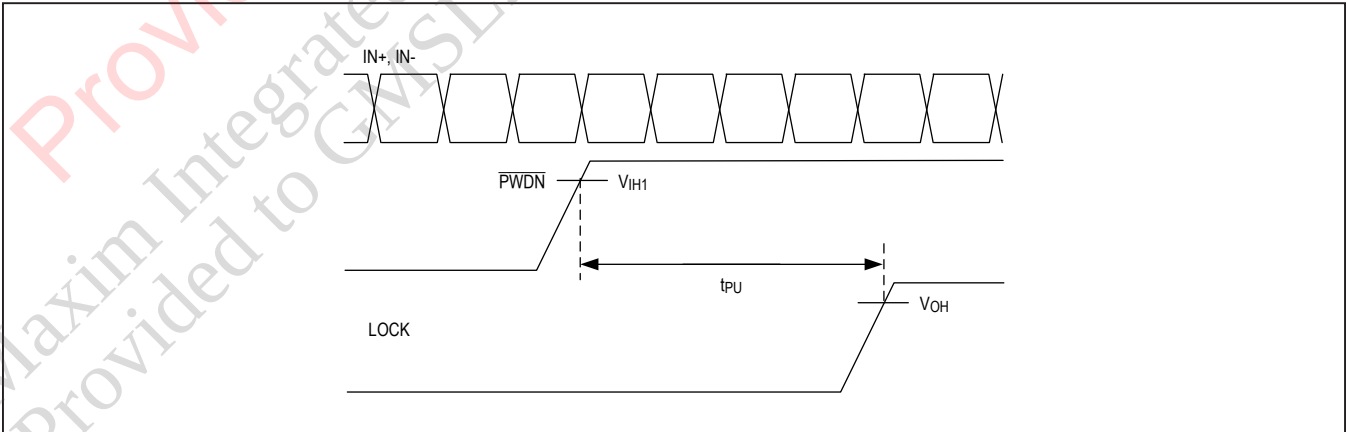


Figure 9. GMSL1 Power-Up Delay

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Dual GMSL2/GMSL1 to CSI-2 Deserializer

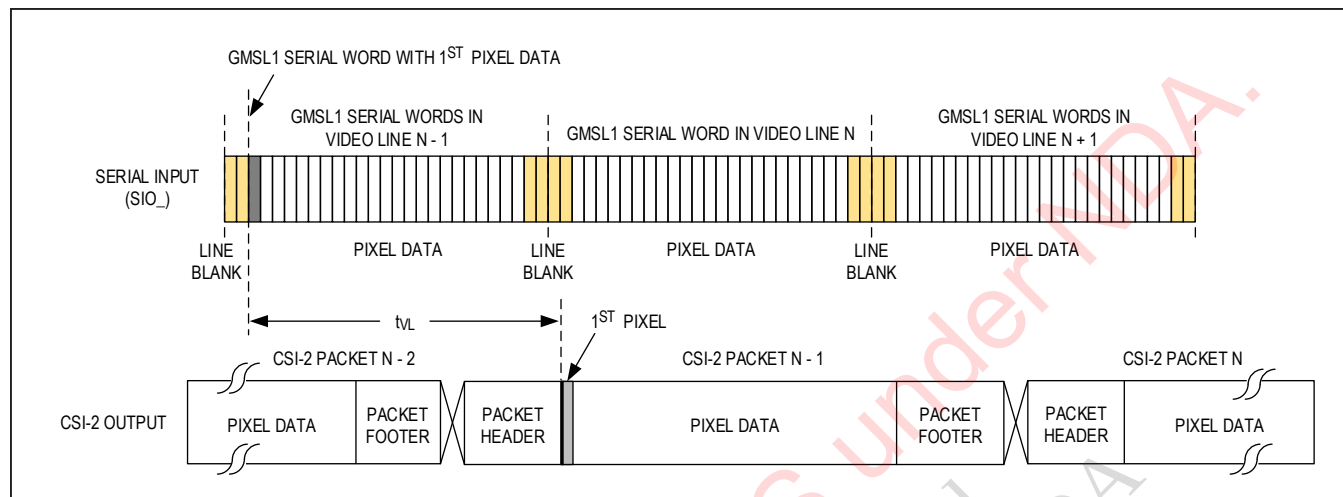


Figure 10. GMSL1 Video Latency

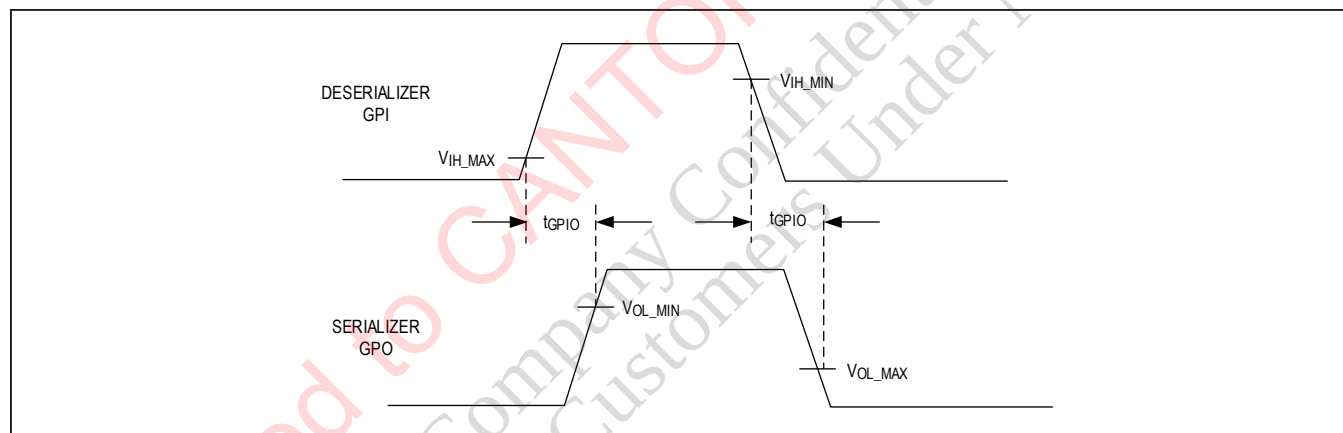
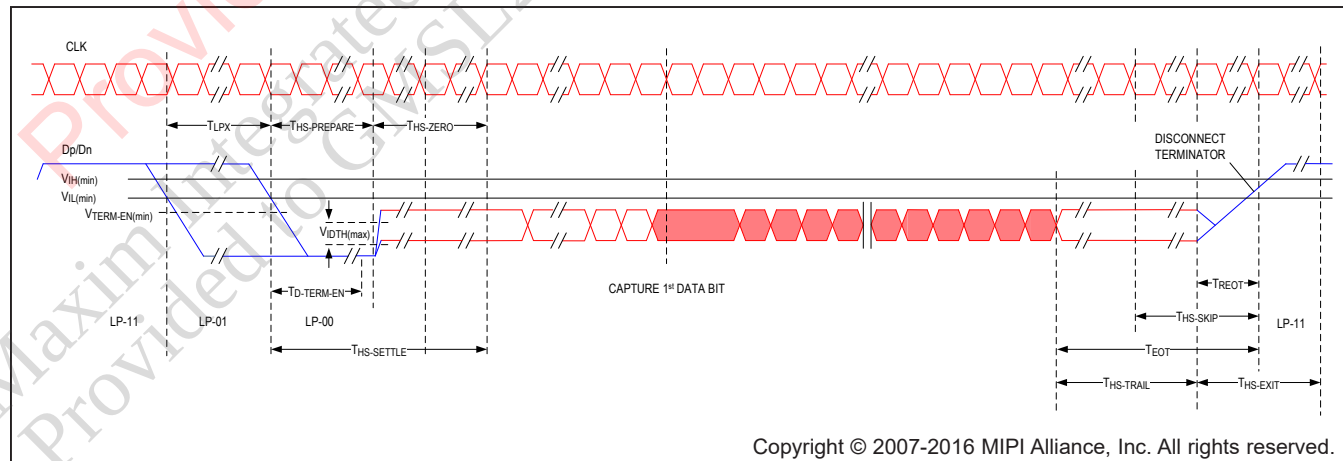
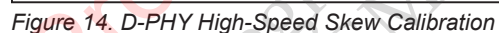
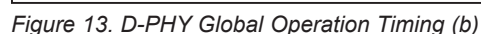


Figure 11. GSML1 GPI-to-GPO Delay



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Figure 12. D-PHY Global Operation Timing (a)



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Dual GMSL2/GMSL1 to CSI-2 Deserializer

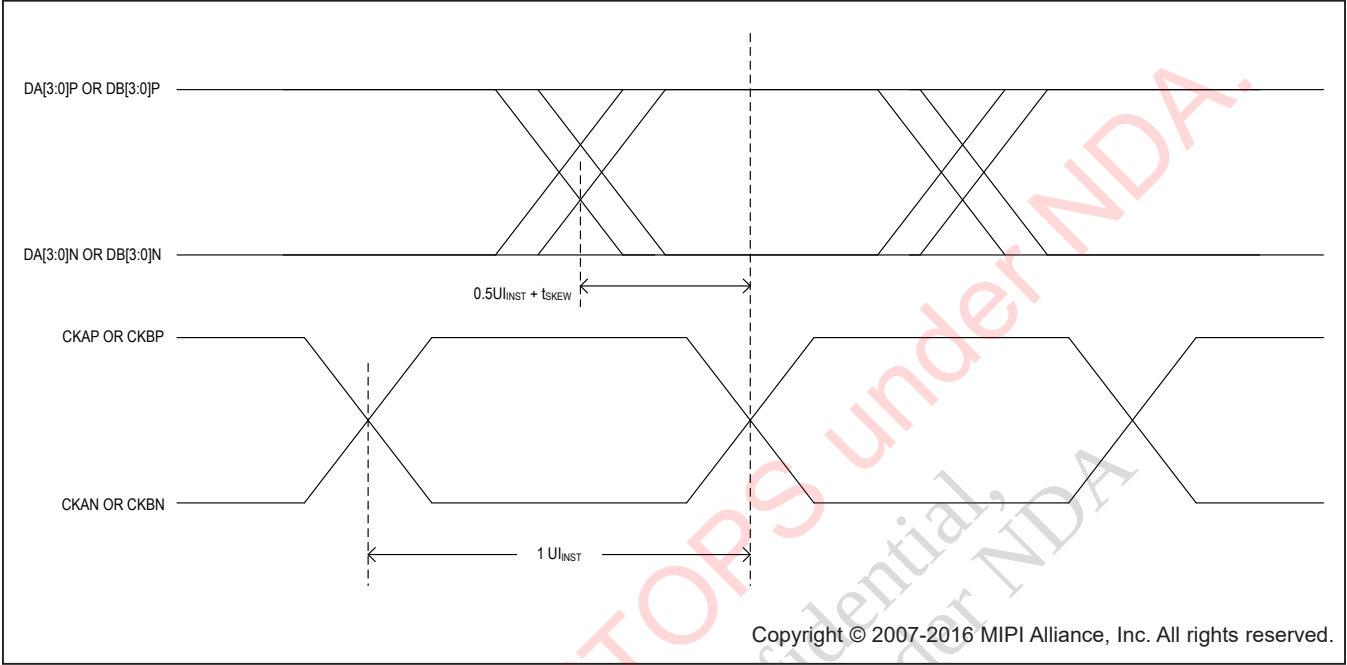


Figure 15. D-PHY Data Clock Timing

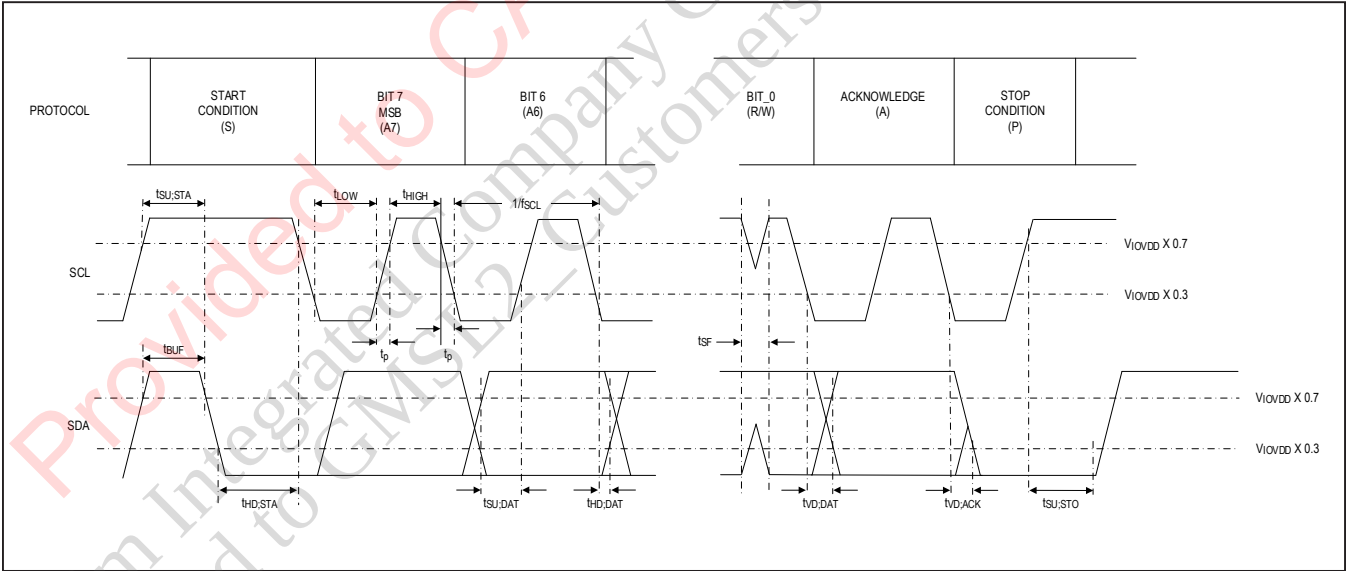


Figure 16. I<sup>2</sup>C Timing Parameters

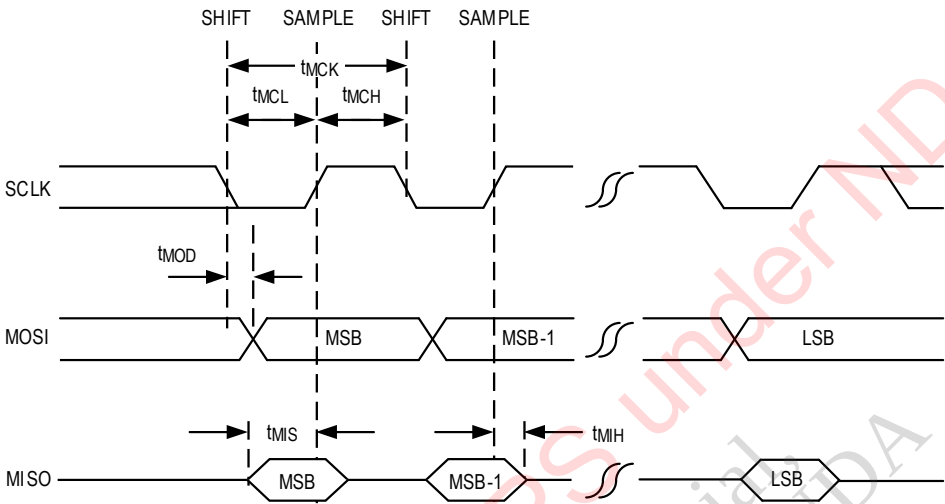


Figure 17. SPI Master-Mode Timing Parameters

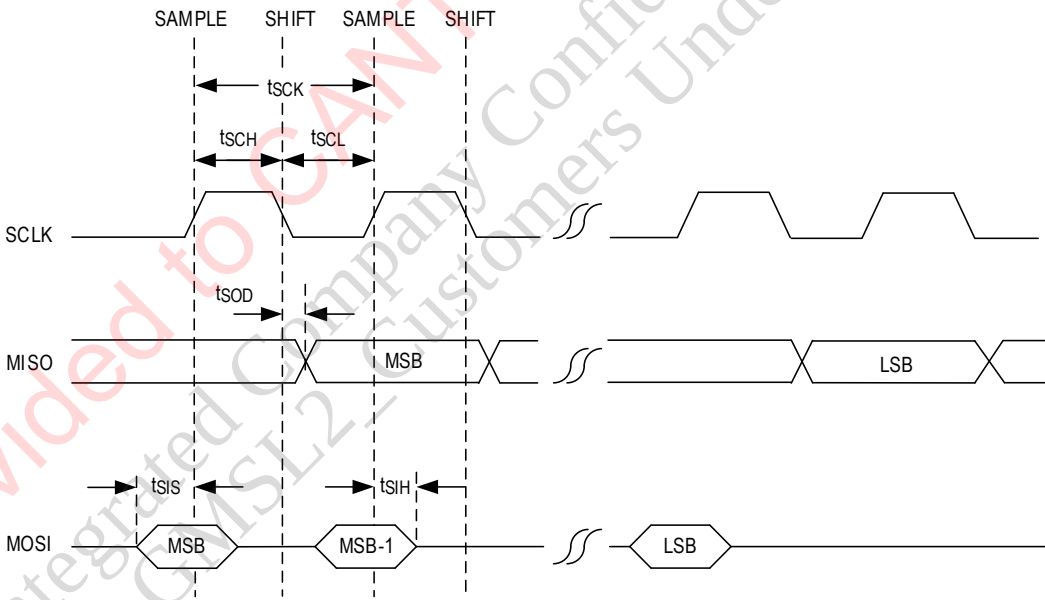


Figure 18. SPI Slave-Mode Timing Parameters

## MAX9296B

## Dual GMSL2/GMSL1 to CSI-2 Deserializer

## Detailed Description

## Introduction

Maxim's GMSL2 serializers and deserializers provide sophisticated link management for high-speed, low bit-error-rate, serial data transport. They support a comprehensive suite of display, camera, and communication interfaces over a single wire.

GMSL2 devices for display applications provide up to 6Gbps forward and 1.5Gbps reverse packetized data transmission over each fixed-speed link. GMSL2 devices for camera, radar, and lidar applications provide a reverse channel rate of 187.5Mbps. Most devices with two GMSL2 links provide a total capacity of 12Gbps and 3Gbps reverse in specific configurations.

The following sections provide a brief overview of the device functions and features. Contact factory for additional information and details on the configuration of each function and feature.

## Product Overview

The MAX9296B converts data on single or dual GMSL2/GMSL1 serial inputs to CSI-2 packets on one or two 4-Lane CSI-2 output ports, one 4-Lane and two 2-Lane CSI-2 output ports, or four 2-Lane CSI-2 output ports.

Data from the serial inputs can be aggregated and output on a single CSI-2 port, replicated on a second port, or output on either port.

The dual-link mode functions logically as a single link, but doubles the forward channel bandwidth by using two serial links.

The MAX9296B pairs with any GMSL2 or GMSL1 serializer.

Figure 19 shows two serial links transporting video data for capture by an SoC. The forward channels operate at a fixed 3Gbps or 6Gbps rate. The reverse channels operate at a fixed 187.5Mbps rate. Idle packets are substituted to fill any unused link bandwidth in both forward and reverse channels.

The MAX9296B is able to aggregate and replicate data from both input ports and output the data combined onto a MIPI CSI-2 port as well as output the data separately on individual ports. This example shows the combined data on a 1x4 port and the separate data each on a 1x2 port.

Forward- and reverse-channel bandwidth can be allocated in any proportion to video or control data. For this link, and the following examples, the maximum payload data rate in the forward direction for a single link is approximately 5.2Gbps for a 6Gbps GMSL2 link and 2.6Gbps for a 3Gbps GMSL2 link. The maximum payload data rate in the reverse direction is approximately 162Mbps for a 6Gbps or 3Gbps GMSL2 link.

The number of D-PHY v1.2 data lanes in a CSI-2 port is programmable as one, two, three, or four lanes when in 4-Lane mode and one or two lanes in 2-Lane mode. The maximum bit rate per lane is 2.0Gbps.

Figure 20 shows two video sources operating independently. The sources can have different video timing, resolutions, frame rates, and data types. Both serial links must be set to the same forward rate (both 3Gbps or both 6Gbps).

Data from link A and link B are output on separate dedicated CSI-2 ports for capture by the SoC. Dedicated ports allow an SoC with slower D-PHY lanes to capture data from two high-bandwidth sources.

Figure 21 shows two video sources operating independently. The sources can have different video timing, resolutions, frame rates, and data types. Both serial links must be set to the same forward rate (both 3Gbps or both 6Gbps).

In this example, data from link A and link B are aggregated before being output for capture by the SoC's using the 2-Lane mode. Aggregating reduces the number of CSI-2 ports required on the SoC.

The SoC identifies the video source by reading the packet's virtual channel or data type. If both sources use the same virtual channel or data type, the MAX9296B can be programmed to assign a different virtual channel and/or data type (16 of each are available).

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Dual GMSL2/GMSL1 to CSI-2 Deserializer

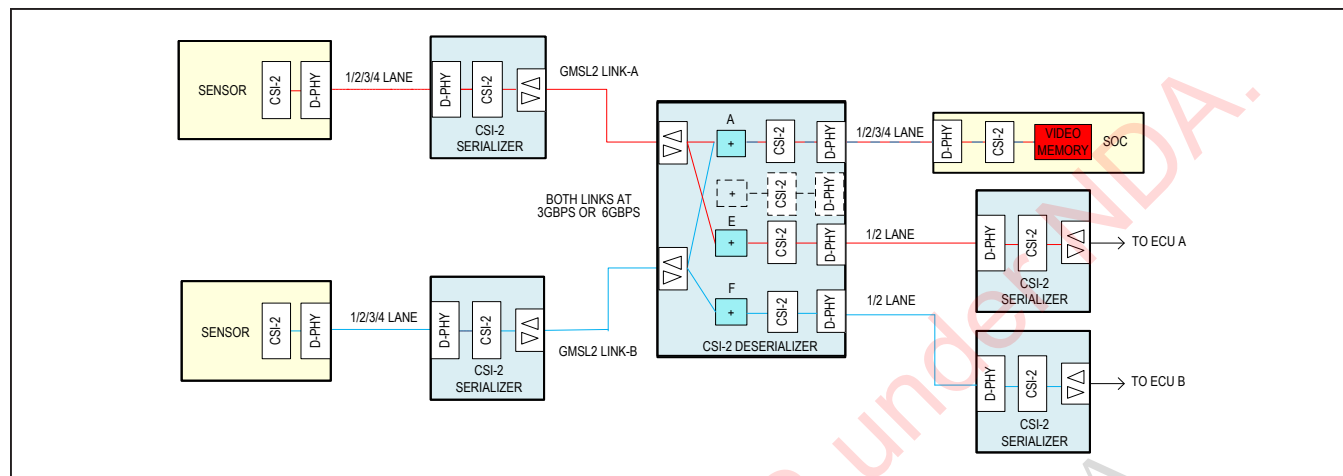


Figure 19. Single GMSL2 Link

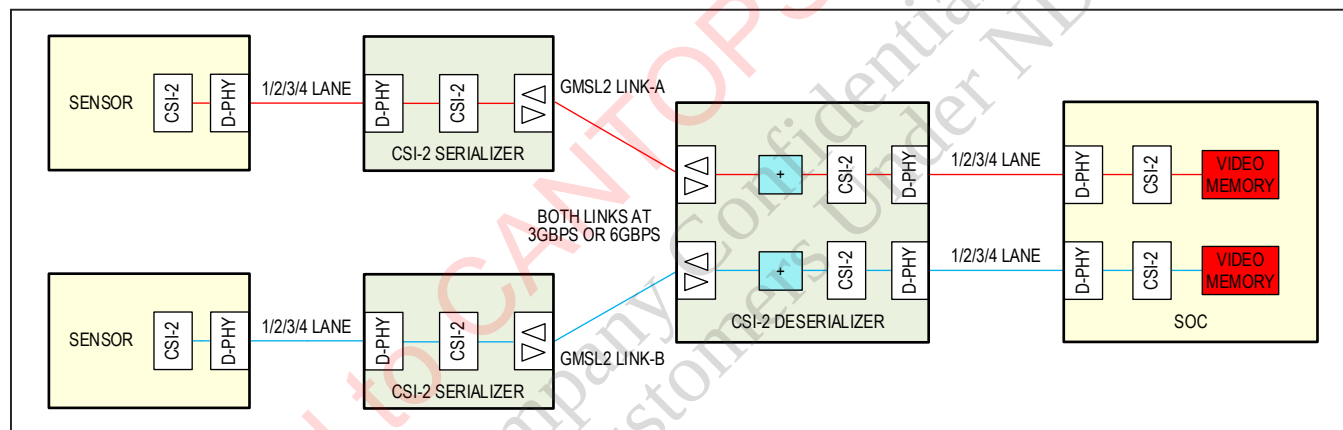


Figure 20. Independent GMSL2 Links with Dedicated Outputs

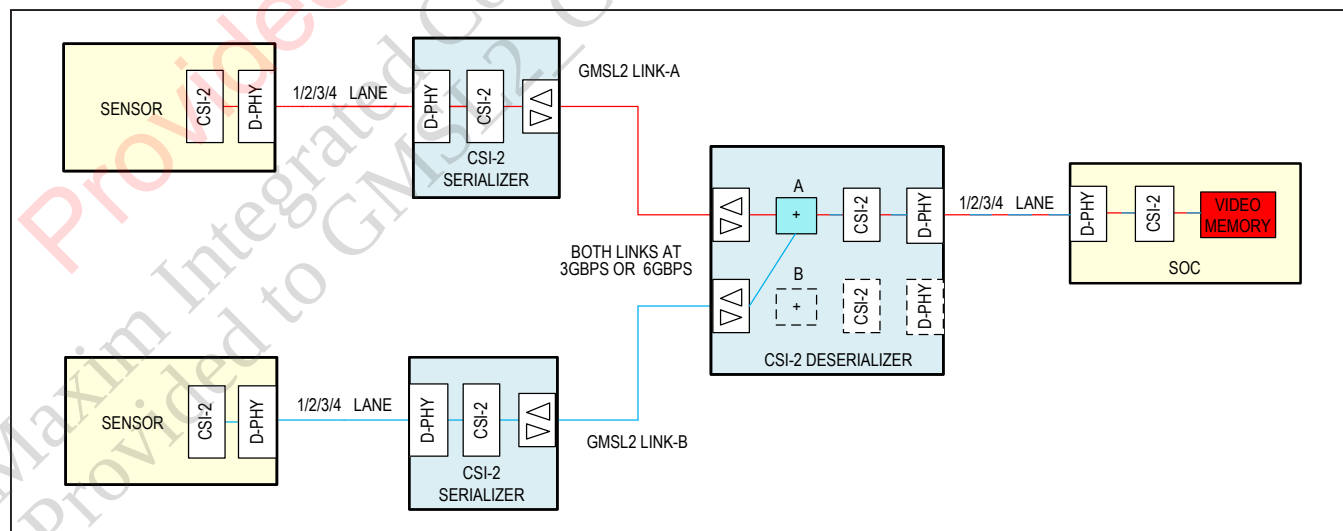


Figure 21. Independent GMSL2 Links with Aggregated Output

## MAX9296B

## Dual GMSL2/GMSL1 to CSI-2 Deserializer

Figure 22 shows two video sources operating independently. The sources can have different video timing, resolutions, frame rates, and data types. Both serial links must be set to the same forward rate (both 3Gbps or both 6Gbps).

In this example, data from link A and link B are aggregated and output on one CSI-2 port and replicated on the other CSI-2 port. Data on the replicated output is captured by a second SoC for dual processing.

Figure 23 shows two video sources operating independently over GMSL1 and GMSL2 links. The sources can have different video timing, resolutions, frame rates, and data types. Typically, video on the GMSL1 link is a single data type. Unlike a CSI-2 interface, the parallel interface

of the source on the GMSL1 link does not provide a virtual channel or data type to forward to the MAX9296B. Instead, the virtual channel and data type for the GMSL1 link can be assigned by programming MAX9296B registers.

The GMSL1 link maximum payload data rate is 3.6Gbps when paired with a GMSL2 serializer operating in GMSL1 compatible mode at 4.5Gbps. When paired with legacy GMSL1 serializers, the forward channel rate is limited to the serializer's highest data rate. The maximum control-channel bit rate for all GMSL1 devices is 1Mbps.

In this example, data from GMSL2 link A and GMSL1 link B are aggregated before being output for capture by the SoC. Aggregating reduces the number of CSI-2 ports required on the SoC.

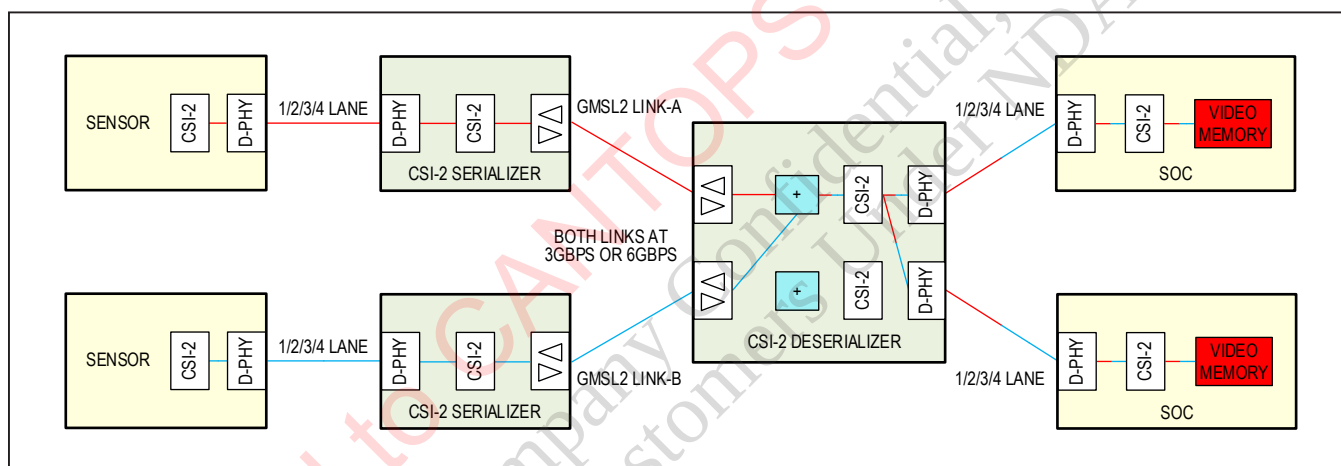


Figure 22. Independent GMSL2 Links with Aggregated and Replicated Output

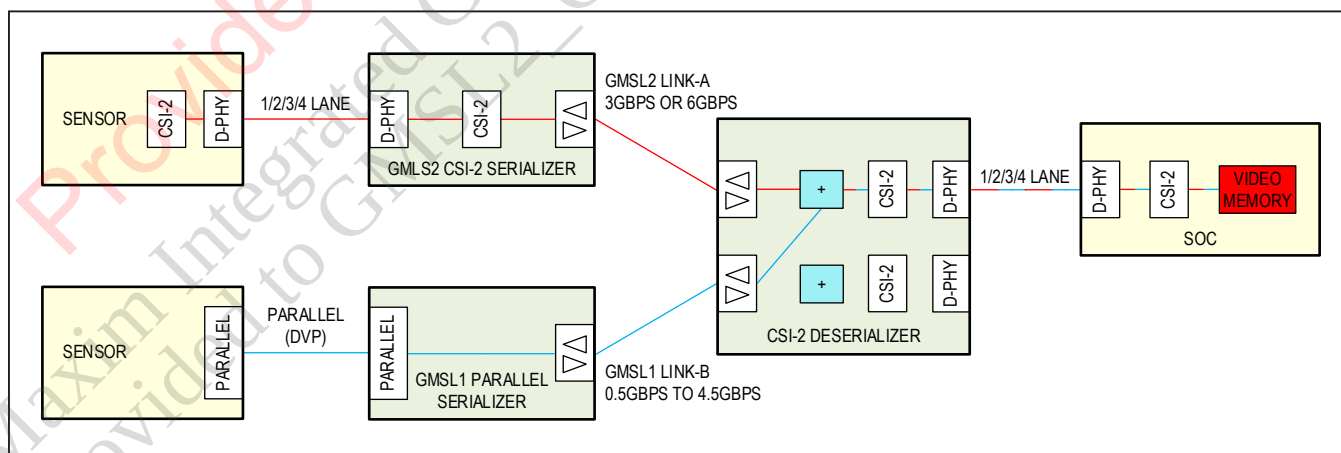


Figure 23. Independent GMSL1 and GMSL2 Links with Aggregated Output



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Figure 24 shows a dual link transporting video for capture by an SoC. The logical function of dual link is the same as the single link in Figure 19, but bandwidth is shared equally on two links instead of one, doubling the data rate compared to the single link. In dual link mode, circuitry in the MAX9296B compensates for typical mismatches in serial link interconnect delay. The maximum payload data rate in the forward direction is approximately 10.4Gbps for a 2x 6Gbps dual link and 5.2Gbps for a 2x 3Gbps Dual Link. The data rate in the reverse direction is approximately 364Mbps (2x 162Mbps) for both forward rates.

### Other Functions

GMSL2 serializers and deserializers have a main I<sup>2</sup>C/UART control-channel interface that an ECU uses to access serializer and deserializer registers and peripheral devices from either end of the link. Each device also has two pass-through I<sup>2</sup>C/UART channels available for local or remote peripheral control. The pass-through I<sup>2</sup>C/UART channels do not have access to serializer and deserializer registers.

For peripheral control, the MAX9296B includes an SPI master/slave interface with two slave select pins. The SPI interface enables a host SPI master on one side of the GMSL2 link to control a peripheral SPI slave on the opposite side. The host can be located at either end of the link, or can swap ends by reprogramming the GMSL2 devices. A GMSL2 device can be configured as an SPI master or slave.

The multifunction pins (MFPs) on the MAX9296B can be configured to provide a variety of general-purpose input, output, and input/output functions (GPIOs). GPIOs are typically used to pass low speed (< 100Kbps) signals over the GMSL2 link, although rates in excess of 1MHz are also supported. A pass-through GPIO can be set up in the forward or reverse direction.

The MAX9296B includes a video crossbar and watermark generation or detection. The crossbar can be used to reorder the color and sync signals. Watermark generation and detection is used to verify that the video image is not frozen.

GMSL2 devices incorporate numerous link-margin optimization and monitoring functions to aid in maximizing link-margin. Continuous (1Hz) adaptive equalization optimizes link-margin to adapt to environmental changes and cable aging. An eye-opening monitor with various threshold alarm levels provides continuous link-margin diagnosis. The monitor issues a runtime alert whenever link degradation is detected. PRBS checking verifies correct link and video-channel operation.

### GMSL2 Protocol

GMSL2 is a fixed-rate transmission medium designed to carry multiple types of communication channels concurrently. The link bit rate is based on either a constant-frequency link clock generated from the 25MHz crystal oscillator or on an external reference frequency. The link clock is completely independent of the video pixel clock.

GMSL2 uses a packet-based protocol to seamlessly share the link bandwidth between communication channels. Bandwidth allocation is dynamic. An inactive channel consumes no link bandwidth to allow other active channels to share the full link bandwidth. Maximum packet size is limited, preventing a single channel from consuming link bandwidth for an extended time. In most cases, available link bandwidth exceeds the bandwidth requirement. Idle packets are used to fill the unused link bandwidth.

The same data protocol is used on forward and reverse channels and for both video and control-channel data.

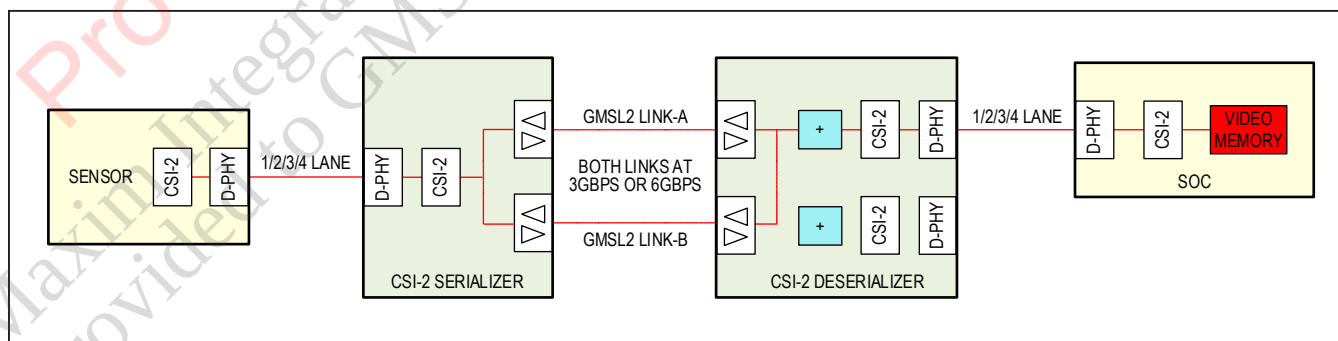


Figure 24. Dual Link GMSL2

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**GMSL2 Physical Layer**

Maxim's GMSL2 serial links family has transmitter and receiver capabilities enabled simultaneously, allowing full-duplex operation on a single wire. A single cable between the serializer and deserializer delivers data transmitted from each end of the link. Forward transmission refers to data sent from the serializer to the deserializer. Reverse transmission refers to data sent from the deserializer to the serializer.

Fixed forward rate options of 3Gbps or 6Gbps are available for both coax and STP. The defaults are 6Gbps in coax mode and 3Gbps in STP mode. The reverse rate is fixed at 187.5Mbps for devices intended for camera, radar, and lidar applications. Both forward and reverse rates are doubled in dual link configurations, in which a serializer and paired deserializer are connected using two links. Dual link mode is not available in all GMSL2 device configurations.

**Cabling Options**

GMSL2 accommodates either 50Ω coaxial or 100Ω shielded-twisted pair (STP) cabling. Applications using two links can also use shielded-twisted quad (STQ) cabling.

Cables must have sufficient return and insertion loss characteristics for best full-duplex link performance. The available link rates and GMSL2 dynamic link optimization enable support of a wide range of cabling options. Contact the factory for insertion and return loss guidelines.

CFG1's level at power-up determines which cabling option—coax or STP—applies. See [Table 11](#).

In coax mode, use only the noninverted SIO pin. In STP configurations, both the noninverted and inverted SIO pins are enabled by default. Any unused SIO pins must be AC-terminated with 50Ω to ground.

The GMSL2 design guideline allows cable lengths of up to 15m for coax or up to 7.5m for STP commonly used in the automotive industry. The guideline assumes two in-line connectors (in addition to the endpoint connectors) and 25mm of PCB trace at each end.

A 100nF AC-coupling capacitor is normally used for GMSL2 links.

**GMSL2 Bandwidth Sharing**

The GMSL forward bandwidth is shared among video, the I<sup>2</sup>C/UART control channel, pass-through I<sup>2</sup>C/UARTs, SPI, and GPIOs, as well as various protocol-specific data exchanges (i.e., info frames, sync, and acknowledgements). The reverse-channel bandwidth is similarly shared, with the exception of video packets.

The total link bandwidth used by all communication channels cannot exceed the fixed available link bandwidth.

A packet-based transmission format and dynamic bandwidth allocation allow link bandwidth to be shared flexibly among the various communication channels that request the link for packet transmissions. With dynamic bandwidth allocation, an inactive channel consumes no link bandwidth, leaving the full link bandwidth available for all active communication channels to share. The packet-based protocol limits maximum packet size to prevent a single channel from monopolizing the link bandwidth at the expense of other channels.

The video and control-channel packets can be assigned a priority level of low, normal, high, or urgent. The scheduler transmits the packet with the highest priority among the pending requests. Packets with maximum latency requirements can be assigned an increased priority.

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**GMSL2 Bandwidth Calculations**

The GMSL2 forward link has a fixed link rate of 3Gbps or 6Gbps. The reverse link rate is also fixed at 187.5Mbps for devices intended for camera, radar, and lidar applications. The GMSL2 protocol overhead is approximately 14%. This leaves approximately 2.6Gbps or 5.2Gbps of data throughput in the forward direction and 162Mbps in the reverse direction.

Worst-case applications must not exceed the available throughput of the forward and reverse links. Maxim's evaluation kit (EV kit) GUI includes a bandwidth (BW) calculator that estimates initial bandwidth requirements. Maxim also offers other tools useful for calculating link bandwidth utilization. Consult the factory for high-bandwidth use cases to ensure error-free performance.

[Table 3](#) provides rough estimates of the bandwidth utilization for each of the communication channels.

**GMSL1 Backwards Compatibility**

The MAX9296B is designed to pair with any GMSL1 serializer. However, some GMSL1 serializer features are not supported. GMSL1 backwards compatibility is only supported with forward link rates from 500Mbps to 4.5Gbps and a reverse link rate of 1Mbps. The full 4.5Gbps GMSL1 forward link bandwidth is available only when the MAX9296B is paired with GMSL2 serializers that include GMSL1 support. In this case, both devices must be configured to use GMSL1 compatibility mode. When the MAX9296B is paired with a legacy GMSL1 only

serializer, the MAX9296B must be configured for GMSL1 compatibility mode, and the available forward link rate is reduced to comply with the limitations of the specified GMSL1 serializer.

[Table 4](#) specifies the availability of common GMSL2 features in GMSL2 devices that are operated in GMSL1 mode. The MAX9296B might not support all GMSL1 features in [Table 4](#). Some GMSL2 features, such as SPI, are only available in GMSL2 mode, and as a result, they are never available in GMSL1 mode regardless of the devices used. Features not supported by both devices in a link must be disabled. All utilized features must be enabled and configured consistently, with settings appropriate to both ends of the link.

Most features specified in [Table 4](#) can be enabled/disabled by appropriate register configuration and/or configuration input pin power-up state. Serializer and deserializer subsystems are responsible for implementing the configuration pin connections needed to achieve the desired settings. Interfacing coax interconnects to non-coax capable GMSL1 devices requires special hardware considerations to ensure reliable functionality. Contact the factory for additional information regarding general GMSL1 operation or GMSL1/GMSL2 device interoperability.

GMSL1 mode designs can increase activity detector noise tolerance by writing to the activity detector threshold register LOSVTH<6:0> = 0x08 (0x143C<6:0> and 0x153C<6:0> for SIOA\_ input and SIOB\_ input, respectively).

**Table 3. Forward and Reverse Link Bandwidth Utilization**

| DATA             | APPROXIMATE BANDWIDTH UTILIZATION MAX9296B                                                    |
|------------------|-----------------------------------------------------------------------------------------------|
| Forward Video    | $H \times V \times \text{fps} \times \text{bpp} \times (1 + \% \text{ blanking}) \times 1.14$ |
| I <sup>2</sup> C | 18 to 60 x I <sup>2</sup> C clock rate, depending on available link bandwidth                 |
| UART             | 6 x UART bit rate, 5.5 x when parity bit enabled                                              |
| SPI              | 2.5 x SPI rate                                                                                |
| GPIO             | 60 x GPIO transition rate without delay compensation                                          |
|                  | 80 x GPIO transition rate with delay compensation enabled                                     |

Definitions:

H = horizontal resolution

V = vertical resolution

fps = frames per second

bpp = bits per pixel

blanking = total horizontal and vertical blanking

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Table 4. Feature Availability in GMSL1 Mode

| FEATURE NAME                            | GMSL2 SERIALIZER IN GMSL1 MODE | GMSL2 DESERIALIZER IN GMSL1 MODE |
|-----------------------------------------|--------------------------------|----------------------------------|
| Coax                                    | Yes                            | Yes                              |
| Bus Width Select (BWS)                  | Yes                            | Yes                              |
| High-Bandwidth Mode (HIBW)              | Yes                            | Yes                              |
| Data Rate Select (DRS) (Low speed mode) | Yes                            | Yes                              |
| DBL (Double Mode)                       | Yes                            | Yes                              |
| HSYNC/VSYNC Encoding                    | Yes                            | Yes                              |
| Pixel CRC (6 bits per pixel)            | Yes                            | Yes                              |
| Video Line CRC (32 bits per line)       | Yes                            | Yes                              |
| Hamming Error Correction                | No                             | No                               |
| I <sup>2</sup> C to I <sup>2</sup> C    | Yes                            | Yes                              |
| UART to UART                            | Yes                            | Yes                              |
| UART to I <sup>2</sup> C                | No                             | No                               |
| Pass-Through I <sup>2</sup> C Channels  | No                             | No                               |
| I <sup>2</sup> C Address Translation    | Yes                            | Yes                              |
| SPI Control Channel                     | No                             | No                               |
| High-Immunity Mode                      | Yes                            | Yes                              |
| REV_FAST with HIBW Mode                 | Yes                            | Yes                              |
| Packet Control Channel with CRC         | Yes                            | Yes                              |
| Packet CC Retransmission                | Yes                            | Yes                              |
| Configuration Link                      | Yes                            | Yes                              |
| I <sup>2</sup> S on Forward Link        | Yes                            | No                               |
| I <sup>2</sup> S on Reverse Link        | No                             | No                               |
| 7.1 TDM                                 | Yes                            | No                               |
| GPI to GPO on Reverse Channel           | Yes                            | Yes                              |
| Frame Sync                              | Yes                            | Yes                              |
| Delay Compensated GPI/GPO               | No                             | No                               |
| Line Fault                              | Yes                            | Yes                              |
| UART Base Mode                          | Yes                            | Yes                              |
| UART Bypass Mode                        | Yes                            | Yes                              |
| Spread Spectrum                         | Yes                            | Yes                              |

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Table 4. Feature Availability in GMSL1 Mode (continued)

| FEATURE NAME                             | GMSL2 SERIALIZER IN GMSL1 MODE | GMSL2 DESERIALIZER IN GMSL1 MODE |
|------------------------------------------|--------------------------------|----------------------------------|
| Pre/Deemphasis                           | No                             | N/A                              |
| Legacy Programmable Equalization         | N/A                            | No                               |
| Adaptive Equalization                    | N/A                            | Yes                              |
| Twisted Pair Splitter Mode               | No                             | N/A                              |
| Auto HDCP                                | No                             | No                               |
| Watermark                                | Yes                            | No                               |
| Video Timing Generator                   | Yes                            | No                               |
| Video Crossbar                           | Yes                            | Yes                              |
| PRBS                                     | Yes                            | Yes                              |
| CNTL0, 1, 2, 3 on Forward Channel        | Yes                            | Yes                              |
| A/V Status Register Interrupt            | No                             | No                               |
| Serializer Audio Output from HDMI Source | Yes                            | N/A                              |
| HS/VS/DE Inversion                       | Yes                            | Yes                              |
| WS/SCK Inversion                         | Yes                            | N/A                              |
| Jitter-Filtering PLL                     | No                             | No                               |
| Sleep Mode                               | Yes                            | Yes                              |

**Power Supplies**

The MAX9296B offers an array of power supply configuration options.

The 1V core supply can be provided directly or by an internal regulator. For optimum power efficiency, connect  $1.0V \pm 5\%$  to  $V_{DD}$ . If using the internal regulator, connect  $1.2V \pm 5\%$  to  $V_{DD}$ , and first write  $REG\_ENABLE = 1$ , then write  $REG\_MNL = 1$ .

The  $V_{DDIO}$  supply for the GPIO pins can be 1.8V to 3.3V for flexibility in accommodating devices interfacing to the part. The allowable supply voltage range is 1.7V ( $1.8V - 5\%$ ) to 3.6V ( $3.3V + 9\%$ ).

$V_{DD18}$  is the primary analog supply. It requires a  $1.8V \pm 5\%$  connection.

$V_{TERM}$  is the D-PHY termination supply. It requires a  $1.2V \pm 5\%$  connection.

See [Table 1](#) for power supply tolerances and noise requirements. Properly bypassing all power supplies is essential to ensuring high frequency circuit stability. See [Table 2](#). Contact the factory for guidance on sharing supplies and optimizing supply decoupling.

**Thermal Management**

Power consumption of GMSL2 devices varies depending on how the device is used. Care must be taken to provide sufficient heat dissipation with proper board and cooling design techniques. The package's exposed pad must be connected to the PCB ground plane by an array of vias. This approach simultaneously provides the lowest electrical and thermal impedances.

System thermal management must keep the operating junction temperature below  $+125^{\circ}C$  to meet electrical specifications and avoid impacting device reliability.

Refer to Application Note 4083: *Thermal Characterization of IC Packages* for guidance.



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**Control Channel and Side Channels**

A  $\mu$ C or other controller can send and receive control and side-channel data over the GMSL2 serial link simultaneously with high-speed video data. All GMSL2 devices support the following interfaces:

- Main I<sup>2</sup>C/UART (internal access)
- Pass-through I<sup>2</sup>C/UART
- SPI
- GPIO

Some GMSL2 devices also support the following interfaces:

- I<sup>2</sup>S/TDM audio
- RGMII/RMII

All of the above interfaces can pass data through the GMSL2 link, but the GMSL2 device registers can be accessed and configured only through the main I<sup>2</sup>C/UART interface.

The side channel, with its various interfaces, is accessed using multifunction pins. Multifunction pins have a default function and can be programmed to an alternate function after power-up. Due to a practical limit in the number of pins available on a given device, not all interfaces can be simultaneously supported. See [Table 7](#) and [Table 8](#) and the [Pin Description](#) section for default and alternate multifunction pin functions, as well as available combinations of interfaces.

**Main I<sup>2</sup>C/UART**

The main I<sup>2</sup>C/UART is located on the SDA<sub>RX</sub> and SCL<sub>TX</sub> pins of each GMSL2 device. The I<sup>2</sup>C (SDA, SCL) or UART (Tx, Rx) interface is selected by the CFG0 pin voltage at power-up (see [Table 10](#)). The selected interface provides master access to both GMSL2 registers and device registers from either end of the link.

The master  $\mu$ C can reside on either end of the link (usually the serializer side for display applications and the deserializer side for camera applications). The MAX9296B supports dual master microcontrollers provided that software arbitration (such as token passing) is used to prevent packet collisions. The control channel allows only one master  $\mu$ C to communicate at a time.

To configure peripheral devices over the link, the GMSL2 serializer and deserializer must use the same control-channel interface (both I<sup>2</sup>C or both UART). Unlike GMSL1 devices, there is no I<sup>2</sup>C-to-UART conversion capability. I<sup>2</sup>C/UART outputs are open-drain and require appropriately sized external pullup resistors for proper operation.

**Pass-Through I<sup>2</sup>C/UART**

GMSL2 devices have two pass-through I<sup>2</sup>C/UART channels. These channels do not have access to registers in either the GMSL2 serializer or the deserializer; they simply tunnel the I<sup>2</sup>C or UART signal across the GMSL2 link. This allows I<sup>2</sup>C channels to be separated so that no multi-master conflicts occur. I<sup>2</sup>C/UART outputs are open drain and require appropriately sized external pullup resistors for proper operation.

**SPI**

GMSL2 enables a host SPI master on one side of the GMSL2 link to control a peripheral SPI slave on the opposite side. Communication can be in either direction across the GMSL2 link.

The SPI clock range is 600kHz to 50MHz. Care must be taken to meet setup and hold time requirements when using at speeds higher than 20MHz.

**General-Purpose Inputs and Outputs (GPIO)**

GPIOs are typically used to pass low-speed (< 100Kbps) signals over the GMSL2 link. A pass-through GPIO can be set up in the forward (serializer to deserializer) or reverse (deserializer to serializer) direction. GPIO transmissions are transition-based; a GPIO packet is created and transmitted on the GMSL2 link when a rising- or falling-edge transition is detected at a GPI pin. The transition is regenerated at a GPO on the other end of the link.

The multifunction pin GPIO can be programmed as GPI (input), GPO (push-pull output or open-drain output), or GPIO (bidirectional input/output). Each GPIO can also be programmed for 1M $\Omega$  or 40k $\Omega$  pullup or pulldown (or none). Although an internal pullup is provided, high-speed open-drain outputs require an appropriate value external pullup resistor to V<sub>DDIO</sub>. Inputs cannot be left floating. Always ensure that every pin configured as an input has a pullup or pulldown programmed or is driven by another IC.

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A GPI on one side of the serial link can be mapped to a single GPO or multiple GPOs on the other side of the link. Each GPI is assigned a pin ID, with the destination GPO(s) on the other side of the link set to the same pin ID. By default, the ID mapping is GPIO0-GPIO0, GPIO1-GPIO1, GPIO2-GPIO2, etc. However, the GPIO mapping can be arbitrarily changed through register settings.

GPI transitions can be transmitted in two modes: delay-compensated and non-delay-compensated. When delay compensation is enabled, the GPI-to-GPO delay across the link is a precise, fixed value. Latency increases, but jitter and skew decrease.

The state of each GPIO can be read or written by register either locally or remotely over the GMSL2 link by a  $\mu$ C using the control-channel I<sup>2</sup>C/UART interface.

In non-delay-compensated mode, channel latency is not fixed. The GPI transition is sent as soon as possible based on priority and available link bandwidth. This variable delay is a result of multiple communication channels sharing the link. Non-delay-compensated mode should be used with signals tolerant to delay variation (i.e.,  $\mu$ C interrupts).

The MAX9296B register settings allow priorities to be assigned to GPI pins. If no priorities are set, GPI transitions are transmitted in the order they occur. However, when priorities are set, transitions on GPIs with higher priority are transmitted earlier.

Typical GMSL2-only device delays for 6Gbps forward and 187.5Mbps reverse-link rates are shown in [Table 6](#).

### MFP Function Map and Speed Groups

Side-channel functions, such as pass-through I<sup>2</sup>C/UART and SPI, are enabled by programming Multi-Function Pins (MFPs). Each MFP has several possible functions, but only one can be used at a time.

Some functions require only a single MFP, but most are implemented across a group of MFPs. For example, LOCK is a single MFP, but SPI takes several MFPs. A user selects MFP functions to suit their use-case by programming the appropriate registers.

Each MFP is placed in one of four speed groups (A-C and NA) with each speed group having a default output transition time. Register programming allows the transition time of most speed groups to be changed from their default values. However, for the SDA\_, SCL\_, TX\_, LOCK, ERB and ODO\_ functions, the open-drain driver's fall time is not adjustable and the rise time is determined by the value of the pull-up resistor. When a speed group's transition time is changed, the transition time of all applicable functions in an MFP's speed group are changed.

Transition times depend on the transition time setting and V<sub>DDIO</sub> supply voltage. Refer to [Table 9](#) for typical transition times.

**Table 5. SPI Latching Edge and Speed**

| FREQUENCY (MHz) | V <sub>DDIO</sub> (V) | LATCHING EDGE       | SPI_SPEED[1:0] |
|-----------------|-----------------------|---------------------|----------------|
| < 12.5          | 1.7–2.24              | Opposite from shift | 01             |
|                 | 2.25–3.6              |                     | 10             |
| 12.5–25         | 1.7–2.24              | Opposite from shift | 00             |
|                 | 2.25–3.6              |                     | 01             |
| 25–50           | 1.7–2.24              | Same as shift       | 00             |
|                 | 2.25–3.6              |                     | 01             |

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Table 6. Typical GPIO Delays for Forward and Reverse Link Transmission

|                                                                             | DELAY COMPENSATION | DELAY |
|-----------------------------------------------------------------------------|--------------------|-------|
| GPIO forwarding from serializer to deserializer (6Gbps forward channel)     | 0                  | 720ns |
|                                                                             | 1                  | 3.5µs |
| GPIO forwarding from deserializer to serializer (187.5Mbps reverse channel) | 0                  | 6µs   |
|                                                                             | 1                  | 15µs  |

Table 7. GMSL2 Pin Function Map

| PIN   | LATCH ON POWER-UP | I <sup>2</sup> C/UART                                                                          | SPI     | OTHER FUNCTIONS                    | GPIO            | POWER-UP DEFAULT | SPEED GROUP | TRANSITION TIME DEFAULT |
|-------|-------------------|------------------------------------------------------------------------------------------------|---------|------------------------------------|-----------------|------------------|-------------|-------------------------|
| MFP0  |                   | Pass-through I <sup>2</sup> C/UART#1 SDA1/RX1 or pass-through I <sup>2</sup> C/UART#2 SDA2/RX2 | SCLK    | FRSYNC_OUT<br>VS2<br>DE1/DV1<br>MS | GPIO0           | GPIO             | A           | "10"                    |
| MFP1  |                   | Pass-through I <sup>2</sup> C/UART#1 SCL1/TX1 or pass-through I <sup>2</sup> C/UART#2 SCL2/TX2 |         | LOCK                               | GPIO1           | LOCK             | C           | "11"                    |
| MFP2  | CFG0              |                                                                                                | BNE/SS1 |                                    | GPO2            | GPO2             | B           | "11"                    |
| MFP3  | CFG1              |                                                                                                | SS2     | VS1<br>DE1/DV1                     | GPO3            | GPO3             | B           | "11"                    |
| MFP4  |                   |                                                                                                | RO(Alt) | ERRB                               | GPIO4           | ERRB             | C           | "11"                    |
| MFP5  |                   | Pass-through I <sup>2</sup> C/UART#2 SDA2/RX2 or pass-through I <sup>2</sup> C/UART#1 SDA1/RX1 | MOSI    |                                    | GPIO5           | RX1              | A           | "10"                    |
| MFP6  |                   | Pass-through I <sup>2</sup> C/UART#2 SCL2/TX2 or pass-through I <sup>2</sup> C/UART#1 SCL1/TX1 | MISO    |                                    | GPIO6           | TX1              | A           | "10"                    |
| MFP11 |                   | Main-channel SDA/Rx                                                                            |         |                                    | GPI11/<br>ODO11 | SDA or Rx        | NA          | NA                      |
| MFP12 |                   | Main-channel SCL/Tx                                                                            |         |                                    | GPI12/<br>ODO12 | SCL or Tx        | NA          | NA                      |

NA = Not applicable



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**Table 8. GMSL1 Pin Function Map**

| PIN   | LATCH ON POWER-UP | I <sup>2</sup> C/UART  | OTHER FUNCTIONS             | GPIO        | POWER-UP DEFAULT | SPEED GROUP | TRANSITION TIME DEFAULT |
|-------|-------------------|------------------------|-----------------------------|-------------|------------------|-------------|-------------------------|
| MFP0  |                   |                        | FRSYNC_OUT<br>MS<br>CNTL1   | GPIO0       | GPIO             | A           | "10"                    |
| MFP1  |                   |                        | FRSYNC_IN<br>LOCK           | GPIO1       | LOCK             | C           | "11"                    |
| MFP2  | CFG0              |                        | CNTL0                       | GPO2        | GPO2             | B           | "11"                    |
| MFP3  | CFG1              |                        | CNTL3                       | GPO3        | GPO3             | B           | "11"                    |
| MFP4  |                   |                        | ERRB                        | GPIO4       | ERRB             | C           | "11"                    |
| MFP5  |                   |                        | CNTL2<br>GPI_1              | GPIO5       | GPIO5            | A           | "10"                    |
| MFP6  |                   |                        | FRSYNC_IN<br>CNTL4<br>GPI_0 | GPIO6       | GPIO6            | A           | "10"                    |
| MFP11 |                   | Main Channel<br>SDA/RX |                             | GPI11/ODO11 | SDA or RX        | NA          | NA                      |
| MFP12 |                   | Main Channel<br>SCL/TX |                             | GPI12/ODO12 | SCL or TX        | NA          | NA                      |

NA = Not applicable

**Table 9. Control- and Side-Channel Typical Rise and Fall Times**

| PIN OR SPEED GROUP<br>TRANSITION TIMES | RISE TIME*               |                          | FALL TIME**              |                          |
|----------------------------------------|--------------------------|--------------------------|--------------------------|--------------------------|
|                                        | V <sub>DDIO</sub> = 1.8V | V <sub>DDIO</sub> = 3.3V | V <sub>DDIO</sub> = 1.8V | V <sub>DDIO</sub> = 3.3V |
| LOCK, ERRB                             | 2.1ns                    | 1.1ns                    | 2.0ns                    | 1.1ns                    |
| "00"                                   | 1.0ns                    | 0.7ns                    | 1.0ns                    | 0.7ns                    |
| "01"                                   | 3.0ns                    | 1.0ns                    | 3.0ns                    | 1.0ns                    |
| "10"                                   | 5.0ns                    | 3.0ns                    | 5.0ns                    | 3.0ns                    |
| "11"                                   | 10.0ns                   | 5.0ns                    | 10.0ns                   | 5.0ns                    |
| SDA_, SCL_, TX_, ODO_                  | ***                      | ***                      | 40.0ns                   | 30.0ns                   |

\*20% to 80%, 10pF load.

\*\*80% to 20%, 10pF load.

\*\*\*Rise time for open-drain outputs depends on external pullup resistor value.

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**CFG Latch at Power-Up Pins**

Voltage levels at the CFG0 and CFG1 pins are latched at power-up, or upon a low-to-high transition of PWDNB. These levels set initial register values and functional modes that may not be easily programmed through I<sup>2</sup>C or UART after the IC powers up. The CFG pins select device address, I<sup>2</sup>C or UART main control channel, GMSL2 serial rate, and coax or STP cable. See [Table 10](#) and [Table 11](#).

The voltage level for each pin is set by an external precision resistor divider connected between VDDIO and ground ([Figure 5](#)), or for some configurations, by a single resistor connected to VDDIO or ground. [Table 10](#) and

[Table 11](#) show the recommended resistor values to select each configuration. The voltage level at the CFG pins is typically latched 1ms after all MAX9296B supplies reach minimum levels required by the power-on-reset (POR) circuit. CFG pins must not be loaded with more than 10pF at power-up to ensure the proper voltage level.

If the requirements described in the [Table 10](#) and [Table 11](#) notes are met, the CFG pin's secondary functions can be used after the CFG pin voltage levels are latched at power-up. See the [Pin Description](#) section and [Table 7](#) and [Table 8](#) for secondary functions). CFG pins cannot be used as general-purpose inputs.

**Table 10. CFG0 Input Map**

| CFG0 INPUT VOLTAGE<br>(PERCENTAGE OF V <sub>DDIO</sub> ) (Notes a, b) |         |         | SUGGESTED RESISTOR VALUES<br>(±1% TOLERANCE) (Note c) |        | MAPPED CONFIGURATION<br>(Note d) |                |
|-----------------------------------------------------------------------|---------|---------|-------------------------------------------------------|--------|----------------------------------|----------------|
| MIN (%)                                                               | TYP (%) | MAX (%) | R1 (Ω)                                                | R2 (Ω) | I2CSEL                           | DEVICE ADDRESS |
| 0.0                                                                   | 0.0     | 11.7    | OPEN                                                  | 10000  | I <sup>2</sup> C                 | 0x90           |
| 16.9                                                                  | 20.2    | 23.6    | 80600                                                 | 20500  |                                  | 0x94           |
| 28.8                                                                  | 32.1    | 35.5    | 68100                                                 | 32400  |                                  | 0xD0           |
| 40.7                                                                  | 44.0    | 47.4    | 56200                                                 | 44200  |                                  | 0xD8           |
| 52.6                                                                  | 56.0    | 59.3    | 44200                                                 | 56200  | UART                             | 0xD8           |
| 64.5                                                                  | 67.9    | 71.2    | 32400                                                 | 68100  |                                  | 0xD0           |
| 76.4                                                                  | 79.8    | 83.1    | 20500                                                 | 80600  |                                  | 0x94           |
| 88.3                                                                  | 100     | 100     | 10000                                                 | OPEN   |                                  | 0x90           |

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Table 11. CFG1 Input Map

| CFG1 INPUT VOLTAGE<br>(PERCENTAGE OF $V_{DDIO}$ ) (Notes a, b) |         |         | SUGGESTED RESISTOR VALUES<br>( $\pm 1\%$ TOLERANCE) (Note c) |                 | MAPPED CONFIGURATION<br>(Notes e, f, g) |                 |                    |
|----------------------------------------------------------------|---------|---------|--------------------------------------------------------------|-----------------|-----------------------------------------|-----------------|--------------------|
| MIN (%)                                                        | TYP (%) | MAX (%) | R1 ( $\Omega$ )                                              | R2 ( $\Omega$ ) | CXTP                                    | GMSL1/<br>GMSL2 | HIM/<br>GMSL2 RATE |
| 0.0                                                            | 0.0     | 11.7    | OPEN                                                         | 10000           | COAX                                    | GMSL2           | 6Gbps              |
| 16.9                                                           | 20.2    | 23.6    | 80600                                                        | 20500           |                                         | GMSL1           | HIM Enabled        |
| 28.8                                                           | 32.1    | 35.5    | 68100                                                        | 32400           |                                         |                 | HIM Disabled       |
| 40.7                                                           | 44.0    | 47.4    | 56200                                                        | 44200           | STP                                     | GMSL2           | 6Gbps              |
| 52.6                                                           | 56.0    | 59.3    | 44200                                                        | 56200           |                                         | GMSL1           | 3Gbps              |
| 64.5                                                           | 67.9    | 71.2    | 32400                                                        | 68100           |                                         |                 | HIM Enabled        |
| 76.4                                                           | 79.8    | 83.1    | 20500                                                        | 80600           |                                         |                 | HIM Disabled       |
| 88.3                                                           | 100     | 100     | 10000                                                        | OPEN            | COAX                                    | GMSL2           | 3Gbps              |

Notes:

- Voltage divider resistor tolerance,  $V_{DDIO}$  supply ripple, and external loading must not cause the CFG0 or CFG1 input voltage to exceed the maximum or minimum limits.
- Until the input voltage is latched, any load on CFG0 or CFG1 (other than R1 and R2) must be  $\geq 25 \times (R1 + R2)$ . Load capacitance (including R1 and R2) must be lumped-load  $\leq 10\text{pF}$ .
- Each resistor in the voltage divider must be  $\leq 100\text{k}\Omega$ .
- I<sup>2</sup>CSEL: I<sup>2</sup>C or UART interface for SDA\_RX and SCL\_TX  
DEVICE ADDRESS: device address
- CXTP: Shielded twisted-pair (SIO\_P, SIO\_N) or coax (SIO\_P) serial link.  
GMSL1/GMSL2: GMSL1 or GMSL2 operating mode.  
HIM applies when GMSL1 operating mode is selected. High Immunity Mode for reverse control channel.  
GMSL2 RATE applies when GMSL2 operating mode is selected. 3Gbps or 6Gbps GMSL2 serial link bit rate.
- Serial links SIOA\_ and SIOB\_ are set to the same forward rate.
- GMSL1 default BWS = 0 (24 bit).

## Clocking

### GMSL Reference Clock

The GMSL2 devices require a reference clock source to generate the 6GHz line rate clock and associated internal clocks. Both the serializer and deserializer can be clocked with an external 25MHz crystal or an external clock source with a frequency accuracy of  $\pm 200\text{ppm}$ .

### Spread-Spectrum Clocking

Maxim's GMSL2 links provide exceptional EMI performance. Optional spread-spectrum clocking (SSC) is available to mitigate electromagnetic interference emitted from devices and interconnections and to provide additional margin.

SSC reduces peaks in the frequency spectrum by spreading the signal over a wider bandwidth. The spread has a 25kHz sawtooth modulation profile, programmable to deviate up to  $\pm 2500\text{ppm}$  from the center frequency.

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**Power-Up and Link Startup**

GMSL2 ICs are in power-down mode when PWDNB pin is low or when any of the power supplies are down. Register and configurations are set to default reset conditions.

The serializer and deserializer can power up in any order. After PWDNB is released and all power supplies are up, each device starts its power-up sequence and performs the following actions in sequence:

- 1) Latch at power-up pins register set. Set internal registers according to the selected configuration (selected by CFG0, CFG1).
- 2) Control channel (I<sup>2</sup>C or UART) is functional on local side. Device registers are writable and readable.
- 3) Link is established based on the following settings:
  - a) Single-link automatic selection mode (AUTO\_LINK = 1 and LINK\_CFG = 1 or 2): Automatically select which PHY to use to establish GMSL2 link by periodically attempting to handshake using PHY A and PHY B.
  - b) Single-link manual selection mode (AUTO\_LINK = 0 and LINK\_CFG = 1 or 2): If LINK\_CFG = 1, establish link using PHY A. If LINK\_CFG = 2, establish link using PHY B.
  - c) Dual-link mode (LINK\_CFG = 0): Establish link using both PHYs.
  - d) Splitter (serializer)/aggregator (deserializer) mode (LINK\_CFG = 3): Establish link using both PHYs (for specific applications only).
- 4) Each enabled PHY performs link calibration, equalizer adaptation, and data-channel locking. Both chips set their LOCK pin high.
- 5) Control channel is available from the remote side.

This entire link-up process, from the time that the last part's PWDNB input is brought high, takes approximately 20ms nominally and 100ms maximum for any channels that meet the GMSL2 channel specification.

After the devices are linked, they can be configured. This can be done locally or over the control channel, by a microcontroller on either the serializer side or the deserializer side.

**Device Reset**

There are three general-reset options available through register writes:

- 1) RESET\_ALL resets all blocks, including all registers, digital blocks, and analog blocks. This process is similar to driving the PWDNB pin low and then high.
- 2) Setting RESET\_LINK resets all GMSL PHY-related digital logic and all data pipelines. After this bit is set, all control registers are still accessible through the local control channel. The link remains in RESET until RESET\_LINK is cleared.
- 3) RESET\_ONESHOT resets all GMSL PHY-related digital logic and all data pipelines, and then automatically clears itself. This is similar to setting and clearing RESET\_LINK.

Program the registers that affect GMSL-2 link operation (i.e., TX\_RATE, RX\_RATE, CXTP\_A/B, AUTO\_LINK, LINK\_CFG, GMSL2) first, followed by RESET\_LINK or RESET\_ONESHOT.

**Link and Video Lock****Link Lock**

Link lock happens automatically on power-up, and is an indication that the cable is plugged in and the system is running. Lock is obtained with no interaction or commands from the microcontroller to the GMSL2 parts.

Link lock indicates that the PLLs for the GMSL2 link are locked to each other and the data receive paths are locked (forward channel in serializer, reverse channel in deserializer). Video and control-channel functions (I<sup>2</sup>C/UART, SPI, GPIO) can be used immediately after link lock is asserted.

In dual link, only one link locks automatically. The user must then program the serializer to the desired operating mode.

The GMSL2 link uses the crystal or external reference input as the reference clock for the GMSL2 link, so a valid video input (pixel clock) is not needed for the GMSL2 link to lock.

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## Dual GMSL2/GMSL1 to CSI-2 Deserializer

**Video Lock**

The video-lock bit indicates that the deserializer is receiving valid video data. After the GMSL2 link has locked, the deserializer video output PLL begins its locking sequence. The deserializer normally begins outputting video data several milliseconds after it asserts line lock, provided that it is receiving video packets from the serializer. The video lock status is typically read from a register. However, the deserializer LOCK pin behavior can be changed by a register setting so that the LOCK pin is asserted only when the deserializer is outputting video.

**Power Standby and Sleep Mode**

A power-manager block is present in all GMSL2 products. Its primary function is to monitor supply voltages and control power-down (standby) and sleep modes.

There are two ways to enter low-power mode while all power supplies are active: asserting the PWDNB pin or invoking the sleep state. Both states offer very low power supply currents.

Asserting the PWDNB pin (active-low) places the device in standby power mode and resets the digital registers and configurations to their default power-up condition. Any supply that drops below its internal threshold settings also places the device in power-down mode.

The sleep state offers the preservation of all critical register settings and configurations. The device can be put into the sleep state through an I<sup>2</sup>C/UART command. The resume state restores the device to the pre-sleep condition, without the need for additional register writes. Resume is invoked by an I<sup>2</sup>C command or a low-frequency clock beacon transmitted from the master device over the GMSL2 link.

**Error and Fault-Condition Monitoring**

Both the serializer and deserializer have an open-drain, multipurpose error reporting and interrupt status output. The active-low ERRB pin is driven by the logical OR of a wide variety of error and event status indicators. The ability of each error condition to drive ERRB is maskable by register settings. Each error and event that can drive ERRB has a status flag within a subblock of registers, so the reason for assertion of ERRB can be determined by reading the register status.

**AEQ (Adaptive Equalization)**

GMSL2 devices automatically adapt receiver characteristics to compensate for the insertion and return loss characteristics of the channel, which consist of the cables, connectors and PCBs. This optimizes performance on any channel that meets the GMSL2 channel spec.

The equalizer architecture makes GMSL2 links robust against noise, crosstalk, and reflections. Initial adaptation is performed during link lock and is then invoked at a rate of approximately 1Hz to track temperature and voltage variations. The adaptation process optimizes the equalizer coefficients to maximize the eye opening using the built-in eye-opening monitor.

**Video Pipes, Aggregation and Replication**

See [Figure 25](#), and [Figure 26](#).

In GMSL2 pixel mode, the transportation of video data is based on the concept of video pipes. Carrying data in pipes allows GMSL2 to bridge different digital video interfaces (i.e., HDMI source to CSI-2 sink) and perform watermark generation and detection.

A pipe carries a video stream (or streams) and associated video synchronization data.

A pipe operates in one of three modes. In all modes, a pipe can carry multiple concurrent video streams, with each stream having different virtual channels and data types, as follows:

Mode 1: Streams with constant bpp of up to 24bpp. The bpp of the streams must be the same.

Mode 2: Streams with 16, 14, 12, 10, or 8bpp. Streams less than 16bpp are padded with zeros.

Mode 3: Streams with two different bpp values. The bpp of one stream must be twice the bpp of the other stream. The higher bpp stream maximum is 24bpp.

Modes 1 and 3 carry data at full bandwidth but put more restrictions on bpp than Mode 2. Mode 2 allows streams with different bpp, but streams of less than 16bpp are carried using more bandwidth than necessary on the GMSL2 link (because of zero-padding). Mode 1 or Mode 3 are sufficient for most applications. Mode 2 requires less programming and is more convenient if the application does not require maximum link bandwidth.

The MAX9296B has four video pipes. If a rear-view camera with YUV422 output is connected (for example), one pipe is sufficient to carry the YUV422 video stream. High bandwidth, high dynamic range (HDR) cameras that output multiple-exposure data usually require more than one pipe. In an HDR camera application with multiexposure data of RAW16 and RAW12 per pixel along and two lines of embedded data per frame would, in Mode 3, use one pipe for RAW16 and embedded 8-bit data and, in Mode 1, another pipe for RAW12 data. The total number of pipes used by all serializers connected to the MAX9296B can not exceed four pipes.



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## Dual GMSL2/GMSL1 to CSI-2 Deserializer

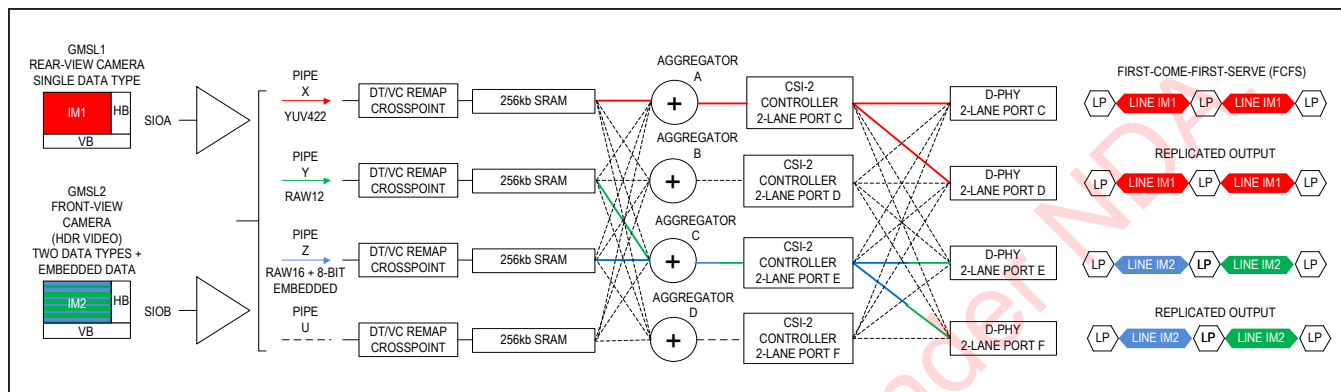


Figure 25. GMSL1 and GMSL2 Camera Video Output on Separate Ports

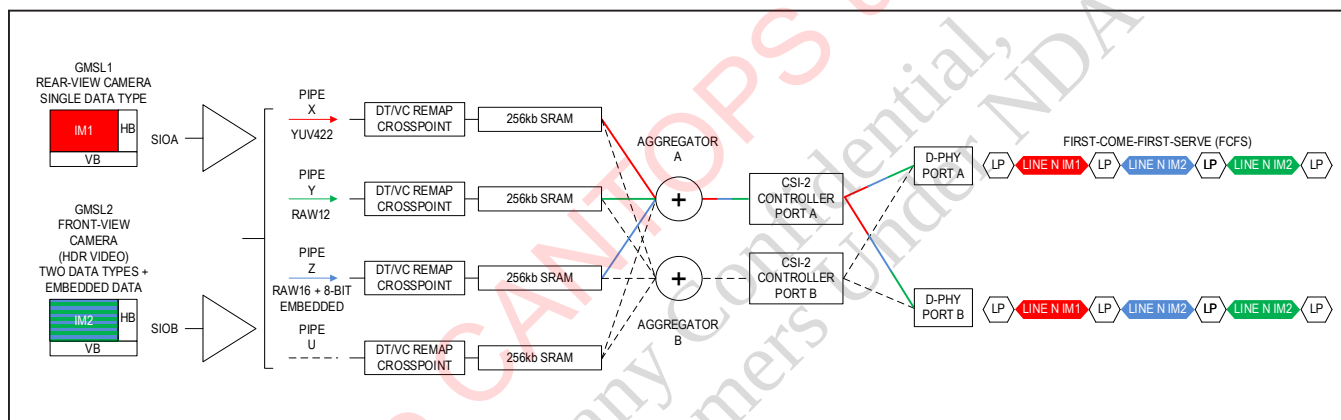


Figure 26. GMSL1 and GMSL2 Camera Video Aggregated and Replicated

Each of the four pipes in the MAX9296B has a dedicated video-line buffer. Each 32k byte (256kb) video-line buffer has capacity for a line length of up to 8191 24-bit pixels (8191 pixels includes line-blanking pixels).

Before data enters a line buffer, it goes through a cross-point switch and a data type (DT) and virtual channel (VC) reassignment stage. If the video source has a CSI-2 output, packet DT and VC can be left as-is or reassigned through register programming. Up to 16 DT/VC incoming pairs can be mapped to 16 DT/VC outgoing pairs.

If the video source does not have a CSI-2 output, a DT and VC can be programmed in the MAX9296B for insertion into the CSI-2 output packet. The cross-point switch can change the order of the bits in the incoming video pixel data to any order, if desired.

A line buffer stores a complete line of video data before the data is available for read-out by an aggregator. Each buffer connects to two aggregators, but only one aggrega-

tor (as programmed by the user) can read a line of data out of the buffer at a time. Once data is read out, it cannot be read out a second time by the same aggregator or by the other aggregator. However, in a time-domain multiplexed fashion, both aggregators can be programmed to read from same buffer. Up to four pipes can be aggregated by one aggregator.

Data is read out from line memory on a first-come, first-served basis. When a complete line of video data has filled a line memory, it is read out. The order in which the line memories reach filled status is the order in which they are read out.

Video data can be routed according to DT or VC, based on the source CSI-2 packet's DT/VC, or by a DT/VC assigned or reassigned by the MAX9296B. For example, data in Pipe X with VC0 can be programmed to be read out by Aggregator A, while data in Pipe X with VC1 can be programmed to be read out by Aggregator B.

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## Dual GMSL2/GMSL1 to CSI-2 Deserializer

The aggregator forwards the line of video from the line buffer and video synchronization data to its associated CSI-2 controller for packet generation. Packets can be output on the controller's port or replicated by routing them to the other port. For example, as shown in [Figure 25](#), when in 4x2 mode the CSI-2 controller for Port C can output packets on D-PHY Port C and also route them for output on Ports D with the controller for Port E sending video to Ports E and F. See [Table 12](#) for 2x4 mode and 4x2 mode mapping.

A D-PHY port can only accept packets from one controller; it cannot aggregate packets from the two controllers.

**Table 12. Controller to PHY Mapping**

| CONFIGURATION | CONTROLLER | PORT |
|---------------|------------|------|
| phy_4x2       | 0          | C    |
|               | 1          | D    |
|               | 2          | E    |
|               | 3          | F    |
| phy_2x4       | NA         | A    |
|               | 1          | A    |
|               | 2          | B    |
|               | NA         | B    |
| phy_1x4a_22   | NA         | A    |
|               | 1          | A    |
|               | 2          | E    |
|               | 3          | F    |
| phy_1x4b_22   | 0          | C    |
|               | 1          | D    |
|               | 2          | B    |
|               | NA         | B    |

To prevent buffer overflow, the CSI-2 port data rate needs to be programmed to a value that is greater than or equal to the incoming data rate. Programming the output rate to be faster than the bandwidth of the incoming video increases packet spacing (LP time between packets).

### Video Line CRC

A CRC32 polynomial is used to generate a 32-bit code at the end of each DE (data enable) or DV (data valid) pulse (DE is default; DV can be selected instead by register). The code is transferred to the receiver (deserializer) side using info frames. The CRC checker generates the same code on the receiver side and checks if the generated and received CRC codes are the same. If not, it asserts an error. The CRC check is done at every falling edge of DE (or DV when selected) on the receiver side, even if the info frame is not received.

### Eye-Opening Monitor

The eye-opening monitor (EOM) enables GMSL2 parts to monitor the link margin on an active link and generate an interrupt if it falls below an acceptable level. For example, if a cable is damaged, the link can run error-free, but have less link margin than desired. This allows the customer to proactively react to deteriorating cable performance before any link errors occur.

GMSL2 devices can measure the horizontal or vertical eye-opening of the equalizer's output. The measurement is activated automatically at a rate of approximately 1Hz once a link is active. The EOM block compares the data sampled at the center of the eye with a sample that is offset in phase for the horizontal EOM or offset in voltage for the vertical EOM. The eye-opening is then reported, and the EOM can trigger an interrupt or a reset if the opening falls below user-defined thresholds.



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## Dual GMSL2/GMSL1 to CSI-2 Deserializer

**Watermarking**

The watermarking block allows users to detect a frozen-frame failure in a frame-based processing system between the generator and detector. This feature is specifically targeted to detect frozen frames caused by SoCs in safety-critical applications. It does not detect frozen frames that occur before the watermark generator or after the watermark detector.

GMSL2 devices contain both a watermark generator and watermark detector. This allows both serializers and deserializers to insert a watermark or detect a watermark in a safety-relevant video stream. The watermark generator inserts a time-varying watermark that is highly redundant and robust to image processing and display stream compression. The watermark detector looks for this time-varying watermark, and failure to see all of the generated watermarks indicates a frozen-frame failure in a frame-based processing system between the generator and detector. Upon detection of this error condition, the watermark detector can generate an interrupt and/or blank the output video, returning the display to a safe state in less than 500ms.

**Vertical and Data Enable or Data Valid Sync Outputs**

The MAX9296B can output the vertical sync (VS) and data enable/data valid (DE/DV) of a video stream for monitoring of video timing by a processor. This feature provides access to VS and DE/DV signals not available directly at the CSI-2 output. VS and/or DE/DV output are alternate functions of MFP2 and MFP3.

**PRBS****Video PRBS**

The video channel has a PRBS generator in the serializer and a PRBS checker in the deserializer for testing the video channel operation. The video PRBS generator can work with the PCLK recovered by the serializer from an HDMI, parallel, or MIPI input. An external PCLK can also be provided to a designated GPIO pin to run the video PRBS generator.

Note that all link bandwidth is not used by the video channel alone in GMSL2 mode, so it is possible to have a bit error on the link that does not cause a video PRBS error.

**Link PRBS**

The GMSL PHY has a link PRBS feature for testing link integrity for every link bit. In this mode, the link is not established as in normal operation, but instead the link integrity test is performed in both directions of the link simultaneously using a PRBS-7 pattern. This feature is feasible only when the user has simultaneous local control-channel access to both ends of the link. Contact the factory for programming procedures regarding link PRBS mode.

**CSI-2 Port**

Packets are generated according to the MIPI CSI-2 v1.3 protocol specification and the MIPI D-PHY v1.2 transmitter physical layer specification (refer to these MIPI standards for details). The CSI-2 controllers in the MAX9296B have been upgraded to include the CSI-2 v2.0 virtual channel extension for D-PHY to allow 16 virtual channels instead of the 4 virtual channels specified in the CSI-2 v1.3 specification.

Within a port, any number of data lanes can be enabled (one, two, three, or four lanes). To ease PCB layout, swapping of data lanes and swapping of the polarity of the differential signals within a lane are supported. For example, the position of data lane 3 can be exchanged with the position of data lane 2 (lane swap), and the inverting and noninverting outputs of data lane 1 can be swapped (polarity swap).

Initial and periodic deskew pattern generation by the D-PHY port is supported (deskew is required for data-lane bit rates > 1.5Gbps and optional for bit rates ≤ 1.5Gbps).

The default D-PHY lane rate is 1.5Gbps/lane. The lane rate can be changed in 100Mbps steps with simple register programming. More involved register programming allows smaller step changes in the lane rate.

An unused D-PHY port or CSI-2 controller can be individually powered down to save power.

When a port is programmed to use fewer than four data lanes, the unused lanes should be left open.

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## Dual GMSL2/GMSL1 to CSI-2 Deserializer

## Applications Information

### Control Channel Programming

GMSL2 device registers are accessed and configured only through the main I<sup>2</sup>C/UART interface.

### Host-to-Peripheral Main I<sup>2</sup>C and Pass-Through I<sup>2</sup>C Communication

When communicating between a host and peripheral, main and pass-through I<sup>2</sup>C operation is the same. An I<sup>2</sup>C tunnel across the GMSL2 link connects the host's I<sup>2</sup>C master to the peripheral's I<sup>2</sup>C slave. This logically connects separated I<sup>2</sup>C buses, enabling I<sup>2</sup>C transactions across the serial link to occur (with some delay) as if performed on the same physical I<sup>2</sup>C bus. The GMSL2 serializer and deserializer are intermediary devices; the host I<sup>2</sup>C master connects to a GMSL2 device I<sup>2</sup>C slave, and the peripheral I<sup>2</sup>C slave connects to a GMSL2 device I<sup>2</sup>C master.

For example, when the host I<sup>2</sup>C master transacts on one side of the link (local side), data is forwarded to the other side (remote side) by the I<sup>2</sup>C slave of the local-side GMSL2 device. Data is then received by the I<sup>2</sup>C master of the remote-side GMSL2 device, which in turn generates the same I<sup>2</sup>C transaction with the peripheral slave I<sup>2</sup>C. The remote-side GMSL2 device sends back any I<sup>2</sup>C data expected by the local side.

The I<sup>2</sup>C interface uses clock stretching (holding SCL low) to account for timing differences between master and slave and to allow time for data to be forwarded and received across the serial link. The host I<sup>2</sup>C master and peripheral I<sup>2</sup>C slave must support clock stretching by the GMSL2 device.

SDA and SCL lines operate as both an input and an open-drain output. Pullup resistors are required on SDA and SCL.

Each transmission consists of a START condition sent by a master, followed by the device's 7-bit slave address plus a R/W bit, register address bytes, one or more data bytes, and finally a STOP condition.

Register addresses are 16-bits wide. Single or multiple data bytes can be written or read (by address auto-increments).

### Device Address

Each device on the I<sup>2</sup>C UART control channel must have a unique address. The GMSL2 device address is set to one of several 7-bit addresses according to the voltage level of the ADD or CFG pins at power up. See the [CFG Latch at Power-Up Pins](#) section. Note that device address can be changed after power-up by writing to the DEV\_ADDR register.

### Main I<sup>2</sup>C Host-to-GMSL2 Device Communication

The host I<sup>2</sup>C master has access to GMSL2 serializer and deserializer registers. The host can program GMSL2 device registers to change the pass-through I<sup>2</sup>C/UART interface from I<sup>2</sup>C to UART.

### Main UART

When the main I<sup>2</sup>C/UART is configured as UART, there are two operating modes: base and bypass.

### UART Base Mode

Base mode is the means by which the microcontroller communicates with the serializer and deserializer where registers in these and peripheral devices can be accessed. Base mode is typically enabled by default at power-up. In base mode, the  $\mu$ C is the host and can access the registers of both the serializer and deserializer from either side of the link using the GMSL2 UART packet protocol. The  $\mu$ C can also program the peripherals on the remote side by sending the UART packets to the serializer or deserializer. The  $\mu$ C communicates with a UART peripheral in base mode (through INTTYPE register settings). The device addresses of the serializer and deserializer in this mode are programmable.

In base mode, the serializer, deserializer, and peripheral registers can be written and read using the half-duplex GMSL2 UART protocol. Base mode is enabled by default at power-up.

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Figure 29 shows the UART protocol for writing and reading in base mode.

UART Bypass Mode

In bypass mode, the serializer/deserializer ignore UART commands from the  $\mu$ C, and the  $\mu$ C communicates only with the peripherals using its own defined UART protocol. The  $\mu$ C cannot access the serializer/deserializer's registers in this mode. The UART transitions are simply sent over the GMSL link. Ignoring UART transactions prevents inadvertent misprogramming of serializer and deserializer registers. The device addresses of the serializer and deserializer in this mode are not programmable.

Switching between UART Base and Bypass Modes

There are two ways to switch between base mode and bypass mode: by register programming or by using the mode select (MS) pin.

When register is programmed, bypass mode is active only as long as there is UART activity. When there is no UART activity for a selected timeout, both devices exit bypass mode and the bit is automatically cleared.

When set by the MS pin, a high level puts the device into bypass mode. A low level puts the device in base mode. MS is set on-the-fly and is not latched on power-up.

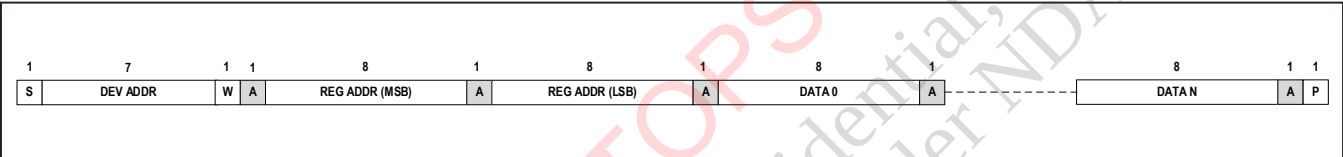


Figure 27. I2C Write Packet Format

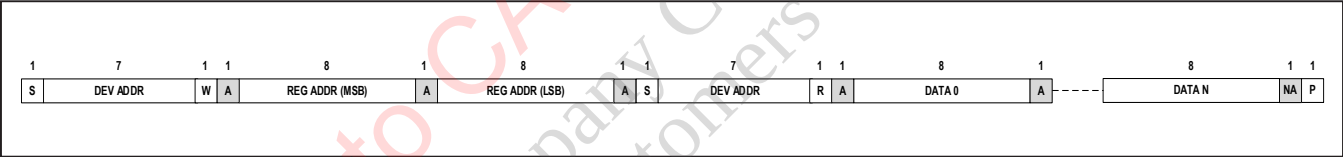


Figure 28. I2C Read Packet Format

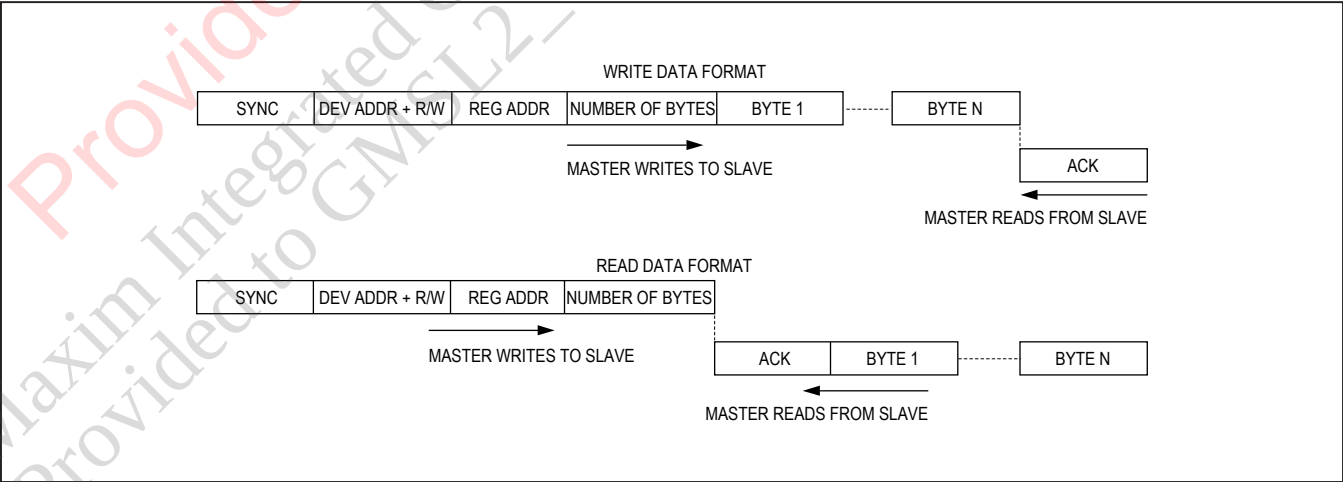


Figure 29. UART Protocol for Base Mode

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## Dual GMSL2/GMSL1 to CSI-2 Deserializer

**UART Frame Format**

Regular UART frames with an even parity bit are used to carry 1 byte of data each. A frame consists of a low start bit followed by 8 data bits, a parity bit, and a high stop bit. The parity bit is high if the number of ones in 8-bit data is odd, otherwise, it is low. There must be at least 1 high stop bit. If the next frame is in the same packet, there can be at most 4 high bits from the end of the stop bit to the beginning of the next start bit. When a parity bit error occurs, the packet, starting from the frame with the error, is discarded. The start of each frame is always a high-to-low transition (i.e., the stop bit is high and the start bit is low). The phase of the internal UART bit clock is adjusted using the start bit of each frame. The framer calibrates the length of one UART bit in terms of the internal oscillator clock using the synchronization frame (i.e., the first frame of a UART packet transmission). In bypass mode, the parity bit is enabled by default, but the frames are not checked for parity errors. Either even or odd parity can be used. The parity bit is passed along with UART data transmissions; the data recipient must perform error checking. The parity bit can optionally be disabled before entering bypass mode. Note that the bit rate in bypass mode must be the same bit rate last used in base mode.

**Synchronization Frame**

The serializer/deserializer must calibrate internal bit length counters with the UART bit rate for proper recovery of UART frames. The  $\mu$ C sends a sync frame (a regular UART frame with the value 0x79) as the first frame of each data packet. The sync frame allows the addressed device to calibrate bit lengths in terms of the device's internal 150 MHz clock. Sync frames must be properly

detected before the subsequent frames of the packet can be correctly received. When the line stays high for at least 32 bits, the packet boundary is reset and the framer begins waiting for the next sync frame.

**Acknowledge Frame**

When a packet is successfully received, the addressed device responds with an acknowledge frame to inform the  $\mu$ C that no errors were detected in the transmitted packet. The acknowledge frame is sent after the last bit of the valid packet has been received. The acknowledge frame is a regular UART frame (value 0xC3). Data written to the serializer/deserializer registers do not take effect until after the acknowledge byte is sent.

**Write Packet**

A write packet consists of a 5-byte packet header followed by 1 or more data bytes. A packet is recognized as a write packet when the LSB of the device address frame is 0. The addressed device responds with an acknowledge frame if no errors were detected while receiving the packet. Byte Count indicates the number of data bytes to be written and this number cannot be zero.

See [Figure 33](#).

**Read Packet**

A read packet consists of 5 bytes. The LSB of the device address frame is 1 for read packets. If no errors were detected while receiving a valid read packet, the addressed device responds with an acknowledge frame followed by one or more data bytes. Byte Count indicates the number of data bytes to be read and this number cannot be zero.

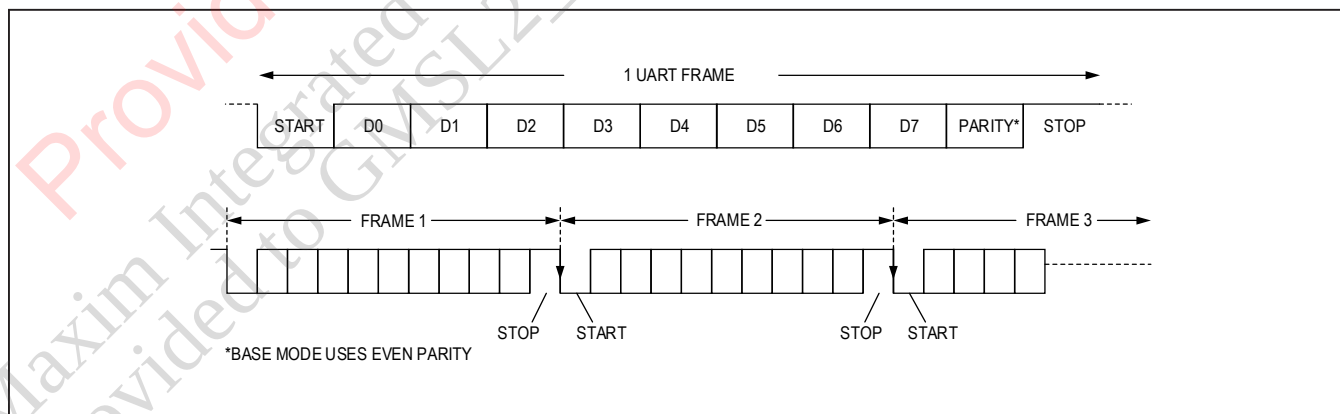


Figure 30. UART Data Format for Base Mode

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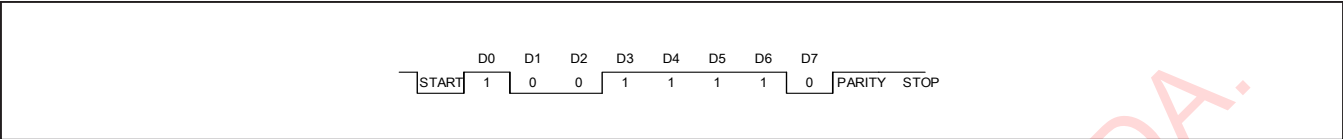


Figure 31. UART Synchronization Frame

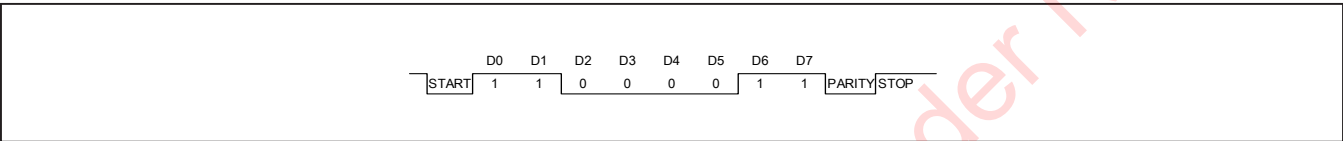


Figure 32. UART Acknowledge Frame

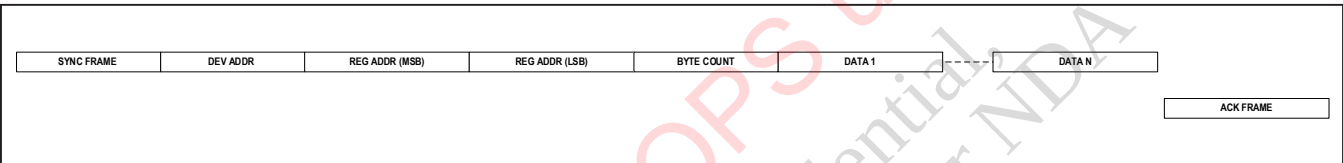


Figure 33. UART Write Packet Format

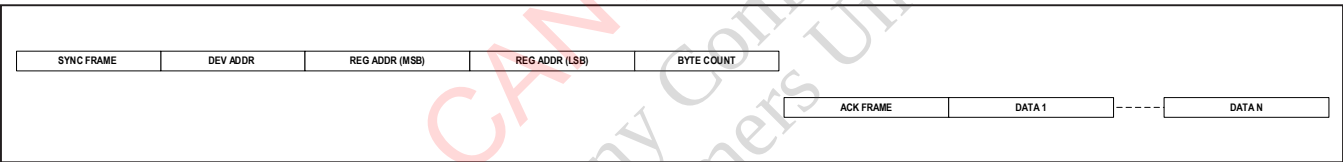


Figure 34. UART Read Packet Format

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## Ordering Information

| PART NUMBER      | TEMP RANGE      | PIN-PACKAGE    | TOP MARKING |
|------------------|-----------------|----------------|-------------|
| MAX9296BGTM/V+   | -40°C to +105°C | 48 TQFN-EP*    | MAXIM       |
| MAX9296BGTM/V+T  | -40°C to +105°C | 48 TQFN-EP*    | MAXIM       |
| MAX9296BGTM/VY+  | -40°C to +105°C | 48 TQFN-SW-EP* | MAXIM       |
| MAX9296BGTM/VY+T | -40°C to +105°C | 48 TQFN-SW-EP* | MAXIM       |

+Denotes a lead(Pb)-free/RoHS-compliant package.

T = Tape and reel.

/V Denotes an automotive qualified product.

Y Denotes wettable flank.

\*EP = Exposed pad.

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## Revision History

| REVISION NUMBER | REVISION DATE | DESCRIPTION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | PAGES CHANGED |
|-----------------|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| 0               | 4/19          | Initial release                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | —             |
| 1               | 6/19          | Updated TOCs 1–8 and <i>Pin Description</i> section                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 15, 17        |
| 2               | 12/19         | Removed Single-Layer Board information from <i>Absolute Maximum Ratings</i> and from <i>Package Information</i> . Updated <i>AC Electrical Characteristics</i> GMSL2 Master MOSI Data Output Delay and GMSL2 SPI Slave MISO Output Delay rows. Updated Figure 17 and Figure 16. Updated <i>Switching Between UART Base and Bypass Modes</i> section.                                                                                                                                                                                                                                                                                           | 7, 13, 31, 52 |
| 3               | 5/21          | Updated GMSL2 MAX values for Power Supply Currents: <ul style="list-style-type: none"> <li>• Single 6Gbps link: <math>V_{DD} = 1.05V</math> from 460mA to 550mA and <math>V_{DD} = 1.26V</math> from 460mA to 550mA</li> <li>• Two 6Gbps links: <math>V_{DD} = 1.05V</math> from 640mA to 690mA and <math>V_{DD} = 1.26V</math> from 640mA to 690mA</li> </ul> Updated GMSL2 MAX values for Power Supply Currents: <ul style="list-style-type: none"> <li>• <math>V_{DD18} = 1.9V</math> from 100mA to 110mA</li> <li>• <math>V_{DD} = 1.05V</math> from 410mA to 490mA</li> <li>• <math>V_{DD} = 1.26V</math> from 410mA to 490mA.</li> </ul> | 11            |

For pricing, delivery, and ordering information, please visit Maxim Integrated's online storefront at <https://www.maximintegrated.com/en/storefront/storefront.html>.

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Click [here](#) for production status of specific part numbers.

## MAX9296B Dual 1 x 4 or 2 x 2 CSI-2 GMSL2 Deserializer

## Appendix A: Register Map and Tables

### Reserved, Unused, and Read-Only Register Bits

Not all register bits in the register space are shown in the register table. Any bit not explicitly defined in the register table should be treated as reserved and should not be modified. When a write is required to a register with both defined and undefined register bits, first read the register's contents, then create a new register value by only changing the defined bits, and finally write the new byte to the register (Read/Replace/Write).

In this document, default values are provided for read-only register bits. Read-only bit states are changed at powerup according to the actual state of the device. To avoid overwriting these bits, treat read-only bits as undefined.

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Appendix A: Register Map and Tables

Register Map

MAX9296B Dual 1 x 4 or 2 x 2 CSI-2 GMSL2 Deserializer

| ADDRESS | NAME       | MSB              |                 |                 |                 |                 |                  |               | LSB           |
|---------|------------|------------------|-----------------|-----------------|-----------------|-----------------|------------------|---------------|---------------|
| DEV     |            |                  |                 |                 |                 |                 |                  |               |               |
| 0x00    | REG0[7:0]  | DEV_ADDR[6:0]    |                 |                 |                 |                 |                  |               | CFG_BLK_OCK   |
| 0x01    | REG1[7:0]  | IIC_2_EN_N       | IIC_1_EN_N      | DIS_LO_CAL_CC   | DIS_REM_CC      | TX_RATE[1:0]    |                  | RX_RATE[1:0]  |               |
| 0x02    | REG2[7:0]  | VID_EN_U         | VID_EN_Z        | VID_EN_Y        | VID_EN_X        | -               | -                | -             | -             |
| 0x03    | REG3[7:0]  | LOCK_CFG         | PT_SWAP_P       | UART_2_EN       | UART_1_EN       | -               | -                | -             | -             |
| 0x05    | REG5[7:0]  | LOCK_EN_N        | ERRB_EN_N       | -               | -               | -               | -                | -             | -             |
| 0x06    | REG6[7:0]  | -                | -               | -               | I2CSEL          | -               | -                | -             | -             |
| 0x0D    | REG13[7:0] | DEV_ID[7:0]      |                 |                 |                 |                 |                  |               |               |
| 0x0E    | REG14[7:0] | -                | -               | -               | -               | DEV_REV[3:0]    |                  |               |               |
| 0x0F    | REG15[7:0] | -                | -               | SPEED_CPBL[1:0] |                 | DV_CPBL_L       | DUAL_CPBL        | SPLTR_CPBL    | HDCP_CPBL     |
| OVERLAP |            |                  |                 |                 |                 |                 |                  |               |               |
| TCTRL   |            |                  |                 |                 |                 |                 |                  |               |               |
| 0x09    | PWR1[7:0]  | OVERTEMP         | VDD_OV_LIVE     | -               | -               | -               | -                | -             | -             |
| 0x0C    | PWR4[7:0]  | -                | DIS_LO_CAL_WAKE | WAKE_EN_N_B     | WAKE_EN_N_A     | -               | -                | -             | -             |
| 0x10    | CTRL0[7:0] | RESET_ALL        | RESET_LINK      | RESET_ONESHOT   | AUTO_LINK       | SLEEP           | REG_ENABLE       | LINK_CFG[1:0] |               |
| 0x11    | CTRL1[7:0] | -                | -               | -               | BACK_COMP_SPLTR | -               | CXTP_B           | -             | CXTP_A        |
| 0x12    | CTRL2[7:0] | -                | -               | -               | REG_MNL         | -               | -                | -             | -             |
| 0x13    | CTRL3[7:0] | -                | -               | LINK_MODE[1:0]  |                 | LOCKED          | ERROR            | CMU_LOCKED    | -             |
| 0x18    | INTR0[7:0] | -                | -               | -               | -               | AUTO_ERR_RST_EN | DEC_ERR_THR[2:0] |               |               |
| 0x19    | INTR1[7:0] | PKT_CNT_EXP[3:0] |                 |                 |                 | AUTO_CNT_RST_EN | PKT_CNT_THR[2:0] |               |               |
| 0x1A    | INTR2[7:0] | PHY_INT_OEN_B    | PHY_INT_OEN_A   | REM_ERR_OEN     | MEM_INT_ERR_OEN | -               | IDLE_ERR_OEN     | DEC_ERR_OEN_B | DEC_ERR_OEN_A |

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Appendix A: Register Map and Tables

| ADDRESS | NAME                                 | MSB                      |                |                    |                   |                  |                |                 | LSB                 |
|---------|--------------------------------------|--------------------------|----------------|--------------------|-------------------|------------------|----------------|-----------------|---------------------|
| 0x1B    | <a href="#">INTR3[7:0]</a>           | PHY_IN_T_B               | PHY_IN_T_A     | REM_ER_R_FLAG      | MEM_IN_T_ERR_FLAG | -                | IDLE_ER_R_FLAG | DEC_ER_R_FLAG_B | DEC_ER_R_FLAG_A     |
| 0x1C    | <a href="#">INTR4[7:0]</a>           | EOM_ERR_OEN_B            | EOM_ERR_OEN_A  | -                  | -                 | MAX_RT_OEN       | RT_CNT_OEN     | PKT_CNT_OEN     | WM_ERR_OEN          |
| 0x1D    | <a href="#">INTR5[7:0]</a>           | EOM_ERR_FLAG_B           | EOM_ERR_FLAG_A | -                  | -                 | MAX_RT_FLAG      | RT_CNT_FLAG    | PKT_CNT_FLAG    | WM_ERR_FLAG         |
| 0x1E    | <a href="#">INTR6[7:0]</a>           | VDDCMP_INT_OEN           | PORZ_INT_OEN   | VDDBAD_INT_OEN     | FSYNC_ERR_OEN     | LCRC_ERR_OEN     | VPRBS_ERR_OEN  | -               | VID_PXL_CRC_ERR_OEN |
| 0x1F    | <a href="#">INTR7[7:0]</a>           | VDDCMP_INT_FLAG          | PORZ_INT_FLAG  | VDDBAD_INT_FLAG    | -                 | LCRC_ERR_FLAG    | VPRBS_ERR_FLAG | -               | VID_PXL_CRC_ERR     |
| 0x20    | <a href="#">INTR8[7:0]</a>           | ERR_TX_EN                | -              | -                  | ERR_TX_ID[4:0]    |                  |                |                 |                     |
| 0x21    | <a href="#">INTR9[7:0]</a>           | ERR_RX_EN                | ERR_RX_RECVD   | -                  | ERR_RX_ID[4:0]    |                  |                |                 |                     |
| 0x22    | <a href="#">CNT0[7:0]</a>            | DEC_ERR_A[7:0]           |                |                    |                   |                  |                |                 |                     |
| 0x23    | <a href="#">CNT1[7:0]</a>            | DEC_ERR_B[7:0]           |                |                    |                   |                  |                |                 |                     |
| 0x24    | <del><a href="#">CNT2[7:0]</a></del> | <del>IDLE_ERR[7:0]</del> |                |                    |                   |                  |                |                 |                     |
| 0x25    | <a href="#">CNT3[7:0]</a>            | PKT_CNT[7:0]             |                |                    |                   |                  |                |                 |                     |
| GMSL    |                                      |                          |                |                    |                   |                  |                |                 |                     |
| 0x2A    | <a href="#">TX2[7:0]</a>             | ERRG_CNT[1:0]            |                | ERRG_RATE[1:0]     |                   | ERRG_BURST[2:0]  |                |                 | ERRG_PERR           |
| 0x2B    | <a href="#">TX3[7:0]</a>             | -                        | -              | -                  | -                 | -                | TIMEOUT[2:0]   |                 |                     |
| 0x2C    | <a href="#">RX0[7:0]</a>             | PKT_CNT_LBW[1:0]         |                | -                  | -                 | PKT_CNT_SEL[3:0] |                |                 |                     |
| 0x2F    | <a href="#">RX3[7:0]</a>             | PRBS_SYNCED_B            | SYNC_LOCKED_B  | WBLOCK_B           | FAILLOCK_B        | PRBS_SYNCED_A    | SYNC_LOCKED_A  | WBLOCK_A        | FAILLOCK_A          |
| 0x30    | <a href="#">GPIOA[7:0]</a>           | GPIO_RX_FAST_BIDIR_EN    | -              | GPIO_FWD_CDLY[5:0] |                   |                  |                |                 |                     |
| 0x31    | <a href="#">GPIOB[7:0]</a>           | GPIO_TX_WNDW[1:0]        |                | GPIO_REV_CDLY[5:0] |                   |                  |                |                 |                     |
| CC      |                                      |                          |                |                    |                   |                  |                |                 |                     |
| 0x40    | <a href="#">I2C_0[7:0]</a>           | -                        | -              | SLV_SH[1:0]        |                   | -                | SLV_TO[2:0]    |                 |                     |
| 0x41    | <a href="#">I2C_1[7:0]</a>           | -                        | MST_BT[2:0]    |                    |                   | -                | MST_TO[2:0]    |                 |                     |
| 0x42    | <a href="#">I2C_2[7:0]</a>           | SRC_A[6:0]               |                |                    |                   |                  |                |                 | -                   |
| 0x43    | <a href="#">I2C_3[7:0]</a>           | DST_A[6:0]               |                |                    |                   |                  |                |                 | -                   |
| 0x44    | <a href="#">I2C_4[7:0]</a>           | SRC_B[6:0]               |                |                    |                   |                  |                |                 | -                   |
| 0x45    | <a href="#">I2C_5[7:0]</a>           | DST_B[6:0]               |                |                    |                   |                  |                |                 | -                   |
| 0x46    | <a href="#">I2C_6[7:0]</a>           | -                        | -              | -                  | -                 | I2C_AUTO_CFG     | -              | -               | -                   |

HIDE

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Appendix A: Register Map and Tables

| ADDRESS      | NAME                           | MSB                      |                          |                            |                            |                          |                         |                            | LSB                        |
|--------------|--------------------------------|--------------------------|--------------------------|----------------------------|----------------------------|--------------------------|-------------------------|----------------------------|----------------------------|
| 0x47         | <a href="#">I2C_7[7:0]</a>     | UART_R<br>X_OVER<br>FLOW | UART_T<br>X_OVER<br>FLOW | –                          | –                          | –                        | I2C_TIM<br>ED_OUT       | REM_AC<br>K_ACKE<br>D      | REM_AC<br>K_RECV<br>ED     |
| 0x48         | <a href="#">UART_0[7:0]</a>    | ARB_TO_LEN[1:0]          |                          | REM_M<br>S_EN              | LOC_MS<br>_EN              | BYPASS<br>_DIS_PA<br>R   | BYPASS_TO[1:0]          |                            | BYPASS<br>_EN              |
| 0x49         | <a href="#">UART_1[7:0]</a>    | BITLEN_LSB[7:0]          |                          |                            |                            |                          |                         |                            |                            |
| 0x4A         | <a href="#">UART_2[7:0]</a>    | OUT_DELAY[1:0]           |                          | BITLEN_MSB[5:0]            |                            |                          |                         |                            |                            |
| 0x4C         | <a href="#">I2C_PT_0[7:0]</a>  | –                        | –                        | SLV_SH_PT[1:0]             |                            | –                        | SLV_TO_PT[2:0]          |                            |                            |
| 0x4D         | <a href="#">I2C_PT_1[7:0]</a>  | –                        | MST_BT_PT[2:0]           |                            |                            | –                        | MST_TO_PT[2:0]          |                            |                            |
| 0x4E         | <a href="#">I2C_PT_2[7:0]</a>  | XOVER_<br>EN_2           | I2C_TIM<br>ED_OUT<br>_2  | REM_AC<br>K_ACKE<br>D_2    | REM_AC<br>K_RECV<br>ED_2   | XOVER_<br>EN_1           | I2C_TIM<br>ED_OUT<br>_1 | REM_AC<br>K_ACKE<br>D_1    | REM_AC<br>K_RECV<br>ED_1   |
| 0x4F         | <a href="#">UART_PT_0[7:0]</a> | BITLEN_<br>MAN_CF<br>G_2 | DIS_PA<br>R_2            | UART_R<br>X_OVER<br>FLOW_2 | UART_T<br>X_OVER<br>FLOW_2 | BITLEN_<br>MAN_CF<br>G_1 | DIS_PA<br>R_1           | UART_R<br>X_OVER<br>FLOW_1 | UART_T<br>X_OVER<br>FLOW_1 |
| CFGH VIDEO_X |                                |                          |                          |                            |                            |                          |                         |                            |                            |
| 0x50         | <a href="#">RX0[7:0]</a>       | RX_CRC<br>_EN            | –                        | –                          | –                          | –                        | –                       | STR_SEL[1:0]               |                            |
| CFGH VIDEO_Y |                                |                          |                          |                            |                            |                          |                         |                            |                            |
| 0x51         | <a href="#">RX0[7:0]</a>       | RX_CRC<br>_EN            | –                        | –                          | –                          | –                        | –                       | STR_SEL[1:0]               |                            |
| CFGH VIDEO_Z |                                |                          |                          |                            |                            |                          |                         |                            |                            |
| 0x52         | <a href="#">RX0[7:0]</a>       | RX_CRC<br>_EN            | –                        | –                          | –                          | –                        | –                       | STR_SEL[1:0]               |                            |
| CFGH VIDEO_U |                                |                          |                          |                            |                            |                          |                         |                            |                            |
| 0x53         | <a href="#">RX0[7:0]</a>       | RX_CRC<br>_EN            | –                        | –                          | –                          | –                        | –                       | STR_SEL[1:0]               |                            |
| CFGH INFOFR  |                                |                          |                          |                            |                            |                          |                         |                            |                            |
| 0x60         | <a href="#">TR0[7:0]</a>       | TX_CRC<br>_EN            | RX_CRC<br>_EN            | –                          | –                          | PRIO_VAL[1:0]            |                         | PRIO_CFG[1:0]              |                            |
| 0x61         | <a href="#">TR1[7:0]</a>       | BW_MULT[1:0]             |                          |                            | BW_VAL[5:0]                |                          |                         |                            |                            |
| 0x63         | <a href="#">TR3[7:0]</a>       | –                        | –                        | TX_SPL<br>T_MASK<br>_B     | TX_SPL<br>T_MASK<br>_A     | –                        | TX_SRC_ID[2:0]          |                            |                            |
| 0x64         | <a href="#">TR4[7:0]</a>       | RX_SRC_SEL[7:0]          |                          |                            |                            |                          |                         |                            |                            |
| CFGH SPI     |                                |                          |                          |                            |                            |                          |                         |                            |                            |
| 0x68         | <a href="#">TR0[7:0]</a>       | TX_CRC<br>_EN            | RX_CRC<br>_EN            | –                          | –                          | PRIO_VAL[1:0]            |                         | PRIO_CFG[1:0]              |                            |
| 0x69         | <a href="#">TR1[7:0]</a>       | BW_MULT[1:0]             |                          |                            | BW_VAL[5:0]                |                          |                         |                            |                            |
| 0x6B         | <a href="#">TR3[7:0]</a>       | –                        | –                        | TX_SPL<br>T_MASK<br>_B     | TX_SPL<br>T_MASK<br>_A     | –                        | TX_SRC_ID[2:0]          |                            |                            |
| 0x6C         | <a href="#">TR4[7:0]</a>       | RX_SRC_SEL[7:0]          |                          |                            |                            |                          |                         |                            |                            |
| 0x6D         | <a href="#">ARQ0[7:0]</a>      | ARQ_AU<br>TO_CFG         | ACK_CN<br>T              | MATCH_<br>SRC_ID           | ACK_SR<br>C_ID             | EN                       | –                       | –                          | –                          |

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| ADDRESS     | NAME      | MSB             |             |                 |                 |               |                |                | LSB        |
|-------------|-----------|-----------------|-------------|-----------------|-----------------|---------------|----------------|----------------|------------|
| 0x6E        | ARQ1[7:0] | –               | MAX_RT[2:0] |                 |                 | –             | –              | MAX_RT_ERR_OEN | RT_CNT_OEN |
| 0x6F        | ARQ2[7:0] | MAX_RT_ERR      | RT_CNT[6:0] |                 |                 |               |                |                |            |
| CFG_C CC    |           |                 |             |                 |                 |               |                |                |            |
| 0x70        | TR0[7:0]  | TX_CRC_EN       | RX_CRC_EN   | –               | –               | PRIO_VAL[1:0] |                | PRIO_CFG[1:0]  |            |
| 0x71        | TR1[7:0]  | BW_MULT[1:0]    |             | BW_VAL[5:0]     |                 |               |                |                |            |
| 0x73        | TR3[7:0]  | –               | –           | TX_SPL_T_MASK_B | TX_SPL_T_MASK_A | –             | TX_SRC_ID[2:0] |                |            |
| 0x74        | TR4[7:0]  | RX_SRC_SEL[7:0] |             |                 |                 |               |                |                |            |
| 0x75        | ARQ0[7:0] | ARQ_AU_TO_CFG   | ACK_CNT     | MATCH_SRC_ID    | ACK_SRC_ID      | EN            | –              | –              | –          |
| 0x76        | ARQ1[7:0] | –               | MAX_RT[2:0] |                 |                 | –             | –              | MAX_RT_ERR_OEN | RT_CNT_OEN |
| 0x77        | ARQ2[7:0] | MAX_RT_ERR      | RT_CNT[6:0] |                 |                 |               |                |                |            |
| CFG_L GPIO  |           |                 |             |                 |                 |               |                |                |            |
| 0x78        | TR0[7:0]  | TX_CRC_EN       | RX_CRC_EN   | –               | –               | PRIO_VAL[1:0] |                | PRIO_CFG[1:0]  |            |
| 0x79        | TR1[7:0]  | BW_MULT[1:0]    |             | BW_VAL[5:0]     |                 |               |                |                |            |
| 0x7B        | TR3[7:0]  | –               | –           | TX_SPL_T_MASK_B | TX_SPL_T_MASK_A | –             | TX_SRC_ID[2:0] |                |            |
| 0x7C        | TR4[7:0]  | RX_SRC_SEL[7:0] |             |                 |                 |               |                |                |            |
| 0x7D        | ARQ0[7:0] | ARQ_AU_TO_CFG   | ACK_CNT     | MATCH_SRC_ID    | ACK_SRC_ID      | EN            | –              | –              | –          |
| 0x7E        | ARQ1[7:0] | –               | MAX_RT[2:0] |                 |                 | –             | –              | MAX_RT_ERR_OEN | RT_CNT_OEN |
| 0x7F        | ARQ2[7:0] | MAX_RT_ERR      | RT_CNT[6:0] |                 |                 |               |                |                |            |
| CFG_C IIC_X |           |                 |             |                 |                 |               |                |                |            |
| 0x80        | TR0[7:0]  | TX_CRC_EN       | RX_CRC_EN   | –               | –               | PRIO_VAL[1:0] |                | PRIO_CFG[1:0]  |            |
| 0x81        | TR1[7:0]  | BW_MULT[1:0]    |             | BW_VAL[5:0]     |                 |               |                |                |            |
| 0x83        | TR3[7:0]  | –               | –           | TX_SPL_T_MASK_B | TX_SPL_T_MASK_A | –             | TX_SRC_ID[2:0] |                |            |
| 0x84        | TR4[7:0]  | RX_SRC_SEL[7:0] |             |                 |                 |               |                |                |            |
| 0x85        | ARQ0[7:0] | ARQ_AU_TO_CFG   | ACK_CNT     | MATCH_SRC_ID    | ACK_SRC_ID      | EN            | –              | –              | –          |

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| ADDRESS    | NAME                            | MSB             |               |                |                |               |                |                | LSB         |
|------------|---------------------------------|-----------------|---------------|----------------|----------------|---------------|----------------|----------------|-------------|
| 0x86       | <a href="#">ARQ1[7:0]</a>       | –               | MAX_RT[2:0]   |                |                | –             | –              | MAX_RT_ERR_OEN | RT_CNT_OEN  |
| 0x87       | <a href="#">ARQ2[7:0]</a>       | MAX_RT_ERR      | RT_CNT[6:0]   |                |                |               |                |                |             |
| CFGC IIC_Y |                                 |                 |               |                |                |               |                |                |             |
| 0x88       | <a href="#">TR0[7:0]</a>        | TX_CRC_EN       | RX_CRC_EN     | –              | –              | PRIO_VAL[1:0] |                | PRIO_CFG[1:0]  |             |
| 0x89       | <a href="#">TR1[7:0]</a>        | BW_MULT[1:0]    |               |                | BW_VAL[5:0]    |               |                |                |             |
| 0x8B       | <a href="#">TR3[7:0]</a>        | –               | –             | TX_SPLT_MASK_B | TX_SPLT_MASK_A | –             | TX_SRC_ID[2:0] |                |             |
| 0x8C       | <a href="#">TR4[7:0]</a>        | RX_SRC_SEL[7:0] |               |                |                |               |                |                |             |
| 0x8D       | <a href="#">ARQ0[7:0]</a>       | ARQ_AU_TO_CFG   | ACK_CNT       | MATCH_SRC_ID   | ACK_SRC_ID     | EN            | –              | –              | –           |
| 0x8E       | <a href="#">ARQ1[7:0]</a>       | –               | MAX_RT[2:0]   |                |                | –             | –              | MAX_RT_ERR_OEN | RT_CNT_OEN  |
| 0x8F       | <a href="#">ARQ2[7:0]</a>       | MAX_RT_ERR      | RT_CNT[6:0]   |                |                |               |                |                |             |
| VID_RX X   |                                 |                 |               |                |                |               |                |                |             |
| 0x100      | <a href="#">VIDEO_RX0[7:0]</a>  | LCRC_ERR        | –             | –              | –              | –             | LINE_CRC_SEL   | LINE_CRC_EN    | DIS_PKT_DET |
| 0x103      | <a href="#">VIDEO_RX3[7:0]</a>  | –               | HD_TR_MODE    | DLOCKED        | VLOCKED        | HLOCKED       | DTRACKEN       | VTRACKEN       | HTRACKEN    |
| 0x108      | <a href="#">VIDEO_RX8[7:0]</a>  | VID_BLK_LEN_ERR | VID_LOCK      | VID_PKT_DET    | VID_SEQ_ERR    | –             | –              | –              | –           |
| 0x10A      | <a href="#">VIDEO_RX10[7:0]</a> | –               | MASK_VIDEO_DE | –              | –              | –             | –              | –              | –           |
| 0x10E      | <a href="#">INFO_RX1[7:0]</a>   | SPEED_LIM[1:0]  |               | –              | –              | –             | –              | –              | –           |
| VID_RX Y   |                                 |                 |               |                |                |               |                |                |             |
| 0x112      | <a href="#">VIDEO_RX0[7:0]</a>  | LCRC_ERR        | –             | –              | –              | –             | LINE_CRC_SEL   | LINE_CRC_EN    | DIS_PKT_DET |
| 0x115      | <a href="#">VIDEO_RX3[7:0]</a>  | –               | HD_TR_MODE    | DLOCKED        | VLOCKED        | HLOCKED       | DTRACKEN       | VTRACKEN       | HTRACKEN    |
| 0x11A      | <a href="#">VIDEO_RX8[7:0]</a>  | VID_BLK_LEN_ERR | VID_LOCK      | VID_PKT_DET    | VID_SEQ_ERR    | –             | –              | –              | –           |
| 0x11C      | <a href="#">VIDEO_RX10[7:0]</a> | –               | MASK_VIDEO_DE | –              | –              | –             | –              | –              | –           |
| 0x120      | <a href="#">INFO_RX1[7:0]</a>   | SPEED_LIM[1:0]  |               | –              | –              | –             | –              | –              | –           |
| VID_RX Z   |                                 |                 |               |                |                |               |                |                |             |
| 0x124      | <a href="#">VIDEO_RX0[7:0]</a>  | LCRC_ERR        | –             | –              | –              | –             | LINE_CRC_SEL   | LINE_CRC_EN    | DIS_PKT_DET |



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| ADDRESS  | NAME                            | MSB                   |               |                    |                    |              |              |                    | LSB           |
|----------|---------------------------------|-----------------------|---------------|--------------------|--------------------|--------------|--------------|--------------------|---------------|
| 0x127    | <a href="#">VIDEO_RX3[7:0]</a>  | –                     | HD_TR_MODE    | DLOCKE D           | VLOCKE D           | HLOCKE D     | DTRACK EN    | VTRACK EN          | HTRACK EN     |
| 0x12C    | <a href="#">VIDEO_RX8[7:0]</a>  | VID_BLK_LEN_ERR       | VID_LOCK      | VID_PKT_DET        | VID_SEQ_ERR        | –            | –            | –                  | –             |
| 0x12E    | <a href="#">VIDEO_RX10[7:0]</a> | –                     | MASK_VIDEO_DE | –                  | –                  | –            | –            | –                  | –             |
| 0x132    | <a href="#">INFO_RX1[7:0]</a>   | SPEED_LIM[1:0]        |               | –                  | –                  | –            | –            | –                  | –             |
| VID_RX U |                                 |                       |               |                    |                    |              |              |                    |               |
| 0x136    | <a href="#">VIDEO_RX0[7:0]</a>  | LCRC_ERR              | –             | –                  | –                  | –            | LINE_CRC_SEL | LINE_CRC_EN        | DIS_PKT_DET   |
| 0x139    | <a href="#">VIDEO_RX3[7:0]</a>  | –                     | HD_TR_MODE    | DLOCKE D           | VLOCKE D           | HLOCKE D     | DTRACK EN    | VTRACK EN          | HTRACK EN     |
| 0x13E    | <a href="#">VIDEO_RX8[7:0]</a>  | VID_BLK_LEN_ERR       | VID_LOCK      | VID_PKT_DET        | VID_SEQ_ERR        | –            | –            | –                  | –             |
| 0x140    | <a href="#">VIDEO_RX10[7:0]</a> | –                     | MASK_VIDEO_DE | –                  | –                  | –            | –            | –                  | –             |
| 0x144    | <a href="#">INFO_RX1[7:0]</a>   | SPEED_LIM[1:0]        |               | –                  | –                  | –            | –            | –                  | –             |
| SPI      |                                 |                       |               |                    |                    |              |              |                    |               |
| 0x170    | <a href="#">SPI_0[7:0]</a>      | SPI_LOC_ID[1:0]       |               | SPI_CC_TRG_ID[1:0] |                    | SPI_IGN_R_ID | SPI_CC_EN    | MST_SLVN           | SPI_EN        |
| 0x171    | <a href="#">SPI_1[7:0]</a>      | SPI_LOC_N[5:0]        |               |                    |                    |              |              | SPI_BASE_PRIO[1:0] |               |
| 0x172    | <a href="#">SPI_2[7:0]</a>      | REQ_HOLD_OFF[2:0]     |               |                    | FULL_SCK_SETUP     | SPI_MO_D3_F  | SPI_MO_D3    | SPIM_S2_ACT_H      | SPIM_S1_ACT_H |
| 0x173    | <a href="#">SPI_3[7:0]</a>      | SPIM_SS_DLY_CLKS[7:0] |               |                    |                    |              |              |                    |               |
| 0x174    | <a href="#">SPI_4[7:0]</a>      | SPIM_SCK_LO_CLKS[7:0] |               |                    |                    |              |              |                    |               |
| 0x175    | <a href="#">SPI_5[7:0]</a>      | SPIM_SCK_HI_CLKS[7:0] |               |                    |                    |              |              |                    |               |
| 0x176    | <a href="#">SPI_6[7:0]</a>      | –                     | –             | BNE                | SPIS_RWN           | SS_IO_EN_2   | SS_IO_EN_1   | BNE_IO_EN          | RWN_IO_EN     |
| 0x177    | <a href="#">SPI_7[7:0]</a>      | SPI_RX_OVRFLW         | SPI_TX_OVRFLW | –                  | SPIS_BYTE_CNT[4:0] |              |              |                    |               |
| 0x178    | <a href="#">SPI_8[7:0]</a>      | REQ_HOLD_OFF_TO[7:0]  |               |                    |                    |              |              |                    |               |
| WM       |                                 |                       |               |                    |                    |              |              |                    |               |
| 0x190    | <a href="#">WM_0[7:0]</a>       | WM_LEN                | WM_MODE[2:0]  |                    |                    | WM_DET[1:0]  |              | –                  | WM_EN         |
| 0x192    | <a href="#">WM_2[7:0]</a>       | –                     | –             | –                  | –                  | HsyncPol     | VsyncPol     | WM_NPFILT[1:0]     |               |
| 0x193    | <a href="#">WM_3[7:0]</a>       | –                     | WM_TH[6:0]    |                    |                    |              |              |                    |               |
| 0x194    | <a href="#">WM_4[7:0]</a>       | –                     | –             | WM_YUV_IN[1:0]     |                    | WM_CO_LORADJ | –            | WM_MASKMODE[1:0]   |               |

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|---------|--------------------------------|----------------|---------------|---------------|--------------|---|---|---------------|--------------|
| 0x195   | <a href="#">WM_5[7:0]</a>      | –              | –             | –             | –            | – | – | WM_DE<br>TOUT | WM_ER<br>ROR |
| 0x196   | <a href="#">WM_6[7:0]</a>      | WM_TIMER[7:0]  |               |               |              |   |   |               |              |
| 0x1AE   | <a href="#">WM_WREN_0[7:0]</a> | WM_WREN_L[7:0] |               |               |              |   |   |               |              |
| 0x1AF   | <a href="#">WM_WREN_1[7:0]</a> | WM_WREN_H[7:0] |               |               |              |   |   |               |              |
| VRX X   |                                |                |               |               |              |   |   |               |              |
| 0x1C0   | <a href="#">CROSS_0[7:0]</a>   | –              | CROSS0<br>_I  | CROSS0<br>_F  | CROSS0[4:0]  |   |   |               |              |
| 0x1C1   | <a href="#">CROSS_1[7:0]</a>   | –              | CROSS1<br>_I  | CROSS1<br>_F  | CROSS1[4:0]  |   |   |               |              |
| 0x1C2   | <a href="#">CROSS_2[7:0]</a>   | –              | CROSS2<br>_I  | CROSS2<br>_F  | CROSS2[4:0]  |   |   |               |              |
| 0x1C3   | <a href="#">CROSS_3[7:0]</a>   | –              | CROSS3<br>_I  | CROSS3<br>_F  | CROSS3[4:0]  |   |   |               |              |
| 0x1C4   | <a href="#">CROSS_4[7:0]</a>   | –              | CROSS4<br>_I  | CROSS4<br>_F  | CROSS4[4:0]  |   |   |               |              |
| 0x1C5   | <a href="#">CROSS_5[7:0]</a>   | –              | CROSS5<br>_I  | CROSS5<br>_F  | CROSS5[4:0]  |   |   |               |              |
| 0x1C6   | <a href="#">CROSS_6[7:0]</a>   | –              | CROSS6<br>_I  | CROSS6<br>_F  | CROSS6[4:0]  |   |   |               |              |
| 0x1C7   | <a href="#">CROSS_7[7:0]</a>   | –              | CROSS7<br>_I  | CROSS7<br>_F  | CROSS7[4:0]  |   |   |               |              |
| 0x1C8   | <a href="#">CROSS_8[7:0]</a>   | –              | CROSS8<br>_I  | CROSS8<br>_F  | CROSS8[4:0]  |   |   |               |              |
| 0x1C9   | <a href="#">CROSS_9[7:0]</a>   | –              | CROSS9<br>_I  | CROSS9<br>_F  | CROSS9[4:0]  |   |   |               |              |
| 0x1CA   | <a href="#">CROSS_10[7:0]</a>  | –              | CROSS1<br>0_I | CROSS1<br>0_F | CROSS10[4:0] |   |   |               |              |
| 0x1CB   | <a href="#">CROSS_11[7:0]</a>  | –              | CROSS1<br>1_I | CROSS1<br>1_F | CROSS11[4:0] |   |   |               |              |
| 0x1CC   | <a href="#">CROSS_12[7:0]</a>  | –              | CROSS1<br>2_I | CROSS1<br>2_F | CROSS12[4:0] |   |   |               |              |
| 0x1CD   | <a href="#">CROSS_13[7:0]</a>  | –              | CROSS1<br>3_I | CROSS1<br>3_F | CROSS13[4:0] |   |   |               |              |
| 0x1CE   | <a href="#">CROSS_14[7:0]</a>  | –              | CROSS1<br>4_I | CROSS1<br>4_F | CROSS14[4:0] |   |   |               |              |
| 0x1CF   | <a href="#">CROSS_15[7:0]</a>  | –              | CROSS1<br>5_I | CROSS1<br>5_F | CROSS15[4:0] |   |   |               |              |
| 0x1D0   | <a href="#">CROSS_16[7:0]</a>  | –              | CROSS1<br>6_I | CROSS1<br>6_F | CROSS16[4:0] |   |   |               |              |
| 0x1D1   | <a href="#">CROSS_17[7:0]</a>  | –              | CROSS1<br>7_I | CROSS1<br>7_F | CROSS17[4:0] |   |   |               |              |
| 0x1D2   | <a href="#">CROSS_18[7:0]</a>  | –              | CROSS1<br>8_I | CROSS1<br>8_F | CROSS18[4:0] |   |   |               |              |
| 0x1D3   | <a href="#">CROSS_19[7:0]</a>  | –              | CROSS1<br>9_I | CROSS1<br>9_F | CROSS19[4:0] |   |   |               |              |

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|---------|-------------------------------|----------------|------------|------------|---------------|---|---|---|------------|
| 0x1D4   | <a href="#">CROSS_20[7:0]</a> | –              | CROSS2_0_I | CROSS2_0_F | CROSS20[4:0]  |   |   |   |            |
| 0x1D5   | <a href="#">CROSS_21[7:0]</a> | –              | CROSS2_1_I | CROSS2_1_F | CROSS21[4:0]  |   |   |   |            |
| 0x1D6   | <a href="#">CROSS_22[7:0]</a> | –              | CROSS2_2_I | CROSS2_2_F | CROSS22[4:0]  |   |   |   |            |
| 0x1D7   | <a href="#">CROSS_23[7:0]</a> | –              | CROSS2_3_I | CROSS2_3_F | CROSS23[4:0]  |   |   |   |            |
| 0x1D8   | <a href="#">CROSS_HS[7:0]</a> | –              | CROSS_HS_I | CROSS_HS_F | CROSS_HS[4:0] |   |   |   |            |
| 0x1D9   | <a href="#">CROSS_VS[7:0]</a> | –              | CROSS_VS_I | CROSS_VS_F | CROSS_VS[4:0] |   |   |   |            |
| 0x1DA   | <a href="#">CROSS_DE[7:0]</a> | –              | CROSS_DE_I | CROSS_DE_F | CROSS_DE[4:0] |   |   |   |            |
| 0x1DB   | <a href="#">PRBS_ERR[7:0]</a> | VPRBS_ERR[7:0] |            |            |               |   |   |   |            |
| 0x1DC   | <a href="#">VPRBS[7:0]</a>    | –              | –          | VPRBS_FAIL | VPRBS_CHK_EN  | – | – | – | VIDEO_LOCK |
| 0x1DD   | <a href="#">CROSS_27[7:0]</a> | ALT_CROSSBAR   | CROSS2_7_I | CROSS2_7_F | CROSS27[4:0]  |   |   |   |            |
| 0x1DE   | <a href="#">CROSS_28[7:0]</a> | –              | CROSS2_8_I | CROSS2_8_F | CROSS28[4:0]  |   |   |   |            |
| 0x1DF   | <a href="#">CROSS_29[7:0]</a> | –              | CROSS2_9_I | CROSS2_9_F | CROSS29[4:0]  |   |   |   |            |
| VRX Y   |                               |                |            |            |               |   |   |   |            |
| 0x1E0   | <a href="#">CROSS_0[7:0]</a>  | –              | CROSS0_I   | CROSS0_F   | CROSS0[4:0]   |   |   |   |            |
| 0x1E1   | <a href="#">CROSS_1[7:0]</a>  | –              | CROSS1_I   | CROSS1_F   | CROSS1[4:0]   |   |   |   |            |
| 0x1E2   | <a href="#">CROSS_2[7:0]</a>  | –              | CROSS2_I   | CROSS2_F   | CROSS2[4:0]   |   |   |   |            |
| 0x1E3   | <a href="#">CROSS_3[7:0]</a>  | –              | CROSS3_I   | CROSS3_F   | CROSS3[4:0]   |   |   |   |            |
| 0x1E4   | <a href="#">CROSS_4[7:0]</a>  | –              | CROSS4_I   | CROSS4_F   | CROSS4[4:0]   |   |   |   |            |
| 0x1E5   | <a href="#">CROSS_5[7:0]</a>  | –              | CROSS5_I   | CROSS5_F   | CROSS5[4:0]   |   |   |   |            |
| 0x1E6   | <a href="#">CROSS_6[7:0]</a>  | –              | CROSS6_I   | CROSS6_F   | CROSS6[4:0]   |   |   |   |            |
| 0x1E7   | <a href="#">CROSS_7[7:0]</a>  | –              | CROSS7_I   | CROSS7_F   | CROSS7[4:0]   |   |   |   |            |
| 0x1E8   | <a href="#">CROSS_8[7:0]</a>  | –              | CROSS8_I   | CROSS8_F   | CROSS8[4:0]   |   |   |   |            |
| 0x1E9   | <a href="#">CROSS_9[7:0]</a>  | –              | CROSS9_I   | CROSS9_F   | CROSS9[4:0]   |   |   |   |            |
| 0x1EA   | <a href="#">CROSS_10[7:0]</a> | –              | CROSS10_I  | CROSS10_F  | CROSS10[4:0]  |   |   |   |            |

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|---------|-------------------------------|----------------|------------|------------|---------------|---|---|---|------------|
| 0x1EB   | <a href="#">CROSS_11[7:0]</a> | –              | CROSS1_1_I | CROSS1_1_F | CROSS11[4:0]  |   |   |   |            |
| 0x1EC   | <a href="#">CROSS_12[7:0]</a> | –              | CROSS1_2_I | CROSS1_2_F | CROSS12[4:0]  |   |   |   |            |
| 0x1ED   | <a href="#">CROSS_13[7:0]</a> | –              | CROSS1_3_I | CROSS1_3_F | CROSS13[4:0]  |   |   |   |            |
| 0x1EE   | <a href="#">CROSS_14[7:0]</a> | –              | CROSS1_4_I | CROSS1_4_F | CROSS14[4:0]  |   |   |   |            |
| 0x1EF   | <a href="#">CROSS_15[7:0]</a> | –              | CROSS1_5_I | CROSS1_5_F | CROSS15[4:0]  |   |   |   |            |
| 0x1F0   | <a href="#">CROSS_16[7:0]</a> | –              | CROSS1_6_I | CROSS1_6_F | CROSS16[4:0]  |   |   |   |            |
| 0x1F1   | <a href="#">CROSS_17[7:0]</a> | –              | CROSS1_7_I | CROSS1_7_F | CROSS17[4:0]  |   |   |   |            |
| 0x1F2   | <a href="#">CROSS_18[7:0]</a> | –              | CROSS1_8_I | CROSS1_8_F | CROSS18[4:0]  |   |   |   |            |
| 0x1F3   | <a href="#">CROSS_19[7:0]</a> | –              | CROSS1_9_I | CROSS1_9_F | CROSS19[4:0]  |   |   |   |            |
| 0x1F4   | <a href="#">CROSS_20[7:0]</a> | –              | CROSS2_0_I | CROSS2_0_F | CROSS20[4:0]  |   |   |   |            |
| 0x1F5   | <a href="#">CROSS_21[7:0]</a> | –              | CROSS2_1_I | CROSS2_1_F | CROSS21[4:0]  |   |   |   |            |
| 0x1F6   | <a href="#">CROSS_22[7:0]</a> | –              | CROSS2_2_I | CROSS2_2_F | CROSS22[4:0]  |   |   |   |            |
| 0x1F7   | <a href="#">CROSS_23[7:0]</a> | –              | CROSS2_3_I | CROSS2_3_F | CROSS23[4:0]  |   |   |   |            |
| 0x1F8   | <a href="#">CROSS_HS[7:0]</a> | –              | CROSS_HS_I | CROSS_HS_F | CROSS_HS[4:0] |   |   |   |            |
| 0x1F9   | <a href="#">CROSS_VS[7:0]</a> | –              | CROSS_VS_I | CROSS_VS_F | CROSS_VS[4:0] |   |   |   |            |
| 0x1FA   | <a href="#">CROSS_DE[7:0]</a> | –              | CROSS_DE_I | CROSS_DE_F | CROSS_DE[4:0] |   |   |   |            |
| 0x1FB   | <a href="#">PRBS_ERR[7:0]</a> | VPRBS_ERR[7:0] |            |            |               |   |   |   |            |
| 0x1FC   | <a href="#">VPRBS[7:0]</a>    | –              | –          | VPRBS_FAIL | VPRBS_CHK_EN  | – | – | – | VIDEO_LOCK |
| 0x1FD   | <a href="#">CROSS_27[7:0]</a> | ALT_CR OSSBAR  | CROSS2_7_I | CROSS2_7_F | CROSS27[4:0]  |   |   |   |            |
| 0x1FE   | <a href="#">CROSS_28[7:0]</a> | –              | CROSS2_8_I | CROSS2_8_F | CROSS28[4:0]  |   |   |   |            |
| 0x1FF   | <a href="#">CROSS_29[7:0]</a> | –              | CROSS2_9_I | CROSS2_9_F | CROSS29[4:0]  |   |   |   |            |
| VRX Z   |                               |                |            |            |               |   |   |   |            |
| 0x200   | <a href="#">CROSS_0[7:0]</a>  | –              | CROSS0_I   | CROSS0_F   | CROSS0[4:0]   |   |   |   |            |
| 0x201   | <a href="#">CROSS_1[7:0]</a>  | –              | CROSS1_I   | CROSS1_F   | CROSS1[4:0]   |   |   |   |            |

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| ADDRESS | NAME                          | MSB |            |            |               |  |  | LSB |
|---------|-------------------------------|-----|------------|------------|---------------|--|--|-----|
| 0x202   | <a href="#">CROSS_2[7:0]</a>  | –   | CROSS2_I   | CROSS2_F   | CROSS2[4:0]   |  |  |     |
| 0x203   | <a href="#">CROSS_3[7:0]</a>  | –   | CROSS3_I   | CROSS3_F   | CROSS3[4:0]   |  |  |     |
| 0x204   | <a href="#">CROSS_4[7:0]</a>  | –   | CROSS4_I   | CROSS4_F   | CROSS4[4:0]   |  |  |     |
| 0x205   | <a href="#">CROSS_5[7:0]</a>  | –   | CROSS5_I   | CROSS5_F   | CROSS5[4:0]   |  |  |     |
| 0x206   | <a href="#">CROSS_6[7:0]</a>  | –   | CROSS6_I   | CROSS6_F   | CROSS6[4:0]   |  |  |     |
| 0x207   | <a href="#">CROSS_7[7:0]</a>  | –   | CROSS7_I   | CROSS7_F   | CROSS7[4:0]   |  |  |     |
| 0x208   | <a href="#">CROSS_8[7:0]</a>  | –   | CROSS8_I   | CROSS8_F   | CROSS8[4:0]   |  |  |     |
| 0x209   | <a href="#">CROSS_9[7:0]</a>  | –   | CROSS9_I   | CROSS9_F   | CROSS9[4:0]   |  |  |     |
| 0x20A   | <a href="#">CROSS_10[7:0]</a> | –   | CROSS10_I  | CROSS10_F  | CROSS10[4:0]  |  |  |     |
| 0x20B   | <a href="#">CROSS_11[7:0]</a> | –   | CROSS11_I  | CROSS11_F  | CROSS11[4:0]  |  |  |     |
| 0x20C   | <a href="#">CROSS_12[7:0]</a> | –   | CROSS12_I  | CROSS12_F  | CROSS12[4:0]  |  |  |     |
| 0x20D   | <a href="#">CROSS_13[7:0]</a> | –   | CROSS13_I  | CROSS13_F  | CROSS13[4:0]  |  |  |     |
| 0x20E   | <a href="#">CROSS_14[7:0]</a> | –   | CROSS14_I  | CROSS14_F  | CROSS14[4:0]  |  |  |     |
| 0x20F   | <a href="#">CROSS_15[7:0]</a> | –   | CROSS15_I  | CROSS15_F  | CROSS15[4:0]  |  |  |     |
| 0x210   | <a href="#">CROSS_16[7:0]</a> | –   | CROSS16_I  | CROSS16_F  | CROSS16[4:0]  |  |  |     |
| 0x211   | <a href="#">CROSS_17[7:0]</a> | –   | CROSS17_I  | CROSS17_F  | CROSS17[4:0]  |  |  |     |
| 0x212   | <a href="#">CROSS_18[7:0]</a> | –   | CROSS18_I  | CROSS18_F  | CROSS18[4:0]  |  |  |     |
| 0x213   | <a href="#">CROSS_19[7:0]</a> | –   | CROSS19_I  | CROSS19_F  | CROSS19[4:0]  |  |  |     |
| 0x214   | <a href="#">CROSS_20[7:0]</a> | –   | CROSS20_I  | CROSS20_F  | CROSS20[4:0]  |  |  |     |
| 0x215   | <a href="#">CROSS_21[7:0]</a> | –   | CROSS21_I  | CROSS21_F  | CROSS21[4:0]  |  |  |     |
| 0x216   | <a href="#">CROSS_22[7:0]</a> | –   | CROSS22_I  | CROSS22_F  | CROSS22[4:0]  |  |  |     |
| 0x217   | <a href="#">CROSS_23[7:0]</a> | –   | CROSS23_I  | CROSS23_F  | CROSS23[4:0]  |  |  |     |
| 0x218   | <a href="#">CROSS_HS[7:0]</a> | –   | CROSS_HS_I | CROSS_HS_F | CROSS_HS[4:0] |  |  |     |
| 0x219   | <a href="#">CROSS_VS[7:0]</a> | –   | CROSS_VS_I | CROSS_VS_F | CROSS_VS[4:0] |  |  |     |

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| ADDRESS | NAME                          | MSB            |            |            |               |   |   |   | LSB        |
|---------|-------------------------------|----------------|------------|------------|---------------|---|---|---|------------|
| 0x21A   | <a href="#">CROSS_DE[7:0]</a> | –              | CROSS_DE_I | CROSS_DE_F | CROSS_DE[4:0] |   |   |   |            |
| 0x21B   | <a href="#">PRBS_ERR[7:0]</a> | VPRBS_ERR[7:0] |            |            |               |   |   |   |            |
| 0x21C   | <a href="#">VPRBS[7:0]</a>    | –              | –          | VPRBS_FAIL | VPRBS_CHK_EN  | – | – | – | VIDEO_LOCK |
| 0x21D   | <a href="#">CROSS_27[7:0]</a> | ALT_CROSSBAR   | CROSS27_I  | CROSS27_F  | CROSS27[4:0]  |   |   |   |            |
| 0x21E   | <a href="#">CROSS_28[7:0]</a> | –              | CROSS28_I  | CROSS28_F  | CROSS28[4:0]  |   |   |   |            |
| 0x21F   | <a href="#">CROSS_29[7:0]</a> | –              | CROSS29_I  | CROSS29_F  | CROSS29[4:0]  |   |   |   |            |
| VRX U   |                               |                |            |            |               |   |   |   |            |
| 0x220   | <a href="#">CROSS_0[7:0]</a>  | –              | CROSS0_I   | CROSS0_F   | CROSS0[4:0]   |   |   |   |            |
| 0x221   | <a href="#">CROSS_1[7:0]</a>  | –              | CROSS1_I   | CROSS1_F   | CROSS1[4:0]   |   |   |   |            |
| 0x222   | <a href="#">CROSS_2[7:0]</a>  | –              | CROSS2_I   | CROSS2_F   | CROSS2[4:0]   |   |   |   |            |
| 0x223   | <a href="#">CROSS_3[7:0]</a>  | –              | CROSS3_I   | CROSS3_F   | CROSS3[4:0]   |   |   |   |            |
| 0x224   | <a href="#">CROSS_4[7:0]</a>  | –              | CROSS4_I   | CROSS4_F   | CROSS4[4:0]   |   |   |   |            |
| 0x225   | <a href="#">CROSS_5[7:0]</a>  | –              | CROSS5_I   | CROSS5_F   | CROSS5[4:0]   |   |   |   |            |
| 0x226   | <a href="#">CROSS_6[7:0]</a>  | –              | CROSS6_I   | CROSS6_F   | CROSS6[4:0]   |   |   |   |            |
| 0x227   | <a href="#">CROSS_7[7:0]</a>  | –              | CROSS7_I   | CROSS7_F   | CROSS7[4:0]   |   |   |   |            |
| 0x228   | <a href="#">CROSS_8[7:0]</a>  | –              | CROSS8_I   | CROSS8_F   | CROSS8[4:0]   |   |   |   |            |
| 0x229   | <a href="#">CROSS_9[7:0]</a>  | –              | CROSS9_I   | CROSS9_F   | CROSS9[4:0]   |   |   |   |            |
| 0x22A   | <a href="#">CROSS_10[7:0]</a> | –              | CROSS10_I  | CROSS10_F  | CROSS10[4:0]  |   |   |   |            |
| 0x22B   | <a href="#">CROSS_11[7:0]</a> | –              | CROSS11_I  | CROSS11_F  | CROSS11[4:0]  |   |   |   |            |
| 0x22C   | <a href="#">CROSS_12[7:0]</a> | –              | CROSS12_I  | CROSS12_F  | CROSS12[4:0]  |   |   |   |            |
| 0x22D   | <a href="#">CROSS_13[7:0]</a> | –              | CROSS13_I  | CROSS13_F  | CROSS13[4:0]  |   |   |   |            |
| 0x22E   | <a href="#">CROSS_14[7:0]</a> | –              | CROSS14_I  | CROSS14_F  | CROSS14[4:0]  |   |   |   |            |
| 0x22F   | <a href="#">CROSS_15[7:0]</a> | –              | CROSS15_I  | CROSS15_F  | CROSS15[4:0]  |   |   |   |            |
| 0x230   | <a href="#">CROSS_16[7:0]</a> | –              | CROSS16_I  | CROSS16_F  | CROSS16[4:0]  |   |   |   |            |

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| ADDRESS | NAME                          | MSB                |              |             |                 |         |            |            | LSB          |
|---------|-------------------------------|--------------------|--------------|-------------|-----------------|---------|------------|------------|--------------|
| 0x231   | <a href="#">CROSS_17[7:0]</a> | –                  | CROSS1_7_I   | CROSS1_7_F  | CROSS17[4:0]    |         |            |            |              |
| 0x232   | <a href="#">CROSS_18[7:0]</a> | –                  | CROSS1_8_I   | CROSS1_8_F  | CROSS18[4:0]    |         |            |            |              |
| 0x233   | <a href="#">CROSS_19[7:0]</a> | –                  | CROSS1_9_I   | CROSS1_9_F  | CROSS19[4:0]    |         |            |            |              |
| 0x234   | <a href="#">CROSS_20[7:0]</a> | –                  | CROSS2_0_I   | CROSS2_0_F  | CROSS20[4:0]    |         |            |            |              |
| 0x235   | <a href="#">CROSS_21[7:0]</a> | –                  | CROSS2_1_I   | CROSS2_1_F  | CROSS21[4:0]    |         |            |            |              |
| 0x236   | <a href="#">CROSS_22[7:0]</a> | –                  | CROSS2_2_I   | CROSS2_2_F  | CROSS22[4:0]    |         |            |            |              |
| 0x237   | <a href="#">CROSS_23[7:0]</a> | –                  | CROSS2_3_I   | CROSS2_3_F  | CROSS23[4:0]    |         |            |            |              |
| 0x238   | <a href="#">CROSS_HS[7:0]</a> | –                  | CROSS_HS_I   | CROSS_HS_F  | CROSS_HS[4:0]   |         |            |            |              |
| 0x239   | <a href="#">CROSS_VS[7:0]</a> | –                  | CROSS_VS_I   | CROSS_VS_F  | CROSS_VS[4:0]   |         |            |            |              |
| 0x23A   | <a href="#">CROSS_DE[7:0]</a> | –                  | CROSS_DE_I   | CROSS_DE_F  | CROSS_DE[4:0]   |         |            |            |              |
| 0x23B   | <a href="#">PRBS_ERR[7:0]</a> | VPRBS_ERR[7:0]     |              |             |                 |         |            |            |              |
| 0x23C   | <a href="#">VPRBS[7:0]</a>    | –                  | –            | VPRBS_FAIL  | VPRBS_CHK_EN    | –       | –          | –          | VIDEO_LOCK   |
| 0x23D   | <a href="#">CROSS_27[7:0]</a> | ALT_CROSSBAR       | CROSS2_7_I   | CROSS2_7_F  | CROSS27[4:0]    |         |            |            |              |
| 0x23E   | <a href="#">CROSS_28[7:0]</a> | –                  | CROSS2_8_I   | CROSS2_8_F  | CROSS28[4:0]    |         |            |            |              |
| 0x23F   | <a href="#">CROSS_29[7:0]</a> | –                  | CROSS2_9_I   | CROSS2_9_F  | CROSS29[4:0]    |         |            |            |              |
| GPIO0 0 |                               |                    |              |             |                 |         |            |            |              |
| 0x2B0   | <a href="#">GPIO_A[7:0]</a>   | RES_CFG            | TX_PIO       | TX_COM_P_EN | GPIO_OUT        | GPIO_IN | GPIO_RX_EN | GPIO_TX_EN | GPIO_OUT_DIS |
| 0x2B1   | <a href="#">GPIO_B[7:0]</a>   | PULL_UPDN_SEL[1:0] |              | OUT_TYPE    | GPIO_TX_ID[4:0] |         |            |            |              |
| 0x2B2   | <a href="#">GPIO_C[7:0]</a>   | OVR_RESCFG         | GPIO_RECEVED | –           | GPIO_RX_ID[4:0] |         |            |            |              |
| GPIO1 1 |                               |                    |              |             |                 |         |            |            |              |
| 0x2B3   | <a href="#">GPIO_A[7:0]</a>   | RES_CFG            | TX_PIO       | TX_COM_P_EN | GPIO_OUT        | GPIO_IN | GPIO_RX_EN | GPIO_TX_EN | GPIO_OUT_DIS |
| 0x2B4   | <a href="#">GPIO_B[7:0]</a>   | PULL_UPDN_SEL[1:0] |              | OUT_TYPE    | GPIO_TX_ID[4:0] |         |            |            |              |
| 0x2B5   | <a href="#">GPIO_C[7:0]</a>   | OVR_RESCFG         | –            | –           | GPIO_RX_ID[4:0] |         |            |            |              |
| GPIO2 2 |                               |                    |              |             |                 |         |            |            |              |
| 0x2B6   | <a href="#">GPIO_A[7:0]</a>   | RES_CFG            | TX_PIO       | TX_COM_P_EN | GPIO_OUT        | GPIO_IN | GPIO_RX_EN | GPIO_TX_EN | GPIO_OUT_DIS |



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| ADDRESS | NAME                        | MSB                |             |                |                 |         |                |                | LSB              |
|---------|-----------------------------|--------------------|-------------|----------------|-----------------|---------|----------------|----------------|------------------|
| 0x2B7   | <a href="#">GPIO_B[7:0]</a> | PULL_UPDN_SEL[1:0] |             | OUT_TY<br>PE   | GPIO_TX_ID[4:0] |         |                |                |                  |
| 0x2B8   | <a href="#">GPIO_C[7:0]</a> | OVR_RE<br>S_CFG    | –           | –              | GPIO_RX_ID[4:0] |         |                |                |                  |
| GPIO3 3 |                             |                    |             |                |                 |         |                |                |                  |
| 0x2B9   | <a href="#">GPIO_A[7:0]</a> | RES_CF<br>G        | TX_PRI<br>O | –              | GPIO_O<br>UT    | GPIO_IN | GPIO_R<br>X_EN | GPIO_T<br>X_EN | GPIO_O<br>UT_DIS |
| 0x2BA   | <a href="#">GPIO_B[7:0]</a> | PULL_UPDN_SEL[1:0] |             | OUT_TY<br>PE   | GPIO_TX_ID[4:0] |         |                |                |                  |
| 0x2BB   | <a href="#">GPIO_C[7:0]</a> | OVR_RE<br>S_CFG    | –           | –              | GPIO_RX_ID[4:0] |         |                |                |                  |
| GPIO4 4 |                             |                    |             |                |                 |         |                |                |                  |
| 0x2BC   | <a href="#">GPIO_A[7:0]</a> | RES_CF<br>G        | TX_PRI<br>O | TX_COM<br>P_EN | GPIO_O<br>UT    | GPIO_IN | GPIO_R<br>X_EN | GPIO_T<br>X_EN | GPIO_O<br>UT_DIS |
| 0x2BD   | <a href="#">GPIO_B[7:0]</a> | PULL_UPDN_SEL[1:0] |             | OUT_TY<br>PE   | –               | –       | –              | –              | –                |
| 0x2BE   | <a href="#">GPIO_C[7:0]</a> | OVR_RE<br>S_CFG    | –           | –              | GPIO_RX_ID[4:0] |         |                |                |                  |
| GPIO5 5 |                             |                    |             |                |                 |         |                |                |                  |
| 0x2BF   | <a href="#">GPIO_A[7:0]</a> | RES_CF<br>G        | TX_PRI<br>O | TX_COM<br>P_EN | GPIO_O<br>UT    | GPIO_IN | GPIO_R<br>X_EN | GPIO_T<br>X_EN | GPIO_O<br>UT_DIS |
| 0x2C0   | <a href="#">GPIO_B[7:0]</a> | PULL_UPDN_SEL[1:0] |             | OUT_TY<br>PE   | GPIO_TX_ID[4:0] |         |                |                |                  |
| 0x2C1   | <a href="#">GPIO_C[7:0]</a> | OVR_RE<br>S_CFG    | –           | –              | GPIO_RX_ID[4:0] |         |                |                |                  |
| GPIO6 6 |                             |                    |             |                |                 |         |                |                |                  |
| 0x2C2   | <a href="#">GPIO_A[7:0]</a> | RES_CF<br>G        | TX_PRI<br>O | TX_COM<br>P_EN | GPIO_O<br>UT    | GPIO_IN | GPIO_R<br>X_EN | GPIO_T<br>X_EN | GPIO_O<br>UT_DIS |
| 0x2C3   | <a href="#">GPIO_B[7:0]</a> | PULL_UPDN_SEL[1:0] |             | OUT_TY<br>PE   | GPIO_TX_ID[4:0] |         |                |                |                  |
| 0x2C4   | <a href="#">GPIO_C[7:0]</a> | OVR_RE<br>S_CFG    | –           | –              | GPIO_RX_ID[4:0] |         |                |                |                  |
| GPIO7 7 |                             |                    |             |                |                 |         |                |                |                  |
| 0x2C5   | <a href="#">GPIO_A[7:0]</a> | RES_CF<br>G        | TX_PRI<br>O | TX_COM<br>P_EN | GPIO_O<br>UT    | GPIO_IN | GPIO_R<br>X_EN | GPIO_T<br>X_EN | GPIO_O<br>UT_DIS |
| 0x2C6   | <a href="#">GPIO_B[7:0]</a> | PULL_UPDN_SEL[1:0] |             | OUT_TY<br>PE   | GPIO_TX_ID[4:0] |         |                |                |                  |
| 0x2C7   | <a href="#">GPIO_C[7:0]</a> | OVR_RE<br>S_CFG    | –           | –              | GPIO_RX_ID[4:0] |         |                |                |                  |
| GPIO8 8 |                             |                    |             |                |                 |         |                |                |                  |
| 0x2C8   | <a href="#">GPIO_A[7:0]</a> | RES_CF<br>G        | TX_PRI<br>O | TX_COM<br>P_EN | GPIO_O<br>UT    | GPIO_IN | GPIO_R<br>X_EN | GPIO_T<br>X_EN | GPIO_O<br>UT_DIS |
| 0x2C9   | <a href="#">GPIO_B[7:0]</a> | PULL_UPDN_SEL[1:0] |             | OUT_TY<br>PE   | GPIO_TX_ID[4:0] |         |                |                |                  |
| 0x2CA   | <a href="#">GPIO_C[7:0]</a> | OVR_RE<br>S_CFG    | –           | –              | GPIO_RX_ID[4:0] |         |                |                |                  |

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| ADDRESS   | NAME           | MSB                |                   |                   |                   |                  |                                |                   | LSB             |
|-----------|----------------|--------------------|-------------------|-------------------|-------------------|------------------|--------------------------------|-------------------|-----------------|
| GPIO9 9   |                |                    |                   |                   |                   |                  |                                |                   |                 |
| 0x2CB     | GPIO_A[7:0]    | RES_CFG            | TX_PRIORITY       | TX_COM_P_EN       | GPIO_OUTPUT       | GPIO_IN          | GPIO_RX_EN                     | GPIO_TX_EN        | GPIO_OUTPUT_DIS |
| 0x2CC     | GPIO_B[7:0]    | PULL_UPDN_SEL[1:0] |                   | OUT_TYPE          | GPIO_TX_ID[4:0]   |                  |                                |                   |                 |
| 0x2CD     | GPIO_C[7:0]    | OVR_RESCFG         | –                 | –                 | GPIO_RX_ID[4:0]   |                  |                                |                   |                 |
| GPIO10 10 |                |                    |                   |                   |                   |                  |                                |                   |                 |
| 0x2CE     | GPIO_A[7:0]    | RES_CFG            | TX_PRIORITY       | TX_COM_P_EN       | GPIO_OUTPUT       | GPIO_IN          | GPIO_RX_EN                     | GPIO_TX_EN        | GPIO_OUTPUT_DIS |
| 0x2CF     | GPIO_B[7:0]    | PULL_UPDN_SEL[1:0] |                   | –                 | GPIO_TX_ID[4:0]   |                  |                                |                   |                 |
| 0x2D0     | GPIO_C[7:0]    | OVR_RESCFG         | –                 | –                 | GPIO_RX_ID[4:0]   |                  |                                |                   |                 |
| GPIO11 11 |                |                    |                   |                   |                   |                  |                                |                   |                 |
| 0x2D1     | GPIO_A[7:0]    | RES_CFG            | TX_PRIORITY       | TX_COM_P_EN       | GPIO_OUTPUT       | GPIO_IN          | GPIO_RX_EN                     | GPIO_TX_EN        | GPIO_OUTPUT_DIS |
| 0x2D2     | GPIO_B[7:0]    | PULL_UPDN_SEL[1:0] |                   | OUT_TYPE          | GPIO_TX_ID[4:0]   |                  |                                |                   |                 |
| 0x2D3     | GPIO_C[7:0]    | OVR_RESCFG         | –                 | –                 | GPIO_RX_ID[4:0]   |                  |                                |                   |                 |
| GPIO12 12 |                |                    |                   |                   |                   |                  |                                |                   |                 |
| 0x2D4     | GPIO_A[7:0]    | RES_CFG            | TX_PRIORITY       | TX_COM_P_EN       | GPIO_OUTPUT       | GPIO_IN          | GPIO_RX_EN                     | GPIO_TX_EN        | GPIO_OUTPUT_DIS |
| 0x2D5     | GPIO_B[7:0]    | PULL_UPDN_SEL[1:0] |                   | OUT_TYPE          | GPIO_TX_ID[4:0]   |                  |                                |                   |                 |
| 0x2D6     | GPIO_C[7:0]    | OVR_RESCFG         | –                 | –                 | GPIO_RX_ID[4:0]   |                  |                                |                   |                 |
| CMU       |                |                    |                   |                   |                   |                  |                                |                   |                 |
| 0x304     | CMU4[7:0]      | A_SPEED[1:0]       |                   | B_SPEED[1:0]      |                   | –                | –                              | C_SPEED[1:0]      |                 |
| BACKTOP   |                |                    |                   |                   |                   |                  |                                |                   |                 |
| 0x308     | BACKTOP1[7:0]  | CSIPLL_UNLOCK      | CSIPLLZ_UNLOCK    | CSIPLLY_UNLOCK    | CSIPLLX_UNLOCK    | LINE_SPL2        | LINE_SPL1                      | –                 | BACKTOP_P_EN    |
| 0x312     | BACKTOP11[7:0] | cmd_ove<br>rflow4  | cmd_ove<br>rflow3 | cmd_ove<br>rflow2 | cmd_ove<br>rflow1 | –                | LMO_Z                          | LMO_Y             | –               |
| 0x313     | BACKTOP12[7:0] | soft_bpp_x[4:0]    |                   |                   |                   |                  | BACKTO<br>P_MEM<br>CRC_ER<br>R | CSI_OU<br>T_EN    | –               |
| 0x314     | BACKTOP13[7:0] | soft_vc_y[3:0]     |                   |                   |                   | soft_vc_x[3:0]   |                                |                   |                 |
| 0x315     | BACKTOP14[7:0] | soft_vc_u[3:0]     |                   |                   |                   | soft_vc_z[3:0]   |                                |                   |                 |
| 0x316     | BACKTOP15[7:0] | soft_dt_y_h[1:0]   |                   | soft_dt_x[5:0]    |                   |                  |                                |                   |                 |
| 0x317     | BACKTOP16[7:0] | soft_dt_z_h[3:0]   |                   |                   |                   | soft_dt_y_l[3:0] |                                |                   |                 |
| 0x318     | BACKTOP17[7:0] | soft_dt_u[5:0]     |                   |                   |                   |                  |                                | soft_dt_z_l[1:0]  |                 |
| 0x319     | BACKTOP18[7:0] | soft_bpp_z_h[2:0]  |                   |                   | soft_bpp_y[4:0]   |                  |                                |                   |                 |
| 0x31A     | BACKTOP19[7:0] | –                  | soft_bpp_u[4:0]   |                   |                   |                  |                                | soft_bpp_z_l[1:0] |                 |

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| ADDRESS  | NAME                           | MSB                                    |                     |                                        |                                   |                                        |                   |                   | LSB       |
|----------|--------------------------------|----------------------------------------|---------------------|----------------------------------------|-----------------------------------|----------------------------------------|-------------------|-------------------|-----------|
| 0x31B    | <a href="#">BACKTOP20[7:0]</a> | phy0_csi_tx_dpll_fb_fraction_in_l[7:0] |                     |                                        |                                   |                                        |                   |                   |           |
| 0x31C    | <a href="#">BACKTOP21[7:0]</a> | bpp8dblu                               | bpp8dblz            | bpp8dbly                               | bpp8dblx                          | phy0_csi_tx_dpll_fb_fraction_in_h[3:0] |                   |                   |           |
| 0x31D    | <a href="#">BACKTOP22[7:0]</a> | override_bpp_vc_dty                    | override_bpp_vc_dtx | phy0_csi_tx_dpll_fb_fraction_predef_en | phy0_csi_tx_dpll_predef_freq[4:0] |                                        |                   |                   |           |
| 0x31E    | <a href="#">BACKTOP23[7:0]</a> | phy1_csi_tx_dpll_fb_fraction_in_l[7:0] |                     |                                        |                                   |                                        |                   |                   |           |
| 0x31F    | <a href="#">BACKTOP24[7:0]</a> | bpp8dblu_mode                          | bpp8dblz_mode       | bpp8dbly_mode                          | bpp8dblx_mode                     | phy1_csi_tx_dpll_fb_fraction_in_h[3:0] |                   |                   |           |
| 0x320    | <a href="#">BACKTOP25[7:0]</a> | override_bpp_vc_dtu                    | override_bpp_vc_dtz | phy1_csi_tx_dpll_fb_fraction_predef_en | phy1_csi_tx_dpll_predef_freq[4:0] |                                        |                   |                   |           |
| 0x321    | <a href="#">BACKTOP26[7:0]</a> | phy2_csi_tx_dpll_fb_fraction_in_l[7:0] |                     |                                        |                                   |                                        |                   |                   |           |
| 0x322    | <a href="#">BACKTOP27[7:0]</a> | yuv_8_10_mux_mode4                     | yuv_8_10_mux_mode3  | yuv_8_10_mux_mode2                     | yuv_8_10_mux_mode1                | phy2_csi_tx_dpll_fb_fraction_in_h[3:0] |                   |                   |           |
| 0x323    | <a href="#">BACKTOP28[7:0]</a> | –                                      | –                   | phy2_csi_tx_dpll_fb_fraction_predef_en | phy2_csi_tx_dpll_predef_freq[4:0] |                                        |                   |                   |           |
| 0x324    | <a href="#">BACKTOP29[7:0]</a> | phy3_csi_tx_dpll_fb_fraction_in_l[7:0] |                     |                                        |                                   |                                        |                   |                   |           |
| 0x325    | <a href="#">BACKTOP30[7:0]</a> | BACKTOP_W_FRAME                        | –                   | –                                      | –                                 | phy3_csi_tx_dpll_fb_fraction_in_h[3:0] |                   |                   |           |
| 0x326    | <a href="#">BACKTOP31[7:0]</a> | –                                      | –                   | phy3_csi_tx_dpll_fb_fraction_predef_en | phy3_csi_tx_dpll_predef_freq[4:0] |                                        |                   |                   |           |
| 0x327    | <a href="#">BACKTOP32[7:0]</a> | bpp10dblu_mode                         | bpp10dblz_mode      | bpp10dbly_mode                         | bpp10dblx_mode                    | bpp10dblu                              | bpp10dblz         | bpp10dbly         | bpp10dblx |
| 0x328    | <a href="#">BACKTOP33[7:0]</a> | –                                      | –                   | –                                      | –                                 | bpp12dblu                              | bpp12dblz         | bpp12dbly         | bpp12dblx |
| MIPI_PHY |                                |                                        |                     |                                        |                                   |                                        |                   |                   |           |
| 0x330    | <a href="#">MIPI_PHY0[7:0]</a> | force_csi_out_en                       | force_clk_3_en      | force_clk_0_en                         | phy_1x4_b_22                      | phy_1x4_a_22                           | phy_2x4           | –                 | phy_4x2   |
| 0x331    | <a href="#">MIPI_PHY1[7:0]</a> | t_hs_przero[1:0]                       |                     | t_hs_prep[1:0]                         |                                   | t_clk_trail[1:0]                       |                   | t_clk_przero[1:0] |           |
| 0x332    | <a href="#">MIPI_PHY2[7:0]</a> | phy_Stdbyn[3:0]                        |                     |                                        |                                   | t_lpx[1:0]                             |                   | t_hs_trail[1:0]   |           |
| 0x333    | <a href="#">MIPI_PHY3[7:0]</a> | phy1_lane_map[3:0]                     |                     |                                        |                                   | phy0_lane_map[3:0]                     |                   |                   |           |
| 0x334    | <a href="#">MIPI_PHY4[7:0]</a> | phy3_lane_map[3:0]                     |                     |                                        |                                   | phy2_lane_map[3:0]                     |                   |                   |           |
| 0x335    | <a href="#">MIPI_PHY5[7:0]</a> | t_clk_prep[1:0]                        |                     | phy1_pol_map[2:0]                      |                                   |                                        | phy0_pol_map[2:0] |                   |           |
| 0x336    | <a href="#">MIPI_PHY6[7:0]</a> | phy_cp1                                | phy_cp0             | phy3_pol_map[2:0]                      |                                   |                                        | phy2_pol_map[2:0] |                   |           |
| 0x338    | <a href="#">MIPI_PHY8[7:0]</a> | t_lpxesc[2:0]                          |                     |                                        | –                                 |                                        | –                 | –                 | –         |

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| ADDRESS   | NAME                            | MSB                     |              |                 |                         |                    |                    | LSB               |
|-----------|---------------------------------|-------------------------|--------------|-----------------|-------------------------|--------------------|--------------------|-------------------|
| 0x339     | <a href="#">MIPI_PHY9[7:0]</a>  | phy_cp0_dst[1:0]        |              | –               | –                       | –                  | –                  | phy_cp0_overflow  |
| 0x33A     | <a href="#">MIPI_PHY10[7:0]</a> | phy_cp0_src[1:0]        |              | –               | –                       | –                  | –                  | phy_cp0_underflow |
| 0x33B     | <a href="#">MIPI_PHY11[7:0]</a> | phy_cp1_dst[1:0]        |              | –               | –                       | –                  | –                  | phy_cp1_overflow  |
| 0x33C     | <a href="#">MIPI_PHY12[7:0]</a> | phy_cp1_src[1:0]        |              | –               | –                       | –                  | –                  | phy_cp1_underflow |
| FSYNC     |                                 |                         |              |                 |                         |                    |                    |                   |
| 0x3E0     | <a href="#">FSYNC_0[7:0]</a>    | EN_OFLOW_RST_FS         | –            | FSYNC_OUT_PIN   | EN_VSGEN                | FSYNC_MODE[1:0]    |                    | FSYNC_METH[1:0]   |
| 0x3E1     | <a href="#">FSYNC_1[7:0]</a>    | –                       | –            | –               | –                       | FSYNC_PER_DIV[3:0] |                    |                   |
| 0x3E2     | <a href="#">FSYNC_2[7:0]</a>    | MST_LINK_SEL[2:0]       |              |                 | K_VAL_SIGN              | K_VAL[3:0]         |                    |                   |
| 0x3E3     | <a href="#">FSYNC_3[7:0]</a>    | P_VAL_L[7:0]            |              |                 |                         |                    |                    |                   |
| 0x3E4     | <a href="#">FSYNC_4[7:0]</a>    | –                       | –            | P_VAL_SIGN      | P_VAL_H[4:0]            |                    |                    |                   |
| 0x3E5     | <a href="#">FSYNC_5[7:0]</a>    | FSYNC_PERIOD_L[7:0]     |              |                 |                         |                    |                    |                   |
| 0x3E6     | <a href="#">FSYNC_6[7:0]</a>    | FSYNC_PERIOD_M[7:0]     |              |                 |                         |                    |                    |                   |
| 0x3E7     | <a href="#">FSYNC_7[7:0]</a>    | FSYNC_PERIOD_H[7:0]     |              |                 |                         |                    |                    |                   |
| 0x3E8     | <a href="#">FSYNC_8[7:0]</a>    | FRM_DIFF_ERR_THR_L[7:0] |              |                 |                         |                    |                    |                   |
| 0x3E9     | <a href="#">FSYNC_9[7:0]</a>    | –                       | –            | –               | FRM_DIFF_ERR_THR_H[4:0] |                    |                    |                   |
| 0x3EA     | <a href="#">FSYNC_10[7:0]</a>   | OVLP_WINDOW_L[7:0]      |              |                 |                         |                    |                    |                   |
| 0x3EB     | <a href="#">FSYNC_11[7:0]</a>   | EN_FSN_LAST             | –            | –               | OVLP_WINDOW_H[4:0]      |                    |                    |                   |
| 0x3F0     | <a href="#">FSYNC_16[7:0]</a>   | FSYNC_ERR_CNT[7:0]      |              |                 |                         |                    |                    |                   |
| 0x3F1     | <a href="#">FSYNC_17[7:0]</a>   | FSYNC_TX_ID[4:0]        |              |                 |                         |                    | FSYNC_ERR_THR[2:0] |                   |
| 0x3F2     | <a href="#">FSYNC_18[7:0]</a>   | CALC_FRM_LEN_L[7:0]     |              |                 |                         |                    |                    |                   |
| 0x3F3     | <a href="#">FSYNC_19[7:0]</a>   | CALC_FRM_LEN_M[7:0]     |              |                 |                         |                    |                    |                   |
| 0x3F4     | <a href="#">FSYNC_20[7:0]</a>   | CALC_FRM_LEN_H[7:0]     |              |                 |                         |                    |                    |                   |
| 0x3F5     | <a href="#">FSYNC_21[7:0]</a>   | FRM_DIFF_L[7:0]         |              |                 |                         |                    |                    |                   |
| 0x3F6     | <a href="#">FSYNC_22[7:0]</a>   | FSYNC_LOSS_OF_LOCK      | FSYNC_LOCKED | FRM_DIFF_H[5:0] |                         |                    |                    |                   |
| MIPI_TX 0 |                                 |                         |              |                 |                         |                    |                    |                   |
| 0x401     | <a href="#">MIPI_TX1[7:0]</a>   | MODE[7:0]               |              |                 |                         |                    |                    |                   |
| 0x402     | <a href="#">MIPI_TX2[7:0]</a>   | STATUS[7:0]             |              |                 |                         |                    |                    |                   |
| 0x403     | <a href="#">MIPI_TX3[7:0]</a>   | DESKEW_INIT[7:0]        |              |                 |                         |                    |                    |                   |
| 0x404     | <a href="#">MIPI_TX4[7:0]</a>   | DESKEW_PER[7:0]         |              |                 |                         |                    |                    |                   |
| 0x405     | <a href="#">MIPI_TX5[7:0]</a>   | CSI2_T_PRE[7:0]         |              |                 |                         |                    |                    |                   |

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| ADDRESS | NAME                           | MSB                  |                      |   |                      |                     |                      | LSB |
|---------|--------------------------------|----------------------|----------------------|---|----------------------|---------------------|----------------------|-----|
| 0x406   | <a href="#">MIPI_TX6[7:0]</a>  | CSI2_T_POST[7:0]     |                      |   |                      |                     |                      |     |
| 0x407   | <a href="#">MIPI_TX7[7:0]</a>  | CSI2_TX_GAP[7:0]     |                      |   |                      |                     |                      |     |
| 0x408   | <a href="#">MIPI_TX8[7:0]</a>  | CSI2_TWAKEUP_L[7:0]  |                      |   |                      |                     |                      |     |
| 0x409   | <a href="#">MIPI_TX9[7:0]</a>  | CSI2_TWAKEUP_M[7:0]  |                      |   |                      |                     |                      |     |
| 0x40A   | <a href="#">MIPI_TX10[7:0]</a> | CSI2_LANE_CNT[1:0]   | –                    | – | CSI_VC_X_EN          | CSI2_TWAKEUP_H[2:0] |                      |     |
| 0x40B   | <a href="#">MIPI_TX11[7:0]</a> | MAP_EN_L[7:0]        |                      |   |                      |                     |                      |     |
| 0x40C   | <a href="#">MIPI_TX12[7:0]</a> | MAP_EN_H[7:0]        |                      |   |                      |                     |                      |     |
| 0x40D   | <a href="#">MIPI_TX13[7:0]</a> | MAP_SRC_0[7:0]       |                      |   |                      |                     |                      |     |
| 0x40E   | <a href="#">MIPI_TX14[7:0]</a> | MAP_DST_0[7:0]       |                      |   |                      |                     |                      |     |
| 0x40F   | <a href="#">MIPI_TX15[7:0]</a> | MAP_SRC_1[7:0]       |                      |   |                      |                     |                      |     |
| 0x410   | <a href="#">MIPI_TX16[7:0]</a> | MAP_DST_1[7:0]       |                      |   |                      |                     |                      |     |
| 0x411   | <a href="#">MIPI_TX17[7:0]</a> | MAP_SRC_2[7:0]       |                      |   |                      |                     |                      |     |
| 0x412   | <a href="#">MIPI_TX18[7:0]</a> | MAP_DST_2[7:0]       |                      |   |                      |                     |                      |     |
| 0x413   | <a href="#">MIPI_TX19[7:0]</a> | MAP_SRC_3[7:0]       |                      |   |                      |                     |                      |     |
| 0x414   | <a href="#">MIPI_TX20[7:0]</a> | MAP_DST_3[7:0]       |                      |   |                      |                     |                      |     |
| 0x415   | <a href="#">MIPI_TX21[7:0]</a> | MAP_SRC_4[7:0]       |                      |   |                      |                     |                      |     |
| 0x416   | <a href="#">MIPI_TX22[7:0]</a> | MAP_DST_4[7:0]       |                      |   |                      |                     |                      |     |
| 0x417   | <a href="#">MIPI_TX23[7:0]</a> | MAP_SRC_5[7:0]       |                      |   |                      |                     |                      |     |
| 0x418   | <a href="#">MIPI_TX24[7:0]</a> | MAP_DST_5[7:0]       |                      |   |                      |                     |                      |     |
| 0x419   | <a href="#">MIPI_TX25[7:0]</a> | MAP_SRC_6[7:0]       |                      |   |                      |                     |                      |     |
| 0x41A   | <a href="#">MIPI_TX26[7:0]</a> | MAP_DST_6[7:0]       |                      |   |                      |                     |                      |     |
| 0x41B   | <a href="#">MIPI_TX27[7:0]</a> | MAP_SRC_7[7:0]       |                      |   |                      |                     |                      |     |
| 0x41C   | <a href="#">MIPI_TX28[7:0]</a> | MAP_DST_7[7:0]       |                      |   |                      |                     |                      |     |
| 0x41D   | <a href="#">MIPI_TX29[7:0]</a> | MAP_SRC_8[7:0]       |                      |   |                      |                     |                      |     |
| 0x41E   | <a href="#">MIPI_TX30[7:0]</a> | MAP_DST_8[7:0]       |                      |   |                      |                     |                      |     |
| 0x41F   | <a href="#">MIPI_TX31[7:0]</a> | MAP_SRC_9[7:0]       |                      |   |                      |                     |                      |     |
| 0x420   | <a href="#">MIPI_TX32[7:0]</a> | MAP_DST_9[7:0]       |                      |   |                      |                     |                      |     |
| 0x421   | <a href="#">MIPI_TX33[7:0]</a> | MAP_SRC_10[7:0]      |                      |   |                      |                     |                      |     |
| 0x422   | <a href="#">MIPI_TX34[7:0]</a> | MAP_DST_10[7:0]      |                      |   |                      |                     |                      |     |
| 0x423   | <a href="#">MIPI_TX35[7:0]</a> | MAP_SRC_11[7:0]      |                      |   |                      |                     |                      |     |
| 0x424   | <a href="#">MIPI_TX36[7:0]</a> | MAP_DST_11[7:0]      |                      |   |                      |                     |                      |     |
| 0x425   | <a href="#">MIPI_TX37[7:0]</a> | MAP_SRC_12[7:0]      |                      |   |                      |                     |                      |     |
| 0x426   | <a href="#">MIPI_TX38[7:0]</a> | MAP_DST_12[7:0]      |                      |   |                      |                     |                      |     |
| 0x427   | <a href="#">MIPI_TX39[7:0]</a> | MAP_SRC_13[7:0]      |                      |   |                      |                     |                      |     |
| 0x428   | <a href="#">MIPI_TX40[7:0]</a> | MAP_DST_13[7:0]      |                      |   |                      |                     |                      |     |
| 0x429   | <a href="#">MIPI_TX41[7:0]</a> | MAP_SRC_14[7:0]      |                      |   |                      |                     |                      |     |
| 0x42A   | <a href="#">MIPI_TX42[7:0]</a> | MAP_DST_14[7:0]      |                      |   |                      |                     |                      |     |
| 0x42B   | <a href="#">MIPI_TX43[7:0]</a> | MAP_SRC_15[7:0]      |                      |   |                      |                     |                      |     |
| 0x42C   | <a href="#">MIPI_TX44[7:0]</a> | MAP_DST_15[7:0]      |                      |   |                      |                     |                      |     |
| 0x42D   | <a href="#">MIPI_TX45[7:0]</a> | MAP_DPHY_DEST_3[1:0] | MAP_DPHY_DEST_2[1:0] |   | MAP_DPHY_DEST_1[1:0] |                     | MAP_DPHY_DEST_0[1:0] |     |

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| ADDRESS   | NAME                           | MSB                    |   |                       |                       |                       |                       |                       | LSB                   |
|-----------|--------------------------------|------------------------|---|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| 0x42E     | <a href="#">MIPI_TX46[7:0]</a> | MAP_DPHY_DEST_7[1:0]   |   | MAP_DPHY_DEST_6[1:0]  |                       | MAP_DPHY_DEST_5[1:0]  |                       | MAP_DPHY_DEST_4[1:0]  |                       |
| 0x42F     | <a href="#">MIPI_TX47[7:0]</a> | MAP_DPHY_DEST_11[1:0]  |   | MAP_DPHY_DEST_10[1:0] |                       | MAP_DPHY_DEST_9[1:0]  |                       | MAP_DPHY_DEST_8[1:0]  |                       |
| 0x430     | <a href="#">MIPI_TX48[7:0]</a> | MAP_DPHY_DEST_15[1:0]  |   | MAP_DPHY_DEST_14[1:0] |                       | MAP_DPHY_DEST_13[1:0] |                       | MAP_DPHY_DEST_12[1:0] |                       |
| 0x431     | <a href="#">MIPI_TX49[7:0]</a> | MAP_CON[7:0]           |   |                       |                       |                       |                       |                       |                       |
| 0x433     | <a href="#">MIPI_TX51[7:0]</a> | -                      | - | -                     | ALT2_M<br>EM_MA<br>P8 | MODE_<br>DT           | ALT_ME<br>M_MAP1<br>0 | ALT_ME<br>M_MAP8      | ALT_ME<br>M_MAP1<br>2 |
| MIPI_TX 1 |                                |                        |   |                       |                       |                       |                       |                       |                       |
| 0x441     | <a href="#">MIPI_TX1[7:0]</a>  | MODE[7:0]              |   |                       |                       |                       |                       |                       |                       |
| 0x442     | <a href="#">MIPI_TX2[7:0]</a>  | STATUS[7:0]            |   |                       |                       |                       |                       |                       |                       |
| 0x443     | <a href="#">MIPI_TX3[7:0]</a>  | DESKEW_INIT[7:0]       |   |                       |                       |                       |                       |                       |                       |
| 0x444     | <a href="#">MIPI_TX4[7:0]</a>  | DESKEW_PER[7:0]        |   |                       |                       |                       |                       |                       |                       |
| 0x445     | <a href="#">MIPI_TX5[7:0]</a>  | CSI2_T_PRE[7:0]        |   |                       |                       |                       |                       |                       |                       |
| 0x446     | <a href="#">MIPI_TX6[7:0]</a>  | CSI2_T_POST[7:0]       |   |                       |                       |                       |                       |                       |                       |
| 0x447     | <a href="#">MIPI_TX7[7:0]</a>  | CSI2_TX_GAP[7:0]       |   |                       |                       |                       |                       |                       |                       |
| 0x448     | <a href="#">MIPI_TX8[7:0]</a>  | CSI2_TWAKEUP_L[7:0]    |   |                       |                       |                       |                       |                       |                       |
| 0x449     | <a href="#">MIPI_TX9[7:0]</a>  | CSI2_TWAKEUP_M[7:0]    |   |                       |                       |                       |                       |                       |                       |
| 0x44A     | <a href="#">MIPI_TX10[7:0]</a> | CSI2_LANE_CNT[1:<br>0] | - | -                     | CSI_VC<br>X_EN        | CSI2_TWAKEUP_H[2:0]   |                       |                       |                       |
| 0x44B     | <a href="#">MIPI_TX11[7:0]</a> | MAP_EN_L[7:0]          |   |                       |                       |                       |                       |                       |                       |
| 0x44C     | <a href="#">MIPI_TX12[7:0]</a> | MAP_EN_H[7:0]          |   |                       |                       |                       |                       |                       |                       |
| 0x44D     | <a href="#">MIPI_TX13[7:0]</a> | MAP_SRC_0[7:0]         |   |                       |                       |                       |                       |                       |                       |
| 0x44E     | <a href="#">MIPI_TX14[7:0]</a> | MAP_DST_0[7:0]         |   |                       |                       |                       |                       |                       |                       |
| 0x44F     | <a href="#">MIPI_TX15[7:0]</a> | MAP_SRC_1[7:0]         |   |                       |                       |                       |                       |                       |                       |
| 0x450     | <a href="#">MIPI_TX16[7:0]</a> | MAP_DST_1[7:0]         |   |                       |                       |                       |                       |                       |                       |
| 0x451     | <a href="#">MIPI_TX17[7:0]</a> | MAP_SRC_2[7:0]         |   |                       |                       |                       |                       |                       |                       |
| 0x452     | <a href="#">MIPI_TX18[7:0]</a> | MAP_DST_2[7:0]         |   |                       |                       |                       |                       |                       |                       |
| 0x453     | <a href="#">MIPI_TX19[7:0]</a> | MAP_SRC_3[7:0]         |   |                       |                       |                       |                       |                       |                       |
| 0x454     | <a href="#">MIPI_TX20[7:0]</a> | MAP_DST_3[7:0]         |   |                       |                       |                       |                       |                       |                       |
| 0x455     | <a href="#">MIPI_TX21[7:0]</a> | MAP_SRC_4[7:0]         |   |                       |                       |                       |                       |                       |                       |
| 0x456     | <a href="#">MIPI_TX22[7:0]</a> | MAP_DST_4[7:0]         |   |                       |                       |                       |                       |                       |                       |
| 0x457     | <a href="#">MIPI_TX23[7:0]</a> | MAP_SRC_5[7:0]         |   |                       |                       |                       |                       |                       |                       |
| 0x458     | <a href="#">MIPI_TX24[7:0]</a> | MAP_DST_5[7:0]         |   |                       |                       |                       |                       |                       |                       |
| 0x459     | <a href="#">MIPI_TX25[7:0]</a> | MAP_SRC_6[7:0]         |   |                       |                       |                       |                       |                       |                       |
| 0x45A     | <a href="#">MIPI_TX26[7:0]</a> | MAP_DST_6[7:0]         |   |                       |                       |                       |                       |                       |                       |
| 0x45B     | <a href="#">MIPI_TX27[7:0]</a> | MAP_SRC_7[7:0]         |   |                       |                       |                       |                       |                       |                       |
| 0x45C     | <a href="#">MIPI_TX28[7:0]</a> | MAP_DST_7[7:0]         |   |                       |                       |                       |                       |                       |                       |
| 0x45D     | <a href="#">MIPI_TX29[7:0]</a> | MAP_SRC_8[7:0]         |   |                       |                       |                       |                       |                       |                       |
| 0x45E     | <a href="#">MIPI_TX30[7:0]</a> | MAP_DST_8[7:0]         |   |                       |                       |                       |                       |                       |                       |
| 0x45F     | <a href="#">MIPI_TX31[7:0]</a> | MAP_SRC_9[7:0]         |   |                       |                       |                       |                       |                       |                       |

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| ADDRESS   | NAME                           | MSB                   |                       |                       |                       |                     |                       |                  | LSB                   |
|-----------|--------------------------------|-----------------------|-----------------------|-----------------------|-----------------------|---------------------|-----------------------|------------------|-----------------------|
| 0x460     | <a href="#">MIPI_TX32[7:0]</a> | MAP_DST_9[7:0]        |                       |                       |                       |                     |                       |                  |                       |
| 0x461     | <a href="#">MIPI_TX33[7:0]</a> | MAP_SRC_10[7:0]       |                       |                       |                       |                     |                       |                  |                       |
| 0x462     | <a href="#">MIPI_TX34[7:0]</a> | MAP_DST_10[7:0]       |                       |                       |                       |                     |                       |                  |                       |
| 0x463     | <a href="#">MIPI_TX35[7:0]</a> | MAP_SRC_11[7:0]       |                       |                       |                       |                     |                       |                  |                       |
| 0x464     | <a href="#">MIPI_TX36[7:0]</a> | MAP_DST_11[7:0]       |                       |                       |                       |                     |                       |                  |                       |
| 0x465     | <a href="#">MIPI_TX37[7:0]</a> | MAP_SRC_12[7:0]       |                       |                       |                       |                     |                       |                  |                       |
| 0x466     | <a href="#">MIPI_TX38[7:0]</a> | MAP_DST_12[7:0]       |                       |                       |                       |                     |                       |                  |                       |
| 0x467     | <a href="#">MIPI_TX39[7:0]</a> | MAP_SRC_13[7:0]       |                       |                       |                       |                     |                       |                  |                       |
| 0x468     | <a href="#">MIPI_TX40[7:0]</a> | MAP_DST_13[7:0]       |                       |                       |                       |                     |                       |                  |                       |
| 0x469     | <a href="#">MIPI_TX41[7:0]</a> | MAP_SRC_14[7:0]       |                       |                       |                       |                     |                       |                  |                       |
| 0x46A     | <a href="#">MIPI_TX42[7:0]</a> | MAP_DST_14[7:0]       |                       |                       |                       |                     |                       |                  |                       |
| 0x46B     | <a href="#">MIPI_TX43[7:0]</a> | MAP_SRC_15[7:0]       |                       |                       |                       |                     |                       |                  |                       |
| 0x46C     | <a href="#">MIPI_TX44[7:0]</a> | MAP_DST_15[7:0]       |                       |                       |                       |                     |                       |                  |                       |
| 0x46D     | <a href="#">MIPI_TX45[7:0]</a> | MAP_DPHY_DEST_3[1:0]  | MAP_DPHY_DEST_2[1:0]  | MAP_DPHY_DEST_1[1:0]  | MAP_DPHY_DEST_0[1:0]  |                     |                       |                  |                       |
| 0x46E     | <a href="#">MIPI_TX46[7:0]</a> | MAP_DPHY_DEST_7[1:0]  | MAP_DPHY_DEST_6[1:0]  | MAP_DPHY_DEST_5[1:0]  | MAP_DPHY_DEST_4[1:0]  |                     |                       |                  |                       |
| 0x46F     | <a href="#">MIPI_TX47[7:0]</a> | MAP_DPHY_DEST_11[1:0] | MAP_DPHY_DEST_10[1:0] | MAP_DPHY_DEST_9[1:0]  | MAP_DPHY_DEST_8[1:0]  |                     |                       |                  |                       |
| 0x470     | <a href="#">MIPI_TX48[7:0]</a> | MAP_DPHY_DEST_15[1:0] | MAP_DPHY_DEST_14[1:0] | MAP_DPHY_DEST_13[1:0] | MAP_DPHY_DEST_12[1:0] |                     |                       |                  |                       |
| 0x471     | <a href="#">MIPI_TX49[7:0]</a> | MAP_CON[7:0]          |                       |                       |                       |                     |                       |                  |                       |
| 0x472     | <a href="#">MIPI_TX50[7:0]</a> | SKEW_PER_SEL[7:0]     |                       |                       |                       |                     |                       |                  |                       |
| 0x473     | <a href="#">MIPI_TX51[7:0]</a> | -                     | -                     | -                     | ALT2_M<br>EM_MA<br>P8 | MODE_<br>DT         | ALT_ME<br>M_MAP1<br>0 | ALT_ME<br>M_MAP8 | ALT_ME<br>M_MAP1<br>2 |
| MIPI_TX 2 |                                |                       |                       |                       |                       |                     |                       |                  |                       |
| 0x481     | <a href="#">MIPI_TX1[7:0]</a>  | MODE[7:0]             |                       |                       |                       |                     |                       |                  |                       |
| 0x482     | <a href="#">MIPI_TX2[7:0]</a>  | STATUS[7:0]           |                       |                       |                       |                     |                       |                  |                       |
| 0x483     | <a href="#">MIPI_TX3[7:0]</a>  | DESKEW_INIT[7:0]      |                       |                       |                       |                     |                       |                  |                       |
| 0x484     | <a href="#">MIPI_TX4[7:0]</a>  | DESKEW_PER[7:0]       |                       |                       |                       |                     |                       |                  |                       |
| 0x485     | <a href="#">MIPI_TX5[7:0]</a>  | CSI2_T_PRE[7:0]       |                       |                       |                       |                     |                       |                  |                       |
| 0x486     | <a href="#">MIPI_TX6[7:0]</a>  | CSI2_T_POST[7:0]      |                       |                       |                       |                     |                       |                  |                       |
| 0x487     | <a href="#">MIPI_TX7[7:0]</a>  | CSI2_TX_GAP[7:0]      |                       |                       |                       |                     |                       |                  |                       |
| 0x488     | <a href="#">MIPI_TX8[7:0]</a>  | CSI2_TWAKEUP_L[7:0]   |                       |                       |                       |                     |                       |                  |                       |
| 0x489     | <a href="#">MIPI_TX9[7:0]</a>  | CSI2_TWAKEUP_M[7:0]   |                       |                       |                       |                     |                       |                  |                       |
| 0x48A     | <a href="#">MIPI_TX10[7:0]</a> | CSI2_LANE_CNT[1:0]    | -                     | -                     | CSI_VC<br>X_EN        | CSI2_TWAKEUP_H[2:0] |                       |                  |                       |
| 0x48B     | <a href="#">MIPI_TX11[7:0]</a> | MAP_EN_L[7:0]         |                       |                       |                       |                     |                       |                  |                       |
| 0x48C     | <a href="#">MIPI_TX12[7:0]</a> | MAP_EN_H[7:0]         |                       |                       |                       |                     |                       |                  |                       |
| 0x48D     | <a href="#">MIPI_TX13[7:0]</a> | MAP_SRC_0[7:0]        |                       |                       |                       |                     |                       |                  |                       |
| 0x48E     | <a href="#">MIPI_TX14[7:0]</a> | MAP_DST_0[7:0]        |                       |                       |                       |                     |                       |                  |                       |
| 0x48F     | <a href="#">MIPI_TX15[7:0]</a> | MAP_SRC_1[7:0]        |                       |                       |                       |                     |                       |                  |                       |



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| ADDRESS | NAME                           | MSB                   |                       |                       |                       |             |                       |                  | LSB                   |
|---------|--------------------------------|-----------------------|-----------------------|-----------------------|-----------------------|-------------|-----------------------|------------------|-----------------------|
| 0x490   | <a href="#">MIPI_TX16[7:0]</a> | MAP_DST_1[7:0]        |                       |                       |                       |             |                       |                  |                       |
| 0x491   | <a href="#">MIPI_TX17[7:0]</a> | MAP_SRC_2[7:0]        |                       |                       |                       |             |                       |                  |                       |
| 0x492   | <a href="#">MIPI_TX18[7:0]</a> | MAP_DST_2[7:0]        |                       |                       |                       |             |                       |                  |                       |
| 0x493   | <a href="#">MIPI_TX19[7:0]</a> | MAP_SRC_3[7:0]        |                       |                       |                       |             |                       |                  |                       |
| 0x494   | <a href="#">MIPI_TX20[7:0]</a> | MAP_DST_3[7:0]        |                       |                       |                       |             |                       |                  |                       |
| 0x495   | <a href="#">MIPI_TX21[7:0]</a> | MAP_SRC_4[7:0]        |                       |                       |                       |             |                       |                  |                       |
| 0x496   | <a href="#">MIPI_TX22[7:0]</a> | MAP_DST_4[7:0]        |                       |                       |                       |             |                       |                  |                       |
| 0x497   | <a href="#">MIPI_TX23[7:0]</a> | MAP_SRC_5[7:0]        |                       |                       |                       |             |                       |                  |                       |
| 0x498   | <a href="#">MIPI_TX24[7:0]</a> | MAP_DST_5[7:0]        |                       |                       |                       |             |                       |                  |                       |
| 0x499   | <a href="#">MIPI_TX25[7:0]</a> | MAP_SRC_6[7:0]        |                       |                       |                       |             |                       |                  |                       |
| 0x49A   | <a href="#">MIPI_TX26[7:0]</a> | MAP_DST_6[7:0]        |                       |                       |                       |             |                       |                  |                       |
| 0x49B   | <a href="#">MIPI_TX27[7:0]</a> | MAP_SRC_7[7:0]        |                       |                       |                       |             |                       |                  |                       |
| 0x49C   | <a href="#">MIPI_TX28[7:0]</a> | MAP_DST_7[7:0]        |                       |                       |                       |             |                       |                  |                       |
| 0x49D   | <a href="#">MIPI_TX29[7:0]</a> | MAP_SRC_8[7:0]        |                       |                       |                       |             |                       |                  |                       |
| 0x49E   | <a href="#">MIPI_TX30[7:0]</a> | MAP_DST_8[7:0]        |                       |                       |                       |             |                       |                  |                       |
| 0x49F   | <a href="#">MIPI_TX31[7:0]</a> | MAP_SRC_9[7:0]        |                       |                       |                       |             |                       |                  |                       |
| 0x4A0   | <a href="#">MIPI_TX32[7:0]</a> | MAP_DST_9[7:0]        |                       |                       |                       |             |                       |                  |                       |
| 0x4A1   | <a href="#">MIPI_TX33[7:0]</a> | MAP_SRC_10[7:0]       |                       |                       |                       |             |                       |                  |                       |
| 0x4A2   | <a href="#">MIPI_TX34[7:0]</a> | MAP_DST_10[7:0]       |                       |                       |                       |             |                       |                  |                       |
| 0x4A3   | <a href="#">MIPI_TX35[7:0]</a> | MAP_SRC_11[7:0]       |                       |                       |                       |             |                       |                  |                       |
| 0x4A4   | <a href="#">MIPI_TX36[7:0]</a> | MAP_DST_11[7:0]       |                       |                       |                       |             |                       |                  |                       |
| 0x4A5   | <a href="#">MIPI_TX37[7:0]</a> | MAP_SRC_12[7:0]       |                       |                       |                       |             |                       |                  |                       |
| 0x4A6   | <a href="#">MIPI_TX38[7:0]</a> | MAP_DST_12[7:0]       |                       |                       |                       |             |                       |                  |                       |
| 0x4A7   | <a href="#">MIPI_TX39[7:0]</a> | MAP_SRC_13[7:0]       |                       |                       |                       |             |                       |                  |                       |
| 0x4A8   | <a href="#">MIPI_TX40[7:0]</a> | MAP_DST_13[7:0]       |                       |                       |                       |             |                       |                  |                       |
| 0x4A9   | <a href="#">MIPI_TX41[7:0]</a> | MAP_SRC_14[7:0]       |                       |                       |                       |             |                       |                  |                       |
| 0x4AA   | <a href="#">MIPI_TX42[7:0]</a> | MAP_DST_14[7:0]       |                       |                       |                       |             |                       |                  |                       |
| 0x4AB   | <a href="#">MIPI_TX43[7:0]</a> | MAP_SRC_15[7:0]       |                       |                       |                       |             |                       |                  |                       |
| 0x4AC   | <a href="#">MIPI_TX44[7:0]</a> | MAP_DST_15[7:0]       |                       |                       |                       |             |                       |                  |                       |
| 0x4AD   | <a href="#">MIPI_TX45[7:0]</a> | MAP_DPHY_DEST_3[1:0]  | MAP_DPHY_DEST_2[1:0]  | MAP_DPHY_DEST_1[1:0]  | MAP_DPHY_DEST_0[1:0]  |             |                       |                  |                       |
| 0x4AE   | <a href="#">MIPI_TX46[7:0]</a> | MAP_DPHY_DEST_7[1:0]  | MAP_DPHY_DEST_6[1:0]  | MAP_DPHY_DEST_5[1:0]  | MAP_DPHY_DEST_4[1:0]  |             |                       |                  |                       |
| 0x4AF   | <a href="#">MIPI_TX47[7:0]</a> | MAP_DPHY_DEST_11[1:0] | MAP_DPHY_DEST_10[1:0] | MAP_DPHY_DEST_9[1:0]  | MAP_DPHY_DEST_8[1:0]  |             |                       |                  |                       |
| 0x4B0   | <a href="#">MIPI_TX48[7:0]</a> | MAP_DPHY_DEST_15[1:0] | MAP_DPHY_DEST_14[1:0] | MAP_DPHY_DEST_13[1:0] | MAP_DPHY_DEST_12[1:0] |             |                       |                  |                       |
| 0x4B1   | <a href="#">MIPI_TX49[7:0]</a> | MAP_CON[7:0]          |                       |                       |                       |             |                       |                  |                       |
| 0x4B2   | <a href="#">MIPI_TX50[7:0]</a> | SKEW_PER_SEL[7:0]     |                       |                       |                       |             |                       |                  |                       |
| 0x4B3   | <a href="#">MIPI_TX51[7:0]</a> | –                     | –                     | –                     | ALT2_M<br>EM_MA<br>P8 | MODE_<br>DT | ALT_ME<br>M_MAP1<br>0 | ALT_ME<br>M_MAP8 | ALT_ME<br>M_MAP1<br>2 |

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| ADDRESS   | NAME           | MSB                 |   |   |   |                |                     | LSB |
|-----------|----------------|---------------------|---|---|---|----------------|---------------------|-----|
| MIPI_TX 3 |                |                     |   |   |   |                |                     |     |
| 0x4C1     | MIPI_TX1[7:0]  | MODE[7:0]           |   |   |   |                |                     |     |
| 0x4C2     | MIPI_TX2[7:0]  | STATUS[7:0]         |   |   |   |                |                     |     |
| 0x4C3     | MIPI_TX3[7:0]  | DESKEW_INIT[7:0]    |   |   |   |                |                     |     |
| 0x4C4     | MIPI_TX4[7:0]  | DESKEW_PER[7:0]     |   |   |   |                |                     |     |
| 0x4C5     | MIPI_TX5[7:0]  | CSI2_T_PRE[7:0]     |   |   |   |                |                     |     |
| 0x4C6     | MIPI_TX6[7:0]  | CSI2_T_POST[7:0]    |   |   |   |                |                     |     |
| 0x4C8     | MIPI_TX8[7:0]  | CSI2_TWAKEUP_L[7:0] |   |   |   |                |                     |     |
| 0x4C9     | MIPI_TX9[7:0]  | CSI2_TWAKEUP_M[7:0] |   |   |   |                |                     |     |
| 0x4CA     | MIPI_TX10[7:0] | -                   | - | - | - | CSI_VC<br>X_EN | CSI2_TWAKEUP_H[2:0] |     |
| 0x4CB     | MIPI_TX11[7:0] | MAP_EN_L[7:0]       |   |   |   |                |                     |     |
| 0x4CC     | MIPI_TX12[7:0] | MAP_EN_H[7:0]       |   |   |   |                |                     |     |
| 0x4CD     | MIPI_TX13[7:0] | MAP_SRC_0[7:0]      |   |   |   |                |                     |     |
| 0x4CE     | MIPI_TX14[7:0] | MAP_DST_0[7:0]      |   |   |   |                |                     |     |
| 0x4CF     | MIPI_TX15[7:0] | MAP_SRC_1[7:0]      |   |   |   |                |                     |     |
| 0x4D0     | MIPI_TX16[7:0] | MAP_DST_1[7:0]      |   |   |   |                |                     |     |
| 0x4D1     | MIPI_TX17[7:0] | MAP_SRC_2[7:0]      |   |   |   |                |                     |     |
| 0x4D2     | MIPI_TX18[7:0] | MAP_DST_2[7:0]      |   |   |   |                |                     |     |
| 0x4D3     | MIPI_TX19[7:0] | MAP_SRC_3[7:0]      |   |   |   |                |                     |     |
| 0x4D4     | MIPI_TX20[7:0] | MAP_DST_3[7:0]      |   |   |   |                |                     |     |
| 0x4D5     | MIPI_TX21[7:0] | MAP_SRC_4[7:0]      |   |   |   |                |                     |     |
| 0x4D6     | MIPI_TX22[7:0] | MAP_DST_4[7:0]      |   |   |   |                |                     |     |
| 0x4D7     | MIPI_TX23[7:0] | MAP_SRC_5[7:0]      |   |   |   |                |                     |     |
| 0x4D8     | MIPI_TX24[7:0] | MAP_DST_5[7:0]      |   |   |   |                |                     |     |
| 0x4D9     | MIPI_TX25[7:0] | MAP_SRC_6[7:0]      |   |   |   |                |                     |     |
| 0x4DA     | MIPI_TX26[7:0] | MAP_DST_6[7:0]      |   |   |   |                |                     |     |
| 0x4DB     | MIPI_TX27[7:0] | MAP_SRC_7[7:0]      |   |   |   |                |                     |     |
| 0x4DC     | MIPI_TX28[7:0] | MAP_DST_7[7:0]      |   |   |   |                |                     |     |
| 0x4DD     | MIPI_TX29[7:0] | MAP_SRC_8[7:0]      |   |   |   |                |                     |     |
| 0x4DE     | MIPI_TX30[7:0] | MAP_DST_8[7:0]      |   |   |   |                |                     |     |
| 0x4DF     | MIPI_TX31[7:0] | MAP_SRC_9[7:0]      |   |   |   |                |                     |     |
| 0x4E0     | MIPI_TX32[7:0] | MAP_DST_9[7:0]      |   |   |   |                |                     |     |
| 0x4E1     | MIPI_TX33[7:0] | MAP_SRC_10[7:0]     |   |   |   |                |                     |     |
| 0x4E2     | MIPI_TX34[7:0] | MAP_DST_10[7:0]     |   |   |   |                |                     |     |
| 0x4E3     | MIPI_TX35[7:0] | MAP_SRC_11[7:0]     |   |   |   |                |                     |     |
| 0x4E4     | MIPI_TX36[7:0] | MAP_DST_11[7:0]     |   |   |   |                |                     |     |
| 0x4E5     | MIPI_TX37[7:0] | MAP_SRC_12[7:0]     |   |   |   |                |                     |     |
| 0x4E6     | MIPI_TX38[7:0] | MAP_DST_12[7:0]     |   |   |   |                |                     |     |
| 0x4E7     | MIPI_TX39[7:0] | MAP_SRC_13[7:0]     |   |   |   |                |                     |     |
| 0x4E8     | MIPI_TX40[7:0] | MAP_DST_13[7:0]     |   |   |   |                |                     |     |

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| ADDRESS       | NAME                               | MSB                   |                       |                       |                       |             |                       |                  | LSB                   |
|---------------|------------------------------------|-----------------------|-----------------------|-----------------------|-----------------------|-------------|-----------------------|------------------|-----------------------|
| 0x4E9         | <a href="#">MIPI_TX41[7:0]</a>     | MAP_SRC_14[7:0]       |                       |                       |                       |             |                       |                  |                       |
| 0x4EA         | <a href="#">MIPI_TX42[7:0]</a>     | MAP_DST_14[7:0]       |                       |                       |                       |             |                       |                  |                       |
| 0x4EB         | <a href="#">MIPI_TX43[7:0]</a>     | MAP_SRC_15[7:0]       |                       |                       |                       |             |                       |                  |                       |
| 0x4EC         | <a href="#">MIPI_TX44[7:0]</a>     | MAP_DST_15[7:0]       |                       |                       |                       |             |                       |                  |                       |
| 0x4ED         | <a href="#">MIPI_TX45[7:0]</a>     | MAP_DPHY_DEST_3[1:0]  | MAP_DPHY_DEST_2[1:0]  | MAP_DPHY_DEST_1[1:0]  | MAP_DPHY_DEST_0[1:0]  |             |                       |                  |                       |
| 0x4EE         | <a href="#">MIPI_TX46[7:0]</a>     | MAP_DPHY_DEST_7[1:0]  | MAP_DPHY_DEST_6[1:0]  | MAP_DPHY_DEST_5[1:0]  | MAP_DPHY_DEST_4[1:0]  |             |                       |                  |                       |
| 0x4EF         | <a href="#">MIPI_TX47[7:0]</a>     | MAP_DPHY_DEST_11[1:0] | MAP_DPHY_DEST_10[1:0] | MAP_DPHY_DEST_9[1:0]  | MAP_DPHY_DEST_8[1:0]  |             |                       |                  |                       |
| 0x4F0         | <a href="#">MIPI_TX48[7:0]</a>     | MAP_DPHY_DEST_15[1:0] | MAP_DPHY_DEST_14[1:0] | MAP_DPHY_DEST_13[1:0] | MAP_DPHY_DEST_12[1:0] |             |                       |                  |                       |
| 0x4F1         | <a href="#">MIPI_TX49[7:0]</a>     | MAP_CON[7:0]          |                       |                       |                       |             |                       |                  |                       |
| 0x4F3         | <a href="#">MIPI_TX51[7:0]</a>     | –                     | –                     | –                     | ALT2_M<br>EM_MA<br>P8 | MODE_<br>DT | ALT_ME<br>M_MAP1<br>0 | ALT_ME<br>M_MAP8 | ALT_ME<br>M_MAP1<br>2 |
| MIPI_TX_EXT 0 |                                    |                       |                       |                       |                       |             |                       |                  |                       |
| 0x500         | <a href="#">MIPI_TX_EXT0[7:0]</a>  | MAP_SRC_0_H[2:0]      |                       | MAP_DST_0_H[2:0]      |                       | –           |                       | –                |                       |
| 0x501         | <a href="#">MIPI_TX_EXT1[7:0]</a>  | MAP_SRC_1_H[2:0]      |                       | MAP_DST_1_H[2:0]      |                       | –           |                       | –                |                       |
| 0x502         | <a href="#">MIPI_TX_EXT2[7:0]</a>  | MAP_SRC_2_H[2:0]      |                       | MAP_DST_2_H[2:0]      |                       | –           |                       | –                |                       |
| 0x503         | <a href="#">MIPI_TX_EXT3[7:0]</a>  | MAP_SRC_3_H[2:0]      |                       | MAP_DST_3_H[2:0]      |                       | –           |                       | –                |                       |
| 0x504         | <a href="#">MIPI_TX_EXT4[7:0]</a>  | MAP_SRC_4_H[2:0]      |                       | MAP_DST_4_H[2:0]      |                       | –           |                       | –                |                       |
| 0x505         | <a href="#">MIPI_TX_EXT5[7:0]</a>  | MAP_SRC_5_H[2:0]      |                       | MAP_DST_5_H[2:0]      |                       | –           |                       | –                |                       |
| 0x506         | <a href="#">MIPI_TX_EXT6[7:0]</a>  | MAP_SRC_6_H[2:0]      |                       | MAP_DST_6_H[2:0]      |                       | –           |                       | –                |                       |
| 0x507         | <a href="#">MIPI_TX_EXT7[7:0]</a>  | MAP_SRC_7_H[2:0]      |                       | MAP_DST_7_H[2:0]      |                       | –           |                       | –                |                       |
| 0x508         | <a href="#">MIPI_TX_EXT8[7:0]</a>  | MAP_SRC_8_H[2:0]      |                       | MAP_DST_8_H[2:0]      |                       | –           |                       | –                |                       |
| 0x509         | <a href="#">MIPI_TX_EXT9[7:0]</a>  | MAP_SRC_9_H[2:0]      |                       | MAP_DST_9_H[2:0]      |                       | –           |                       | –                |                       |
| 0x50A         | <a href="#">MIPI_TX_EXT10[7:0]</a> | MAP_SRC_10_H[2:0]     |                       | MAP_DST_10_H[2:0]     |                       | –           |                       | –                |                       |
| 0x50B         | <a href="#">MIPI_TX_EXT11[7:0]</a> | MAP_SRC_11_H[2:0]     |                       | MAP_DST_11_H[2:0]     |                       | –           |                       | –                |                       |
| 0x50C         | <a href="#">MIPI_TX_EXT12[7:0]</a> | MAP_SRC_12_H[2:0]     |                       | MAP_DST_12_H[2:0]     |                       | –           |                       | –                |                       |
| 0x50D         | <a href="#">MIPI_TX_EXT13[7:0]</a> | MAP_SRC_13_H[2:0]     |                       | MAP_DST_13_H[2:0]     |                       | –           |                       | –                |                       |
| 0x50E         | <a href="#">MIPI_TX_EXT14[7:0]</a> | MAP_SRC_14_H[2:0]     |                       | MAP_DST_14_H[2:0]     |                       | –           |                       | –                |                       |
| 0x50F         | <a href="#">MIPI_TX_EXT15[7:0]</a> | MAP_SRC_15_H[2:0]     |                       | MAP_DST_15_H[2:0]     |                       | –           |                       | –                |                       |
| MIPI_TX_EXT 1 |                                    |                       |                       |                       |                       |             |                       |                  |                       |
| 0x510         | <a href="#">MIPI_TX_EXT0[7:0]</a>  | MAP_SRC_0_H[2:0]      |                       | MAP_DST_0_H[2:0]      |                       | –           |                       | –                |                       |
| 0x511         | <a href="#">MIPI_TX_EXT1[7:0]</a>  | MAP_SRC_1_H[2:0]      |                       | MAP_DST_1_H[2:0]      |                       | –           |                       | –                |                       |
| 0x512         | <a href="#">MIPI_TX_EXT2[7:0]</a>  | MAP_SRC_2_H[2:0]      |                       | MAP_DST_2_H[2:0]      |                       | –           |                       | –                |                       |
| 0x513         | <a href="#">MIPI_TX_EXT3[7:0]</a>  | MAP_SRC_3_H[2:0]      |                       | MAP_DST_3_H[2:0]      |                       | –           |                       | –                |                       |
| 0x514         | <a href="#">MIPI_TX_EXT4[7:0]</a>  | MAP_SRC_4_H[2:0]      |                       | MAP_DST_4_H[2:0]      |                       | –           |                       | –                |                       |
| 0x515         | <a href="#">MIPI_TX_EXT5[7:0]</a>  | MAP_SRC_5_H[2:0]      |                       | MAP_DST_5_H[2:0]      |                       | –           |                       | –                |                       |
| 0x516         | <a href="#">MIPI_TX_EXT6[7:0]</a>  | MAP_SRC_6_H[2:0]      |                       | MAP_DST_6_H[2:0]      |                       | –           |                       | –                |                       |
| 0x517         | <a href="#">MIPI_TX_EXT7[7:0]</a>  | MAP_SRC_7_H[2:0]      |                       | MAP_DST_7_H[2:0]      |                       | –           |                       | –                |                       |
| 0x518         | <a href="#">MIPI_TX_EXT8[7:0]</a>  | MAP_SRC_8_H[2:0]      |                       | MAP_DST_8_H[2:0]      |                       | –           |                       | –                |                       |

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| ADDRESS              | NAME                               | MSB               |  |  |                   |   | LSB |
|----------------------|------------------------------------|-------------------|--|--|-------------------|---|-----|
| 0x519                | <a href="#">MIPI_TX_EXT9[7:0]</a>  | MAP_SRC_9_H[2:0]  |  |  | MAP_DST_9_H[2:0]  | – | –   |
| 0x51A                | <a href="#">MIPI_TX_EXT10[7:0]</a> | MAP_SRC_10_H[2:0] |  |  | MAP_DST_10_H[2:0] | – | –   |
| 0x51B                | <a href="#">MIPI_TX_EXT11[7:0]</a> | MAP_SRC_11_H[2:0] |  |  | MAP_DST_11_H[2:0] | – | –   |
| 0x51C                | <a href="#">MIPI_TX_EXT12[7:0]</a> | MAP_SRC_12_H[2:0] |  |  | MAP_DST_12_H[2:0] | – | –   |
| 0x51D                | <a href="#">MIPI_TX_EXT13[7:0]</a> | MAP_SRC_13_H[2:0] |  |  | MAP_DST_13_H[2:0] | – | –   |
| 0x51E                | <a href="#">MIPI_TX_EXT14[7:0]</a> | MAP_SRC_14_H[2:0] |  |  | MAP_DST_14_H[2:0] | – | –   |
| 0x51F                | <a href="#">MIPI_TX_EXT15[7:0]</a> | MAP_SRC_15_H[2:0] |  |  | MAP_DST_15_H[2:0] | – | –   |
| <b>MIPI_TX_EXT 2</b> |                                    |                   |  |  |                   |   |     |
| 0x520                | <a href="#">MIPI_TX_EXT0[7:0]</a>  | MAP_SRC_0_H[2:0]  |  |  | MAP_DST_0_H[2:0]  | – | –   |
| 0x521                | <a href="#">MIPI_TX_EXT1[7:0]</a>  | MAP_SRC_1_H[2:0]  |  |  | MAP_DST_1_H[2:0]  | – | –   |
| 0x522                | <a href="#">MIPI_TX_EXT2[7:0]</a>  | MAP_SRC_2_H[2:0]  |  |  | MAP_DST_2_H[2:0]  | – | –   |
| 0x523                | <a href="#">MIPI_TX_EXT3[7:0]</a>  | MAP_SRC_3_H[2:0]  |  |  | MAP_DST_3_H[2:0]  | – | –   |
| 0x524                | <a href="#">MIPI_TX_EXT4[7:0]</a>  | MAP_SRC_4_H[2:0]  |  |  | MAP_DST_4_H[2:0]  | – | –   |
| 0x525                | <a href="#">MIPI_TX_EXT5[7:0]</a>  | MAP_SRC_5_H[2:0]  |  |  | MAP_DST_5_H[2:0]  | – | –   |
| 0x526                | <a href="#">MIPI_TX_EXT6[7:0]</a>  | MAP_SRC_6_H[2:0]  |  |  | MAP_DST_6_H[2:0]  | – | –   |
| 0x527                | <a href="#">MIPI_TX_EXT7[7:0]</a>  | MAP_SRC_7_H[2:0]  |  |  | MAP_DST_7_H[2:0]  | – | –   |
| 0x528                | <a href="#">MIPI_TX_EXT8[7:0]</a>  | MAP_SRC_8_H[2:0]  |  |  | MAP_DST_8_H[2:0]  | – | –   |
| 0x529                | <a href="#">MIPI_TX_EXT9[7:0]</a>  | MAP_SRC_9_H[2:0]  |  |  | MAP_DST_9_H[2:0]  | – | –   |
| 0x52A                | <a href="#">MIPI_TX_EXT10[7:0]</a> | MAP_SRC_10_H[2:0] |  |  | MAP_DST_10_H[2:0] | – | –   |
| 0x52B                | <a href="#">MIPI_TX_EXT11[7:0]</a> | MAP_SRC_11_H[2:0] |  |  | MAP_DST_11_H[2:0] | – | –   |
| 0x52C                | <a href="#">MIPI_TX_EXT12[7:0]</a> | MAP_SRC_12_H[2:0] |  |  | MAP_DST_12_H[2:0] | – | –   |
| 0x52D                | <a href="#">MIPI_TX_EXT13[7:0]</a> | MAP_SRC_13_H[2:0] |  |  | MAP_DST_13_H[2:0] | – | –   |
| 0x52E                | <a href="#">MIPI_TX_EXT14[7:0]</a> | MAP_SRC_14_H[2:0] |  |  | MAP_DST_14_H[2:0] | – | –   |
| 0x52F                | <a href="#">MIPI_TX_EXT15[7:0]</a> | MAP_SRC_15_H[2:0] |  |  | MAP_DST_15_H[2:0] | – | –   |
| <b>MIPI_TX_EXT 3</b> |                                    |                   |  |  |                   |   |     |
| 0x530                | <a href="#">MIPI_TX_EXT0[7:0]</a>  | MAP_SRC_0_H[2:0]  |  |  | MAP_DST_0_H[2:0]  | – | –   |
| 0x531                | <a href="#">MIPI_TX_EXT1[7:0]</a>  | MAP_SRC_1_H[2:0]  |  |  | MAP_DST_1_H[2:0]  | – | –   |
| 0x532                | <a href="#">MIPI_TX_EXT2[7:0]</a>  | MAP_SRC_2_H[2:0]  |  |  | MAP_DST_2_H[2:0]  | – | –   |
| 0x533                | <a href="#">MIPI_TX_EXT3[7:0]</a>  | MAP_SRC_3_H[2:0]  |  |  | MAP_DST_3_H[2:0]  | – | –   |
| 0x534                | <a href="#">MIPI_TX_EXT4[7:0]</a>  | MAP_SRC_4_H[2:0]  |  |  | MAP_DST_4_H[2:0]  | – | –   |
| 0x535                | <a href="#">MIPI_TX_EXT5[7:0]</a>  | MAP_SRC_5_H[2:0]  |  |  | MAP_DST_5_H[2:0]  | – | –   |
| 0x536                | <a href="#">MIPI_TX_EXT6[7:0]</a>  | MAP_SRC_6_H[2:0]  |  |  | MAP_DST_6_H[2:0]  | – | –   |
| 0x537                | <a href="#">MIPI_TX_EXT7[7:0]</a>  | MAP_SRC_7_H[2:0]  |  |  | MAP_DST_7_H[2:0]  | – | –   |
| 0x538                | <a href="#">MIPI_TX_EXT8[7:0]</a>  | MAP_SRC_8_H[2:0]  |  |  | MAP_DST_8_H[2:0]  | – | –   |
| 0x539                | <a href="#">MIPI_TX_EXT9[7:0]</a>  | MAP_SRC_9_H[2:0]  |  |  | MAP_DST_9_H[2:0]  | – | –   |
| 0x53A                | <a href="#">MIPI_TX_EXT10[7:0]</a> | MAP_SRC_10_H[2:0] |  |  | MAP_DST_10_H[2:0] | – | –   |
| 0x53B                | <a href="#">MIPI_TX_EXT11[7:0]</a> | MAP_SRC_11_H[2:0] |  |  | MAP_DST_11_H[2:0] | – | –   |
| 0x53C                | <a href="#">MIPI_TX_EXT12[7:0]</a> | MAP_SRC_12_H[2:0] |  |  | MAP_DST_12_H[2:0] | – | –   |
| 0x53D                | <a href="#">MIPI_TX_EXT13[7:0]</a> | MAP_SRC_13_H[2:0] |  |  | MAP_DST_13_H[2:0] | – | –   |
| 0x53E                | <a href="#">MIPI_TX_EXT14[7:0]</a> | MAP_SRC_14_H[2:0] |  |  | MAP_DST_14_H[2:0] | – | –   |
| 0x53F                | <a href="#">MIPI_TX_EXT15[7:0]</a> | MAP_SRC_15_H[2:0] |  |  | MAP_DST_15_H[2:0] | – | –   |

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| ADDRESS | NAME           | MSB                   |                       |                       |                   |                           |                        |                          | LSB                     |
|---------|----------------|-----------------------|-----------------------|-----------------------|-------------------|---------------------------|------------------------|--------------------------|-------------------------|
| MISC    |                |                       |                       |                       |                   |                           |                        |                          |                         |
| 0x548   | UART_PT_0[7:0] | BITLEN_PT_1_L[7:0]    |                       |                       |                   |                           |                        |                          |                         |
| 0x549   | UART_PT_1[7:0] | –                     | –                     | BITLEN_PT_1_H[5:0]    |                   |                           |                        |                          |                         |
| 0x54A   | UART_PT_2[7:0] | BITLEN_PT_2_L[7:0]    |                       |                       |                   |                           |                        |                          |                         |
| 0x54B   | UART_PT_3[7:0] | –                     | –                     | BITLEN_PT_2_H[5:0]    |                   |                           |                        |                          |                         |
| 0x550   | I2C_PT_4[7:0]  | SRC_A_1[6:0]          |                       |                       |                   |                           |                        |                          | –                       |
| 0x551   | I2C_PT_5[7:0]  | DST_A_1[6:0]          |                       |                       |                   |                           |                        |                          | –                       |
| 0x552   | I2C_PT_6[7:0]  | SRC_B_1[6:0]          |                       |                       |                   |                           |                        |                          | –                       |
| 0x553   | I2C_PT_7[7:0]  | DST_B_1[6:0]          |                       |                       |                   |                           |                        |                          | –                       |
| 0x554   | I2C_PT_8[7:0]  | SRC_A_2[6:0]          |                       |                       |                   |                           |                        |                          | –                       |
| 0x555   | I2C_PT_9[7:0]  | DST_A_2[6:0]          |                       |                       |                   |                           |                        |                          | –                       |
| 0x557   | I2C_PT_11[7:0] | DST_B_2[6:0]          |                       |                       |                   |                           |                        |                          | –                       |
| 0x559   | BW_MEAS_0[7:0] | BW_USE_THR[7:0]       |                       |                       |                   |                           |                        |                          |                         |
| 0x55A   | BW_MEAS_1[7:0] | –                     | –                     | –                     | –                 | BW_US<br>E_OEN            | BW_INT                 | BW_US<br>E_LINK          | BW_US<br>E_LT           |
| 0x55C   | CNT4[7:0]      | VID_PXL_CRC_ERR0[7:0] |                       |                       |                   |                           |                        |                          |                         |
| 0x55D   | CNT5[7:0]      | VID_PXL_CRC_ERR1[7:0] |                       |                       |                   |                           |                        |                          |                         |
| 0x55E   | CNT6[7:0]      | VID_PXL_CRC_ERR2[7:0] |                       |                       |                   |                           |                        |                          |                         |
| 0x55F   | CNT7[7:0]      | VID_PXL_CRC_ERR3[7:0] |                       |                       |                   |                           |                        |                          |                         |
| GMSL1 A |                |                       |                       |                       |                   |                           |                        |                          |                         |
| 0xB02   | GMSL1_2[7:0]   | –                     | –                     | –                     | –                 | –                         | –                      | SRNG[1:0]                |                         |
| 0xB04   | GMSL1_4[7:0]   | –                     | OUTENB                | PRBSEN                | –                 | CC_POR<br>T_SEL           | –                      | REVCCE<br>N              | FWDCC<br>EN             |
| 0xB05   | GMSL1_5[7:0]   | I2CMET<br>HOD         | NO_RE<br>M_MST        | HVTR_M<br>ODE         | EN_EQ             | EQTUNE[3:0]               |                        |                          |                         |
| 0xB06   | GMSL1_6[7:0]   | HIGHIM<br>M           | MAX_RT<br>_EN         | I2C_RT<br>_EN         | GPI_CO<br>MP_EN   | GPI_RT<br>_EN             | HV_SRC[2:0]            |                          |                         |
| 0xB07   | GMSL1_7[7:0]   | DBL                   | DRS                   | BWS                   | –                 | HIBW                      | HVEN                   | –                        | PXL_CR<br>C             |
| 0xB08   | GMSL1_8[7:0]   | –                     | –                     | GPI_EN                | EN_FSY<br>NC_TX   | –                         | PKTCC<br>_EN           | CC_CRC_LENGTH[<br>1:0]   |                         |
| 0xB0D   | GMSL1_D[7:0]   | I2C_LOC<br>_ACK       | –                     | –                     | –                 | –                         | HS_TRA<br>CK_FSY<br>NC | –                        | –                       |
| 0xB0E   | GMSL1_E[7:0]   | DET_THR[7:0]          |                       |                       |                   |                           |                        |                          |                         |
| 0xB0F   | GMSL1_F[7:0]   | –                     | EN_DE<br>_FILT        | EN_HS<br>_FILT        | EN_VS<br>_FILT    | DE_EN                     | –                      | –                        | PRBS_T<br>YPE           |
| 0xB10   | GMSL1_10[7:0]  | RCEG_TYPE[1:0]        |                       | RCEG_B<br>OUND        | RCEG_ERR_NUM[3:0] |                           |                        |                          | RCEG_E<br>N             |
| 0xB11   | GMSL1_11[7:0]  | RCEG_ERR_RATE[3:0]    |                       |                       |                   | RCEG_LO_BST_PR<br>B[1:0]  |                        | RCEG_LO_BST_LE<br>N[1:0] |                         |
| 0xB12   | GMSL1_12[7:0]  | –                     | CC_CRC<br>_ERR_E<br>N | LINE_CRC_LOC[1:0<br>] |                   | LINE_C<br>RC_EN_<br>GMSL1 | –                      | MAX_RT<br>_ERR_E<br>N    | RCEG_E<br>RR_PER<br>_EN |

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|---------|-------------------------------|--------------------------|------------------|--------------------|---------------------|----------------|--------------|--------------------|--------------|
| 0xB13   | <a href="#">GMSL1_13[7:0]</a> | EOM_EN_G1                | EOM_PER_MOD_E_G1 | EOM_MAN_TRG_REQ_G1 | EOM_MIN_THR_G1[4:0] |                |              |                    |              |
| 0xB14   | <a href="#">GMSL1_14[7:0]</a> | AEQ_EN                   | AEQ_PER_MOD_E    | AEQ_MAN_TRG_REQ    | EOM_PER_THR[4:0]    |                |              |                    |              |
| 0xB15   | <a href="#">GMSL1_15[7:0]</a> | DET_ERR[7:0]             |                  |                    |                     |                |              |                    |              |
| 0xB16   | <a href="#">GMSL1_16[7:0]</a> | PRBS_ERR[7:0]            |                  |                    |                     |                |              |                    |              |
| 0xB17   | <a href="#">GMSL1_17[7:0]</a> | I2C_TIMED_OUT_GMSL1      | MAX_RT_ERR_I2C   | PRBS_OK            | GPI_IN              | MAX_RT_ERR_GPI | –            | –                  | –            |
| 0xB18   | <a href="#">GMSL1_18[7:0]</a> | CC_RETR_CNT[7:0]         |                  |                    |                     |                |              |                    |              |
| 0xB19   | <a href="#">GMSL1_19[7:0]</a> | CC_CRC_ERRCNT[7:0]       |                  |                    |                     |                |              |                    |              |
| 0xB1A   | <a href="#">GMSL1_1A[7:0]</a> | RCEG_ERR_CNT[7:0]        |                  |                    |                     |                |              |                    |              |
| 0xB1B   | <a href="#">GMSL1_1B[7:0]</a> | –                        | –                | –                  | –                   | –              | LINE_CRC_ERR | –                  | –            |
| 0xB1C   | <a href="#">GMSL1_1C[7:0]</a> | –                        | –                | EOM_EYE_WIDTH[5:0] |                     |                |              |                    |              |
| 0xB1D   | <a href="#">GMSL1_1D[7:0]</a> | –                        | –                | –                  | UNDERBOOST_DET      | AEQ_BST[3:0]   |              |                    |              |
| 0xB20   | <a href="#">GMSL1_20[7:0]</a> | CRC_VALUE_0[7:0]         |                  |                    |                     |                |              |                    |              |
| 0xB21   | <a href="#">GMSL1_21[7:0]</a> | CRC_VALUE_1[7:0]         |                  |                    |                     |                |              |                    |              |
| 0xB22   | <a href="#">GMSL1_22[7:0]</a> | CRC_VALUE_2[7:0]         |                  |                    |                     |                |              |                    |              |
| 0xB23   | <a href="#">GMSL1_23[7:0]</a> | CRC_VALUE_3[7:0]         |                  |                    |                     |                |              |                    |              |
| 0xB96   | <a href="#">GMSL1_96[7:0]</a> | CONV_GMSL1_DATATYPE[4:0] |                  |                    |                     |                | DIS_HIBW_E   | CONV_GMSL1_EN      | DBL_ALIGN_TO |
| 0xBCA   | <a href="#">GMSL1_CA[7:0]</a> | –                        | –                | –                  | –                   | –              | PHASELOCK    | WBLOCK_G1          | DATAOK       |
| 0xBCB   | <a href="#">GMSL1_CB[7:0]</a> | –                        | –                | –                  | –                   | –              | –            | –                  | LOCKED_G1    |
| 0xBD1   | <a href="#">GMSL1_D1[7:0]</a> | CNTL_OUT_ORD[2:0]        |                  |                    | CNTL_OUT_EN[4:0]    |                |              |                    |              |
| GMSL1 B |                               |                          |                  |                    |                     |                |              |                    |              |
| 0xC02   | <a href="#">GMSL1_2[7:0]</a>  | –                        | –                | –                  | –                   | –              | –            | SRNG[1:0]          |              |
| 0xC04   | <a href="#">GMSL1_4[7:0]</a>  | –                        | OUTENB           | PRBSEN             | –                   | CC_PORT_SEL    | –            | REVCCE_N           | FWDCC_EN     |
| 0xC05   | <a href="#">GMSL1_5[7:0]</a>  | I2CMETHOD                | NO_REM_MST       | HVTR_MODE          | EN_EQ               | EQTUNE[3:0]    |              |                    |              |
| 0xC06   | <a href="#">GMSL1_6[7:0]</a>  | HIGHIMM                  | MAX_RT_EN        | I2C_RT_EN          | GPI_COMP_EN         | GPI_RT_EN      | HV_SRC[2:0]  |                    |              |
| 0xC07   | <a href="#">GMSL1_7[7:0]</a>  | DBL                      | DRS              | BWS                | –                   | HIBW           | HVEN         | –                  | PXL_CRC      |
| 0xC08   | <a href="#">GMSL1_8[7:0]</a>  | –                        | –                | GPI_EN             | EN_FSYNC_TX         | –              | PKTCC_EN     | CC_CRC_LENGTH[1:0] |              |

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| ADDRESS      | NAME                          | MSB                         |                         |                                |                        |                           |                        |                          | LSB                     |
|--------------|-------------------------------|-----------------------------|-------------------------|--------------------------------|------------------------|---------------------------|------------------------|--------------------------|-------------------------|
| 0xC0D        | <a href="#">GMSL1_D[7:0]</a>  | I2C_LOC_ACK                 | -                       | -                              | -                      | -                         | HS_TRA<br>CK_FSY<br>NC | -                        | -                       |
| 0xC0E        | <a href="#">GMSL1_E[7:0]</a>  | DET_THR[7:0]                |                         |                                |                        |                           |                        |                          |                         |
| 0xC0F        | <a href="#">GMSL1_F[7:0]</a>  | -                           | EN_DE_FILT              | EN_HS_FILT                     | EN_VS_FILT             | DE_EN                     | -                      | -                        | PRBS_T<br>YPE           |
| 0xC10        | <a href="#">GMSL1_10[7:0]</a> | RCEG_TYPE[1:0]              |                         | RCEG_B<br>OUND                 | RCEG_ERR_NUM[3:0]      |                           |                        |                          | RCEG_E<br>N             |
| 0xC11        | <a href="#">GMSL1_11[7:0]</a> | RCEG_ERR_RATE[3:0]          |                         |                                |                        | RCEG_LO_BST_PR<br>B[1:0]  |                        | RCEG_LO_BST_LE<br>N[1:0] |                         |
| 0xC12        | <a href="#">GMSL1_12[7:0]</a> | -                           | CC_CRC<br>_ERR_E<br>N   | LINE_CRC_LOC[1:0]              |                        | LINE_C<br>RC_EN<br>_GMSL1 | -                      | MAX_RT<br>_ERR_E<br>N    | RCEG_E<br>RR_PER<br>_EN |
| 0xC13        | <a href="#">GMSL1_13[7:0]</a> | EOM_E<br>N_G1               | EOM_PE<br>R_MOD<br>E_G1 | EOM_M<br>AN_TRG<br>_REQ_G<br>1 | EOM_MIN_THR_G1[4:0]    |                           |                        |                          |                         |
| 0xC14        | <a href="#">GMSL1_14[7:0]</a> | AEQ_EN                      | AEQ_PE<br>R_MOD<br>E    | AEQ_MA<br>N_TRG<br>_REQ        | EOM_PER_THR[4:0]       |                           |                        |                          |                         |
| 0xC15        | <a href="#">GMSL1_15[7:0]</a> | DET_ERR[7:0]                |                         |                                |                        |                           |                        |                          |                         |
| 0xC16        | <a href="#">GMSL1_16[7:0]</a> | PRBS_ERR[7:0]               |                         |                                |                        |                           |                        |                          |                         |
| 0xC17        | <a href="#">GMSL1_17[7:0]</a> | I2C_TIM<br>ED_OUT<br>_GMSL1 | MAX_RT<br>_ERR_I2<br>C  | PRBS_O<br>K                    | GPI_IN                 | MAX_RT<br>_ERR_G<br>PI    | -                      | -                        | -                       |
| 0xC18        | <a href="#">GMSL1_18[7:0]</a> | CC_RETR_CNT[7:0]            |                         |                                |                        |                           |                        |                          |                         |
| 0xC19        | <a href="#">GMSL1_19[7:0]</a> | CC_CRC_ERRCNT[7:0]          |                         |                                |                        |                           |                        |                          |                         |
| 0xC1A        | <a href="#">GMSL1_1A[7:0]</a> | RCEG_ERR_CNT[7:0]           |                         |                                |                        |                           |                        |                          |                         |
| 0xC1B        | <a href="#">GMSL1_1B[7:0]</a> | -                           | -                       | -                              | -                      | -                         | LINE_C<br>RC_ERR       | -                        | -                       |
| 0xC1C        | <a href="#">GMSL1_1C[7:0]</a> | -                           | -                       | EOM_EYE_WIDTH[5:0]             |                        |                           |                        |                          |                         |
| 0xC1D        | <a href="#">GMSL1_1D[7:0]</a> | -                           | -                       | -                              | UNDER<br>BOOST_<br>DET | AEQ_BST[3:0]              |                        |                          |                         |
| 0xC20        | <a href="#">GMSL1_20[7:0]</a> | CRC_VALUE_0[7:0]            |                         |                                |                        |                           |                        |                          |                         |
| 0xC21        | <a href="#">GMSL1_21[7:0]</a> | CRC_VALUE_1[7:0]            |                         |                                |                        |                           |                        |                          |                         |
| 0xC22        | <a href="#">GMSL1_22[7:0]</a> | CRC_VALUE_2[7:0]            |                         |                                |                        |                           |                        |                          |                         |
| 0xC23        | <a href="#">GMSL1_23[7:0]</a> | CRC_VALUE_3[7:0]            |                         |                                |                        |                           |                        |                          |                         |
| 0xC96        | <a href="#">GMSL1_96[7:0]</a> | CONV_GMSL1_DATATYPE[4:0]    |                         |                                |                        |                           | DIS_HIB<br>W_E         | CONV_<br>GMSL1_<br>EN    | DBL_ALI<br>GN_TO        |
| 0xCCA        | <a href="#">GMSL1_CA[7:0]</a> | -                           | -                       | -                              | -                      | -                         | PHASEL<br>OCK          | WBLOC<br>K_G1            | DATAOK                  |
| 0xCD1        | <a href="#">GMSL1_D1[7:0]</a> | CNTL_OUT_ORD[2:0]           |                         |                                | CNTL_OUT_EN[4:0]       |                           |                        |                          |                         |
| GMSL1_COMMON |                               |                             |                         |                                |                        |                           |                        |                          |                         |
| 0xF00        | <a href="#">GMSL1_EN[7:0]</a> | -                           | -                       | -                              | -                      | -                         | -                      | LINK_EN<br>_B            | LINK_EN<br>_A           |



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| ADDRESS   | NAME                                | MSB                 |                  |                |                |                  |                 |                       | LSB                   |
|-----------|-------------------------------------|---------------------|------------------|----------------|----------------|------------------|-----------------|-----------------------|-----------------------|
| 0xF01     | <a href="#">GMSL1_UART[7:0]</a>     | –                   | –                | –              | –              | –                | –               | GMSL1_UART_A<br>RB_TO | GMSL1_UART_A<br>RB_EN |
| 0xF02     | <a href="#">COMMON1[7:0]</a>        | –                   | –                | –              | –              | –                | –               | REM_ACK<br>D_G1_B     | REM_ACK<br>D_G1_A     |
| 0xF03     | <a href="#">GMSL1_ERR_OEN[7:0]</a>  | –                   | –                | –              | –              | –                | –               | G1_BE<br>RR_OEN       | G1_A_E<br>RR_OEN      |
| 0xF04     | <a href="#">GMSL1_ERR_FLAG[7:0]</a> | –                   | –                | –              | –              | –                | –               | G1_BE<br>RR_FLAG      | G1_A_E<br>RR_FLAG     |
| 0xF05     | <a href="#">I2C_0[7:0]</a>          | –                   | –                | G1_SLV_SH[1:0] |                | –                | G1_SLV_TO[2:0]  |                       |                       |
| 0xF06     | <a href="#">I2C_1[7:0]</a>          | EN_I2C<br>LOOPBACK  | G1_MST_BT[2:0]   |                |                | –                | G1_MST_TO[2:0]  |                       |                       |
| 0xF07     | <a href="#">I2C_2[7:0]</a>          | G1_SRC_A[6:0]       |                  |                |                |                  |                 |                       | –                     |
| 0xF08     | <a href="#">I2C_3[7:0]</a>          | G1_DST_A[6:0]       |                  |                |                |                  |                 |                       | –                     |
| 0xF09     | <a href="#">I2C_4[7:0]</a>          | G1_SRC_B[6:0]       |                  |                |                |                  |                 |                       | –                     |
| 0xF0A     | <a href="#">I2C_5[7:0]</a>          | G1_DST_B[6:0]       |                  |                |                |                  |                 |                       | –                     |
| SPI_CC_WR |                                     |                     |                  |                |                |                  |                 |                       |                       |
| 0x1300    | <a href="#">SPI_CC_WR [7:0]</a>     | –                   | –                | –              | –              | –                | –               | –                     | –                     |
| SPI_CC_RD |                                     |                     |                  |                |                |                  |                 |                       |                       |
| 0x1380    | <a href="#">SPI_CC_RD [7:0]</a>     | –                   | –                | –              | –              | –                | –               | –                     | –                     |
| RLMS A    |                                     |                     |                  |                |                |                  |                 |                       |                       |
| 0x1403    | <a href="#">RLMS3[7:0]</a>          | AdaptEn             | –                | –              | –              | –                | –               | –                     | –                     |
| 0x1404    | <a href="#">RLMS4[7:0]</a>          | EOM_CHK_AMOUNT[3:0] |                  |                |                | EOM_CHK_THR[1:0] |                 | EOM_PERR_MOD<br>E     | EOM_EN                |
| 0x1405    | <a href="#">RLMS5[7:0]</a>          | EOM_MAN_TRG<br>REQ  | EOM_MIN_THR[6:0] |                |                |                  |                 |                       |                       |
| 0x1406    | <a href="#">RLMS6[7:0]</a>          | EOM_PV_MODE         | EOM_RST_THR[6:0] |                |                |                  |                 |                       |                       |
| 0x1407    | <a href="#">RLMS7[7:0]</a>          | EOM_DONE            | EOM[6:0]         |                |                |                  |                 |                       |                       |
| 0x1434    | <a href="#">RLMS34[7:0]</a>         | EyeMonPerCntL[7:0]  |                  |                |                |                  |                 |                       |                       |
| 0x1435    | <a href="#">RLMS35[7:0]</a>         | EyeMonPerCntH[7:0]  |                  |                |                |                  |                 |                       |                       |
| 0x1437    | <a href="#">RLMS37[7:0]</a>         | –                   | –                | –              | EyeMon<br>Done | EyeMon<br>CntClr | EyeMon<br>Start | EyeMon<br>Ph          | EyeMon<br>DPol        |
| 0x1439    | <a href="#">RLMS39[7:0]</a>         | EyeMonErrCntH[7:0]  |                  |                |                |                  |                 |                       |                       |
| 0x143A    | <a href="#">RLMS3A[7:0]</a>         | EyeMonValCntL[7:0]  |                  |                |                |                  |                 |                       |                       |
| 0x143B    | <a href="#">RLMS3B[7:0]</a>         | EyeMonValCntH[7:0]  |                  |                |                |                  |                 |                       |                       |
| 0x143D    | <a href="#">RLMS3D[7:0]</a>         | ErrChPh[6:0]        |                  |                |                |                  |                 |                       | ErrChPh<br>TogEn      |
| 0x143E    | <a href="#">RLMS3E[7:0]</a>         | ErrChPh<br>SecTA    | ErrChPhSec[6:0]  |                |                |                  |                 |                       |                       |

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| ADDRESS | NAME                        | MSB                |                       |               |   |   |                  |                  | LSB |         |
|---------|-----------------------------|--------------------|-----------------------|---------------|---|---|------------------|------------------|-----|---------|
| 0x143F  | <a href="#">RLMS3F[7:0]</a> | ErrChPhPriTA       | ErrChPhPri[6:0]       |               |   |   |                  |                  |     |         |
| 0x1458  | <a href="#">RLMS58[7:0]</a> | –                  | ErrChVTh1[6:0]        |               |   |   |                  |                  |     |         |
| 0x1459  | <a href="#">RLMS59[7:0]</a> | –                  | ErrChVTh0[6:0]        |               |   |   |                  |                  |     |         |
| 0x145B  | <a href="#">RLMS5B[7:0]</a> | –                  | ErrChVTh[6:0]         |               |   |   |                  |                  |     |         |
| 0x1464  | <a href="#">RLMS64[7:0]</a> | –                  | –                     | –             | – | – | –                | TxSSCMode[1:0]   |     |         |
| 0x1470  | <a href="#">RLMS70[7:0]</a> | –                  | TxSSCFrqCtrl[6:0]     |               |   |   |                  |                  |     |         |
| 0x1471  | <a href="#">RLMS71[7:0]</a> | –                  | TxSSCCenSprSt[5:0]    |               |   |   |                  |                  |     | TxSSCEn |
| 0x1472  | <a href="#">RLMS72[7:0]</a> | TxSSCPreScL[7:0]   |                       |               |   |   |                  |                  |     |         |
| 0x1473  | <a href="#">RLMS73[7:0]</a> | –                  | –                     | –             | – | – | TxSSCPreScH[2:0] |                  |     |         |
| 0x1474  | <a href="#">RLMS74[7:0]</a> | TxSSCPhL[7:0]      |                       |               |   |   |                  |                  |     |         |
| 0x1475  | <a href="#">RLMS75[7:0]</a> | –                  | TxSSCPhH[6:0]         |               |   |   |                  |                  |     |         |
| 0x1476  | <a href="#">RLMS76[7:0]</a> | –                  | –                     | –             | – | – | –                | TxSSCPhQuad[1:0] |     |         |
| 0x1495  | <a href="#">RLMS95[7:0]</a> | TxAmpManEn         | –                     | TxAmpMan[5:0] |   |   |                  |                  |     |         |
| 0x14A4  | <a href="#">RLMSA4[7:0]</a> | AEQ_PER_MULT[1:0]  | AEQ_PER[5:0]          |               |   |   |                  |                  |     |         |
| 0x14AC  | <a href="#">RLMSAC[7:0]</a> | ErrChPhSecTAFR3G   | ErrChPhSecFR3G[6:0]   |               |   |   |                  |                  |     |         |
| 0x14AD  | <a href="#">RLMSAD[7:0]</a> | ErrChPhPriTAFR3G   | ErrChPhPriFR3G[6:0]   |               |   |   |                  |                  |     |         |
| 0x14AE  | <a href="#">RLMSAE[7:0]</a> | ErrChPhSecTAFR1G5  | ErrChPhSecFR1G5[6:0]  |               |   |   |                  |                  |     |         |
| 0x14AF  | <a href="#">RLMSAF[7:0]</a> | ErrChPhPriTAFR1G5  | ErrChPhPriFR1G5[6:0]  |               |   |   |                  |                  |     |         |
| 0x14B0  | <a href="#">RLMSB0[7:0]</a> | ErrChPhSecTASR1G5  | ErrChPhSecSR1G5[6:0]  |               |   |   |                  |                  |     |         |
| 0x14B1  | <a href="#">RLMSB1[7:0]</a> | ErrChPhPriTASR1G5  | ErrChPhPriSR1G5[6:0]  |               |   |   |                  |                  |     |         |
| 0x14B2  | <a href="#">RLMSB2[7:0]</a> | ErrChPhSecTASRG75  | ErrChPhSecSRG75[6:0]  |               |   |   |                  |                  |     |         |
| 0x14B3  | <a href="#">RLMSB3[7:0]</a> | ErrChPhPriTASRG75  | ErrChPhPriSRG75[6:0]  |               |   |   |                  |                  |     |         |
| 0x14B4  | <a href="#">RLMSB4[7:0]</a> | ErrChPhSecTASRG375 | ErrChPhSecSRG375[6:0] |               |   |   |                  |                  |     |         |
| 0x14B5  | <a href="#">RLMSB5[7:0]</a> | ErrChPhPriTASRG375 | ErrChPhPriSRG375[6:0] |               |   |   |                  |                  |     |         |

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|---------|-----------------------------|-----------------------------|------------------------|----------------|----------------|----------------------|-------------------|------------------|------------------|
| 0x14B6  | <a href="#">RLMSB6[7:0]</a> | ErrChPh<br>SecTAS<br>RG1875 | ErrChPhSecSRG1875[6:0] |                |                |                      |                   |                  |                  |
| 0x14B7  | <a href="#">RLMSB7[7:0]</a> | ErrChPh<br>PriTASR<br>G1875 | ErrChPhPriSRG1875[6:0] |                |                |                      |                   |                  |                  |
| 0x14C4  | <a href="#">RLMSC4[7:0]</a> | –                           | –                      | –              | –              | –                    | RevFast           | –                | –                |
| RLMS B  |                             |                             |                        |                |                |                      |                   |                  |                  |
| 0x1503  | <a href="#">RLMS3[7:0]</a>  | AdaptEn                     | –                      | –              | –              | –                    | –                 | –                | –                |
| 0x1504  | <a href="#">RLMS4[7:0]</a>  | EOM_CHK_AMOUNT[3:0]         |                        |                |                | EOM_CHK_THR[1:0<br>] | EOM_PERR_MOD<br>E | EOM_EN           |                  |
| 0x1505  | <a href="#">RLMS5[7:0]</a>  | EOM_MAN_TRG<br>_REQ         | EOM_MIN_THR[6:0]       |                |                |                      |                   |                  |                  |
| 0x1506  | <a href="#">RLMS6[7:0]</a>  | EOM_PV<br>_MODE             | EOM_RST_THR[6:0]       |                |                |                      |                   |                  |                  |
| 0x1507  | <a href="#">RLMS7[7:0]</a>  | EOM_D<br>ONE                | EOM[6:0]               |                |                |                      |                   |                  |                  |
| 0x1534  | <a href="#">RLMS34[7:0]</a> | EyeMonPerCntL[7:0]          |                        |                |                |                      |                   |                  |                  |
| 0x1535  | <a href="#">RLMS35[7:0]</a> | EyeMonPerCntH[7:0]          |                        |                |                |                      |                   |                  |                  |
| 0x1537  | <a href="#">RLMS37[7:0]</a> | –                           | –                      | –              | EyeMon<br>Done | EyeMon<br>CntClr     | EyeMon<br>Start   | EyeMon<br>Ph     | EyeMon<br>DPol   |
| 0x1539  | <a href="#">RLMS39[7:0]</a> | EyeMonErrCntH[7:0]          |                        |                |                |                      |                   |                  |                  |
| 0x153A  | <a href="#">RLMS3A[7:0]</a> | EyeMonValCntL[7:0]          |                        |                |                |                      |                   |                  |                  |
| 0x153B  | <a href="#">RLMS3B[7:0]</a> | EyeMonValCntH[7:0]          |                        |                |                |                      |                   |                  |                  |
| 0x153D  | <a href="#">RLMS3D[7:0]</a> | ErrChPh[6:0]                |                        |                |                |                      |                   |                  | ErrChPh<br>TogEn |
| 0x153E  | <a href="#">RLMS3E[7:0]</a> | ErrChPh<br>SecTA            | ErrChPhSec[6:0]        |                |                |                      |                   |                  |                  |
| 0x153F  | <a href="#">RLMS3F[7:0]</a> | ErrChPh<br>PriTA            | ErrChPhPri[6:0]        |                |                |                      |                   |                  |                  |
| 0x1558  | <a href="#">RLMS58[7:0]</a> | –                           | ErrChVTh1[6:0]         |                |                |                      |                   |                  |                  |
| 0x1559  | <a href="#">RLMS59[7:0]</a> | –                           | ErrChVTh0[6:0]         |                |                |                      |                   |                  |                  |
| 0x155B  | <a href="#">RLMS5B[7:0]</a> | –                           | ErrChVTh[6:0]          |                |                |                      |                   |                  |                  |
| 0x1564  | <a href="#">RLMS64[7:0]</a> | –                           | –                      | –              | –              | –                    | –                 | TxSSCMode[1:0]   |                  |
| 0x1570  | <a href="#">RLMS70[7:0]</a> | –                           | TxSSCFrqCtrl[6:0]      |                |                |                      |                   |                  |                  |
| 0x1571  | <a href="#">RLMS71[7:0]</a> | –                           | TxSSCCenSprSt[5:0]     |                |                |                      |                   |                  | TxSSCE<br>n      |
| 0x1572  | <a href="#">RLMS72[7:0]</a> | TxSSCPreScL[7:0]            |                        |                |                |                      |                   |                  |                  |
| 0x1573  | <a href="#">RLMS73[7:0]</a> | –                           | –                      | –              | –              | –                    | TxSSCPreScH[2:0]  |                  |                  |
| 0x1574  | <a href="#">RLMS74[7:0]</a> | TxSSCPhL[7:0]               |                        |                |                |                      |                   |                  |                  |
| 0x1575  | <a href="#">RLMS75[7:0]</a> | –                           | TxSSCPhH[6:0]          |                |                |                      |                   |                  |                  |
| 0x1576  | <a href="#">RLMS76[7:0]</a> | –                           | –                      | –              | –              | –                    | –                 | TxSSCPhQuad[1:0] |                  |
| 0x1595  | <a href="#">RLMS95[7:0]</a> | TxAmp<br>ManEn              | –                      | TxAmpIMan[5:0] |                |                      |                   |                  |                  |

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|------------------|-----------------------------|----------------------|------------------------|---|---|---|---------|---|-----|
| 0x15A4           | <a href="#">RLMSA4[7:0]</a> | AEQ_PER_MULT[1:0]    | AEQ_PER[5:0]           |   |   |   |         |   |     |
| 0x15AC           | <a href="#">RLMSAC[7:0]</a> | ErrChPhSecTAF R3G    | ErrChPhSecFR3G[6:0]    |   |   |   |         |   |     |
| 0x15AD           | <a href="#">RLMSAD[7:0]</a> | ErrChPhPriTAF R3G    | ErrChPhPriFR3G[6:0]    |   |   |   |         |   |     |
| 0x15AE           | <a href="#">RLMSAE[7:0]</a> | ErrChPhSecTAF R1G5   | ErrChPhSecFR1G5[6:0]   |   |   |   |         |   |     |
| 0x15AF           | <a href="#">RLMSAF[7:0]</a> | ErrChPhPriTAF R1G5   | ErrChPhPriFR1G5[6:0]   |   |   |   |         |   |     |
| 0x15B0           | <a href="#">RLMSB0[7:0]</a> | ErrChPhSecTAS R1G5   | ErrChPhSecSR1G5[6:0]   |   |   |   |         |   |     |
| 0x15B1           | <a href="#">RLMSB1[7:0]</a> | ErrChPhPriTAS R1G5   | ErrChPhPriSR1G5[6:0]   |   |   |   |         |   |     |
| 0x15B2           | <a href="#">RLMSB2[7:0]</a> | ErrChPhSecTAS RG75   | ErrChPhSecSRG75[6:0]   |   |   |   |         |   |     |
| 0x15B3           | <a href="#">RLMSB3[7:0]</a> | ErrChPhPriTAS RG75   | ErrChPhPriSRG75[6:0]   |   |   |   |         |   |     |
| 0x15B4           | <a href="#">RLMSB4[7:0]</a> | ErrChPhSecTAS RG375  | ErrChPhSecSRG375[6:0]  |   |   |   |         |   |     |
| 0x15B5           | <a href="#">RLMSB5[7:0]</a> | ErrChPhPriTAS RG375  | ErrChPhPriSRG375[6:0]  |   |   |   |         |   |     |
| 0x15B6           | <a href="#">RLMSB6[7:0]</a> | ErrChPhSecTAS RG1875 | ErrChPhSecSRG1875[6:0] |   |   |   |         |   |     |
| 0x15B7           | <a href="#">RLMSB7[7:0]</a> | ErrChPhPriTAS RG1875 | ErrChPhPriSRG1875[6:0] |   |   |   |         |   |     |
| 0x15C4           | <a href="#">RLMSC4[7:0]</a> | –                    | –                      | – | – | – | RevFast | – | –   |
| <b>DPLL CSI2</b> |                             |                      |                        |   |   |   |         |   |     |
| <b>DPLL CSI3</b> |                             |                      |                        |   |   |   |         |   |     |

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Appendix A: Register Map and Tables

Register Details

**REG0 (0x0)**

| BIT         | 7             | 6 | 5 | 4 | 3 | 2 | 1 | 0           |
|-------------|---------------|---|---|---|---|---|---|-------------|
| Field       | DEV_ADDR[6:0] |   |   |   |   |   |   | CFG_BLOCK   |
| Reset       | 0b1001000     |   |   |   |   |   |   | 0b0         |
| Access Type | Write, Read   |   |   |   |   |   |   | Write, Read |

| BITFIELD  | BITS | DESCRIPTION                                                                                                                                | DECODE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-----------|------|--------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DEV_ADDR  | 7:1  | Device I <sup>2</sup> C Address<br><br>Default address is specified by CFG0 pin as described in the device data sheet.                     | 0b0000000: I <sup>2</sup> C write/read address is 0x00/0x01<br>0b0000001: I <sup>2</sup> C write/read address is 0x02/0x03<br>...<br>...<br>0b1001000: I <sup>2</sup> C write/read address is 0x90/0x91<br>0b1001010: I <sup>2</sup> C write/read address is 0x94/0x95<br>0b1001100: I <sup>2</sup> C write/read address is 0x98/0x99<br>0b1101000: I <sup>2</sup> C write/read address is 0xD0/0xD1<br>0b1101010: I <sup>2</sup> C write/read address is 0xD4/0xD5<br>0b1101100: I <sup>2</sup> C write/read address is 0xD8/0xD9<br>0b0101000: I <sup>2</sup> C write/read address is 0x50/0x51<br>0b0101010: I <sup>2</sup> C write/read address is 0x54/0x55<br>...<br>0b1111111: I <sup>2</sup> C write/read address is 0xFE/0xFF |
| CFG_BLOCK | 0    | Configuration Block<br><br>When set, all registers become non-writable (read-only). This bit can be used to freeze the chip configuration. | 0b0: Not Blocked<br>0b1: Blocked                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

**REG1 (0x1)**

| BIT         | 7           | 6           | 5            | 4           | 3            | 2 | 1            | 0 |
|-------------|-------------|-------------|--------------|-------------|--------------|---|--------------|---|
| Field       | IIC_2_EN    | IIC_1_EN    | DIS_LOCAL_CC | DIS_REM_CC  | TX_RATE[1:0] |   | RX_RATE[1:0] |   |
| Reset       | 0x0         | 0x0         | 0b0          | 0b0         | 0x0          |   | 0x2          |   |
| Access Type | Write, Read | Write, Read | Write, Read  | Write, Read | Write, Read  |   | Write, Read  |   |

| BITFIELD     | BITS | DESCRIPTION                                                         | DECODE                                                                                                          |
|--------------|------|---------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| IIC_2_EN     | 7    | Enable pass-through I <sup>2</sup> C Channel 2 (SDA2/Rx2, SCL2/Tx2) | 0b0: I <sup>2</sup> C pass-through Channel 2 disabled<br>0b1: I <sup>2</sup> C pass-through Channel 2 enabled   |
| IIC_1_EN     | 6    | Enable pass-through I <sup>2</sup> C Channel 1 (SDA1/Rx1, SCL1/Tx1) | 0b0: I <sup>2</sup> C pass-through Channel 1 disabled<br>0b1: I <sup>2</sup> C pass-through Channel 1 enabled   |
| DIS_LOCAL_CC | 5    | Disable control-channel connection to Rx/SDA and Tx/SCL pins        | 0b0: Rx/SDA and Tx/XCL connected to control channel<br>0b1: Rx/SDA and Tx/SCL disconnected from control channel |
| DIS_REM_C C  | 4    | Disable remote control-channel link over GMSL2 connection           | 0b0: Remote control channel enabled<br>0b1: Remote control channel disabled                                     |

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| BITFIELD | BITS | DESCRIPTION                                                                                                                                                                                           | DECODE                                                                |
|----------|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| TX_RATE  | 3:2  | Transmitter rate (when changed, becomes active after next link reset)                                                                                                                                 | 0b00: 187.5Mbps<br>0b01: Reserved<br>0b10: Reserved<br>0b11: Reserved |
| RX_RATE  | 1:0  | Receiver rate (when changed, becomes active after next link reset)<br><br>Default value is set by CXTP pin at power-up: 6Gbps when CXTP = 1 (Coax cable) and 3Gbps when CXTP = 0 (Twisted-pair cable) | 0b00: Reserved<br>0b01: 3Gbps<br>0b10: 6Gbps<br>0b11: Reserved        |

**REG2 (0x2)**

| BIT         | 7           | 6           | 5           | 4           | 3 | 2 | 1 | 0 |
|-------------|-------------|-------------|-------------|-------------|---|---|---|---|
| Field       | VID_EN_U    | VID_EN_Z    | VID_EN_Y    | VID_EN_X    | – | – | – | – |
| Reset       | 0b1         | 0b1         | 0b1         | 0b1         | – | – | – | – |
| Access Type | Write, Read | Write, Read | Write, Read | Write, Read | – | – | – | – |

| BITFIELD | BITS | DESCRIPTION  | DECODE                                                                          |
|----------|------|--------------|---------------------------------------------------------------------------------|
| VID_EN_U | 7    | Video Enable | 0b0: Video transmit Channel U disabled<br>0b1: Video transmit Channel U enabled |
| VID_EN_Z | 6    | Video Enable | 0b0: Video transmit Channel Z disabled<br>0b1: Video transmit Channel Z enabled |
| VID_EN_Y | 5    | Video Enable | 0b0: Video transmit Channel Y disabled<br>0b1: Video transmit Channel Y enabled |
| VID_EN_X | 4    | Video Enable | 0b0: Video transmit Channel X disabled<br>0b1: Video transmit Channel X enabled |

**REG3 (0x3)**

| BIT         | 7           | 6           | 5              | 4              | 3 | 2 | 1 | 0 |
|-------------|-------------|-------------|----------------|----------------|---|---|---|---|
| Field       | LOCK_CFG    | PT_SWAP     | UART_2_EN<br>N | UART_1_EN<br>N | – | – | – | – |
| Reset       | 0b0         | 0x1         | 0b0            | 0b1            | – | – | – | – |
| Access Type | Write, Read | Write, Read | Write, Read    | Write, Read    | – | – | – | – |

| BITFIELD  | BITS | DESCRIPTION                                                    | DECODE                                                                                |
|-----------|------|----------------------------------------------------------------|---------------------------------------------------------------------------------------|
| LOCK_CFG  | 7    | Configure LOCK pin behavior                                    | 0b0: GMSL2 link locked<br>0b1: GMSL2 link locked and MIPI output started              |
| PT_SWAP   | 6    | Swap I <sup>2</sup> C/UART pass-through device pin assignments | 0b0: Do not swap pin assignments<br>0b1: Swap pin assignments                         |
| UART_2_EN | 5    | Enable pass-through UART Channel 2 (SDA2/Rx2, SCL2/Tx2)        | 0b0: Pass-through UART Channel 2 disabled<br>0b1: Pass-through UART Channel 2 enabled |
| UART_1_EN | 4    | Enable pass-through UART Channel 1 (SDA1/Rx1, SCL1/Tx1)        | 0b0: Pass-through UART Channel 1 disabled<br>0b1: Pass-through UART Channel 1 enabled |

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**REG5 (0x5)**

| BIT         | 7           | 6           | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|-------------|-------------|---|---|---|---|---|---|
| Field       | LOCK_EN     | ERRB_EN     | – | – | – | – | – | – |
| Reset       | 0x1         | 0x1         | – | – | – | – | – | – |
| Access Type | Write, Read | Write, Read | – | – | – | – | – | – |

| BITFIELD | BITS | DESCRIPTION        | DECODE                                                |
|----------|------|--------------------|-------------------------------------------------------|
| LOCK_EN  | 7    | Enable LOCK Output | 0b0: LOCK output disabled<br>0b1: LOCK output enabled |
| ERRB_EN  | 6    | Enable ERRB Output | 0b0: ERRB output disabled<br>0b1: ERRB output enabled |

**REG6 (0x6)**

| BIT         | 7 | 6 | 5 | 4           | 3 | 2 | 1 | 0 |
|-------------|---|---|---|-------------|---|---|---|---|
| Field       | – | – | – | I2CSEL      | – | – | – | – |
| Reset       | – | – | – | 0b0         | – | – | – | – |
| Access Type | – | – | – | Write, Read | – | – | – | – |

| BITFIELD | BITS | DESCRIPTION                                                                                        | DECODE                             |
|----------|------|----------------------------------------------------------------------------------------------------|------------------------------------|
| I2CSEL   | 4    | I <sup>2</sup> C/UART selection<br>Bit is set according to the latched CFG0 pin value at power-up. | 0b0: UART<br>0b1: I <sup>2</sup> C |

**REG13 (0xD)**

| BIT         | 7           | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|-------------|---|---|---|---|---|---|---|
| Field       | DEV_ID[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x90        |   |   |   |   |   |   |   |
| Access Type | Read Only   |   |   |   |   |   |   |   |

| BITFIELD | BITS | DESCRIPTION       | DECODE         |
|----------|------|-------------------|----------------|
| DEV_ID   | 7:0  | Device Identifier | 0x96: MAX9296B |

**REG14 (0xE)**

| BIT         | 7 | 6 | 5 | 4 | 3            | 2 | 1 | 0 |
|-------------|---|---|---|---|--------------|---|---|---|
| Field       | – | – | – | – | DEV_REV[3:0] |   |   |   |
| Reset       | – | – | – | – | 0x00         |   |   |   |
| Access Type | – | – | – | – | Read Only    |   |   |   |

| BITFIELD | BITS | DESCRIPTION     | DECODE               |
|----------|------|-----------------|----------------------|
| DEV_REV  | 3:0  | Device Revision | 0xX: Revision number |



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**REG15 (0xF)**

| BIT         | 7 | 6 | 5                | 4 | 3         | 2          | 1           | 0          |
|-------------|---|---|------------------|---|-----------|------------|-------------|------------|
| Field       | – | – | SPEED_CPB L[1:0] |   | DV_CPB L  | DUAL_CPB L | SPLTR_CPB L | HDCP_CPB L |
| Reset       | – | – | 0x0              |   | 0b0       | 0b0        | 0b0         | 0b0        |
| Access Type | – | – | Read Only        |   | Read Only | Read Only  | Read Only   | Read Only  |

| BITFIELD    | BITS | DESCRIPTION                 | DECODE                                                                                                                                 |
|-------------|------|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| SPEED_CPB L | 5:4  | Video resolution capability | 0b00: No PCLK frequency limit<br>0b01: PCLK is limited to 240MHz<br>0b10: PCLK is limited to 160MHz<br>0b11: PCLK is limited to 104MHz |
| DV_CPB L    | 3    | Dual-View Video Capability  | 0b0: Dual-view video splitting is not available on this device<br>0b1: Dual-view video splitting is available on this device           |
| DUAL_CPB L  | 2    | Dual-Link Capability        | 0b0: Dual-link mode is not available on this device<br>0b1: Dual-link mode is available on this device                                 |
| SPLTR_CPB L | 1    | Splitter-Mode Capability    | 0b0: Splitter mode is not available on this device<br>0b1: Splitter mode is available on this device                                   |
| HDCP_CPB L  | 0    | HDCP Capability             | 0b0: HDCP not enabled on this device<br>0b1: HDCP enabled on this device                                                               |

**PWR1 (0x9)**

| BIT         | 7         | 6           | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|-----------|-------------|---|---|---|---|---|---|
| Field       | OVERTEMP  | VDD_OV_LIVE | – | – | – | – | – | – |
| Reset       | 0b0       | 0x0         | – | – | – | – | – | – |
| Access Type | Read Only | Read Only   | – | – | – | – | – | – |

| BITFIELD    | BITS | DESCRIPTION                                                                                                                         | DECODE                                                               |
|-------------|------|-------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| OVERTEMP    | 7    | Temperature Monitor Overtemp Indicator<br><br>Trip temperature $T_{TRIP}$ set by SET_TEMP1P0[1:0] in CMU6.                          | 0x0: $T < T_{TRIP}$<br>0x1: $T > T_{TRIP}$                           |
| VDD_OV_LIVE | 6    | Live $V_{DDSW}$ overvoltage status<br><br>$T_{TRIP}$ and hysteresis levels set by OV_LEVEL and OV_HYSTERESIS in register PM_OV_STAT | 0x0: $V_{DD\_sw} < V_{trip\_ov}$<br>0x1: $V_{DD\_sw} > V_{trip\_ov}$ |

**PWR4 (0xC)**

| BIT         | 7 | 6              | 5           | 4           | 3 | 2 | 1 | 0 |
|-------------|---|----------------|-------------|-------------|---|---|---|---|
| Field       | – | DIS_LOCAL_WAKE | WAKE_EN_B   | WAKE_EN_A   | – | – | – | – |
| Reset       | – | 0b0            | 0b0         | 0b1         | – | – | – | – |
| Access Type | – | Write, Read    | Write, Read | Write, Read | – | – | – | – |

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| BITFIELD       | BITS | DESCRIPTION                                       | DECODE                                                                    |
|----------------|------|---------------------------------------------------|---------------------------------------------------------------------------|
| DIS_LOCAL_WAKE | 6    | Disable Wake-Up by Local $\mu$ C from SDA_RX Pin  | 0b0: Local wake-up enabled<br>0b1: Local wake-up disabled                 |
| WAKE_EN_B      | 5    | Enable Wake-Up by Remote Chip Connected to Link B | 0b0: Link B remote wake-up disabled<br>0b1: Link B remote wake-up enabled |
| WAKE_EN_A      | 4    | Enable Wake-Up by Remote Chip Connected to Link A | 0b0: Link A remote wake-up disabled<br>0b1: Link A remote wake-up enabled |

**CTRL0 (0x10)**

| BIT         | 7             | 6              | 5                      | 4             | 3           | 2              | 1             | 0 |
|-------------|---------------|----------------|------------------------|---------------|-------------|----------------|---------------|---|
| Field       | RESET_AL<br>L | RESET_LIN<br>K | RESET_ON<br>ESHOT      | AUTO_LIN<br>K | SLEEP       | REG_ENAB<br>LE | LINK_CFG[1:0] |   |
| Reset       | 0b0           | 0b0            | 0b0                    | 0b1           | 0b0         | 0x0            | 0b01          |   |
| Access Type | Write, Read   | Write, Read    | Write Clears All, Read | Write, Read   | Write, Read | Write, Read    | Write, Read   |   |

| BITFIELD          | BITS | DESCRIPTION                                                                                                                                                                                              | DECODE                                                                                                                                                                                                                                       |
|-------------------|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| RESET_ALL         | 7    | Writing a 1 to this bit resets the device (including all blocks) and resets registers to their default values.<br><br>This is equivalent to toggling the PWDNB pin. The bit self clears when written to. | 0b0: No action<br>0b1: Activate chip reset                                                                                                                                                                                                   |
| RESET_LIN<br>K    | 6    | Reset whole data path (keep register settings)<br><br>Write 1 to activate reset, write 0 to release reset.                                                                                               | 0b0: Release link reset<br>0b1: Activate link reset                                                                                                                                                                                          |
| RESET_ONE<br>SHOT | 5    | Reset whole data path (keep register settings) one shot<br><br>Write 1 to activate reset, bit self clears and automatically releases reset.                                                              | 0b0: No action<br>0b1: Reset data path                                                                                                                                                                                                       |
| AUTO_LINK         | 4    | Automatically select link configuration (single A, single B or dual)<br><br>Splitter mode is not automatic. For splitter mode, set LINK_CFG = 0b11.                                                      | 0b0: Disable automatic link configuration<br>0b1: Enable automatic link configuration                                                                                                                                                        |
| SLEEP             | 3    | Activate Sleep Mode                                                                                                                                                                                      | 0b0: Sleep mode disabled<br>0b1: Sleep mode enabled                                                                                                                                                                                          |
| REG_ENABL<br>E    | 2    | V <sub>DD</sub> LDO Regulator Enable                                                                                                                                                                     | 0b0: V <sub>DD</sub> LDO regulator disabled (bypassed) when REG_MNL = 1<br>0b1: V <sub>DD</sub> LDO regulator enabled when REG_ENABLE = 1 and REG_MNL = 1. When V <sub>DD</sub> = 1.2V, first set REG_ENABLE = 1, and then write REG_MNL = 1 |

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| BITFIELD | BITS | DESCRIPTION                                                                                       | DECODE                                                                                                                                                                                                                                                                                                                             |
|----------|------|---------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LINK_CFG | 1:0  | In conjunction with AUTO_LINK, this bitfield selects the link configuration per the decode table. | 0b00: If AUTO_LINK = 0, Dual link selected. If AUTO_LINK = 1, Link mode is automatically selected<br>0b01: If AUTO_LINK = 0, Link A is selected. If AUTO_LINK = 1, Link mode is automatically selected<br>0b10: If AUTO_LINK = 0, Link B is selected. If AUTO_LINK = 1, Link mode is automatically selected<br>0b11: Splitter mode |

**CTRL1 (0x11)**

| BIT         | 7 | 6 | 5 | 4                    | 3 | 2           | 1 | 0           |
|-------------|---|---|---|----------------------|---|-------------|---|-------------|
| Field       | – | – | – | BACK_COM<br>MP_SPLTR | – | CXTP_B      | – | CXTP_A      |
| Reset       | – | – | – | 0x0                  | – | 0b0         | – | 0b0         |
| Access Type | – | – | – | Write, Read          | – | Write, Read | – | Write, Read |

| BITFIELD            | BITS | DESCRIPTION                                                                                                 | DECODE                                                    |
|---------------------|------|-------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| BACK_COM<br>P_SPLTR | 4    | Enable Splitter-Mode Compatibility                                                                          | 0b0: Compatibility disabled<br>0b1: Compatibility enabled |
| CXTP_B              | 2    | Coax/twisted-pair cable select for Link B<br>Bit is set according to the latched CXTP pin value at power-up | 0b0: Shielded twisted-pair drive<br>0b1: Coax drive       |
| CXTP_A              | 0    | Coax/twisted-pair cable select for Link A<br>Bit is set according to the latched CXTP pin value at power-up | 0b0: Shielded twisted-pair drive<br>0b1: Coax drive       |

**CTRL2 (0x12)**

| BIT         | 7 | 6 | 5 | 4           | 3 | 2 | 1 | 0 |
|-------------|---|---|---|-------------|---|---|---|---|
| Field       | – | – | – | REG_MNL     | – | – | – | – |
| Reset       | – | – | – | 0b0         | – | – | – | – |
| Access Type | – | – | – | Write, Read | – | – | – | – |

| BITFIELD | BITS | DESCRIPTION                                                                                  | DECODE                                         |
|----------|------|----------------------------------------------------------------------------------------------|------------------------------------------------|
| REG_MNL  | 4    | Enable regulator manual mode to allow regulator to be forced on or off via REG_ENABLE (0x10) | 0b0: Normal mode<br>0b1: Regulator manual mode |

**CTRL3 (0x13)**

| BIT         | 7 | 6 | 5              | 4 | 3         | 2         | 1          | 0 |
|-------------|---|---|----------------|---|-----------|-----------|------------|---|
| Field       | – | – | LINK_MODE[1:0] |   | LOCKED    | ERROR     | CMU_LOCKED | – |
| Reset       | – | – | 0x1            |   | 0b0       | 0b0       | 0b0        | – |
| Access Type | – | – | Read Only      |   | Read Only | Read Only | Read Only  | – |

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| BITFIELD   | BITS | DESCRIPTION                                       | DECODE                                                                     |
|------------|------|---------------------------------------------------|----------------------------------------------------------------------------|
| LINK_MODE  | 5:4  | Active link mode                                  | 0b00: Dual link<br>0b01: Link A<br>0b10: Link B<br>0b11: Splitter mode     |
| LOCKED     | 3    | GMSL2 link locked (bidirectional)                 | 0b0: GMSL2 link not locked<br>0b1: GMSL2 link locked                       |
| ERROR      | 2    | Reflects error status (inverse of ERRB pin value) | 0b0: ERRB not asserted (ERRB pin = 1)<br>0b1: ERRB asserted (ERRB pin = 0) |
| CMU_LOCKED | 1    | Clock multiplier unit (CMU) locked                | 0b0: CMU not locked<br>0b1: CMU locked                                     |

**INTR0 (0x18)**

| BIT         | 7 | 6 | 5 | 4 | 3               | 2                | 1 | 0 |
|-------------|---|---|---|---|-----------------|------------------|---|---|
| Field       | – | – | – | – | AUTO_ERR_RST_EN | DEC_ERR_THR[2:0] |   |   |
| Reset       | – | – | – | – | 0b0             | 0x0              |   |   |
| Access Type | – | – | – | – | Write, Read     | Write, Read      |   |   |

| BITFIELD        | BITS | DESCRIPTION                                                                                                                                                                                                                               | DECODE                                                                                                                                                   |
|-----------------|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| AUTO_ERR_RST_EN | 3    | Automatically resets DEC_ERR_A (0x22), DEC_ERR_B (0x23), and <del>IDLE_ERR (0x24)</del> bitfields after ERRB pin is asserted for 1µs                                                                                                      | 0b0: Auto reset disabled<br>0b1: Auto reset enabled                                                                                                      |
| DEC_ERR_THR     | 2:0  | Decoding and idle-error reporting threshold.<br>DEC_ERR_FLAG_A is asserted when DEC_ERR_A ≥ DEC_ERR_THR.<br>DEC_ERR_FLAG_B is asserted when DEC_ERR_B ≥ DEC_ERR_THR.<br><del>IDLE_ERR_FLAG is asserted when IDLE_ERR ≥ DEC_ERR_THR.</del> | 0b000: 1 error<br>0b001: 2 errors<br>0b010: 4 errors<br>0b011: 8 errors<br>0b100: 16 errors<br>0b101: 32 errors<br>0b110: 64 errors<br>0b111: 128 errors |

**INTR1 (0x19)**

| BIT         | 7                | 6 | 5 | 4 | 3               | 2                | 1 | 0 |
|-------------|------------------|---|---|---|-----------------|------------------|---|---|
| Field       | PKT_CNT_EXP[3:0] |   |   |   | AUTO_CNT_RST_EN | PKT_CNT_THR[2:0] |   |   |
| Reset       | 0x0              |   |   |   | 0b0             | 0x0              |   |   |
| Access Type | Write, Read      |   |   |   | Write, Read     | Write, Read      |   |   |

| BITFIELD        | BITS | DESCRIPTION                                                                                      | DECODE                                              |
|-----------------|------|--------------------------------------------------------------------------------------------------|-----------------------------------------------------|
| PKT_CNT_EXP     | 7:4  | Packet Count Multiplier Exponent<br><br>See the description of PKT_CNT bitfield (register CNT3). | 0bXXX: PKT_CNT exponent                             |
| AUTO_CNT_RST_EN | 3    | Automatically reset PKT_CNT bitfield (register CNT3) after ERRB pin is asserted for 1µs          | 0b0: Auto reset disabled<br>0b1: Auto reset enabled |

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| BITFIELD        | BITS | DESCRIPTION                                                                                                                                              | DECODE                                                                                                                                                           |
|-----------------|------|----------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PKT_CNT_T<br>HR | 2:0  | Packet Count Reporting Threshold<br><br>See PKT_CNT bitfield description (register CNT3).<br><br>PKT_CNT_FLAG is asserted when<br>PKT_CNT ≥ PKT_CNT_THR. | 0b000: 1 packet<br>0b001: 2 packets<br>0b010: 4 packets<br>0b011: 8 packets<br>0b100: 16 packets<br>0b101: 32 packets<br>0b110: 64 packets<br>0b111: 128 packets |

**INTR2 (0x1A)**

HIDE

| BIT         | 7             | 6             | 5           | 4               | 3 | 2            | 1             | 0             |
|-------------|---------------|---------------|-------------|-----------------|---|--------------|---------------|---------------|
| Field       | PHY_INT_OEN_B | PHY_INT_OEN_A | REM_ERR_OEN | MEM_INT_ERR_OEN | – | IDLE_ERR_OEN | DEC_ERR_OEN_B | DEC_ERR_OEN_A |
| Reset       | 0b0           | 0b0           | 0b0         | 0x0             | – | 0b0          | 0b1           | 0b1           |
| Access Type | Write, Read   | Write, Read   | Write, Read | Write, Read     | – | Write, Read  | Write, Read   | Write, Read   |

| BITFIELD        | BITS | DESCRIPTION                                                             | DECODE                                            |
|-----------------|------|-------------------------------------------------------------------------|---------------------------------------------------|
| PHY_INT_OEN_B   | 7    | Enable reporting of PHY interrupt (PHY_INT_B - 0x1B) at ERRB pin        | 0b0: Reporting disabled<br>0b1: Reporting enabled |
| PHY_INT_OEN_A   | 6    | Enable reporting of PHY interrupt (PHY_INT_A - 0x1B) at ERRB pin        | 0b0: Reporting disabled<br>0b1: Reporting enabled |
| REM_ERR_OEN     | 5    | Enable reporting of remote error status (REM_ERR - 0x1B) at ERRB pin    | 0b0: Reporting disabled<br>0b1: Reporting enabled |
| MEM_INT_ERR_OEN | 4    | Enable memory error status reporting on ERRB pin                        | 0b0: Reporting disabled<br>0b1: Reporting enabled |
| IDLE_ERR_OEN    | 2    | Enable reporting of idle-word errors (IDLE_ERR_FLAG - 0x1B) at ERRB pin | 0b0: Reporting disabled<br>0b1: Reporting enabled |
| DEC_ERR_OEN_B   | 1    | Enable reporting of decoding errors (DEC_ERR_FLAG_B - 0x1B) at ERRB pin | 0b0: Reporting disabled<br>0b1: Reporting enabled |
| DEC_ERR_OEN_A   | 0    | Enable reporting of decoding errors (DEC_ERR_FLAG_A - 0x1B) at ERRB pin | 0b0: Reporting disabled<br>0b1: Reporting enabled |

**INTR3 (0x1B)**

HIDE

| BIT         | 7         | 6         | 5            | 4                | 3 | 2             | 1              | 0              |
|-------------|-----------|-----------|--------------|------------------|---|---------------|----------------|----------------|
| Field       | PHY_INT_B | PHY_INT_A | REM_ERR_FLAG | MEM_INT_ERR_FLAG | – | IDLE_ERR_FLAG | DEC_ERR_FLAG_B | DEC_ERR_FLAG_A |
| Reset       | 0b0       | 0b0       | 0b0          | 0x0              | – | 0b0           | 0b0            | 0b0            |
| Access Type | Read Only | Read Only | Read Only    | Read Only        | – | Read Only     | Read Only      | Read Only      |

| BITFIELD  | BITS | DESCRIPTION                                                                                                           | DECODE                                                    |
|-----------|------|-----------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| PHY_INT_B | 7    | PHY Interrupt of Link B<br><br>See bitfields I2C_PT_0 (0x4C), I2C_PT_1 (0x4D), I2C_PT_2 (0x4E), and UART_PT_0 (0x4F). | 0b0: No Link B PHY interrupt<br>0b1: Link B PHY interrupt |
| PHY_INT_A | 6    | PHY Interrupt of Link A. See registers I2C_PT_0 (0x4C), I2C_PT_1 (0x4D), I2C_PT_2 (0x4E), and UART_PT_0 (0x4F).       | 0b0: No Link A PHY interrupt<br>0b1: Link A PHY interrupt |

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| BITLEVEL             | BITS | DESCRIPTION                                                                                  | DECODE                                                      |
|----------------------|------|----------------------------------------------------------------------------------------------|-------------------------------------------------------------|
| REM_ERR_F<br>LAG     | 5    | Received remote side error status (inverse of remote side ERRB pin level)                    | 0b0: No remote side error<br>0b1: Remote side error         |
| MEM_INT_E<br>RR_FLAG | 4    | Memory error flag                                                                            | 0x0: No memory error occurred<br>0x1: Memory error occurred |
| IDE_ERR_F<br>LAG     | 2    | Idle word error flag. Asserted when<br>IDLE_ERR ≥ DEC_ERR_THR                                | 0b0: Flag not asserted<br>0b1: Flag asserted                |
| DEC_ERR_F<br>LAG_B   | 1    | Decoding Error Flag for Link B<br><br>Asserted when DEC_ERR_B (0x23) ≥<br>DEC_ERR_THR (0x18) | 0b0: Flag not asserted<br>0b1: Flag asserted                |
| DEC_ERR_F<br>LAG_A   | 0    | Decoding Error Flag for Link A<br><br>Asserted when DEC_ERR_A (0x22) ≥<br>DEC_ERR_THR (0x18) | 0b0: Flag not asserted<br>0b1: Flag asserted                |

INTR4 (0x1C)

| BIT            | 7                 | 6                 | 5 | 4 | 3              | 2              | 1               | 0              |
|----------------|-------------------|-------------------|---|---|----------------|----------------|-----------------|----------------|
| Field          | EOM_ERR_<br>OEN_B | EOM_ERR_<br>OEN_A | – | – | MAX_RT_O<br>EN | RT_CNT_O<br>EN | PKT_CNT_<br>OEN | WM_ERR_<br>OEN |
| Reset          | 0b0               | 0b0               | – | – | 0b1            | 0b0            | 0b0             | 0b1            |
| Access<br>Type | Write, Read       | Write, Read       | – | – | Write, Read    | Write, Read    | Write, Read     | Write, Read    |

| BITLEVEL          | BITS | DESCRIPTION                                                                                                     | DECODE                                            |
|-------------------|------|-----------------------------------------------------------------------------------------------------------------|---------------------------------------------------|
| EOM_ERR_<br>OEN_B | 7    | Enable reporting of eye-opening monitor error<br>(EOM_ERR_FLAG_B - 0x1D ) for Link B at<br>ERRB pin             | 0b0: Reporting disabled<br>0b1: Reporting enabled |
| EOM_ERR_<br>OEN_A | 6    | Enable reporting of eye-opening monitor error<br>(EOM_ERR_FLAG_A - 0x1D) for Link A at<br>ERRB pin              | 0b0: Reporting disabled<br>0b1: Reporting enabled |
| MAX_RT_OE<br>N    | 3    | Enable reporting of combined ARQ maximum<br>retransmission limit error flag<br>(MAX_RT_FLAG - 0x1D) at ERRB pin | 0b0: Reporting disabled<br>0b1: Reporting enabled |
| RT_CNT_OE<br>N    | 2    | Enable reporting of combined ARQ<br>retransmission event flag (RT_CNT_FLAG -<br>0x1D) at ERRB pin               | 0b0: Reporting disabled<br>0b1: Reporting enabled |
| PKT_CNT_O<br>EN   | 1    | Enable reporting of packet count flag<br>(PKT_CNT_FLAG- 0x1D) at ERRB pin                                       | 0b0: Reporting disabled<br>0b1: Reporting enabled |
| WM_ERR_O<br>EN    | 0    | Enable reporting of watermark errors<br>(WM_ERR_FLAG - 0x1D) at ERRB pin                                        | 0b0: Reporting disabled<br>0b1: Reporting enabled |

INTR5 (0x1D)

| BIT            | 7                  | 6                  | 5 | 4 | 3               | 2               | 1                | 0                  |
|----------------|--------------------|--------------------|---|---|-----------------|-----------------|------------------|--------------------|
| Field          | EOM_ERR_<br>FLAG_B | EOM_ERR_<br>FLAG_A | – | – | MAX_RT_F<br>LAG | RT_CNT_F<br>LAG | PKT_CNT_<br>FLAG | WM_ERR_<br>FLAG    |
| Reset          | 0b0                | 0b0                | – | – | 0b0             | 0b0             | 0b0              | 0b0                |
| Access<br>Type | Read Only          | Read Only          | – | – | Read Only       | Read Only       | Read Only        | Read<br>Clears All |

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| BITFIELD       | BITS | DESCRIPTION                                                                                                                                                                                                     | DECODE                                                  |
|----------------|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|
| EOM_ERR_FLAG_B | 7    | Eye-opening is below configured threshold for Link B                                                                                                                                                            | 0b0: No EOM error on Link B<br>0b1: EOM error on Link B |
| EOM_ERR_FLAG_A | 6    | Eye-opening is below configured threshold for Link A                                                                                                                                                            | 0b0: No EOM error on Link A<br>0b1: EOM error on Link A |
| MAX_RT_FLAG    | 3    | Combined ARQ maximum retransmission limit error flag<br><br>Asserted when any of the selected channels ARQ retransmission limit is reached. Selection is done by each channel's MAX_RT_ERR_OEN (0x1C) bitfield. | 0b0: Flag not asserted<br>0b1: Flag asserted            |
| RT_CNT_FLAG    | 2    | Combined ARQ retransmission event flag<br><br>Asserted when any of the selected channels have done at least one ARQ retransmission. Selection is done by each channel's RT_CNT_OEN (0x1C) bitfield.             | 0b0: Flag not asserted<br>0b1: Flag asserted            |
| PKT_CNT_FLAG   | 1    | Packet Count Flag<br><br>Asserted when PKT_CNT (0x25) ≥ PKT_CNT_THR (0x19)                                                                                                                                      | 0b0: Flag not asserted<br>0b1: Flag asserted            |
| WM_ERR_FLAG    | 0    | Watermark Error Flag<br><br>Asserted when a watermark error is detected.                                                                                                                                        | 0b0: Flag not asserted<br>0b1: Flag asserted            |

**INTR6 (0x1E)**

| BIT         | 7              | 6            | 5              | 4             | 3            | 2             | 1 | 0                   |
|-------------|----------------|--------------|----------------|---------------|--------------|---------------|---|---------------------|
| Field       | VDDCMP_INT_OEN | PORZ_INT_OEN | VDDBAD_INT_OEN | FSYNC_ERR_OEN | LCRC_ERR_OEN | VPRBS_ERR_OEN | – | VID_PXL_CRC_ERR_OEN |
| Reset       | 0b0            | 0b0          | 0b0            | 0b1           | 0b1          | 0b1           | – | 0x0                 |
| Access Type | Write, Read    | Write, Read  | Write, Read    | Write, Read   | Write, Read  | Write, Read   | – | Write, Read         |

| BITFIELD            | BITS | DESCRIPTION                                                                  | DECODE                                                                                         |
|---------------------|------|------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| VDDCMP_INT_OEN      | 7    | Enable reporting of VDDCMP interrupt (VDDCMP_INT_FLAG - 0x1F) at ERRB pin    | 0b0: Reporting disabled<br>0b1: Reporting enabled                                              |
| PORZ_INT_OEN        | 6    | Enable reporting of PORZ interrupt (PORZ_INT_FLAG - 0x1F) at ERRB pin        | 0b0: Reporting disabled<br>0b1: Reporting enabled                                              |
| VDDBAD_INT_OEN      | 5    | Enable reporting of VDDBAD interrupt (VDDBAD_INT_FLAG - 0x1F) at ERRB pin    | 0b0: Reporting disabled<br>0b1: Reporting enabled                                              |
| FSYNC_ERR_OEN       | 4    | Enable reporting of frame sync errors (FSYNC_ERR_FLAG - 0x1F) at ERRB pin    | 0b0: Reporting disabled<br>0b1: Reporting enabled                                              |
| LCRC_ERR_OEN        | 3    | Enable reporting of video line CRC errors (LCRC_ERR_FLAG - 0x1F) at ERRB pin | 0b0: Reporting disabled<br>0b1: Reporting enabled                                              |
| VPRBS_ERR_OEN       | 2    | Enable reporting of video PRBS errors (VPRBS_ERR_FLAG - 0x1F) at ERRB pin    | 0b0: Reporting disabled<br>0b1: Reporting enabled                                              |
| VID_PXL_CRC_ERR_OEN | 0    | Video Pixel CRC Error Counter Interrupt Output Enable                        | 0x0: Error counter interrupt output not enabled<br>0x1: Error counter interrupt output enabled |



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**INTR7 (0x1F)**

| BIT         | 7               | 6             | 5               | 4 | 3             | 2              | 1 | 0               |
|-------------|-----------------|---------------|-----------------|---|---------------|----------------|---|-----------------|
| Field       | VDDCMP_INT_FLAG | PORZ_INT_FLAG | VDDBAD_INT_FLAG | – | LCRC_ERR_FLAG | VPRBS_ERR_FLAG | – | VID_PXL_CRC_ERR |
| Reset       | 0b0             | 0b0           | 0b0             | – | 0b0           | 0b0            | – | 0x0             |
| Access Type | Read Only       | Read Only     | Read Only       | – | Read Only     | Read Only      | – | Read Only       |

| BITFIELD        | BITS | DESCRIPTION                                                                    | DECODE                                                                                         |
|-----------------|------|--------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| VDDCMP_INT_FLAG | 7    | VDDCMP Interrupt Flag                                                          | 0b0: Flag not asserted<br>0b1: Flag asserted                                                   |
| PORZ_INT_FLAG   | 6    | PORZ Interrupt Flag                                                            | 0b0: Flag not asserted<br>0b1: Flag asserted                                                   |
| VDDBAD_INT_FLAG | 5    | VDDBAD Status Interrupt                                                        | 0b0: Flag not asserted<br>0b1: Flag asserted                                                   |
| LCRC_ERR_FLAG   | 3    | Video Line CRC Error Flag<br>Asserted when a video line CRC error is detected. | 0b0: Flag not asserted<br>0b1: Flag asserted                                                   |
| VPRBS_ERR_FLAG  | 2    | Video PRBS Error Flag<br>Asserted when VPRBS_ERR (0x1E) > 0.                   | 0b0: Flag not asserted<br>0b1: Flag asserted                                                   |
| VID_PXL_CRC_ERR | 0    | Video Pixel CRC Error Counter Interrupt                                        | 0x0: Error counter interrupt output not enabled<br>0x1: Error counter interrupt output enabled |

**INTR8 (0x20)**

| BIT         | 7           | 6 | 5 | 4 | 3 | 2              | 1 | 0 |
|-------------|-------------|---|---|---|---|----------------|---|---|
| Field       | ERR_TX_EN   | – | – | – | – | ERR_TX_ID[4:0] | – | – |
| Reset       | 0b1         | – | – | – | – | 0x1F           | – | – |
| Access Type | Write, Read | – | – | – | – | Write, Read    | – | – |

| BITFIELD  | BITS | DESCRIPTION                                                                                 | DECODE                                                                    |
|-----------|------|---------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|
| ERR_TX_EN | 7    | Transmit local error status (inverse of ERRB pin level) to remote side through GPIO channel | 0b0: Transmit error status disabled<br>0b1: Transmit error status enabled |
| ERR_TX_ID | 4:0  | GPIO ID used for transmitting ERR_TX                                                        | 0bXXXXX: Value of GPIO ID for transmitting ERR_TX                         |

**INTR9 (0x21)**

| BIT         | 7           | 6               | 5 | 4 | 3 | 2              | 1 | 0 |
|-------------|-------------|-----------------|---|---|---|----------------|---|---|
| Field       | ERR_RX_EN   | ERR_RX_RECEIVED | – | – | – | ERR_RX_ID[4:0] | – | – |
| Reset       | 0b1         | 0b1             | – | – | – | 0x1F           | – | – |
| Access Type | Write, Read | Write, Read     | – | – | – | Write, Read    | – | – |

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| BITFIELD        | BITS | DESCRIPTION                                                                   | DECODE                                                                  |
|-----------------|------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------|
| ERR_RX_EN       | 7    | Received remote error status (inverse of ERRB pin level) through GPIO channel | 0b0: Receive error status disabled<br>0b1: Receive error status enabled |
| ERR_RX_RECEIVED | 6    | Received ERR_RX value                                                         | 0b0: ERR_RX value not received<br>0b1: ERR_RX value received            |
| ERR_RX_ID       | 4:0  | GPIO ID used for receiving ERR_RX                                             | 0bXXXXX: Value of GPIO ID for receiving ERR_TX                          |

**CNT0 (0x22)**

| BIT         | 7               | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|-----------------|---|---|---|---|---|---|---|
| Field       | DEC_ERR_A[7:0]  |   |   |   |   |   |   |   |
| Reset       | 0x00            |   |   |   |   |   |   |   |
| Access Type | Read Clears All |   |   |   |   |   |   |   |

| BITFIELD  | BITS | DESCRIPTION                                                                                                           | DECODE                                          |
|-----------|------|-----------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|
| DEC_ERR_A | 7:0  | Number of decoding (disparity) errors detected at Link A<br><br>Resets after reading or with the rising edge of LOCK. | 0xXX: Number of Link A decoding errors detected |

**CNT1 (0x23)**

| BIT         | 7               | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|-----------------|---|---|---|---|---|---|---|
| Field       | DEC_ERR_B[7:0]  |   |   |   |   |   |   |   |
| Reset       | 0x00            |   |   |   |   |   |   |   |
| Access Type | Read Clears All |   |   |   |   |   |   |   |

| BITFIELD  | BITS | DESCRIPTION                                                                                                           | DECODE                                          |
|-----------|------|-----------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|
| DEC_ERR_B | 7:0  | Number of decoding (disparity) errors detected at Link B<br><br>Resets after reading or with the rising edge of LOCK. | 0xXX: Number of Link B decoding errors detected |

**CNT2 (0x24)**

HIDE

| BIT         | 7               | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|-----------------|---|---|---|---|---|---|---|
| Field       | IDLE_ERR[7:0]   |   |   |   |   |   |   |   |
| Reset       | 0x00            |   |   |   |   |   |   |   |
| Access Type | Read Clears All |   |   |   |   |   |   |   |

| BITFIELD | BITS | DESCRIPTION                                                                                      | DECODE                                    |
|----------|------|--------------------------------------------------------------------------------------------------|-------------------------------------------|
| IDLE_ERR | 7:0  | Number of idle-word errors detected.<br><br>Reset after reading or with the rising edge of LOCK. | 0xXX: Number of idle word errors detected |

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**CNT3 (0x25)**

| BIT         | 7               | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|-----------------|---|---|---|---|---|---|---|
| Field       | PKT_CNT[7:0]    |   |   |   |   |   |   |   |
| Reset       | 0x00            |   |   |   |   |   |   |   |
| Access Type | Read Clears All |   |   |   |   |   |   |   |

| BITFIELD | BITS | DESCRIPTION                                                                                                                                                                                                                                                                                                                                                                                                                                           | DECODE                                  |
|----------|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
| PKT_CNT  | 7:0  | <p>Number of received packets of a selected type</p> <p>Packet type is selected with PKT_CNT_SEL (0x2C) bitfield. Reported packet count is a scaled value, such that actual packet count is <math>\geq \text{PKT\_CNT} \times (2^{\text{PKT\_CNT\_EXP}})</math> and <math>&lt; (\text{PKT\_CNT} + 1) \times (2^{\text{PKT\_CNT\_EXP}})</math>.</p> <p>When maximum value is reported, the packet count is greater or equal to the reported value.</p> | 0xXX: Scaled number of received packets |

**TX2 (0x2A)**

| BIT         | 7             | 6 | 5              | 4 | 3               | 2 | 1           | 0 |
|-------------|---------------|---|----------------|---|-----------------|---|-------------|---|
| Field       | ERRG_CNT[1:0] |   | ERRG_RATE[1:0] |   | ERRG_BURST[2:0] |   | ERRG_PER    |   |
| Reset       | 0b00          |   | 0b10           |   | 0b000           |   | 0b0         |   |
| Access Type | Write, Read   |   | Write, Read    |   | Write, Read     |   | Write, Read |   |

| BITFIELD   | BITS | DESCRIPTION                                  | DECODE                                                                                                                                                 |
|------------|------|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| ERRG_CNT   | 7:6  | Number of errors to be generated             | 0b00: Continuous<br>0b01: 16<br>0b10: 128<br>0b11: 1024                                                                                                |
| ERRG_RATE  | 5:4  | Error generator average bit error rate       | 0b00: 1 in 5120 bits<br>0b01: 1 in 81920 bits<br>0b10: 1 in 1310720 bits<br>0b11: 1 in 20971520 bits                                                   |
| ERRG_BURST | 3:1  | Error generator burst error length           | 0b000: 1 error<br>0b001: 2 errors<br>0b010: 3 errors<br>0b011: 4 errors<br>0b100: 8 errors<br>0b101: 12 errors<br>0b110: 16 errors<br>0b111: 20 errors |
| ERRG_PER   | 0    | Error generator error distribution selection | 0b0: Pseudorandom<br>0b1: Periodic                                                                                                                     |

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**TX3 (0x2B)**

| BIT         | 7 | 6 | 5 | 4 | 3 | 2            | 1 | 0 |
|-------------|---|---|---|---|---|--------------|---|---|
| Field       | – | – | – | – | – | TIMEOUT[2:0] |   |   |
| Reset       | – | – | – | – | – | 0b100        |   |   |
| Access Type | – | – | – | – | – | Write, Read  |   |   |

| BITFIELD | BITS | DESCRIPTION                                                                                                                                                                                                                       | DECODE                                                                                                                                                                                                                               |
|----------|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TIMEOUT  | 2:0  | ARQ Timeout duration multiplier<br><br>Multiplies a timeout base constant to set the ARQ timeout. The timeout base is set by the reverse channel link rate (RX_RATE) as follows:<br>RX_RATE      Timeout Base<br>187.5Mbps    8µs | 0b000: 0.5 x Timeout Base<br>0b001: 1.0 x Timeout Base<br>0b010: 1.5 x Timeout Base<br>0b011: 2.0 x Timeout Base<br>0b100: 2.5 x Timeout Base<br>0b101: 3.0 x Timeout Base<br>0b110: 3.5 x Timeout Base<br>0b111: 4.0 x Timeout Base |

**RX0 (0x2C)**

| BIT         | 7                | 6 | 5 | 4 | 3                | 2 | 1 | 0 |
|-------------|------------------|---|---|---|------------------|---|---|---|
| Field       | PKT_CNT_LBW[1:0] |   | – | – | PKT_CNT_SEL[3:0] |   |   |   |
| Reset       | 0b00             |   | – | – | 0x0              |   |   |   |
| Access Type | Write, Read      |   | – | – | Write, Read      |   |   |   |

| BITFIELD    | BITS | DESCRIPTION                                                                                                                                                 | DECODE                                                                                                                                                                                                                                                                     |
|-------------|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PKT_CNT_LBW | 7:6  | Selects the sub-type of low-bandwidth packets to count at PKT_CNT bitfield in register CNT3 (0x25)                                                          | 0b00: Count LBW data packets only<br>0b01: Count LBW acknowledge packets only<br>0b10: Count LBW data and acknowledge packets<br>0b11: Reserved                                                                                                                            |
| PKT_CNT_SEL | 3:0  | Selects the type of received packets to count at PKT_CNT bitfield in register CNT3 (0x25).<br><br>The selected packet type must be supported by the device. | 0x0: None<br>0x1: VIDEO<br>0x2: AUDIO<br>0x3: INFO Frame<br>0x4: SPI<br>0x5: I <sup>2</sup> C<br>0x6: UART<br>0x7: GPIO<br>0x8: AHDCP<br>0x9: RGMII<br>0xA: Reserved<br>0xB: Reserved<br>0xC: Reserved<br>0xD: Reserved<br>0xE: All<br>0xF: Unknown and packets with error |

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RX3 (0x2F)

| BIT         | 7             | 6             | 5         | 4               | 3             | 2             | 1         | 0               |
|-------------|---------------|---------------|-----------|-----------------|---------------|---------------|-----------|-----------------|
| Field       | PRBS_SYNCED_B | SYNC_LOCKED_B | WBLOCK_B  | FAILLOCK_B      | PRBS_SYNCED_A | SYNC_LOCKED_A | WBLOCK_A  | FAILLOCK_A      |
| Reset       | 0b0           | 0b0           | 0b0       | 0b0             | 0b0           | 0b0           | 0b0       | 0b0             |
| Access Type | Read Only     | Read Only     | Read Only | Read Clears All | Read Only     | Read Only     | Read Only | Read Clears All |

| BITFIELD      | BITS | DESCRIPTION                                   | DECODE                                                                               |
|---------------|------|-----------------------------------------------|--------------------------------------------------------------------------------------|
| PRBS_SYNCED_B | 7    | Link PRBS is synced to incoming data (Link B) | 0b0: Link PRBS not synced to incoming data<br>0b1: Link PRBS synced to incoming data |
| SYNC_LOCKED_B | 6    | Sync tracking locked (Link B)                 | 0b0: Sync tracking is not locked<br>0b1: Sync tracking is locked                     |
| WBLOCK_B      | 5    | Word boundary locked (Link B)                 | 0b0: Word boundary not locked<br>0b1: Word boundary locked                           |
| FAILLOCK_B    | 4    | Failed word boundary lock (latched) (Link B)  | 0b0: Word boundary lock did not fail<br>0b1: Word boundary lock failed               |
| PRBS_SYNCED_A | 3    | Link PRBS is synced to incoming data          | 0b0: Link PRBS not synced to incoming data<br>0b1: Link PRBS synced to incoming data |
| SYNC_LOCKED_A | 2    | Sync tracking locked (Link A)                 | 0b0: Sync tracking is not locked<br>0b1: Sync tracking is locked                     |
| WBLOCK_A      | 1    | Word boundary locked (Link A)                 | 0b0: Word boundary not locked<br>0b1: Word boundary locked                           |
| FAILLOCK_A    | 0    | Failed word boundary lock (latched) (Link A)  | 0b0: Word boundary lock did not fail<br>0b1: Word boundary lock failed               |

GPIOA (0x30)

| BIT         | 7                     | 6 | 5                  | 4 | 3 | 2 | 1 | 0 |
|-------------|-----------------------|---|--------------------|---|---|---|---|---|
| Field       | GPIO_RX_FAST_BIDIR_EN | — | GPIO_FWD_CDLY[5:0] |   |   |   |   |   |
| Reset       | 0b0                   | — | 0b000001           |   |   |   |   |   |
| Access Type | Write, Read           | — | Write, Read        |   |   |   |   |   |

| BITFIELD              | BITS | DESCRIPTION                                                                                                                                                                                                                           | DECODE                                                                    |
|-----------------------|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|
| GPIO_RX_FAST_BIDIR_EN | 7    | GPIO fast-direction switch for bidirectional IO                                                                                                                                                                                       | 0b0: Fast-direction switch disabled<br>0b1: Fast-direction switch enabled |
| GPIO_FWD_CDLY         | 5:0  | Compensation delay multiplier for the forward direction<br><br>This must be same value as GPIO_FWD_CDLY of the chip on the other side of the link.<br><br>Total delay is the (value + 1) multiplied by 1.7µs. Default delay is 3.4µs. | 0bXXXXXX: Forward compensation delay multiplier value                     |

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**GPIOB (0x31)**

| BIT         | 7                 | 6 | 5                  | 4 | 3 | 2 | 1 | 0 |
|-------------|-------------------|---|--------------------|---|---|---|---|---|
| Field       | GPIO_TX_WNDW[1:0] |   | GPIO_REV_CDLY[5:0] |   |   |   |   |   |
| Reset       | 0b10              |   | 0b001000           |   |   |   |   |   |
| Access Type | Write, Read       |   | Write, Read        |   |   |   |   |   |

| BITLED        | BITS | DESCRIPTION                                                                                                                                                                                                                            | DECODE                                                       |
|---------------|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|
| GPIO_TX_WNDW  | 7:6  | Wait time to create a packet after a GPIO transition<br><br>This allows different GPIO input transitions to be grouped into a single packet, thereby increasing GPIO bandwidth efficiency.                                             | 0b00: Disabled<br>0b01: 200ns<br>0b10: 500ns<br>0b11: 1000ns |
| GPIO_REV_CDLY | 5:0  | Compensation delay multiplier for the reverse direction<br><br>This must be same value as GPIO_REV_CDLY of the chip on the other side of the link.<br><br>Total delay is the (value + 1) multiplied by 1.7µs. Default delay is 15.3µs. | 0bXXXXXX: Reverse compensation delay multiplier value        |

**I2C\_0 (0x40)**

| BIT         | 7 | 6 | 5           | 4 | 3 | 2           | 1 | 0 |
|-------------|---|---|-------------|---|---|-------------|---|---|
| Field       | – | – | SLV_SH[1:0] |   | – | SLV_TO[2:0] |   |   |
| Reset       | – | – | 0x2         |   | – | 0x6         |   |   |
| Access Type | – | – | Write, Read |   | – | Write, Read |   |   |

| BITLED | BITS | DESCRIPTION                                                                                                                                                                                                                                                                                       | DECODE                                                                                                                                                                        |
|--------|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SLV_SH | 5:4  | I <sup>2</sup> C-to-I <sup>2</sup> C master bit rate setting<br><br>Configures the I <sup>2</sup> C bit rate used by the internal I <sup>2</sup> C master (in the device on remote side from the external I <sup>2</sup> C master).<br><br>Set this according to the I <sup>2</sup> C speed mode. | 0b00: Set for I <sup>2</sup> C Fast-mode Plus speed<br>0b01: Set for I <sup>2</sup> C Fast-mode speed<br>0b10: Set for I <sup>2</sup> C Standard-mode speed<br>0b11: Reserved |
| SLV_TO | 2:0  | I <sup>2</sup> C-to-I <sup>2</sup> C slave timeout setting<br><br>Internal GMSL2 I <sup>2</sup> C slave times out after the configured duration if it does not receive any response while waiting for a packet from remote device.                                                                | 0b000: 16µs<br>0b001: 1ms<br>0b010: 2ms<br>0b011: 4ms<br>0b100: 8ms<br>0b101: 16ms<br>0b110: 32ms<br>0b111: Disabled                                                          |

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I2C 1 (0x41)

| BIT         | 7 | 6           | 5 | 4 | 3 | 2           | 1 | 0 |
|-------------|---|-------------|---|---|---|-------------|---|---|
| Field       | – | MST_BT[2:0] |   |   | – | MST_TO[2:0] |   |   |
| Reset       | – | 0x5         |   |   | – | 0x6         |   |   |
| Access Type | – | Write, Read |   |   | – | Write, Read |   |   |

| BITFIELD | BITS | DESCRIPTION                                                                                                                                                                                                                                                                            | DECODE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|----------|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MST_BT   | 6:4  | I <sup>2</sup> C-to-I <sup>2</sup> C master bit rate setting. Configures the I <sup>2</sup> C bit rate used by the internal I <sup>2</sup> C master (in the device on remote side from the external I <sup>2</sup> C master)<br>Set this according to the I <sup>2</sup> C speed mode. | 0b000: 9.92Kbps - Set for I <sup>2</sup> C Standard-mode speed<br>0b001: 33.2Kbps - Set for I <sup>2</sup> C Standard-mode speed<br>0b010: 99.2Kbps - Set for I <sup>2</sup> C Standard or Fast-mode speed<br>0b011: 123Kbps - Set for I <sup>2</sup> C Fast-mode speed<br>0b100: 203Kbps - Set for I <sup>2</sup> C Fast-mode speed<br>0b101: 397Kbps - Set for I <sup>2</sup> C Fast or Fast-mode Plus speed<br>0b110: 625Kbps - Set for I <sup>2</sup> C Fast-mode Plus speed<br>0b111: 980Kbps - Set for I <sup>2</sup> C Fast-mode Plus speed |
| MST_TO   | 2:0  | I <sup>2</sup> C-to-I <sup>2</sup> C master timeout setting. Internal GMSL2 I <sup>2</sup> C master times out after the configured duration if it does not receive any response while waiting for a packet from remote device.                                                         | 0b000: 16μs<br>0b001: 1ms<br>0b010: 2ms<br>0b011: 4ms<br>0b100: 8ms<br>0b101: 16ms<br>0b110: 32ms<br>0b111: Disabled                                                                                                                                                                                                                                                                                                                                                                                                                               |

I2C 2 (0x42)

| BIT         | 7           | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|-------------|---|---|---|---|---|---|---|
| Field       | SRC_A[6:0]  |   |   |   |   |   |   | – |
| Reset       | 0x0         |   |   |   |   |   |   | – |
| Access Type | Write, Read |   |   |   |   |   |   | – |

| BITFIELD | BITS | DESCRIPTION                                                                                                                                                                                                                       | DECODE                                     |
|----------|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|
| SRC_A    | 7:1  | I <sup>2</sup> C address translator source A<br><br>When I <sup>2</sup> C device address matches I <sup>2</sup> C SRC_A, internal I <sup>2</sup> C master (on remote side) replaces the device address by I <sup>2</sup> C DST_A. | 0bXXXXXXX: Value of I <sup>2</sup> C SRC_A |

I2C 3 (0x43)

| BIT         | 7           | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|-------------|---|---|---|---|---|---|---|
| Field       | DST_A[6:0]  |   |   |   |   |   |   | – |
| Reset       | 0x0         |   |   |   |   |   |   | – |
| Access Type | Write, Read |   |   |   |   |   |   | – |



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| BITFIELD | BITS | DESCRIPTION                                                                                                                               | DECODE                                     |
|----------|------|-------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|
| DST_A    | 7:1  | I <sup>2</sup> C address translator destination A<br><br>See the description of I <sup>2</sup> C SRC_A bitfield in register I2C_2 (0x42). | 0bXXXXXXX: Value of I <sup>2</sup> C DST_A |

I2C\_4 (0x44)

| BIT         | 7           | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|-------------|---|---|---|---|---|---|---|
| Field       | SRC_B[6:0]  |   |   |   |   |   |   | – |
| Reset       | 0x0         |   |   |   |   |   |   | – |
| Access Type | Write, Read |   |   |   |   |   |   | – |

| BITFIELD | BITS | DESCRIPTION                                                                                                                                                                                                                       | DECODE                                     |
|----------|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|
| SRC_B    | 7:1  | I <sup>2</sup> C address translator source B<br><br>When I <sup>2</sup> C device address matches I <sup>2</sup> C SRC_B, internal I <sup>2</sup> C master (on remote side) replaces the device address by I <sup>2</sup> C DST_B. | 0bXXXXXXX: Value of I <sup>2</sup> C SRC_B |

I2C\_5 (0x45)

| BIT         | 7           | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|-------------|---|---|---|---|---|---|---|
| Field       | DST_B[6:0]  |   |   |   |   |   |   | – |
| Reset       | 0x0         |   |   |   |   |   |   | – |
| Access Type | Write, Read |   |   |   |   |   |   | – |

| BITFIELD | BITS | DESCRIPTION                                                                                                                               | DECODE                                     |
|----------|------|-------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|
| DST_B    | 7:1  | I <sup>2</sup> C address translator destination B<br><br>See the description of I <sup>2</sup> C SRC_B bitfield in register I2C_4 (0x44). | 0bXXXXXXX: Value of I <sup>2</sup> C DST_B |

I2C\_6 (0x46)

| BIT         | 7 | 6 | 5 | 4 | 3            | 2 | 1 | 0 |
|-------------|---|---|---|---|--------------|---|---|---|
| Field       | – | – | – | – | I2C_AUTO_CFG | – | – | – |
| Reset       | – | – | – | – | 0x1          | – | – | – |
| Access Type | – | – | – | – | Write, Read  | – | – | – |

| BITFIELD     | BITS | DESCRIPTION                                                                                                                                                                                                                                                                  | DECODE                                                                                                                                                                                         |
|--------------|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| I2C_AUTO_CFG | 3    | When set to 1, I <sup>2</sup> C-to-I <sup>2</sup> C number of links is determined based on splitter mode automatically.<br><br>In splitter mode, response from two I <sup>2</sup> C channels are expected, otherwise response from one I <sup>2</sup> C channel is expected. | 0b0: Number of I <sup>2</sup> C-to-I <sup>2</sup> C links set by I2C_SRC_CNT[2:0] bits<br>0b1: Splitter mode automatically determines the number of I <sup>2</sup> C-to-I <sup>2</sup> C links |

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I2C 7 (0x47)

| BIT         | 7                  | 6                  | 5 | 4 | 3 | 2             | 1             | 0             |
|-------------|--------------------|--------------------|---|---|---|---------------|---------------|---------------|
| Field       | UART_RX_OVERFLOW   | UART_TX_OVERFLOW   | – | – | – | I2C_TIMED_OUT | REM_ACK_ACKED | REM_ACK_RECVD |
| Reset       | 0x0                | 0x0                | – | – | – | 0x0           | 0x0           | 0x0           |
| Access Type | Read<br>Clears All | Read<br>Clears All | – | – | – | Read Only     | Read Only     | Read Only     |

| BITFIELD         | BITS | DESCRIPTION                                                                                    | DECODE                                                                           |
|------------------|------|------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| UART_RX_OVERFLOW | 7    | UART Rx FIFO overflow                                                                          | 0b0: No overflow occurred<br>0b1: Overflow occurred                              |
| UART_TX_OVERFLOW | 6    | UART Tx FIFO overflow                                                                          | 0b0: No overflow occurred<br>0b1: Overflow occurred                              |
| I2C_TIMED_OUT    | 2    | Internal I2C-to-I2C slave or master has timed out while receiving packet from remote device    | 0b0: Timeout has not occurred<br>0b1: Timeout has occurred                       |
| REM_ACK_ACKED    | 1    | Inverse of the I2C acknowledge bit received from remote side                                   | 0b0: I2C acknowledge bit received as 1<br>0b1: I2C acknowledge bit received as 0 |
| REM_ACK_RECVD    | 0    | I2C acknowledge bit for any I2C byte is received from remote side for the previous I2C packet. | 0b0: I2C acknowledge bit not received<br>0b1: I2C acknowledge bit received       |

UART 0 (0x48)

| BIT         | 7               | 6 | 5           | 4           | 3              | 2              | 1 | 0           |
|-------------|-----------------|---|-------------|-------------|----------------|----------------|---|-------------|
| Field       | ARB_TO_LEN[1:0] |   | REM_MS_EN   | LOC_MS_EN   | BYPASS_DIS_PAR | BYPASS_TO[1:0] |   | BYPASS_EN   |
| Reset       | 0x1             |   | 0x0         | 0x0         | 0x0            | 0x1            |   | 0x0         |
| Access Type | Write, Read     |   | Write, Read | Write, Read | Write, Read    | Write, Read    |   | Write, Read |

| BITFIELD   | BITS | DESCRIPTION                                                                                                                                                                                                             | DECODE                                                                                                     |
|------------|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| ARB_TO_LEN | 7:6  | UART Rx source arbitration timeout duration<br><br>UART Rx processes packets from a single UART source at a time. If UART Rx receives no UART packages during the specified duration, it selects the next UART source.. | 0b00: 1ms<br>0b01: 2ms<br>0b10: 8ms<br>0b11: 32ms                                                          |
| REM_MS_EN  | 5    | Enables UART bypass mode control by remote GPIO pin<br><br>When set, remote chip's GPIO is used as MS pin (UART Mode Select). When MS is high, chip is in bypass mode, otherwise chip is in base mode.                  | 0b0: UART bypass mode not controlled by remote MS pin<br>0b1: UART bypass mode controlled by remote MS pin |
| LOC_MS_EN  | 4    | Enables UART bypass mode control by local GPIO pin<br><br>Set to use GPIO2 pin as MS pin (UART Mode Select). When MS is high, chip is in bypass mode, otherwise chip is in base mode.                                   | 0b0: UART bypass mode not controlled by local MS pin<br>0b1: UART bypass mode controlled by local MS pin   |

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| BITFIELD           | BITS | DESCRIPTION                                                                                                                                                                                                                                                | DECODE                                                                                                    |
|--------------------|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| BYPASS_DI<br>S_PAR | 3    | Selects whether to receive and send parity bit in bypass mode                                                                                                                                                                                              | 0b0: Receive and send parity bit in bypass mode<br>0b1: Do not receive and send parity bit in bypass mode |
| BYPASS_TO          | 2:1  | UART soft-bypass timeout duration<br><br>When set to 0b11, BYPASS_EN is never cleared, so the device stays in bypass mode until next power down.                                                                                                           | 0b00: 2ms<br>0b01: 8ms<br>0b10: 32ms<br>0b11: Disabled                                                    |
| BYPASS_EN          | 0    | Enables UART soft-bypass mode<br><br>Bypass mode remains active as long as there is UART activity. When there is no UART activity for selected duration (configured by BYPASS_TO register), device exits bypass mode and the bit is automatically cleared. | 0b0: UART soft-bypass mode disabled<br>0b1: UART soft-bypass mode enabled                                 |

UART\_1 (0x49)

| BIT         | 7               | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|-----------------|---|---|---|---|---|---|---|
| Field       | BITLEN_LSB[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x96            |   |   |   |   |   |   |   |
| Access Type | Read Only       |   |   |   |   |   |   |   |

| BITFIELD   | BITS | DESCRIPTION                          | DECODE                                        |
|------------|------|--------------------------------------|-----------------------------------------------|
| BITLEN_LSB | 7:0  | UART detected bit length, low 8 bits | 0xXX: UART detected bit length (lower 8 bits) |

UART\_2 (0x4A)

| BIT         | 7              | 6 | 5               | 4 | 3 | 2 | 1 | 0 |
|-------------|----------------|---|-----------------|---|---|---|---|---|
| Field       | OUT_DELAY[1:0] |   | BITLEN_MSB[5:0] |   |   |   |   |   |
| Reset       | 0x2            |   | 0x0             |   |   |   |   |   |
| Access Type | Write, Read    |   | Read Only       |   |   |   |   |   |

| BITFIELD   | BITS | DESCRIPTION                                                                                                                                                                                                                                             | DECODE                                                      |
|------------|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|
| OUT_DELAY  | 7:6  | UART initial output delay<br><br>In base mode. The first received UART byte of a packet (sync or acknowledge frame) is delayed by the configured number of bit times in order to output the UART frames of the same packet back to back on remote side. | 0b00: 0 bits<br>0b01: 4 bits<br>0b10: 8 bits<br>0b11: 1 bit |
| BITLEN_MSB | 5:0  | UART detected bit length, high 6 bits                                                                                                                                                                                                                   | 0bXXXXXX: UART detected bit length (upper 6 bits)           |

I2C\_PT\_0 (0x4C)

| BIT         | 7 | 6 | 5              | 4 | 3 | 2              | 1 | 0 |
|-------------|---|---|----------------|---|---|----------------|---|---|
| Field       | – | – | SLV_SH_PT[1:0] |   | – | SLV_TO_PT[2:0] |   |   |
| Reset       | – | – | 0x2            |   | – | 0x6            |   |   |
| Access Type | – | – | Write, Read    |   | – | Write, Read    |   |   |

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| BITFIELD  | BITS | DESCRIPTION                                                                                                                                                                                                                                                                 | DECODE                                                                                                                                                                        |
|-----------|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SLV_SH_PT | 5:4  | Pass-through I <sup>2</sup> C-to-I <sup>2</sup> C slave setup and hold time setting (setup, hold)<br><br>Configures the interval between SDA and SCL transitions when driven by the internal I <sup>2</sup> C slave. Set this according to the I <sup>2</sup> C speed mode: | 0b00: Set for I <sup>2</sup> C Fast-mode Plus speed<br>0b01: Set for I <sup>2</sup> C Fast-mode speed<br>0b10: Set for I <sup>2</sup> C Standard-mode speed<br>0b11: Reserved |
| SLV_TO_PT | 2:0  | Pass-through I <sup>2</sup> C-to-I <sup>2</sup> C slave timeout setting<br><br>Internal GMSL2 I <sup>2</sup> C slave times out after the configured duration if it does not receive any response while waiting for a packet from remote device.                             | 0b000: 16μs<br>0b001: 1ms<br>0b010: 2ms<br>0b011: 4ms<br>0b100: 8ms<br>0b101: 16ms<br>0b110: 32ms<br>0b111: Disabled                                                          |

I<sup>2</sup>C\_PT\_1 (0x4D)

| BIT         | 7 | 6              | 5 | 4 | 3 | 2              | 1 | 0 |
|-------------|---|----------------|---|---|---|----------------|---|---|
| Field       | – | MST_BT_PT[2:0] |   |   | – | MST_TO_PT[2:0] |   |   |
| Reset       | – | 0x5            |   |   | – | 0x6            |   |   |
| Access Type | – | Write, Read    |   |   | – | Write, Read    |   |   |

| BITFIELD  | BITS | DESCRIPTION                                                                                                                                                                                                                                                                                             | DECODE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-----------|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MST_BT_PT | 6:4  | Pass-through I <sup>2</sup> C-to-I <sup>2</sup> C master bit rate setting<br><br>Configures the I <sup>2</sup> C bit rate used by the internal I <sup>2</sup> C master (in the device on remote side from the external I <sup>2</sup> C master). Set this according to the I <sup>2</sup> C speed mode. | 0b000: 9.92Kbps - Set for I <sup>2</sup> C Standard-mode speed<br>0b001: 33.2Kbps - Set for I <sup>2</sup> C Standard-mode speed<br>0b010: 99.2Kbps - Set for I <sup>2</sup> C Standard or Fast-mode speed<br>0b011: 123Kbps - Set for I <sup>2</sup> C Fast-mode speed<br>0b100: 203Kbps - Set for I <sup>2</sup> C Fast-mode speed<br>0b101: 397Kbps - Set for I <sup>2</sup> C Fast or Fast-mode Plus speed<br>0b110: 625Kbps - Set for I <sup>2</sup> C Fast-mode Plus speed<br>0b111: 980Kbps - Set for I <sup>2</sup> C Fast-mode Plus speed |
| MST_TO_PT | 2:0  | Pass-through I <sup>2</sup> C-to-I <sup>2</sup> C master timeout setting<br><br>Internal GMSL2 I <sup>2</sup> C master times out after the configured duration if it does not receive any response while waiting for a packet from the remote device.                                                   | 0b000: 16μs<br>0b001: 1ms<br>0b010: 2ms<br>0b011: 4ms<br>0b100: 8ms<br>0b101: 16ms<br>0b110: 32ms<br>0b111: Disabled                                                                                                                                                                                                                                                                                                                                                                                                                               |

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I2C\_PT\_2 (0x4E)

| BIT         | 7           | 6               | 5               | 4               | 3           | 2               | 1               | 0               |
|-------------|-------------|-----------------|-----------------|-----------------|-------------|-----------------|-----------------|-----------------|
| Field       | XOVER_EN_2  | I2C_TIMED_OUT_2 | REM_ACK_ACKED_2 | REM_ACK_RECVD_2 | XOVER_EN_1  | I2C_TIMED_OUT_1 | REM_ACK_ACKED_1 | REM_ACK_RECVD_1 |
| Reset       | 0x0         | 0x0             | 0x0             | 0x0             | 0x0         | 0x0             | 0x0             | 0x0             |
| Access Type | Write, Read | Read Only       | Read Only       | Read Only       | Write, Read | Read Only       | Read Only       | Read Only       |

| BITFIELD        | BITS | DESCRIPTION                                                                                                                                                                       | DECODE                                                                                                                                                                        |
|-----------------|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| XOVER_EN_2      | 7    | Connects pass-through I <sup>2</sup> C/UART Channel 2 to main control channel on remote side                                                                                      | 0x0: Do not connect pass-through I <sup>2</sup> C/UART Channel 2 to main control channel<br>0x1: Connect pass-through I <sup>2</sup> C/UART Channel 2 to main control channel |
| I2C_TIMED_OUT_2 | 6    | In pass-through I <sup>2</sup> C Channel 2, internal I <sup>2</sup> C-to-I <sup>2</sup> C slave or master has timed out while receiving packet from remote device                 | 0x0: Timeout has not occurred<br>0x1: Timeout has occurred                                                                                                                    |
| REM_ACK_ACKED_2 | 5    | In pass-through I <sup>2</sup> C Channel 2, inverse of the I <sup>2</sup> C acknowledge bit received from remote side                                                             | 0x0: I <sup>2</sup> C acknowledge bit received as 1<br>0x1: Inverse I <sup>2</sup> C acknowledge bit received as 0                                                            |
| REM_ACK_RECVD_2 | 4    | In pass-through I <sup>2</sup> C Channel 2, I <sup>2</sup> C acknowledge bit for any I <sup>2</sup> C byte is received from remote side for the previous I <sup>2</sup> C packet. | 0b0: I <sup>2</sup> C acknowledge bit not received<br>0b1: I <sup>2</sup> C acknowledge bit received                                                                          |
| XOVER_EN_1      | 3    | Connects pass-through I <sup>2</sup> C/UART Channel 1 to main control channel on remote side                                                                                      | 0x0: Do not connect pass-through I <sup>2</sup> C/UART Channel 1 to main control channel<br>0x1: Connect pass-through I <sup>2</sup> C/UART Channel 1 to main control channel |
| I2C_TIMED_OUT_1 | 2    | In pass-through I <sup>2</sup> C Channel 1, internal I <sup>2</sup> C-to-I <sup>2</sup> C slave or master has timed out while receiving packet from remote device                 | 0x0: Timeout has not occurred<br>0x1: Timeout has occurred                                                                                                                    |
| REM_ACK_ACKED_1 | 1    | In pass-through I <sup>2</sup> C Channel 1, inverse of the I <sup>2</sup> C acknowledge bit received from remote side.                                                            | 0x0: I <sup>2</sup> C acknowledge bit received as 1<br>0x1: Inverse I <sup>2</sup> C acknowledge bit received as 0                                                            |
| REM_ACK_RECVD_1 | 0    | In pass-through I <sup>2</sup> C Channel 1, I <sup>2</sup> C acknowledge bit for any I <sup>2</sup> C byte is received from remote side for the previous I <sup>2</sup> C packet. | 0b0: I <sup>2</sup> C acknowledge bit not received<br>0b1: I <sup>2</sup> C acknowledge bit received                                                                          |

UART\_PT\_0 (0x4F)

| BIT         | 7                | 6           | 5                  | 4                  | 3                | 2           | 1                  | 0                  |
|-------------|------------------|-------------|--------------------|--------------------|------------------|-------------|--------------------|--------------------|
| Field       | BITLEN_MAN_CFG_2 | DIS_PAR_2   | UART_RX_OVERFLOW_2 | UART_TX_OVERFLOW_2 | BITLEN_MAN_CFG_1 | DIS_PAR_1   | UART_RX_OVERFLOW_1 | UART_TX_OVERFLOW_1 |
| Reset       | 0b1              | 0b0         | 0b0                | 0b0                | 0b1              | 0b0         | 0b0                | 0b0                |
| Access Type | Write, Read      | Write, Read | Read Clears All    | Read Clears All    | Write, Read      | Write, Read | Read Clears All    | Read Clears All    |

| BITFIELD         | BITS | DESCRIPTION                                                                                                                              | DECODE                                                 |
|------------------|------|------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|
| BITLEN_MAN_CFG_2 | 7    | Uses the custom UART bit rate (selected by the BITLEN_PT_2_L - 0x548 and BITLEN_PT_2_H - 0x549 bitfields) in pass-through UART Channel 1 | 0b0: Use standard bit rate<br>0b1: Use custom bit rate |

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| BITFIELD           | BITS | DESCRIPTION                                                                                                              | DECODE                                                 |
|--------------------|------|--------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|
| DIS_PAR_2          | 6    | Disables parity bit in pass-through UART Channel 2                                                                       | 0b0: Parity bit enabled<br>0b1: Parity bit disabled    |
| UART_RX_OVERFLOW_2 | 5    | Pass-through UART Rx FIFO overflow                                                                                       | 0x0: No overflow occurred<br>0x1: Overflow occurred    |
| UART_TX_OVERFLOW_2 | 4    | Pass-through UART Tx FIFO overflow                                                                                       | 0x0: No overflow occurred<br>0x1: Overflow occurred    |
| BITLEN_MAN_CFG_1   | 3    | Uses the custom UART bit rate (selected by the BITLEN_PT_1_L and BITLEN_PT_1_H registers) in pass-through UART Channel 1 | 0b0: Use standard bit rate<br>0b1: Use custom bit rate |
| DIS_PAR_1          | 2    | Disables parity bit in pass-through UART Channel 1                                                                       | 0b0: Parity bit enabled<br>0b1: Parity bit disabled    |
| UART_RX_OVERFLOW_1 | 1    | Pass-through UART Rx FIFO overflow                                                                                       | 0x0: No overflow occurred<br>0x1: Overflow occurred    |
| UART_TX_OVERFLOW_1 | 0    | Pass-through UART Tx FIFO overflow                                                                                       | 0x0: No overflow occurred<br>0x1: Overflow occurred    |

**RX0 (0x50)**

| BIT         | 7           | 6 | 5 | 4 | 3 | 2 | 1            | 0 |
|-------------|-------------|---|---|---|---|---|--------------|---|
| Field       | RX_CRC_EN   | – | – | – | – | – | STR_SEL[1:0] |   |
| Reset       | 0b0         | – | – | – | – | – | 0x0          |   |
| Access Type | Write, Read | – | – | – | – | – | Write, Read  |   |

| BITFIELD  | BITS | DESCRIPTION                             | DECODE                                                                                    |
|-----------|------|-----------------------------------------|-------------------------------------------------------------------------------------------|
| RX_CRC_EN | 7    | Receive packet CRC enable               | 0b0: No CRC on received packets<br>0b1: Received packets have CRC and checking is enabled |
| STR_SEL   | 1:0  | Receive packets with selected stream ID | 0bXX: Receive packets with this stream ID                                                 |

**RX0 (0x51)**

| BIT         | 7           | 6 | 5 | 4 | 3 | 2 | 1            | 0 |
|-------------|-------------|---|---|---|---|---|--------------|---|
| Field       | RX_CRC_EN   | – | – | – | – | – | STR_SEL[1:0] |   |
| Reset       | 0b0         | – | – | – | – | – | 0x1          |   |
| Access Type | Write, Read | – | – | – | – | – | Write, Read  |   |

| BITFIELD  | BITS | DESCRIPTION                             | DECODE                                                                                    |
|-----------|------|-----------------------------------------|-------------------------------------------------------------------------------------------|
| RX_CRC_EN | 7    | Receive packet CRC enable               | 0b0: No CRC on received packets<br>0b1: Received packets have CRC and checking is enabled |
| STR_SEL   | 1:0  | Receive packets with selected stream ID | 0bXX: Receive packets with this stream ID                                                 |

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RX0 (0x52)

| BIT         | 7           | 6 | 5 | 4 | 3 | 2 | 1            | 0 |
|-------------|-------------|---|---|---|---|---|--------------|---|
| Field       | RX_CRC_EN   | – | – | – | – | – | STR_SEL[1:0] |   |
| Reset       | 0b0         | – | – | – | – | – | 0x2          |   |
| Access Type | Write, Read | – | – | – | – | – | Write, Read  |   |

| BITFIELD  | BITS | DESCRIPTION                             | DECODE                                                                                    |
|-----------|------|-----------------------------------------|-------------------------------------------------------------------------------------------|
| RX_CRC_EN | 7    | Receive packet CRC enable               | 0b0: No CRC on received packets<br>0b1: Received packets have CRC and checking is enabled |
| STR_SEL   | 1:0  | Receive packets with selected stream ID | 0bXX: Receive packets with this stream ID                                                 |

RX0 (0x53)

| BIT         | 7           | 6 | 5 | 4 | 3 | 2 | 1            | 0 |
|-------------|-------------|---|---|---|---|---|--------------|---|
| Field       | RX_CRC_EN   | – | – | – | – | – | STR_SEL[1:0] |   |
| Reset       | 0b0         | – | – | – | – | – | 0x3          |   |
| Access Type | Write, Read | – | – | – | – | – | Write, Read  |   |

| BITFIELD  | BITS | DESCRIPTION                             | DECODE                                                                                    |
|-----------|------|-----------------------------------------|-------------------------------------------------------------------------------------------|
| RX_CRC_EN | 7    | Receive packet CRC enable               | 0b0: No CRC on received packets<br>0b1: Received packets have CRC and checking is enabled |
| STR_SEL   | 1:0  | Receive packets with selected stream ID | 0bXX: Receive packets with this stream ID                                                 |

TR0 (0x60, 0x68, 0x78)

| BIT         | 7           | 6           | 5 | 4 | 3             | 2 | 1             | 0 |
|-------------|-------------|-------------|---|---|---------------|---|---------------|---|
| Field       | TX_CRC_EN   | RX_CRC_EN   | – | – | PRIO_VAL[1:0] |   | PRIO_CFG[1:0] |   |
| Reset       | 0b1         | 0b1         | – | – | 0x0           |   | 0x0           |   |
| Access Type | Write, Read | Write, Read | – | – | Write, Read   |   | Write, Read   |   |

| BITFIELD  | BITS | DESCRIPTION                                                                                                                   | DECODE                                                                                      |
|-----------|------|-------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| TX_CRC_EN | 7    | When set, calculate and append CRC to each packet transmitted from this port                                                  | 0b0: Transmit CRC disabled<br>0b1: Transmit CRC enabled                                     |
| RX_CRC_EN | 6    | When set, indicates that packets received at this port have appended CRC and CRC checking should be performed at each packet. | 0b0: Receive CRC disabled<br>0b1: Receive CRC enabled                                       |
| PRIO_VAL  | 3:2  | Sets the priority for this channel's packet requests                                                                          | 0b00: Low priority<br>0b01: Normal priority<br>0b10: High priority<br>0b11: Urgent priority |



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| BITFIELD | BITS | DESCRIPTION                               | DECODE                                                                                                                                                                       |
|----------|------|-------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PRIO_CFG | 1:0  | Adjust the priority used for this channel | 0b00: Current channel priority is used<br>0b01: Increase current channel priority<br>0b10: Decrease current channel priority<br>0b11: Reset to channel PRIO_VAL[3:2] setting |

**TR1 (0x61, 0x69, 0x71, 0x79, 0x81, 0x89)**

| BIT         | 7            | 6 | 5           | 4 | 3 | 2 | 1 | 0 |
|-------------|--------------|---|-------------|---|---|---|---|---|
| Field       | BW_MULT[1:0] |   | BW_VAL[5:0] |   |   |   |   |   |
| Reset       | 0x2          |   | 0x30        |   |   |   |   |   |
| Access Type | Write, Read  |   | Write, Read |   |   |   |   |   |

| BITFIELD | BITS | DESCRIPTION                                                                                                                      | DECODE                                                                                                                 |
|----------|------|----------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| BW_MULT  | 7:6  | Channel bandwidth-allocation multiplication factor                                                                               | 0b00: Multiply BW_VAL by 1<br>0b01: Multiply BW_VAL by 4<br>0b10: Multiply BW_VAL by 16<br>0b11: Multiply BW_VAL by 16 |
| BW_VAL   | 5:0  | Channel bandwidth-allocation base<br><br>Fair bandwidth use ratio = BW_VAL x BW_MULT/10 as a percentage of total link bandwidth. | 0bXXXXXX: Channel base bandwidth value                                                                                 |

**TR3 (0x63, 0x6B, 0x73, 0x7B, 0x83, 0x8B)**

| BIT         | 7 | 6 | 5              | 4              | 3 | 2              | 1 | 0 |
|-------------|---|---|----------------|----------------|---|----------------|---|---|
| Field       | – | – | TX_SPLT_MASK_B | TX_SPLT_MASK_A | – | TX_SRC_ID[2:0] |   |   |
| Reset       | – | – | 0x1            | 0x1            | – | 0x0            |   |   |
| Access Type | – | – | Write, Read    | Write, Read    | – | Write, Read    |   |   |

| BITFIELD       | BITS | DESCRIPTION                                                                                                                                                 | DECODE                                                                                                                   |
|----------------|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| TX_SPLT_MASK_B | 5    | In splitter mode:<br>When 0, packets from this port are NOT transmitted from Port B, otherwise packets are transmitted from port B.                         | 0b0: Packets are not transmitted over GMSLB in splitter mode<br>0b1: Packets are transmitted over GMSLB in splitter mode |
| TX_SPLT_MASK_A | 4    | In splitter mode:<br>When 0, packets from this port are NOT transmitted from Port A, otherwise packets are transmitted from port A.                         | 0b0: Packets are not transmitted over GMSLA in splitter mode<br>0b1: Packets are transmitted over GMSLA in splitter mode |
| TX_SRC_ID      | 2:0  | Source identifier used in packets transmitted from this channel<br><br>Default values of the 2 MSBs are set by the CFG pins. Default value of the LSB is 0. | 0bXXX: Source ID for packets from this channel                                                                           |

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**TR4 (0x64, 0x6C, 0x74, 0x7C, 0x84, 0x8C)**

| BIT         | 7               | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|-----------------|---|---|---|---|---|---|---|
| Field       | RX_SRC_SEL[7:0] |   |   |   |   |   |   |   |
| Reset       | 0xFF            |   |   |   |   |   |   |   |
| Access Type | Write, Read     |   |   |   |   |   |   |   |

| BITLENGTH  | BITS | DESCRIPTION                                                                                                                                                                                                                      | DECODE                                                                                                                                                                                                                                                                  |
|------------|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| RX_SRC_SEL | 7:0  | Receives packets from selected sources<br>Each bit indicates whether packets with that source ID should be received or not. For example, when SRC_SEL = 00001001, then packets with source ID equal to 0 and 3 will be received. | 0x00: No packets received<br>0x01: Packets from source ID 0 received<br>0x02: Packets from source ID 1 received<br>0x03: Packets from source ID 0 and 1 received<br>0x04: Packets from source ID 2 received<br>...<br>...<br>0xFF: Packets from all source IDs received |

**ARQ0 (0x6D, 0x75, 0x7D, 0x85, 0x8D)**

| BIT         | 7            | 6           | 5            | 4           | 3           | 2 | 1 | 0 |
|-------------|--------------|-------------|--------------|-------------|-------------|---|---|---|
| Field       | ARQ_AUTO_CFG | ACK_CNT     | MATCH_SRC_ID | ACK_SRC_ID  | EN          | — | — | — |
| Reset       | 0b1          | 0b0         | 0b0          | 0b1         | 0b1         | — | — | — |
| Access Type | Write, Read  | Write, Read | Write, Read  | Write, Read | Write, Read | — | — | — |

| BITLENGTH    | BITS | DESCRIPTION                                                                                                                                     | DECODE                                                                                                                                                        |
|--------------|------|-------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ARQ_AUTO_CFG | 7    | When this bit = 1, ARQ settings are automatically selected based on splitter mode.                                                              | 0b0: ARQ settings are based on selections in ACK_SRC_ID and MATCH_SRC_ID bitfields<br>0b1: ARQ settings are automatically selected                            |
| ACK_CNT      | 6    | When ARQ_AUTO_CFG = 0, this bit selects whether to wait for one acknowledge packet or two acknowledge packets for each transmitted data packet. | 0b0: Wait for one acknowledge packet<br>0b1: Wait for two acknowledge packets                                                                                 |
| MATCH_SRC_ID | 5    | Acknowledge packet source ID checking method                                                                                                    | 0: All received acknowledge packets are accepted<br>1: Acknowledge packet is accepted if SRC_ID of the received acknowledge packet matches TX_SRC_ID register |
| ACK_SRC_ID   | 4    | Selects what to use as SRC_ID in transmitted acknowledge packets                                                                                | 0: Use SRC_ID of the received data packet<br>1: Use TX_SRC_ID register                                                                                        |
| EN           | 3    | Enables ARQ                                                                                                                                     | 0b0: ARQ disabled<br>0b1: ARQ enabled                                                                                                                         |

**ARQ1 (0x6E, 0x76, 0x7E, 0x86, 0x8E)**

| BIT         | 7 | 6           | 5 | 4 | 3 | 2 | 1              | 0           |
|-------------|---|-------------|---|---|---|---|----------------|-------------|
| Field       | — | MAX_RT[2:0] |   |   | — | — | MAX_RT_ERR_OEN | RT_CNT_OEN  |
| Reset       | — | 0x7         |   |   | — | — | 0b1            | 0b0         |
| Access Type | — | Write, Read |   |   | — | — | Write, Read    | Write, Read |

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| BITFIELD       | BITS | DESCRIPTION                                                                                                                                                    | DECODE                                                                                                                                            |
|----------------|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| MAX_RT         | 6:4  | Maximum retransmit limit<br>ARQ will stop retransmitting after this many attempts for a single packet.                                                         | 0bXXX: Maximum retransmit limit                                                                                                                   |
| MAX_RT_ERR_OEN | 1    | Enables reporting of ARQ maximum retransmission limit errors (MAX_RT_ERR) for this channel at ERRB pin                                                         | 0b0: ARQ maximum retransmit limit errors reporting at ERRB pin disabled<br>0b1: ARQ maximum retransmit limit errors reporting at ERRB pin enabled |
| RT_CNT_OEN     | 0    | Enables reporting of ARQ retransmission event for this channel at ERRB pin<br><br>When enabled, ERRB is asserted when RT_CNT of this channel is greater than 0 | 0b0: ARQ retransmission count reporting at ERRB pin disabled<br>0b1: ARQ retransmission count reporting at ERRB pin enabled                       |

**ARQ2 (0x6F, 0x77, 0x7F, 0x87, 0x8F)**

| BIT         | 7                  | 6               | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|--------------------|-----------------|---|---|---|---|---|---|
| Field       | MAX_RT_ERR         | RT_CNT[6:0]     |   |   |   |   |   |   |
| Reset       | 0b0                | 0x0             |   |   |   |   |   |   |
| Access Type | Read<br>Clears All | Read Clears All |   |   |   |   |   |   |

| BITFIELD   | BITS | DESCRIPTION                                                                  | DECODE                                                                                     |
|------------|------|------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| MAX_RT_ERR | 7    | Reached maximum retransmission limit (MAX_RT) for one packet in this channel | 0b0: Maximum retransmission limit not reached<br>0b1: Maximum retransmission limit reached |
| RT_CNT     | 6:0  | Total retransmission count in this channel                                   | 0bXX: Count of retransmissions for this channel                                            |

**TR0 (0x70, 0x80, 0x88)**

| BIT         | 7           | 6           | 5 | 4 | 3             | 2 | 1             | 0 |
|-------------|-------------|-------------|---|---|---------------|---|---------------|---|
| Field       | TX_CRC_EN   | RX_CRC_EN   | – | – | PRIO_VAL[1:0] |   | PRIO_CFG[1:0] |   |
| Reset       | 0b1         | 0b1         | – | – | 0x0           |   | 0x0           |   |
| Access Type | Write, Read | Write, Read | – | – | Write, Read   |   | Write, Read   |   |

| BITFIELD  | BITS | DESCRIPTION                                                                                                                  | DECODE                                                                                                                                                                       |
|-----------|------|------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TX_CRC_EN | 7    | When set, calculates and appends CRC to each packet transmitted from this port                                               | 0b0: Transmit CRC disabled<br>0b1: Transmit CRC enabled                                                                                                                      |
| RX_CRC_EN | 6    | When set, indicates that packets received at this port have appended CRC and CRC checking should be performed at each packet | 0b0: Receive CRC disabled<br>0b1: Receive CRC enabled                                                                                                                        |
| PRIO_VAL  | 3:2  | Sets the priority for this channel's packet requests                                                                         | 0b00: Low priority<br>0b01: Normal priority<br>0b10: High priority<br>0b11: Urgent priority                                                                                  |
| PRIO_CFG  | 1:0  | Adjusts the priority used for this channel                                                                                   | 0b00: Current channel priority is used<br>0b01: Increase current channel priority<br>0b10: Decrease current channel priority<br>0b11: Reset to channel PRIO_VAL[3:2] setting |

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VIDEO\_RX0 (0x100, 0x112, 0x124, 0x136)

| BIT         | 7                  | 6 | 5 | 4 | 3 | 2            | 1           | 0           |
|-------------|--------------------|---|---|---|---|--------------|-------------|-------------|
| Field       | LCRC_ERR           | – | – | – | – | LINE_CRC_SEL | LINE_CRC_EN | DIS_PKT_DET |
| Reset       | 0x0                | – | – | – | – | 0b0          | 0b1         | 0b0         |
| Access Type | Read<br>Clears All | – | – | – | – | Write, Read  | Write, Read | Write, Read |

| BITFIELD     | BITS | DESCRIPTION                                                                                                                                                                                                  | DECODE                                                          |
|--------------|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|
| LCRC_ERR     | 7    | Video Line CRC Error flag                                                                                                                                                                                    | 0b0: No video line CRC detected<br>0b1: Video line CRC detected |
| LINE_CRC_SEL | 2    | Line CRC trigger selection                                                                                                                                                                                   | 0b0: Use DE<br>0b1: Use HS                                      |
| LINE_CRC_EN  | 1    | Video line CRC enable                                                                                                                                                                                        | 0b0: Video line CRC disabled<br>0b1: Video line CRC enabled     |
| DIS_PKT_DET  | 0    | Disable packet detector. If the video is restarted with a different BPP when the packet detector is disabled, toggle this register or the video receive enable register to make sure the video link restarts | 0b0: Packet detector enabled<br>0b1: Packet detector disabled   |

VIDEO\_RX3 (0x103, 0x115, 0x127, 0x139)

| BIT         | 7 | 6           | 5         | 4         | 3         | 2           | 1           | 0           |
|-------------|---|-------------|-----------|-----------|-----------|-------------|-------------|-------------|
| Field       | – | HD_TR_MODE  | DLOCKED   | VLOCKED   | HLOCKED   | DTRACKEN    | VTRACKEN    | HTRACKEN    |
| Reset       | – | 0b1         | 0b0       | 0b0       | 0b0       | 0b0         | 0b0         | 0b0         |
| Access Type | – | Write, Read | Read Only | Read Only | Read Only | Write, Read | Write, Read | Write, Read |

| BITFIELD   | BITS | DESCRIPTION                                                                                                                                                                          | DECODE                                                                                          |
|------------|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| HD_TR_MODE | 6    | HS, DE tracking enable full periodic pattern                                                                                                                                         | 0b0: Allow only partial periodic HS, DE<br>0b1: Allow partial periodic and full periodic HS, DE |
| DLOCKED    | 5    | DE tracking locked                                                                                                                                                                   | 0b0: DE tracking not locked<br>0b1: DE tracking locked                                          |
| VLOCKED    | 4    | VS tracking locked                                                                                                                                                                   | 0b0: VS tracking not locked<br>0b1: VS tracking locked                                          |
| HLOCKED    | 3    | HS tracking locked                                                                                                                                                                   | 0b0: HS tracking not locked<br>0b1: HS tracking locked                                          |
| DTRACKEN   | 2    | DE tracking enable (Disable if FSYNC = 1). The system observes DE pulses and, when it locks on the pattern, compensates for a limited number of missing pulses or suppress glitches. | 0b0: DE tracking disabled<br>0b1: DE tracking enabled                                           |
| VTRACKEN   | 1    | VS tracking enable (Disable if FSYNC = 1). The system observes VS pulses and, when it locks on the pattern, compensates for a limited number of missing pulses or suppress glitches. | 0b0: VS tracking disabled<br>0b1: VS tracking enabled                                           |

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| BITFIELD | BITS | DESCRIPTION                                                                                                                                                                          | DECODE                                                   |
|----------|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|
| HTRACKEN | 0    | HS tracking enable (Disable if FSYNC = 1). The system observes HS pulses and, when it locks on the pattern, compensates for a limited number of missing pulses or suppress glitches. | 0b0: HS tracking disabled<br>0b1: HSYNC tracking enabled |

**VIDEO\_RX8 (0x108, 0x11A, 0x12C, 0x13E)**

| BIT         | 7               | 6         | 5           | 4               | 3 | 2 | 1 | 0 |
|-------------|-----------------|-----------|-------------|-----------------|---|---|---|---|
| Field       | VID_BLK_LEN_ERR | VID_LOCK  | VID_PKT_DET | VID_SEQ_ERR     | – | – | – | – |
| Reset       | 0b0             | 0b0       | 0b0         | 0b0             | – | – | – | – |
| Access Type | Read Clears All | Read Only | Read Only   | Read Clears All | – | – | – | – |

| BITFIELD        | BITS | DESCRIPTION                                    | DECODE                                                              |
|-----------------|------|------------------------------------------------|---------------------------------------------------------------------|
| VID_BLK_LEN_ERR | 7    | Video Rx block length error detected           | 0b0: No error detected<br>0b1: Video Rx block length error detected |
| VID_LOCK        | 6    | Video pipeline locked                          | 0b0: Video pipeline not locked<br>0b1: Video pipeline locked        |
| VID_PKT_DET     | 5    | Sufficient video Rx packet throughput detected | 0b0: Not enough throughput<br>0b1: Sufficient throughput detected   |
| VID_SEQ_ERR     | 4    | Video Rx sequence error happened               | 0b0: No error detected<br>0b1: Error detected                       |

**VIDEO\_RX10 (0x10A, 0x11C, 0x12E, 0x140)**

| BIT         | 7 | 6             | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|---|---------------|---|---|---|---|---|---|
| Field       | – | MASK_VIDEO_DE | – | – | – | – | – | – |
| Reset       | – | 0x0           | – | – | – | – | – | – |
| Access Type | – | Write, Read   | – | – | – | – | – | – |

| BITFIELD      | BITS | DESCRIPTION        | DECODE                                                    |
|---------------|------|--------------------|-----------------------------------------------------------|
| MASK_VIDEO_DE | 6    | Mask video with DE | 0x0: Do not mask video with DE<br>0x1: Mask video with DE |

**INFO\_RX1 (0x10E, 0x120, 0x132, 0x144)**

| BIT         | 7              | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|----------------|---|---|---|---|---|---|---|
| Field       | SPEED_LIM[1:0] |   | – | – | – | – | – | – |
| Reset       | 0x0            |   | – | – | – | – | – | – |
| Access Type | Read Only      |   | – | – | – | – | – | – |

| BITFIELD  | BITS | DESCRIPTION               |
|-----------|------|---------------------------|
| SPEED_LIM | 7:6  | Received speed limitation |

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SPI\_0 (0x170)

| BIT         | 7               | 6 | 5                  | 4 | 3           | 2           | 1           | 0           |
|-------------|-----------------|---|--------------------|---|-------------|-------------|-------------|-------------|
| Field       | SPI_LOC_ID[1:0] |   | SPI_CC_TRG_ID[1:0] |   | SPI_IGNR_ID | SPI_CC_EN   | MST_SLVN    | SPI_EN      |
| Reset       | 0x0             |   | 0x0                |   | 0x1         | 0x0         | 0x0         | 0x0         |
| Access Type | Write, Read     |   | Write, Read        |   | Write, Read | Write, Read | Write, Read | Write, Read |

| BITFIELD      | BITS | DESCRIPTION                                                                  | DECODE                                                                           |
|---------------|------|------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| SPI_LOC_ID    | 7:6  | Program to local ID if filtering packets based on header ID                  | 0b00: ID = 0<br>0b01: ID = 1<br>0b10: ID = 2<br>0b11: ID = 3                     |
| SPI_CC_TRG_ID | 5:4  | ID for GMSL2 header in SPI control channel bridge mode                       | 0b00: ID = 0<br>0b01: ID = 1<br>0b10: ID = 2<br>0b11: ID = 3                     |
| SPI_IGNR_ID   | 3    | Selects if SPI should use or ignore header ID to decide on packet acceptance | 0b0: Accept only packets with proper ID<br>0b1: Ignore ID and accept all packets |
| SPI_CC_EN     | 2    | Enables control channel SPI bridge function                                  | 0b0: SPI bridge disabled<br>0b1: SPI bridge enabled                              |
| MST_SLVN      | 1    | Selects if SPI is master or slave                                            | 0b0: SPI slave<br>0b1: SPI master                                                |
| SPI_EN        | 0    | Enables SPI channel                                                          | 0b0: SPI channel disabled<br>0b1: SPI channel enabled                            |

SPI\_1 (0x171)

| BIT         | 7              | 6 | 5 | 4 | 3 | 2 | 1                  | 0 |
|-------------|----------------|---|---|---|---|---|--------------------|---|
| Field       | SPI_LOC_N[5:0] |   |   |   |   |   | SPI_BASE_PRIO[1:0] |   |
| Reset       | 0x07           |   |   |   |   |   | 0x1                |   |
| Access Type | Write, Read    |   |   |   |   |   | Write, Read        |   |

| BITFIELD      | BITS | DESCRIPTION                                                                                                                                            | DECODE                                                                                                              |
|---------------|------|--------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| SPI_LOC_N     | 7:2  | Sets the packet size ((2N + 1) bytes) for GMSL2 SPI packets. If this is programmed to a value greater than 7, ARQ of the SPI channel must be disabled. | 0b000000: Packet size is 1 byte<br>0b000001: Packet size is 3 bytes<br>.<br>.<br>0b111111: Packet size is 127 bytes |
| SPI_BASE_PRIO | 1:0  | Starting GMSL2 request priority advances by 1 (capacity permitting) if Tx buffer is over half full.                                                    | 0b00: Priority 0 (low)<br>0b01: Priority 1<br>0b10: Priority 2<br>0b11: Priority 3                                  |

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SPI\_2 (0x172)

| BIT         | 7                 | 6 | 5 | 4              | 3           | 2           | 1              | 0              |
|-------------|-------------------|---|---|----------------|-------------|-------------|----------------|----------------|
| Field       | REQ_HOLD_OFF[2:0] |   |   | FULL_SCK_SETUP | SPI_MOD3_F  | SPI_MOD3    | SPIM_SS2_ACT_H | SPIM_SS1_ACT_H |
| Reset       | 0x0               |   |   | 0x0            | 0x0         | 0x0         | 0x1            | 0x1            |
| Access Type | Write, Read       |   |   | Write, Read    | Write, Read | Write, Read | Write, Read    | Write, Read    |

| BITFIELD       | BITS | DESCRIPTION                                                                                | DECODE                                                                                                                                    |
|----------------|------|--------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| REQ_HOLD_OFF   | 7:5  | Holds off GMSL2 request until this number of extra bytes are received on SPI port          | 0b00: No extra bytes<br>0b01: 1 extra byte<br>0b10: 2 extra bytes<br>0b11: 3 extra bytes                                                  |
| FULL_SCK_SETUP | 4    | Samples MISO after half or full SCK period                                                 | 0b0: MISO sampled after half-SCK period<br>0b1: MISO sampled after full-SCK period                                                        |
| SPI_MOD3_F     | 3    | Allows the suppression of an extra SCK prior to SS deassertion when SPI mode 3 is selected | 0b0: Extra SCK present prior to SS deassertion when in SPI mode 3<br>0b1: Extra SCK suppressed prior to SS deassertion when in SPI mode 3 |
| SPI_MOD3       | 2    | Selects SPI mode 0 or 3                                                                    | 0b0: SPI mode 0<br>0b1: SPI mode 3                                                                                                        |
| SPIM_SS2_ACT_H | 1    | Sets the polarity for SS2 when the SPI is a master                                         | 0b0: Slave select 2 is active low<br>0b1: Slave select 2 is active high                                                                   |
| SPIM_SS1_ACT_H | 0    | Sets the polarity for SS1 when the SPI is a master                                         | 0b0: Slave select 1 is active low<br>0b1: Slave select 1 is active high                                                                   |

SPI\_3 (0x173)

| BIT         | 7                     | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|-----------------------|---|---|---|---|---|---|---|
| Field       | SPIM_SS_DLY_CLKS[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00                  |   |   |   |   |   |   |   |
| Access Type | Write, Read           |   |   |   |   |   |   |   |

| BITFIELD         | BITS | DESCRIPTION                                                                                                                                                                                        | DECODE                       |
|------------------|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| SPIM_SS_DLY_CLKS | 7:0  | Number of 300MHz clocks to delay between:<br>1) Assertion of SS and start of SCK pulses<br>2) End of SCK pulses and deassertion of SS<br>3) Deassertion of SS and reassertion of SS (if necessary) | 0xXX: Number of clock delays |

SPI\_4 (0x174)

| BIT         | 7                     | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|-----------------------|---|---|---|---|---|---|---|
| Field       | SPIM_SCK_LO_CLKS[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00                  |   |   |   |   |   |   |   |
| Access Type | Write, Read           |   |   |   |   |   |   |   |

| BITFIELD         | BITS | DESCRIPTION                              | DECODE                                  |
|------------------|------|------------------------------------------|-----------------------------------------|
| SPIM_SCK_LO_CLKS | 7:0  | Number of 300MHz clocks for SCK low time | 0xXX: Number of clocks for SCK low time |



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**SPI\_5 (0x175)**

| BIT         | 7                     | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|-----------------------|---|---|---|---|---|---|---|
| Field       | SPIM_SCK_HI_CLKS[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00                  |   |   |   |   |   |   |   |
| Access Type | Write, Read           |   |   |   |   |   |   |   |

| BITFIELD         | BITS | DESCRIPTION                               | DECODE                                   |
|------------------|------|-------------------------------------------|------------------------------------------|
| SPIM_SCK_HI_CLKS | 7:0  | Number of 300MHz clocks for SCK high time | 0xXX: Number of clocks for SCK high time |

**SPI\_6 (0x176)**

| BIT         | 7 | 6 | 5         | 4           | 3           | 2           | 1           | 0           |
|-------------|---|---|-----------|-------------|-------------|-------------|-------------|-------------|
| Field       | – | – | BNE       | SPIS_RWN    | SS_IO_EN_2  | SS_IO_EN_1  | BNE_IO_EN   | RWN_IO_EN   |
| Reset       | – | – | 0x0       | 0x0         | 0x0         | 0x0         | 0x0         | 0x0         |
| Access Type | – | – | Read Only | Write, Read | Write, Read | Write, Read | Write, Read | Write, Read |

| BITFIELD   | BITS | DESCRIPTION                                                                         | DECODE                                                                         |
|------------|------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| BNE        | 5    | Alternates GPU status register to use for BNE status if GPIO is not available       | 0b0: Buffer empty<br>0b1: Buffer not empty                                     |
| SPIS_RWN   | 4    | Alternates GPU control register to use for Rd/WrN control if GPIO is not available. | 0b0: Write<br>0b1: Read                                                        |
| SS_IO_EN_2 | 3    | Enables GPIO for use as slave select 2 output                                       | 0b0: GPIO not used for SPI SS2 function<br>0b1: GPIO used for SPI SS2 function |
| SS_IO_EN_1 | 2    | Enables GPIO for use as slave select 1 output                                       | 0b0: GPIO not used for SPI SS1 function<br>0b1: GPIO used for SPI SS1 function |
| BNE_IO_EN  | 1    | Enables GPIO for use as BNE output for SPI data available status                    | 0b0: GPIO not used for SPI BNE function<br>0b1: GPIO used for SPI BNE function |
| RWN_IO_EN  | 0    | Enables GPIO for use as RO input for control of SPI data movement                   | 0b0: GPIO not used for SPI RO function<br>0b1: GPIO used for SPI RO function   |

**SPI\_7 (0x177)**

| BIT         | 7                  | 6                  | 5 | 4                  | 3 | 2 | 1 | 0 |
|-------------|--------------------|--------------------|---|--------------------|---|---|---|---|
| Field       | SPI_RX_OVRFLW      | SPI_TX_OVRFLW      | – | SPIS_BYTE_CNT[4:0] |   |   |   |   |
| Reset       | 0b0                | 0b0                | – | 0x0                |   |   |   |   |
| Access Type | Read<br>Clears All | Read<br>Clears All | – | Read Only          |   |   |   |   |

| BITFIELD      | BITS | DESCRIPTION                                                    | DECODE                                                        |
|---------------|------|----------------------------------------------------------------|---------------------------------------------------------------|
| SPI_RX_OVRFLW | 7    | SPI Rx buffer overflow flag                                    | 0b0: No SPI Rx buffer overflow<br>0b1: SPI Rx buffer overflow |
| SPI_TX_OVRFLW | 6    | SPI Tx buffer overflow flag                                    | 0b0: No SPI Tx buffer overflow<br>0b1: SPI Tx buffer overflow |
| SPIS_BYTE_CNT | 4:0  | Number of SPI data bytes available for reading from Rx buffer. | 0bXXXXX: Number of bytes available                            |

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SPI\_8 (0x178)

| BIT         | 7                    | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|----------------------|---|---|---|---|---|---|---|
| Field       | REQ_HOLD_OFF_TO[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00                 |   |   |   |   |   |   |   |
| Access Type | Write, Read          |   |   |   |   |   |   |   |

| BITFIELD        | BITS | DESCRIPTION                                                                    | DECODE                                                            |
|-----------------|------|--------------------------------------------------------------------------------|-------------------------------------------------------------------|
| REQ_HOLD_OFF_TO | 7:0  | Timeout delay (in 100nS increments) for GMSL2 request hold off (0 is disable). | 0xXX: Number of 100nS delay increments for GMSL2 request hold off |

WM\_0 (0x190)

| BIT         | 7           | 6           | 5           | 4           | 3 | 2 | 1 | 0 |
|-------------|-------------|-------------|-------------|-------------|---|---|---|---|
| Field       | WM_LEN      | WM_MODE     | WM_DET      | WM_EN       |   |   |   |   |
| Reset       | 0x0         |             |             | 0x0         |   |   |   |   |
| Access Type | Write, Read | Write, Read | Write, Read | Write, Read |   |   |   |   |

| BITFIELD | BITS | DESCRIPTION                    | DECODE                                                                                                                                                                                                                                                                                                                                                                                                       |
|----------|------|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| WM_LEN   | 7    | Watermark length               | 0b000: Default generator mode - cycle through all 4 watermarks in video stream (default),<br>0b001: Default generator mode - cycle through only 2 watermarks to replicate a frozen frame error condition.<br>0b010-0b111: Reserved                                                                                                                                                                           |
| WM_MODE  | 6:4  | Watermark mode                 | 0b1: 63-bit<br>0b000: Insert WM-/WM+ every frame, temporal dither enabled<br>0b001: Reserved<br>0b010: Insert WM0/WM1 every frame, temporal dither enabled, polarity disabled (Error generation mode)<br>0b011: Insert WM0/WM1 every frame, do not reset WM, temporal dither enabled<br>0b100: Insert WM0/WM1 every frame, temporal dither disabled<br>0b101: Reserved<br>0b110: Reserved<br>0b111: Reserved |
| WM_DET   | 3:2  | Watermark detection/generation | 0b00: Insert watermark in video stream<br>0b01: Detect watermark and remove from outgoing video stream<br>0b10: Reserved<br>0b11: Reserved                                                                                                                                                                                                                                                                   |
| WM_EN    | 0    | Watermark enable               | 0b0: Watermarking disabled<br>0b1: Watermarking enabled                                                                                                                                                                                                                                                                                                                                                      |

WM\_2 (0x192)

| BIT         | 7 | 6 | 5 | 4 | 3           | 2           | 1              | 0 |
|-------------|---|---|---|---|-------------|-------------|----------------|---|
| Field       | – | – | – | – | HsyncPol    | VsyncPol    | WM_NPFILT[1:0] |   |
| Reset       | – | – | – | – | 0b0         | 0b0         | 0x0            |   |
| Access Type | – | – | – | – | Write, Read | Write, Read | Write, Read    |   |

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| BITFIELD  | BITS | DESCRIPTION                      | DECODE                                 |
|-----------|------|----------------------------------|----------------------------------------|
| HsyncPol  | 3    | HS polarity                      | 0b0: Non-inverting<br>0x1: Invert      |
| VsyncPol  | 2    | VS polarity                      | 0b0: Non-inverting<br>0x1: Invert      |
| WM_NPFILT | 1:0  | Phase accumulator terminal count | 0bXX: Phase accumulator terminal count |

**WM\_3 (0x193)**

| BIT         | 7 | 6           | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|---|-------------|---|---|---|---|---|---|
| Field       | – | WM_TH[6:0]  |   |   |   |   |   |   |
| Reset       | – | 0x14        |   |   |   |   |   |   |
| Access Type | – | Write, Read |   |   |   |   |   |   |

| BITFIELD | BITS | DESCRIPTION              | DECODE                              |
|----------|------|--------------------------|-------------------------------------|
| WM_TH    | 6:0  | Matched filter threshold | 0bXXXXXXX: Matched filter threshold |

**WM\_4 (0x194)**

| BIT         | 7 | 6 | 5              | 4 | 3               | 2 | 1                | 0 |
|-------------|---|---|----------------|---|-----------------|---|------------------|---|
| Field       | – | – | WM_YUV_IN[1:0] |   | WM_COLO<br>RADJ | – | WM_MASKMODE[1:0] |   |
| Reset       | – | – | 0x1            |   | 0x0             | – | 0x0              |   |
| Access Type | – | – | Write, Read    |   | Write, Read     | – | Write, Read      |   |

| BITFIELD        | BITS | DESCRIPTION                        | DECODE                                                                                                                                                    |
|-----------------|------|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| WM_YUV_IN       | 5:4  | Input color format                 | 0b0: RGB 8-bit color data<br>0b1: YUV444 8-bits each                                                                                                      |
| WM_COLOR<br>ADJ | 3    | Color adjust                       | 0b0: Masked off LSBs of U.<br>Use if temporal dithering is enabled.<br>0b1: Set U[Ko:0] = 1/2 WM gain.<br>Use this mode if temporal dithering is disabled |
| WM_MASKM<br>ODE | 1:0  | Sets watermark mask for the device | 0b00: Mask if WM is detected<br>0b01: Mask if WM is detected, blank if error is<br>detected<br>0b10: Reserved<br>b011: Reserved                           |

**WM\_5 (0x195)**

| BIT         | 7 | 6 | 5 | 4 | 3 | 2 | 1             | 0            |
|-------------|---|---|---|---|---|---|---------------|--------------|
| Field       | – | – | – | – | – | – | WM_DETO<br>UT | WM_ERRO<br>R |
| Reset       | – | – | – | – | – | – | 0x0           | 0x0          |
| Access Type | – | – | – | – | – | – | Read Only     | Read Only    |

| BITFIELD      | BITS | DESCRIPTION                       | DECODE                                                 |
|---------------|------|-----------------------------------|--------------------------------------------------------|
| WM_DETOU<br>T | 1    | Live frame-based detection output | 0b0: Watermark not detected<br>0b1: Watermark detected |

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| BITFIELD | BITS | DESCRIPTION                      | DECODE                                                                                              |
|----------|------|----------------------------------|-----------------------------------------------------------------------------------------------------|
| WM_ERROR | 0    | Live active-high watermark error | 0b0: No watermark error<br>0b1: Watermark error active, bit automatically clears when error clears. |

**WM\_6 (0x196)**

| BIT         | 7             | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|---------------|---|---|---|---|---|---|---|
| Field       | WM_TIMER[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00          |   |   |   |   |   |   |   |
| Access Type | Write, Read   |   |   |   |   |   |   |   |

| BITFIELD | BITS | DESCRIPTION                                                                                                     | DECODE                       |
|----------|------|-----------------------------------------------------------------------------------------------------------------|------------------------------|
| WM_TIMER | 7:0  | Time in 2 msec steps the frozen frame condition must be observed before an error is generated.<br>0 = No filter | 0xXX: Number of milliseconds |

**WM\_WREN\_0 (0x1AE)**

| BIT         | 7              | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|----------------|---|---|---|---|---|---|---|
| Field       | WM_WREN_L[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00           |   |   |   |   |   |   |   |
| Access Type | Write, Read    |   |   |   |   |   |   |   |

| BITFIELD  | BITS | DESCRIPTION                                                                                                                                        | DECODE                                                                         |
|-----------|------|----------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| WM_WREN_L | 7:0  | Writes 0xBA to WM_WREN_L and 0xDC to WM_WREN_H registers to enable writing to watermark registers.<br>Otherwise watermark registers are read only. | 0xBA: Enables writing to WM registers<br>Others: WM registers remain read only |

**WM\_WREN\_1 (0x1AF)**

| BIT         | 7              | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|----------------|---|---|---|---|---|---|---|
| Field       | WM_WREN_H[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00           |   |   |   |   |   |   |   |
| Access Type | Write, Read    |   |   |   |   |   |   |   |

| BITFIELD  | BITS | DESCRIPTION                                                                                                                                        | DECODE                                                                         |
|-----------|------|----------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| WM_WREN_H | 7:0  | Writes 0xBA to WM_WREN_L and 0xDC to WM_WREN_H registers to enable writing to watermark registers.<br>Otherwise watermark registers are read only. | 0xDC: Enables writing to WM registers<br>Others: WM registers remain read only |

**CROSS\_0 (0x1C0, 0x1E0, 0x200, 0x220)**

| BIT         | 7 | 6           | 5           | 4           | 3 | 2 | 1 | 0 |
|-------------|---|-------------|-------------|-------------|---|---|---|---|
| Field       | – | CROSS0_I    | CROSS0_F    | CROSS0[4:0] |   |   |   |   |
| Reset       | – | 0x0         | 0x0         | 0x00        |   |   |   |   |
| Access Type | – | Write, Read | Write, Read | Write, Read |   |   |   |   |

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| BITFIELD | BITS | DESCRIPTION                                                                 | DECODE                                                  |
|----------|------|-----------------------------------------------------------------------------|---------------------------------------------------------|
| CROSS0_I | 6    | Inverts CrossX                                                              | 0b0: Do not invert bit<br>0b1: Invert bit               |
| CROSS0_F | 5    | Forces CrossX to 0 before inversion                                         | 0b0: Do not force bit to zero<br>0b1: Force bit to zero |
| CROSS0   | 4:0  | Maps incoming bit position set by this field to the outgoing bit position 0 | 0bXXXXX: Incoming bit position                          |

**CROSS 1 (0x1C1, 0x1E1, 0x201, 0x221)**

| BIT         | 7 | 6           | 5           | 4           | 3 | 2 | 1 | 0 |
|-------------|---|-------------|-------------|-------------|---|---|---|---|
| Field       | – | CROSS1_I    | CROSS1_F    | CROSS1[4:0] |   |   |   |   |
| Reset       | – | 0x0         | 0x0         | 0x01        |   |   |   |   |
| Access Type | – | Write, Read | Write, Read | Write, Read |   |   |   |   |

| BITFIELD | BITS | DESCRIPTION                                                                 | DECODE                                                  |
|----------|------|-----------------------------------------------------------------------------|---------------------------------------------------------|
| CROSS1_I | 6    | Inverts CrossX                                                              | 0b0: Do not invert bit<br>0b1: Invert bit               |
| CROSS1_F | 5    | Forces CrossX to 0 before inversion                                         | 0b0: Do not force bit to zero<br>0b1: Force bit to zero |
| CROSS1   | 4:0  | Maps incoming bit position set by this field to the outgoing bit position 1 | 0bXXXXX: Incoming bit position                          |

**CROSS 2 (0x1C2, 0x1E2, 0x202, 0x222)**

| BIT         | 7 | 6           | 5           | 4           | 3 | 2 | 1 | 0 |
|-------------|---|-------------|-------------|-------------|---|---|---|---|
| Field       | – | CROSS2_I    | CROSS2_F    | CROSS2[4:0] |   |   |   |   |
| Reset       | – | 0x0         | 0x0         | 0x02        |   |   |   |   |
| Access Type | – | Write, Read | Write, Read | Write, Read |   |   |   |   |

| BITFIELD | BITS | DESCRIPTION                                                                 | DECODE                                                  |
|----------|------|-----------------------------------------------------------------------------|---------------------------------------------------------|
| CROSS2_I | 6    | Inverts CrossX                                                              | 0b0: Do not invert bit<br>0b1: Invert bit               |
| CROSS2_F | 5    | Forces CrossX to 0 before inversion                                         | 0b0: Do not force bit to zero<br>0b1: Force bit to zero |
| CROSS2   | 4:0  | Maps incoming bit position set by this field to the outgoing bit position 2 | 0bXXXXX: Incoming bit position                          |

**CROSS 3 (0x1C3, 0x1E3, 0x203, 0x223)**

| BIT         | 7 | 6           | 5           | 4           | 3 | 2 | 1 | 0 |
|-------------|---|-------------|-------------|-------------|---|---|---|---|
| Field       | – | CROSS3_I    | CROSS3_F    | CROSS3[4:0] |   |   |   |   |
| Reset       | – | 0x0         | 0x0         | 0x03        |   |   |   |   |
| Access Type | – | Write, Read | Write, Read | Write, Read |   |   |   |   |

| BITFIELD | BITS | DESCRIPTION    | DECODE                                    |
|----------|------|----------------|-------------------------------------------|
| CROSS3_I | 6    | Inverts CrossX | 0b0: Do not invert bit<br>0b1: Invert bit |

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| BITFIELD | BITS | DESCRIPTION                                                                 | DECODE                                                  |
|----------|------|-----------------------------------------------------------------------------|---------------------------------------------------------|
| CROSS3_F | 5    | Forces CrossX to 0 before inversion                                         | 0b0: Do not force bit to zero<br>0b1: Force bit to zero |
| CROSS3   | 4:0  | Maps incoming bit position set by this field to the outgoing bit position 3 | 0bXXXXX: Incoming bit position                          |

**CROSS 4 (0x1C4, 0x1E4, 0x204, 0x224)**

| BIT         | 7 | 6           | 5           | 4           | 3 | 2 | 1 | 0 |
|-------------|---|-------------|-------------|-------------|---|---|---|---|
| Field       | – | CROSS4_I    | CROSS4_F    | CROSS4[4:0] |   |   |   |   |
| Reset       | – | 0x0         | 0x0         | 0x04        |   |   |   |   |
| Access Type | – | Write, Read | Write, Read | Write, Read |   |   |   |   |

| BITFIELD | BITS | DESCRIPTION                                                                 | DECODE                                                  |
|----------|------|-----------------------------------------------------------------------------|---------------------------------------------------------|
| CROSS4_I | 6    | Inverts CrossX                                                              | 0b0: Do not invert bit<br>0b1: Invert bit               |
| CROSS4_F | 5    | Forces CrossX to 0 before inversion                                         | 0b0: Do not force bit to zero<br>0b1: Force bit to zero |
| CROSS4   | 4:0  | Maps incoming bit position set by this field to the outgoing bit position 4 | 0bXXXXX: Incoming bit position                          |

**CROSS 5 (0x1C5, 0x1E5, 0x205, 0x225)**

| BIT         | 7 | 6           | 5           | 4           | 3 | 2 | 1 | 0 |
|-------------|---|-------------|-------------|-------------|---|---|---|---|
| Field       | – | CROSS5_I    | CROSS5_F    | CROSS5[4:0] |   |   |   |   |
| Reset       | – | 0x0         | 0x0         | 0x05        |   |   |   |   |
| Access Type | – | Write, Read | Write, Read | Write, Read |   |   |   |   |

| BITFIELD | BITS | DESCRIPTION                                                                 | DECODE                                                  |
|----------|------|-----------------------------------------------------------------------------|---------------------------------------------------------|
| CROSS5_I | 6    | Inverts CrossX                                                              | 0b0: Do not invert bit<br>0b1: Invert bit               |
| CROSS5_F | 5    | Forces CrossX to 0 before inversion                                         | 0b0: Do not force bit to zero<br>0b1: Force bit to zero |
| CROSS5   | 4:0  | Maps incoming bit position set by this field to the outgoing bit position 5 | 0bXXXXX: Incoming bit position                          |

**CROSS 6 (0x1C6, 0x1E6, 0x206, 0x226)**

| BIT         | 7 | 6           | 5           | 4           | 3 | 2 | 1 | 0 |
|-------------|---|-------------|-------------|-------------|---|---|---|---|
| Field       | – | CROSS6_I    | CROSS6_F    | CROSS6[4:0] |   |   |   |   |
| Reset       | – | 0x0         | 0x0         | 0x06        |   |   |   |   |
| Access Type | – | Write, Read | Write, Read | Write, Read |   |   |   |   |

| BITFIELD | BITS | DESCRIPTION                         | DECODE                                                  |
|----------|------|-------------------------------------|---------------------------------------------------------|
| CROSS6_I | 6    | Inverts CrossX                      | 0b0: Do not invert bit<br>0b1: Invert bit               |
| CROSS6_F | 5    | Forces CrossX to 0 before inversion | 0b0: Do not force bit to zero<br>0b1: Force bit to zero |

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| BITFIELD | BITS | DESCRIPTION                                                                 | DECODE                         |
|----------|------|-----------------------------------------------------------------------------|--------------------------------|
| CROSS6   | 4:0  | Maps incoming bit position set by this field to the outgoing bit position 6 | 0bXXXXX: Incoming bit position |

**CROSS\_7 (0x1C7, 0x1E7, 0x207, 0x227)**

| BIT         | 7 | 6           | 5           | 4           | 3 | 2 | 1 | 0 |
|-------------|---|-------------|-------------|-------------|---|---|---|---|
| Field       | – | CROSS7_I    | CROSS7_F    | CROSS7[4:0] |   |   |   |   |
| Reset       | – | 0x0         | 0x0         | 0x07        |   |   |   |   |
| Access Type | – | Write, Read | Write, Read | Write, Read |   |   |   |   |

| BITFIELD | BITS | DESCRIPTION                                                                 | DECODE                                                  |
|----------|------|-----------------------------------------------------------------------------|---------------------------------------------------------|
| CROSS7_I | 6    | Inverts CrossX                                                              | 0b0: Do not invert bit<br>0b1: Invert bit               |
| CROSS7_F | 5    | Forces CrossX to 0 before inversion                                         | 0b0: Do not force bit to zero<br>0b1: Force bit to zero |
| CROSS7   | 4:0  | Maps incoming bit position set by this field to the outgoing bit position 7 | 0bXXXXX: Incoming bit position                          |

**CROSS\_8 (0x1C8, 0x1E8, 0x208, 0x228)**

| BIT         | 7 | 6           | 5           | 4           | 3 | 2 | 1 | 0 |
|-------------|---|-------------|-------------|-------------|---|---|---|---|
| Field       | – | CROSS8_I    | CROSS8_F    | CROSS8[4:0] |   |   |   |   |
| Reset       | – | 0x0         | 0x0         | 0x08        |   |   |   |   |
| Access Type | – | Write, Read | Write, Read | Write, Read |   |   |   |   |

| BITFIELD | BITS | DESCRIPTION                                                                 | DECODE                                                  |
|----------|------|-----------------------------------------------------------------------------|---------------------------------------------------------|
| CROSS8_I | 6    | Inverts CrossX                                                              | 0b0: Do not invert bit<br>0b1: Invert bit               |
| CROSS8_F | 5    | Forces CrossX to 0 before inversion                                         | 0b0: Do not force bit to zero<br>0b1: Force bit to zero |
| CROSS8   | 4:0  | Maps incoming bit position set by this field to the outgoing bit position 8 | 0bXXXXX: Incoming bit position                          |

**CROSS\_9 (0x1C9, 0x1E9, 0x209, 0x229)**

| BIT         | 7 | 6           | 5           | 4           | 3 | 2 | 1 | 0 |
|-------------|---|-------------|-------------|-------------|---|---|---|---|
| Field       | – | CROSS9_I    | CROSS9_F    | CROSS9[4:0] |   |   |   |   |
| Reset       | – | 0x0         | 0x0         | 0x09        |   |   |   |   |
| Access Type | – | Write, Read | Write, Read | Write, Read |   |   |   |   |

| BITFIELD | BITS | DESCRIPTION                                                                 | DECODE                                                  |
|----------|------|-----------------------------------------------------------------------------|---------------------------------------------------------|
| CROSS9_I | 6    | Inverts CrossX                                                              | 0b0: Do not invert bit<br>0b1: Invert bit               |
| CROSS9_F | 5    | Forces CrossX to 0 before inversion                                         | 0b0: Do not force bit to zero<br>0b1: Force bit to zero |
| CROSS9   | 4:0  | Maps incoming bit position set by this field to the outgoing bit position 9 | 0bXXXXX: Incoming bit position                          |



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CROSS\_10 (0x1CA, 0x1EA, 0x20A, 0x22A)

| BIT         | 7    | 6                                                                            | 5           | 4            | 3                                                       | 2 | 1 | 0 |
|-------------|------|------------------------------------------------------------------------------|-------------|--------------|---------------------------------------------------------|---|---|---|
| Field       | –    | CROSS10_I                                                                    | CROSS10_F   | CROSS10[4:0] |                                                         |   |   |   |
| Reset       | –    | 0x0                                                                          | 0x0         | 0x0A         |                                                         |   |   |   |
| Access Type | –    | Write, Read                                                                  | Write, Read | Write, Read  |                                                         |   |   |   |
| BITFIELD    | BITS | DESCRIPTION                                                                  |             |              | DECODE                                                  |   |   |   |
| CROSS10_I   | 6    | Inverts CrossX                                                               |             |              | 0b0: Do not invert bit<br>0b1: Invert bit               |   |   |   |
| CROSS10_F   | 5    | Forces CrossX to 0 before inversion                                          |             |              | 0b0: Do not force bit to zero<br>0b1: Force bit to zero |   |   |   |
| CROSS10     | 4:0  | Maps incoming bit position set by this field to the outgoing bit position 10 |             |              | 0bXXXXX: Incoming bit position                          |   |   |   |

CROSS\_11 (0x1CB, 0x1EB, 0x20B, 0x22B)

| BIT         | 7    | 6                                                                            | 5           | 4            | 3                                                       | 2 | 1 | 0 |
|-------------|------|------------------------------------------------------------------------------|-------------|--------------|---------------------------------------------------------|---|---|---|
| Field       | –    | CROSS11_I                                                                    | CROSS11_F   | CROSS11[4:0] |                                                         |   |   |   |
| Reset       | –    | 0x0                                                                          | 0x0         | 0x0B         |                                                         |   |   |   |
| Access Type | –    | Write, Read                                                                  | Write, Read | Write, Read  |                                                         |   |   |   |
| BITFIELD    | BITS | DESCRIPTION                                                                  |             |              | DECODE                                                  |   |   |   |
| CROSS11_I   | 6    | Inverts CrossX                                                               |             |              | 0b0: Do not invert bit<br>0b1: Invert bit               |   |   |   |
| CROSS11_F   | 5    | Forces CrossX to 0 before inversion                                          |             |              | 0b0: Do not force bit to zero<br>0b1: Force bit to zero |   |   |   |
| CROSS11     | 4:0  | Maps incoming bit position set by this field to the outgoing bit position 11 |             |              | 0bXXXXX: Incoming bit position                          |   |   |   |

CROSS\_12 (0x1CC, 0x1EC, 0x20C, 0x22C)

| BIT         | 7    | 6                                                                            | 5           | 4            | 3                                                       | 2 | 1 | 0 |
|-------------|------|------------------------------------------------------------------------------|-------------|--------------|---------------------------------------------------------|---|---|---|
| Field       | –    | CROSS12_I                                                                    | CROSS12_F   | CROSS12[4:0] |                                                         |   |   |   |
| Reset       | –    | 0x0                                                                          | 0x0         | 0x0C         |                                                         |   |   |   |
| Access Type | –    | Write, Read                                                                  | Write, Read | Write, Read  |                                                         |   |   |   |
| BITFIELD    | BITS | DESCRIPTION                                                                  |             |              | DECODE                                                  |   |   |   |
| CROSS12_I   | 6    | Inverts CrossX                                                               |             |              | 0b0: Do not invert bit<br>0b1: Invert bit               |   |   |   |
| CROSS12_F   | 5    | Forces CrossX to 0 before inversion                                          |             |              | 0b0: Do not force bit to zero<br>0b1: Force bit to zero |   |   |   |
| CROSS12     | 4:0  | Maps incoming bit position set by this field to the outgoing bit position 12 |             |              | 0bXXXXX: Incoming bit position                          |   |   |   |

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Appendix A: Register Map and Tables

CROSS\_13 (0x1CD, 0x1ED, 0x20D, 0x22D)

| BIT         | 7    | 6                                                                            | 5           | 4            | 3                                                       | 2 | 1 | 0 |
|-------------|------|------------------------------------------------------------------------------|-------------|--------------|---------------------------------------------------------|---|---|---|
| Field       | –    | CROSS13_I                                                                    | CROSS13_F   | CROSS13[4:0] |                                                         |   |   |   |
| Reset       | –    | 0x0                                                                          | 0x0         | 0x0D         |                                                         |   |   |   |
| Access Type | –    | Write, Read                                                                  | Write, Read | Write, Read  |                                                         |   |   |   |
| BITFIELD    | BITS | DESCRIPTION                                                                  |             |              | DECODE                                                  |   |   |   |
| CROSS13_I   | 6    | Inverts CrossX                                                               |             |              | 0b0: Do not invert bit<br>0b1: Invert bit               |   |   |   |
| CROSS13_F   | 5    | Forces CrossX to 0 before inversion                                          |             |              | 0b0: Do not force bit to zero<br>0b1: Force bit to zero |   |   |   |
| CROSS13     | 4:0  | Maps incoming bit position set by this field to the outgoing bit position 13 |             |              | 0bXXXXX: Incoming bit position                          |   |   |   |

CROSS\_14 (0x1CE, 0x1EE, 0x20E, 0x22E)

| BIT         | 7    | 6                                                                            | 5           | 4            | 3                                                       | 2 | 1 | 0 |
|-------------|------|------------------------------------------------------------------------------|-------------|--------------|---------------------------------------------------------|---|---|---|
| Field       | –    | CROSS14_I                                                                    | CROSS14_F   | CROSS14[4:0] |                                                         |   |   |   |
| Reset       | –    | 0x0                                                                          | 0x0         | 0x0E         |                                                         |   |   |   |
| Access Type | –    | Write, Read                                                                  | Write, Read | Write, Read  |                                                         |   |   |   |
| BITFIELD    | BITS | DESCRIPTION                                                                  |             |              | DECODE                                                  |   |   |   |
| CROSS14_I   | 6    | Inverts CrossX                                                               |             |              | 0b0: Do not invert bit<br>0b1: Invert bit               |   |   |   |
| CROSS14_F   | 5    | Forces CrossX to 0 before inversion                                          |             |              | 0b0: Do not force bit to zero<br>0b1: Force bit to zero |   |   |   |
| CROSS14     | 4:0  | Maps incoming bit position set by this field to the outgoing bit position 14 |             |              | 0bXXXXX: Incoming bit position                          |   |   |   |

CROSS\_15 (0x1CF, 0x1EF, 0x20F, 0x22F)

| BIT         | 7    | 6                                                                            | 5           | 4            | 3                                                       | 2 | 1 | 0 |
|-------------|------|------------------------------------------------------------------------------|-------------|--------------|---------------------------------------------------------|---|---|---|
| Field       | –    | CROSS15_I                                                                    | CROSS15_F   | CROSS15[4:0] |                                                         |   |   |   |
| Reset       | –    | 0x0                                                                          | 0x0         | 0x0F         |                                                         |   |   |   |
| Access Type | –    | Write, Read                                                                  | Write, Read | Write, Read  |                                                         |   |   |   |
| BITFIELD    | BITS | DESCRIPTION                                                                  |             |              | DECODE                                                  |   |   |   |
| CROSS15_I   | 6    | Inverts CrossX                                                               |             |              | 0b0: Do not invert bit<br>0b1: Invert bit               |   |   |   |
| CROSS15_F   | 5    | Forces CrossX to 0 before inversion                                          |             |              | 0b0: Do not force bit to zero<br>0b1: Force bit to zero |   |   |   |
| CROSS15     | 4:0  | Maps incoming bit position set by this field to the outgoing bit position 15 |             |              | 0bXXXXX: Incoming bit position                          |   |   |   |

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Appendix A: Register Map and Tables

CROSS\_16 (0x1D0, 0x1F0, 0x210, 0x230)

| BIT         | 7    | 6                                                                            | 5           | 4            | 3                                                       | 2 | 1 | 0 |
|-------------|------|------------------------------------------------------------------------------|-------------|--------------|---------------------------------------------------------|---|---|---|
| Field       | –    | CROSS16_I                                                                    | CROSS16_F   | CROSS16[4:0] |                                                         |   |   |   |
| Reset       | –    | 0x0                                                                          | 0x0         | 0x10         |                                                         |   |   |   |
| Access Type | –    | Write, Read                                                                  | Write, Read | Write, Read  |                                                         |   |   |   |
| BITFIELD    | BITS | DESCRIPTION                                                                  |             |              | DECODE                                                  |   |   |   |
| CROSS16_I   | 6    | Inverts CrossX                                                               |             |              | 0b0: Do not invert bit<br>0b1: Invert bit               |   |   |   |
| CROSS16_F   | 5    | Forces CrossX to 0 before inversion                                          |             |              | 0b0: Do not force bit to zero<br>0b1: Force bit to zero |   |   |   |
| CROSS16     | 4:0  | Maps incoming bit position set by this field to the outgoing bit position 16 |             |              | 0bXXXXX: Incoming bit position                          |   |   |   |

CROSS\_17 (0x1D1, 0x1F1, 0x211, 0x231)

| BIT         | 7    | 6                                                                            | 5           | 4            | 3                                                       | 2 | 1 | 0 |
|-------------|------|------------------------------------------------------------------------------|-------------|--------------|---------------------------------------------------------|---|---|---|
| Field       | –    | CROSS17_I                                                                    | CROSS17_F   | CROSS17[4:0] |                                                         |   |   |   |
| Reset       | –    | 0x0                                                                          | 0x0         | 0x11         |                                                         |   |   |   |
| Access Type | –    | Write, Read                                                                  | Write, Read | Write, Read  |                                                         |   |   |   |
| BITFIELD    | BITS | DESCRIPTION                                                                  |             |              | DECODE                                                  |   |   |   |
| CROSS17_I   | 6    | Inverts CrossX                                                               |             |              | 0b0: Do not invert bit<br>0b1: Invert bit               |   |   |   |
| CROSS17_F   | 5    | Forces CrossX to 0 before inversion                                          |             |              | 0b0: Do not force bit to zero<br>0b1: Force bit to zero |   |   |   |
| CROSS17     | 4:0  | Maps incoming bit position set by this field to the outgoing bit position 17 |             |              | 0bXXXXX: Incoming bit position                          |   |   |   |

CROSS\_18 (0x1D2, 0x1F2, 0x212, 0x232)

| BIT         | 7    | 6                                                                            | 5           | 4            | 3                                                       | 2 | 1 | 0 |
|-------------|------|------------------------------------------------------------------------------|-------------|--------------|---------------------------------------------------------|---|---|---|
| Field       | –    | CROSS18_I                                                                    | CROSS18_F   | CROSS18[4:0] |                                                         |   |   |   |
| Reset       | –    | 0x0                                                                          | 0x0         | 0x12         |                                                         |   |   |   |
| Access Type | –    | Write, Read                                                                  | Write, Read | Write, Read  |                                                         |   |   |   |
| BITFIELD    | BITS | DESCRIPTION                                                                  |             |              | DECODE                                                  |   |   |   |
| CROSS18_I   | 6    | Inverts CrossX                                                               |             |              | 0b0: Do not invert bit<br>0b1: Invert bit               |   |   |   |
| CROSS18_F   | 5    | Forces CrossX to 0 before inversion                                          |             |              | 0b0: Do not force bit to zero<br>0b1: Force bit to zero |   |   |   |
| CROSS18     | 4:0  | Maps incoming bit position set by this field to the outgoing bit position 18 |             |              | 0bXXXXX: Incoming bit position                          |   |   |   |

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Appendix A: Register Map and Tables

**CROSS\_19 (0x1D3, 0x1F3, 0x213, 0x233)**

| BIT         | 7    | 6                                                                            | 5           | 4            | 3                                                       | 2 | 1 | 0 |
|-------------|------|------------------------------------------------------------------------------|-------------|--------------|---------------------------------------------------------|---|---|---|
| Field       | –    | CROSS19_I                                                                    | CROSS19_F   | CROSS19[4:0] |                                                         |   |   |   |
| Reset       | –    | 0x0                                                                          | 0x0         | 0x13         |                                                         |   |   |   |
| Access Type | –    | Write, Read                                                                  | Write, Read | Write, Read  |                                                         |   |   |   |
| BITFIELD    | BITS | DESCRIPTION                                                                  |             |              | DECODE                                                  |   |   |   |
| CROSS19_I   | 6    | Inverts CrossX                                                               |             |              | 0b0: Do not invert bit<br>0b1: Invert bit               |   |   |   |
| CROSS19_F   | 5    | Forces CrossX to 0 before inversion                                          |             |              | 0b0: Do not force bit to zero<br>0b1: Force bit to zero |   |   |   |
| CROSS19     | 4:0  | Maps incoming bit position set by this field to the outgoing bit position 19 |             |              | 0bXXXXX: Incoming bit position                          |   |   |   |

**CROSS\_20 (0x1D4, 0x1F4, 0x214, 0x234)**

| BIT         | 7    | 6                                                                            | 5           | 4            | 3                                                       | 2 | 1 | 0 |
|-------------|------|------------------------------------------------------------------------------|-------------|--------------|---------------------------------------------------------|---|---|---|
| Field       | –    | CROSS20_I                                                                    | CROSS20_F   | CROSS20[4:0] |                                                         |   |   |   |
| Reset       | –    | 0x0                                                                          | 0x0         | 0x14         |                                                         |   |   |   |
| Access Type | –    | Write, Read                                                                  | Write, Read | Write, Read  |                                                         |   |   |   |
| BITFIELD    | BITS | DESCRIPTION                                                                  |             |              | DECODE                                                  |   |   |   |
| CROSS20_I   | 6    | Inverts CrossX                                                               |             |              | 0b0: Do not invert bit<br>0b1: Invert bit               |   |   |   |
| CROSS20_F   | 5    | Forces CrossX to 0 before inversion                                          |             |              | 0b0: Do not force bit to zero<br>0b1: Force bit to zero |   |   |   |
| CROSS20     | 4:0  | Maps incoming bit position set by this field to the outgoing bit position 20 |             |              | 0bXXXXX: Incoming bit position                          |   |   |   |

**CROSS\_21 (0x1D5, 0x1F5, 0x215, 0x235)**

| BIT         | 7    | 6                                                                            | 5           | 4            | 3                                                       | 2 | 1 | 0 |
|-------------|------|------------------------------------------------------------------------------|-------------|--------------|---------------------------------------------------------|---|---|---|
| Field       | –    | CROSS21_I                                                                    | CROSS21_F   | CROSS21[4:0] |                                                         |   |   |   |
| Reset       | –    | 0x0                                                                          | 0x0         | 0x15         |                                                         |   |   |   |
| Access Type | –    | Write, Read                                                                  | Write, Read | Write, Read  |                                                         |   |   |   |
| BITFIELD    | BITS | DESCRIPTION                                                                  |             |              | DECODE                                                  |   |   |   |
| CROSS21_I   | 6    | Inverts CrossX                                                               |             |              | 0b0: Do not invert bit<br>0b1: Invert bit               |   |   |   |
| CROSS21_F   | 5    | Forces CrossX to 0 before inversion                                          |             |              | 0b0: Do not force bit to zero<br>0b1: Force bit to zero |   |   |   |
| CROSS21     | 4:0  | Maps incoming bit position set by this field to the outgoing bit position 21 |             |              | 0bXXXXX: Incoming bit position                          |   |   |   |

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**CROSS\_22 (0x1D6, 0x1F6, 0x216, 0x236)**

| BIT         | 7    | 6                                                                            | 5           | 4            | 3                                                       | 2 | 1 | 0 |
|-------------|------|------------------------------------------------------------------------------|-------------|--------------|---------------------------------------------------------|---|---|---|
| Field       | –    | CROSS22_I                                                                    | CROSS22_F   | CROSS22[4:0] |                                                         |   |   |   |
| Reset       | –    | 0x0                                                                          | 0x0         | 0x16         |                                                         |   |   |   |
| Access Type | –    | Write, Read                                                                  | Write, Read | Write, Read  |                                                         |   |   |   |
| BITFIELD    | BITS | DESCRIPTION                                                                  |             |              | DECODE                                                  |   |   |   |
| CROSS22_I   | 6    | Inverts CrossX                                                               |             |              | 0b0: Do not invert bit<br>0b1: Invert bit               |   |   |   |
| CROSS22_F   | 5    | Forces CrossX to 0 before inversion                                          |             |              | 0b0: Do not force bit to zero<br>0b1: Force bit to zero |   |   |   |
| CROSS22     | 4:0  | Maps incoming bit position set by this field to the outgoing bit position 22 |             |              | 0bXXXXX: Incoming bit position                          |   |   |   |

**CROSS\_23 (0x1D7, 0x1F7, 0x217, 0x237)**

| BIT         | 7    | 6                                                                            | 5           | 4            | 3                                                       | 2 | 1 | 0 |
|-------------|------|------------------------------------------------------------------------------|-------------|--------------|---------------------------------------------------------|---|---|---|
| Field       | –    | CROSS23_I                                                                    | CROSS23_F   | CROSS23[4:0] |                                                         |   |   |   |
| Reset       | –    | 0x0                                                                          | 0x0         | 0x17         |                                                         |   |   |   |
| Access Type | –    | Write, Read                                                                  | Write, Read | Write, Read  |                                                         |   |   |   |
| BITFIELD    | BITS | DESCRIPTION                                                                  |             |              | DECODE                                                  |   |   |   |
| CROSS23_I   | 6    | Inverts CrossX                                                               |             |              | 0b0: Do not invert bit<br>0b1: Invert bit               |   |   |   |
| CROSS23_F   | 5    | Forces CrossX to 0 before inversion                                          |             |              | 0b0: Do not force bit to zero<br>0b1: Force bit to zero |   |   |   |
| CROSS23     | 4:0  | Maps incoming bit position set by this field to the outgoing bit position 23 |             |              | 0bXXXXX: Incoming bit position                          |   |   |   |

**CROSS\_HS (0x1D8, 0x1F8, 0x218, 0x238)**

| BIT         | 7    | 6                                     | 5           | 4             | 3                                                       | 2 | 1 | 0 |
|-------------|------|---------------------------------------|-------------|---------------|---------------------------------------------------------|---|---|---|
| Field       | –    | CROSS_HS_I                            | CROSS_HS_F  | CROSS_HS[4:0] |                                                         |   |   |   |
| Reset       | –    | 0x0                                   | 0x0         | 0x18          |                                                         |   |   |   |
| Access Type | –    | Write, Read                           | Write, Read | Write, Read   |                                                         |   |   |   |
| BITFIELD    | BITS | DESCRIPTION                           |             |               | DECODE                                                  |   |   |   |
| CROSS_HS_I  | 6    | Inverts CROSS_HS                      |             |               | 0b0: Do not invert bit<br>0b1: Invert bit               |   |   |   |
| CROSS_HS_F  | 5    | Forces CROSS_HS to 0 before inversion |             |               | 0b0: Do not force bit to zero<br>0b1: Force bit to zero |   |   |   |
| CROSS_HS    | 4:0  | Maps selected internal signal to HS   |             |               | 0bXXXXX: Incoming bit position                          |   |   |   |

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CROSS\_VS (0x1D9, 0x1F9, 0x219, 0x239)

| BIT         | 7 | 6           | 5           | 4             | 3 | 2 | 1 | 0 |
|-------------|---|-------------|-------------|---------------|---|---|---|---|
| Field       | – | CROSS_VS_I  | CROSS_VS_F  | CROSS_VS[4:0] |   |   |   |   |
| Reset       | – | 0x0         | 0x0         | 0x19          |   |   |   |   |
| Access Type | – | Write, Read | Write, Read | Write, Read   |   |   |   |   |

| BITFIELD   | BITS | DESCRIPTION                           | DECODE                                                  |
|------------|------|---------------------------------------|---------------------------------------------------------|
| CROSS_VS_I | 6    | Inverts CROSS_VS                      | 0b0: Do not invert bit<br>0b1: Invert bit               |
| CROSS_VS_F | 5    | Forces CROSS_VS to 0 before inversion | 0b0: Do not force bit to zero<br>0b1: Force bit to zero |
| CROSS_VS   | 4:0  | Maps selected internal signal to VS   | 0bXXXXX: Incoming bit position                          |

CROSS\_DE (0x1DA, 0x1FA, 0x21A, 0x23A)

| BIT         | 7 | 6           | 5           | 4             | 3 | 2 | 1 | 0 |
|-------------|---|-------------|-------------|---------------|---|---|---|---|
| Field       | – | CROSS_DE_I  | CROSS_DE_F  | CROSS_DE[4:0] |   |   |   |   |
| Reset       | – | 0x0         | 0x0         | 0x1A          |   |   |   |   |
| Access Type | – | Write, Read | Write, Read | Write, Read   |   |   |   |   |

| BITFIELD   | BITS | DESCRIPTION                           | DECODE                                                  |
|------------|------|---------------------------------------|---------------------------------------------------------|
| CROSS_DE_I | 6    | Inverts CROSS_DE                      | 0b0: Do not invert bit<br>0b1: Invert bit               |
| CROSS_DE_F | 5    | Forces Cross_DE to 0 before inversion | 0b0: Do not force bit to zero<br>0b1: Force bit to zero |
| CROSS_DE   | 4:0  | Maps selected internal signal to DE   | 0bXXXXX: Incoming bit position                          |

PRBS\_ERR (0x1DB, 0x1FB, 0x21B, 0x23B)

| BIT         | 7               | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|-----------------|---|---|---|---|---|---|---|
| Field       | VPRBS_ERR[7:0]  |   |   |   |   |   |   |   |
| Reset       | 0x00            |   |   |   |   |   |   |   |
| Access Type | Read Clears All |   |   |   |   |   |   |   |

| BITFIELD  | BITS | DESCRIPTION                               | DECODE                                            |
|-----------|------|-------------------------------------------|---------------------------------------------------|
| VPRBS_ERR | 7:0  | Video PRBS error counter. Clears on read. | 0xXX: Number of video PRBS errors since last read |

VPRBS (0x1DC, 0x1FC, 0x21C, 0x23C)

| BIT         | 7 | 6 | 5          | 4            | 3 | 2 | 1 | 0          |
|-------------|---|---|------------|--------------|---|---|---|------------|
| Field       | – | – | VPRBS_FAIL | VPRBS_CHK_EN | – | – | – | VIDEO_LOCK |
| Reset       | – | – | 0b0        | 0b0          | – | – | – | 0b0        |
| Access Type | – | – | Read Only  | Write, Read  | – | – | – | Read Only  |

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| BITFIELD     | BITS | DESCRIPTION                                             | DECODE                                                              |
|--------------|------|---------------------------------------------------------|---------------------------------------------------------------------|
| VPRBS_FAIL   | 5    | Video PRBS check pass/fail                              | 0b0: Video PRBS check passed<br>0b1: Video check failed             |
| VPRBS_CHK_EN | 4    | Enables video PRBS checker                              | 0b0: Video PRBS checker disabled<br>0b1: Video PRBS checker enabled |
| VIDEO_LOCK   | 0    | Video channel is locked and outputting valid video data | 0b0: Video channel not locked<br>0b1: Video channel locked          |

**CROSS 27 (0x1DD, 0x1FD, 0x21D, 0x23D)**

| BIT         | 7            | 6           | 5           | 4            | 3 | 2 | 1 | 0 |
|-------------|--------------|-------------|-------------|--------------|---|---|---|---|
| Field       | ALT_CROSSBAR | CROSS27_I   | CROSS27_F   | CROSS27[4:0] |   |   |   |   |
| Reset       | 0b0          | 0b0         | 0b0         | 0b11011      |   |   |   |   |
| Access Type | Write, Read  | Write, Read | Write, Read | Write, Read  |   |   |   |   |

| BITFIELD     | BITS | DESCRIPTION                                                                                 | DECODE                                                     |
|--------------|------|---------------------------------------------------------------------------------------------|------------------------------------------------------------|
| ALT_CROSSBAR | 7    | Selects whether to use the crossbar in the VRX block or the alternative crossbar in the RDP | 0x0: Use crossbar in VRX block<br>0x1: Use crossbar in RDP |
| CROSS27_I    | 6    | Inverts CrossX                                                                              | 0b0: Do not invert bit<br>0b1: Invert bit                  |
| CROSS27_F    | 5    | Forces CrossX to 0 before inversion                                                         | 0b0: Do not force bit to zero<br>0b1: Force bit to zero    |
| CROSS27      | 4:0  | Maps selected internal signal to CrossX                                                     | 0bXXXXX: Incoming bit position                             |

**CROSS 28 (0x1DE, 0x1FE, 0x21E, 0x23E)**

| BIT         | 7 | 6           | 5           | 4            | 3 | 2 | 1 | 0 |
|-------------|---|-------------|-------------|--------------|---|---|---|---|
| Field       | – | CROSS28_I   | CROSS28_F   | CROSS28[4:0] |   |   |   |   |
| Reset       | – | 0b0         | 0b0         | 0b11100      |   |   |   |   |
| Access Type | – | Write, Read | Write, Read | Write, Read  |   |   |   |   |

| BITFIELD  | BITS | DESCRIPTION                             | DECODE                                                  |
|-----------|------|-----------------------------------------|---------------------------------------------------------|
| CROSS28_I | 6    | Inverts CrossX                          | 0b0: Do not invert bit<br>0b1: Invert bit               |
| CROSS28_F | 5    | Forces CrossX to 0 before inversion     | 0b0: Do not force bit to zero<br>0b1: Force bit to zero |
| CROSS28   | 4:0  | Maps selected internal signal to CrossX | 0bXXXXX: Incoming bit position                          |

**CROSS 29 (0x1DF, 0x1FF, 0x21F, 0x23F)**

| BIT         | 7 | 6           | 5           | 4            | 3 | 2 | 1 | 0 |
|-------------|---|-------------|-------------|--------------|---|---|---|---|
| Field       | – | CROSS29_I   | CROSS29_F   | CROSS29[4:0] |   |   |   |   |
| Reset       | – | 0b0         | 0b0         | 0b11101      |   |   |   |   |
| Access Type | – | Write, Read | Write, Read | Write, Read  |   |   |   |   |



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| BITFIELD  | BITS | DESCRIPTION                             | DECODE                                                  |
|-----------|------|-----------------------------------------|---------------------------------------------------------|
| CROSS29_I | 6    | Inverts CrossX                          | 0b0: Do not invert bit<br>0b1: Invert bit               |
| CROSS29_F | 5    | Forces CrossX to 0 before inversion     | 0b0: Do not force bit to zero<br>0b1: Force bit to zero |
| CROSS29   | 4:0  | Maps selected internal signal to CrossX | 0bXXXXX: Incoming bit position                          |

**GPIO\_A (0x2B0)**

| BIT         | 7           | 6           | 5           | 4           | 3         | 2           | 1           | 0            |
|-------------|-------------|-------------|-------------|-------------|-----------|-------------|-------------|--------------|
| Field       | RES_CFG     | TX_PRIO     | TX_COMP_EN  | GPIO_OUT    | GPIO_IN   | GPIO_RX_EN  | GPIO_TX_EN  | GPIO_OUT_DIS |
| Reset       | 0b1         | 0x0         | 0x0         | 0b0         | 0b0       | 0b0         | 0b1         | 0b1          |
| Access Type | Write, Read | Write, Read | Write, Read | Write, Read | Read Only | Write, Read | Write, Read | Write, Read  |

| BITFIELD     | BITS | DESCRIPTION                                     | DECODE                                                                                                        |
|--------------|------|-------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| RES_CFG      | 7    | Resistor pullup/pulldown strength               | 0b0: 40kΩ<br>0b1: 1MΩ                                                                                         |
| TX_PRIO      | 6    | Priority for GPIO scheduling                    | 0b0: Low priority<br>0b1: High priority                                                                       |
| TX_COMP_EN   | 5    | Jitter minimization compensation enable         | 0b0: Jitter compensation disabled<br>0b1: Jitter compensation enabled                                         |
| GPIO_OUT     | 4    | GPIO pin output drive value when GPIO_RX_EN = 0 | 0b0: This GPIO pin output is driven to 0<br>0b1: This GPIO pin output is driven to 1                          |
| GPIO_IN      | 3    | GPIO pin input level                            | 0b0: This GPIO pin value is 0<br>0b1: This GPIO pin value is 1                                                |
| GPIO_RX_EN   | 2    | GPIO out source control                         | 0b0: This GPIO source disabled for GMSL2 reception<br>0b1: This GPIO source enabled for GMSL2 reception       |
| GPIO_TX_EN   | 1    | GPIO Tx source control                          | 0b0: This GPIO source disabled for GMSL2 transmission<br>0b1: This GPIO source enabled for GMSL2 transmission |
| GPIO_OUT_DIS | 0    | Disables GPIO output driver                     | 0b0: Output driver enabled<br>0b1: Output driver disabled                                                     |

**GPIO\_B (0x2B1)**

| BIT         | 7                  | 6 | 5           | 4               | 3 | 2 | 1 | 0 |
|-------------|--------------------|---|-------------|-----------------|---|---|---|---|
| Field       | PULL_UPDN_SEL[1:0] |   | OUT_TYPE    | GPIO_TX_ID[4:0] |   |   |   |   |
| Reset       | 0x2                |   | 0b1         | 0x00            |   |   |   |   |
| Access Type | Write, Read        |   | Write, Read | Write, Read     |   |   |   |   |

| BITFIELD      | BITS | DESCRIPTION                          | DECODE                                                         |
|---------------|------|--------------------------------------|----------------------------------------------------------------|
| PULL_UPDN_SEL | 7:6  | Buffer pullup/pulldown configuration | 0b00: None<br>0b01: Pullup<br>0b10: Pulldown<br>0b11: Reserved |

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| BITFIELD   | BITS | DESCRIPTION                        | DECODE                            |
|------------|------|------------------------------------|-----------------------------------|
| OUT_TYPE   | 5    | Driver type selection              | 0b0: Open-drain<br>0b1: Push-pull |
| GPIO_TX_ID | 4:0  | GPIO ID for pin while transmitting | 0bXXXXX: This GPIO transmit ID    |

**GPIO\_C (0x2B2)**

| BIT         | 7           | 6            | 5 | 4               | 3 | 2 | 1 | 0 |
|-------------|-------------|--------------|---|-----------------|---|---|---|---|
| Field       | OVR_RES_CFG | GPIO_REC_VED | – | GPIO_RX_ID[4:0] |   |   |   |   |
| Reset       | 0x0         | 0b1          | – | 0x00            |   |   |   |   |
| Access Type | Write, Read | Write, Read  | – | Write, Read     |   |   |   |   |

| BITFIELD    | BITS | DESCRIPTION                                                                                                                                                                                | DECODE                                                                                                                                                       |
|-------------|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| OVR_RES_CFG | 7    | Overrides non-GPIO port function IO setting. When set, RES_CFG and PULL_UPDN_SEL are effective when pin is configured as non-GPIO. When cleared, non-GPIO pin function determines IO type. | 0b0: Non-GPIO function determines IO type when alternate function is selected<br>0b1: RES_CFG and PULL_UPDN_SEL determine IO type for non-GPIO configuration |
| GPIO_RECVED | 6    | Received GPIO value                                                                                                                                                                        |                                                                                                                                                              |
| GPIO_RX_ID  | 4:0  | GPIO ID for pin while receiving                                                                                                                                                            | 0bXXXXX: This GPIO receive ID                                                                                                                                |

**GPIO\_A (0x2B3)**

| BIT         | 7           | 6           | 5           | 4           | 3         | 2           | 1           | 0            |
|-------------|-------------|-------------|-------------|-------------|-----------|-------------|-------------|--------------|
| Field       | RES_CFG     | TX_PRIO     | TX_COMP_EN  | GPIO_OUT    | GPIO_IN   | GPIO_RX_EN  | GPIO_TX_EN  | GPIO_OUT_DIS |
| Reset       | 0b1         | 0x0         | 0x0         | 0b0         | 0b0       | 0b1         | 0b0         | 0b0          |
| Access Type | Write, Read | Write, Read | Write, Read | Write, Read | Read Only | Write, Read | Write, Read | Write, Read  |

| BITFIELD   | BITS | DESCRIPTION                                     | DECODE                                                                                                        |
|------------|------|-------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| RES_CFG    | 7    | Resistor pullup/pulldown strength               | 0b0: 40kΩ<br>0b1: 1MΩ                                                                                         |
| TX_PRIO    | 6    | Priority for GPIO scheduling                    | 0b0: Low priority<br>0b1: High priority                                                                       |
| TX_COMP_EN | 5    | Jitter minimization compensation enable         | 0b0: Jitter compensation disabled<br>0b1: Jitter compensation enabled                                         |
| GPIO_OUT   | 4    | GPIO pin output drive value when GPIO_RX_EN = 0 | 0b0: This GPIO pin output is driven to 0<br>0b1: This GPIO pin output is driven to 1                          |
| GPIO_IN    | 3    | GPIO pin input level                            | 0b0: This GPIO pin value is 0<br>0b1: This GPIO pin value is 1                                                |
| GPIO_RX_EN | 2    | GPIO out source control                         | 0b0: This GPIO source disabled for GMSL2 reception<br>0b1: This GPIO source enabled for GMSL2 reception       |
| GPIO_TX_EN | 1    | GPIO Tx source control                          | 0b0: This GPIO source disabled for GMSL2 transmission<br>0b1: This GPIO source enabled for GMSL2 transmission |

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| BITFIELD     | BITS | DESCRIPTION                 | DECODE                                                    |
|--------------|------|-----------------------------|-----------------------------------------------------------|
| GPIO_OUT_DIS | 0    | Disables GPIO output driver | 0b0: Output driver enabled<br>0b1: Output driver disabled |

**GPIO\_B (0x2B4)**

| BIT         | 7                  | 6 | 5           | 4               | 3 | 2 | 1 | 0 |
|-------------|--------------------|---|-------------|-----------------|---|---|---|---|
| Field       | PULL_UPDN_SEL[1:0] |   | OUT_TYPE    | GPIO_TX_ID[4:0] |   |   |   |   |
| Reset       | 0x2                |   | 0b1         | 0x01            |   |   |   |   |
| Access Type | Write, Read        |   | Write, Read | Write, Read     |   |   |   |   |

| BITFIELD      | BITS | DESCRIPTION                          | DECODE                                                         |
|---------------|------|--------------------------------------|----------------------------------------------------------------|
| PULL_UPDN_SEL | 7:6  | Buffer pullup/pulldown configuration | 0b00: None<br>0b01: Pullup<br>0b10: Pulldown<br>0b11: Reserved |
| OUT_TYPE      | 5    | Driver type selection                | 0b0: Open-drain<br>0b1: Push-pull                              |
| GPIO_TX_ID    | 4:0  | GPIO ID for pin while transmitting   | 0bXXXXX: This GPIO transmit ID                                 |

**GPIO\_C (0x2B5)**

| BIT         | 7           | 6 | 5 | 4               | 3 | 2 | 1 | 0 |
|-------------|-------------|---|---|-----------------|---|---|---|---|
| Field       | OVR_RES_CFG | – | – | GPIO_RX_ID[4:0] |   |   |   |   |
| Reset       | 0x0         | – | – | 0x01            |   |   |   |   |
| Access Type | Write, Read | – | – | Write, Read     |   |   |   |   |

| BITFIELD    | BITS | DESCRIPTION                                                                                                                                                                                | DECODE                                                                                                                                                       |
|-------------|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| OVR_RES_CFG | 7    | Overrides non-GPIO port function IO setting. When set, res_cfg and pull_updn_sel are effective when pin is configured as non GPIO. When cleared, non-GPIO pin function determines IO type. | 0b0: Non-GPIO function determines IO type when alternate function is selected<br>0b1: RES_CFG and PULL_UPDN_SEL determine IO type for non-GPIO configuration |
| GPIO_RX_ID  | 4:0  | GPIO ID for pin while receiving                                                                                                                                                            | 0bXXXXX: This GPIO receive ID                                                                                                                                |

**GPIO\_A (0x2B6)**

| BIT         | 7           | 6           | 5           | 4           | 3         | 2           | 1           | 0            |
|-------------|-------------|-------------|-------------|-------------|-----------|-------------|-------------|--------------|
| Field       | RES_CFG     | TX_PRIO     | TX_COMP_EN  | GPIO_OUT    | GPIO_IN   | GPIO_RX_EN  | GPIO_TX_EN  | GPIO_OUT_DIS |
| Reset       | 0b1         | 0x0         | 0x0         | 0b0         | 0b0       | 0b0         | 0b0         | 0b1          |
| Access Type | Write, Read | Write, Read | Write, Read | Write, Read | Read Only | Write, Read | Write, Read | Write, Read  |

| BITFIELD | BITS | DESCRIPTION                       | DECODE                                  |
|----------|------|-----------------------------------|-----------------------------------------|
| RES_CFG  | 7    | Resistor pullup/pulldown strength | 0b0: 40kΩ<br>0b1: 1MΩ                   |
| TX_PRIO  | 6    | Priority for GPIO scheduling      | 0b0: Low priority<br>0b1: High priority |

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| BITFIELD     | BITS | DESCRIPTION                                     | DECODE                                                                                                        |
|--------------|------|-------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| TX_COMP_EN   | 5    | Jitter minimization compensation enable         | 0b0: Jitter compensation disabled<br>0b1: Jitter compensation enabled                                         |
| GPIO_OUT     | 4    | GPIO pin output drive value when GPIO_RX_EN = 0 | 0b0: This GPIO pin output is driven to 0<br>0b1: This GPIO pin output is driven to 1                          |
| GPIO_IN      | 3    | GPIO pin input level                            | 0b0: This GPIO pin value is 0<br>0b1: This GPIO pin value is 1                                                |
| GPIO_RX_EN   | 2    | GPIO out source control                         | 0b0: This GPIO source disabled for GMSL2 reception<br>0b1: This GPIO source enabled for GMSL2 reception       |
| GPIO_TX_EN   | 1    | GPIO Tx source control                          | 0b0: This GPIO source disabled for GMSL2 transmission<br>0b1: This GPIO source enabled for GMSL2 transmission |
| GPIO_OUT_DIS | 0    | Disables GPIO output driver                     | 0b0: Output driver enabled<br>0b1: Output driver disabled                                                     |

**GPIO\_B (0x2B7)**

| BIT         | 7                  | 6 | 5           | 4               | 3 | 2 | 1 | 0 |
|-------------|--------------------|---|-------------|-----------------|---|---|---|---|
| Field       | PULL_UPDN_SEL[1:0] |   | OUT_TYPE    | GPIO_TX_ID[4:0] |   |   |   |   |
| Reset       | 0x0                |   | 0b1         | 0x02            |   |   |   |   |
| Access Type | Write, Read        |   | Write, Read | Write, Read     |   |   |   |   |

| BITFIELD      | BITS | DESCRIPTION                          | DECODE                                                         |
|---------------|------|--------------------------------------|----------------------------------------------------------------|
| PULL_UPDN_SEL | 7:6  | Buffer pullup/pulldown configuration | 0b00: None<br>0b01: Pullup<br>0b10: Pulldown<br>0b11: Reserved |
| OUT_TYPE      | 5    | Driver type selection                | 0b0: Open-drain<br>0b1: Push-pull                              |
| GPIO_TX_ID    | 4:0  | GPIO ID for pin while transmitting   | 0bXXXXX: This GPIO transmit ID                                 |

**GPIO\_C (0x2B8)**

| BIT         | 7           | 6 | 5 | 4               | 3 | 2 | 1 | 0 |
|-------------|-------------|---|---|-----------------|---|---|---|---|
| Field       | OVR_RES_CFG | — | — | GPIO_RX_ID[4:0] |   |   |   |   |
| Reset       | 0x0         | — | — | 0x02            |   |   |   |   |
| Access Type | Write, Read | — | — | Write, Read     |   |   |   |   |

| BITFIELD    | BITS | DESCRIPTION                                                                                                                                                                                | DECODE                                                                                                                                                       |
|-------------|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| OVR_RES_CFG | 7    | Overrides non-GPIO port function IO setting. When set, RES_CFG and PULL_UPDN_SEL are effective when pin is configured as non GPIO. When cleared, non-GPIO pin function determines IO type. | 0b0: Non-GPIO function determines IO type when alternate function is selected<br>0b1: RES_CFG and PULL_UPDN_SEL determine IO type for non-GPIO configuration |
| GPIO_RX_ID  | 4:0  | GPIO ID for pin while receiving                                                                                                                                                            | 0bXXXXX: This GPIO receive ID                                                                                                                                |

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GPIO\_A (0x2B9)

| BIT         | 7           | 6           | 5 | 4           | 3         | 2           | 1           | 0            |
|-------------|-------------|-------------|---|-------------|-----------|-------------|-------------|--------------|
| Field       | RES_CFG     | TX_PRIO     | – | GPIO_OUT    | GPIO_IN   | GPIO_RX_EN  | GPIO_TX_EN  | GPIO_OUT_DIS |
| Reset       | 0b1         | 0x0         | – | 0b0         | 0b0       | 0b0         | 0b0         | 0b1          |
| Access Type | Write, Read | Write, Read | – | Write, Read | Read Only | Write, Read | Write, Read | Write, Read  |

| BITFIELD     | BITS | DESCRIPTION                                     | DECODE                                                                                                        |
|--------------|------|-------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| RES_CFG      | 7    | Resistor pullup/pulldown strength               | 0b0: 40kΩ<br>0b1: 1MΩ                                                                                         |
| TX_PRIO      | 6    | Priority for GPIO scheduling                    | 0b0: Low priority<br>0b1: High priority                                                                       |
| GPIO_OUT     | 4    | GPIO pin output drive value when GPIO_RX_EN = 0 | 0b0: This GPIO pin output is driven to 0<br>0b1: This GPIO pin output is driven to 1                          |
| GPIO_IN      | 3    | GPIO pin input level                            | 0b0: This GPIO pin value is 0<br>0b1: This GPIO pin value is 1                                                |
| GPIO_RX_EN   | 2    | GPIO out source control                         | 0b0: This GPIO source disabled for GMSL2 reception<br>0b1: This GPIO source enabled for GMSL2 reception       |
| GPIO_TX_EN   | 1    | GPIO Tx source control                          | 0b0: This GPIO source disabled for GMSL2 transmission<br>0b1: This GPIO source enabled for GMSL2 transmission |
| GPIO_OUT_DIS | 0    | Disables GPIO output driver                     | 0b0: Output driver enabled<br>0b1: Output driver disabled                                                     |

GPIO\_B (0x2BA)

| BIT         | 7                  | 6 | 5           | 4               | 3 | 2 | 1 | 0 |
|-------------|--------------------|---|-------------|-----------------|---|---|---|---|
| Field       | PULL_UPDN_SEL[1:0] |   | OUT_TYPE    | GPIO_TX_ID[4:0] |   |   |   |   |
| Reset       | 0x0                |   | 0b1         | 0x03            |   |   |   |   |
| Access Type | Write, Read        |   | Write, Read | Write, Read     |   |   |   |   |

| BITFIELD      | BITS | DESCRIPTION                          | DECODE                                                         |
|---------------|------|--------------------------------------|----------------------------------------------------------------|
| PULL_UPDN_SEL | 7:6  | Buffer pullup/pulldown configuration | 0b00: None<br>0b01: Pullup<br>0b10: Pulldown<br>0b11: Reserved |
| OUT_TYPE      | 5    | Driver type selection                | 0b0: Open-drain<br>0b1: Push-pull                              |
| GPIO_TX_ID    | 4:0  | GPIO ID for pin while transmitting   | 0bXXXXX: This GPIO transmit ID                                 |

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**GPIO\_C (0x2BB)**

| BIT         | 7           | 6 | 5 | 4               | 3 | 2 | 1 | 0 |
|-------------|-------------|---|---|-----------------|---|---|---|---|
| Field       | OVR_RES_CFG | – | – | GPIO_RX_ID[4:0] |   |   |   |   |
| Reset       | 0x0         | – | – | 0x03            |   |   |   |   |
| Access Type | Write, Read | – | – | Write, Read     |   |   |   |   |

| BITFIELD    | BITS | DESCRIPTION                                                                                                                                                                               | DECODE                                                                                                                                                       |
|-------------|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| OVR_RES_CFG | 7    | Override non-GPIO port function IO setting. When set, RES_CFG and PULL_UPDN_SEL are effective when pin is configured as non GPIO. When cleared, non-GPIO pin function determines IO type. | 0b0: Non-GPIO function determines IO type when alternate function is selected<br>0b1: RES_CFG and PULL_UPDN_SEL determine IO type for non-GPIO configuration |
| GPIO_RX_ID  | 4:0  | GPIO ID for pin while receiving                                                                                                                                                           | 0bXXXXX: This GPIO receive ID                                                                                                                                |

**GPIO\_A (0x2BC)**

| BIT         | 7           | 6           | 5           | 4           | 3         | 2           | 1           | 0            |
|-------------|-------------|-------------|-------------|-------------|-----------|-------------|-------------|--------------|
| Field       | RES_CFG     | TX_PRIO     | TX_COMP_EN  | GPIO_OUT    | GPIO_IN   | GPIO_RX_EN  | GPIO_TX_EN  | GPIO_OUT_DIS |
| Reset       | 0b1         | 0x0         | 0x0         | 0b0         | 0b0       | 0b0         | 0b0         | 0b1          |
| Access Type | Write, Read | Write, Read | Write, Read | Write, Read | Read Only | Write, Read | Write, Read | Write, Read  |

| BITFIELD     | BITS | DESCRIPTION                                     | DECODE                                                                                                        |
|--------------|------|-------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| RES_CFG      | 7    | Resistor pullup/pulldown strength               | 0b0: 40kΩ<br>0b1: 1MΩ                                                                                         |
| TX_PRIO      | 6    | Priority for GPIO scheduling                    | 0b0: Low priority<br>0b1: High priority                                                                       |
| TX_COMP_EN   | 5    | Jitter minimization compensation enable         | 0b0: Jitter compensation disabled<br>0b1: Jitter compensation enabled                                         |
| GPIO_OUT     | 4    | GPIO pin output drive value when GPIO_RX_EN = 0 | 0b0: This GPIO pin output is driven to 0<br>0b1: This GPIO pin output is driven to 1                          |
| GPIO_IN      | 3    | GPIO pin input level                            | 0b0: This GPIO pin value is 0<br>0b1: This GPIO pin value is 1                                                |
| GPIO_RX_EN   | 2    | GPIO out source control                         | 0b0: This GPIO source disabled for GMSL2 reception<br>0b1: This GPIO source enabled for GMSL2 reception       |
| GPIO_TX_EN   | 1    | GPIO Tx source control                          | 0b0: This GPIO source disabled for GMSL2 transmission<br>0b1: This GPIO source enabled for GMSL2 transmission |
| GPIO_OUT_DIS | 0    | Disables GPIO output driver                     | 0b0: Output driver enabled<br>0b1: Output driver disabled                                                     |

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**GPIO\_B (0x2BD)**

| BIT         | 7                  | 6 | 5           | 4 | 3 | 2 | 1 | 0 |
|-------------|--------------------|---|-------------|---|---|---|---|---|
| Field       | PULL_UPDN_SEL[1:0] |   | OUT_TYPE    | – | – | – | – | – |
| Reset       | 0x2                |   | 0b1         | – | – | – | – | – |
| Access Type | Write, Read        |   | Write, Read | – | – | – | – | – |

| BITFIELD      | BITS | DESCRIPTION                          | DECODE                                                         |
|---------------|------|--------------------------------------|----------------------------------------------------------------|
| PULL_UPDN_SEL | 7:6  | Buffer pullup/pulldown configuration | 0b00: None<br>0b01: Pullup<br>0b10: Pulldown<br>0b11: Reserved |
| OUT_TYPE      | 5    | Driver type selection                | 0b0: Open-drain<br>0b1: Push-pull                              |

**GPIO\_C (0x2BE)**

| BIT         | 7           | 6 | 5 | 4               | 3 | 2 | 1 | 0 |
|-------------|-------------|---|---|-----------------|---|---|---|---|
| Field       | OVR_RES_CFG | – | – | GPIO_RX_ID[4:0] |   |   |   |   |
| Reset       | 0x0         | – | – | 0x04            |   |   |   |   |
| Access Type | Write, Read | – | – | Write, Read     |   |   |   |   |

| BITFIELD    | BITS | DESCRIPTION                                                                                                                                                                                | DECODE                                                                                                                                                       |
|-------------|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| OVR_RES_CFG | 7    | Overrides non-GPIO port function IO setting. When set, RES_CFG and PULL_UPDN_SEL are effective when pin is configured as non GPIO. When cleared, non-GPIO pin function determines IO type. | 0b0: Non-GPIO function determines IO type when alternate function is selected<br>0b1: RES_CFG and PULL_UPDN_SEL determine IO type for non-GPIO configuration |
| GPIO_RX_ID  | 4:0  | GPIO ID for pin while receiving                                                                                                                                                            | 0bXXXXX: This GPIO receive ID                                                                                                                                |

**GPIO\_A (0x2BF)**

| BIT         | 7           | 6           | 5           | 4           | 3         | 2           | 1           | 0            |
|-------------|-------------|-------------|-------------|-------------|-----------|-------------|-------------|--------------|
| Field       | RES_CFG     | TX_PRIO     | TX_COMP_EN  | GPIO_OUT    | GPIO_IN   | GPIO_RX_EN  | GPIO_TX_EN  | GPIO_OUT_DIS |
| Reset       | 0b1         | 0x0         | 0x0         | 0b0         | 0b0       | 0b1         | 0b0         | 0b0          |
| Access Type | Write, Read | Write, Read | Write, Read | Write, Read | Read Only | Write, Read | Write, Read | Write, Read  |

| BITFIELD   | BITS | DESCRIPTION                                     | DECODE                                                                               |
|------------|------|-------------------------------------------------|--------------------------------------------------------------------------------------|
| RES_CFG    | 7    | Resistor pullup/pulldown strength               | 0b0: 40kΩ<br>0b1: 1MΩ                                                                |
| TX_PRIO    | 6    | Priority for GPIO scheduling                    | 0b0: Low priority<br>0b1: High priority                                              |
| TX_COMP_EN | 5    | Jitter minimization compensation enable         | 0b0: Jitter compensation disabled<br>0b1: Jitter compensation enabled                |
| GPIO_OUT   | 4    | GPIO pin output drive value when GPIO_RX_EN = 0 | 0b0: This GPIO pin output is driven to 0<br>0b1: This GPIO pin output is driven to 1 |
| GPIO_IN    | 3    | GPIO pin input level                            | 0b0: This GPIO pin value is 0<br>0b1: This GPIO pin value is 1                       |



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| BITFIELD     | BITS | DESCRIPTION                 | DECODE                                                                                                        |
|--------------|------|-----------------------------|---------------------------------------------------------------------------------------------------------------|
| GPIO_RX_EN   | 2    | GPIO out source control     | 0b0: This GPIO source disabled for GMSL2 reception<br>0b1: This GPIO source enabled for GMSL2 reception       |
| GPIO_TX_EN   | 1    | GPIO Tx source control      | 0b0: This GPIO source disabled for GMSL2 transmission<br>0b1: This GPIO source enabled for GMSL2 transmission |
| GPIO_OUT_DIS | 0    | Disables GPIO output driver | 0b0: Output driver enabled<br>0b1: Output driver disabled                                                     |

**GPIO\_B (0x2C0)**

| BIT         | 7                  | 6 | 5           | 4               | 3 | 2 | 1 | 0 |
|-------------|--------------------|---|-------------|-----------------|---|---|---|---|
| Field       | PULL_UPDN_SEL[1:0] |   | OUT_TYPE    | GPIO_TX_ID[4:0] |   |   |   |   |
| Reset       | 0x2                |   | 0b1         | 0x05            |   |   |   |   |
| Access Type | Write, Read        |   | Write, Read | Write, Read     |   |   |   |   |

| BITFIELD      | BITS | DESCRIPTION                          | DECODE                                                         |
|---------------|------|--------------------------------------|----------------------------------------------------------------|
| PULL_UPDN_SEL | 7:6  | Buffer pullup/pulldown configuration | 0b00: None<br>0b01: Pullup<br>0b10: Pulldown<br>0b11: Reserved |
| OUT_TYPE      | 5    | Driver type selection                | 0b0: Open-drain<br>0b1: Push-pull                              |
| GPIO_TX_ID    | 4:0  | GPIO ID for pin while transmitting   | 0bXXXXX: This GPIO transmit ID                                 |

**GPIO\_C (0x2C1)**

| BIT         | 7           | 6 | 5 | 4               | 3 | 2 | 1 | 0 |
|-------------|-------------|---|---|-----------------|---|---|---|---|
| Field       | OVR_RES_CFG | – | – | GPIO_RX_ID[4:0] |   |   |   |   |
| Reset       | 0x0         | – | – | 0x05            |   |   |   |   |
| Access Type | Write, Read | – | – | Write, Read     |   |   |   |   |

| BITFIELD    | BITS | DESCRIPTION                                                                                                                                                                                | DECODE                                                                                                                                                       |
|-------------|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| OVR_RES_CFG | 7    | Overrides non-GPIO port function IO setting. When set, res_cfg and pull_updn_sel are effective when pin is configured as non GPIO. When cleared, non-GPIO pin function determines IO type. | 0b0: Non-GPIO function determines IO type when alternate function is selected<br>0b1: RES_CFG and PULL_UPDN_SEL determine IO type for non-GPIO configuration |
| GPIO_RX_ID  | 4:0  | GPIO ID for pin while receiving                                                                                                                                                            | 0bXXXXX: This GPIO receive ID                                                                                                                                |

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GPIO\_A (0x2C2)

| BIT         | 7           | 6           | 5           | 4           | 3         | 2           | 1           | 0            |
|-------------|-------------|-------------|-------------|-------------|-----------|-------------|-------------|--------------|
| Field       | RES_CFG     | TX_PRIO     | TX_COMP_EN  | GPIO_OUT    | GPIO_IN   | GPIO_RX_EN  | GPIO_TX_EN  | GPIO_OUT_DIS |
| Reset       | 0b1         | 0x0         | 0x0         | 0b0         | 0b0       | 0b0         | 0b1         | 0b1          |
| Access Type | Write, Read | Write, Read | Write, Read | Write, Read | Read Only | Write, Read | Write, Read | Write, Read  |

| BITFIELD     | BITS | DESCRIPTION                                     | DECODE                                                                                                        |
|--------------|------|-------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| RES_CFG      | 7    | Resistor pullup/pulldown strength               | 0b0: 40kΩ<br>0b1: 1MΩ                                                                                         |
| TX_PRIO      | 6    | Priority for GPIO scheduling                    | 0b0: Low priority<br>0b1: High priority                                                                       |
| TX_COMP_EN   | 5    | Jitter minimization compensation enable         | 0b0: Jitter compensation disabled<br>0b1: Jitter compensation enabled                                         |
| GPIO_OUT     | 4    | GPIO pin output drive value when GPIO_RX_EN = 0 | 0b0: This GPIO pin output is driven to 0<br>0b1: This GPIO pin output is driven to 1                          |
| GPIO_IN      | 3    | GPIO pin input level                            | 0b0: This GPIO pin value is 0<br>0b1: This GPIO pin value is 1                                                |
| GPIO_RX_EN   | 2    | GPIO out source control                         | 0b0: This GPIO source disabled for GMSL2 reception<br>0b1: This GPIO source enabled for GMSL2 reception       |
| GPIO_TX_EN   | 1    | GPIO Tx source control                          | 0b0: This GPIO source disabled for GMSL2 transmission<br>0b1: This GPIO source enabled for GMSL2 transmission |
| GPIO_OUT_DIS | 0    | Disables GPIO output driver                     | 0b0: Output driver enabled<br>0b1: Output driver disabled                                                     |

GPIO\_B (0x2C3)

| BIT         | 7                  | 6 | 5           | 4               | 3 | 2 | 1 | 0 |
|-------------|--------------------|---|-------------|-----------------|---|---|---|---|
| Field       | PULL_UPDN_SEL[1:0] |   | OUT_TYPE    | GPIO_TX_ID[4:0] |   |   |   |   |
| Reset       | 0x2                |   | 0b1         | 0x06            |   |   |   |   |
| Access Type | Write, Read        |   | Write, Read | Write, Read     |   |   |   |   |

| BITFIELD      | BITS | DESCRIPTION                          | DECODE                                                         |
|---------------|------|--------------------------------------|----------------------------------------------------------------|
| PULL_UPDN_SEL | 7:6  | Buffer pullup/pulldown configuration | 0b00: None<br>0b01: Pullup<br>0b10: Pulldown<br>0b11: Reserved |
| OUT_TYPE      | 5    | Driver type selection                | 0b0: Open-drain<br>0b1: Push-pull                              |
| GPIO_TX_ID    | 4:0  | GPIO ID for pin while transmitting   | 0bXXXXX: This GPIO transmit ID                                 |

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**GPIO\_C (0x2C4)**

| BIT         | 7           | 6 | 5 | 4               | 3 | 2 | 1 | 0 |
|-------------|-------------|---|---|-----------------|---|---|---|---|
| Field       | OVR_RES_CFG | – | – | GPIO_RX_ID[4:0] |   |   |   |   |
| Reset       | 0x0         | – | – | 0x06            |   |   |   |   |
| Access Type | Write, Read | – | – | Write, Read     |   |   |   |   |

| BITFIELD    | BITS | DESCRIPTION                                                                                                                                                                                | DECODE                                                                                                                                                       |
|-------------|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| OVR_RES_CFG | 7    | Overrides non-GPIO port function IO setting. When set, RES_CFG and PULL_UPDN_SEL are effective when pin is configured as non GPIO. When cleared, non-GPIO pin function determines IO type. | 0b0: Non GPIO function determines IO type when alternate function is selected<br>0b1: RES_CFG and PULL_UPDN_SEL determine IO type for non GPIO configuration |
| GPIO_RX_ID  | 4:0  | GPIO ID for pin while receiving                                                                                                                                                            | 0bXXXXX: This GPIO receive ID                                                                                                                                |

**GPIO\_A (0x2C5)**

| BIT         | 7           | 6           | 5           | 4           | 3         | 2           | 1           | 0            |
|-------------|-------------|-------------|-------------|-------------|-----------|-------------|-------------|--------------|
| Field       | RES_CFG     | TX_PRIO     | TX_COMP_EN  | GPIO_OUT    | GPIO_IN   | GPIO_RX_EN  | GPIO_TX_EN  | GPIO_OUT_DIS |
| Reset       | 0b1         | 0x0         | 0x0         | 0b0         | 0b0       | 0b0         | 0b0         | 0b1          |
| Access Type | Write, Read | Write, Read | Write, Read | Write, Read | Read Only | Write, Read | Write, Read | Write, Read  |

| BITFIELD     | BITS | DESCRIPTION                                     | DECODE                                                                                                        |
|--------------|------|-------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| RES_CFG      | 7    | Resistor pullup/pulldown strength               | 0b0: 40kΩ<br>0b1: 1MΩ                                                                                         |
| TX_PRIO      | 6    | Priority for GPIO scheduling                    | 0b0: Low priority<br>0b1: High priority                                                                       |
| TX_COMP_EN   | 5    | Jitter minimization compensation enable         | 0b0: Jitter compensation disabled<br>0b1: Jitter compensation enabled                                         |
| GPIO_OUT     | 4    | GPIO pin output drive value when GPIO_RX_EN = 0 | 0b0: This GPIO pin output is driven to 0<br>0b1: This GPIO pin output is driven to 1                          |
| GPIO_IN      | 3    | GPIO pin input level                            | 0b0: This GPIO pin value is 0<br>0b1: This GPIO pin value is 1                                                |
| GPIO_RX_EN   | 2    | GPIO out source control                         | 0b0: This GPIO source disabled for GMSL2 reception<br>0b1: This GPIO source enabled for GMSL2 reception       |
| GPIO_TX_EN   | 1    | GPIO Tx source control                          | 0b0: This GPIO source disabled for GMSL2 transmission<br>0b1: This GPIO source enabled for GMSL2 transmission |
| GPIO_OUT_DIS | 0    | Disables GPIO output driver                     | 0b0: Output driver enabled<br>0b1: Output driver disabled                                                     |

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**GPIO\_B (0x2C6)**

| BIT         | 7                  | 6 | 5           | 4               | 3 | 2 | 1 | 0 |
|-------------|--------------------|---|-------------|-----------------|---|---|---|---|
| Field       | PULL_UPDN_SEL[1:0] |   | OUT_TYPE    | GPIO_TX_ID[4:0] |   |   |   |   |
| Reset       | 0x2                |   | 0b1         | 0x07            |   |   |   |   |
| Access Type | Write, Read        |   | Write, Read | Write, Read     |   |   |   |   |

| BITFIELD      | BITS | DESCRIPTION                          | DECODE                                                         |
|---------------|------|--------------------------------------|----------------------------------------------------------------|
| PULL_UPDN_SEL | 7:6  | Buffer pullup/pulldown configuration | 0b00: None<br>0b01: Pullup<br>0b10: Pulldown<br>0b11: Reserved |
| OUT_TYPE      | 5    | Driver type selection                | 0b0: Open-drain<br>0b1: Push-pull                              |
| GPIO_TX_ID    | 4:0  | GPIO ID for pin while transmitting   | 0bXXXXX: This GPIO transmit ID                                 |

**GPIO\_C (0x2C7)**

| BIT         | 7           | 6 | 5 | 4               | 3 | 2 | 1 | 0 |
|-------------|-------------|---|---|-----------------|---|---|---|---|
| Field       | OVR_RES_CFG | – | – | GPIO_RX_ID[4:0] |   |   |   |   |
| Reset       | 0x0         | – | – | 0x07            |   |   |   |   |
| Access Type | Write, Read | – | – | Write, Read     |   |   |   |   |

| BITFIELD    | BITS | DESCRIPTION                                                                                                                                                                               | DECODE                                                                                                                                                       |
|-------------|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| OVR_RES_CFG | 7    | Override non-GPIO port function IO setting. When set, RES_CFG and PULL_UPDN_SEL are effective when pin is configured as non-GPIO. When cleared, non-GPIO pin function determines IO type. | 0b0: Non-GPIO function determines IO type when alternate function is selected<br>0b1: RES_CFG and PULL_UPDN_SEL determine IO type for non-GPIO configuration |
| GPIO_RX_ID  | 4:0  | GPIO ID for pin while receiving                                                                                                                                                           | 0bXXXXX: This GPIO receive ID                                                                                                                                |

**GPIO\_A (0x2C8)**

| BIT         | 7           | 6           | 5           | 4           | 3         | 2           | 1           | 0            |
|-------------|-------------|-------------|-------------|-------------|-----------|-------------|-------------|--------------|
| Field       | RES_CFG     | TX_PRIO     | TX_COMP_EN  | GPIO_OUT    | GPIO_IN   | GPIO_RX_EN  | GPIO_TX_EN  | GPIO_OUT_DIS |
| Reset       | 0b1         | 0x0         | 0x0         | 0b0         | 0b0       | 0b0         | 0b0         | 0b1          |
| Access Type | Write, Read | Write, Read | Write, Read | Write, Read | Read Only | Write, Read | Write, Read | Write, Read  |

| BITFIELD   | BITS | DESCRIPTION                                     | DECODE                                                                               |
|------------|------|-------------------------------------------------|--------------------------------------------------------------------------------------|
| RES_CFG    | 7    | Resistor pullup/pulldown strength               | 0b0: 40kΩ<br>0b1: 1MΩ                                                                |
| TX_PRIO    | 6    | Priority for GPIO scheduling                    | 0b0: Low priority<br>0b1: High priority                                              |
| TX_COMP_EN | 5    | Jitter minimization compensation enable         | 0b0: Jitter compensation disabled<br>0b1: Jitter compensation enabled                |
| GPIO_OUT   | 4    | GPIO pin output drive value when GPIO_RX_EN = 0 | 0b0: This GPIO pin output is driven to 0<br>0b1: This GPIO pin output is driven to 1 |

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| BITFIELD     | BITS | DESCRIPTION                 | DECODE                                                                                                        |
|--------------|------|-----------------------------|---------------------------------------------------------------------------------------------------------------|
| GPIO_IN      | 3    | GPIO pin input level        | 0b0: This GPIO pin value is 0<br>0b1: This GPIO pin value is 1                                                |
| GPIO_RX_EN   | 2    | GPIO out source control     | 0b0: This GPIO source disabled for GMSL2 reception<br>0b1: This GPIO source enabled for GMSL2 reception       |
| GPIO_TX_EN   | 1    | GPIO Tx source control      | 0b0: This GPIO source disabled for GMSL2 transmission<br>0b1: This GPIO source enabled for GMSL2 transmission |
| GPIO_OUT_DIS | 0    | Disables GPIO output driver | 0b0: Output driver enabled<br>0b1: Output driver disabled                                                     |

**GPIO\_B (0x2C9)**

| BIT         | 7                  | 6 | 5           | 4               | 3 | 2 | 1 | 0 |
|-------------|--------------------|---|-------------|-----------------|---|---|---|---|
| Field       | PULL_UPDN_SEL[1:0] |   | OUT_TYPE    | GPIO_TX_ID[4:0] |   |   |   |   |
| Reset       | 0x2                |   | 0b1         | 0x08            |   |   |   |   |
| Access Type | Write, Read        |   | Write, Read | Write, Read     |   |   |   |   |

| BITFIELD      | BITS | DESCRIPTION                          | DECODE                                                         |
|---------------|------|--------------------------------------|----------------------------------------------------------------|
| PULL_UPDN_SEL | 7:6  | Buffer pullup/pulldown configuration | 0b00: None<br>0b01: Pullup<br>0b10: Pulldown<br>0b11: Reserved |
| OUT_TYPE      | 5    | Driver type selection                | 0b0: Open-drain<br>0b1: Push-pull                              |
| GPIO_TX_ID    | 4:0  | GPIO ID for pin while transmitting   | 0bXXXXX: This GPIO transmit ID                                 |

**GPIO\_C (0x2CA)**

| BIT         | 7           | 6 | 5 | 4               | 3 | 2 | 1 | 0 |
|-------------|-------------|---|---|-----------------|---|---|---|---|
| Field       | OVR_RES_CFG | – | – | GPIO_RX_ID[4:0] |   |   |   |   |
| Reset       | 0x0         | – | – | 0x08            |   |   |   |   |
| Access Type | Write, Read | – | – | Write, Read     |   |   |   |   |

| BITFIELD    | BITS | DESCRIPTION                                                                                                                                                                                | DECODE                                                                                                                                                       |
|-------------|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| OVR_RES_CFG | 7    | Overrides non-GPIO port function IO setting. When set, RES_CFG and PULL_UPDN_SEL are effective when pin is configured as non GPIO. When cleared, non-GPIO pin function determines IO type. | 0b0: Non-GPIO function determines IO type when alternate function is selected<br>0b1: RES_CFG and PULL_UPDN_SEL determine IO type for non-GPIO configuration |
| GPIO_RX_ID  | 4:0  | GPIO ID for pin while receiving                                                                                                                                                            | 0bXXXXX: This GPIO receive ID                                                                                                                                |

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**GPIO\_A (0x2CB)**

| BIT         | 7           | 6           | 5           | 4           | 3         | 2           | 1           | 0            |
|-------------|-------------|-------------|-------------|-------------|-----------|-------------|-------------|--------------|
| Field       | RES_CFG     | TX_PRIO     | TX_COMP_EN  | GPIO_OUT    | GPIO_IN   | GPIO_RX_EN  | GPIO_TX_EN  | GPIO_OUT_DIS |
| Reset       | 0b1         | 0x0         | 0x0         | 0b0         | 0b0       | 0b0         | 0b0         | 0b1          |
| Access Type | Write, Read | Write, Read | Write, Read | Write, Read | Read Only | Write, Read | Write, Read | Write, Read  |

| BITFIELD     | BITS | DESCRIPTION                                     | DECODE                                                                                                        |
|--------------|------|-------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| RES_CFG      | 7    | Resistor pullup/pulldown strength               | 0b0: 40kΩ<br>0b1: 1MΩ                                                                                         |
| TX_PRIO      | 6    | Priority for GPIO scheduling                    | 0b0: Low priority<br>0b1: High priority                                                                       |
| TX_COMP_EN   | 5    | Jitter minimization compensation enable         | 0b0: Jitter compensation disabled<br>0b1: Jitter compensation enabled                                         |
| GPIO_OUT     | 4    | GPIO pin output drive value when GPIO_RX_EN = 0 | 0b0: This GPIO pin output is driven to 0<br>0b1: This GPIO pin output is driven to 1                          |
| GPIO_IN      | 3    | GPIO pin input level                            | 0b0: This GPIO pin value is 0<br>0b1: This GPIO pin value is 1                                                |
| GPIO_RX_EN   | 2    | GPIO out source control                         | 0b0: This GPIO source disabled for GMSL2 reception<br>0b1: This GPIO source enabled for GMSL2 reception       |
| GPIO_TX_EN   | 1    | GPIO Tx source control                          | 0b0: This GPIO source disabled for GMSL2 transmission<br>0b1: This GPIO source enabled for GMSL2 transmission |
| GPIO_OUT_DIS | 0    | Disables GPIO output driver                     | 0b0: Output driver enabled<br>0b1: Output driver disabled                                                     |

**GPIO\_B (0x2CC)**

| BIT         | 7                  | 6 | 5           | 4               | 3 | 2 | 1 | 0 |
|-------------|--------------------|---|-------------|-----------------|---|---|---|---|
| Field       | PULL_UPDN_SEL[1:0] |   | OUT_TYPE    | GPIO_TX_ID[4:0] |   |   |   |   |
| Reset       | 0x2                |   | 0b1         | 0x09            |   |   |   |   |
| Access Type | Write, Read        |   | Write, Read | Write, Read     |   |   |   |   |

| BITFIELD      | BITS | DESCRIPTION                          | DECODE                                                         |
|---------------|------|--------------------------------------|----------------------------------------------------------------|
| PULL_UPDN_SEL | 7:6  | Buffer pullup/pulldown configuration | 0b00: None<br>0b01: Pullup<br>0b10: Pulldown<br>0b11: Reserved |
| OUT_TYPE      | 5    | Driver type selection                | 0b0: Open-drain<br>0b1: Push-pull                              |
| GPIO_TX_ID    | 4:0  | GPIO ID for pin while transmitting   | 0bXXXXX: This GPIO transmit ID                                 |

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GPIO\_C (0x2CD)

| BIT         | 7           | 6 | 5 | 4               | 3 | 2 | 1 | 0 |
|-------------|-------------|---|---|-----------------|---|---|---|---|
| Field       | OVR_RES_CFG | – | – | GPIO_RX_ID[4:0] |   |   |   |   |
| Reset       | 0x0         | – | – | 0x09            |   |   |   |   |
| Access Type | Write, Read | – | – | Write, Read     |   |   |   |   |

| BITFIELD    | BITS | DESCRIPTION                                                                                                                                                                                | DECODE                                                                                                                                                       |
|-------------|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| OVR_RES_CFG | 7    | Overrides non-GPIO port function IO setting. When set, RES_CFG and PULL_UPDN_SEL are effective when pin is configured as non GPIO. When cleared, non-GPIO pin function determines IO type. | 0b0: Non-GPIO function determines IO type when alternate function is selected<br>0b1: RES_CFG and PULL_UPDN_SEL determine IO type for non-GPIO configuration |
| GPIO_RX_ID  | 4:0  | GPIO ID for pin while receiving                                                                                                                                                            | 0bXXXXX: This GPIO receive ID                                                                                                                                |

GPIO\_A (0x2CE)

| BIT         | 7           | 6           | 5           | 4           | 3         | 2           | 1           | 0            |
|-------------|-------------|-------------|-------------|-------------|-----------|-------------|-------------|--------------|
| Field       | RES_CFG     | TX_PRIO     | TX_COMP_EN  | GPIO_OUT    | GPIO_IN   | GPIO_RX_EN  | GPIO_TX_EN  | GPIO_OUT_DIS |
| Reset       | 0b1         | 0x0         | 0x0         | 0b0         | 0b0       | 0b0         | 0b0         | 0b1          |
| Access Type | Write, Read | Write, Read | Write, Read | Write, Read | Read Only | Write, Read | Write, Read | Write, Read  |

| BITFIELD     | BITS | DESCRIPTION                                     | DECODE                                                                                                        |
|--------------|------|-------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| RES_CFG      | 7    | Resistor pullup/pulldown strength               | 0b0: 40kΩ<br>0b1: 1MΩ                                                                                         |
| TX_PRIO      | 6    | Priority for GPIO scheduling                    | 0b0: Low priority<br>0b1: High priority                                                                       |
| TX_COMP_EN   | 5    | Jitter minimization compensation enable         | 0b0: Jitter compensation disabled<br>0b1: Jitter compensation enabled                                         |
| GPIO_OUT     | 4    | GPIO pin output drive value when GPIO_RX_EN = 0 | 0b0: This GPIO pin output is driven to 0<br>0b1: This GPIO pin output is driven to 1                          |
| GPIO_IN      | 3    | GPIO pin input level                            | 0b0: This GPIO pin value is 0<br>0b1: This GPIO pin value is 1                                                |
| GPIO_RX_EN   | 2    | GPIO out source control                         | 0b0: This GPIO source disabled for GMSL2 reception<br>0b1: This GPIO source enabled for GMSL2 reception       |
| GPIO_TX_EN   | 1    | GPIO Tx source control                          | 0b0: This GPIO source disabled for GMSL2 transmission<br>0b1: This GPIO source enabled for GMSL2 transmission |
| GPIO_OUT_DIS | 0    | Disables GPIO output driver                     | 0b0: Output driver enabled<br>0b1: Output driver disabled                                                     |



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**GPIO\_B (0x2CF)**

| BIT         | 7                  | 6 | 5 | 4               | 3 | 2 | 1 | 0 |
|-------------|--------------------|---|---|-----------------|---|---|---|---|
| Field       | PULL_UPDN_SEL[1:0] |   | – | GPIO_TX_ID[4:0] |   |   |   |   |
| Reset       | 0x2                |   | – | 0x0A            |   |   |   |   |
| Access Type | Write, Read        |   | – | Write, Read     |   |   |   |   |

| BITFIELD      | BITS | DESCRIPTION                          | DECODE                                                         |
|---------------|------|--------------------------------------|----------------------------------------------------------------|
| PULL_UPDN_SEL | 7:6  | Buffer pullup/pulldown configuration | 0b00: None<br>0b01: Pullup<br>0b10: Pulldown<br>0b11: Reserved |
| GPIO_TX_ID    | 4:0  | GPIO ID for pin while transmitting   | 0bXXXXX: This GPIO transmit ID                                 |

**GPIO\_C (0x2D0)**

| BIT         | 7           | 6 | 5 | 4               | 3 | 2 | 1 | 0 |
|-------------|-------------|---|---|-----------------|---|---|---|---|
| Field       | OVR_RES_CFG | – | – | GPIO_RX_ID[4:0] |   |   |   |   |
| Reset       | 0x0         | – | – | 0x0A            |   |   |   |   |
| Access Type | Write, Read | – | – | Write, Read     |   |   |   |   |

| BITFIELD    | BITS | DESCRIPTION                                                                                                                                                                                | DECODE                                                                                                                                                       |
|-------------|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| OVR_RES_CFG | 7    | Overrides non-GPIO port function IO setting. When set, RES_CFG and PULL_UPDN_SEL are effective when pin is configured as non GPIO. When cleared, non-GPIO pin function determines IO type. | 0b0: Non-GPIO function determines IO type when alternate function is selected<br>0b1: RES_CFG and PULL_UPDN_SEL determine IO type for non-GPIO configuration |
| GPIO_RX_ID  | 4:0  | GPIO ID for pin while receiving                                                                                                                                                            | 0bXXXXX: This GPIO receive ID                                                                                                                                |

**GPIO\_A (0x2D1)**

| BIT         | 7           | 6           | 5           | 4           | 3         | 2           | 1           | 0            |
|-------------|-------------|-------------|-------------|-------------|-----------|-------------|-------------|--------------|
| Field       | RES_CFG     | TX_PRIO     | TX_COMP_EN  | GPIO_OUT    | GPIO_IN   | GPIO_RX_EN  | GPIO_TX_EN  | GPIO_OUT_DIS |
| Reset       | 0b1         | 0x0         | 0x0         | 0b0         | 0b0       | 0b0         | 0b0         | 0b1          |
| Access Type | Write, Read | Write, Read | Write, Read | Write, Read | Read Only | Write, Read | Write, Read | Write, Read  |

| BITFIELD   | BITS | DESCRIPTION                                     | DECODE                                                                               |
|------------|------|-------------------------------------------------|--------------------------------------------------------------------------------------|
| RES_CFG    | 7    | Resistor pullup/pulldown strength               | 0b0: 40kΩ<br>0b1: 1MΩ                                                                |
| TX_PRIO    | 6    | Priority for GPIO scheduling                    | 0b0: Low priority<br>0b1: High priority                                              |
| TX_COMP_EN | 5    | Jitter minimization compensation enable         | 0b0: Jitter compensation disabled<br>0b1: Jitter compensation enabled                |
| GPIO_OUT   | 4    | GPIO pin output drive value when GPIO_RX_EN = 0 | 0b0: This GPIO pin output is driven to 0<br>0b1: This GPIO pin output is driven to 1 |
| GPIO_IN    | 3    | GPIO pin input level                            | 0b0: This GPIO pin value is 0<br>0b1: This GPIO pin value is 1                       |

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| BITFIELD     | BITS | DESCRIPTION                 | DECODE                                                                                                        |
|--------------|------|-----------------------------|---------------------------------------------------------------------------------------------------------------|
| GPIO_RX_EN   | 2    | GPIO out source control     | 0b0: This GPIO source disabled for GMSL2 reception<br>0b1: This GPIO source enabled for GMSL2 reception       |
| GPIO_TX_EN   | 1    | GPIO Tx source control      | 0b0: This GPIO source disabled for GMSL2 transmission<br>0b1: This GPIO source enabled for GMSL2 transmission |
| GPIO_OUT_DIS | 0    | Disables GPIO output driver | 0b0: Output driver enabled<br>0b1: Output driver disabled                                                     |

**GPIO\_B (0x2D2)**

| BIT         | 7                  | 6 | 5           | 4               | 3 | 2 | 1 | 0 |
|-------------|--------------------|---|-------------|-----------------|---|---|---|---|
| Field       | PULL_UPDN_SEL[1:0] |   | OUT_TYPE    | GPIO_TX_ID[4:0] |   |   |   |   |
| Reset       | 0x2                |   | 0b1         | 0x0B            |   |   |   |   |
| Access Type | Write, Read        |   | Write, Read | Write, Read     |   |   |   |   |

| BITFIELD      | BITS | DESCRIPTION                          | DECODE                                                         |
|---------------|------|--------------------------------------|----------------------------------------------------------------|
| PULL_UPDN_SEL | 7:6  | Buffer pullup/pulldown configuration | 0b00: None<br>0b01: Pullup<br>0b10: Pulldown<br>0b11: Reserved |
| OUT_TYPE      | 5    | Driver type selection                | 0b0: Open-drain<br>0b1: Reserved                               |
| GPIO_TX_ID    | 4:0  | GPIO ID for pin while transmitting   | 0bXXXXX: This GPIO transmit ID                                 |

**GPIO\_C (0x2D3)**

| BIT         | 7           | 6 | 5 | 4               | 3 | 2 | 1 | 0 |
|-------------|-------------|---|---|-----------------|---|---|---|---|
| Field       | OVR_RES_CFG | – | – | GPIO_RX_ID[4:0] |   |   |   |   |
| Reset       | 0x0         | – | – | 0x0B            |   |   |   |   |
| Access Type | Write, Read | – | – | Write, Read     |   |   |   |   |

| BITFIELD    | BITS | DESCRIPTION                                                                                                                                                                                | DECODE                                                                                                                                                       |
|-------------|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| OVR_RES_CFG | 7    | Overrides non-GPIO port function IO setting. When set, RES_CFG and PULL_UPDN_SEL are effective when pin is configured as non GPIO. When cleared, non-GPIO pin function determines IO type. | 0b0: Non-GPIO function determines IO type when alternate function is selected<br>0b1: RES_CFG and PULL_UPDN_SEL determine IO type for non-GPIO configuration |
| GPIO_RX_ID  | 4:0  | GPIO ID for pin while receiving                                                                                                                                                            | 0bXXXXX: This GPIO receive ID                                                                                                                                |

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**GPIO\_A (0x2D4)**

| BIT         | 7           | 6           | 5           | 4           | 3         | 2           | 1           | 0            |
|-------------|-------------|-------------|-------------|-------------|-----------|-------------|-------------|--------------|
| Field       | RES_CFG     | TX_PRIO     | TX_COMP_EN  | GPIO_OUT    | GPIO_IN   | GPIO_RX_EN  | GPIO_TX_EN  | GPIO_OUT_DIS |
| Reset       | 0b1         | 0x0         | 0x0         | 0b0         | 0b0       | 0b0         | 0b0         | 0b1          |
| Access Type | Write, Read | Write, Read | Write, Read | Write, Read | Read Only | Write, Read | Write, Read | Write, Read  |

| BITFIELD     | BITS | DESCRIPTION                                     | DECODE                                                                                                        |
|--------------|------|-------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| RES_CFG      | 7    | Resistor pullup/pulldown strength               | 0b0: 40kΩ<br>0b1: 1MΩ                                                                                         |
| TX_PRIO      | 6    | Priority for GPIO scheduling                    | 0b0: Low priority<br>0b1: High priority                                                                       |
| TX_COMP_EN   | 5    | Jitter minimization compensation enable         | 0b0: Jitter compensation disabled<br>0b1: Jitter compensation enabled                                         |
| GPIO_OUT     | 4    | GPIO pin output drive value when GPIO_RX_EN = 0 | 0b0: This GPIO pin output is driven to 0<br>0b1: This GPIO pin output is driven to 1                          |
| GPIO_IN      | 3    | GPIO pin input level                            | 0b0: This GPIO pin value is 0<br>0b1: This GPIO pin value is 1                                                |
| GPIO_RX_EN   | 2    | GPIO out source control                         | 0b0: This GPIO source disabled for GMSL2 reception<br>0b1: This GPIO source enabled for GMSL2 reception       |
| GPIO_TX_EN   | 1    | GPIO Tx source control                          | 0b0: This GPIO source disabled for GMSL2 transmission<br>0b1: This GPIO source enabled for GMSL2 transmission |
| GPIO_OUT_DIS | 0    | Disables GPIO output driver                     | 0b0: Output driver enabled<br>0b1: Output driver disabled                                                     |

**GPIO\_B (0x2D5)**

| BIT         | 7                  | 6 | 5           | 4               | 3 | 2 | 1 | 0 |
|-------------|--------------------|---|-------------|-----------------|---|---|---|---|
| Field       | PULL_UPDN_SEL[1:0] |   | OUT_TYPE    | GPIO_TX_ID[4:0] |   |   |   |   |
| Reset       | 0x2                |   | 0b1         | 0x0C            |   |   |   |   |
| Access Type | Write, Read        |   | Write, Read | Write, Read     |   |   |   |   |

| BITFIELD      | BITS | DESCRIPTION                          | DECODE                                                         |
|---------------|------|--------------------------------------|----------------------------------------------------------------|
| PULL_UPDN_SEL | 7:6  | Buffer pullup/pulldown configuration | 0b00: None<br>0b01: Pullup<br>0b10: Pulldown<br>0b11: Reserved |
| OUT_TYPE      | 5    | Driver type selection                | 0b0: Open-drain<br>0b1: Reserved                               |
| GPIO_TX_ID    | 4:0  | GPIO ID for pin while transmitting   | 0bXXXXX: This GPIO transmit ID                                 |

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**GPIO\_C (0x2D6)**

| BIT         | 7           | 6 | 5 | 4               | 3 | 2 | 1 | 0 |
|-------------|-------------|---|---|-----------------|---|---|---|---|
| Field       | OVR_RES_CFG | – | – | GPIO_RX_ID[4:0] |   |   |   |   |
| Reset       | 0x0         | – | – | 0x0C            |   |   |   |   |
| Access Type | Write, Read | – | – | Write, Read     |   |   |   |   |

| BITFIELD    | BITS | DESCRIPTION                                                                                                                                                                                | DECODE                                                                                                                                                       |
|-------------|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| OVR_RES_CFG | 7    | Overrides non-GPIO port function IO setting. When set, RES_CFG and PULL_UPDN_SEL are effective when pin is configured as non GPIO. When cleared, non-GPIO pin function determines IO type. | 0b0: Non-GPIO function determines IO type when alternate function is selected<br>0b1: RES_CFG and PULL_UPDN_SEL determine IO type for non-GPIO configuration |
| GPIO_RX_ID  | 4:0  | GPIO ID for pin while receiving                                                                                                                                                            | 0bXXXXX: This GPIO receive ID                                                                                                                                |

**CMU4 (0x304)**

| BIT         | 7            | 6 | 5            | 4 | 3 | 2 | 1            | 0 |
|-------------|--------------|---|--------------|---|---|---|--------------|---|
| Field       | A_SPEED[1:0] |   | B_SPEED[1:0] |   | – | – | C_SPEED[1:0] |   |
| Reset       | 0x2          |   | 0x3          |   | – | – | 0x3          |   |
| Access Type | Write, Read  |   | Write, Read  |   | – | – | Write, Read  |   |

| BITFIELD | BITS | DESCRIPTION                                                                                        | DECODE                                                            |
|----------|------|----------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|
| A_SPEED  | 7:6  | Control A_SPEED group edge rate. First value is for VDDIO = 1.8V, second value is for VDDIO = 3.3V | 0x0: 2ns, 1ns<br>0x1: 4ns, 2ns<br>0x2: 8ns, 4ns<br>0x3: 16ns, 8ns |
| B_SPEED  | 5:4  | Control B_SPEED group edge rate. First value is for VDDIO = 1.8V, second value is for VDDIO = 3.3V | 0x0: 2ns, 1ns<br>0x1: 4ns, 2ns<br>0x2: 8ns, 4ns<br>0x3: 16ns, 8ns |
| C_SPEED  | 1:0  | Control C_SPEED group edge rate. First value is for VDDIO = 1.8V, second value is for VDDIO = 3.3V | 0x0: 2ns, 1ns<br>0x1: 4ns, 2ns<br>0x2: 8ns, 4ns<br>0x3: 16ns, 8ns |

**BACKTOP1 (0x308)**

| BIT         | 7             | 6             | 5             | 4             | 3           | 2           | 1 | 0           |
|-------------|---------------|---------------|---------------|---------------|-------------|-------------|---|-------------|
| Field       | CSIPLL_U_LOCK | CSIPLL_Z_LOCK | CSIPLL_Y_LOCK | CSIPLL_X_LOCK | LINE_SPL2   | LINE_SPL1   | – | BACKTOP_EN  |
| Reset       | 0b0           | 0b0           | 0b0           | 0b0           | 0b0         | 0b0         | – | 0b1         |
| Access Type | Read Only     | Read Only     | Read Only     | Read Only     | Write, Read | Write, Read | – | Write, Read |

| BITFIELD      | BITS | DESCRIPTION      | DECODE                                               |
|---------------|------|------------------|------------------------------------------------------|
| CSIPLL_U_LOCK | 7    | CSI U PLL locked | 0b0: CSI2 U PLL not locked<br>0b1: CSI2 U PLL locked |
| CSIPLL_Z_LOCK | 6    | CSI Z PLL locked | 0b0: CSI2 Z PLL not locked<br>0b1: CSI2 Z PLL locked |

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| BITFIELD     | BITS | DESCRIPTION                                               | DECODE                                                                                                                                          |
|--------------|------|-----------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| CSIPLLY_LOCK | 5    | CSI Y PLL locked                                          | 0b0: CSI2 Y PLL not locked<br>0b1: CSI2 Y PLL locked                                                                                            |
| CSIPLLX_LOCK | 4    | CSI X PLL locked                                          | 0b0: CSI2 X PLL not locked<br>0b1: CSI2 X PLL locked                                                                                            |
| LINE_SPL2    | 3    | Line-based distribution to line memories for Video Pipe Z | 0b0: Disable line-based distribution to line memories for Video Pipe Z<br>0b1: Enable line-based distribution to line memories for Video Pipe Z |
| LINE_SPL1    | 2    | Line-based distribution to line memories for Video Pipe X | 0b0: Disable line-based distribution to line memories for Video Pipe X<br>0b1: Enable line-based distribution to line memories for Video Pipe X |
| BACKTOP_EN   | 0    | Backtop write logic enable                                | 0b0: Disable writes to BACKTOP register block<br>0b1: Enable writes to BACKTOP register block                                                   |

**BACKTOP11 (0x312)**

| BIT         | 7             | 6             | 5             | 4             | 3 | 2         | 1         | 0 |
|-------------|---------------|---------------|---------------|---------------|---|-----------|-----------|---|
| Field       | cmd_overflow4 | cmd_overflow3 | cmd_overflow2 | cmd_overflow1 | – | LMO_Z     | LMO_Y     | – |
| Reset       | 0b0           | 0b0           | 0b0           | 0b0           | – | 0b0       | 0b0       | – |
| Access Type | Read Only     | Read Only     | Read Only     | Read Only     | – | Read Only | Read Only | – |

| BITFIELD      | BITS | DESCRIPTION                                     | DECODE                                                                          |
|---------------|------|-------------------------------------------------|---------------------------------------------------------------------------------|
| cmd_overflow4 | 7    | Command FIFO overflow detection                 | 0b0: No overflow detected<br>0b1: Overflow detected                             |
| cmd_overflow3 | 6    | Command FIFO overflow detection                 | 0b0: No overflow detected<br>0b1: Overflow detected                             |
| cmd_overflow2 | 5    | Command FIFO overflow detection                 | 0b0: No overflow detected<br>0b1: Overflow detected                             |
| cmd_overflow1 | 4    | Command FIFO overflow detection                 | 0b0: No overflow detected<br>0b1: Overflow detected                             |
| LMO_Z         | 2    | Pipeline Z line memory overflow sticky register | 0b0: Pipeline Z no line memory overflow<br>0b1: Pipeline Z line memory overflow |
| LMO_Y         | 1    | Pipeline Y line memory overflow sticky register | 0b0: Pipeline Y no line memory overflow<br>0b1: Pipeline Y line memory overflow |

**BACKTOP12 (0x313)**

| BIT         | 7               | 6 | 5 | 4 | 3 | 2                   | 1           | 0 |
|-------------|-----------------|---|---|---|---|---------------------|-------------|---|
| Field       | soft_bpp_x[4:0] |   |   |   |   | BACKTOP_MEM_CRC_ERR | CSI_OUT_EN  | – |
| Reset       | 0x00            |   |   |   |   | 0x0                 | 0b1         | – |
| Access Type | Write, Read     |   |   |   |   | Read Only           | Write, Read | – |

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| BITFIELD                    | BITS | DESCRIPTION                         | DECODE                                                                                                                                                                                                                               |
|-----------------------------|------|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| soft_bpp_x                  | 7:3  | Software defined BPP for pipeline X | 0x8: Datatypes = x2A, x10-12, x31-37<br>0xA: Datatypes = x2B<br>0xC: Datatypes = x2C<br>0xE: Datatypes = x2D<br>0x10: Datatypes = x22, x1E, x2E<br>0x12: Datatypes = x23<br>0x14: Datatypes = x1F, x2F<br>0x18: Datatypes = x24, x30 |
| BACKTOP_<br>MEM_CRC_<br>ERR | 2    | Backtop memory CRC error flag       | 0b0: No CRC error occurred<br>0b1: CRC error occurred                                                                                                                                                                                |
| CSI_OUT_E<br>N              | 1    | Enables CSI output                  | 0b0: CSI output disabled<br>0x1: CSI output enabled                                                                                                                                                                                  |

**BACKTOP13 (0x314)**

| BIT            | 7              | 6 | 5 | 4 | 3              | 2 | 1 | 0 |
|----------------|----------------|---|---|---|----------------|---|---|---|
| Field          | soft_vc_y[3:0] |   |   |   | soft_vc_x[3:0] |   |   |   |
| Reset          | 0x0            |   |   |   | 0x0            |   |   |   |
| Access<br>Type | Write, Read    |   |   |   | Write, Read    |   |   |   |

| BITFIELD  | BITS | DESCRIPTION                                     | DECODE                                               |
|-----------|------|-------------------------------------------------|------------------------------------------------------|
| soft_vc_y | 7:4  | Software defined virtual channel for pipeline Y | 0xX: Software defined virtual channel for pipeline Y |
| soft_vc_x | 3:0  | Software defined virtual channel for pipeline X | 0xX: Software defined virtual channel for pipeline X |

**BACKTOP14 (0x315)**

| BIT            | 7              | 6 | 5 | 4 | 3              | 2 | 1 | 0 |
|----------------|----------------|---|---|---|----------------|---|---|---|
| Field          | soft_vc_u[3:0] |   |   |   | soft_vc_z[3:0] |   |   |   |
| Reset          | 0x0            |   |   |   | 0x0            |   |   |   |
| Access<br>Type | Write, Read    |   |   |   | Write, Read    |   |   |   |

| BITFIELD  | BITS | DESCRIPTION                                     | DECODE                                               |
|-----------|------|-------------------------------------------------|------------------------------------------------------|
| soft_vc_u | 7:4  | Software defined virtual channel for pipeline U | 0xX: Software defined virtual channel for pipeline U |
| soft_vc_z | 3:0  | Software defined virtual channel for pipeline Z | 0xX: Software defined virtual channel for pipeline Z |

**BACKTOP15 (0x316)**

| BIT            | 7                | 6 | 5              | 4 | 3 | 2 | 1 | 0 |
|----------------|------------------|---|----------------|---|---|---|---|---|
| Field          | soft_dt_y_h[1:0] |   | soft_dt_x[5:0] |   |   |   |   |   |
| Reset          | 0x0              |   | 0x00           |   |   |   |   |   |
| Access<br>Type | Write, Read      |   | Write, Read    |   |   |   |   |   |

| BITFIELD    | BITS | DESCRIPTION                                            | DECODE                                                       |
|-------------|------|--------------------------------------------------------|--------------------------------------------------------------|
| soft_dt_y_h | 7:6  | High bits of software defined data type for pipeline Y | 0bXX: High bits of software defined data type for pipeline Y |

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| BITFIELD  | BITS | DESCRIPTION                               | DECODE                                                                                                                                                                                                                                                                                                                                                          |
|-----------|------|-------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| soft_dt_x | 5:0  | Software defined data type for pipeline X | 0x10: GENERIC8<br>0x11: GENERIC8<br>0x12: EMBEDDED<br>0x1E: YUV422 8bit<br>0x1F: YUV422 10bit<br>0x22: RGB565<br>0x23: RGB666<br>0x24: RGB888<br>0x2A: RAW8<br>0x2B: RAW10<br>0x2C: RAW12<br>0x2D: RAW14<br>0x2E: RAW16<br>0x2F: RAW20<br>0x30: YUV422 12 BIT<br>0x31: UDP8<br>0x32: UDP8<br>0x33: UDP8<br>0x34: UDP8<br>0x35: UDP8<br>0x36: UDP8<br>0x37: UDP8 |

**BACKTOP16 (0x317)**

| BIT         | 7                | 6 | 5 | 4 | 3                | 2 | 1 | 0 |
|-------------|------------------|---|---|---|------------------|---|---|---|
| Field       | soft_dt_z_h[3:0] |   |   |   | soft_dt_y_l[3:0] |   |   |   |
| Reset       | 0x0              |   |   |   | 0x0              |   |   |   |
| Access Type | Write, Read      |   |   |   | Write, Read      |   |   |   |

| BITFIELD    | BITS | DESCRIPTION                                            | DECODE                                                      |
|-------------|------|--------------------------------------------------------|-------------------------------------------------------------|
| soft_dt_z_h | 7:4  | High bits of software defined data type for pipeline Z | 0xX: High bits of software defined data type for pipeline Z |
| soft_dt_y_l | 3:0  | Low bits of software defined data type for pipeline Y  | 0xX: Low bits of software defined data type for pipeline Y  |

**BACKTOP17 (0x318)**

| BIT         | 7              | 6 | 5 | 4 | 3 | 2 | 1                | 0 |
|-------------|----------------|---|---|---|---|---|------------------|---|
| Field       | soft_dt_u[5:0] |   |   |   |   |   | soft_dt_z_l[1:0] |   |
| Reset       | 0x00           |   |   |   |   |   | 0x0              |   |
| Access Type | Write, Read    |   |   |   |   |   | Write, Read      |   |



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| BITFIELD    | BITS | DESCRIPTION                                           | DECODE                                                                                                                                                                                                                                                                                                                                                          |
|-------------|------|-------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| soft_dt_u   | 7:2  | Software defined data type for pipeline U             | 0x10: GENERIC8<br>0x11: GENERIC8<br>0x12: EMBEDDED<br>0x1E: YUV422 8bit<br>0x1F: YUV422 10bit<br>0x22: RGB565<br>0x23: RGB666<br>0x24: RGB888<br>0x2A: RAW8<br>0x2B: RAW10<br>0x2C: RAW12<br>0x2D: RAW14<br>0x2E: RAW16<br>0x2F: RAW20<br>0x30: YUV422 12 BIT<br>0x31: UDP8<br>0x32: UDP8<br>0x33: UDP8<br>0x34: UDP8<br>0x35: UDP8<br>0x36: UDP8<br>0x37: UDP8 |
| soft_dt_z_l | 1:0  | Low bits of software defined data type for pipeline Z | 0xX: Low bits of software defined data type for pipeline Z                                                                                                                                                                                                                                                                                                      |

**BACKTOP18 (0x319)**

| BIT         | 7                 | 6 | 5 | 4 | 3               | 2 | 1 | 0 |
|-------------|-------------------|---|---|---|-----------------|---|---|---|
| Field       | soft_bpp_z_h[2:0] |   |   |   | soft_bpp_y[4:0] |   |   |   |
| Reset       | 0x0               |   |   |   | 0x00            |   |   |   |
| Access Type | Write, Read       |   |   |   | Write, Read     |   |   |   |

| BITFIELD     | BITS | DESCRIPTION                                      | DECODE                                                                                                                                                                                                                               |
|--------------|------|--------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| soft_bpp_z_h | 7:5  | High bits of software defined BPP for pipeline Z | 0xX: High bits of software defined BPP for pipeline Z                                                                                                                                                                                |
| soft_bpp_y   | 4:0  | Software defined BPP for pipeline Y              | 0x8: Datatypes = x2A, x10-12, x31-37<br>0xA: Datatypes = x2B<br>0xC: Datatypes = x2C<br>0xE: Datatypes = x2D<br>0x10: Datatypes = x22, x1E, x2E<br>0x12: Datatypes = x23<br>0x14: Datatypes = x1F, x2F<br>0x18: Datatypes = x24, x30 |

**BACKTOP19 (0x31A)**

| BIT         | 7 | 6               | 5 | 4 | 3 | 2                 | 1 | 0 |
|-------------|---|-----------------|---|---|---|-------------------|---|---|
| Field       | — | soft_bpp_u[4:0] |   |   |   | soft_bpp_z_l[1:0] |   |   |
| Reset       | — | 0x00            |   |   |   | 0x0               |   |   |
| Access Type | — | Write, Read     |   |   |   | Write, Read       |   |   |

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| BITFIELD     | BITS | DESCRIPTION                                     | DECODE                                                                                                                                                                                                                               |
|--------------|------|-------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| soft_bpp_u   | 6:2  | Software defined BPP for pipeline U             | 0x8: Datatypes = x2A, x10-12, x31-37<br>0xA: Datatypes = x2B<br>0xC: Datatypes = x2C<br>0xE: Datatypes = x2D<br>0x10: Datatypes = x22, x1E, x2E<br>0x12: Datatypes = x23<br>0x14: Datatypes = x1F, x2F<br>0x18: Datatypes = x24, x30 |
| soft_bpp_z_l | 1:0  | Low bits of software defined BPP for pipeline Z | 0bXX: Low bits of software defined BPP for pipeline Z                                                                                                                                                                                |

**BACKTOP20 (0x31B)**

| BIT         | 7                                      | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|----------------------------------------|---|---|---|---|---|---|---|
| Field       | phy0_csi_tx_dpll_fb_fraction_in_l[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00                                   |   |   |   |   |   |   |   |
| Access Type | Write, Read                            |   |   |   |   |   |   |   |

| BITFIELD                          | BITS | DESCRIPTION                                                            | DECODE                                             |
|-----------------------------------|------|------------------------------------------------------------------------|----------------------------------------------------|
| phy0_csi_tx_dpll_fb_fraction_in_l | 7:0  | Low byte of software override value for CSI PHY0 frequency fine tuning | 0xXX: PHY0 frequency fine tuning override low byte |

**BACKTOP21 (0x31C)**

| BIT         | 7           | 6           | 5           | 4           | 3                                      | 2 | 1 | 0 |
|-------------|-------------|-------------|-------------|-------------|----------------------------------------|---|---|---|
| Field       | bpp8dblu    | bpp8dblz    | bpp8dbly    | bpp8dblx    | phy0_csi_tx_dpll_fb_fraction_in_h[3:0] |   |   |   |
| Reset       | 0b0         | 0b0         | 0b0         | 0b0         | 0x0                                    |   |   |   |
| Access Type | Write, Read | Write, Read | Write, Read | Write, Read | Write, Read                            |   |   |   |

| BITFIELD                          | BITS | DESCRIPTION                                                           | DECODE                                                                              |
|-----------------------------------|------|-----------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| bpp8dblu                          | 7    | BPP = 8 processed as 16-bit color enable for pipeline U               | 0b0: Do not process BPP = 8 as 16-bit color<br>0b1: Process BPP = 8 as 16-bit color |
| bpp8dblz                          | 6    | BPP = 8 processed as 16-bit color enable for pipeline Z               | 0b0: Do not process BPP = 8 as 16-bit color<br>0b1: Process BPP = 8 as 16-bit color |
| bpp8dbly                          | 5    | BPP = 8 processed as 16-bit color enable for pipeline Y               | 0b0: Do not process BPP = 8 as 16-bit color<br>0b1: Process BPP = 8 as 16-bit color |
| bpp8dblx                          | 4    | BPP = 8 processed as 16-bit color enable for pipeline X               | 0b0: Do not process BPP = 8 as 16-bit color<br>0b1: Process BPP = 8 as 16-bit color |
| phy0_csi_tx_dpll_fb_fraction_in_h | 3:0  | High nibble of software override value for PHY0 frequency fine tuning | 0xX: PHY0 frequency fine tuning override high nibble                                |

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**BACKTOP22 (0x31D)**

| BIT            | 7                       | 6                       | 5                                                  | 4                                 | 3 | 2 | 1 | 0 |
|----------------|-------------------------|-------------------------|----------------------------------------------------|-----------------------------------|---|---|---|---|
| Field          | override_bp<br>p_vc_dty | override_bp<br>p_vc_dtx | phy0_csi_tx<br>_dpll_fb_fra<br>ction_predef<br>_en | phy0_csi_tx_dpll_predef_freq[4:0] |   |   |   |   |
| Reset          | 0b0                     | 0b0                     | 0b1                                                | 0x0F                              |   |   |   |   |
| Access<br>Type | Write, Read             | Write, Read             | Write, Read                                        | Write, Read                       |   |   |   |   |

| BITFIELD                                           | BITS | DESCRIPTION                                                  | DECODE                                                                                                                |
|----------------------------------------------------|------|--------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| override_bpp<br>_vc_dty                            | 7    | Software override enable for BPP, VC and DT                  | 0b0: Software override for BPP, VC and DT disabled<br>0x1: Software override for BPP, VC and DT enabled               |
| override_bpp<br>_vc_dtx                            | 6    | Software override enable for BPP, VC and DT                  | 0b0: Software override for BPP, VC and DT disabled<br>0x1: Software override for BPP, VC and DT enabled               |
| phy0_csi_tx_<br>dpll_fb_fra<br>ction_predef_e<br>n | 5    | CSI PHY0 software override disable for frequency fine tuning | 0b0: Software override for frequency fine tuning enabled<br>0x1: Software override for frequency fine tuning disabled |
| phy0_csi_tx_<br>dpll_predef_f<br>req               | 4:0  | CSI PHY0 output frequency in multiples of 100Mhz             | 0bXXXXX: CSI PHY0 output frequency (*100MHz)                                                                          |

**BACKTOP23 (0x31E)**

| BIT            | 7                                      | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|----------------|----------------------------------------|---|---|---|---|---|---|---|
| Field          | phy1_csi_tx_dpll_fb_fraction_in_l[7:0] |   |   |   |   |   |   |   |
| Reset          | 0x00                                   |   |   |   |   |   |   |   |
| Access<br>Type | Write, Read                            |   |   |   |   |   |   |   |

| BITFIELD                                  | BITS | DESCRIPTION                                                            | DECODE                                             |
|-------------------------------------------|------|------------------------------------------------------------------------|----------------------------------------------------|
| phy1_csi_tx_<br>dpll_fb_fra<br>ction_in_l | 7:0  | Low byte of software override value for CSI PHY1 frequency fine tuning | 0xXX: PHY1 frequency fine tuning override low byte |

**BACKTOP24 (0x31F)**

| BIT            | 7                 | 6                 | 5                 | 4                 | 3                                      | 2 | 1 | 0 |
|----------------|-------------------|-------------------|-------------------|-------------------|----------------------------------------|---|---|---|
| Field          | bpp8dblu_m<br>ode | bpp8dblz_m<br>ode | bpp8dbly_m<br>ode | bpp8dblx_m<br>ode | phy1_csi_tx_dpll_fb_fraction_in_h[3:0] |   |   |   |
| Reset          | 0b0               | 0b0               | 0b0               | 0b0               | 0x0                                    |   |   |   |
| Access<br>Type | Write, Read       | Write, Read       | Write, Read       | Write, Read       | Write, Read                            |   |   |   |

| BITFIELD          | BITS | DESCRIPTION                                             | DECODE                                                                    |
|-------------------|------|---------------------------------------------------------|---------------------------------------------------------------------------|
| bpp8dblu_mo<br>de | 7    | Enable 8-bit write alternate map to RAMs for pipeline U | 0b0: Write alternative map disabled<br>0x1: Write alternative map enabled |

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| BITFIELD                          | BITS | DESCRIPTION                                                           | DECODE                                                                    |
|-----------------------------------|------|-----------------------------------------------------------------------|---------------------------------------------------------------------------|
| bpp8dblz_mode                     | 6    | Enable 8-bit write alternate map to RAMs for pipeline Z               | 0b0: Write alternative map disabled<br>0x1: Write alternative map enabled |
| bpp8dbly_mode                     | 5    | Enable 8-bit write alternate map to RAMs for pipeline Y               | 0b0: Write alternative map disabled<br>0x1: Write alternative map enabled |
| bpp8dblx_mode                     | 4    | Enable 8-bit write alternate map to RAMs for pipeline X               | 0b0: Write alternative map disabled<br>0x1: Write alternative map enabled |
| phy1_csi_tx_dpll_fb_fraction_in_h | 3:0  | High nibble of software override value for PHY1 frequency fine tuning | 0xX: PHY1 frequency fine tuning override high nibble                      |

**BACKTOP25 (0x320)**

| BIT         | 7                   | 6                   | 5                                      | 4                                 | 3 | 2 | 1 | 0 |
|-------------|---------------------|---------------------|----------------------------------------|-----------------------------------|---|---|---|---|
| Field       | override_bpp_vc_dtu | override_bpp_vc_dtz | phy1_csi_tx_dpll_fb_fraction_predef_en | phy1_csi_tx_dpll_predef_freq[4:0] |   |   |   |   |
| Reset       | 0b0                 | 0b0                 | 0b1                                    | 0x0F                              |   |   |   |   |
| Access Type | Write, Read         | Write, Read         | Write, Read                            | Write, Read                       |   |   |   |   |

| BITFIELD                               | BITS | DESCRIPTION                                                  | DECODE                                                                                                                |
|----------------------------------------|------|--------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| override_bpp_vc_dtu                    | 7    | Software override enable for BPP, VC and DT                  | 0b0: Software override for BPP, VC and DT disabled<br>0x1: Software override for BPP, VC and DT enabled               |
| override_bpp_vc_dtz                    | 6    | Software override enable for BPP, VC and DT                  | 0b0: Software override for BPP, VC and DT disabled<br>0x1: Software override for BPP, VC and DT enabled               |
| phy1_csi_tx_dpll_fb_fraction_predef_en | 5    | CSI PHY1 software override disable for frequency fine tuning | 0b0: Software override for frequency fine tuning enabled<br>0x1: Software override for frequency fine tuning disabled |
| phy1_csi_tx_dpll_predef_freq           | 4:0  | CSI PHY1 output frequency in multiples of 100Mhz             | 0bXXXXX: CSI PHY1 output frequency (*100MHz)                                                                          |

**BACKTOP26 (0x321)**

| BIT         | 7                                      | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|----------------------------------------|---|---|---|---|---|---|---|
| Field       | phy2_csi_tx_dpll_fb_fraction_in_l[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00                                   |   |   |   |   |   |   |   |
| Access Type | Write, Read                            |   |   |   |   |   |   |   |

| BITFIELD                          | BITS | DESCRIPTION                                                            | DECODE                                             |
|-----------------------------------|------|------------------------------------------------------------------------|----------------------------------------------------|
| phy2_csi_tx_dpll_fb_fraction_in_l | 7:0  | Low byte of software override value for CSI PHY2 frequency fine tuning | 0xXX: PHY2 frequency fine tuning override low byte |

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**BACKTOP27 (0x322)**

| BIT         | 7                  | 6                  | 5                  | 4                  | 3                                       | 2 | 1 | 0 |
|-------------|--------------------|--------------------|--------------------|--------------------|-----------------------------------------|---|---|---|
| Field       | yuv_8_10_mux_mode4 | yuv_8_10_mux_mode3 | yuv_8_10_mux_mode2 | yuv_8_10_mux_mode1 | phy2_csi_tx_dppll_fb_fraction_in_h[3:0] |   |   |   |
| Reset       | 0x0                | 0x0                | 0x0                | 0x0                | 0x0                                     |   |   |   |
| Access Type | Write, Read        | Write, Read        | Write, Read        | Write, Read        | Write, Read                             |   |   |   |

| BITFIELD                           | BITS | DESCRIPTION                                                           | DECODE                                                                                            |
|------------------------------------|------|-----------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| yuv_8_10_mux_mode4                 | 7    | Mode 4 yuv422 8/10 bit muxed mode support                             | 0x0: Disable yuv422 8/10 bit muxed mode support<br>0x1: Enable yuv422 8/10 bit muxed mode support |
| yuv_8_10_mux_mode3                 | 6    | Mode 3 yuv422 8/10 bit muxed mode support                             | 0x0: Disable yuv422 8/10 bit muxed mode support<br>0x1: Enable yuv422 8/10 bit muxed mode support |
| yuv_8_10_mux_mode2                 | 5    | Mode 2 yuv422 8/10 bit muxed mode support                             | 0x0: Disable yuv422 8/10 bit muxed mode support<br>0x1: Enable yuv422 8/10 bit muxed mode support |
| yuv_8_10_mux_mode1                 | 4    | Mode 1 yuv422 8/10 bit muxed mode support                             | 0x0: Disable yuv422 8/10 bit muxed mode support<br>0x1: Enable yuv422 8/10 bit muxed mode support |
| phy2_csi_tx_dppll_fb_fraction_in_h | 3:0  | High nibble of software override value for PHY2 frequency fine tuning | 0xX: PHY2 frequency fine tuning override high nibble                                              |

**BACKTOP28 (0x323)**

| BIT         | 7 | 6 | 5                                       | 4                                  | 3 | 2 | 1 | 0 |
|-------------|---|---|-----------------------------------------|------------------------------------|---|---|---|---|
| Field       | — | — | phy2_csi_tx_dppll_fb_fraction_predef_en | phy2_csi_tx_dppll_predef_freq[4:0] |   |   |   |   |
| Reset       | — | — | 0b1                                     | 0x0F                               |   |   |   |   |
| Access Type | — | — | Write, Read                             | Write, Read                        |   |   |   |   |

| BITFIELD                                | BITS | DESCRIPTION                                                  | DECODE                                                                                                                |
|-----------------------------------------|------|--------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| phy2_csi_tx_dppll_fb_fraction_predef_en | 5    | CSI PHY2 software override disable for frequency fine tuning | 0b0: Software override for frequency fine tuning enabled<br>0x1: Software override for frequency fine tuning disabled |
| phy2_csi_tx_dppll_predef_freq           | 4:0  | CSI PHY2 output frequency in multiples of 100Mhz             | 0bXXXXX: CSI PHY2 output frequency (*100MHz)                                                                          |

**BACKTOP29 (0x324)**

| BIT         | 7                                       | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|-----------------------------------------|---|---|---|---|---|---|---|
| Field       | phy3_csi_tx_dppll_fb_fraction_in_l[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00                                    |   |   |   |   |   |   |   |
| Access Type | Write, Read                             |   |   |   |   |   |   |   |

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| BITFIELD                          | BITS | DESCRIPTION                                                            | DECODE                                             |
|-----------------------------------|------|------------------------------------------------------------------------|----------------------------------------------------|
| phy3_csi_tx_dpll_fb_fraction_in_l | 7:0  | Low byte of software override value for CSI PHY3 frequency fine tuning | 0xXX: PHY3 frequency fine tuning override low byte |

**BACKTOP30 (0x325)**

| BIT         | 7               | 6 | 5 | 4 | 3                                      | 2 | 1 | 0 |
|-------------|-----------------|---|---|---|----------------------------------------|---|---|---|
| Field       | BACKTOP_W_FRAME | – | – | – | phy3_csi_tx_dpll_fb_fraction_in_h[3:0] |   |   |   |
| Reset       | 0x0             | – | – | – | 0x0                                    |   |   |   |
| Access Type | Write, Read     | – | – | – | Write, Read                            |   |   |   |

| BITFIELD                          | BITS | DESCRIPTION                                                                                     | DECODE                                                                                                                        |
|-----------------------------------|------|-------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| BACKTOP_W_FRAME                   | 7    | When this register is set, backtop waits for a new frame before generating MIPI packet requests | 0b0: Do not wait for a new frame before generating a MIPI packet<br>0b1: Wait for a new frame before generating a MIPI packet |
| phy3_csi_tx_dpll_fb_fraction_in_h | 3:0  | High nibble of software override value for PHY3 frequency fine tuning                           | 0xXX: PHY3 frequency fine tuning override low byte                                                                            |

**BACKTOP31 (0x326)**

| BIT         | 7 | 6 | 5                                      | 4                                 | 3 | 2 | 1 | 0 |
|-------------|---|---|----------------------------------------|-----------------------------------|---|---|---|---|
| Field       | – | – | phy3_csi_tx_dpll_fb_fraction_predef_en | phy3_csi_tx_dpll_predef_freq[4:0] |   |   |   |   |
| Reset       | – | – | 0b1                                    | 0x0F                              |   |   |   |   |
| Access Type | – | – | Write, Read                            | Write, Read                       |   |   |   |   |

| BITFIELD                               | BITS | DESCRIPTION                                                  | DECODE                                                                                                                |
|----------------------------------------|------|--------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| phy3_csi_tx_dpll_fb_fraction_predef_en | 5    | CSI PHY3 software override disable for frequency fine tuning | 0b0: Software override for frequency fine tuning enabled<br>0x1: Software override for frequency fine tuning disabled |
| phy3_csi_tx_dpll_predef_freq           | 4:0  | CSI PHY3 output frequency in multiples of 100Mhz             | 0bXXXXX: CSI PHY3 output frequency (*100MHz)                                                                          |

**BACKTOP32 (0x327)**

| BIT         | 7              | 6              | 5              | 4              | 3           | 2           | 1           | 0           |
|-------------|----------------|----------------|----------------|----------------|-------------|-------------|-------------|-------------|
| Field       | bpp10dblu_mode | bpp10dblz_mode | bpp10dbly_mode | bpp10dblx_mode | bpp10dblu   | bpp10dblz   | bpp10dbly   | bpp10dblx   |
| Reset       | 0b0            | 0b0            | 0b0            | 0b0            | 0b0         | 0b0         | 0b0         | 0b0         |
| Access Type | Write, Read    | Write, Read    | Write, Read    | Write, Read    | Write, Read | Write, Read | Write, Read | Write, Read |

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| BITFIELD            | BITS | DESCRIPTION                                             | DECODE                                                                          |
|---------------------|------|---------------------------------------------------------|---------------------------------------------------------------------------------|
| bpp10dblu_m<br>ode  | 7    | Enable 8-bit write alternate map to RAMs for pipeline X | 0b0: Write alternative map disabled<br>0x1: Write alternative map enabled       |
| bpp10dblz_m<br>ode  | 6    | Enable 8-bit write alternate map to RAMs for pipeline X | 0b0: Write alternative map disabled<br>0x1: Write alternative map enabled       |
| bpp10dbly_m<br>ode  | 5    | Enable 8-bit write alternate map to RAMs for pipeline X | 0b0: Write alternative map disabled<br>0x1: Write alternative map enabled       |
| bpp10dbl_x_m<br>ode | 4    | Enable 8-bit write alternate map to RAMs for pipeline X | 0b0: Write alternative map disabled<br>0x1: Write alternative map enabled       |
| bpp10dblu           | 3    | BPP=8 processed as 16-bit color enable for pipeline X   | 0b0: Do not process BPP=8 as 16-bit color<br>0b1: Process BPP=8 as 16-bit color |
| bpp10dblz           | 2    | BPP=8 processed as 16-bit color enable for pipeline X   | 0b0: Do not process BPP=8 as 16-bit color<br>0b1: Process BPP=8 as 16-bit color |
| bpp10dbly           | 1    | BPP=8 processed as 16-bit color enable for pipeline X   | 0b0: Do not process BPP=8 as 16-bit color<br>0b1: Process BPP=8 as 16-bit color |
| bpp10dbl_x          | 0    | BPP=8 processed as 16-bit color enable for pipeline X   | 0b0: Do not process BPP=8 as 16-bit color<br>0b1: Process BPP=8 as 16-bit color |

**BACKTOP33 (0x328)**

| BIT         | 7 | 6 | 5 | 4 | 3           | 2           | 1           | 0           |
|-------------|---|---|---|---|-------------|-------------|-------------|-------------|
| Field       | – | – | – | – | bpp12dblu   | bpp12dblz   | bpp12dbly   | bpp12dbl_x  |
| Reset       | – | – | – | – | 0b0         | 0b0         | 0b0         | 0b0         |
| Access Type | – | – | – | – | Write, Read | Write, Read | Write, Read | Write, Read |

| BITFIELD   | BITS | DESCRIPTION                                           | DECODE                                                                          |
|------------|------|-------------------------------------------------------|---------------------------------------------------------------------------------|
| bpp12dblu  | 3    | BPP=8 processed as 16-bit color enable for pipeline X | 0b0: Do not process BPP=8 as 16-bit color<br>0b1: Process BPP=8 as 16-bit color |
| bpp12dblz  | 2    | BPP=8 processed as 16-bit color enable for pipeline X | 0b0: Do not process BPP=8 as 16-bit color<br>0b1: Process BPP=8 as 16-bit color |
| bpp12dbly  | 1    | BPP=8 processed as 16-bit color enable for pipeline X | 0b0: Do not process BPP=8 as 16-bit color<br>0b1: Process BPP=8 as 16-bit color |
| bpp12dbl_x | 0    | BPP=8 processed as 16-bit color enable for pipeline X | 0b0: Do not process BPP=8 as 16-bit color<br>0b1: Process BPP=8 as 16-bit color |

**MIPI\_PHY0 (0x330)**

| BIT         | 7                    | 6                 | 5                 | 4               | 3               | 2           | 1 | 0           |
|-------------|----------------------|-------------------|-------------------|-----------------|-----------------|-------------|---|-------------|
| Field       | force_csi_o<br>ut_en | force_clk3_<br>en | force_clk0_<br>en | phy_1x4b_2<br>2 | phy_1x4a_2<br>2 | phy_2x4     | – | phy_4x2     |
| Reset       | 0b0                  | 0b0               | 0b0               | 0b0             | 0b0             | 0b1         | – | 0b0         |
| Access Type | Write, Read          | Write, Read       | Write, Read       | Write, Read     | Write, Read     | Write, Read | – | Write, Read |

| BITFIELD              | BITS | DESCRIPTION                                          | DECODE                                                                                                           |
|-----------------------|------|------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| force_csi_out_<br>_en | 7    | Force CSI output clock for use in MIPI loopback test | 0b0: CSI output clock not enabled for MIPI loopback test<br>0b1: CSI output clock enabled for MIPI loopback test |



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| BITFIELD      | BITS | DESCRIPTION                             | DECODE                                                                                                                                                                                                          |
|---------------|------|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| force_clk3_en | 6    | Force DPHY3 clock enable in pin control | 0b0: DPHY3 not enabled as clock<br>0b1: DPHY3 enabled as clock                                                                                                                                                  |
| force_clk0_en | 5    | Force DPHY0 clock enable in pin control | 0x0: DPHY0 not enabled as clock<br>0x1: DPHY0 enabled as clock                                                                                                                                                  |
| phy_1x4b_22   | 4    | MIPI output configured as 2x2 and 1x4   | 0b0: 1x4b and 2x2 configuration not selected<br>0b1: MIPI output configured as three ports. Port A and Port B are two data-lane ports, using PHY0 and PHY1. Port C is a four-lane port using PHY2 and PHY3.     |
| phy_1x4a_22   | 3    | MIPI output configured as 1x4 and 2x2   | 0b0: 1x4a and 2x2 configuration not selected<br>0b1: MIPI output configured as three ports. Port A is a four data-lane port using PHY0 and PHY1. Port C and Port D are two data-lane ports using PHY2 and PHY3. |
| phy_2x4       | 2    | MIPI output configured as 2x4           | 0b0: 2x4 not selected<br>0b1: MIPI output configured as two ports with four data lanes each. Port A uses PHY0 and PHY1, port B uses PHY2 and PHY3                                                               |
| phy_4x2       | 0    | MIPI output configured as 4x2           | 0b0: 4x2 configuration not selected<br>0b1: MIPI output configured as four two-lane MIPI ports                                                                                                                  |

**MIPI\_PHY1 (0x331)**

| BIT         | 7                | 6 | 5              | 4 | 3                | 2 | 1                 | 0 |
|-------------|------------------|---|----------------|---|------------------|---|-------------------|---|
| Field       | t_hs_przero[1:0] |   | t_hs_prep[1:0] |   | t_clk_trail[1:0] |   | t_clk_przero[1:0] |   |
| Reset       | 0b00             |   | 0b00           |   | 0b00             |   | 0b00              |   |
| Access Type | Write, Read      |   | Write, Read    |   | Write, Read      |   | Write, Read       |   |

| BITFIELD     | BITS | DESCRIPTION                                         | DECODE                                                                               |
|--------------|------|-----------------------------------------------------|--------------------------------------------------------------------------------------|
| t_hs_przero  | 7:6  | Typical DPHY data lane HS_prepare + HS_zero timing  | 0b00: 146ns + 24UI<br>0b01: 160ns + 24UI<br>0b10: 173ns + 24UI<br>0b11: 200ns + 24UI |
| t_hs_prep    | 5:4  | Typical DPHY data lane HS_prepare timing            | 0b00: 46.7ns + 4UI<br>0b01: 53.4ns + 4UI<br>0b10: 60.0ns + 4UI<br>0b11: 66.7ns + 4UI |
| t_clk_trail  | 3:2  | Typical DPHY clock HS_trail timing                  | 0b00: 160ns<br>0b01: 167ns<br>0b10: 173ns<br>0b11: 180ns                             |
| t_clk_przero | 1:0  | Typical DPHY clock lane HS_prepare + HS_zero timing | 0b00: 306ns<br>0b01: 600ns<br>0b10: 900ns<br>0b11: 1200ns                            |

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**MIPI\_PHY2 (0x332)**

| BIT         | 7                 | 6 | 5 | 4 | 3           | 2 | 1               | 0 |
|-------------|-------------------|---|---|---|-------------|---|-----------------|---|
| Field       | phy_Stdbby_n[3:0] |   |   |   | t_lpx[1:0]  |   | t_hs_trail[1:0] |   |
| Reset       | 0xF               |   |   |   | 0b00        |   | 0b00            |   |
| Access Type | Write, Read       |   |   |   | Write, Read |   | Write, Read     |   |

| BITFIELD     | BITS | DESCRIPTION                                          | DECODE                                                                                                                                                                          |
|--------------|------|------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| phy_Stdbby_n | 7:4  | Puts PHY into standby mode if not used to save power | 0b1111: All PHYs not in standby<br>0bXXX0: Put PHY0 in standby mode<br>0bXX0X: Put PHY1 in standby mode<br>0bX0XX: Put PHY2 in standby mode<br>0b0XXX: Put PHY3 in standby mode |
| t_lpx        | 3:2  | Typical DPHY Tlpx timing                             | 0b00: 53.4ns<br>0b01: 106.7ns<br>0b10: 160ns<br>0b11: 213.4ns                                                                                                                   |
| t_hs_trail   | 1:0  | Typical DPHY data lane HS_trail timing               | 0b00: 66.7ns + 8UI<br>0b01: 80ns + 8UI<br>0b10: 93.4ns + 8UI<br>0b11: 106.7ns + 8UI                                                                                             |

**MIPI\_PHY3 (0x333)**

| BIT         | 7                  | 6 | 5 | 4 | 3                  | 2 | 1 | 0 |
|-------------|--------------------|---|---|---|--------------------|---|---|---|
| Field       | phy1_lane_map[3:0] |   |   |   | phy0_lane_map[3:0] |   |   |   |
| Reset       | 0x4                |   |   |   | 0xE                |   |   |   |
| Access Type | Write, Read        |   |   |   | Write, Read        |   |   |   |

| BITFIELD      | BITS | DESCRIPTION                                                                                                                                                                                                         | DECODE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|---------------|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| phy1_lane_map | 7:4  | Port A lane mapping configurations for 1x4 mode (phy_1x4a_22 enabled) or 2x4 mode (phy_2x4 enabled)<br><br>Port D lane mapping configurations for 4x2 mode (phy_4x2 enabled) or 2x2-lane mode (phy_1x4b_22 enabled) | Port A Lane 0bXX00: Map DA0 to DA0P/N pins (default) 0bXX01: Map DA1 to DA0P/N pins 0bXX10: Map DA2 to DA0P/N pins 0bXX11: Map DA3 to DA0P/N pins 0b00XX: Map DA0 to DA1P/N pins 0b01XX: Map DA1 to DA1P/N pins (default) 0b10XX: Map DA2 to DA1P/N pins 0b11XX: Map DA3 to DA1P/N pins Port D Lane 0bXX00: Map DD0 to DD0P/N pins (default) 0bXX01: Map DD1 to DD0P/N pins 0bXX10: Map DC0 to DD0P/N pins 0bXX11: Map DC1 to DD0P/N pins 0b00XX: Map DD0 to DD1P/N pins 0b01XX: Map DD1 to DD1P/N pins (default) 0b10XX: Map DC0 to DD1P/N pins 0b11XX: Map DC1 to DD1P/N pins |

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| BITFIELD      | BITS | DESCRIPTION                                                                                                                                                                                                         | DECODE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|---------------|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| phy0_lane_map | 3:0  | Port A lane mapping configurations for 1x4 mode (phy_1x4a_22 enabled) or 2x4 mode (phy_2x4 enabled)<br><br>Port C lane mapping configurations for 4x2 mode (phy_4x2 enabled) or 2x2-lane mode (phy_1x4b_22 enabled) | Port A Lane 0bXX00: Map DA0 to DA2P/N pins 0bXX01: Map DA1 to DA2P/N pins 0bXX10: Map DA2 to DA2P/N pins (default) 0bXX11: Map DA3 to DA2P/N pins 0b00XX: Map DA0 to DA3P/N pins 0b01XX: Map DA1 to DA3P/N pins 0b10XX: Map DA2 to DA3P/N pins 0b11XX: Map DA3 to DA3P/N pins (default) Port C Lane 0bXX00: Map DD0 to DC0P/N pins 0bXX01: Map DD1 to DC0P/N pins 0bXX10: Map DC0 to DC0P/N pins (default) 0bXX11: Map DC1 to DC0P/N pins 0b00XX: Map DD0 to DC1P/N pins 0b01XX: Map DD1 to DC1P/N pins 0b10XX: Map DC0 to DC1P/N pins 0b11XX: Map DC1 to DC1P/N pins (default) |

MIPI\_PHY4 (0x334)

| BIT         | 7                  | 6 | 5 | 4 | 3                  | 2 | 1 | 0 |
|-------------|--------------------|---|---|---|--------------------|---|---|---|
| Field       | phy3_lane_map[3:0] |   |   |   | phy2_lane_map[3:0] |   |   |   |
| Reset       | 0xE                |   |   |   | 0x4                |   |   |   |
| Access Type | Write, Read        |   |   |   | Write, Read        |   |   |   |

| BITFIELD      | BITS | DESCRIPTION                                                                                                                                                                                                     | DECODE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|---------------|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| phy3_lane_map | 7:4  | Port B lane mapping configurations for 1x4 mode (phy_1x4b_22 enabled) or 2x4 mode (phy_2x4 enabled)<br>Port E lane mapping configurations for 4x2 mode (phy_4x2 enabled) or 2x2-lane mode (phy_1x4a_22 enabled) | Port B Lane 0bXX00: Map DB0 to DB0P/N pins (default) 0bXX01: Map DB1 to DB0P/N pins 0bXX10: Map DB2 to DB0P/N pins 0bXX11: Map DB3 to DB0P/N pins 0b00XX: Map DB0 to DB1P/N pins 0b01XX: Map DB1 to DB1P/N pins (default) 0b10XX: Map DB2 to DB1P/N pins 0b11XX: Map DB3 to DB1P/N pins Port E Lane 0bXX00: Map DE0 to DE0P/N pins (default) 0bXX01: Map DE1 to DE0P/N pins 0bXX10: Map DF0 to DE0P/N pins 0bXX11: Map DF1 to DE0P/N pins 0b00XX: Map DE0 to DE1P/N pins 0b01XX: Map DE1 to DE1P/N pins (default) 0b10XX: Map DF0 to DE1P/N pins 0b11XX: Map DF1 to DE1P/N pins |
| phy2_lane_map | 3:0  | Port B lane mapping configurations for 1x4 mode (phy_1x4b_22 enabled) or 2x4 mode (phy_2x4 enabled)<br>Port F lane mapping configurations for 4x2 mode (phy_4x2 enabled) or 2x2-lane mode (phy_1x4a_22 enabled) | Port B Lane 0bXX00: Map DB0 to DB2P/N pins 0bXX01: Map DB1 to DB2P/N pins 0bXX10: Map DB2 to DB2P/N pins (default) 0bXX11: Map DB3 to DB2P/N pins 0b00XX: Map DB0 to DB3P/N pins 0b01XX: Map DB1 to DB3P/N pins 0b10XX: Map DB2 to DB3P/N pins 0b11XX: Map DB3 to DB3P/N pins (default) Port F Lane 0bXX00: Map DE0 to DF0P/N pins 0bXX01: Map DE1 to DF0P/N pins 0bXX10: Map DF0 to DF0P/N pins (default) 0bXX11: Map DF1 to DF0P/N pins 0b00XX: Map DE0 to DF1P/N pins 0b01XX: Map DE1 to DF1P/N pins 0b10XX: Map DF0 to DF1P/N pins 0b11XX: Map DF1 to DF1P/N pins (default) |

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**MIPI\_PHY5 (0x335)**

| BIT         | 7               | 6 | 5                 | 4 | 3 | 2                 | 1 | 0 |
|-------------|-----------------|---|-------------------|---|---|-------------------|---|---|
| Field       | t_clk_prep[1:0] |   | phy1_pol_map[2:0] |   |   | phy0_pol_map[2:0] |   |   |
| Reset       | 0b00            |   | 0b000             |   |   | 0b000             |   |   |
| Access Type | Write, Read     |   | Write, Read       |   |   | Write, Read       |   |   |

| BITFIELD     | BITS | DESCRIPTION                                                                                                                                   | DECODE                                                                                                                                                                                                                                                    |
|--------------|------|-----------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| t_clk_prep   | 7:6  | Typical DPHY clock lane HS_prepare timing                                                                                                     | 0b00: 40ns<br>0b01: 46.7ns<br>0b10: 53.4ns<br>0b11: 60ns                                                                                                                                                                                                  |
| phy1_pol_map | 5:3  | PHY1 lane polarity for clock, D1 and D0. In normal polarity, P is positive, N is negative. In inverse polarity, P is negative, N is positive. | 0bXX0: Normal polarity for data lane 0<br>0bXX1: Inverse polarity for data lane 0<br>0bX0X: Normal polarity for data lane 1<br>0bX1X: Inverse polarity for data lane 1<br>0b0XX: Normal polarity for clock lane<br>0b1XX: Inverse polarity for clock lane |
| phy0_pol_map | 2:0  | PHY0 lane polarity for clock, D1 and D0. In normal polarity, P is positive, N is negative. In inverse polarity, P is negative, N is positive. | 0bXX0: Normal polarity for data lane 0<br>0bXX1: Inverse polarity for data lane 0<br>0bX0X: Normal polarity for data lane 1<br>0bX1X: Inverse polarity for data lane 1<br>0b0XX: Normal polarity for clock lane<br>0b1XX: Inverse polarity for clock lane |

**MIPI\_PHY6 (0x336)**

| BIT         | 7           | 6           | 5                 | 4 | 3 | 2                 | 1 | 0 |
|-------------|-------------|-------------|-------------------|---|---|-------------------|---|---|
| Field       | phy_cp1     | phy_cp0     | phy3_pol_map[2:0] |   |   | phy2_pol_map[2:0] |   |   |
| Reset       | 0b0         | 0b0         | 0b000             |   |   | 0b000             |   |   |
| Access Type | Write, Read | Write, Read | Write, Read       |   |   | Write, Read       |   |   |

| BITFIELD     | BITS | DESCRIPTION                                                                                                                                   | DECODE                                                                                                                                                                                                                                                    |
|--------------|------|-----------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| phy_cp1      | 7    | PHY port copy 1                                                                                                                               | 0x0: Disable copy 1<br>0x1: Enable copy 1                                                                                                                                                                                                                 |
| phy_cp0      | 6    | PHY port copy 0                                                                                                                               | 0x0: Disable copy 0<br>0x1: Enable copy 0                                                                                                                                                                                                                 |
| phy3_pol_map | 5:3  | PHY3 lane polarity for clock, D1 and D0. In normal polarity, P is positive, N is negative. In inverse polarity, P is negative, N is positive. | 0bXX0: Normal polarity for data lane 0<br>0bXX1: Inverse polarity for data lane 0<br>0bX0X: Normal polarity for data lane 1<br>0bX1X: Inverse polarity for data lane 1<br>0b0XX: Normal polarity for clock lane<br>0b1XX: Inverse polarity for clock lane |
| phy2_pol_map | 2:0  | PHY2 lane polarity for clock, D1 and D0. In normal polarity, P is positive, N is negative. In inverse polarity, P is negative, N is positive. | 0bXX0: Normal polarity for data lane 0<br>0bXX1: Inverse polarity for data lane 0<br>0bX0X: Normal polarity for data lane 1<br>0bX1X: Inverse polarity for data lane 1<br>0b0XX: Normal polarity for clock lane<br>0b1XX: Inverse polarity for clock lane |

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MIPI\_PHY8 (0x338)

| BIT         | 7             | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|---------------|---|---|---|---|---|---|---|
| Field       | t_lpxesc[2:0] |   |   | – | – | – | – | – |
| Reset       | 0b000         |   |   | – | – | – | – | – |
| Access Type | Write, Read   |   |   | – | – | – | – | – |

| BITFIELD | BITS | DESCRIPTION                             | DECODE                                                                                                                          |
|----------|------|-----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| t_lpxesc | 7:5  | Typical DPHY Tlpx timing in escape mode | 0b000: 66.67ns<br>0b001: 80ns<br>0b010: 100ns<br>0b011: 133ns<br>0b100: 200ns<br>0b101: 400ns<br>0b110: 1000ns<br>0b111: 2000ns |

MIPI\_PHY9 (0x339)

| BIT         | 7                | 6 | 5 | 4 | 3 | 2 | 1 | 0                |
|-------------|------------------|---|---|---|---|---|---|------------------|
| Field       | phy_cp0_dst[1:0] |   | – | – | – | – | – | phy_cp0_overflow |
| Reset       | 0b00             |   | – | – | – | – | – | 0b0              |
| Access Type | Write, Read      |   | – | – | – | – | – | Read Clears All  |

| BITFIELD         | BITS | DESCRIPTION              | DECODE                                        |
|------------------|------|--------------------------|-----------------------------------------------|
| phy_cp0_dst      | 7:6  | PHY copy 0 destination   | 0bXX: PHY copy 0 destination                  |
| phy_cp0_overflow | 0    | PHY copy 0 FIFO overflow | 0x0: No overflow error<br>0x1: Overflow error |

MIPI\_PHY10 (0x33A)

| BIT         | 7                | 6 | 5 | 4 | 3 | 2 | 1 | 0                 |
|-------------|------------------|---|---|---|---|---|---|-------------------|
| Field       | phy_cp0_src[1:0] |   | – | – | – | – | – | phy_cp0_underflow |
| Reset       | 0b00             |   | – | – | – | – | – | 0b0               |
| Access Type | Write, Read      |   | – | – | – | – | – | Read Clears All   |

| BITFIELD          | BITS | DESCRIPTION               | DECODE                                          |
|-------------------|------|---------------------------|-------------------------------------------------|
| phy_cp0_src       | 7:6  | PHY copy 0 source         | 0bXX: PHY copy 0 source                         |
| phy_cp0_underflow | 0    | PHY copy 0 FIFO underflow | 0x0: No underflow error<br>0x1: Underflow error |

MIPI\_PHY11 (0x33B)

| BIT         | 7                | 6 | 5 | 4 | 3 | 2 | 1 | 0                |
|-------------|------------------|---|---|---|---|---|---|------------------|
| Field       | phy_cp1_dst[1:0] |   | – | – | – | – | – | phy_cp1_overflow |
| Reset       | 0b00             |   | – | – | – | – | – | 0b0              |
| Access Type | Write, Read      |   | – | – | – | – | – | Read Clears All  |

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| BITFIELD         | BITS | DESCRIPTION              | DECODE                                        |
|------------------|------|--------------------------|-----------------------------------------------|
| phy_cp1_dst      | 7:6  | PHY copy 1 destination   | 0bXX: PHY copy 1 destination                  |
| phy_cp1_overflow | 0    | PHY copy 1 FIFO overflow | 0x0: No overflow error<br>0x1: Overflow error |

**MIPI\_PHY12 (0x33C)**

| BIT         | 7                | 6 | 5 | 4 | 3 | 2 | 1 | 0                  |
|-------------|------------------|---|---|---|---|---|---|--------------------|
| Field       | phy_cp1_src[1:0] |   | – | – | – | – | – | phy_cp1_underflow  |
| Reset       | 0b00             |   | – | – | – | – | – | 0b0                |
| Access Type | Write, Read      |   | – | – | – | – | – | Read<br>Clears All |

| BITFIELD          | BITS | DESCRIPTION               | DECODE                                          |
|-------------------|------|---------------------------|-------------------------------------------------|
| phy_cp1_src       | 7:6  | Phy copy 1 source         | 0bXX: Phy copy 1 source                         |
| phy_cp1_underflow | 0    | Phy copy 1 FIFO underflow | 0x0: No underflow error<br>0x1: Underflow error |

**FSYNC\_0 (0x3E0)**

| BIT         | 7               | 6 | 5             | 4           | 3               | 2 | 1               | 0 |
|-------------|-----------------|---|---------------|-------------|-----------------|---|-----------------|---|
| Field       | EN_OFLOW_RST_FS | – | FSYNC_OUT_PIN | EN_VS_GEN   | FSYNC_MODE[1:0] |   | FSYNC_METH[1:0] |   |
| Reset       | 0x0             | – | 0b0           | 0x0         | 0x3             |   | 0x2             |   |
| Access Type | Write, Read     | – | Write, Read   | Write, Read | Write, Read     |   | Write, Read     |   |

| BITFIELD        | BITS | DESCRIPTION                                                                                                          | DECODE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-----------------|------|----------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| EN_OFLOW_RST_FS | 7    | Enables link PRBS-7 generator. When enabled, memory overflow resets frame sync generation                            | 0b0: Link PRBS generator disabled<br>0b1: Link PRBS generator enabled                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| FSYNC_OUT_PIN   | 5    | Selects pin to output frame sync signal (effective only when bitfield FSYNC_METH = 0b01)                             | 0b0: N/A<br>0b1: MFP0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| EN_VS_GEN       | 4    | When enabled, VS is generated internally by the frame sync generator (not effective when bitfield FSYNC_MODE = 0b11) | 0x0: Internal frame sync does not generate VS<br>0x1: Internal frame sync generates VS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| FSYNC_MODE      | 3:2  | Frame synchronization mode                                                                                           | 0b00: Frame sync generation is on.<br>GMSL2 mode: GPIO is not used as FSYNC output<br>GMSL1 mode: FSYNC pin is High-Z<br>0b01: Frame sync generation is on.<br>GMSL2 mode: GPIO is used as FSYNC output and drives a slave device<br>GMSL1 mode: FSYNC pin drives a slave device<br>0b10: Frame sync generation is off.<br>GMSL2 mode: GPIO is not used as FSYNC output.<br>GMSL1 mode: FSYNC pin is driven by a master device<br>0b11: Frame sync generation is off.<br>GMSL2 mode: GPIO is not used as FSYNC output<br>GMSL1 mode: GPI-to-GPO channel can be used for frame sync |

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| BITFIELD       | BITS | DESCRIPTION                  | DECODE                                                          |
|----------------|------|------------------------------|-----------------------------------------------------------------|
| FSYNC_MET<br>H | 1:0  | Frame synchronization method | 0b00: Manual<br>0b01: Semi-auto<br>0b10: Auto<br>0b11: Reserved |

**FSYNC\_1 (0x3E1)**

| BIT            | 7 | 6 | 5 | 4 | 3                  | 2 | 1 | 0 |
|----------------|---|---|---|---|--------------------|---|---|---|
| Field          | – | – | – | – | FSYNC_PER_DIV[3:0] |   |   |   |
| Reset          | – | – | – | – | 0x0                |   |   |   |
| Access<br>Type | – | – | – | – | Write, Read        |   |   |   |

| BITFIELD          | BITS | DESCRIPTION                                              | DECODE                                                                                                                                                                   |
|-------------------|------|----------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FSYNC_PER<br>_DIV | 3:0  | Frame sync transmission period in terms of VSYNC periods | 0x0: 1<br>0x1: 2<br>0x2: 3<br>0x3: 6<br>0x4: 8<br>0x5: 10<br>0x6: 12<br>0x7: 16<br>0x8: 20<br>0x9: 24<br>0xA: 32<br>0xB: 48<br>0xC: 64<br>0xD: 80<br>0xE: 96<br>0xF: 128 |

**FSYNC\_2 (0x3E2)**

| BIT            | 7                 | 6 | 5 | 4              | 3           | 2 | 1 | 0 |
|----------------|-------------------|---|---|----------------|-------------|---|---|---|
| Field          | MST_LINK_SEL[2:0] |   |   | K_VAL_SIG<br>N | K_VAL[3:0]  |   |   |   |
| Reset          | 0x4               |   |   | 0x0            | 0x1         |   |   |   |
| Access<br>Type | Write, Read       |   |   | Write, Read    | Write, Read |   |   |   |

| BITFIELD         | BITS | DESCRIPTION                                  | DECODE                                                                                                                                                       |
|------------------|------|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MST_LINK_<br>SEL | 7:5  | Master link select for frame sync generation | 0b000: Video X<br>0b001: Video Y<br>0b010: Video Z<br>0b011: Video U<br>0b100: Auto select<br>0b101: Auto select<br>0b110: Auto select<br>0b111: Auto select |
| K_VAL_SIG<br>N   | 4    | Sign bit of K_VAL                            | 0b0: K_VAL is positive<br>0b1: K_VAL is negative                                                                                                             |



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| BITFIELD | BITS | DESCRIPTION                                                                                                                                                                     | DECODE                                                                                                                                                               |
|----------|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| K_VAL    | 3:0  | Desired frame sync margin (microseconds, typical) with respect to the VSYNC of the slowest link in auto mode or with respect to the VSYNC of the master link in semi-auto mode. | 0x0: 1<br>0x1: 2<br>0x2: 3<br>0x3: 4<br>0x4: 5<br>0x5: 6<br>0x6: 7<br>0x7: 8<br>0x8: 10<br>0x9: 12<br>0xA: 14<br>0xB: 16<br>0xC: 20<br>0xD: 24<br>0xE: 28<br>0xF: 32 |

**FSYNC\_3 (0x3E3)**

| BIT         | 7            | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|--------------|---|---|---|---|---|---|---|
| Field       | P_VAL_L[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00         |   |   |   |   |   |   |   |
| Access Type | Write, Read  |   |   |   |   |   |   |   |

| BITFIELD | BITS | DESCRIPTION                                                                                                                                                                                | DECODE                                          |
|----------|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|
| P_VAL_L  | 7:0  | Low byte of desired frame sync margin in terms of PCLK cycles with respect to the VSYNC of the slowest link in auto mode or with respect to the VSYNC of the master link in semi-auto mode | 0xXX: Low byte of PCLK cycles frame sync margin |

**FSYNC\_4 (0x3E4)**

| BIT         | 7 | 6 | 5           | 4            | 3 | 2 | 1 | 0 |
|-------------|---|---|-------------|--------------|---|---|---|---|
| Field       | – | – | P_VAL_SIG N | P_VAL_H[4:0] |   |   |   |   |
| Reset       | – | – | 0x0         | 0x00         |   |   |   |   |
| Access Type | – | – | Write, Read | Write, Read  |   |   |   |   |

| BITFIELD    | BITS | DESCRIPTION                                                                                                                                                                                 | DECODE                                              |
|-------------|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|
| P_VAL_SIG N | 5    | Sign bit of P_VAL                                                                                                                                                                           | 0b0: P_VAL is positive<br>0b1: P_VAL is negative    |
| P_VAL_H     | 4:0  | High bits of desired frame sync margin in terms of PCLK cycles with respect to the VSYNC of the slowest link in auto mode or with respect to the VSYNC of the master link in semi-auto mode | 0b00000: High bits of PCLK cycles frame sync margin |

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**FSYNC\_5 (0x3E5)**

| BIT         | 7                   | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|---------------------|---|---|---|---|---|---|---|
| Field       | FSYNC_PERIOD_L[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00                |   |   |   |   |   |   |   |
| Access Type | Write, Read         |   |   |   |   |   |   |   |

| BITFIELD       | BITS | DESCRIPTION                                                                                                                              | DECODE                                        |
|----------------|------|------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|
| FSYNC_PERIOD_L | 7:0  | Low byte of frame sync period in terms of pixel clock (effective when FSYNC_0 (0x3E0) bitfields FSYNC_METH = 0b00 and FSYNC_MODE = 0b0X) | 0xXX: Low byte of PCLK cycles in frame period |

**FSYNC\_6 (0x3E6)**

| BIT         | 7                   | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|---------------------|---|---|---|---|---|---|---|
| Field       | FSYNC_PERIOD_M[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00                |   |   |   |   |   |   |   |
| Access Type | Write, Read         |   |   |   |   |   |   |   |

| BITFIELD       | BITS | DESCRIPTION                                                                                                                                 | DECODE                                           |
|----------------|------|---------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|
| FSYNC_PERIOD_M | 7:0  | Middle byte of frame sync period in terms of pixel clock (effective when FSYNC_0 (0x3E0) bitfields FSYNC_METH = 0b00 and FSYNC_MODE = 0b0X) | 0xXX: Middle byte of PCLK cycles in frame period |

**FSYNC\_7 (0x3E7)**

| BIT         | 7                   | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|---------------------|---|---|---|---|---|---|---|
| Field       | FSYNC_PERIOD_H[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00                |   |   |   |   |   |   |   |
| Access Type | Write, Read         |   |   |   |   |   |   |   |

| BITFIELD       | BITS | DESCRIPTION                                                                                                                               | DECODE                                         |
|----------------|------|-------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|
| FSYNC_PERIOD_H | 7:0  | High byte of frame sync period in terms of pixel clock (effective when FSYNC_0 (0x3E0) bitfields FSYNC_METH = 0b00 and FSYNC_MODE = 0b0X) | 0xXX: High byte of PCLK cycles in frame period |

**FSYNC\_8 (0x3E8)**

| BIT         | 7                       | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|-------------------------|---|---|---|---|---|---|---|
| Field       | FRM_DIFF_ERR_THR_L[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00                    |   |   |   |   |   |   |   |
| Access Type | Write, Read             |   |   |   |   |   |   |   |

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| BITFIELD           | BITS | DESCRIPTION                                                                                                                                                                                           | DECODE                                                     |
|--------------------|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|
| FRM_DIFF_ERR_THR_L | 7:0  | Low byte of the error threshold for difference between the earliest and latest VSYNCs in terms of PCLK cycles. The default is 40μs for 96MHz PCLK. The function is disabled when all 13 bits are 0's. | 0xXX: Low byte of PCLK cycles in the VSYNC error threshold |

**FSYNC\_9 (0x3E9)**

| BIT         | 7 | 6 | 5 | 4                       | 3 | 2 | 1 | 0 |
|-------------|---|---|---|-------------------------|---|---|---|---|
| Field       | – | – | – | FRM_DIFF_ERR_THR_H[4:0] |   |   |   |   |
| Reset       | – | – | – | 0x0F                    |   |   |   |   |
| Access Type | – | – | – | Write, Read             |   |   |   |   |

| BITFIELD           | BITS | DESCRIPTION                                                                                                                                                                                            | DECODE                                                         |
|--------------------|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|
| FRM_DIFF_ERR_THR_H | 4:0  | High bits of the error threshold for difference between the earliest and latest VSYNCs in terms of PCLK cycles. The default is 40μs for 96MHz PCLK. The function is disabled when all 13 bits are 0's. | 0bXXXXX: High bits of PCLK cycles in the VSYNC error threshold |

**FSYNC\_10 (0x3EA)**

| BIT         | 7                  | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|--------------------|---|---|---|---|---|---|---|
| Field       | OVLP_WINDOW_L[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x80               |   |   |   |   |   |   |   |
| Access Type | Write, Read        |   |   |   |   |   |   |   |

| BITFIELD      | BITS | DESCRIPTION                                                                                                                                          | DECODE                                                     |
|---------------|------|------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|
| OVLP_WINDOW_L | 7:0  | Low byte of the overlap window value in terms of PCLK cycles. The default is 60μs for 96MHz PCLK. The function is disabled when all 13 bits are 0's. | 0xXX: Low byte of PCLK cycles in the VSYNC overlap window. |

**FSYNC\_11 (0x3EB)**

| BIT         | 7            | 6 | 5 | 4                  | 3 | 2 | 1 | 0 |
|-------------|--------------|---|---|--------------------|---|---|---|---|
| Field       | EN_FSIN_LAST | – | – | OVLP_WINDOW_H[4:0] |   |   |   |   |
| Reset       | 0x0          | – | – | 0x16               |   |   |   |   |
| Access Type | Write, Read  | – | – | Write, Read        |   |   |   |   |

| BITFIELD      | BITS | DESCRIPTION                                                                                                                                          | DECODE                                                                                                  |
|---------------|------|------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| EN_FSIN_LAST  | 7    | FSIN position                                                                                                                                        | 0b0: FSIN can occur anywhere with respect to VS rising edges<br>0b1: FSIN occurs after all rising edges |
| OVLP_WINDOW_H | 4:0  | Low byte of the overlap window value in terms of PCLK cycles. The default is 60μs for 96MHz PCLK. The function is disabled when all 13 bits are 0's. | 0bXXXXX: High bits of PCLK cycles in the VSYNC overlap window.                                          |

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**FSYNC\_16 (0x3F0)**

| BIT         | 7                  | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|--------------------|---|---|---|---|---|---|---|
| Field       | FSYNC_ERR_CNT[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00               |   |   |   |   |   |   |   |
| Access Type | Read Clears All    |   |   |   |   |   |   |   |

| BITFIELD      | BITS | DESCRIPTION                                                                     | DECODE                         |
|---------------|------|---------------------------------------------------------------------------------|--------------------------------|
| FSYNC_ERR_CNT | 7:0  | Frame sync error counter. Resets to 0 when read or when FSYNC_LOCKED goes high. | 0xXX: Frame sync error counter |

**FSYNC\_17 (0x3F1)**

| BIT         | 7                | 6 | 5 | 4 | 3                  | 2 | 1 | 0 |
|-------------|------------------|---|---|---|--------------------|---|---|---|
| Field       | FSYNC_TX_ID[4:0] |   |   |   | FSYNC_ERR_THR[2:0] |   |   |   |
| Reset       | 0x1E             |   |   |   | 0x0                |   |   |   |
| Access Type | Write, Read      |   |   |   | Write, Read        |   |   |   |

| BITFIELD      | BITS | DESCRIPTION                                                                                          | DECODE                                      |
|---------------|------|------------------------------------------------------------------------------------------------------|---------------------------------------------|
| FSYNC_TX_ID   | 7:3  | GPIO ID used for transmitting FSYNC signal                                                           | 0bXXXXX: GPIO ID                            |
| FSYNC_ERR_THR | 2:0  | Frame sync error reporting threshold. FSYNC_ERR_FLAG is asserted when FSYNC_ERR_CNT ≥ FSYNC_ERR_THR. | 0bXXX: Frame sync error reporting threshold |

**FSYNC\_18 (0x3F2)**

| BIT         | 7                   | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|---------------------|---|---|---|---|---|---|---|
| Field       | CALC_FRM_LEN_L[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00                |   |   |   |   |   |   |   |
| Access Type | Read Only           |   |   |   |   |   |   |   |

| BITFIELD       | BITS | DESCRIPTION                                                                                                                                                                              | DECODE                                                                                      |
|----------------|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| CALC_FRM_LEN_L | 7:0  | Low byte of calculated VS period (number of PCLKs) of master link in auto or semi-auto synchronization mode. Use when FSYNC_0 (0x3E0) bitfields FSYNC_METH = 0b10 and FSYNC_MODE = 0b0X. | 0xXX: Low byte of PCLKs in VS period in master link auto or semi-auto synchronization mode. |

**FSYNC\_19 (0x3F3)**

| BIT         | 7                   | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|---------------------|---|---|---|---|---|---|---|
| Field       | CALC_FRM_LEN_M[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00                |   |   |   |   |   |   |   |
| Access Type | Read Only           |   |   |   |   |   |   |   |

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| BITFIELD       | BITS | DESCRIPTION                                                                                                                                                                                 | DECODE                                                                                         |
|----------------|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| CALC_FRM_LEN_M | 7:0  | Middle byte of calculated VS period (number of PCLKs) of master link in auto or semi-auto synchronization mode. Use when FSYNC_0 (0x3E0) bitfields FSYNC_METH = 0b10 and FSYNC_MODE = 0b01. | 0xXX: Middle byte of PCLKs in VS period in master link auto or semi-auto synchronization mode. |

**FSYNC\_20 (0x3F4)**

| BIT         | 7                   | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|---------------------|---|---|---|---|---|---|---|
| Field       | CALC_FRM_LEN_H[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00                |   |   |   |   |   |   |   |
| Access Type | Read Only           |   |   |   |   |   |   |   |

| BITFIELD       | BITS | DESCRIPTION                                                                                                                                                                               | DECODE                                                                                       |
|----------------|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| CALC_FRM_LEN_H | 7:0  | High byte of calculated VS period (number of PCLKs) of master link in auto or semi-auto synchronization mode. Use when FSYNC_0 (0x3E0) bitfields FSYNC_METH = 0b10 and FSYNC_MODE = 0b0X. | 0xXX: High byte of PCLKs in VS period in master link auto or semi-auto synchronization mode. |

**FSYNC\_21 (0x3F5)**

| BIT         | 7               | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|-----------------|---|---|---|---|---|---|---|
| Field       | FRM_DIFF_L[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00            |   |   |   |   |   |   |   |
| Access Type | Read Only       |   |   |   |   |   |   |   |

| BITFIELD   | BITS | DESCRIPTION                                                                                         | DECODE                                                                                                     |
|------------|------|-----------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| FRM_DIFF_L | 7:0  | Low byte of the difference between the fastest and the slowest frame in terms of master PCLK cycles | 0xXX: Low byte of the difference between the fastest and slowest frame in terms of the master PCLK cycles. |

**FSYNC\_22 (0x3F6)**

| BIT         | 7                  | 6            | 5               | 4 | 3 | 2 | 1 | 0 |
|-------------|--------------------|--------------|-----------------|---|---|---|---|---|
| Field       | FSYNC_LOSS_OF_LOCK | FSYNC_LOCKED | FRM_DIFF_H[5:0] |   |   |   |   |   |
| Reset       | 0x0                | 0x0          | 0x00            |   |   |   |   |   |
| Access Type | Read Clears All    | Read Only    | Read Only       |   |   |   |   |   |

| BITFIELD           | BITS | DESCRIPTION                                                                                          | DECODE                                                                                          |
|--------------------|------|------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| FSYNC_LOSS_OF_LOCK | 7    | Frame synchronization lost lock flag. Cleared when read                                              | 0b0: Frame sync lock not lost<br>0b1: Frame sync lock lost                                      |
| FSYNC_LOCKED       | 6    | Frame synchronization lock flag                                                                      | 0b0: Frame sync not locked<br>0b1: Frame sync locked                                            |
| FRM_DIFF_H         | 5:0  | High bits of the difference between the fastest and the slowest frame in terms of master PCLK cycles | 0bXXXXXX: High bits of number of PCLKs in the difference between the fastest and slowest frame. |

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MIPI\_TX1 (0x401)

| BIT         | 7           | 6                                 | 5 | 4 | 3                                                                                                   | 2 | 1 | 0 |
|-------------|-------------|-----------------------------------|---|---|-----------------------------------------------------------------------------------------------------|---|---|---|
| Field       | MODE[7:0]   |                                   |   |   |                                                                                                     |   |   |   |
| Reset       | 0x00        |                                   |   |   |                                                                                                     |   |   |   |
| Access Type | Write, Read |                                   |   |   |                                                                                                     |   |   |   |
| BITFIELD    | BITS        | DESCRIPTION                       |   |   | DECODE                                                                                              |   |   |   |
| MODE        | 7:0         | MIPI VS short packet counter mode |   |   | 0bXXXXXXX0: Disable MIPI VS short packet counter<br>0bXXXXXXX1: Enable MIPI VS short packet counter |   |   |   |

MIPI\_TX2 (0x402)

| BIT         | 7           | 6                                                                                                                                                            | 5 | 4 | 3                                                                                                                                                                                                                                                                  | 2 | 1 | 0 |
|-------------|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|
| Field       | STATUS[7:0] |                                                                                                                                                              |   |   |                                                                                                                                                                                                                                                                    |   |   |   |
| Reset       | 0x00        |                                                                                                                                                              |   |   |                                                                                                                                                                                                                                                                    |   |   |   |
| Access Type | Read Only   |                                                                                                                                                              |   |   |                                                                                                                                                                                                                                                                    |   |   |   |
| BITFIELD    | BITS        | DESCRIPTION                                                                                                                                                  |   |   | DECODE                                                                                                                                                                                                                                                             |   |   |   |
| STATUS      | 7:0         | MIPI Tx status. The register is split into three decode segments:<br>bit [0] SYNC mode enable.<br>bit [1] Video sync flag<br>bit [2] Loss of video sync flag |   |   | 0bXXXXXXX0: SYNC mode disabled<br>0bXXXXXXX1: SYNC mode enabled<br>0bXXXXXXX0X: Video channels not in-sync<br>0bXXXXXXX1X: Video channels in-sync<br>0bXXXXXX0XX: No loss of video sync<br>0bXXXXXX1XX: Video sync lost after last read of this register or RESET. |   |   |   |

MIPI\_TX3 (0x403)

| BIT         | 7                | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|------------------|---|---|---|---|---|---|---|
| Field       | DESKEW_INIT[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00             |   |   |   |   |   |   |   |
| Access Type | Write, Read      |   |   |   |   |   |   |   |

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| BITFIELD    | BITS | DESCRIPTION                                                                                                                                                                                                                                                                                                                                                   | DECODE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-------------|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DESKEW_INIT | 7:0  | DPHY deskew initial calibration control. The register is split into six decode segments:<br>bit [7] Selects auto-initial deskew calibration on or off<br>bit [6] Reserved<br>bit [5] Any bit change initiates an initial calibration if bit [4] = 1<br>bit [4] Selects manual initial on or off<br>bit [3] Reserved<br>bit [2:0] Selects initial deskew width | 0bXXXXXX000: Initial deskew width = 1*32k UI<br>0bXXXXXX001: Initial deskew width = 2*32k UI<br>0bXXXXXX010: Initial deskew width = 3*32k UI<br>0bXXXXXX011: Initial deskew width = 4*32k UI<br>0bXXXXXX100: Initial deskew width = 5*32k UI<br>0bXXXXXX101: Initial deskew width = 6*32k UI<br>0bXXXXXX110: Initial deskew width = 7*32k UI<br>0bXXXXXX111: Initial deskew width = 8*32k UI<br>0bXXXXX0XXX: Reserved<br>0bXXXXX1XXX: Reserved<br>0bXXX0XXXX: Manual initial off<br>0bXXX1XXXX: Manual initial on<br>0bXX0XXXXX: If bit 4 = 1, triggers one time immediate initial skew calibration<br>0bXX1XXXXX: If bit 4 = 1, triggers one time immediate initial skew calibration<br>0bX0XXXXXX: Reserved<br>0bX1XXXXXX: Reserved<br>0b0XXXXXXX: Auto initial deskew off<br>0b1XXXXXXX: Auto initial deskew on |

MIPI\_TX4 (0x404)

| BIT         | 7               | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|-----------------|---|---|---|---|---|---|---|
| Field       | DESKEW_PER[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00            |   |   |   |   |   |   |   |
| Access Type | Write, Read     |   |   |   |   |   |   |   |



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| BITFIELD       | BITS | DESCRIPTION                                                                                                                                                                                                                                                                                            | DECODE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|----------------|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DESKEW_P<br>ER | 7:0  | DPHY periodic deskew calibration control.<br>The register is split into four decode segments:<br>bit [7] Selects periodic deskew calibration on or off<br>bit [6] Selects generation on rising or falling edge of VS<br>bit [5:3] Selects periodic interval<br>bit [2:0] Selects periodic deskew width | 0bXXXXXX000: Periodic deskew width = 1k UI<br>0bXXXXXX001: Periodic deskew width = 2k UI<br>0bXXXXXX010: Periodic deskew width = 3k UI<br>0bXXXXXX011: Periodic deskew width = 4k UI<br>0bXXXXXX100: Periodic deskew width = 5k UI<br>0bXXXXXX101: Periodic deskew width = 6k UI<br>0bXXXXXX110: Periodic deskew width = 7k UI<br>0bXXXXXX111: Periodic deskew width = 8k UI<br>0bXX000XXX: Periodic deskew calibration generated every frame<br>0bXX001XXX: Periodic deskew calibration generated every 2 frames<br>0bXX010XXX: Periodic deskew calibration generated every 4 frames<br>0bXX011XXX: Periodic deskew calibration generated every 8 frames<br>0bXX100XXX: Periodic deskew calibration generated every 16 frames<br>0bXX101XXX: Periodic deskew calibration generated every 32 frames<br>0bXX110XXX: Periodic deskew calibration generated every 64 frames<br>0bXX111XXX: Periodic deskew calibration generated every 128 frames<br>0bX0XXXXXXX: Periodic deskew calibration generated at rising edge of VS<br>0bX1XXXXXXX: Periodic deskew calibration generated at falling edge of VS<br>0b0XXXXXXX: Periodic deskew calibration off<br>0b1XXXXXXX: Periodic deskew calibration on |

**MIPI\_TX5 (0x405)**

| BIT         | 7               | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|-----------------|---|---|---|---|---|---|---|
| Field       | CSI2_T_PRE[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x71            |   |   |   |   |   |   |   |
| Access Type | Write, Read     |   |   |   |   |   |   |   |

| BITFIELD   | BITS | DESCRIPTION                                                | DECODE                           |
|------------|------|------------------------------------------------------------|----------------------------------|
| CSI2_T_PRE | 7:0  | Number of MIPI byte clocks to wait before enabling HS data | 0xXX: Number of MIPI byte clocks |

**MIPI\_TX6 (0x406)**

| BIT         | 7                | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|------------------|---|---|---|---|---|---|---|
| Field       | CSI2_T_POST[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x19             |   |   |   |   |   |   |   |
| Access Type | Write, Read      |   |   |   |   |   |   |   |

| BITFIELD        | BITS | DESCRIPTION                                                | DECODE                           |
|-----------------|------|------------------------------------------------------------|----------------------------------|
| CSI2_T_POS<br>T | 7:0  | Number of MIPI byte clocks to hold clock active after data | 0xXX: Number of MIPI byte clocks |

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**MIPI\_TX7 (0x407)**

| BIT         | 7                | 6                                                                                                                       | 5 | 4 | 3                                | 2 | 1 | 0 |
|-------------|------------------|-------------------------------------------------------------------------------------------------------------------------|---|---|----------------------------------|---|---|---|
| Field       | CSI2_TX_GAP[7:0] |                                                                                                                         |   |   |                                  |   |   |   |
| Reset       | 0x1C             |                                                                                                                         |   |   |                                  |   |   |   |
| Access Type | Write, Read      |                                                                                                                         |   |   |                                  |   |   |   |
| BITFIELD    | BITS             | DESCRIPTION                                                                                                             |   |   | DECODE                           |   |   |   |
| CSI2_TX_GAP | 7:0              | Number of MIPI byte clocks to wait after the HS clock has entered LP before enabling it again for the next transmission |   |   | 0xXX: Number of MIPI byte clocks |   |   |   |

**MIPI\_TX8 (0x408)**

| BIT            | 7                   | 6                                 | 5 | 4 | 3                                | 2 | 1 | 0 |
|----------------|---------------------|-----------------------------------|---|---|----------------------------------|---|---|---|
| Field          | CSI2_TWAKEUP_L[7:0] |                                   |   |   |                                  |   |   |   |
| Reset          | 0x00                |                                   |   |   |                                  |   |   |   |
| Access Type    | Write, Read         |                                   |   |   |                                  |   |   |   |
| BITFIELD       | BITS                | DESCRIPTION                       |   |   | DECODE                           |   |   |   |
| CSI2_TWAKEUP_L | 7:0                 | Low byte of CSI2 transmit wake-up |   |   | 0xXX: Number of MIPI byte clocks |   |   |   |

**MIPI\_TX9 (0x409)**

| BIT            | 7                   | 6                                    | 5 | 4 | 3                                | 2 | 1 | 0 |
|----------------|---------------------|--------------------------------------|---|---|----------------------------------|---|---|---|
| Field          | CSI2_TWAKEUP_M[7:0] |                                      |   |   |                                  |   |   |   |
| Reset          | 0x01                |                                      |   |   |                                  |   |   |   |
| Access Type    | Write, Read         |                                      |   |   |                                  |   |   |   |
| BITFIELD       | BITS                | DESCRIPTION                          |   |   | DECODE                           |   |   |   |
| CSI2_TWAKEUP_M | 7:0                 | Middle byte of CSI2 transmit wake-up |   |   | 0xXX: Number of MIPI byte clocks |   |   |   |

**MIPI\_TX10 (0x40A)**

| BIT         | 7                  | 6 | 5 | 4 | 3           | 2                   | 1 | 0 |
|-------------|--------------------|---|---|---|-------------|---------------------|---|---|
| Field       | CSI2_LANE_CNT[1:0] |   | – | – | CSI_VCX_EN  | CSI2_TWAKEUP_H[2:0] |   |   |
| Reset       | 0b11               |   | – | – | 0b0         | 0b000               |   |   |
| Access Type | Write, Read        |   | – | – | Write, Read | Write, Read         |   |   |

| BITFIELD      | BITS | DESCRIPTION                   | DECODE                                                                                         |
|---------------|------|-------------------------------|------------------------------------------------------------------------------------------------|
| CSI2_LANE_CNT | 7:6  | Set number of MIPI data lanes | 0b00: One data lane<br>0b01: Two data lanes<br>0b10: Three data lanes<br>0b11: Four data lanes |
| CSI_VCX_EN    | 3    | Enables VC extension          | 0b0: Disable VC extension<br>0b1: Enable VC extension                                          |

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| BITFIELD           | BITS | DESCRIPTION                        | DECODE                            |
|--------------------|------|------------------------------------|-----------------------------------|
| CSI2_TWAK<br>EUP_H | 2:0  | High bits of CSI2 transmit wake-up | 0bXXX: Number of MIPI byte clocks |

**MIPI\_TX11 (0x40B)**

| BIT         | 7             | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|---------------|---|---|---|---|---|---|---|
| Field       | MAP_EN_L[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00          |   |   |   |   |   |   |   |
| Access Type | Write, Read   |   |   |   |   |   |   |   |

| BITFIELD | BITS | DESCRIPTION                                                                                                                                                                                                                                                                | DECODE                                                                                                                                                                                                                                                                                                                                                                                                       |
|----------|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MAP_EN_L | 7:0  | Mapping enable low byte [7:0]. Each bit enables 1 of 8 mapping and distribution entries (defined in MAP_SRC_x, MAP_DST_x and MAP_DPHY_DST_x) for the current video stream. Non-matched virtual channel (VC) and data types (DT) pass to the corresponding CSI2 controller. | 0xX1: Enable MAP_SRC_0 and MAP_DST_0 registers<br>0xX2: Enable MAP_SRC_1 and MAP_DST_1 registers<br>0xX4: Enable MAP_SRC_2 and MAP_DST_2 registers<br>0xX8: Enable MAP_SRC_3 and MAP_DST_3 registers<br>0x1X: Enable MAP_SRC_4 and MAP_DST_4 registers<br>0x2X: Enable MAP_SRC_5 and MAP_DST_5 registers<br>0x4X: Enable MAP_SRC_6 and MAP_DST_6 registers<br>0x8X: Enable MAP_SRC_7 and MAP_DST_7 registers |

**MIPI\_TX12 (0x40C)**

| BIT         | 7             | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|---------------|---|---|---|---|---|---|---|
| Field       | MAP_EN_H[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00          |   |   |   |   |   |   |   |
| Access Type | Write, Read   |   |   |   |   |   |   |   |

| BITFIELD | BITS | DESCRIPTION                                                                                                                                                                                                                                                                  | DECODE                                                                                                                                                                                                                                                                                                                                                                                                                   |
|----------|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MAP_EN_H | 7:0  | Mapping enable high byte [15:8]. Each bit enables 1 of 8 mapping and distribution entries (defined in MAP_SRC_x, MAP_DST_x and MAP_DPHY_DST_x) for the current video stream. Non-matched virtual channel (VC) and data types (DT) pass to the corresponding CSI2 controller. | 0xX1: Enable MAP_SRC_8 and MAP_DST_8 registers<br>0xX2: Enable MAP_SRC_9 and MAP_DST_9 registers<br>0xX4: Enable MAP_SRC_10 and MAP_DST_10 registers<br>0xX8: Enable MAP_SRC_11 and MAP_DST_11 registers<br>0x1X: Enable MAP_SRC_12 and MAP_DST_12 registers<br>0x2X: Enable MAP_SRC_13 and MAP_DST_13 registers<br>0x4X: Enable MAP_SRC_14 and MAP_DST_14 registers<br>0x8X: Enable MAP_SRC_15 and MAP_DST_15 registers |

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MIPI\_TX13 (0x40D)

| BIT         | 7              | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|----------------|---|---|---|---|---|---|---|
| Field       | MAP_SRC_0[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00           |   |   |   |   |   |   |   |
| Access Type | Write, Read    |   |   |   |   |   |   |   |

| BITFIELD  | BITS | DESCRIPTION                                                                                                                              | DECODE                           |
|-----------|------|------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| MAP_SRC_0 | 7:0  | Mapping source register 0. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT source for map 0 |

MIPI\_TX14 (0x40E)

| BIT         | 7              | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|----------------|---|---|---|---|---|---|---|
| Field       | MAP_DST_0[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00           |   |   |   |   |   |   |   |
| Access Type | Write, Read    |   |   |   |   |   |   |   |

| BITFIELD  | BITS | DESCRIPTION                                                                                                                                   | DECODE                                |
|-----------|------|-----------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| MAP_DST_0 | 7:0  | Mapping destination register 0. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT destination for map 0 |

MIPI\_TX15 (0x40F)

| BIT         | 7              | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|----------------|---|---|---|---|---|---|---|
| Field       | MAP_SRC_1[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00           |   |   |   |   |   |   |   |
| Access Type | Write, Read    |   |   |   |   |   |   |   |

| BITFIELD  | BITS | DESCRIPTION                                                                                                                              | DECODE                           |
|-----------|------|------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| MAP_SRC_1 | 7:0  | Mapping source register 1. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT source for map 1 |

MIPI\_TX16 (0x410)

| BIT         | 7              | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|----------------|---|---|---|---|---|---|---|
| Field       | MAP_DST_1[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00           |   |   |   |   |   |   |   |
| Access Type | Write, Read    |   |   |   |   |   |   |   |

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| BITFIELD  | BITS | DESCRIPTION                                                                                                                                   | DECODE                                |
|-----------|------|-----------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| MAP_DST_1 | 7:0  | Mapping destination register 1. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT destination for map 1 |

**MIPI\_TX17 (0x411)**

| BIT         | 7              | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|----------------|---|---|---|---|---|---|---|
| Field       | MAP_SRC_2[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00           |   |   |   |   |   |   |   |
| Access Type | Write, Read    |   |   |   |   |   |   |   |

| BITFIELD  | BITS | DESCRIPTION                                                                                                                              | DECODE                           |
|-----------|------|------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| MAP_SRC_2 | 7:0  | Mapping source register 2. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT source for map 2 |

**MIPI\_TX18 (0x412)**

| BIT         | 7              | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|----------------|---|---|---|---|---|---|---|
| Field       | MAP_DST_2[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00           |   |   |   |   |   |   |   |
| Access Type | Write, Read    |   |   |   |   |   |   |   |

| BITFIELD  | BITS | DESCRIPTION                                                                                                                                   | DECODE                                |
|-----------|------|-----------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| MAP_DST_2 | 7:0  | Mapping destination register 2. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT destination for map 2 |

**MIPI\_TX19 (0x413)**

| BIT         | 7              | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|----------------|---|---|---|---|---|---|---|
| Field       | MAP_SRC_3[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00           |   |   |   |   |   |   |   |
| Access Type | Write, Read    |   |   |   |   |   |   |   |

| BITFIELD  | BITS | DESCRIPTION                                                                                                                              | DECODE                           |
|-----------|------|------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| MAP_SRC_3 | 7:0  | Mapping source register 3. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT source for map 3 |

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MIPI\_TX20 (0x414)

| BIT         | 7              | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|----------------|---|---|---|---|---|---|---|
| Field       | MAP_DST_3[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00           |   |   |   |   |   |   |   |
| Access Type | Write, Read    |   |   |   |   |   |   |   |

| BITFIELD  | BITS | DESCRIPTION                                                                                                                                   | DECODE                                |
|-----------|------|-----------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| MAP_DST_3 | 7:0  | Mapping destination register 3. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT destination for map 3 |

MIPI\_TX21 (0x415)

| BIT         | 7              | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|----------------|---|---|---|---|---|---|---|
| Field       | MAP_SRC_4[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00           |   |   |   |   |   |   |   |
| Access Type | Write, Read    |   |   |   |   |   |   |   |

| BITFIELD  | BITS | DESCRIPTION                                                                                                                              | DECODE                           |
|-----------|------|------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| MAP_SRC_4 | 7:0  | Mapping source register 4. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT source for map 4 |

MIPI\_TX22 (0x416)

| BIT         | 7              | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|----------------|---|---|---|---|---|---|---|
| Field       | MAP_DST_4[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00           |   |   |   |   |   |   |   |
| Access Type | Write, Read    |   |   |   |   |   |   |   |

| BITFIELD  | BITS | DESCRIPTION                                                                                                                                   | DECODE                                |
|-----------|------|-----------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| MAP_DST_4 | 7:0  | Mapping destination register 4. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT destination for map 4 |

MIPI\_TX23 (0x417)

| BIT         | 7              | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|----------------|---|---|---|---|---|---|---|
| Field       | MAP_SRC_5[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00           |   |   |   |   |   |   |   |
| Access Type | Write, Read    |   |   |   |   |   |   |   |

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| BITFIELD  | BITS | DESCRIPTION                                                                                                                              | DECODE                           |
|-----------|------|------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| MAP_SRC_5 | 7:0  | Mapping source register 5. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT source for map 5 |

MIPI\_TX24 (0x418)

| BIT         | 7              | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|----------------|---|---|---|---|---|---|---|
| Field       | MAP_DST_5[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00           |   |   |   |   |   |   |   |
| Access Type | Write, Read    |   |   |   |   |   |   |   |

| BITFIELD  | BITS | DESCRIPTION                                                                                                                                   | DECODE                                |
|-----------|------|-----------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| MAP_DST_5 | 7:0  | Mapping destination register 5. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT destination for map 5 |

MIPI\_TX25 (0x419)

| BIT         | 7              | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|----------------|---|---|---|---|---|---|---|
| Field       | MAP_SRC_6[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00           |   |   |   |   |   |   |   |
| Access Type | Write, Read    |   |   |   |   |   |   |   |

| BITFIELD  | BITS | DESCRIPTION                                                                                                                              | DECODE                           |
|-----------|------|------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| MAP_SRC_6 | 7:0  | Mapping source register 6. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT source for map 6 |

MIPI\_TX26 (0x41A)

| BIT         | 7              | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|----------------|---|---|---|---|---|---|---|
| Field       | MAP_DST_6[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00           |   |   |   |   |   |   |   |
| Access Type | Write, Read    |   |   |   |   |   |   |   |

| BITFIELD  | BITS | DESCRIPTION                                                                                                                                   | DECODE                                |
|-----------|------|-----------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| MAP_DST_6 | 7:0  | Mapping destination register 6. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT destination for map 6 |



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Appendix A: Register Map and Tables

MIPI\_TX27 (0x41B)

| BIT         | 7              | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|----------------|---|---|---|---|---|---|---|
| Field       | MAP_SRC_7[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00           |   |   |   |   |   |   |   |
| Access Type | Write, Read    |   |   |   |   |   |   |   |

| BITFIELD  | BITS | DESCRIPTION                                                                                                                              | DECODE                           |
|-----------|------|------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| MAP_SRC_7 | 7:0  | Mapping source register 7. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT source for map 7 |

MIPI\_TX28 (0x41C)

| BIT         | 7              | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|----------------|---|---|---|---|---|---|---|
| Field       | MAP_DST_7[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00           |   |   |   |   |   |   |   |
| Access Type | Write, Read    |   |   |   |   |   |   |   |

| BITFIELD  | BITS | DESCRIPTION                                                                                                                                   | DECODE                                |
|-----------|------|-----------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| MAP_DST_7 | 7:0  | Mapping destination register 7. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT destination for map 7 |

MIPI\_TX29 (0x41D)

| BIT         | 7              | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|----------------|---|---|---|---|---|---|---|
| Field       | MAP_SRC_8[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00           |   |   |   |   |   |   |   |
| Access Type | Write, Read    |   |   |   |   |   |   |   |

| BITFIELD  | BITS | DESCRIPTION                                                                                                                              | DECODE                           |
|-----------|------|------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| MAP_SRC_8 | 7:0  | Mapping source register 8. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT source for map 8 |

MIPI\_TX30 (0x41E)

| BIT         | 7              | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|----------------|---|---|---|---|---|---|---|
| Field       | MAP_DST_8[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00           |   |   |   |   |   |   |   |
| Access Type | Write, Read    |   |   |   |   |   |   |   |

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| BITFIELD  | BITS | DESCRIPTION                                                                                                                                   | DECODE                                |
|-----------|------|-----------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| MAP_DST_8 | 7:0  | Mapping destination register 8. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT destination for map 8 |

**MIPI\_TX31 (0x41F)**

| BIT         | 7              | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|----------------|---|---|---|---|---|---|---|
| Field       | MAP_SRC_9[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00           |   |   |   |   |   |   |   |
| Access Type | Write, Read    |   |   |   |   |   |   |   |

| BITFIELD  | BITS | DESCRIPTION                                                                                                                              | DECODE                           |
|-----------|------|------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| MAP_SRC_9 | 7:0  | Mapping source register 9. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT source for map 9 |

**MIPI\_TX32 (0x420)**

| BIT         | 7              | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|----------------|---|---|---|---|---|---|---|
| Field       | MAP_DST_9[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00           |   |   |   |   |   |   |   |
| Access Type | Write, Read    |   |   |   |   |   |   |   |

| BITFIELD  | BITS | DESCRIPTION                                                                                                                                   | DECODE                                |
|-----------|------|-----------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| MAP_DST_9 | 7:0  | Mapping destination register 9. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT destination for map 9 |

**MIPI\_TX33 (0x421)**

| BIT         | 7               | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|-----------------|---|---|---|---|---|---|---|
| Field       | MAP_SRC_10[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00            |   |   |   |   |   |   |   |
| Access Type | Write, Read     |   |   |   |   |   |   |   |

| BITFIELD   | BITS | DESCRIPTION                                                                                                                               | DECODE                            |
|------------|------|-------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
| MAP_SRC_10 | 7:0  | Mapping source register 10. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT source for map 10 |

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MIPI\_TX34 (0x422)

| BIT         | 7               | 6                                                                                                                                              | 5 | 4 | 3                                      | 2 | 1 | 0 |
|-------------|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------|---|---|----------------------------------------|---|---|---|
| Field       | MAP_DST_10[7:0] |                                                                                                                                                |   |   |                                        |   |   |   |
| Reset       | 0x00            |                                                                                                                                                |   |   |                                        |   |   |   |
| Access Type | Write, Read     |                                                                                                                                                |   |   |                                        |   |   |   |
| BITFIELD    | BITS            | DESCRIPTION                                                                                                                                    |   |   | DECODE                                 |   |   |   |
| MAP_DST_10  | 7:0             | Mapping destination register 10. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type |   |   | 0xXX: VC and DT destination for map 10 |   |   |   |

MIPI\_TX35 (0x423)

| BIT         | 7               | 6                                                                                                                                         | 5 | 4 | 3                                 | 2 | 1 | 0 |
|-------------|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------|---|---|-----------------------------------|---|---|---|
| Field       | MAP_SRC_11[7:0] |                                                                                                                                           |   |   |                                   |   |   |   |
| Reset       | 0x00            |                                                                                                                                           |   |   |                                   |   |   |   |
| Access Type | Write, Read     |                                                                                                                                           |   |   |                                   |   |   |   |
| BITFIELD    | BITS            | DESCRIPTION                                                                                                                               |   |   | DECODE                            |   |   |   |
| MAP_SRC_11  | 7:0             | Mapping source register 11. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type |   |   | 0xXX: VC and DT source for map 11 |   |   |   |

MIPI\_TX36 (0x424)

| BIT         | 7               | 6                                                                                                                                              | 5 | 4 | 3                                      | 2 | 1 | 0 |
|-------------|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------|---|---|----------------------------------------|---|---|---|
| Field       | MAP_DST_11[7:0] |                                                                                                                                                |   |   |                                        |   |   |   |
| Reset       | 0x00            |                                                                                                                                                |   |   |                                        |   |   |   |
| Access Type | Write, Read     |                                                                                                                                                |   |   |                                        |   |   |   |
| BITFIELD    | BITS            | DESCRIPTION                                                                                                                                    |   |   | DECODE                                 |   |   |   |
| MAP_DST_11  | 7:0             | Mapping destination register 11. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type |   |   | 0xXX: VC and DT destination for map 11 |   |   |   |

MIPI\_TX37 (0x425)

| BIT         | 7               | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|-----------------|---|---|---|---|---|---|---|
| Field       | MAP_SRC_12[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00            |   |   |   |   |   |   |   |
| Access Type | Write, Read     |   |   |   |   |   |   |   |

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| BITFIELD   | BITS | DESCRIPTION                                                                                                                               | DECODE                            |
|------------|------|-------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
| MAP_SRC_12 | 7:0  | Mapping source register 12. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT source for map 12 |

**MIPI\_TX38 (0x426)**

| BIT         | 7               | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|-----------------|---|---|---|---|---|---|---|
| Field       | MAP_DST_12[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00            |   |   |   |   |   |   |   |
| Access Type | Write, Read     |   |   |   |   |   |   |   |

| BITFIELD   | BITS | DESCRIPTION                                                                                                                                    | DECODE                                 |
|------------|------|------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| MAP_DST_12 | 7:0  | Mapping destination register 12. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT destination for map 12 |

**MIPI\_TX39 (0x427)**

| BIT         | 7               | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|-----------------|---|---|---|---|---|---|---|
| Field       | MAP_SRC_13[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00            |   |   |   |   |   |   |   |
| Access Type | Write, Read     |   |   |   |   |   |   |   |

| BITFIELD   | BITS | DESCRIPTION                                                                                                                               | DECODE                            |
|------------|------|-------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
| MAP_SRC_13 | 7:0  | Mapping source register 13. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT source for map 13 |

**MIPI\_TX40 (0x428)**

| BIT         | 7               | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|-----------------|---|---|---|---|---|---|---|
| Field       | MAP_DST_13[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00            |   |   |   |   |   |   |   |
| Access Type | Write, Read     |   |   |   |   |   |   |   |

| BITFIELD   | BITS | DESCRIPTION                                                                                                                                    | DECODE                                 |
|------------|------|------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| MAP_DST_13 | 7:0  | Mapping destination register 13. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT destination for map 13 |

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MIPI\_TX41 (0x429)

| BIT         | 7               | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|-----------------|---|---|---|---|---|---|---|
| Field       | MAP_SRC_14[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00            |   |   |   |   |   |   |   |
| Access Type | Write, Read     |   |   |   |   |   |   |   |

| BITFIELD   | BITS | DESCRIPTION                                                                                                                               | DECODE                            |
|------------|------|-------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
| MAP_SRC_14 | 7:0  | Mapping source register 14. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT source for map 14 |

MIPI\_TX42 (0x42A)

| BIT         | 7               | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|-----------------|---|---|---|---|---|---|---|
| Field       | MAP_DST_14[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00            |   |   |   |   |   |   |   |
| Access Type | Write, Read     |   |   |   |   |   |   |   |

| BITFIELD   | BITS | DESCRIPTION                                                                                                                                    | DECODE                                 |
|------------|------|------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| MAP_DST_14 | 7:0  | Mapping destination register 14. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT destination for map 14 |

MIPI\_TX43 (0x42B)

| BIT         | 7               | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|-----------------|---|---|---|---|---|---|---|
| Field       | MAP_SRC_15[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00            |   |   |   |   |   |   |   |
| Access Type | Write, Read     |   |   |   |   |   |   |   |

| BITFIELD   | BITS | DESCRIPTION                                                                                                                               | DECODE                            |
|------------|------|-------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
| MAP_SRC_15 | 7:0  | Mapping source register 15. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT source for map 15 |

MIPI\_TX44 (0x42C)

| BIT         | 7               | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|-----------------|---|---|---|---|---|---|---|
| Field       | MAP_DST_15[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00            |   |   |   |   |   |   |   |
| Access Type | Write, Read     |   |   |   |   |   |   |   |

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| BITFIELD   | BITS | DESCRIPTION                                                                                                                                    | DECODE                                 |
|------------|------|------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| MAP_DST_15 | 7:0  | Mapping destination register 15. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT destination for map 15 |

**MIPI\_TX45 (0x42D)**

| BIT         | 7                    | 6 | 5                    | 4 | 3                    | 2 | 1                    | 0 |
|-------------|----------------------|---|----------------------|---|----------------------|---|----------------------|---|
| Field       | MAP_DPHY_DEST_3[1:0] |   | MAP_DPHY_DEST_2[1:0] |   | MAP_DPHY_DEST_1[1:0] |   | MAP_DPHY_DEST_0[1:0] |   |
| Reset       | 0b00                 |   | 0b00                 |   | 0b00                 |   | 0b00                 |   |
| Access Type | Write, Read          |   | Write, Read          |   | Write, Read          |   | Write, Read          |   |

| BITFIELD        | BITS | DESCRIPTION                                                          | DECODE                                                                               |
|-----------------|------|----------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| MAP_DPHY_DEST_3 | 7:6  | CSI2 controller destination for MAP_SRC_3 and MAP_DST_0 mapping regs | 0b00: Map to DPHY0<br>0b01: Map to DPHY1<br>0b10: Map to DPHY2<br>0b11: Map to DPHY3 |
| MAP_DPHY_DEST_2 | 5:4  | CSI2 controller destination for MAP_SRC_2 and MAP_DST_0 mapping regs | 0b00: Map to DPHY0<br>0b01: Map to DPHY1<br>0b10: Map to DPHY2<br>0b11: Map to DPHY3 |
| MAP_DPHY_DEST_1 | 3:2  | CSI2 controller destination for MAP_SRC_1 and MAP_DST_0 mapping regs | 0b00: Map to DPHY0<br>0b01: Map to DPHY1<br>0b10: Map to DPHY2<br>0b11: Map to DPHY3 |
| MAP_DPHY_DEST_0 | 1:0  | CSI2 controller destination for MAP_SRC_0 and MAP_DST_0 mapping regs | 0b00: Map to DPHY0<br>0b01: Map to DPHY1<br>0b10: Map to DPHY2<br>0b11: Map to DPHY3 |

**MIPI\_TX46 (0x42E)**

| BIT         | 7                    | 6 | 5                    | 4 | 3                    | 2 | 1                    | 0 |
|-------------|----------------------|---|----------------------|---|----------------------|---|----------------------|---|
| Field       | MAP_DPHY_DEST_7[1:0] |   | MAP_DPHY_DEST_6[1:0] |   | MAP_DPHY_DEST_5[1:0] |   | MAP_DPHY_DEST_4[1:0] |   |
| Reset       | 0b00                 |   | 0b00                 |   | 0b00                 |   | 0b00                 |   |
| Access Type | Write, Read          |   | Write, Read          |   | Write, Read          |   | Write, Read          |   |

| BITFIELD        | BITS | DESCRIPTION                                                          | DECODE                                                                               |
|-----------------|------|----------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| MAP_DPHY_DEST_7 | 7:6  | CSI2 controller destination for MAP_SRC_7 and MAP_DST_0 mapping regs | 0b00: Map to DPHY0<br>0b01: Map to DPHY1<br>0b10: Map to DPHY2<br>0b11: Map to DPHY3 |
| MAP_DPHY_DEST_6 | 5:4  | CSI2 controller destination for MAP_SRC_6 and MAP_DST_0 mapping regs | 0b00: Map to DPHY0<br>0b01: Map to DPHY1<br>0b10: Map to DPHY2<br>0b11: Map to DPHY3 |
| MAP_DPHY_DEST_5 | 3:2  | CSI2 controller destination for MAP_SRC_5 and MAP_DST_0 mapping regs | 0b00: Map to DPHY0<br>0b01: Map to DPHY1<br>0b10: Map to DPHY2<br>0b11: Map to DPHY3 |

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| BITFIELD        | BITS | DESCRIPTION                                                          | DECODE                                                                               |
|-----------------|------|----------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| MAP_DPHY_DEST_4 | 1:0  | CSI2 controller destination for MAP_SRC_4 and MAP_DST_0 mapping regs | 0b00: Map to DPHY0<br>0b01: Map to DPHY1<br>0b10: Map to DPHY2<br>0b11: Map to DPHY3 |

**MIPI\_TX47 (0x42F)**

| BIT         | 7                     | 6 | 5                     | 4 | 3                    | 2 | 1                    | 0 |
|-------------|-----------------------|---|-----------------------|---|----------------------|---|----------------------|---|
| Field       | MAP_DPHY_DEST_11[1:0] |   | MAP_DPHY_DEST_10[1:0] |   | MAP_DPHY_DEST_9[1:0] |   | MAP_DPHY_DEST_8[1:0] |   |
| Reset       | 0b00                  |   | 0b00                  |   | 0b00                 |   | 0b00                 |   |
| Access Type | Write, Read           |   | Write, Read           |   | Write, Read          |   | Write, Read          |   |

| BITFIELD         | BITS | DESCRIPTION                                                           | DECODE                                                                               |
|------------------|------|-----------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| MAP_DPHY_DEST_11 | 7:6  | CSI2 controller destination for MAP_SRC_11 and MAP_DST_0 mapping regs | 0b00: Map to DPHY0<br>0b01: Map to DPHY1<br>0b10: Map to DPHY2<br>0b11: Map to DPHY3 |
| MAP_DPHY_DEST_10 | 5:4  | CSI2 controller destination for MAP_SRC_10 and MAP_DST_0 mapping regs | 0b00: Map to DPHY0<br>0b01: Map to DPHY1<br>0b10: Map to DPHY2<br>0b11: Map to DPHY3 |
| MAP_DPHY_DEST_9  | 3:2  | CSI2 controller destination for MAP_SRC_9 and MAP_DST_0 mapping regs  | 0b00: Map to DPHY0<br>0b01: Map to DPHY1<br>0b10: Map to DPHY2<br>0b11: Map to DPHY3 |
| MAP_DPHY_DEST_8  | 1:0  | CSI2 controller destination for MAP_SRC_8 and MAP_DST_0 mapping regs  | 0b00: Map to DPHY0<br>0b01: Map to DPHY1<br>0b10: Map to DPHY2<br>0b11: Map to DPHY3 |

**MIPI\_TX48 (0x430)**

| BIT         | 7                     | 6 | 5                     | 4 | 3                     | 2 | 1                     | 0 |
|-------------|-----------------------|---|-----------------------|---|-----------------------|---|-----------------------|---|
| Field       | MAP_DPHY_DEST_15[1:0] |   | MAP_DPHY_DEST_14[1:0] |   | MAP_DPHY_DEST_13[1:0] |   | MAP_DPHY_DEST_12[1:0] |   |
| Reset       | 0b00                  |   | 0b00                  |   | 0b00                  |   | 0b00                  |   |
| Access Type | Write, Read           |   | Write, Read           |   | Write, Read           |   | Write, Read           |   |

| BITFIELD         | BITS | DESCRIPTION                                                           | DECODE                                                                               |
|------------------|------|-----------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| MAP_DPHY_DEST_15 | 7:6  | CSI2 controller destination for MAP_SRC_15 and MAP_DST_0 mapping regs | 0b00: Map to DPHY0<br>0b01: Map to DPHY1<br>0b10: Map to DPHY2<br>0b11: Map to DPHY3 |
| MAP_DPHY_DEST_14 | 5:4  | CSI2 controller destination for MAP_SRC_14 and MAP_DST_0 mapping regs | 0b00: Map to DPHY0<br>0b01: Map to DPHY1<br>0b10: Map to DPHY2<br>0b11: Map to DPHY3 |



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| BITFIELD         | BITS | DESCRIPTION                                                           | DECODE                                                                               |
|------------------|------|-----------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| MAP_DPHY_DEST_13 | 3:2  | CSI2 controller destination for MAP_SRC_13 and MAP_DST_0 mapping regs | 0b00: Map to DPHY0<br>0b01: Map to DPHY1<br>0b10: Map to DPHY2<br>0b11: Map to DPHY3 |
| MAP_DPHY_DEST_12 | 1:0  | CSI2 controller destination for MAP_SRC_12 and MAP_DST_0 mapping regs | 0b00: Map to DPHY0<br>0b01: Map to DPHY1<br>0b10: Map to DPHY2<br>0b11: Map to DPHY3 |

**MIPI\_TX49 (0x431)**

| BIT         | 7            | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|--------------|---|---|---|---|---|---|---|
| Field       | MAP_CON[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00         |   |   |   |   |   |   |   |
| Access Type | Write, Read  |   |   |   |   |   |   |   |

| BITFIELD | BITS | DESCRIPTION                                                                                                                                                                                                                                                                                                               | DECODE                                                                                                                                                                                                                                                                                                                                                                                                                               |
|----------|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MAP_CON  | 7:0  | Concatenation register for sync mode settings. The register is split into four decode segments:<br>bit [7] Enable 4WxH or Wx4H<br>bit [6] Reserved<br>bit [5:4] Master video stream number. This is the first line to concatenate. All others follow in order<br>bit [3:0] Each bit selects a video stream to concatenate | 0bXXXXXXXX1: Concatenate video stream 0<br>0bXXXXXXXX1X: Concatenate video stream 1<br>0bXXXXX1XX: Concatenate video stream 2<br>0bXXXXX1XXX: Concatenate video stream 3<br>0bXX00XXXX: Select video stream 0 as master<br>0bXX01XXXX: Select video stream 1 as master<br>0bXX10XXXX: Select video stream 2 as master<br>0bXX11XXXX: Select video stream 3 as master<br>0b0XXXXXXX: Enable Wx4H mode<br>0b1XXXXXXX: Enable 4WxH mode |

**MIPI\_TX51 (0x433)**

| BIT         | 7 | 6 | 5 | 4             | 3           | 2             | 1            | 0             |
|-------------|---|---|---|---------------|-------------|---------------|--------------|---------------|
| Field       | – | – | – | ALT2_MEM_MAP8 | MODE_DT     | ALT_MEM_MAP10 | ALT_MEM_MAP8 | ALT_MEM_MAP12 |
| Reset       | – | – | – | 0b0           | 0b0         | 0b0           | 0b0          | 0b0           |
| Access Type | – | – | – | Write, Read   | Write, Read | Write, Read   | Write, Read  | Write, Read   |

| BITFIELD      | BITS | DESCRIPTION                                     | DECODE                                                                                        |
|---------------|------|-------------------------------------------------|-----------------------------------------------------------------------------------------------|
| ALT2_MEM_MAP8 | 4    | Alternative memory map enable for 8-bit DT      | 0b0: Alternate memory map disabled<br>0b1: Alternate memory map enabled                       |
| MODE_DT       | 3    | MIPI Tx enable 24-bit packing of 8-bit MIPI UDP | 0x0: Disable 24-bit packing of 8-bit MIPI UDP<br>0x1: Enable 24-bit packing of 8-bit MIPI UDP |
| ALT_MEM_MAP10 | 2    | Alternative memory map enable for 10-bit DT     | 0b0: Alternate memory map disabled<br>0b1: Alternate memory map enabled                       |
| ALT_MEM_MAP8  | 1    | Alternative memory map enable for 8-bit DT      | 0b0: Alternate memory map disabled<br>0b1: Alternate memory map enabled                       |
| ALT_MEM_MAP12 | 0    | Alternative memory map enable for 12-bit DT     | 0b0: Alternate memory map disabled<br>0b1: Alternate memory map enabled                       |

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Appendix A: Register Map and Tables

MIPI\_TX1 (0x441)

| BIT         | 7           | 6                                 | 5 | 4 | 3                                                                                                   | 2 | 1 | 0 |
|-------------|-------------|-----------------------------------|---|---|-----------------------------------------------------------------------------------------------------|---|---|---|
| Field       | MODE[7:0]   |                                   |   |   |                                                                                                     |   |   |   |
| Reset       | 0x00        |                                   |   |   |                                                                                                     |   |   |   |
| Access Type | Write, Read |                                   |   |   |                                                                                                     |   |   |   |
| BITFIELD    | BITS        | DESCRIPTION                       |   |   | DECODE                                                                                              |   |   |   |
| MODE        | 7:0         | MIPI VS short packet counter mode |   |   | 0bXXXXXXX0: Disable MIPI VS short packet counter<br>0bXXXXXXX1: Enable MIPI VS short packet counter |   |   |   |

MIPI\_TX2 (0x442)

| BIT         | 7           | 6                                                                                                                                                            | 5 | 4 | 3                                                                                                                                                                                                                                                                  | 2 | 1 | 0 |
|-------------|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|
| Field       | STATUS[7:0] |                                                                                                                                                              |   |   |                                                                                                                                                                                                                                                                    |   |   |   |
| Reset       | 0x00        |                                                                                                                                                              |   |   |                                                                                                                                                                                                                                                                    |   |   |   |
| Access Type | Read Only   |                                                                                                                                                              |   |   |                                                                                                                                                                                                                                                                    |   |   |   |
| BITFIELD    | BITS        | DESCRIPTION                                                                                                                                                  |   |   | DECODE                                                                                                                                                                                                                                                             |   |   |   |
| STATUS      | 7:0         | MIPI Tx status. The register is split into three decode segments:<br>bit [0] SYNC mode enable.<br>bit [1] Video sync flag<br>bit [2] Loss of video sync flag |   |   | 0bXXXXXXX0: SYNC mode disabled<br>0bXXXXXXX1: SYNC mode enabled<br>0bXXXXXXX0X: Video channels not in-sync<br>0bXXXXXXX1X: Video channels in-sync<br>0bXXXXXX0XX: No loss of video sync<br>0bXXXXXX1XX: Video sync lost after last read of this register or RESET. |   |   |   |

MIPI\_TX3 (0x443)

| BIT         | 7                | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|------------------|---|---|---|---|---|---|---|
| Field       | DESKEW_INIT[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00             |   |   |   |   |   |   |   |
| Access Type | Write, Read      |   |   |   |   |   |   |   |

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Appendix A: Register Map and Tables

| BITFIELD    | BITS | DESCRIPTION                                                                                                                                                                                                                                                                                                                                                   | DECODE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-------------|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DESKEW_INIT | 7:0  | DPHY deskew initial calibration control. The register is split into six decode segments:<br>bit [7] Selects auto-initial deskew calibration on or off<br>bit [6] Reserved<br>bit [5] Any bit change initiates an initial calibration if bit [4] = 1<br>bit [4] Selects manual initial on or off<br>bit [3] Reserved<br>bit [2:0] Selects initial deskew width | 0bXXXXXX000: Initial deskew width = 1*32k UI<br>0bXXXXXX001: Initial deskew width = 2*32k UI<br>0bXXXXXX010: Initial deskew width = 3*32k UI<br>0bXXXXXX011: Initial deskew width = 4*32k UI<br>0bXXXXXX100: Initial deskew width = 5*32k UI<br>0bXXXXXX101: Initial deskew width = 6*32k UI<br>0bXXXXXX110: Initial deskew width = 7*32k UI<br>0bXXXXXX111: Initial deskew width = 8*32k UI<br>0bXXXXX0XXX: Reserved<br>0bXXXXX1XXX: Reserved<br>0bXXX0XXXX: Manual initial off<br>0bXXX1XXXX: Manual initial on<br>0bXX0XXXXX: If bit 4 = 1, triggers one time immediate initial skew calibration<br>0bXX1XXXXX: If bit 4 = 1, triggers one time immediate initial skew calibration<br>0bX0XXXXXX: Reserved<br>0bX1XXXXXX: Reserved<br>0b0XXXXXXX: Auto initial deskew off<br>0b1XXXXXXX: Auto initial deskew on |

MIPI\_TX4 (0x444)

| BIT         | 7               | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|-----------------|---|---|---|---|---|---|---|
| Field       | DESKEW_PER[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00            |   |   |   |   |   |   |   |
| Access Type | Write, Read     |   |   |   |   |   |   |   |

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| BITFIELD       | BITS | DESCRIPTION                                                                                                                                                                                                                                                                                            | DECODE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|----------------|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DESKEW_P<br>ER | 7:0  | DPHY periodic deskew calibration control.<br>The register is split into four decode segments:<br>bit [7] Selects periodic deskew calibration on or off<br>bit [6] Selects generation on rising or falling edge of VS<br>bit [5:3] Selects periodic interval<br>bit [2:0] Selects periodic deskew width | 0bXXXXXX000: Periodic deskew width = 1k UI<br>0bXXXXXX001: Periodic deskew width = 2k UI<br>0bXXXXXX010: Periodic deskew width = 3k UI<br>0bXXXXXX011: Periodic deskew width = 4k UI<br>0bXXXXXX100: Periodic deskew width = 5k UI<br>0bXXXXXX101: Periodic deskew width = 6k UI<br>0bXXXXXX110: Periodic deskew width = 7k UI<br>0bXXXXXX111: Periodic deskew width = 8k UI<br>0bXX000XXX: Periodic deskew calibration generated every frame<br>0bXX001XXX: Periodic deskew calibration generated every 2 frames<br>0bXX010XXX: Periodic deskew calibration generated every 4 frames<br>0bXX011XXX: Periodic deskew calibration generated every 8 frames<br>0bXX100XXX: Periodic deskew calibration generated every 16 frames<br>0bXX101XXX: Periodic deskew calibration generated every 32 frames<br>0bXX110XXX: Periodic deskew calibration generated every 64 frames<br>0bXX111XXX: Periodic deskew calibration generated every 128 frames<br>0bX0XXXXXXX: Periodic deskew calibration generated at rising edge of VS<br>0bX1XXXXXXX: Periodic deskew calibration generated at falling edge of VS<br>0b0XXXXXXX: Periodic deskew calibration off<br>0b1XXXXXXX: Periodic deskew calibration on |

**MIPI\_TX5 (0x445)**

| BIT         | 7               | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|-----------------|---|---|---|---|---|---|---|
| Field       | CSI2_T_PRE[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x71            |   |   |   |   |   |   |   |
| Access Type | Write, Read     |   |   |   |   |   |   |   |

| BITFIELD   | BITS | DESCRIPTION                                                | DECODE                           |
|------------|------|------------------------------------------------------------|----------------------------------|
| CSI2_T_PRE | 7:0  | Number of MIPI byte clocks to wait before enabling HS data | 0xXX: Number of MIPI byte clocks |

**MIPI\_TX6 (0x446)**

| BIT         | 7                | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|------------------|---|---|---|---|---|---|---|
| Field       | CSI2_T_POST[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x19             |   |   |   |   |   |   |   |
| Access Type | Write, Read      |   |   |   |   |   |   |   |

| BITFIELD        | BITS | DESCRIPTION                                                | DECODE                           |
|-----------------|------|------------------------------------------------------------|----------------------------------|
| CSI2_T_POS<br>T | 7:0  | Number of MIPI byte clocks to hold clock active after data | 0xXX: Number of MIPI byte clocks |

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Appendix A: Register Map and Tables

MIPI\_TX7 (0x447)

| BIT         | 7                | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|------------------|---|---|---|---|---|---|---|
| Field       | CSI2_TX_GAP[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x1C             |   |   |   |   |   |   |   |
| Access Type | Write, Read      |   |   |   |   |   |   |   |

| BITFIELD    | BITS | DESCRIPTION                                                                                                                              | DECODE                           |
|-------------|------|------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| CSI2_TX_GAP | 7:0  | Number of MIPI byte clocks to wait after the HS clock has entered low-power (LP) mode before enabling it again for the next transmission | 0xXX: Number of MIPI byte clocks |

MIPI\_TX8 (0x448)

| BIT         | 7                   | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|---------------------|---|---|---|---|---|---|---|
| Field       | CSI2_TWAKEUP_L[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00                |   |   |   |   |   |   |   |
| Access Type | Write, Read         |   |   |   |   |   |   |   |

| BITFIELD       | BITS | DESCRIPTION                       | DECODE                           |
|----------------|------|-----------------------------------|----------------------------------|
| CSI2_TWAKEUP_L | 7:0  | Low byte of CSI2 transmit wake-up | 0xXX: Number of MIPI byte clocks |

MIPI\_TX9 (0x449)

| BIT         | 7                   | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|---------------------|---|---|---|---|---|---|---|
| Field       | CSI2_TWAKEUP_M[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x01                |   |   |   |   |   |   |   |
| Access Type | Write, Read         |   |   |   |   |   |   |   |

| BITFIELD       | BITS | DESCRIPTION                          | DECODE                           |
|----------------|------|--------------------------------------|----------------------------------|
| CSI2_TWAKEUP_M | 7:0  | Middle byte of CSI2 transmit wake-up | 0xXX: Number of MIPI byte clocks |

MIPI\_TX10 (0x44A)

| BIT         | 7                  | 6 | 5 | 4 | 3           | 2                   | 1 | 0 |
|-------------|--------------------|---|---|---|-------------|---------------------|---|---|
| Field       | CSI2_LANE_CNT[1:0] |   | – | – | CSI_VCX_EN  | CSI2_TWAKEUP_H[2:0] |   |   |
| Reset       | 0b11               |   | – | – | 0b0         | 0b000               |   |   |
| Access Type | Write, Read        |   | – | – | Write, Read | Write, Read         |   |   |

| BITFIELD      | BITS | DESCRIPTION                   | DECODE                                                                                         |
|---------------|------|-------------------------------|------------------------------------------------------------------------------------------------|
| CSI2_LANE_CNT | 7:6  | Set number of MIPI data lanes | 0b00: One data lane<br>0b01: Two data lanes<br>0b10: Three data lanes<br>0b11: Four data lanes |
| CSI_VCX_EN    | 3    | Enables VC extension          | 0x0: Disable VC extension<br>0x1: Enable VC extension                                          |

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| BITFIELD           | BITS | DESCRIPTION                        | DECODE                           |
|--------------------|------|------------------------------------|----------------------------------|
| CSI2_TWAK<br>EUP_H | 2:0  | High bits of CSI2 transmit wake-up | 0xXX: Number of MIPI byte clocks |

**MIPI\_TX11 (0x44B)**

| BIT         | 7             | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|---------------|---|---|---|---|---|---|---|
| Field       | MAP_EN_L[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00          |   |   |   |   |   |   |   |
| Access Type | Write, Read   |   |   |   |   |   |   |   |

| BITFIELD | BITS | DESCRIPTION                                                                                                                                                                                                                                                                | DECODE                                                                                                                                                                                                                                                                                                                                                                                                       |
|----------|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MAP_EN_L | 7:0  | Mapping enable low byte [7:0]. Each bit enables 1 of 8 mapping and distribution entries (defined in MAP_SRC_x, MAP_DST_x and MAP_DPHY_DST_x) for the current video stream. Non-matched virtual channel (VC) and data types (DT) pass to the corresponding CSI2 controller. | 0xX1: Enable Map_SRC_0 and Map_DST_0 registers<br>0xX2: Enable MAP_SRC_1 and MAP_DST_1 registers<br>0xX4: Enable MAP_SRC_2 and MAP_DST_2 registers<br>0xX8: Enable MAP_SRC_3 and MAP_DST_3 registers<br>0x1X: Enable MAP_SRC_4 and MAP_DST_4 registers<br>0x2X: Enable MAP_SRC_5 and MAP_DST_5 registers<br>0x4X: Enable MAP_SRC_6 and MAP_DST_6 registers<br>0x8X: Enable MAP_SRC_7 and MAP_DST_7 registers |

**MIPI\_TX12 (0x44C)**

| BIT         | 7             | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|---------------|---|---|---|---|---|---|---|
| Field       | MAP_EN_H[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00          |   |   |   |   |   |   |   |
| Access Type | Write, Read   |   |   |   |   |   |   |   |

| BITFIELD | BITS | DESCRIPTION                                                                                                                                                                                                                                                                  | DECODE                                                                                                                                                                                                                                                                                                                                                                                                                   |
|----------|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MAP_EN_H | 7:0  | Mapping enable high byte [15:8]. Each bit enables 1 of 8 mapping and distribution entries (defined in MAP_SRC_x, MAP_DST_x and MAP_DPHY_DST_x) for the current video stream. Non-matched virtual channel (VC) and data types (DT) pass to the corresponding CSI2 controller. | 0xX1: Enable MAP_SRC_8 and MAP_DST_8 registers<br>0xX2: Enable MAP_SRC_9 and MAP_DST_9 registers<br>0xX4: Enable MAP_SRC_10 and MAP_DST_10 registers<br>0xX8: Enable MAP_SRC_11 and MAP_DST_11 registers<br>0x1X: Enable MAP_SRC_12 and MAP_DST_12 registers<br>0x2X: Enable MAP_SRC_13 and MAP_DST_13 registers<br>0x4X: Enable MAP_SRC_14 and MAP_DST_14 registers<br>0x8X: Enable MAP_SRC_15 and MAP_DST_15 registers |

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MIPI\_TX13 (0x44D)

| BIT         | 7              | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|----------------|---|---|---|---|---|---|---|
| Field       | MAP_SRC_0[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00           |   |   |   |   |   |   |   |
| Access Type | Write, Read    |   |   |   |   |   |   |   |

| BITFIELD  | BITS | DESCRIPTION                                                                                                                              | DECODE                           |
|-----------|------|------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| MAP_SRC_0 | 7:0  | Mapping source register 0. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT source for map 0 |

MIPI\_TX14 (0x44E)

| BIT         | 7              | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|----------------|---|---|---|---|---|---|---|
| Field       | MAP_DST_0[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00           |   |   |   |   |   |   |   |
| Access Type | Write, Read    |   |   |   |   |   |   |   |

| BITFIELD  | BITS | DESCRIPTION                                                                                                                                   | DECODE                                |
|-----------|------|-----------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| MAP_DST_0 | 7:0  | Mapping destination register 0. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT destination for map 0 |

MIPI\_TX15 (0x44F)

| BIT         | 7              | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|----------------|---|---|---|---|---|---|---|
| Field       | MAP_SRC_1[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00           |   |   |   |   |   |   |   |
| Access Type | Write, Read    |   |   |   |   |   |   |   |

| BITFIELD  | BITS | DESCRIPTION                                                                                                                              | DECODE                           |
|-----------|------|------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| MAP_SRC_1 | 7:0  | Mapping source register 1. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT source for map 1 |

MIPI\_TX16 (0x450)

| BIT         | 7              | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|----------------|---|---|---|---|---|---|---|
| Field       | MAP_DST_1[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00           |   |   |   |   |   |   |   |
| Access Type | Write, Read    |   |   |   |   |   |   |   |



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| BITFIELD  | BITS | DESCRIPTION                                                                                                                                   | DECODE                                |
|-----------|------|-----------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| MAP_DST_1 | 7:0  | Mapping destination register 1. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT destination for map 1 |

**MIPI\_TX17 (0x451)**

| BIT         | 7              | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|----------------|---|---|---|---|---|---|---|
| Field       | MAP_SRC_2[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00           |   |   |   |   |   |   |   |
| Access Type | Write, Read    |   |   |   |   |   |   |   |

| BITFIELD  | BITS | DESCRIPTION                                                                                                                              | DECODE                           |
|-----------|------|------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| MAP_SRC_2 | 7:0  | Mapping source register 2. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT source for map 2 |

**MIPI\_TX18 (0x452)**

| BIT         | 7              | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|----------------|---|---|---|---|---|---|---|
| Field       | MAP_DST_2[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00           |   |   |   |   |   |   |   |
| Access Type | Write, Read    |   |   |   |   |   |   |   |

| BITFIELD  | BITS | DESCRIPTION                                                                                                                                   | DECODE                                |
|-----------|------|-----------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| MAP_DST_2 | 7:0  | Mapping destination register 2. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT destination for map 2 |

**MIPI\_TX19 (0x453)**

| BIT         | 7              | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|----------------|---|---|---|---|---|---|---|
| Field       | MAP_SRC_3[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00           |   |   |   |   |   |   |   |
| Access Type | Write, Read    |   |   |   |   |   |   |   |

| BITFIELD  | BITS | DESCRIPTION                                                                                                                              | DECODE                           |
|-----------|------|------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| MAP_SRC_3 | 7:0  | Mapping source register 3. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT source for map 3 |

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MIPI\_TX20 (0x454)

| BIT         | 7              | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|----------------|---|---|---|---|---|---|---|
| Field       | MAP_DST_3[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00           |   |   |   |   |   |   |   |
| Access Type | Write, Read    |   |   |   |   |   |   |   |

| BITFIELD  | BITS | DESCRIPTION                                                                                                                                   | DECODE                                |
|-----------|------|-----------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| MAP_DST_3 | 7:0  | Mapping destination register 3. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT destination for map 3 |

MIPI\_TX21 (0x455)

| BIT         | 7              | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|----------------|---|---|---|---|---|---|---|
| Field       | MAP_SRC_4[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00           |   |   |   |   |   |   |   |
| Access Type | Write, Read    |   |   |   |   |   |   |   |

| BITFIELD  | BITS | DESCRIPTION                                                                                                                              | DECODE                           |
|-----------|------|------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| MAP_SRC_4 | 7:0  | Mapping source register 4. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT source for map 4 |

MIPI\_TX22 (0x456)

| BIT         | 7              | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|----------------|---|---|---|---|---|---|---|
| Field       | MAP_DST_4[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00           |   |   |   |   |   |   |   |
| Access Type | Write, Read    |   |   |   |   |   |   |   |

| BITFIELD  | BITS | DESCRIPTION                                                                                                                                   | DECODE                                |
|-----------|------|-----------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| MAP_DST_4 | 7:0  | Mapping destination register 4. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT destination for map 4 |

MIPI\_TX23 (0x457)

| BIT         | 7              | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|----------------|---|---|---|---|---|---|---|
| Field       | MAP_SRC_5[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00           |   |   |   |   |   |   |   |
| Access Type | Write, Read    |   |   |   |   |   |   |   |

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| BITFIELD  | BITS | DESCRIPTION                                                                                                                              | DECODE                           |
|-----------|------|------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| MAP_SRC_5 | 7:0  | Mapping source register 5. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT source for map 5 |

MIPI\_TX24 (0x458)

| BIT         | 7              | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|----------------|---|---|---|---|---|---|---|
| Field       | MAP_DST_5[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00           |   |   |   |   |   |   |   |
| Access Type | Write, Read    |   |   |   |   |   |   |   |

| BITFIELD  | BITS | DESCRIPTION                                                                                                                                   | DECODE                                |
|-----------|------|-----------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| MAP_DST_5 | 7:0  | Mapping destination register 5. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT destination for map 5 |

MIPI\_TX25 (0x459)

| BIT         | 7              | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|----------------|---|---|---|---|---|---|---|
| Field       | MAP_SRC_6[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00           |   |   |   |   |   |   |   |
| Access Type | Write, Read    |   |   |   |   |   |   |   |

| BITFIELD  | BITS | DESCRIPTION                                                                                                                              | DECODE                           |
|-----------|------|------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| MAP_SRC_6 | 7:0  | Mapping source register 6. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT source for map 6 |

MIPI\_TX26 (0x45A)

| BIT         | 7              | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|----------------|---|---|---|---|---|---|---|
| Field       | MAP_DST_6[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00           |   |   |   |   |   |   |   |
| Access Type | Write, Read    |   |   |   |   |   |   |   |

| BITFIELD  | BITS | DESCRIPTION                                                                                                                                   | DECODE                                |
|-----------|------|-----------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| MAP_DST_6 | 7:0  | Mapping destination register 6. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT destination for map 6 |

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Appendix A: Register Map and Tables

MIPI\_TX27 (0x45B)

| BIT         | 7              | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|----------------|---|---|---|---|---|---|---|
| Field       | MAP_SRC_7[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00           |   |   |   |   |   |   |   |
| Access Type | Write, Read    |   |   |   |   |   |   |   |

| BITFIELD  | BITS | DESCRIPTION                                                                                                                              | DECODE                           |
|-----------|------|------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| MAP_SRC_7 | 7:0  | Mapping source register 7. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT source for map 7 |

MIPI\_TX28 (0x45C)

| BIT         | 7              | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|----------------|---|---|---|---|---|---|---|
| Field       | MAP_DST_7[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00           |   |   |   |   |   |   |   |
| Access Type | Write, Read    |   |   |   |   |   |   |   |

| BITFIELD  | BITS | DESCRIPTION                                                                                                                                   | DECODE                                |
|-----------|------|-----------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| MAP_DST_7 | 7:0  | Mapping destination register 7. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT destination for map 7 |

MIPI\_TX29 (0x45D)

| BIT         | 7              | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|----------------|---|---|---|---|---|---|---|
| Field       | MAP_SRC_8[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00           |   |   |   |   |   |   |   |
| Access Type | Write, Read    |   |   |   |   |   |   |   |

| BITFIELD  | BITS | DESCRIPTION                                                                                                                              | DECODE                           |
|-----------|------|------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| MAP_SRC_8 | 7:0  | Mapping source register 8. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT source for map 8 |

MIPI\_TX30 (0x45E)

| BIT         | 7              | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|----------------|---|---|---|---|---|---|---|
| Field       | MAP_DST_8[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00           |   |   |   |   |   |   |   |
| Access Type | Write, Read    |   |   |   |   |   |   |   |

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| BITFIELD  | BITS | DESCRIPTION                                                                                                                                   | DECODE                                |
|-----------|------|-----------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| MAP_DST_8 | 7:0  | Mapping destination register 8. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT destination for map 8 |

**MIPI\_TX31 (0x45F)**

| BIT         | 7              | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|----------------|---|---|---|---|---|---|---|
| Field       | MAP_SRC_9[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00           |   |   |   |   |   |   |   |
| Access Type | Write, Read    |   |   |   |   |   |   |   |

| BITFIELD  | BITS | DESCRIPTION                                                                                                                              | DECODE                           |
|-----------|------|------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| MAP_SRC_9 | 7:0  | Mapping source register 9. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT source for map 9 |

**MIPI\_TX32 (0x460)**

| BIT         | 7              | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|----------------|---|---|---|---|---|---|---|
| Field       | MAP_DST_9[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00           |   |   |   |   |   |   |   |
| Access Type | Write, Read    |   |   |   |   |   |   |   |

| BITFIELD  | BITS | DESCRIPTION                                                                                                                                   | DECODE                                |
|-----------|------|-----------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| MAP_DST_9 | 7:0  | Mapping destination register 9. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT destination for map 9 |

**MIPI\_TX33 (0x461)**

| BIT         | 7               | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|-----------------|---|---|---|---|---|---|---|
| Field       | MAP_SRC_10[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00            |   |   |   |   |   |   |   |
| Access Type | Write, Read     |   |   |   |   |   |   |   |

| BITFIELD   | BITS | DESCRIPTION                                                                                                                               | DECODE                            |
|------------|------|-------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
| MAP_SRC_10 | 7:0  | Mapping source register 10. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT source for map 10 |

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MIPI\_TX34 (0x462)

| BIT         | 7               | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|-----------------|---|---|---|---|---|---|---|
| Field       | MAP_DST_10[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00            |   |   |   |   |   |   |   |
| Access Type | Write, Read     |   |   |   |   |   |   |   |

| BITFIELD   | BITS | DESCRIPTION                                                                                                                                    | DECODE                                 |
|------------|------|------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| MAP_DST_10 | 7:0  | Mapping destination register 10. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT destination for map 10 |

MIPI\_TX35 (0x463)

| BIT         | 7               | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|-----------------|---|---|---|---|---|---|---|
| Field       | MAP_SRC_11[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00            |   |   |   |   |   |   |   |
| Access Type | Write, Read     |   |   |   |   |   |   |   |

| BITFIELD   | BITS | DESCRIPTION                                                                                                                               | DECODE                            |
|------------|------|-------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
| MAP_SRC_11 | 7:0  | Mapping source register 11. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT source for map 11 |

MIPI\_TX36 (0x464)

| BIT         | 7               | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|-----------------|---|---|---|---|---|---|---|
| Field       | MAP_DST_11[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00            |   |   |   |   |   |   |   |
| Access Type | Write, Read     |   |   |   |   |   |   |   |

| BITFIELD   | BITS | DESCRIPTION                                                                                                                                    | DECODE                                 |
|------------|------|------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| MAP_DST_11 | 7:0  | Mapping destination register 11. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT destination for map 11 |

MIPI\_TX37 (0x465)

| BIT         | 7               | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|-----------------|---|---|---|---|---|---|---|
| Field       | MAP_SRC_12[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00            |   |   |   |   |   |   |   |
| Access Type | Write, Read     |   |   |   |   |   |   |   |

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Appendix A: Register Map and Tables

| BITFIELD   | BITS | DESCRIPTION                                                                                                                               | DECODE                            |
|------------|------|-------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
| MAP_SRC_12 | 7:0  | Mapping source register 12. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT source for map 12 |

MIPI\_TX38 (0x466)

| BIT         | 7               | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|-----------------|---|---|---|---|---|---|---|
| Field       | MAP_DST_12[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00            |   |   |   |   |   |   |   |
| Access Type | Write, Read     |   |   |   |   |   |   |   |

| BITFIELD   | BITS | DESCRIPTION                                                                                                                                    | DECODE                                 |
|------------|------|------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| MAP_DST_12 | 7:0  | Mapping destination register 12. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT destination for map 12 |

MIPI\_TX39 (0x467)

| BIT         | 7               | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|-----------------|---|---|---|---|---|---|---|
| Field       | MAP_SRC_13[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00            |   |   |   |   |   |   |   |
| Access Type | Write, Read     |   |   |   |   |   |   |   |

| BITFIELD   | BITS | DESCRIPTION                                                                                                                               | DECODE                            |
|------------|------|-------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
| MAP_SRC_13 | 7:0  | Mapping source register 13. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT source for map 13 |

MIPI\_TX40 (0x468)

| BIT         | 7               | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|-----------------|---|---|---|---|---|---|---|
| Field       | MAP_DST_13[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00            |   |   |   |   |   |   |   |
| Access Type | Write, Read     |   |   |   |   |   |   |   |

| BITFIELD   | BITS | DESCRIPTION                                                                                                                                    | DECODE                                 |
|------------|------|------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| MAP_DST_13 | 7:0  | Mapping destination register 13. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT destination for map 13 |



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MIPI\_TX41 (0x469)

| BIT         | 7               | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|-----------------|---|---|---|---|---|---|---|
| Field       | MAP_SRC_14[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00            |   |   |   |   |   |   |   |
| Access Type | Write, Read     |   |   |   |   |   |   |   |

| BITFIELD   | BITS | DESCRIPTION                                                                                                                               | DECODE                            |
|------------|------|-------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
| MAP_SRC_14 | 7:0  | Mapping source register 14. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT source for map 14 |

MIPI\_TX42 (0x46A)

| BIT         | 7               | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|-----------------|---|---|---|---|---|---|---|
| Field       | MAP_DST_14[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00            |   |   |   |   |   |   |   |
| Access Type | Write, Read     |   |   |   |   |   |   |   |

| BITFIELD   | BITS | DESCRIPTION                                                                                                                                    | DECODE                                 |
|------------|------|------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| MAP_DST_14 | 7:0  | Mapping destination register 14. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT destination for map 14 |

MIPI\_TX43 (0x46B)

| BIT         | 7               | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|-----------------|---|---|---|---|---|---|---|
| Field       | MAP_SRC_15[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00            |   |   |   |   |   |   |   |
| Access Type | Write, Read     |   |   |   |   |   |   |   |

| BITFIELD   | BITS | DESCRIPTION                                                                                                                               | DECODE                            |
|------------|------|-------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
| MAP_SRC_15 | 7:0  | Mapping source register 15. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT source for map 15 |

MIPI\_TX44 (0x46C)

| BIT         | 7               | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|-----------------|---|---|---|---|---|---|---|
| Field       | MAP_DST_15[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00            |   |   |   |   |   |   |   |
| Access Type | Write, Read     |   |   |   |   |   |   |   |

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| BITFIELD   | BITS | DESCRIPTION                                                                                                                                    | DECODE                                 |
|------------|------|------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| MAP_DST_15 | 7:0  | Mapping destination register 15. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT destination for map 15 |

**MIPI\_TX45 (0x46D)**

| BIT         | 7                    | 6 | 5                    | 4 | 3                    | 2 | 1                    | 0 |
|-------------|----------------------|---|----------------------|---|----------------------|---|----------------------|---|
| Field       | MAP_DPHY_DEST_3[1:0] |   | MAP_DPHY_DEST_2[1:0] |   | MAP_DPHY_DEST_1[1:0] |   | MAP_DPHY_DEST_0[1:0] |   |
| Reset       | 0b00                 |   | 0b00                 |   | 0b00                 |   | 0b00                 |   |
| Access Type | Write, Read          |   | Write, Read          |   | Write, Read          |   | Write, Read          |   |

| BITFIELD        | BITS | DESCRIPTION                                                          | DECODE                                                                               |
|-----------------|------|----------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| MAP_DPHY_DEST_3 | 7:6  | CSI2 controller destination for MAP_SRC_3 and MAP_DST_0 mapping regs | 0b00: Map to DPHY0<br>0b01: Map to DPHY1<br>0b10: Map to DPHY2<br>0b11: Map to DPHY3 |
| MAP_DPHY_DEST_2 | 5:4  | CSI2 controller destination for MAP_SRC_2 and MAP_DST_0 mapping regs | 0b00: Map to DPHY0<br>0b01: Map to DPHY1<br>0b10: Map to DPHY2<br>0b11: Map to DPHY3 |
| MAP_DPHY_DEST_1 | 3:2  | CSI2 controller destination for MAP_SRC_1 and MAP_DST_0 mapping regs | 0b00: Map to DPHY0<br>0b01: Map to DPHY1<br>0b10: Map to DPHY2<br>0b11: Map to DPHY3 |
| MAP_DPHY_DEST_0 | 1:0  | CSI2 controller destination for MAP_SRC_0 and MAP_DST_0 mapping regs | 0b00: Map to DPHY0<br>0b01: Map to DPHY1<br>0b10: Map to DPHY2<br>0b11: Map to DPHY3 |

**MIPI\_TX46 (0x46E)**

| BIT         | 7                    | 6 | 5                    | 4 | 3                    | 2 | 1                    | 0 |
|-------------|----------------------|---|----------------------|---|----------------------|---|----------------------|---|
| Field       | MAP_DPHY_DEST_7[1:0] |   | MAP_DPHY_DEST_6[1:0] |   | MAP_DPHY_DEST_5[1:0] |   | MAP_DPHY_DEST_4[1:0] |   |
| Reset       | 0b00                 |   | 0b00                 |   | 0b00                 |   | 0b00                 |   |
| Access Type | Write, Read          |   | Write, Read          |   | Write, Read          |   | Write, Read          |   |

| BITFIELD        | BITS | DESCRIPTION                                                          | DECODE                                                                               |
|-----------------|------|----------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| MAP_DPHY_DEST_7 | 7:6  | CSI2 controller destination for MAP_SRC_7 and MAP_DST_0 mapping regs | 0b00: Map to DPHY0<br>0b01: Map to DPHY1<br>0b10: Map to DPHY2<br>0b11: Map to DPHY3 |
| MAP_DPHY_DEST_6 | 5:4  | CSI2 controller destination for MAP_SRC_6 and MAP_DST_0 mapping regs | 0b00: Map to DPHY0<br>0b01: Map to DPHY1<br>0b10: Map to DPHY2<br>0b11: Map to DPHY3 |
| MAP_DPHY_DEST_5 | 3:2  | CSI2 controller destination for MAP_SRC_5 and MAP_DST_0 mapping regs | 0b00: Map to DPHY0<br>0b01: Map to DPHY1<br>0b10: Map to DPHY2<br>0b11: Map to DPHY3 |

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| BITFIELD        | BITS | DESCRIPTION                                                          | DECODE                                                                               |
|-----------------|------|----------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| MAP_DPHY_DEST_4 | 1:0  | CSI2 controller destination for MAP_SRC_4 and MAP_DST_0 mapping regs | 0b00: Map to DPHY0<br>0b01: Map to DPHY1<br>0b10: Map to DPHY2<br>0b11: Map to DPHY3 |

**MIPI\_TX47 (0x46F)**

| BIT         | 7                     | 6 | 5                     | 4 | 3                    | 2 | 1                    | 0 |
|-------------|-----------------------|---|-----------------------|---|----------------------|---|----------------------|---|
| Field       | MAP_DPHY_DEST_11[1:0] |   | MAP_DPHY_DEST_10[1:0] |   | MAP_DPHY_DEST_9[1:0] |   | MAP_DPHY_DEST_8[1:0] |   |
| Reset       | 0b00                  |   | 0b00                  |   | 0b00                 |   | 0b00                 |   |
| Access Type | Write, Read           |   | Write, Read           |   | Write, Read          |   | Write, Read          |   |

| BITFIELD         | BITS | DESCRIPTION                                                           | DECODE                                                                               |
|------------------|------|-----------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| MAP_DPHY_DEST_11 | 7:6  | CSI2 controller destination for MAP_SRC_11 and MAP_DST_0 mapping regs | 0b00: Map to DPHY0<br>0b01: Map to DPHY1<br>0b10: Map to DPHY2<br>0b11: Map to DPHY3 |
| MAP_DPHY_DEST_10 | 5:4  | CSI2 controller destination for MAP_SRC_10 and MAP_DST_0 mapping regs | 0b00: Map to DPHY0<br>0b01: Map to DPHY1<br>0b10: Map to DPHY2<br>0b11: Map to DPHY3 |
| MAP_DPHY_DEST_9  | 3:2  | CSI2 controller destination for MAP_SRC_9 and MAP_DST_0 mapping regs  | 0b00: Map to DPHY0<br>0b01: Map to DPHY1<br>0b10: Map to DPHY2<br>0b11: Map to DPHY3 |
| MAP_DPHY_DEST_8  | 1:0  | CSI2 controller destination for MAP_SRC_8 and MAP_DST_0 mapping regs  | 0b00: Map to DPHY0<br>0b01: Map to DPHY1<br>0b10: Map to DPHY2<br>0b11: Map to DPHY3 |

**MIPI\_TX48 (0x470)**

| BIT         | 7                     | 6 | 5                     | 4 | 3                     | 2 | 1                     | 0 |
|-------------|-----------------------|---|-----------------------|---|-----------------------|---|-----------------------|---|
| Field       | MAP_DPHY_DEST_15[1:0] |   | MAP_DPHY_DEST_14[1:0] |   | MAP_DPHY_DEST_13[1:0] |   | MAP_DPHY_DEST_12[1:0] |   |
| Reset       | 0b00                  |   | 0b00                  |   | 0b00                  |   | 0b00                  |   |
| Access Type | Write, Read           |   | Write, Read           |   | Write, Read           |   | Write, Read           |   |

| BITFIELD         | BITS | DESCRIPTION                                                           | DECODE                                                                               |
|------------------|------|-----------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| MAP_DPHY_DEST_15 | 7:6  | CSI2 controller destination for MAP_SRC_15 and MAP_DST_0 mapping regs | 0b00: Map to DPHY0<br>0b01: Map to DPHY1<br>0b10: Map to DPHY2<br>0b11: Map to DPHY3 |
| MAP_DPHY_DEST_14 | 5:4  | CSI2 controller destination for MAP_SRC_14 and MAP_DST_0 mapping regs | 0b00: Map to DPHY0<br>0b01: Map to DPHY1<br>0b10: Map to DPHY2<br>0b11: Map to DPHY3 |

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| BITFIELD         | BITS | DESCRIPTION                                                           | DECODE                                                                               |
|------------------|------|-----------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| MAP_DPHY_DEST_13 | 3:2  | CSI2 controller destination for MAP_SRC_13 and MAP_DST_0 mapping regs | 0b00: Map to DPHY0<br>0b01: Map to DPHY1<br>0b10: Map to DPHY2<br>0b11: Map to DPHY3 |
| MAP_DPHY_DEST_12 | 1:0  | CSI2 controller destination for MAP_SRC_12 and MAP_DST_0 mapping regs | 0b00: Map to DPHY0<br>0b01: Map to DPHY1<br>0b10: Map to DPHY2<br>0b11: Map to DPHY3 |

**MIPI\_TX49 (0x471)**

| BIT         | 7            | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|--------------|---|---|---|---|---|---|---|
| Field       | MAP_CON[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00         |   |   |   |   |   |   |   |
| Access Type | Write, Read  |   |   |   |   |   |   |   |

| BITFIELD | BITS | DESCRIPTION                                                                                                                                                                                                                                                                                                               | DECODE                                                                                                                                                                                                                                                                                                                                                                                                                               |
|----------|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MAP_CON  | 7:0  | Concatenation register for sync mode settings. The register is split into four decode segments:<br>bit [7] Enable 4WxH or Wx4H<br>bit [6] Reserved<br>bit [5:4] Master video stream number. This is the first line to concatenate. All others follow in order<br>bit [3:0] Each bit selects a video stream to concatenate | 0bXXXXXXXX1: Concatenate video stream 0<br>0bXXXXXXXX1X: Concatenate video stream 1<br>0bXXXXX1XX: Concatenate video stream 2<br>0bXXXXX1XXX: Concatenate video stream 3<br>0bXX00XXXX: Select video stream 0 as master<br>0bXX01XXXX: Select video stream 1 as master<br>0bXX10XXXX: Select video stream 2 as master<br>0bXX11XXXX: Select video stream 3 as master<br>0b0XXXXXXX: Enable Wx4H mode<br>0b1XXXXXXX: Enable 4WxH mode |

**MIPI\_TX50 (0x472)**

| BIT         | 7                 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|-------------------|---|---|---|---|---|---|---|
| Field       | SKEW_PER_SEL[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00              |   |   |   |   |   |   |   |
| Access Type | Write, Read       |   |   |   |   |   |   |   |

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| BITFIELD     | BITS | DESCRIPTION                                                                                                                                                                                                                                                                                              | DECODE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|--------------|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SKEW_PER_SEL | 7:0  | Periodic deskew select register. The register is split into three decode segments:<br>bit [7] Select periodic deskew calibration for one or all virtual channels<br>bit [6:5] Reserved<br>bit [4:0] Virtual channel to generate periodic deskew calibration when only one channel is selected by bit [7] | 0b0XXXXXXX: Generate periodic calibration deskew calibration on all virtual channels<br>0b1XX00000: Periodic deskew calibration generated by virtual channel 0<br>0b1XX00001: Periodic deskew calibration generated by virtual channel 1<br>0b1XX00010: Periodic deskew calibration generated by virtual channel 2<br>0b1XX00011: Periodic deskew calibration generated by virtual channel 3<br>0b1XX00100: Periodic deskew calibration generated by virtual channel 4<br>0b1XX00101: Periodic deskew calibration generated by virtual channel 5<br>0b1XX00110: Periodic deskew calibration generated by virtual channel 6<br>0b1XX00111: Periodic deskew calibration generated by virtual channel 7<br>0b1XX01000: Periodic deskew calibration generated by virtual channel 8<br>0b1XX01001: Periodic deskew calibration generated by virtual channel 9<br>0b1XX01010: Periodic deskew calibration generated by virtual channel 10<br>0b1XX01011: Periodic deskew calibration generated by virtual channel 11<br>0b1XX01100: Periodic deskew calibration generated by virtual channel 12<br>0b1XX01101: Periodic deskew calibration generated by virtual channel 13<br>0b1XX01110: Periodic deskew calibration generated by virtual channel 14<br>0b1XX01111: Periodic deskew calibration generated by virtual channel 15 |

**MIPI\_TX51 (0x473)**

| BIT         | 7 | 6 | 5 | 4             | 3           | 2             | 1            | 0             |
|-------------|---|---|---|---------------|-------------|---------------|--------------|---------------|
| Field       | — | — | — | ALT2_MEM_MAP8 | MODE_DT     | ALT_MEM_MAP10 | ALT_MEM_MAP8 | ALT_MEM_MAP12 |
| Reset       | — | — | — | 0b0           | 0b0         | 0b0           | 0b0          | 0b0           |
| Access Type | — | — | — | Write, Read   | Write, Read | Write, Read   | Write, Read  | Write, Read   |

| BITFIELD      | BITS | DESCRIPTION                                     | DECODE                                                                                        |
|---------------|------|-------------------------------------------------|-----------------------------------------------------------------------------------------------|
| ALT2_MEM_MAP8 | 4    | Alternative memory map enable for 8-bit DT      | 0b0: Alternate memory map disabled<br>0b1: Alternate memory map enabled                       |
| MODE_DT       | 3    | MIPI Tx enable 24-bit packing of 8-bit MIPI UDP | 0x0: Disable 24-bit packing of 8-bit MIPI UDP<br>0x1: Enable 24-bit packing of 8-bit MIPI UDP |
| ALT_MEM_MAP10 | 2    | Alternative memory map enable for 10-bit DT     | 0b0: Alternate memory map disabled<br>0b1: Alternate memory map enabled                       |
| ALT_MEM_MAP8  | 1    | Alternative memory map enable for 8-bit DT      | 0b0: Alternate memory map disabled<br>0b1: Alternate memory map enabled                       |

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| BITFIELD      | BITS | DESCRIPTION                                 | DECODE                                                                  |
|---------------|------|---------------------------------------------|-------------------------------------------------------------------------|
| ALT_MEM_MAP12 | 0    | Alternative memory map enable for 12-bit DT | 0b0: Alternate memory map disabled<br>0b1: Alternate memory map enabled |

**MIPI\_TX1 (0x481)**

| BIT         | 7           | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|-------------|---|---|---|---|---|---|---|
| Field       | MODE[7:0]   |   |   |   |   |   |   |   |
| Reset       | 0x00        |   |   |   |   |   |   |   |
| Access Type | Write, Read |   |   |   |   |   |   |   |

| BITFIELD | BITS | DESCRIPTION                       | DECODE                                                                                              |
|----------|------|-----------------------------------|-----------------------------------------------------------------------------------------------------|
| MODE     | 7:0  | MIPI VS short packet counter mode | 0bXXXXXXX0: Disable MIPI VS short packet counter<br>0bXXXXXXX1: Enable MIPI VS short packet counter |

**MIPI\_TX2 (0x482)**

| BIT         | 7           | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|-------------|---|---|---|---|---|---|---|
| Field       | STATUS[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00        |   |   |   |   |   |   |   |
| Access Type | Read Only   |   |   |   |   |   |   |   |

| BITFIELD | BITS | DESCRIPTION                                                                                                                                                  | DECODE                                                                                                                                                                                                                                                             |
|----------|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| STATUS   | 7:0  | MIPI TX status. The register is split into three decode segments:<br>bit [0] SYNC mode enable.<br>bit [1] Video sync flag<br>bit [2] Loss of video sync flag | 0bXXXXXXX0: SYNC mode disabled<br>0bXXXXXXX1: SYNC mode enabled<br>0bXXXXXXX0X: Video channels not in-sync<br>0bXXXXXXX1X: Video channels in-sync<br>0bXXXXXX0XX: No loss of video sync<br>0bXXXXXX1XX: Video sync lost after last read of this register or RESET. |

**MIPI\_TX3 (0x483)**

| BIT         | 7                | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|------------------|---|---|---|---|---|---|---|
| Field       | DESKEW_INIT[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00             |   |   |   |   |   |   |   |
| Access Type | Write, Read      |   |   |   |   |   |   |   |

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| BITFIELD    | BITS | DESCRIPTION                                                                                                                                                                                                                                                                                                                                                   | DECODE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-------------|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DESKEW_INIT | 7:0  | DPHY deskew initial calibration control. The register is split into six decode segments:<br>bit [7] Selects auto-initial deskew calibration on or off<br>bit [6] Reserved<br>bit [5] Any bit change initiates an initial calibration if bit [4] = 1<br>bit [4] Selects manual initial on or off<br>bit [3] Reserved<br>bit [2:0] Selects initial deskew width | 0bXXXXX000: Initial deskew width = 1*32k UI<br>0bXXXXX001: Initial deskew width = 2*32k UI<br>0bXXXXX010: Initial deskew width = 3*32k UI<br>0bXXXXX011: Initial deskew width = 4*32k UI<br>0bXXXXX100: Initial deskew width = 5*32k UI<br>0bXXXXX101: Initial deskew width = 6*32k UI<br>0bXXXXX110: Initial deskew width = 7*32k UI<br>0bXXXXX111: Initial deskew width = 8*32k UI<br>0bXXXX0XXX: Reserved<br>0bXXXX1XXX: Reserved<br>0bXXX0XXXX: Manual initial off<br>0bXXX1XXXX: Manual initial on<br>0bXX0XXXXX: If bit 4 = 1, triggers one time immediate initial skew calibration<br>0bXX1XXXXX: If bit 4 = 1, triggers one time immediate initial skew calibration<br>0bX0XXXXXX: Reserved<br>0bX1XXXXXX: Reserved<br>0b0XXXXXXX: Auto initial deskew off<br>0b1XXXXXXX: Auto initial deskew on |

MIPI\_TX4 (0x484)

| BIT         | 7               | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|-----------------|---|---|---|---|---|---|---|
| Field       | DESKEW_PER[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00            |   |   |   |   |   |   |   |
| Access Type | Write, Read     |   |   |   |   |   |   |   |



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| BITFIELD       | BITS | DESCRIPTION                                                                                                                                                                                                                                                                                            | DECODE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|----------------|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DESKEW_P<br>ER | 7:0  | DPHY periodic deskew calibration control.<br>The register is split into four decode segments:<br>bit [7] Selects periodic deskew calibration on or off<br>bit [6] Selects generation on rising or falling edge of VS<br>bit [5:3] Selects periodic interval<br>bit [2:0] Selects periodic deskew width | 0bXXXXXX000: Periodic deskew width = 1k UI<br>0bXXXXXX001: Periodic deskew width = 2k UI<br>0bXXXXXX010: Periodic deskew width = 3k UI<br>0bXXXXXX011: Periodic deskew width = 4k UI<br>0bXXXXXX100: Periodic deskew width = 5k UI<br>0bXXXXXX101: Periodic deskew width = 6k UI<br>0bXXXXXX110: Periodic deskew width = 7k UI<br>0bXXXXXX111: Periodic deskew width = 8k UI<br>0bXX000XXX: Periodic deskew calibration generated every frame<br>0bXX001XXX: Periodic deskew calibration generated every 2 frames<br>0bXX010XXX: Periodic deskew calibration generated every 4 frames<br>0bXX011XXX: Periodic deskew calibration generated every 8 frames<br>0bXX100XXX: Periodic deskew calibration generated every 16 frames<br>0bXX101XXX: Periodic deskew calibration generated every 32 frames<br>0bXX110XXX: Periodic deskew calibration generated every 64 frames<br>0bXX111XXX: Periodic deskew calibration generated every 128 frames<br>0bX0XXXXXXX: Periodic deskew calibration generated at rising edge of VS<br>0bX1XXXXXXX: Periodic deskew calibration generated at falling edge of VS<br>0b0XXXXXXX: Periodic deskew calibration off<br>0b1XXXXXXX: Periodic deskew calibration on |

**MIPI\_TX5 (0x485)**

| BIT         | 7               | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|-----------------|---|---|---|---|---|---|---|
| Field       | CSI2_T_PRE[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x71            |   |   |   |   |   |   |   |
| Access Type | Write, Read     |   |   |   |   |   |   |   |

| BITFIELD   | BITS | DESCRIPTION                                                | DECODE                           |
|------------|------|------------------------------------------------------------|----------------------------------|
| CSI2_T_PRE | 7:0  | Number of MIPI byte clocks to wait before enabling HS data | 0xXX: Number of MIPI byte clocks |

**MIPI\_TX6 (0x486)**

| BIT         | 7                | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|------------------|---|---|---|---|---|---|---|
| Field       | CSI2_T_POST[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x19             |   |   |   |   |   |   |   |
| Access Type | Write, Read      |   |   |   |   |   |   |   |

| BITFIELD        | BITS | DESCRIPTION                                                | DECODE                           |
|-----------------|------|------------------------------------------------------------|----------------------------------|
| CSI2_T_POS<br>T | 7:0  | Number of MIPI byte clocks to hold clock active after data | 0xXX: Number of MIPI byte clocks |

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MIPI\_TX7 (0x487)

| BIT         | 7                | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|------------------|---|---|---|---|---|---|---|
| Field       | CSI2_TX_GAP[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x1C             |   |   |   |   |   |   |   |
| Access Type | Write, Read      |   |   |   |   |   |   |   |

| BITFIELD    | BITS | DESCRIPTION                                                                                                                              | DECODE                           |
|-------------|------|------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| CSI2_TX_GAP | 7:0  | Number of MIPI byte clocks to wait after the HS clock has entered low-power (LP) mode before enabling it again for the next transmission | 0xXX: Number of MIPI byte clocks |

MIPI\_TX8 (0x488)

| BIT         | 7                   | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|---------------------|---|---|---|---|---|---|---|
| Field       | CSI2_TWAKEUP_L[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00                |   |   |   |   |   |   |   |
| Access Type | Write, Read         |   |   |   |   |   |   |   |

| BITFIELD       | BITS | DESCRIPTION                       | DECODE                           |
|----------------|------|-----------------------------------|----------------------------------|
| CSI2_TWAKEUP_L | 7:0  | Low byte of CSI2 transmit wake-up | 0xXX: Number of MIPI byte clocks |

MIPI\_TX9 (0x489)

| BIT         | 7                   | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|---------------------|---|---|---|---|---|---|---|
| Field       | CSI2_TWAKEUP_M[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x01                |   |   |   |   |   |   |   |
| Access Type | Write, Read         |   |   |   |   |   |   |   |

| BITFIELD       | BITS | DESCRIPTION                          | DECODE                           |
|----------------|------|--------------------------------------|----------------------------------|
| CSI2_TWAKEUP_M | 7:0  | Middle byte of CSI2 transmit wake-up | 0xXX: Number of MIPI byte clocks |

MIPI\_TX10 (0x48A)

| BIT         | 7                  | 6 | 5 | 4 | 3           | 2                   | 1 | 0 |
|-------------|--------------------|---|---|---|-------------|---------------------|---|---|
| Field       | CSI2_LANE_CNT[1:0] |   | – | – | CSI_VCX_EN  | CSI2_TWAKEUP_H[2:0] |   |   |
| Reset       | 0b11               |   | – | – | 0b0         | 0b000               |   |   |
| Access Type | Write, Read        |   | – | – | Write, Read | Write, Read         |   |   |

| BITFIELD      | BITS | DESCRIPTION                   | DECODE                                                                                         |
|---------------|------|-------------------------------|------------------------------------------------------------------------------------------------|
| CSI2_LANE_CNT | 7:6  | Set number of MIPI data lanes | 0b00: One data lane<br>0b01: Two data lanes<br>0b10: Three data lanes<br>0b11: Four data lanes |
| CSI_VCX_EN    | 3    | Enables VC extension          | 0x0: Disable VC extension<br>0x1: Enable VC extension                                          |

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| BITFIELD           | BITS | DESCRIPTION                        | DECODE                           |
|--------------------|------|------------------------------------|----------------------------------|
| CSI2_TWAK<br>EUP_H | 2:0  | High bits of CSI2 transmit wake-up | 0xXX: Number of MIPI byte clocks |

**MIPI\_TX11 (0x48B)**

| BIT         | 7             | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|---------------|---|---|---|---|---|---|---|
| Field       | MAP_EN_L[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00          |   |   |   |   |   |   |   |
| Access Type | Write, Read   |   |   |   |   |   |   |   |

| BITFIELD | BITS | DESCRIPTION                                                                                                                                                                                                                                                                | DECODE                                                                                                                                                                                                                                                                                                                                                                                                       |
|----------|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MAP_EN_L | 7:0  | Mapping enable low byte [7:0]. Each bit enables 1 of 8 mapping and distribution entries (defined in MAP_SRC_x, MAP_DST_x and MAP_DPHY_DST_x) for the current video stream. Non-matched virtual channel (VC) and data types (DT) pass to the corresponding CSI2 controller. | 0xX1: Enable MAP_SRC_0 and MAP_DST_0 registers<br>0xX2: Enable MAP_SRC_1 and MAP_DST_1 registers<br>0xX4: Enable MAP_SRC_2 and MAP_DST_2 registers<br>0xX8: Enable MAP_SRC_3 and MAP_DST_3 registers<br>0x1X: Enable MAP_SRC_4 and MAP_DST_4 registers<br>0x2X: Enable MAP_SRC_5 and MAP_DST_5 registers<br>0x4X: Enable MAP_SRC_6 and MAP_DST_6 registers<br>0x8X: Enable MAP_SRC_7 and MAP_DST_7 registers |

**MIPI\_TX12 (0x48C)**

| BIT         | 7             | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|---------------|---|---|---|---|---|---|---|
| Field       | MAP_EN_H[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00          |   |   |   |   |   |   |   |
| Access Type | Write, Read   |   |   |   |   |   |   |   |

| BITFIELD | BITS | DESCRIPTION                                                                                                                                                                                                                                                                  | DECODE                                                                                                                                                                                                                                                                                                                                                                                                                   |
|----------|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MAP_EN_H | 7:0  | Mapping enable high byte [15:8]. Each bit enables 1 of 8 mapping and distribution entries (defined in MAP_SRC_x, MAP_DST_x and MAP_DPHY_DST_x) for the current video stream. Non-matched virtual channel (VC) and data types (DT) pass to the corresponding CSI2 controller. | 0xX1: Enable MAP_SRC_8 and MAP_DST_8 registers<br>0xX2: Enable MAP_SRC_9 and MAP_DST_9 registers<br>0xX4: Enable MAP_SRC_10 and MAP_DST_10 registers<br>0xX8: Enable MAP_SRC_11 and MAP_DST_11 registers<br>0x1X: Enable MAP_SRC_12 and MAP_DST_12 registers<br>0x2X: Enable MAP_SRC_13 and MAP_DST_13 registers<br>0x4X: Enable MAP_SRC_14 and MAP_DST_14 registers<br>0x8X: Enable MAP_SRC_15 and MAP_DST_15 registers |

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MIPI\_TX13 (0x48D)

| BIT         | 7              | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|----------------|---|---|---|---|---|---|---|
| Field       | MAP_SRC_0[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00           |   |   |   |   |   |   |   |
| Access Type | Write, Read    |   |   |   |   |   |   |   |

| BITFIELD  | BITS | DESCRIPTION                                                                                                                              | DECODE                           |
|-----------|------|------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| MAP_SRC_0 | 7:0  | Mapping source register 0. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT source for map 0 |

MIPI\_TX14 (0x48E)

| BIT         | 7              | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|----------------|---|---|---|---|---|---|---|
| Field       | MAP_DST_0[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00           |   |   |   |   |   |   |   |
| Access Type | Write, Read    |   |   |   |   |   |   |   |

| BITFIELD  | BITS | DESCRIPTION                                                                                                                                   | DECODE                                |
|-----------|------|-----------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| MAP_DST_0 | 7:0  | Mapping destination register 0. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT destination for map 0 |

MIPI\_TX15 (0x48F)

| BIT         | 7              | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|----------------|---|---|---|---|---|---|---|
| Field       | MAP_SRC_1[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00           |   |   |   |   |   |   |   |
| Access Type | Write, Read    |   |   |   |   |   |   |   |

| BITFIELD  | BITS | DESCRIPTION                                                                                                                              | DECODE                           |
|-----------|------|------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| MAP_SRC_1 | 7:0  | Mapping source register 1. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT source for map 1 |

MIPI\_TX16 (0x490)

| BIT         | 7              | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|----------------|---|---|---|---|---|---|---|
| Field       | MAP_DST_1[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00           |   |   |   |   |   |   |   |
| Access Type | Write, Read    |   |   |   |   |   |   |   |

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| BITFIELD  | BITS | DESCRIPTION                                                                                                                                   | DECODE                                |
|-----------|------|-----------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| MAP_DST_1 | 7:0  | Mapping destination register 1. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT destination for map 1 |

**MIPI\_TX17 (0x491)**

| BIT         | 7              | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|----------------|---|---|---|---|---|---|---|
| Field       | MAP_SRC_2[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00           |   |   |   |   |   |   |   |
| Access Type | Write, Read    |   |   |   |   |   |   |   |

| BITFIELD  | BITS | DESCRIPTION                                                                                                                              | DECODE                           |
|-----------|------|------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| MAP_SRC_2 | 7:0  | Mapping source register 2. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT source for map 2 |

**MIPI\_TX18 (0x492)**

| BIT         | 7              | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|----------------|---|---|---|---|---|---|---|
| Field       | MAP_DST_2[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00           |   |   |   |   |   |   |   |
| Access Type | Write, Read    |   |   |   |   |   |   |   |

| BITFIELD  | BITS | DESCRIPTION                                                                                                                                   | DECODE                                |
|-----------|------|-----------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| MAP_DST_2 | 7:0  | Mapping destination register 2. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT destination for map 2 |

**MIPI\_TX19 (0x493)**

| BIT         | 7              | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|----------------|---|---|---|---|---|---|---|
| Field       | MAP_SRC_3[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00           |   |   |   |   |   |   |   |
| Access Type | Write, Read    |   |   |   |   |   |   |   |

| BITFIELD  | BITS | DESCRIPTION                                                                                                                              | DECODE                           |
|-----------|------|------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| MAP_SRC_3 | 7:0  | Mapping source register 3. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT source for map 3 |

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MIPI\_TX20 (0x494)

| BIT         | 7              | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|----------------|---|---|---|---|---|---|---|
| Field       | MAP_DST_3[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00           |   |   |   |   |   |   |   |
| Access Type | Write, Read    |   |   |   |   |   |   |   |

| BITFIELD  | BITS | DESCRIPTION                                                                                                                                   | DECODE                                |
|-----------|------|-----------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| MAP_DST_3 | 7:0  | Mapping destination register 3. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT destination for map 3 |

MIPI\_TX21 (0x495)

| BIT         | 7              | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|----------------|---|---|---|---|---|---|---|
| Field       | MAP_SRC_4[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00           |   |   |   |   |   |   |   |
| Access Type | Write, Read    |   |   |   |   |   |   |   |

| BITFIELD  | BITS | DESCRIPTION                                                                                                                              | DECODE                           |
|-----------|------|------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| MAP_SRC_4 | 7:0  | Mapping source register 4. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT source for map 4 |

MIPI\_TX22 (0x496)

| BIT         | 7              | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|----------------|---|---|---|---|---|---|---|
| Field       | MAP_DST_4[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00           |   |   |   |   |   |   |   |
| Access Type | Write, Read    |   |   |   |   |   |   |   |

| BITFIELD  | BITS | DESCRIPTION                                                                                                                                   | DECODE                                |
|-----------|------|-----------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| MAP_DST_4 | 7:0  | Mapping destination register 4. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT destination for map 4 |

MIPI\_TX23 (0x497)

| BIT         | 7              | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|----------------|---|---|---|---|---|---|---|
| Field       | MAP_SRC_5[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00           |   |   |   |   |   |   |   |
| Access Type | Write, Read    |   |   |   |   |   |   |   |

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| BITFIELD  | BITS | DESCRIPTION                                                                                                                              | DECODE                           |
|-----------|------|------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| MAP_SRC_5 | 7:0  | Mapping source register 5. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT source for map 5 |

MIPI\_TX24 (0x498)

| BIT         | 7              | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|----------------|---|---|---|---|---|---|---|
| Field       | MAP_DST_5[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00           |   |   |   |   |   |   |   |
| Access Type | Write, Read    |   |   |   |   |   |   |   |

| BITFIELD  | BITS | DESCRIPTION                                                                                                                                   | DECODE                                |
|-----------|------|-----------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| MAP_DST_5 | 7:0  | Mapping destination register 5. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT destination for map 5 |

MIPI\_TX25 (0x499)

| BIT         | 7              | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|----------------|---|---|---|---|---|---|---|
| Field       | MAP_SRC_6[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00           |   |   |   |   |   |   |   |
| Access Type | Write, Read    |   |   |   |   |   |   |   |

| BITFIELD  | BITS | DESCRIPTION                                                                                                                              | DECODE                           |
|-----------|------|------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| MAP_SRC_6 | 7:0  | Mapping source register 6. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT source for map 6 |

MIPI\_TX26 (0x49A)

| BIT         | 7              | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|----------------|---|---|---|---|---|---|---|
| Field       | MAP_DST_6[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00           |   |   |   |   |   |   |   |
| Access Type | Write, Read    |   |   |   |   |   |   |   |

| BITFIELD  | BITS | DESCRIPTION                                                                                                                                   | DECODE                                |
|-----------|------|-----------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| MAP_DST_6 | 7:0  | Mapping destination register 6. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT destination for map 6 |



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MIPI\_TX27 (0x49B)

| BIT         | 7              | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|----------------|---|---|---|---|---|---|---|
| Field       | MAP_SRC_7[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00           |   |   |   |   |   |   |   |
| Access Type | Write, Read    |   |   |   |   |   |   |   |

| BITFIELD  | BITS | DESCRIPTION                                                                                                                              | DECODE                           |
|-----------|------|------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| MAP_SRC_7 | 7:0  | Mapping source register 7. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT source for map 7 |

MIPI\_TX28 (0x49C)

| BIT         | 7              | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|----------------|---|---|---|---|---|---|---|
| Field       | MAP_DST_7[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00           |   |   |   |   |   |   |   |
| Access Type | Write, Read    |   |   |   |   |   |   |   |

| BITFIELD  | BITS | DESCRIPTION                                                                                                                                   | DECODE                                |
|-----------|------|-----------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| MAP_DST_7 | 7:0  | Mapping destination register 7. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT destination for map 7 |

MIPI\_TX29 (0x49D)

| BIT         | 7              | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|----------------|---|---|---|---|---|---|---|
| Field       | MAP_SRC_8[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00           |   |   |   |   |   |   |   |
| Access Type | Write, Read    |   |   |   |   |   |   |   |

| BITFIELD  | BITS | DESCRIPTION                                                                                                                              | DECODE                           |
|-----------|------|------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| MAP_SRC_8 | 7:0  | Mapping source register 8. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT source for map 8 |

MIPI\_TX30 (0x49E)

| BIT         | 7              | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|----------------|---|---|---|---|---|---|---|
| Field       | MAP_DST_8[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00           |   |   |   |   |   |   |   |
| Access Type | Write, Read    |   |   |   |   |   |   |   |

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| BITFIELD  | BITS | DESCRIPTION                                                                                                                                   | DECODE                                |
|-----------|------|-----------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| MAP_DST_8 | 7:0  | Mapping destination register 8. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT destination for map 8 |

**MIPI\_TX31 (0x49F)**

| BIT         | 7              | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|----------------|---|---|---|---|---|---|---|
| Field       | MAP_SRC_9[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00           |   |   |   |   |   |   |   |
| Access Type | Write, Read    |   |   |   |   |   |   |   |

| BITFIELD  | BITS | DESCRIPTION                                                                                                                              | DECODE                           |
|-----------|------|------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| MAP_SRC_9 | 7:0  | Mapping source register 9. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT source for map 9 |

**MIPI\_TX32 (0x4A0)**

| BIT         | 7              | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|----------------|---|---|---|---|---|---|---|
| Field       | MAP_DST_9[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00           |   |   |   |   |   |   |   |
| Access Type | Write, Read    |   |   |   |   |   |   |   |

| BITFIELD  | BITS | DESCRIPTION                                                                                                                                   | DECODE                                |
|-----------|------|-----------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| MAP_DST_9 | 7:0  | Mapping destination register 9. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT destination for map 9 |

**MIPI\_TX33 (0x4A1)**

| BIT         | 7               | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|-----------------|---|---|---|---|---|---|---|
| Field       | MAP_SRC_10[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00            |   |   |   |   |   |   |   |
| Access Type | Write, Read     |   |   |   |   |   |   |   |

| BITFIELD   | BITS | DESCRIPTION                                                                                                                               | DECODE                            |
|------------|------|-------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
| MAP_SRC_10 | 7:0  | Mapping source register 10. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT source for map 10 |

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MIPI\_TX34 (0x4A2)

| BIT         | 7               | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|-----------------|---|---|---|---|---|---|---|
| Field       | MAP_DST_10[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00            |   |   |   |   |   |   |   |
| Access Type | Write, Read     |   |   |   |   |   |   |   |

| BITFIELD   | BITS | DESCRIPTION                                                                                                                                    | DECODE                                 |
|------------|------|------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| MAP_DST_10 | 7:0  | Mapping destination register 10. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT destination for map 10 |

MIPI\_TX35 (0x4A3)

| BIT         | 7               | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|-----------------|---|---|---|---|---|---|---|
| Field       | MAP_SRC_11[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00            |   |   |   |   |   |   |   |
| Access Type | Write, Read     |   |   |   |   |   |   |   |

| BITFIELD   | BITS | DESCRIPTION                                                                                                                               | DECODE                            |
|------------|------|-------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
| MAP_SRC_11 | 7:0  | Mapping source register 11. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT source for map 11 |

MIPI\_TX36 (0x4A4)

| BIT         | 7               | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|-----------------|---|---|---|---|---|---|---|
| Field       | MAP_DST_11[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00            |   |   |   |   |   |   |   |
| Access Type | Write, Read     |   |   |   |   |   |   |   |

| BITFIELD   | BITS | DESCRIPTION                                                                                                                                    | DECODE                                 |
|------------|------|------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| MAP_DST_11 | 7:0  | Mapping destination register 11. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT destination for map 11 |

MIPI\_TX37 (0x4A5)

| BIT         | 7               | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|-----------------|---|---|---|---|---|---|---|
| Field       | MAP_SRC_12[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00            |   |   |   |   |   |   |   |
| Access Type | Write, Read     |   |   |   |   |   |   |   |

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| BITFIELD   | BITS | DESCRIPTION                                                                                                                               | DECODE                            |
|------------|------|-------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
| MAP_SRC_12 | 7:0  | Mapping source register 12. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT source for map 12 |

MIPI\_TX38 (0x4A6)

| BIT         | 7               | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|-----------------|---|---|---|---|---|---|---|
| Field       | MAP_DST_12[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00            |   |   |   |   |   |   |   |
| Access Type | Write, Read     |   |   |   |   |   |   |   |

| BITFIELD   | BITS | DESCRIPTION                                                                                                                                    | DECODE                                 |
|------------|------|------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| MAP_DST_12 | 7:0  | Mapping destination register 12. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT destination for map 12 |

MIPI\_TX39 (0x4A7)

| BIT         | 7               | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|-----------------|---|---|---|---|---|---|---|
| Field       | MAP_SRC_13[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00            |   |   |   |   |   |   |   |
| Access Type | Write, Read     |   |   |   |   |   |   |   |

| BITFIELD   | BITS | DESCRIPTION                                                                                                                               | DECODE                            |
|------------|------|-------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
| MAP_SRC_13 | 7:0  | Mapping source register 13. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT source for map 13 |

MIPI\_TX40 (0x4A8)

| BIT         | 7               | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|-----------------|---|---|---|---|---|---|---|
| Field       | MAP_DST_13[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00            |   |   |   |   |   |   |   |
| Access Type | Write, Read     |   |   |   |   |   |   |   |

| BITFIELD   | BITS | DESCRIPTION                                                                                                                                    | DECODE                                 |
|------------|------|------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| MAP_DST_13 | 7:0  | Mapping destination register 13. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT destination for map 13 |

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MIPI\_TX41 (0x4A9)

| BIT         | 7               | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|-----------------|---|---|---|---|---|---|---|
| Field       | MAP_SRC_14[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00            |   |   |   |   |   |   |   |
| Access Type | Write, Read     |   |   |   |   |   |   |   |

| BITFIELD   | BITS | DESCRIPTION                                                                                                                               | DECODE                            |
|------------|------|-------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
| MAP_SRC_14 | 7:0  | Mapping source register 14. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT source for map 14 |

MIPI\_TX42 (0x4AA)

| BIT         | 7               | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|-----------------|---|---|---|---|---|---|---|
| Field       | MAP_DST_14[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00            |   |   |   |   |   |   |   |
| Access Type | Write, Read     |   |   |   |   |   |   |   |

| BITFIELD   | BITS | DESCRIPTION                                                                                                                                    | DECODE                                 |
|------------|------|------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| MAP_DST_14 | 7:0  | Mapping destination register 14. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT destination for map 14 |

MIPI\_TX43 (0x4AB)

| BIT         | 7               | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|-----------------|---|---|---|---|---|---|---|
| Field       | MAP_SRC_15[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00            |   |   |   |   |   |   |   |
| Access Type | Write, Read     |   |   |   |   |   |   |   |

| BITFIELD   | BITS | DESCRIPTION                                                                                                                               | DECODE                            |
|------------|------|-------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
| MAP_SRC_15 | 7:0  | Mapping source register 15. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT source for map 15 |

MIPI\_TX44 (0x4AC)

| BIT         | 7               | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|-----------------|---|---|---|---|---|---|---|
| Field       | MAP_DST_15[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00            |   |   |   |   |   |   |   |
| Access Type | Write, Read     |   |   |   |   |   |   |   |

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| BITFIELD   | BITS | DESCRIPTION                                                                                                                                    | DECODE                                 |
|------------|------|------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| MAP_DST_15 | 7:0  | Mapping destination register 15. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT destination for map 15 |

**MIPI\_TX45 (0x4AD)**

| BIT         | 7                    | 6 | 5                    | 4 | 3                    | 2 | 1                    | 0 |
|-------------|----------------------|---|----------------------|---|----------------------|---|----------------------|---|
| Field       | MAP_DPHY_DEST_3[1:0] |   | MAP_DPHY_DEST_2[1:0] |   | MAP_DPHY_DEST_1[1:0] |   | MAP_DPHY_DEST_0[1:0] |   |
| Reset       | 0b00                 |   | 0b00                 |   | 0b00                 |   | 0b00                 |   |
| Access Type | Write, Read          |   | Write, Read          |   | Write, Read          |   | Write, Read          |   |

| BITFIELD        | BITS | DESCRIPTION                                                          | DECODE                                                                               |
|-----------------|------|----------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| MAP_DPHY_DEST_3 | 7:6  | CSI2 controller destination for MAP_SRC_3 and MAP_DST_0 mapping regs | 0b00: Map to DPHY0<br>0b01: Map to DPHY1<br>0b10: Map to DPHY2<br>0b11: Map to DPHY3 |
| MAP_DPHY_DEST_2 | 5:4  | CSI2 controller destination for MAP_SRC_2 and MAP_DST_0 mapping regs | 0b00: Map to DPHY0<br>0b01: Map to DPHY1<br>0b10: Map to DPHY2<br>0b11: Map to DPHY3 |
| MAP_DPHY_DEST_1 | 3:2  | CSI2 controller destination for MAP_SRC_1 and MAP_DST_0 mapping regs | 0b00: Map to DPHY0<br>0b01: Map to DPHY1<br>0b10: Map to DPHY2<br>0b11: Map to DPHY3 |
| MAP_DPHY_DEST_0 | 1:0  | CSI2 controller destination for MAP_SRC_0 and MAP_DST_0 mapping regs | 0b00: Map to DPHY0<br>0b01: Map to DPHY1<br>0b10: Map to DPHY2<br>0b11: Map to DPHY3 |

**MIPI\_TX46 (0x4AE)**

| BIT         | 7                    | 6 | 5                    | 4 | 3                    | 2 | 1                    | 0 |
|-------------|----------------------|---|----------------------|---|----------------------|---|----------------------|---|
| Field       | MAP_DPHY_DEST_7[1:0] |   | MAP_DPHY_DEST_6[1:0] |   | MAP_DPHY_DEST_5[1:0] |   | MAP_DPHY_DEST_4[1:0] |   |
| Reset       | 0b00                 |   | 0b00                 |   | 0b00                 |   | 0b00                 |   |
| Access Type | Write, Read          |   | Write, Read          |   | Write, Read          |   | Write, Read          |   |

| BITFIELD        | BITS | DESCRIPTION                                                          | DECODE                                                                               |
|-----------------|------|----------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| MAP_DPHY_DEST_7 | 7:6  | CSI2 controller destination for MAP_SRC_7 and MAP_DST_0 mapping regs | 0b00: Map to DPHY0<br>0b01: Map to DPHY1<br>0b10: Map to DPHY2<br>0b11: Map to DPHY3 |
| MAP_DPHY_DEST_6 | 5:4  | CSI2 controller destination for MAP_SRC_6 and MAP_DST_0 mapping regs | 0b00: Map to DPHY0<br>0b01: Map to DPHY1<br>0b10: Map to DPHY2<br>0b11: Map to DPHY3 |
| MAP_DPHY_DEST_5 | 3:2  | CSI2 controller destination for MAP_SRC_5 and MAP_DST_0 mapping regs | 0b00: Map to DPHY0<br>0b01: Map to DPHY1<br>0b10: Map to DPHY2<br>0b11: Map to DPHY3 |

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| BITFIELD        | BITS | DESCRIPTION                                                          | DECODE                                                                               |
|-----------------|------|----------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| MAP_DPHY_DEST_4 | 1:0  | CSI2 controller destination for MAP_SRC_4 and MAP_DST_0 mapping regs | 0b00: Map to DPHY0<br>0b01: Map to DPHY1<br>0b10: Map to DPHY2<br>0b11: Map to DPHY3 |

**MIPI\_TX47 (0x4AF)**

| BIT         | 7                     | 6 | 5                     | 4 | 3                    | 2 | 1                    | 0 |
|-------------|-----------------------|---|-----------------------|---|----------------------|---|----------------------|---|
| Field       | MAP_DPHY_DEST_11[1:0] |   | MAP_DPHY_DEST_10[1:0] |   | MAP_DPHY_DEST_9[1:0] |   | MAP_DPHY_DEST_8[1:0] |   |
| Reset       | 0b00                  |   | 0b00                  |   | 0b00                 |   | 0b00                 |   |
| Access Type | Write, Read           |   | Write, Read           |   | Write, Read          |   | Write, Read          |   |

| BITFIELD         | BITS | DESCRIPTION                                                           | DECODE                                                                               |
|------------------|------|-----------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| MAP_DPHY_DEST_11 | 7:6  | CSI2 controller destination for MAP_SRC_11 and MAP_DST_0 mapping regs | 0b00: Map to DPHY0<br>0b01: Map to DPHY1<br>0b10: Map to DPHY2<br>0b11: Map to DPHY3 |
| MAP_DPHY_DEST_10 | 5:4  | CSI2 controller destination for MAP_SRC_10 and MAP_DST_0 mapping regs | 0b00: Map to DPHY0<br>0b01: Map to DPHY1<br>0b10: Map to DPHY2<br>0b11: Map to DPHY3 |
| MAP_DPHY_DEST_9  | 3:2  | CSI2 controller destination for MAP_SRC_9 and MAP_DST_0 mapping regs  | 0b00: Map to DPHY0<br>0b01: Map to DPHY1<br>0b10: Map to DPHY2<br>0b11: Map to DPHY3 |
| MAP_DPHY_DEST_8  | 1:0  | CSI2 controller destination for MAP_SRC_8 and MAP_DST_0 mapping regs  | 0b00: Map to DPHY0<br>0b01: Map to DPHY1<br>0b10: Map to DPHY2<br>0b11: Map to DPHY3 |

**MIPI\_TX48 (0x4B0)**

| BIT         | 7                     | 6 | 5                     | 4 | 3                     | 2 | 1                     | 0 |
|-------------|-----------------------|---|-----------------------|---|-----------------------|---|-----------------------|---|
| Field       | MAP_DPHY_DEST_15[1:0] |   | MAP_DPHY_DEST_14[1:0] |   | MAP_DPHY_DEST_13[1:0] |   | MAP_DPHY_DEST_12[1:0] |   |
| Reset       | 0b00                  |   | 0b00                  |   | 0b00                  |   | 0b00                  |   |
| Access Type | Write, Read           |   | Write, Read           |   | Write, Read           |   | Write, Read           |   |

| BITFIELD         | BITS | DESCRIPTION                                                           | DECODE                                                                               |
|------------------|------|-----------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| MAP_DPHY_DEST_15 | 7:6  | CSI2 controller destination for MAP_SRC_15 and MAP_DST_0 mapping regs | 0b00: Map to DPHY0<br>0b01: Map to DPHY1<br>0b10: Map to DPHY2<br>0b11: Map to DPHY3 |
| MAP_DPHY_DEST_14 | 5:4  | CSI2 controller destination for MAP_SRC_14 and MAP_DST_0 mapping regs | 0b00: Map to DPHY0<br>0b01: Map to DPHY1<br>0b10: Map to DPHY2<br>0b11: Map to DPHY3 |

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| BITFIELD         | BITS | DESCRIPTION                                                           | DECODE                                                                               |
|------------------|------|-----------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| MAP_DPHY_DEST_13 | 3:2  | CSI2 controller destination for MAP_SRC_13 and MAP_DST_0 mapping regs | 0b00: Map to DPHY0<br>0b01: Map to DPHY1<br>0b10: Map to DPHY2<br>0b11: Map to DPHY3 |
| MAP_DPHY_DEST_12 | 1:0  | CSI2 controller destination for MAP_SRC_12 and MAP_DST_0 mapping regs | 0b00: Map to DPHY0<br>0b01: Map to DPHY1<br>0b10: Map to DPHY2<br>0b11: Map to DPHY3 |

**MIPI\_TX49 (0x4B1)**

| BIT         | 7            | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|--------------|---|---|---|---|---|---|---|
| Field       | MAP_CON[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00         |   |   |   |   |   |   |   |
| Access Type | Write, Read  |   |   |   |   |   |   |   |

| BITFIELD | BITS | DESCRIPTION                                                                                                                                                                                                                                                                                                               | DECODE                                                                                                                                                                                                                                                                                                                                                                                                                               |
|----------|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MAP_CON  | 7:0  | Concatenation register for sync mode settings. The register is split into four decode segments:<br>bit [7] Enable 4WxH or Wx4H<br>bit [6] Reserved<br>bit [5:4] Master video stream number. This is the first line to concatenate. All others follow in order<br>bit [3:0] Each bit selects a video stream to concatenate | 0bXXXXXXXX1: Concatenate video stream 0<br>0bXXXXXXXX1X: Concatenate video stream 1<br>0bXXXXX1XX: Concatenate video stream 2<br>0bXXXXX1XXX: Concatenate video stream 3<br>0bXX00XXXX: Select video stream 0 as master<br>0bXX01XXXX: Select video stream 1 as master<br>0bXX10XXXX: Select video stream 2 as master<br>0bXX11XXXX: Select video stream 3 as master<br>0b0XXXXXXX: Enable Wx4H mode<br>0b1XXXXXXX: Enable 4WxH mode |

**MIPI\_TX50 (0x4B2)**

| BIT         | 7                 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|-------------------|---|---|---|---|---|---|---|
| Field       | SKEW_PER_SEL[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00              |   |   |   |   |   |   |   |
| Access Type | Write, Read       |   |   |   |   |   |   |   |



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| BITFIELD     | BITS | DESCRIPTION                                                                                                                                                                                                                                                                                              | DECODE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|--------------|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SKEW_PER_SEL | 7:0  | Periodic deskew select register. The register is split into three decode segments:<br>bit [7] Select periodic deskew calibration for one or all virtual channels<br>bit [6:5] Reserved<br>bit [4:0] Virtual channel to generate periodic deskew calibration when only one channel is selected by bit [7] | 0b0XXXXXXX: Generate periodic calibration deskew calibration on all virtual channels<br>0b1XX00000: Periodic deskew calibration generated by virtual channel 0<br>0b1XX00001: Periodic deskew calibration generated by virtual channel 1<br>0b1XX00010: Periodic deskew calibration generated by virtual channel 2<br>0b1XX00011: Periodic deskew calibration generated by virtual channel 3<br>0b1XX00100: Periodic deskew calibration generated by virtual channel 4<br>0b1XX00101: Periodic deskew calibration generated by virtual channel 5<br>0b1XX00110: Periodic deskew calibration generated by virtual channel 6<br>0b1XX00111: Periodic deskew calibration generated by virtual channel 7<br>0b1XX01000: Periodic deskew calibration generated by virtual channel 8<br>0b1XX01001: Periodic deskew calibration generated by virtual channel 9<br>0b1XX01010: Periodic deskew calibration generated by virtual channel 10<br>0b1XX01011: Periodic deskew calibration generated by virtual channel 11<br>0b1XX01100: Periodic deskew calibration generated by virtual channel 12<br>0b1XX01101: Periodic deskew calibration generated by virtual channel 13<br>0b1XX01110: Periodic deskew calibration generated by virtual channel 14<br>0b1XX01111: Periodic deskew calibration generated by virtual channel 15 |

**MIPI\_TX51 (0x4B3)**

| BIT         | 7 | 6 | 5 | 4             | 3           | 2             | 1            | 0             |
|-------------|---|---|---|---------------|-------------|---------------|--------------|---------------|
| Field       | — | — | — | ALT2_MEM_MAP8 | MODE_DT     | ALT_MEM_MAP10 | ALT_MEM_MAP8 | ALT_MEM_MAP12 |
| Reset       | — | — | — | 0b0           | 0b0         | 0b0           | 0b0          | 0b0           |
| Access Type | — | — | — | Write, Read   | Write, Read | Write, Read   | Write, Read  | Write, Read   |

| BITFIELD      | BITS | DESCRIPTION                                     | DECODE                                                                                        |
|---------------|------|-------------------------------------------------|-----------------------------------------------------------------------------------------------|
| ALT2_MEM_MAP8 | 4    | Alternative memory map enable for 8-bit DT      | 0b0: Alternate memory map disabled<br>0b1: Alternate memory map enabled                       |
| MODE_DT       | 3    | MIPI TX enable 24-bit packing of 8-bit MIPI UDP | 0x0: Disable 24-bit packing of 8-bit MIPI UDP<br>0x1: Enable 24-bit packing of 8-bit MIPI UDP |
| ALT_MEM_MAP10 | 2    | Alternative memory map enable for 10-bit DT     | 0b0: Alternate memory map disabled<br>0b1: Alternate memory map enabled                       |
| ALT_MEM_MAP8  | 1    | Alternative memory map enable for 8-bit DT      | 0b0: Alternate memory map disabled<br>0b1: Alternate memory map enabled                       |

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| BITFIELD      | BITS | DESCRIPTION                                 | DECODE                                                                  |
|---------------|------|---------------------------------------------|-------------------------------------------------------------------------|
| ALT_MEM_MAP12 | 0    | Alternative memory map enable for 12-bit DT | 0b0: Alternate memory map disabled<br>0b1: Alternate memory map enabled |

**MIPI\_TX1 (0x4C1)**

| BIT         | 7           | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|-------------|---|---|---|---|---|---|---|
| Field       | MODE[7:0]   |   |   |   |   |   |   |   |
| Reset       | 0x00        |   |   |   |   |   |   |   |
| Access Type | Write, Read |   |   |   |   |   |   |   |

| BITFIELD | BITS | DESCRIPTION                       | DECODE                                                                                              |
|----------|------|-----------------------------------|-----------------------------------------------------------------------------------------------------|
| MODE     | 7:0  | MIPI VS short packet counter mode | 0bXXXXXXX0: Disable MIPI VS short packet counter<br>0bXXXXXXX1: Enable MIPI VS short packet counter |

**MIPI\_TX2 (0x4C2)**

| BIT         | 7           | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|-------------|---|---|---|---|---|---|---|
| Field       | STATUS[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00        |   |   |   |   |   |   |   |
| Access Type | Read Only   |   |   |   |   |   |   |   |

| BITFIELD | BITS | DESCRIPTION                                                                                                                                                  | DECODE                                                                                                                                                                                                                                                             |
|----------|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| STATUS   | 7:0  | MIPI Tx status. The register is split into three decode segments:<br>bit [0] SYNC mode enable.<br>bit [1] Video sync flag<br>bit [2] Loss of video sync flag | 0bXXXXXXX0: SYNC mode disabled<br>0bXXXXXXX1: SYNC mode enabled<br>0bXXXXXXX0X: Video channels not in-sync<br>0bXXXXXXX1X: Video channels in-sync<br>0bXXXXXX0XX: No loss of video sync<br>0bXXXXXX1XX: Video sync lost after last read of this register or RESET. |

**MIPI\_TX3 (0x4C3)**

| BIT         | 7                | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|------------------|---|---|---|---|---|---|---|
| Field       | DESKEW_INIT[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00             |   |   |   |   |   |   |   |
| Access Type | Write, Read      |   |   |   |   |   |   |   |

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| BITFIELD    | BITS | DESCRIPTION                                                                                                                                                                                                                                                                                                                                                   | DECODE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-------------|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DESKEW_INIT | 7:0  | DPHY deskew initial calibration control. The register is split into six decode segments:<br>bit [7] Selects auto-initial deskew calibration on or off<br>bit [6] Reserved<br>bit [5] Any bit change initiates an initial calibration if bit [4] = 1<br>bit [4] Selects manual initial on or off<br>bit [3] Reserved<br>bit [2:0] Selects initial deskew width | 0bXXXXXX000: Initial deskew width = 1*32k UI<br>0bXXXXXX001: Initial deskew width = 2*32k UI<br>0bXXXXXX010: Initial deskew width = 3*32k UI<br>0bXXXXXX011: Initial deskew width = 4*32k UI<br>0bXXXXXX100: Initial deskew width = 5*32k UI<br>0bXXXXXX101: Initial deskew width = 6*32k UI<br>0bXXXXXX110: Initial deskew width = 7*32k UI<br>0bXXXXXX111: Initial deskew width = 8*32k UI<br>0bXXXXX0XXX: Reserved<br>0bXXXXX1XXX: Reserved<br>0bXXX0XXXX: Manual initial off<br>0bXXX1XXXX: Manual initial on<br>0bXX0XXXXX: If bit 4 = 1, triggers one time immediate initial skew calibration<br>0bXX1XXXXX: If bit 4 = 1, triggers one time immediate initial skew calibration<br>0bX0XXXXXX: Reserved<br>0bX1XXXXXX: Reserved<br>0b0XXXXXXX: Auto initial deskew off<br>0b1XXXXXXX: Auto initial deskew on |

MIPI\_TX4 (0x4C4)

| BIT         | 7               | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|-----------------|---|---|---|---|---|---|---|
| Field       | DESKEW_PER[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00            |   |   |   |   |   |   |   |
| Access Type | Write, Read     |   |   |   |   |   |   |   |

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| BITFIELD       | BITS | DESCRIPTION                                                                                                                                                                                                                                                                                            | DECODE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|----------------|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DESKEW_P<br>ER | 7:0  | DPHY periodic deskew calibration control.<br>The register is split into four decode segments:<br>bit [7] Selects periodic deskew calibration on or off<br>bit [6] Selects generation on rising or falling edge of VS<br>bit [5:3] Selects periodic interval<br>bit [2:0] Selects periodic deskew width | 0bXXXXXX000: Periodic deskew width = 1k UI<br>0bXXXXXX001: Periodic deskew width = 2k UI<br>0bXXXXXX010: Periodic deskew width = 3k UI<br>0bXXXXXX011: Periodic deskew width = 4k UI<br>0bXXXXXX100: Periodic deskew width = 5k UI<br>0bXXXXXX101: Periodic deskew width = 6k UI<br>0bXXXXXX110: Periodic deskew width = 7k UI<br>0bXXXXXX111: Periodic deskew width = 8k UI<br>0bXX000XXX: Periodic deskew calibration generated every frame<br>0bXX001XXX: Periodic deskew calibration generated every 2 frames<br>0bXX010XXX: Periodic deskew calibration generated every 4 frames<br>0bXX011XXX: Periodic deskew calibration generated every 8 frames<br>0bXX100XXX: Periodic deskew calibration generated every 16 frames<br>0bXX101XXX: Periodic deskew calibration generated every 32 frames<br>0bXX110XXX: Periodic deskew calibration generated every 64 frames<br>0bXX111XXX: Periodic deskew calibration generated every 128 frames<br>0bX0XXXXXXX: Periodic deskew calibration generated at rising edge of VS<br>0bX1XXXXXXX: Periodic deskew calibration generated at falling edge of VS<br>0b0XXXXXXX: Periodic deskew calibration off<br>0b1XXXXXXX: Periodic deskew calibration on |

**MIPI\_TX5 (0x4C5)**

| BIT         | 7               | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|-----------------|---|---|---|---|---|---|---|
| Field       | CSI2_T_PRE[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x71            |   |   |   |   |   |   |   |
| Access Type | Write, Read     |   |   |   |   |   |   |   |

| BITFIELD   | BITS | DESCRIPTION                                                | DECODE                           |
|------------|------|------------------------------------------------------------|----------------------------------|
| CSI2_T_PRE | 7:0  | Number of MIPI byte clocks to wait before enabling HS data | 0xXX: Number of MIPI byte clocks |

**MIPI\_TX6 (0x4C6)**

| BIT         | 7                | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|------------------|---|---|---|---|---|---|---|
| Field       | CSI2_T_POST[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x19             |   |   |   |   |   |   |   |
| Access Type | Write, Read      |   |   |   |   |   |   |   |

| BITFIELD        | BITS | DESCRIPTION                                                | DECODE                           |
|-----------------|------|------------------------------------------------------------|----------------------------------|
| CSI2_T_POS<br>T | 7:0  | Number of MIPI byte clocks to hold clock active after data | 0xXX: Number of MIPI byte clocks |

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**MIPI\_TX8 (0x4C8)**

| BIT            | 7                   | 6                                 | 5 | 4 | 3                                | 2 | 1 | 0 |
|----------------|---------------------|-----------------------------------|---|---|----------------------------------|---|---|---|
| Field          | CSI2_TWAKEUP_L[7:0] |                                   |   |   |                                  |   |   |   |
| Reset          | 0x00                |                                   |   |   |                                  |   |   |   |
| Access Type    | Write, Read         |                                   |   |   |                                  |   |   |   |
| BITFIELD       | BITS                | DESCRIPTION                       |   |   | DECODE                           |   |   |   |
| CSI2_TWAKEUP_L | 7:0                 | Low byte of CSI2 transmit wake-up |   |   | 0xXX: Number of MIPI byte clocks |   |   |   |

**MIPI\_TX9 (0x4C9)**

| BIT            | 7                   | 6                                    | 5 | 4 | 3                                | 2 | 1 | 0 |
|----------------|---------------------|--------------------------------------|---|---|----------------------------------|---|---|---|
| Field          | CSI2_TWAKEUP_M[7:0] |                                      |   |   |                                  |   |   |   |
| Reset          | 0x01                |                                      |   |   |                                  |   |   |   |
| Access Type    | Write, Read         |                                      |   |   |                                  |   |   |   |
| BITFIELD       | BITS                | DESCRIPTION                          |   |   | DECODE                           |   |   |   |
| CSI2_TWAKEUP_M | 7:0                 | Middle byte of CSI2 transmit wake-up |   |   | 0xXX: Number of MIPI byte clocks |   |   |   |

**MIPI\_TX10 (0x4CA)**

| BIT         | 7 | 6 | 5 | 4 | 3           | 2                   | 1 | 0 |
|-------------|---|---|---|---|-------------|---------------------|---|---|
| Field       | – | – | – | – | CSI_VCX_EN  | CSI2_TWAKEUP_H[2:0] |   |   |
| Reset       | – | – | – | – | 0b0         | 0b000               |   |   |
| Access Type | – | – | – | – | Write, Read | Write, Read         |   |   |

| BITFIELD       | BITS | DESCRIPTION                        | DECODE                                                |
|----------------|------|------------------------------------|-------------------------------------------------------|
| CSI_VCX_EN     | 3    | Enables VC extension               | 0x0: Disable VC extension<br>0x1: Enable VC extension |
| CSI2_TWAKEUP_H | 2:0  | High bits of CSI2 transmit wake-up | 0xXX: Number of MIPI byte clocks                      |

**MIPI\_TX11 (0x4CB)**

| BIT         | 7             | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|---------------|---|---|---|---|---|---|---|
| Field       | MAP_EN_L[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00          |   |   |   |   |   |   |   |
| Access Type | Write, Read   |   |   |   |   |   |   |   |

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| BITFIELD | BITS | DESCRIPTION                                                                                                                                                                                                                                                                | DECODE                                                                                                                                                                                                                                                                                                                                                                                                       |
|----------|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MAP_EN_L | 7:0  | Mapping enable low byte [7:0]. Each bit enables 1 of 8 mapping and distribution entries (defined in MAP_SRC_x, MAP_DST_x and MAP_DPHY_DST_x) for the current video stream. Non-matched virtual channel (VC) and data types (DT) pass to the corresponding CSI2 controller. | 0xX1: Enable MAP_SRC_0 and MAP_DST_0 registers<br>0xX2: Enable MAP_SRC_1 and MAP_DST_1 registers<br>0xX4: Enable MAP_SRC_2 and MAP_DST_2 registers<br>0xX8: Enable MAP_SRC_3 and MAP_DST_3 registers<br>0x1X: Enable MAP_SRC_4 and MAP_DST_4 registers<br>0x2X: Enable MAP_SRC_5 and MAP_DST_5 registers<br>0x4X: Enable MAP_SRC_6 and MAP_DST_6 registers<br>0x8X: Enable MAP_SRC_7 and MAP_DST_7 registers |

**MIPI\_TX12 (0x4CC)**

| BIT         | 7             | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|---------------|---|---|---|---|---|---|---|
| Field       | MAP_EN_H[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00          |   |   |   |   |   |   |   |
| Access Type | Write, Read   |   |   |   |   |   |   |   |

| BITFIELD | BITS | DESCRIPTION                                                                                                                                                                                                                                                                  | DECODE                                                                                                                                                                                                                                                                                                                                                                                                                   |
|----------|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MAP_EN_H | 7:0  | Mapping enable high byte [15:8]. Each bit enables 1 of 8 mapping and distribution entries (defined in MAP_SRC_x, MAP_DST_x and MAP_DPHY_DST_x) for the current video stream. Non matched virtual channel (VC) and data types (DT) pass to the corresponding CSI2 controller. | 0xX1: Enable MAP_SRC_8 and MAP_DST_8 registers<br>0xX2: Enable MAP_SRC_9 and MAP_DST_9 registers<br>0xX4: Enable MAP_SRC_10 and MAP_DST_10 registers<br>0xX8: Enable MAP_SRC_11 and MAP_DST_11 registers<br>0x1X: Enable MAP_SRC_12 and MAP_DST_12 registers<br>0x2X: Enable MAP_SRC_13 and MAP_DST_13 registers<br>0x4X: Enable MAP_SRC_14 and MAP_DST_14 registers<br>0x8X: Enable MAP_SRC_15 and MAP_DST_15 registers |

**MIPI\_TX13 (0x4CD)**

| BIT         | 7              | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|----------------|---|---|---|---|---|---|---|
| Field       | MAP_SRC_0[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00           |   |   |   |   |   |   |   |
| Access Type | Write, Read    |   |   |   |   |   |   |   |

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| BITFIELD  | BITS | DESCRIPTION                                                                                                                              | DECODE                           |
|-----------|------|------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| MAP_SRC_0 | 7:0  | Mapping source register 0. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT source for map 0 |

**MIPI\_TX14 (0x4CE)**

| BIT         | 7              | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|----------------|---|---|---|---|---|---|---|
| Field       | MAP_DST_0[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00           |   |   |   |   |   |   |   |
| Access Type | Write, Read    |   |   |   |   |   |   |   |

| BITFIELD  | BITS | DESCRIPTION                                                                                                                                   | DECODE                                |
|-----------|------|-----------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| MAP_DST_0 | 7:0  | Mapping destination register 0. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT destination for map 0 |

**MIPI\_TX15 (0x4CF)**

| BIT         | 7              | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|----------------|---|---|---|---|---|---|---|
| Field       | MAP_SRC_1[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00           |   |   |   |   |   |   |   |
| Access Type | Write, Read    |   |   |   |   |   |   |   |

| BITFIELD  | BITS | DESCRIPTION                                                                                                                              | DECODE                           |
|-----------|------|------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| MAP_SRC_1 | 7:0  | Mapping source register 1. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT source for map 1 |

**MIPI\_TX16 (0x4D0)**

| BIT         | 7              | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|----------------|---|---|---|---|---|---|---|
| Field       | MAP_DST_1[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00           |   |   |   |   |   |   |   |
| Access Type | Write, Read    |   |   |   |   |   |   |   |

| BITFIELD  | BITS | DESCRIPTION                                                                                                                                   | DECODE                                |
|-----------|------|-----------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| MAP_DST_1 | 7:0  | Mapping destination register 1. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT destination for map 1 |

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MIPI\_TX17 (0x4D1)

| BIT         | 7              | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|----------------|---|---|---|---|---|---|---|
| Field       | MAP_SRC_2[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00           |   |   |   |   |   |   |   |
| Access Type | Write, Read    |   |   |   |   |   |   |   |

| BITFIELD  | BITS | DESCRIPTION                                                                                                                              | DECODE                           |
|-----------|------|------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| MAP_SRC_2 | 7:0  | Mapping source register 2. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT source for map 2 |

MIPI\_TX18 (0x4D2)

| BIT         | 7              | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|----------------|---|---|---|---|---|---|---|
| Field       | MAP_DST_2[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00           |   |   |   |   |   |   |   |
| Access Type | Write, Read    |   |   |   |   |   |   |   |

| BITFIELD  | BITS | DESCRIPTION                                                                                                                                   | DECODE                                |
|-----------|------|-----------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| MAP_DST_2 | 7:0  | Mapping destination register 2. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT destination for map 2 |

MIPI\_TX19 (0x4D3)

| BIT         | 7              | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|----------------|---|---|---|---|---|---|---|
| Field       | MAP_SRC_3[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00           |   |   |   |   |   |   |   |
| Access Type | Write, Read    |   |   |   |   |   |   |   |

| BITFIELD  | BITS | DESCRIPTION                                                                                                                              | DECODE                           |
|-----------|------|------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| MAP_SRC_3 | 7:0  | Mapping source register 3. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT source for map 3 |

MIPI\_TX20 (0x4D4)

| BIT         | 7              | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|----------------|---|---|---|---|---|---|---|
| Field       | MAP_DST_3[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00           |   |   |   |   |   |   |   |
| Access Type | Write, Read    |   |   |   |   |   |   |   |



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| BITFIELD  | BITS | DESCRIPTION                                                                                                                                   | DECODE                                |
|-----------|------|-----------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| MAP_DST_3 | 7:0  | Mapping destination register 3. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT destination for map 3 |

**MIPI\_TX21 (0x4D5)**

| BIT         | 7              | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|----------------|---|---|---|---|---|---|---|
| Field       | MAP_SRC_4[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00           |   |   |   |   |   |   |   |
| Access Type | Write, Read    |   |   |   |   |   |   |   |

| BITFIELD  | BITS | DESCRIPTION                                                                                                                              | DECODE                           |
|-----------|------|------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| MAP_SRC_4 | 7:0  | Mapping source register 4. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT source for map 4 |

**MIPI\_TX22 (0x4D6)**

| BIT         | 7              | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|----------------|---|---|---|---|---|---|---|
| Field       | MAP_DST_4[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00           |   |   |   |   |   |   |   |
| Access Type | Write, Read    |   |   |   |   |   |   |   |

| BITFIELD  | BITS | DESCRIPTION                                                                                                                                   | DECODE                                |
|-----------|------|-----------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| MAP_DST_4 | 7:0  | Mapping destination register 4. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT destination for map 4 |

**MIPI\_TX23 (0x4D7)**

| BIT         | 7              | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|----------------|---|---|---|---|---|---|---|
| Field       | MAP_SRC_5[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00           |   |   |   |   |   |   |   |
| Access Type | Write, Read    |   |   |   |   |   |   |   |

| BITFIELD  | BITS | DESCRIPTION                                                                                                                              | DECODE                           |
|-----------|------|------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| MAP_SRC_5 | 7:0  | Mapping source register 5. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT source for map 5 |

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Appendix A: Register Map and Tables

MIPI\_TX24 (0x4D8)

| BIT         | 7              | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|----------------|---|---|---|---|---|---|---|
| Field       | MAP_DST_5[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00           |   |   |   |   |   |   |   |
| Access Type | Write, Read    |   |   |   |   |   |   |   |

| BITFIELD  | BITS | DESCRIPTION                                                                                                                                   | DECODE                                |
|-----------|------|-----------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| MAP_DST_5 | 7:0  | Mapping destination register 5. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT destination for map 5 |

MIPI\_TX25 (0x4D9)

| BIT         | 7              | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|----------------|---|---|---|---|---|---|---|
| Field       | MAP_SRC_6[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00           |   |   |   |   |   |   |   |
| Access Type | Write, Read    |   |   |   |   |   |   |   |

| BITFIELD  | BITS | DESCRIPTION                                                                                                                              | DECODE                           |
|-----------|------|------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| MAP_SRC_6 | 7:0  | Mapping source register 6. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT source for map 6 |

MIPI\_TX26 (0x4DA)

| BIT         | 7              | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|----------------|---|---|---|---|---|---|---|
| Field       | MAP_DST_6[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00           |   |   |   |   |   |   |   |
| Access Type | Write, Read    |   |   |   |   |   |   |   |

| BITFIELD  | BITS | DESCRIPTION                                                                                                                                   | DECODE                                |
|-----------|------|-----------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| MAP_DST_6 | 7:0  | Mapping destination register 6. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT destination for map 6 |

MIPI\_TX27 (0x4DB)

| BIT         | 7              | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|----------------|---|---|---|---|---|---|---|
| Field       | MAP_SRC_7[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00           |   |   |   |   |   |   |   |
| Access Type | Write, Read    |   |   |   |   |   |   |   |

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| BITFIELD  | BITS | DESCRIPTION                                                                                                                              | DECODE                           |
|-----------|------|------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| MAP_SRC_7 | 7:0  | Mapping source register 7. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT source for map 7 |

**MIPI\_TX28 (0x4DC)**

| BIT         | 7              | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|----------------|---|---|---|---|---|---|---|
| Field       | MAP_DST_7[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00           |   |   |   |   |   |   |   |
| Access Type | Write, Read    |   |   |   |   |   |   |   |

| BITFIELD  | BITS | DESCRIPTION                                                                                                                                   | DECODE                                |
|-----------|------|-----------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| MAP_DST_7 | 7:0  | Mapping destination register 7. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT destination for map 7 |

**MIPI\_TX29 (0x4DD)**

| BIT         | 7              | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|----------------|---|---|---|---|---|---|---|
| Field       | MAP_SRC_8[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00           |   |   |   |   |   |   |   |
| Access Type | Write, Read    |   |   |   |   |   |   |   |

| BITFIELD  | BITS | DESCRIPTION                                                                                                                              | DECODE                           |
|-----------|------|------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| MAP_SRC_8 | 7:0  | Mapping source register 8. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT source for map 8 |

**MIPI\_TX30 (0x4DE)**

| BIT         | 7              | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|----------------|---|---|---|---|---|---|---|
| Field       | MAP_DST_8[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00           |   |   |   |   |   |   |   |
| Access Type | Write, Read    |   |   |   |   |   |   |   |

| BITFIELD  | BITS | DESCRIPTION                                                                                                                                   | DECODE                                |
|-----------|------|-----------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| MAP_DST_8 | 7:0  | Mapping destination register 8. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT destination for map 8 |

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MIPI\_TX31 (0x4DF)

| BIT         | 7              | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|----------------|---|---|---|---|---|---|---|
| Field       | MAP_SRC_9[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00           |   |   |   |   |   |   |   |
| Access Type | Write, Read    |   |   |   |   |   |   |   |

| BITFIELD  | BITS | DESCRIPTION                                                                                                                              | DECODE                           |
|-----------|------|------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| MAP_SRC_9 | 7:0  | Mapping source register 9. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT source for map 9 |

MIPI\_TX32 (0x4E0)

| BIT         | 7              | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|----------------|---|---|---|---|---|---|---|
| Field       | MAP_DST_9[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00           |   |   |   |   |   |   |   |
| Access Type | Write, Read    |   |   |   |   |   |   |   |

| BITFIELD  | BITS | DESCRIPTION                                                                                                                                   | DECODE                                |
|-----------|------|-----------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| MAP_DST_9 | 7:0  | Mapping destination register 9. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT destination for map 9 |

MIPI\_TX33 (0x4E1)

| BIT         | 7               | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|-----------------|---|---|---|---|---|---|---|
| Field       | MAP_SRC_10[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00            |   |   |   |   |   |   |   |
| Access Type | Write, Read     |   |   |   |   |   |   |   |

| BITFIELD   | BITS | DESCRIPTION                                                                                                                               | DECODE                            |
|------------|------|-------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
| MAP_SRC_10 | 7:0  | Mapping source register 10. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT source for map 10 |

MIPI\_TX34 (0x4E2)

| BIT         | 7               | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|-----------------|---|---|---|---|---|---|---|
| Field       | MAP_DST_10[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00            |   |   |   |   |   |   |   |
| Access Type | Write, Read     |   |   |   |   |   |   |   |

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Appendix A: Register Map and Tables

| BITFIELD       | BITS | DESCRIPTION                                                                                                                                    | DECODE                                 |
|----------------|------|------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| MAP_DST_1<br>0 | 7:0  | Mapping destination register 10. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT destination for map 10 |

**MIPI\_TX35 (0x4E3)**

| BIT         | 7               | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|-----------------|---|---|---|---|---|---|---|
| Field       | MAP_SRC_11[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00            |   |   |   |   |   |   |   |
| Access Type | Write, Read     |   |   |   |   |   |   |   |

| BITFIELD       | BITS | DESCRIPTION                                                                                                                               | DECODE                            |
|----------------|------|-------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
| MAP_SRC_1<br>1 | 7:0  | Mapping source register 11. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT source for map 11 |

**MIPI\_TX36 (0x4E4)**

| BIT         | 7               | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|-----------------|---|---|---|---|---|---|---|
| Field       | MAP_DST_11[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00            |   |   |   |   |   |   |   |
| Access Type | Write, Read     |   |   |   |   |   |   |   |

| BITFIELD       | BITS | DESCRIPTION                                                                                                                                    | DECODE                                 |
|----------------|------|------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| MAP_DST_1<br>1 | 7:0  | Mapping destination register 11. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT destination for map 11 |

**MIPI\_TX37 (0x4E5)**

| BIT         | 7               | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|-----------------|---|---|---|---|---|---|---|
| Field       | MAP_SRC_12[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00            |   |   |   |   |   |   |   |
| Access Type | Write, Read     |   |   |   |   |   |   |   |

| BITFIELD       | BITS | DESCRIPTION                                                                                                                               | DECODE                            |
|----------------|------|-------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
| MAP_SRC_1<br>2 | 7:0  | Mapping source register 12. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT source for map 12 |

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MIPI\_TX38 (0x4E6)

| BIT         | 7               | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|-----------------|---|---|---|---|---|---|---|
| Field       | MAP_DST_12[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00            |   |   |   |   |   |   |   |
| Access Type | Write, Read     |   |   |   |   |   |   |   |

| BITFIELD   | BITS | DESCRIPTION                                                                                                                                    | DECODE                                 |
|------------|------|------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| MAP_DST_12 | 7:0  | Mapping destination register 12. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT destination for map 12 |

MIPI\_TX39 (0x4E7)

| BIT         | 7               | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|-----------------|---|---|---|---|---|---|---|
| Field       | MAP_SRC_13[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00            |   |   |   |   |   |   |   |
| Access Type | Write, Read     |   |   |   |   |   |   |   |

| BITFIELD   | BITS | DESCRIPTION                                                                                                                               | DECODE                            |
|------------|------|-------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
| MAP_SRC_13 | 7:0  | Mapping source register 13. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT source for map 13 |

MIPI\_TX40 (0x4E8)

| BIT         | 7               | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|-----------------|---|---|---|---|---|---|---|
| Field       | MAP_DST_13[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00            |   |   |   |   |   |   |   |
| Access Type | Write, Read     |   |   |   |   |   |   |   |

| BITFIELD   | BITS | DESCRIPTION                                                                                                                                    | DECODE                                 |
|------------|------|------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| MAP_DST_13 | 7:0  | Mapping destination register 13. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT destination for map 13 |

MIPI\_TX41 (0x4E9)

| BIT         | 7               | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|-----------------|---|---|---|---|---|---|---|
| Field       | MAP_SRC_14[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00            |   |   |   |   |   |   |   |
| Access Type | Write, Read     |   |   |   |   |   |   |   |

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| BITFIELD   | BITS | DESCRIPTION                                                                                                                               | DECODE                            |
|------------|------|-------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
| MAP_SRC_14 | 7:0  | Mapping source register 14. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT source for map 14 |

**MIPI\_TX42 (0x4EA)**

| BIT         | 7               | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|-----------------|---|---|---|---|---|---|---|
| Field       | MAP_DST_14[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00            |   |   |   |   |   |   |   |
| Access Type | Write, Read     |   |   |   |   |   |   |   |

| BITFIELD   | BITS | DESCRIPTION                                                                                                                                    | DECODE                                 |
|------------|------|------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| MAP_DST_14 | 7:0  | Mapping destination register 14. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT destination for map 14 |

**MIPI\_TX43 (0x4EB)**

| BIT         | 7               | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|-----------------|---|---|---|---|---|---|---|
| Field       | MAP_SRC_15[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00            |   |   |   |   |   |   |   |
| Access Type | Write, Read     |   |   |   |   |   |   |   |

| BITFIELD   | BITS | DESCRIPTION                                                                                                                               | DECODE                            |
|------------|------|-------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
| MAP_SRC_15 | 7:0  | Mapping source register 15. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT source for map 15 |

**MIPI\_TX44 (0x4EC)**

| BIT         | 7               | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|-----------------|---|---|---|---|---|---|---|
| Field       | MAP_DST_15[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00            |   |   |   |   |   |   |   |
| Access Type | Write, Read     |   |   |   |   |   |   |   |

| BITFIELD   | BITS | DESCRIPTION                                                                                                                                    | DECODE                                 |
|------------|------|------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| MAP_DST_15 | 7:0  | Mapping destination register 15. The register is split into two decode segments:<br>bit [7:6] VC - Virtual channel<br>bit [5:0] DT - Data type | 0xXX: VC and DT destination for map 15 |

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**MIPI\_TX45 (0x4ED)**

| BIT         | 7                    | 6 | 5                    | 4 | 3                    | 2 | 1                    | 0 |
|-------------|----------------------|---|----------------------|---|----------------------|---|----------------------|---|
| Field       | MAP_DPHY_DEST_3[1:0] |   | MAP_DPHY_DEST_2[1:0] |   | MAP_DPHY_DEST_1[1:0] |   | MAP_DPHY_DEST_0[1:0] |   |
| Reset       | 0b00                 |   | 0b00                 |   | 0b00                 |   | 0b00                 |   |
| Access Type | Write, Read          |   | Write, Read          |   | Write, Read          |   | Write, Read          |   |

| BITFIELD        | BITS | DESCRIPTION                                                          | DECODE                                                                               |
|-----------------|------|----------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| MAP_DPHY_DEST_3 | 7:6  | CSI2 controller destination for MAP_SRC_3 and MAP_DST_0 mapping regs | 0b00: Map to DPHY0<br>0b01: Map to DPHY1<br>0b10: Map to DPHY2<br>0b11: Map to DPHY3 |
| MAP_DPHY_DEST_2 | 5:4  | CSI2 controller destination for MAP_SRC_2 and MAP_DST_0 mapping regs | 0b00: Map to DPHY0<br>0b01: Map to DPHY1<br>0b10: Map to DPHY2<br>0b11: Map to DPHY3 |
| MAP_DPHY_DEST_1 | 3:2  | CSI2 controller destination for MAP_SRC_1 and MAP_DST_0 mapping regs | 0b00: Map to DPHY0<br>0b01: Map to DPHY1<br>0b10: Map to DPHY2<br>0b11: Map to DPHY3 |
| MAP_DPHY_DEST_0 | 1:0  | CSI2 controller destination for MAP_SRC_0 and MAP_DST_0 mapping regs | 0b00: Map to DPHY0<br>0b01: Map to DPHY1<br>0b10: Map to DPHY2<br>0b11: Map to DPHY3 |

**MIPI\_TX46 (0x4EE)**

| BIT         | 7                    | 6 | 5                    | 4 | 3                    | 2 | 1                    | 0 |
|-------------|----------------------|---|----------------------|---|----------------------|---|----------------------|---|
| Field       | MAP_DPHY_DEST_7[1:0] |   | MAP_DPHY_DEST_6[1:0] |   | MAP_DPHY_DEST_5[1:0] |   | MAP_DPHY_DEST_4[1:0] |   |
| Reset       | 0b00                 |   | 0b00                 |   | 0b00                 |   | 0b00                 |   |
| Access Type | Write, Read          |   | Write, Read          |   | Write, Read          |   | Write, Read          |   |

| BITFIELD        | BITS | DESCRIPTION                                                          | DECODE                                                                               |
|-----------------|------|----------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| MAP_DPHY_DEST_7 | 7:6  | CSI2 controller destination for MAP_SRC_7 and MAP_DST_0 mapping regs | 0b00: Map to DPHY0<br>0b01: Map to DPHY1<br>0b10: Map to DPHY2<br>0b11: Map to DPHY3 |
| MAP_DPHY_DEST_6 | 5:4  | CSI2 controller destination for MAP_SRC_6 and MAP_DST_0 mapping regs | 0b00: Map to DPHY0<br>0b01: Map to DPHY1<br>0b10: Map to DPHY2<br>0b11: Map to DPHY3 |
| MAP_DPHY_DEST_5 | 3:2  | CSI2 controller destination for MAP_SRC_5 and MAP_DST_0 mapping regs | 0b00: Map to DPHY0<br>0b01: Map to DPHY1<br>0b10: Map to DPHY2<br>0b11: Map to DPHY3 |
| MAP_DPHY_DEST_4 | 1:0  | CSI2 controller destination for MAP_SRC_4 and MAP_DST_0 mapping regs | 0b00: Map to DPHY0<br>0b01: Map to DPHY1<br>0b10: Map to DPHY2<br>0b11: Map to DPHY3 |



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MIPI\_TX47 (0x4EF)

| BIT         | 7                     | 6 | 5                     | 4 | 3                    | 2 | 1                    | 0 |
|-------------|-----------------------|---|-----------------------|---|----------------------|---|----------------------|---|
| Field       | MAP_DPHY_DEST_11[1:0] |   | MAP_DPHY_DEST_10[1:0] |   | MAP_DPHY_DEST_9[1:0] |   | MAP_DPHY_DEST_8[1:0] |   |
| Reset       | 0b00                  |   | 0b00                  |   | 0b00                 |   | 0b00                 |   |
| Access Type | Write, Read           |   | Write, Read           |   | Write, Read          |   | Write, Read          |   |

| BITFIELD         | BITS | DESCRIPTION                                                           | DECODE                                                                               |
|------------------|------|-----------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| MAP_DPHY_DEST_11 | 7:6  | CSI2 controller destination for MAP_SRC_11 and MAP_DST_0 mapping regs | 0b00: Map to DPHY0<br>0b01: Map to DPHY1<br>0b10: Map to DPHY2<br>0b11: Map to DPHY3 |
| MAP_DPHY_DEST_10 | 5:4  | CSI2 controller destination for MAP_SRC_10 and MAP_DST_0 mapping regs | 0b00: Map to DPHY0<br>0b01: Map to DPHY1<br>0b10: Map to DPHY2<br>0b11: Map to DPHY3 |
| MAP_DPHY_DEST_9  | 3:2  | CSI2 controller destination for MAP_SRC_9 and MAP_DST_0 mapping regs  | 0b00: Map to DPHY0<br>0b01: Map to DPHY1<br>0b10: Map to DPHY2<br>0b11: Map to DPHY3 |
| MAP_DPHY_DEST_8  | 1:0  | CSI2 controller destination for MAP_SRC_8 and MAP_DST_0 mapping regs  | 0b00: Map to DPHY0<br>0b01: Map to DPHY1<br>0b10: Map to DPHY2<br>0b11: Map to DPHY3 |

MIPI\_TX48 (0x4F0)

| BIT         | 7                     | 6 | 5                     | 4 | 3                     | 2 | 1                     | 0 |
|-------------|-----------------------|---|-----------------------|---|-----------------------|---|-----------------------|---|
| Field       | MAP_DPHY_DEST_15[1:0] |   | MAP_DPHY_DEST_14[1:0] |   | MAP_DPHY_DEST_13[1:0] |   | MAP_DPHY_DEST_12[1:0] |   |
| Reset       | 0b00                  |   | 0b00                  |   | 0b00                  |   | 0b00                  |   |
| Access Type | Write, Read           |   | Write, Read           |   | Write, Read           |   | Write, Read           |   |

| BITFIELD         | BITS | DESCRIPTION                                                           | DECODE                                                                               |
|------------------|------|-----------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| MAP_DPHY_DEST_15 | 7:6  | CSI2 controller destination for MAP_SRC_15 and MAP_DST_0 mapping regs | 0b00: Map to DPHY0<br>0b01: Map to DPHY1<br>0b10: Map to DPHY2<br>0b11: Map to DPHY3 |
| MAP_DPHY_DEST_14 | 5:4  | CSI2 controller destination for MAP_SRC_14 and MAP_DST_0 mapping regs | 0b00: Map to DPHY0<br>0b01: Map to DPHY1<br>0b10: Map to DPHY2<br>0b11: Map to DPHY3 |
| MAP_DPHY_DEST_13 | 3:2  | CSI2 controller destination for MAP_SRC_13 and MAP_DST_0 mapping regs | 0b00: Map to DPHY0<br>0b01: Map to DPHY1<br>0b10: Map to DPHY2<br>0b11: Map to DPHY3 |
| MAP_DPHY_DEST_12 | 1:0  | CSI2 controller destination for MAP_SRC_12 and MAP_DST_0 mapping regs | 0b00: Map to DPHY0<br>0b01: Map to DPHY1<br>0b10: Map to DPHY2<br>0b11: Map to DPHY3 |

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MIPI\_TX49 (0x4F1)

| BIT         | 7            | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|--------------|---|---|---|---|---|---|---|
| Field       | MAP_CON[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00         |   |   |   |   |   |   |   |
| Access Type | Write, Read  |   |   |   |   |   |   |   |

| BITFIELD | BITS | DESCRIPTION                                                                                                                                                                                                                                                                                                               | DECODE                                                                                                                                                                                                                                                                                                                                                                                                                               |
|----------|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MAP_CON  | 7:0  | Concatenation register for sync mode settings. The register is split into four decode segments:<br>bit [7] Enable 4WxH or Wx4H<br>bit [6] Reserved<br>bit [5:4] Master video stream number. This is the first line to concatenate. All others follow in order<br>bit [3:0] Each bit selects a video stream to concatenate | 0bXXXXXXXX1: Concatenate video stream 0<br>0bXXXXXXXX1X: Concatenate video stream 1<br>0bXXXXX1XX: Concatenate video stream 2<br>0bXXXXX1XXX: Concatenate video stream 3<br>0bXX00XXXX: Select video stream 0 as master<br>0bXX01XXXX: Select video stream 1 as master<br>0bXX10XXXX: Select video stream 2 as master<br>0bXX11XXXX: Select video stream 3 as master<br>0b0XXXXXXX: Enable Wx4H mode<br>0b1XXXXXXX: Enable 4WxH mode |

MIPI\_TX51 (0x4F3)

| BIT         | 7 | 6 | 5 | 4             | 3           | 2             | 1            | 0             |
|-------------|---|---|---|---------------|-------------|---------------|--------------|---------------|
| Field       | – | – | – | ALT2_MEM_MAP8 | MODE_DT     | ALT_MEM_MAP10 | ALT_MEM_MAP8 | ALT_MEM_MAP12 |
| Reset       | – | – | – | 0b0           | 0b0         | 0b0           | 0b0          | 0b0           |
| Access Type | – | – | – | Write, Read   | Write, Read | Write, Read   | Write, Read  | Write, Read   |

| BITFIELD      | BITS | DESCRIPTION                                     | DECODE                                                                                        |
|---------------|------|-------------------------------------------------|-----------------------------------------------------------------------------------------------|
| ALT2_MEM_MAP8 | 4    | Alternative memory map enable for 8-bit DT      | 0b0: Alternate memory map disabled<br>0b1: Alternate memory map enabled                       |
| MODE_DT       | 3    | MIPI TX enable 24-bit packing of 8-bit MIPI UDP | 0x0: Disable 24-bit packing of 8-bit MIPI UDP<br>0x1: Enable 24-bit packing of 8-bit MIPI UDP |
| ALT_MEM_MAP10 | 2    | Alternative memory map enable for 10-bit DT     | 0b0: Alternate memory map disabled<br>0b1: Alternate memory map enabled                       |
| ALT_MEM_MAP8  | 1    | Alternative memory map enable for 8-bit DT      | 0b0: Alternate memory map disabled<br>0b1: Alternate memory map enabled                       |
| ALT_MEM_MAP12 | 0    | Alternative memory map enable for 12-bit DT     | 0b0: Alternate memory map disabled<br>0b1: Alternate memory map enabled                       |

MIPI\_TX\_EXT0 (0x500)

| BIT         | 7                | 6 | 5 | 4                | 3 | 2 | 1 | 0 |
|-------------|------------------|---|---|------------------|---|---|---|---|
| Field       | MAP_SRC_0_H[2:0] |   |   | MAP_DST_0_H[2:0] |   |   | – | – |
| Reset       | 0b000            |   |   | 0b000            |   |   | – | – |
| Access Type | Write, Read      |   |   | Write, Read      |   |   | – | – |

| BITFIELD    | BITS | DESCRIPTION                                              | DECODE                                        |
|-------------|------|----------------------------------------------------------|-----------------------------------------------|
| MAP_SRC_0_H | 7:5  | Mapping register source VC high 3 bits for extended mode | 0bXXX: Mapping register source VC high 3 bits |

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Appendix A: Register Map and Tables

| BITFIELD    | BITS | DESCRIPTION                                                   | DECODE                                             |
|-------------|------|---------------------------------------------------------------|----------------------------------------------------|
| MAP_DST_0_H | 4:2  | Mapping register destination VC high 3 bits for extended mode | 0bXXX: Mapping register destination VC high 3 bits |

MIPI\_TX\_EXT1 (0x501)

| BIT         | 7                | 6 | 5 | 4                | 3 | 2 | 1 | 0 |
|-------------|------------------|---|---|------------------|---|---|---|---|
| Field       | MAP_SRC_1_H[2:0] |   |   | MAP_DST_1_H[2:0] |   |   | – | – |
| Reset       | 0b000            |   |   | 0b000            |   |   | – | – |
| Access Type | Write, Read      |   |   | Write, Read      |   |   | – | – |

| BITFIELD    | BITS | DESCRIPTION                                                   | DECODE                                             |
|-------------|------|---------------------------------------------------------------|----------------------------------------------------|
| MAP_SRC_1_H | 7:5  | Mapping register source VC high 3 bits for extended mode      | 0bXXX: Mapping register source VC high 3 bits      |
| MAP_DST_1_H | 4:2  | Mapping register destination VC high 3 bits for extended mode | 0bXXX: Mapping register destination VC high 3 bits |

MIPI\_TX\_EXT2 (0x502)

| BIT         | 7                | 6 | 5 | 4                | 3 | 2 | 1 | 0 |
|-------------|------------------|---|---|------------------|---|---|---|---|
| Field       | MAP_SRC_2_H[2:0] |   |   | MAP_DST_2_H[2:0] |   |   | – | – |
| Reset       | 0b000            |   |   | 0b000            |   |   | – | – |
| Access Type | Write, Read      |   |   | Write, Read      |   |   | – | – |

| BITFIELD    | BITS | DESCRIPTION                                                   | DECODE                                             |
|-------------|------|---------------------------------------------------------------|----------------------------------------------------|
| MAP_SRC_2_H | 7:5  | Mapping register source VC high 3 bits for extended mode      | 0bXXX: Mapping register source VC high 3 bits      |
| MAP_DST_2_H | 4:2  | Mapping register destination VC high 3 bits for extended mode | 0bXXX: Mapping register destination VC high 3 bits |

MIPI\_TX\_EXT3 (0x503)

| BIT         | 7                | 6 | 5 | 4                | 3 | 2 | 1 | 0 |
|-------------|------------------|---|---|------------------|---|---|---|---|
| Field       | MAP_SRC_3_H[2:0] |   |   | MAP_DST_3_H[2:0] |   |   | – | – |
| Reset       | 0b000            |   |   | 0b000            |   |   | – | – |
| Access Type | Write, Read      |   |   | Write, Read      |   |   | – | – |

| BITFIELD    | BITS | DESCRIPTION                                                   | DECODE                                             |
|-------------|------|---------------------------------------------------------------|----------------------------------------------------|
| MAP_SRC_3_H | 7:5  | Mapping register source VC high 3 bits for extended mode      | 0bXXX: Mapping register source VC high 3 bits      |
| MAP_DST_3_H | 4:2  | Mapping register destination VC high 3 bits for extended mode | 0bXXX: Mapping register destination VC high 3 bits |

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Appendix A: Register Map and Tables

MIPI\_TX\_EXT4 (0x504)

| BIT         | 7                | 6 | 5 | 4                | 3 | 2 | 1 | 0 |
|-------------|------------------|---|---|------------------|---|---|---|---|
| Field       | MAP_SRC_4_H[2:0] |   |   | MAP_DST_4_H[2:0] |   |   | – | – |
| Reset       | 0b000            |   |   | 0b000            |   |   | – | – |
| Access Type | Write, Read      |   |   | Write, Read      |   |   | – | – |

| BITFIELD    | BITS | DESCRIPTION                                                   | DECODE                                             |
|-------------|------|---------------------------------------------------------------|----------------------------------------------------|
| MAP_SRC_4_H | 7:5  | Mapping register source VC high 3 bits for extended mode      | 0bXXX: Mapping register source VC high 3 bits      |
| MAP_DST_4_H | 4:2  | Mapping register destination VC high 3 bits for extended mode | 0bXXX: Mapping register destination VC high 3 bits |

MIPI\_TX\_EXT5 (0x505)

| BIT         | 7                | 6 | 5 | 4                | 3 | 2 | 1 | 0 |
|-------------|------------------|---|---|------------------|---|---|---|---|
| Field       | MAP_SRC_5_H[2:0] |   |   | MAP_DST_5_H[2:0] |   |   | – | – |
| Reset       | 0b000            |   |   | 0b000            |   |   | – | – |
| Access Type | Write, Read      |   |   | Write, Read      |   |   | – | – |

| BITFIELD    | BITS | DESCRIPTION                                                   | DECODE                                             |
|-------------|------|---------------------------------------------------------------|----------------------------------------------------|
| MAP_SRC_5_H | 7:5  | Mapping register source VC high 3 bits for extended mode      | 0bXXX: Mapping register source VC high 3 bits      |
| MAP_DST_5_H | 4:2  | Mapping register destination VC high 3 bits for extended mode | 0bXXX: Mapping register destination VC high 3 bits |

MIPI\_TX\_EXT6 (0x506)

| BIT         | 7                | 6 | 5 | 4                | 3 | 2 | 1 | 0 |
|-------------|------------------|---|---|------------------|---|---|---|---|
| Field       | MAP_SRC_6_H[2:0] |   |   | MAP_DST_6_H[2:0] |   |   | – | – |
| Reset       | 0b000            |   |   | 0b000            |   |   | – | – |
| Access Type | Write, Read      |   |   | Write, Read      |   |   | – | – |

| BITFIELD    | BITS | DESCRIPTION                                                   | DECODE                                             |
|-------------|------|---------------------------------------------------------------|----------------------------------------------------|
| MAP_SRC_6_H | 7:5  | Mapping register source VC High 3 bits for extended mode      | 0bXXX: Mapping register source VC High 3 bits      |
| MAP_DST_6_H | 4:2  | Mapping register destination VC high 3 bits for extended mode | 0bXXX: Mapping register destination VC high 3 bits |

MIPI\_TX\_EXT7 (0x507)

| BIT         | 7                | 6 | 5 | 4                | 3 | 2 | 1 | 0 |
|-------------|------------------|---|---|------------------|---|---|---|---|
| Field       | MAP_SRC_7_H[2:0] |   |   | MAP_DST_7_H[2:0] |   |   | – | – |
| Reset       | 0b000            |   |   | 0b000            |   |   | – | – |
| Access Type | Write, Read      |   |   | Write, Read      |   |   | – | – |

| BITFIELD    | BITS | DESCRIPTION                                              | DECODE                                        |
|-------------|------|----------------------------------------------------------|-----------------------------------------------|
| MAP_SRC_7_H | 7:5  | Mapping register source VC high 3 bits for extended mode | 0bXXX: Mapping register source VC high 3 bits |

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Appendix A: Register Map and Tables

| BITFIELD    | BITS | DESCRIPTION                                                   | DECODE                                             |
|-------------|------|---------------------------------------------------------------|----------------------------------------------------|
| MAP_DST_7_H | 4:2  | Mapping register destination VC high 3 bits for extended mode | 0bXXX: Mapping register destination VC high 3 bits |

**MIPI\_TX\_EXT8 (0x508)**

| BIT         | 7                | 6 | 5 | 4                | 3 | 2 | 1 | 0 |
|-------------|------------------|---|---|------------------|---|---|---|---|
| Field       | MAP_SRC_8_H[2:0] |   |   | MAP_DST_8_H[2:0] |   |   | – | – |
| Reset       | 0b000            |   |   | 0b000            |   |   | – | – |
| Access Type | Write, Read      |   |   | Write, Read      |   |   | – | – |

| BITFIELD    | BITS | DESCRIPTION                                                   | DECODE                                             |
|-------------|------|---------------------------------------------------------------|----------------------------------------------------|
| MAP_SRC_8_H | 7:5  | Mapping register source VC high 3 bits for extended mode      | 0bXXX: Mapping register source VC high 3 bits      |
| MAP_DST_8_H | 4:2  | Mapping register destination VC high 3 bits for extended mode | 0bXXX: Mapping register destination VC high 3 bits |

**MIPI\_TX\_EXT9 (0x509)**

| BIT         | 7                | 6 | 5 | 4                | 3 | 2 | 1 | 0 |
|-------------|------------------|---|---|------------------|---|---|---|---|
| Field       | MAP_SRC_9_H[2:0] |   |   | MAP_DST_9_H[2:0] |   |   | – | – |
| Reset       | 0b000            |   |   | 0b000            |   |   | – | – |
| Access Type | Write, Read      |   |   | Write, Read      |   |   | – | – |

| BITFIELD    | BITS | DESCRIPTION                                                   | DECODE                                             |
|-------------|------|---------------------------------------------------------------|----------------------------------------------------|
| MAP_SRC_9_H | 7:5  | Mapping register source VC high 3 bits for extended mode      | 0bXXX: Mapping register source VC high 3 bits      |
| MAP_DST_9_H | 4:2  | Mapping register destination VC high 3 bits for extended mode | 0bXXX: Mapping register destination VC high 3 bits |

**MIPI\_TX\_EXT10 (0x50A)**

| BIT         | 7                 | 6 | 5 | 4                 | 3 | 2 | 1 | 0 |
|-------------|-------------------|---|---|-------------------|---|---|---|---|
| Field       | MAP_SRC_10_H[2:0] |   |   | MAP_DST_10_H[2:0] |   |   | – | – |
| Reset       | 0b000             |   |   | 0b000             |   |   | – | – |
| Access Type | Write, Read       |   |   | Write, Read       |   |   | – | – |

| BITFIELD     | BITS | DESCRIPTION                                                   | DECODE                                             |
|--------------|------|---------------------------------------------------------------|----------------------------------------------------|
| MAP_SRC_10_H | 7:5  | Mapping register source VC high 3 bits for extended mode      | 0bXXX: Mapping register source VC high 3 bits      |
| MAP_DST_10_H | 4:2  | Mapping register destination VC high 3 bits for extended mode | 0bXXX: Mapping register destination VC high 3 bits |

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MIPI\_TX\_EXT11 (0x50B)

| BIT         | 7                 | 6 | 5 | 4                 | 3 | 2 | 1 | 0 |
|-------------|-------------------|---|---|-------------------|---|---|---|---|
| Field       | MAP_SRC_11_H[2:0] |   |   | MAP_DST_11_H[2:0] |   |   | – | – |
| Reset       | 0b000             |   |   | 0b000             |   |   | – | – |
| Access Type | Write, Read       |   |   | Write, Read       |   |   | – | – |

| BITFIELD     | BITS | DESCRIPTION                                                   | DECODE                                             |
|--------------|------|---------------------------------------------------------------|----------------------------------------------------|
| MAP_SRC_11_H | 7:5  | Mapping register source VC high 3 bits for extended mode      | 0bXXX: Mapping register source VC high 3 bits      |
| MAP_DST_11_H | 4:2  | Mapping register destination VC High 3 bits for extended mode | 0bXXX: Mapping register destination VC High 3 bits |

MIPI\_TX\_EXT12 (0x50C)

| BIT         | 7                 | 6 | 5 | 4                 | 3 | 2 | 1 | 0 |
|-------------|-------------------|---|---|-------------------|---|---|---|---|
| Field       | MAP_SRC_12_H[2:0] |   |   | MAP_DST_12_H[2:0] |   |   | – | – |
| Reset       | 0b000             |   |   | 0b000             |   |   | – | – |
| Access Type | Write, Read       |   |   | Write, Read       |   |   | – | – |

| BITFIELD     | BITS | DESCRIPTION                                                   | DECODE                                             |
|--------------|------|---------------------------------------------------------------|----------------------------------------------------|
| MAP_SRC_12_H | 7:5  | Mapping register source VC high 3 bits for extended mode      | 0bXXX: Mapping register source VC high 3 bits      |
| MAP_DST_12_H | 4:2  | Mapping register destination VC high 3 bits for extended mode | 0bXXX: Mapping register destination VC high 3 bits |

MIPI\_TX\_EXT13 (0x50D)

| BIT         | 7                 | 6 | 5 | 4                 | 3 | 2 | 1 | 0 |
|-------------|-------------------|---|---|-------------------|---|---|---|---|
| Field       | MAP_SRC_13_H[2:0] |   |   | MAP_DST_13_H[2:0] |   |   | – | – |
| Reset       | 0b000             |   |   | 0b000             |   |   | – | – |
| Access Type | Write, Read       |   |   | Write, Read       |   |   | – | – |

| BITFIELD     | BITS | DESCRIPTION                                                   | DECODE                                             |
|--------------|------|---------------------------------------------------------------|----------------------------------------------------|
| MAP_SRC_13_H | 7:5  | Mapping register source VC high 3 bits for extended mode      | 0bXXX: Mapping register source VC high 3 bits      |
| MAP_DST_13_H | 4:2  | Mapping register destination VC high 3 bits for extended mode | 0bXXX: Mapping register destination VC high 3 bits |

MIPI\_TX\_EXT14 (0x50E)

| BIT         | 7                 | 6 | 5 | 4                 | 3 | 2 | 1 | 0 |
|-------------|-------------------|---|---|-------------------|---|---|---|---|
| Field       | MAP_SRC_14_H[2:0] |   |   | MAP_DST_14_H[2:0] |   |   | – | – |
| Reset       | 0b000             |   |   | 0b000             |   |   | – | – |
| Access Type | Write, Read       |   |   | Write, Read       |   |   | – | – |

| BITFIELD     | BITS | DESCRIPTION                                              | DECODE                                        |
|--------------|------|----------------------------------------------------------|-----------------------------------------------|
| MAP_SRC_14_H | 7:5  | Mapping register source VC high 3 bits for extended mode | 0bXXX: Mapping register source VC high 3 bits |

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| BITFIELD     | BITS | DESCRIPTION                                                   | DECODE                                             |
|--------------|------|---------------------------------------------------------------|----------------------------------------------------|
| MAP_DST_14_H | 4:2  | Mapping register destination VC high 3 bits for extended mode | 0bXXX: Mapping register destination VC high 3 bits |

MIPI\_TX\_EXT15 (0x50F)

| BIT         | 7                 | 6 | 5 | 4                 | 3 | 2 | 1 | 0 |
|-------------|-------------------|---|---|-------------------|---|---|---|---|
| Field       | MAP_SRC_15_H[2:0] |   |   | MAP_DST_15_H[2:0] |   |   | – | – |
| Reset       | 0b000             |   |   | 0b000             |   |   | – | – |
| Access Type | Write, Read       |   |   | Write, Read       |   |   | – | – |

| BITFIELD     | BITS | DESCRIPTION                                                   | DECODE                                             |
|--------------|------|---------------------------------------------------------------|----------------------------------------------------|
| MAP_SRC_15_H | 7:5  | Mapping register source VC high 3 bits for extended mode      | 0bXXX: Mapping register source VC high 3 bits      |
| MAP_DST_15_H | 4:2  | Mapping register destination VC high 3 bits for extended mode | 0bXXX: Mapping register destination VC high 3 bits |

MIPI\_TX\_EXT0 (0x510)

| BIT         | 7                | 6 | 5 | 4                | 3 | 2 | 1 | 0 |
|-------------|------------------|---|---|------------------|---|---|---|---|
| Field       | MAP_SRC_0_H[2:0] |   |   | MAP_DST_0_H[2:0] |   |   | – | – |
| Reset       | 0b000            |   |   | 0b000            |   |   | – | – |
| Access Type | Write, Read      |   |   | Write, Read      |   |   | – | – |

| BITFIELD    | BITS | DESCRIPTION                                                   | DECODE                                             |
|-------------|------|---------------------------------------------------------------|----------------------------------------------------|
| MAP_SRC_0_H | 7:5  | Mapping register source VC high 3 bits for extended mode      | 0bXXX: Mapping register source VC high 3 bits      |
| MAP_DST_0_H | 4:2  | Mapping register destination VC high 3 bits for extended mode | 0bXXX: Mapping register destination VC high 3 bits |

MIPI\_TX\_EXT1 (0x511)

| BIT         | 7                | 6 | 5 | 4                | 3 | 2 | 1 | 0 |
|-------------|------------------|---|---|------------------|---|---|---|---|
| Field       | MAP_SRC_1_H[2:0] |   |   | MAP_DST_1_H[2:0] |   |   | – | – |
| Reset       | 0b000            |   |   | 0b000            |   |   | – | – |
| Access Type | Write, Read      |   |   | Write, Read      |   |   | – | – |

| BITFIELD    | BITS | DESCRIPTION                                                   | DECODE                                             |
|-------------|------|---------------------------------------------------------------|----------------------------------------------------|
| MAP_SRC_1_H | 7:5  | Mapping register source VC high 3 bits for extended mode      | 0bXXX: Mapping register source VC high 3 bits      |
| MAP_DST_1_H | 4:2  | Mapping register destination VC high 3 bits for extended mode | 0bXXX: Mapping register destination VC high 3 bits |

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MIPI\_TX\_EXT2 (0x512)

| BIT         | 7                | 6 | 5 | 4                | 3 | 2 | 1 | 0 |
|-------------|------------------|---|---|------------------|---|---|---|---|
| Field       | MAP_SRC_2_H[2:0] |   |   | MAP_DST_2_H[2:0] |   |   | – | – |
| Reset       | 0b000            |   |   | 0b000            |   |   | – | – |
| Access Type | Write, Read      |   |   | Write, Read      |   |   | – | – |

| BITFIELD    | BITS | DESCRIPTION                                                   | DECODE                                             |
|-------------|------|---------------------------------------------------------------|----------------------------------------------------|
| MAP_SRC_2_H | 7:5  | Mapping register source VC high 3 bits for extended mode      | 0bXXX: Mapping register source VC high 3 bits      |
| MAP_DST_2_H | 4:2  | Mapping register destination VC high 3 bits for extended mode | 0bXXX: Mapping register destination VC high 3 bits |

MIPI\_TX\_EXT3 (0x513)

| BIT         | 7                | 6 | 5 | 4                | 3 | 2 | 1 | 0 |
|-------------|------------------|---|---|------------------|---|---|---|---|
| Field       | MAP_SRC_3_H[2:0] |   |   | MAP_DST_3_H[2:0] |   |   | – | – |
| Reset       | 0b000            |   |   | 0b000            |   |   | – | – |
| Access Type | Write, Read      |   |   | Write, Read      |   |   | – | – |

| BITFIELD    | BITS | DESCRIPTION                                                   | DECODE                                             |
|-------------|------|---------------------------------------------------------------|----------------------------------------------------|
| MAP_SRC_3_H | 7:5  | Mapping register source VC high 3 bits for extended mode      | 0bXXX: Mapping register source VC high 3 bits      |
| MAP_DST_3_H | 4:2  | Mapping register destination VC high 3 bits for extended mode | 0bXXX: Mapping register destination VC high 3 bits |

MIPI\_TX\_EXT4 (0x514)

| BIT         | 7                | 6 | 5 | 4                | 3 | 2 | 1 | 0 |
|-------------|------------------|---|---|------------------|---|---|---|---|
| Field       | MAP_SRC_4_H[2:0] |   |   | MAP_DST_4_H[2:0] |   |   | – | – |
| Reset       | 0b000            |   |   | 0b000            |   |   | – | – |
| Access Type | Write, Read      |   |   | Write, Read      |   |   | – | – |

| BITFIELD    | BITS | DESCRIPTION                                                   | DECODE                                             |
|-------------|------|---------------------------------------------------------------|----------------------------------------------------|
| MAP_SRC_4_H | 7:5  | Mapping register source VC high 3 bits for extended mode      | 0bXXX: Mapping register source VC high 3 bits      |
| MAP_DST_4_H | 4:2  | Mapping register destination VC high 3 bits for extended mode | 0bXXX: Mapping register destination VC high 3 bits |

MIPI\_TX\_EXT5 (0x515)

| BIT         | 7                | 6 | 5 | 4                | 3 | 2 | 1 | 0 |
|-------------|------------------|---|---|------------------|---|---|---|---|
| Field       | MAP_SRC_5_H[2:0] |   |   | MAP_DST_5_H[2:0] |   |   | – | – |
| Reset       | 0b000            |   |   | 0b000            |   |   | – | – |
| Access Type | Write, Read      |   |   | Write, Read      |   |   | – | – |

| BITFIELD    | BITS | DESCRIPTION                                              | DECODE                                        |
|-------------|------|----------------------------------------------------------|-----------------------------------------------|
| MAP_SRC_5_H | 7:5  | Mapping register source VC high 3 bits for extended mode | 0bXXX: Mapping register source VC high 3 bits |



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Appendix A: Register Map and Tables

| BITFIELD    | BITS | DESCRIPTION                                                   | DECODE                                             |
|-------------|------|---------------------------------------------------------------|----------------------------------------------------|
| MAP_DST_5_H | 4:2  | Mapping register destination VC high 3 bits for extended mode | 0bXXX: Mapping register destination VC high 3 bits |

MIPI\_TX\_EXT6 (0x516)

| BIT         | 7                | 6 | 5 | 4                | 3 | 2 | 1 | 0 |
|-------------|------------------|---|---|------------------|---|---|---|---|
| Field       | MAP_SRC_6_H[2:0] |   |   | MAP_DST_6_H[2:0] |   |   | – | – |
| Reset       | 0b000            |   |   | 0b000            |   |   | – | – |
| Access Type | Write, Read      |   |   | Write, Read      |   |   | – | – |

| BITFIELD    | BITS | DESCRIPTION                                                   | DECODE                                             |
|-------------|------|---------------------------------------------------------------|----------------------------------------------------|
| MAP_SRC_6_H | 7:5  | Mapping register source VC high 3 bits for extended mode      | 0bXXX: Mapping register source VC high 3 bits      |
| MAP_DST_6_H | 4:2  | Mapping register destination VC High 3 bits for extended mode | 0bXXX: Mapping register destination VC High 3 bits |

MIPI\_TX\_EXT7 (0x517)

| BIT         | 7                | 6 | 5 | 4                | 3 | 2 | 1 | 0 |
|-------------|------------------|---|---|------------------|---|---|---|---|
| Field       | MAP_SRC_7_H[2:0] |   |   | MAP_DST_7_H[2:0] |   |   | – | – |
| Reset       | 0b000            |   |   | 0b000            |   |   | – | – |
| Access Type | Write, Read      |   |   | Write, Read      |   |   | – | – |

| BITFIELD    | BITS | DESCRIPTION                                                   | DECODE                                             |
|-------------|------|---------------------------------------------------------------|----------------------------------------------------|
| MAP_SRC_7_H | 7:5  | Mapping register source VC high 3 bits for extended mode      | 0bXXX: Mapping register source VC high 3 bits      |
| MAP_DST_7_H | 4:2  | Mapping register destination VC high 3 bits for extended mode | 0bXXX: Mapping register destination VC high 3 bits |

MIPI\_TX\_EXT8 (0x518)

| BIT         | 7                | 6 | 5 | 4                | 3 | 2 | 1 | 0 |
|-------------|------------------|---|---|------------------|---|---|---|---|
| Field       | MAP_SRC_8_H[2:0] |   |   | MAP_DST_8_H[2:0] |   |   | – | – |
| Reset       | 0b000            |   |   | 0b000            |   |   | – | – |
| Access Type | Write, Read      |   |   | Write, Read      |   |   | – | – |

| BITFIELD    | BITS | DESCRIPTION                                                   | DECODE                                             |
|-------------|------|---------------------------------------------------------------|----------------------------------------------------|
| MAP_SRC_8_H | 7:5  | Mapping register source VC high 3 bits for extended mode      | 0bXXX: Mapping register source VC high 3 bits      |
| MAP_DST_8_H | 4:2  | Mapping register destination VC high 3 bits for extended mode | 0bXXX: Mapping register destination VC high 3 bits |

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Appendix A: Register Map and Tables

MIPI\_TX\_EXT9 (0x519)

| BIT         | 7                | 6 | 5 | 4                | 3 | 2 | 1 | 0 |
|-------------|------------------|---|---|------------------|---|---|---|---|
| Field       | MAP_SRC_9_H[2:0] |   |   | MAP_DST_9_H[2:0] |   |   | – | – |
| Reset       | 0b000            |   |   | 0b000            |   |   | – | – |
| Access Type | Write, Read      |   |   | Write, Read      |   |   | – | – |

| BITFIELD    | BITS | DESCRIPTION                                                   | DECODE                                             |
|-------------|------|---------------------------------------------------------------|----------------------------------------------------|
| MAP_SRC_9_H | 7:5  | Mapping register source VC high 3 bits for extended mode      | 0bXXX: Mapping register source VC high 3 bits      |
| MAP_DST_9_H | 4:2  | Mapping register destination VC high 3 bits for extended mode | 0bXXX: Mapping register destination VC high 3 bits |

MIPI\_TX\_EXT10 (0x51A)

| BIT         | 7                 | 6 | 5 | 4                 | 3 | 2 | 1 | 0 |
|-------------|-------------------|---|---|-------------------|---|---|---|---|
| Field       | MAP_SRC_10_H[2:0] |   |   | MAP_DST_10_H[2:0] |   |   | – | – |
| Reset       | 0b000             |   |   | 0b000             |   |   | – | – |
| Access Type | Write, Read       |   |   | Write, Read       |   |   | – | – |

| BITFIELD     | BITS | DESCRIPTION                                                   | DECODE                                             |
|--------------|------|---------------------------------------------------------------|----------------------------------------------------|
| MAP_SRC_10_H | 7:5  | Mapping register source VC high 3 bits for extended mode      | 0bXXX: Mapping register source VC high 3 bits      |
| MAP_DST_10_H | 4:2  | Mapping register destination VC high 3 bits for extended mode | 0bXXX: Mapping register destination VC high 3 bits |

MIPI\_TX\_EXT11 (0x51B)

| BIT         | 7                 | 6 | 5 | 4                 | 3 | 2 | 1 | 0 |
|-------------|-------------------|---|---|-------------------|---|---|---|---|
| Field       | MAP_SRC_11_H[2:0] |   |   | MAP_DST_11_H[2:0] |   |   | – | – |
| Reset       | 0b000             |   |   | 0b000             |   |   | – | – |
| Access Type | Write, Read       |   |   | Write, Read       |   |   | – | – |

| BITFIELD     | BITS | DESCRIPTION                                                   | DECODE                                             |
|--------------|------|---------------------------------------------------------------|----------------------------------------------------|
| MAP_SRC_11_H | 7:5  | Mapping register source VC high 3 bits for extended mode      | 0bXXX: Mapping register source VC high 3 bits      |
| MAP_DST_11_H | 4:2  | Mapping register destination VC high 3 bits for extended mode | 0bXXX: Mapping register destination VC high 3 bits |

MIPI\_TX\_EXT12 (0x51C)

| BIT         | 7                 | 6 | 5 | 4                 | 3 | 2 | 1 | 0 |
|-------------|-------------------|---|---|-------------------|---|---|---|---|
| Field       | MAP_SRC_12_H[2:0] |   |   | MAP_DST_12_H[2:0] |   |   | – | – |
| Reset       | 0b000             |   |   | 0b000             |   |   | – | – |
| Access Type | Write, Read       |   |   | Write, Read       |   |   | – | – |

| BITFIELD     | BITS | DESCRIPTION                                              | DECODE                                        |
|--------------|------|----------------------------------------------------------|-----------------------------------------------|
| MAP_SRC_12_H | 7:5  | Mapping register source VC high 3 bits for extended mode | 0bXXX: Mapping register source VC high 3 bits |

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Appendix A: Register Map and Tables

| BITFIELD     | BITS | DESCRIPTION                                                   | DECODE                                             |
|--------------|------|---------------------------------------------------------------|----------------------------------------------------|
| MAP_DST_12_H | 4:2  | Mapping register destination VC high 3 bits for extended mode | 0bXXX: Mapping register destination VC high 3 bits |

MIPI\_TX\_EXT13 (0x51D)

| BIT         | 7                 | 6 | 5 | 4                 | 3 | 2 | 1 | 0 |
|-------------|-------------------|---|---|-------------------|---|---|---|---|
| Field       | MAP_SRC_13_H[2:0] |   |   | MAP_DST_13_H[2:0] |   |   | – | – |
| Reset       | 0b000             |   |   | 0b000             |   |   | – | – |
| Access Type | Write, Read       |   |   | Write, Read       |   |   | – | – |

| BITFIELD     | BITS | DESCRIPTION                                                   | DECODE                                             |
|--------------|------|---------------------------------------------------------------|----------------------------------------------------|
| MAP_SRC_13_H | 7:5  | Mapping register source VC high 3 bits for extended mode      | 0bXXX: Mapping register source VC high 3 bits      |
| MAP_DST_13_H | 4:2  | Mapping register destination VC high 3 bits for extended mode | 0bXXX: Mapping register destination VC high 3 bits |

MIPI\_TX\_EXT14 (0x51E)

| BIT         | 7                 | 6 | 5 | 4                 | 3 | 2 | 1 | 0 |
|-------------|-------------------|---|---|-------------------|---|---|---|---|
| Field       | MAP_SRC_14_H[2:0] |   |   | MAP_DST_14_H[2:0] |   |   | – | – |
| Reset       | 0b000             |   |   | 0b000             |   |   | – | – |
| Access Type | Write, Read       |   |   | Write, Read       |   |   | – | – |

| BITFIELD     | BITS | DESCRIPTION                                                   | DECODE                                             |
|--------------|------|---------------------------------------------------------------|----------------------------------------------------|
| MAP_SRC_14_H | 7:5  | Mapping register source VC high 3 bits for extended mode      | 0bXXX: Mapping register source VC high 3 bits      |
| MAP_DST_14_H | 4:2  | Mapping register destination VC high 3 bits for extended mode | 0bXXX: Mapping register destination VC high 3 bits |

MIPI\_TX\_EXT15 (0x51F)

| BIT         | 7                 | 6 | 5 | 4                 | 3 | 2 | 1 | 0 |
|-------------|-------------------|---|---|-------------------|---|---|---|---|
| Field       | MAP_SRC_15_H[2:0] |   |   | MAP_DST_15_H[2:0] |   |   | – | – |
| Reset       | 0b000             |   |   | 0b000             |   |   | – | – |
| Access Type | Write, Read       |   |   | Write, Read       |   |   | – | – |

| BITFIELD     | BITS | DESCRIPTION                                                   | DECODE                                             |
|--------------|------|---------------------------------------------------------------|----------------------------------------------------|
| MAP_SRC_15_H | 7:5  | Mapping register source VC high 3 bits for extended mode      | 0bXXX: Mapping register source VC high 3 bits      |
| MAP_DST_15_H | 4:2  | Mapping register destination VC high 3 bits for extended mode | 0bXXX: Mapping register destination VC high 3 bits |

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Appendix A: Register Map and Tables

MIPI\_TX\_EXT0 (0x520)

| BIT         | 7                | 6 | 5 | 4                | 3 | 2 | 1 | 0 |
|-------------|------------------|---|---|------------------|---|---|---|---|
| Field       | MAP_SRC_0_H[2:0] |   |   | MAP_DST_0_H[2:0] |   |   | – | – |
| Reset       | 0b000            |   |   | 0b000            |   |   | – | – |
| Access Type | Write, Read      |   |   | Write, Read      |   |   | – | – |

| BITFIELD    | BITS | DESCRIPTION                                                   | DECODE                                             |
|-------------|------|---------------------------------------------------------------|----------------------------------------------------|
| MAP_SRC_0_H | 7:5  | Mapping register source VC high 3 bits for extended mode      | 0bXXX: Mapping register source VC high 3 bits      |
| MAP_DST_0_H | 4:2  | Mapping register destination VC high 3 bits for extended mode | 0bXXX: Mapping register destination VC high 3 bits |

MIPI\_TX\_EXT1 (0x521)

| BIT         | 7                | 6 | 5 | 4                | 3 | 2 | 1 | 0 |
|-------------|------------------|---|---|------------------|---|---|---|---|
| Field       | MAP_SRC_1_H[2:0] |   |   | MAP_DST_1_H[2:0] |   |   | – | – |
| Reset       | 0b000            |   |   | 0b000            |   |   | – | – |
| Access Type | Write, Read      |   |   | Write, Read      |   |   | – | – |

| BITFIELD    | BITS | DESCRIPTION                                                   | DECODE                                             |
|-------------|------|---------------------------------------------------------------|----------------------------------------------------|
| MAP_SRC_1_H | 7:5  | Mapping register source VC high 3 bits for extended mode      | 0bXXX: Mapping register source VC high 3 bits      |
| MAP_DST_1_H | 4:2  | Mapping register destination VC high 3 bits for extended mode | 0bXXX: Mapping register destination VC high 3 bits |

MIPI\_TX\_EXT2 (0x522)

| BIT         | 7                | 6 | 5 | 4                | 3 | 2 | 1 | 0 |
|-------------|------------------|---|---|------------------|---|---|---|---|
| Field       | MAP_SRC_2_H[2:0] |   |   | MAP_DST_2_H[2:0] |   |   | – | – |
| Reset       | 0b000            |   |   | 0b000            |   |   | – | – |
| Access Type | Write, Read      |   |   | Write, Read      |   |   | – | – |

| BITFIELD    | BITS | DESCRIPTION                                                   | DECODE                                             |
|-------------|------|---------------------------------------------------------------|----------------------------------------------------|
| MAP_SRC_2_H | 7:5  | Mapping register source VC high 3 bits for extended mode      | 0bXXX: Mapping register source VC high 3 bits      |
| MAP_DST_2_H | 4:2  | Mapping register destination VC high 3 bits for extended mode | 0bXXX: Mapping register destination VC high 3 bits |

MIPI\_TX\_EXT3 (0x523)

| BIT         | 7                | 6 | 5 | 4                | 3 | 2 | 1 | 0 |
|-------------|------------------|---|---|------------------|---|---|---|---|
| Field       | MAP_SRC_3_H[2:0] |   |   | MAP_DST_3_H[2:0] |   |   | – | – |
| Reset       | 0b000            |   |   | 0b000            |   |   | – | – |
| Access Type | Write, Read      |   |   | Write, Read      |   |   | – | – |

| BITFIELD    | BITS | DESCRIPTION                                              | DECODE                                        |
|-------------|------|----------------------------------------------------------|-----------------------------------------------|
| MAP_SRC_3_H | 7:5  | Mapping register source VC high 3 bits for extended mode | 0bXXX: Mapping register source VC high 3 bits |

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Appendix A: Register Map and Tables

| BITFIELD    | BITS | DESCRIPTION                                                   | DECODE                                             |
|-------------|------|---------------------------------------------------------------|----------------------------------------------------|
| MAP_DST_3_H | 4:2  | Mapping register destination VC high 3 bits for extended mode | 0bXXX: Mapping register destination VC high 3 bits |

MIPI\_TX\_EXT4 (0x524)

| BIT         | 7                | 6 | 5 | 4                | 3 | 2 | 1 | 0 |
|-------------|------------------|---|---|------------------|---|---|---|---|
| Field       | MAP_SRC_4_H[2:0] |   |   | MAP_DST_4_H[2:0] |   |   | – | – |
| Reset       | 0b000            |   |   | 0b000            |   |   | – | – |
| Access Type | Write, Read      |   |   | Write, Read      |   |   | – | – |

| BITFIELD    | BITS | DESCRIPTION                                                   | DECODE                                             |
|-------------|------|---------------------------------------------------------------|----------------------------------------------------|
| MAP_SRC_4_H | 7:5  | Mapping register source VC high 3 bits for extended mode      | 0bXXX: Mapping register source VC high 3 bits      |
| MAP_DST_4_H | 4:2  | Mapping register destination VC high 3 bits for extended mode | 0bXXX: Mapping register destination VC high 3 bits |

MIPI\_TX\_EXT5 (0x525)

| BIT         | 7                | 6 | 5 | 4                | 3 | 2 | 1 | 0 |
|-------------|------------------|---|---|------------------|---|---|---|---|
| Field       | MAP_SRC_5_H[2:0] |   |   | MAP_DST_5_H[2:0] |   |   | – | – |
| Reset       | 0b000            |   |   | 0b000            |   |   | – | – |
| Access Type | Write, Read      |   |   | Write, Read      |   |   | – | – |

| BITFIELD    | BITS | DESCRIPTION                                                   | DECODE                                             |
|-------------|------|---------------------------------------------------------------|----------------------------------------------------|
| MAP_SRC_5_H | 7:5  | Mapping register source VC high 3 bits for extended mode      | 0bXXX: Mapping register source VC high 3 bits      |
| MAP_DST_5_H | 4:2  | Mapping register destination VC high 3 bits for extended mode | 0bXXX: Mapping register destination VC high 3 bits |

MIPI\_TX\_EXT6 (0x526)

| BIT         | 7                | 6 | 5 | 4                | 3 | 2 | 1 | 0 |
|-------------|------------------|---|---|------------------|---|---|---|---|
| Field       | MAP_SRC_6_H[2:0] |   |   | MAP_DST_6_H[2:0] |   |   | – | – |
| Reset       | 0b000            |   |   | 0b000            |   |   | – | – |
| Access Type | Write, Read      |   |   | Write, Read      |   |   | – | – |

| BITFIELD    | BITS | DESCRIPTION                                                   | DECODE                                             |
|-------------|------|---------------------------------------------------------------|----------------------------------------------------|
| MAP_SRC_6_H | 7:5  | Mapping register source VC high 3 bits for extended mode      | 0bXXX: Mapping register source VC high 3 bits      |
| MAP_DST_6_H | 4:2  | Mapping register destination VC high 3 bits for extended mode | 0bXXX: Mapping register destination VC high 3 bits |

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Deserializer

Appendix A: Register Map and Tables

MIPI\_TX\_EXT7 (0x527)

| BIT         | 7                | 6 | 5 | 4                | 3 | 2 | 1 | 0 |
|-------------|------------------|---|---|------------------|---|---|---|---|
| Field       | MAP_SRC_7_H[2:0] |   |   | MAP_DST_7_H[2:0] |   |   | – | – |
| Reset       | 0b000            |   |   | 0b000            |   |   | – | – |
| Access Type | Write, Read      |   |   | Write, Read      |   |   | – | – |

| BITFIELD    | BITS | DESCRIPTION                                                   | DECODE                                             |
|-------------|------|---------------------------------------------------------------|----------------------------------------------------|
| MAP_SRC_7_H | 7:5  | Mapping register source VC high 3 bits for extended mode      | 0bXXX: Mapping register source VC high 3 bits      |
| MAP_DST_7_H | 4:2  | Mapping register destination VC high 3 bits for extended mode | 0bXXX: Mapping register destination VC high 3 bits |

MIPI\_TX\_EXT8 (0x528)

| BIT         | 7                | 6 | 5 | 4                | 3 | 2 | 1 | 0 |
|-------------|------------------|---|---|------------------|---|---|---|---|
| Field       | MAP_SRC_8_H[2:0] |   |   | MAP_DST_8_H[2:0] |   |   | – | – |
| Reset       | 0b000            |   |   | 0b000            |   |   | – | – |
| Access Type | Write, Read      |   |   | Write, Read      |   |   | – | – |

| BITFIELD    | BITS | DESCRIPTION                                                   | DECODE                                             |
|-------------|------|---------------------------------------------------------------|----------------------------------------------------|
| MAP_SRC_8_H | 7:5  | Mapping register source VC high 3 bits for extended mode      | 0bXXX: Mapping register source VC high 3 bits      |
| MAP_DST_8_H | 4:2  | Mapping register destination VC high 3 bits for extended mode | 0bXXX: Mapping register destination VC high 3 bits |

MIPI\_TX\_EXT9 (0x529)

| BIT         | 7                | 6 | 5 | 4                | 3 | 2 | 1 | 0 |
|-------------|------------------|---|---|------------------|---|---|---|---|
| Field       | MAP_SRC_9_H[2:0] |   |   | MAP_DST_9_H[2:0] |   |   | – | – |
| Reset       | 0b000            |   |   | 0b000            |   |   | – | – |
| Access Type | Write, Read      |   |   | Write, Read      |   |   | – | – |

| BITFIELD    | BITS | DESCRIPTION                                                   | DECODE                                             |
|-------------|------|---------------------------------------------------------------|----------------------------------------------------|
| MAP_SRC_9_H | 7:5  | Mapping register source VC high 3 bits for extended mode      | 0bXXX: Mapping register source VC high 3 bits      |
| MAP_DST_9_H | 4:2  | Mapping register destination VC high 3 bits for extended mode | 0bXXX: Mapping register destination VC high 3 bits |

MIPI\_TX\_EXT10 (0x52A)

| BIT         | 7                 | 6 | 5 | 4                 | 3 | 2 | 1 | 0 |
|-------------|-------------------|---|---|-------------------|---|---|---|---|
| Field       | MAP_SRC_10_H[2:0] |   |   | MAP_DST_10_H[2:0] |   |   | – | – |
| Reset       | 0b000             |   |   | 0b000             |   |   | – | – |
| Access Type | Write, Read       |   |   | Write, Read       |   |   | – | – |

| BITFIELD     | BITS | DESCRIPTION                                              | DECODE                                        |
|--------------|------|----------------------------------------------------------|-----------------------------------------------|
| MAP_SRC_10_H | 7:5  | Mapping register source VC high 3 bits for extended mode | 0bXXX: Mapping register source VC high 3 bits |

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Appendix A: Register Map and Tables

| BITFIELD     | BITS | DESCRIPTION                                                   | DECODE                                             |
|--------------|------|---------------------------------------------------------------|----------------------------------------------------|
| MAP_DST_10_H | 4:2  | Mapping register destination VC high 3 bits for extended mode | 0bXXX: Mapping register destination VC high 3 bits |

MIPI\_TX\_EXT11 (0x52B)

| BIT         | 7                 | 6 | 5 | 4                 | 3 | 2 | 1 | 0 |
|-------------|-------------------|---|---|-------------------|---|---|---|---|
| Field       | MAP_SRC_11_H[2:0] |   |   | MAP_DST_11_H[2:0] |   |   | – | – |
| Reset       | 0b000             |   |   | 0b000             |   |   | – | – |
| Access Type | Write, Read       |   |   | Write, Read       |   |   | – | – |

| BITFIELD     | BITS | DESCRIPTION                                                   | DECODE                                             |
|--------------|------|---------------------------------------------------------------|----------------------------------------------------|
| MAP_SRC_11_H | 7:5  | Mapping register source VC high 3 bits for extended mode      | 0bXXX: Mapping register source VC high 3 bits      |
| MAP_DST_11_H | 4:2  | Mapping register destination VC high 3 bits for extended mode | 0bXXX: Mapping register destination VC high 3 bits |

MIPI\_TX\_EXT12 (0x52C)

| BIT         | 7                 | 6 | 5 | 4                 | 3 | 2 | 1 | 0 |
|-------------|-------------------|---|---|-------------------|---|---|---|---|
| Field       | MAP_SRC_12_H[2:0] |   |   | MAP_DST_12_H[2:0] |   |   | – | – |
| Reset       | 0b000             |   |   | 0b000             |   |   | – | – |
| Access Type | Write, Read       |   |   | Write, Read       |   |   | – | – |

| BITFIELD     | BITS | DESCRIPTION                                                   | DECODE                                             |
|--------------|------|---------------------------------------------------------------|----------------------------------------------------|
| MAP_SRC_12_H | 7:5  | Mapping register source VC high 3 bits for extended mode      | 0bXXX: Mapping register source VC high 3 bits      |
| MAP_DST_12_H | 4:2  | Mapping register destination VC high 3 bits for extended mode | 0bXXX: Mapping register destination VC high 3 bits |

MIPI\_TX\_EXT13 (0x52D)

| BIT         | 7                 | 6 | 5 | 4                 | 3 | 2 | 1 | 0 |
|-------------|-------------------|---|---|-------------------|---|---|---|---|
| Field       | MAP_SRC_13_H[2:0] |   |   | MAP_DST_13_H[2:0] |   |   | – | – |
| Reset       | 0b000             |   |   | 0b000             |   |   | – | – |
| Access Type | Write, Read       |   |   | Write, Read       |   |   | – | – |

| BITFIELD     | BITS | DESCRIPTION                                                   | DECODE                                             |
|--------------|------|---------------------------------------------------------------|----------------------------------------------------|
| MAP_SRC_13_H | 7:5  | Mapping register source VC high 3 bits for extended mode      | 0bXXX: Mapping register source VC high 3 bits      |
| MAP_DST_13_H | 4:2  | Mapping register destination VC high 3 bits for extended mode | 0bXXX: Mapping register destination VC high 3 bits |

MAX9296B Dual 1 x 4 or 2  
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Appendix A: Register Map and Tables

MIPI\_TX\_EXT14 (0x52E)

| BIT         | 7                 | 6 | 5 | 4                 | 3 | 2 | 1 | 0 |
|-------------|-------------------|---|---|-------------------|---|---|---|---|
| Field       | MAP_SRC_14_H[2:0] |   |   | MAP_DST_14_H[2:0] |   |   | – | – |
| Reset       | 0b000             |   |   | 0b000             |   |   | – | – |
| Access Type | Write, Read       |   |   | Write, Read       |   |   | – | – |

| BITFIELD     | BITS | DESCRIPTION                                                   | DECODE                                             |
|--------------|------|---------------------------------------------------------------|----------------------------------------------------|
| MAP_SRC_14_H | 7:5  | Mapping register source VC high 3 bits for extended mode      | 0bXXX: Mapping register source VC high 3 bits      |
| MAP_DST_14_H | 4:2  | Mapping register destination VC high 3 bits for extended mode | 0bXXX: Mapping register destination VC high 3 bits |

MIPI\_TX\_EXT15 (0x52F)

| BIT         | 7                 | 6 | 5 | 4                 | 3 | 2 | 1 | 0 |
|-------------|-------------------|---|---|-------------------|---|---|---|---|
| Field       | MAP_SRC_15_H[2:0] |   |   | MAP_DST_15_H[2:0] |   |   | – | – |
| Reset       | 0b000             |   |   | 0b000             |   |   | – | – |
| Access Type | Write, Read       |   |   | Write, Read       |   |   | – | – |

| BITFIELD     | BITS | DESCRIPTION                                                   | DECODE                                             |
|--------------|------|---------------------------------------------------------------|----------------------------------------------------|
| MAP_SRC_15_H | 7:5  | Mapping register source VC high 3 bits for extended mode      | 0bXXX: Mapping register source VC high 3 bits      |
| MAP_DST_15_H | 4:2  | Mapping register destination VC high 3 bits for extended mode | 0bXXX: Mapping register destination VC high 3 bits |

MIPI\_TX\_EXT0 (0x530)

| BIT         | 7                | 6 | 5 | 4                | 3 | 2 | 1 | 0 |
|-------------|------------------|---|---|------------------|---|---|---|---|
| Field       | MAP_SRC_0_H[2:0] |   |   | MAP_DST_0_H[2:0] |   |   | – | – |
| Reset       | 0b000            |   |   | 0b000            |   |   | – | – |
| Access Type | Write, Read      |   |   | Write, Read      |   |   | – | – |

| BITFIELD    | BITS | DESCRIPTION                                                   | DECODE                                             |
|-------------|------|---------------------------------------------------------------|----------------------------------------------------|
| MAP_SRC_0_H | 7:5  | Mapping register source VC high 3 bits for extended mode      | 0bXXX: Mapping register source VC high 3 bits      |
| MAP_DST_0_H | 4:2  | Mapping register destination VC high 3 bits for extended mode | 0bXXX: Mapping register destination VC high 3 bits |

MIPI\_TX\_EXT1 (0x531)

| BIT         | 7                | 6 | 5 | 4                | 3 | 2 | 1 | 0 |
|-------------|------------------|---|---|------------------|---|---|---|---|
| Field       | MAP_SRC_1_H[2:0] |   |   | MAP_DST_1_H[2:0] |   |   | – | – |
| Reset       | 0b000            |   |   | 0b000            |   |   | – | – |
| Access Type | Write, Read      |   |   | Write, Read      |   |   | – | – |

| BITFIELD    | BITS | DESCRIPTION                                              | DECODE                                        |
|-------------|------|----------------------------------------------------------|-----------------------------------------------|
| MAP_SRC_1_H | 7:5  | Mapping register source VC high 3 bits for extended mode | 0bXXX: Mapping register source VC high 3 bits |



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Appendix A: Register Map and Tables

| BITFIELD    | BITS | DESCRIPTION                                                   | DECODE                                             |
|-------------|------|---------------------------------------------------------------|----------------------------------------------------|
| MAP_DST_1_H | 4:2  | Mapping register destination VC High 3 bits for extended mode | 0bXXX: Mapping register destination VC High 3 bits |

**MIPI\_TX\_EXT2 (0x532)**

| BIT         | 7                | 6 | 5 | 4                | 3 | 2 | 1 | 0 |
|-------------|------------------|---|---|------------------|---|---|---|---|
| Field       | MAP_SRC_2_H[2:0] |   |   | MAP_DST_2_H[2:0] |   |   | – | – |
| Reset       | 0b000            |   |   | 0b000            |   |   | – | – |
| Access Type | Write, Read      |   |   | Write, Read      |   |   | – | – |

| BITFIELD    | BITS | DESCRIPTION                                                   | DECODE                                             |
|-------------|------|---------------------------------------------------------------|----------------------------------------------------|
| MAP_SRC_2_H | 7:5  | Mapping register source VC high 3 bits for extended mode      | 0bXXX: Mapping register source VC high 3 bits      |
| MAP_DST_2_H | 4:2  | Mapping register destination VC high 3 bits for extended mode | 0bXXX: Mapping register destination VC high 3 bits |

**MIPI\_TX\_EXT3 (0x533)**

| BIT         | 7                | 6 | 5 | 4                | 3 | 2 | 1 | 0 |
|-------------|------------------|---|---|------------------|---|---|---|---|
| Field       | MAP_SRC_3_H[2:0] |   |   | MAP_DST_3_H[2:0] |   |   | – | – |
| Reset       | 0b000            |   |   | 0b000            |   |   | – | – |
| Access Type | Write, Read      |   |   | Write, Read      |   |   | – | – |

| BITFIELD    | BITS | DESCRIPTION                                                   | DECODE                                             |
|-------------|------|---------------------------------------------------------------|----------------------------------------------------|
| MAP_SRC_3_H | 7:5  | Mapping register source VC high 3 bits for extended mode      | 0bXXX: Mapping register source VC high 3 bits      |
| MAP_DST_3_H | 4:2  | Mapping register destination VC high 3 bits for extended mode | 0bXXX: Mapping register destination VC high 3 bits |

**MIPI\_TX\_EXT4 (0x534)**

| BIT         | 7                | 6 | 5 | 4                | 3 | 2 | 1 | 0 |
|-------------|------------------|---|---|------------------|---|---|---|---|
| Field       | MAP_SRC_4_H[2:0] |   |   | MAP_DST_4_H[2:0] |   |   | – | – |
| Reset       | 0b000            |   |   | 0b000            |   |   | – | – |
| Access Type | Write, Read      |   |   | Write, Read      |   |   | – | – |

| BITFIELD    | BITS | DESCRIPTION                                                   | DECODE                                             |
|-------------|------|---------------------------------------------------------------|----------------------------------------------------|
| MAP_SRC_4_H | 7:5  | Mapping register source VC high 3 bits for extended mode      | 0bXXX: Mapping register source VC high 3 bits      |
| MAP_DST_4_H | 4:2  | Mapping register destination VC high 3 bits for extended mode | 0bXXX: Mapping register destination VC high 3 bits |

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Appendix A: Register Map and Tables

MIPI\_TX\_EXT5 (0x535)

| BIT         | 7                | 6 | 5 | 4                | 3 | 2 | 1 | 0 |
|-------------|------------------|---|---|------------------|---|---|---|---|
| Field       | MAP_SRC_5_H[2:0] |   |   | MAP_DST_5_H[2:0] |   |   | – | – |
| Reset       | 0b000            |   |   | 0b000            |   |   | – | – |
| Access Type | Write, Read      |   |   | Write, Read      |   |   | – | – |

| BITFIELD    | BITS | DESCRIPTION                                                   | DECODE                                             |
|-------------|------|---------------------------------------------------------------|----------------------------------------------------|
| MAP_SRC_5_H | 7:5  | Mapping register source VC high 3 bits for extended mode      | 0bXXX: Mapping register source VC high 3 bits      |
| MAP_DST_5_H | 4:2  | Mapping register destination VC High 3 bits for extended mode | 0bXXX: Mapping register destination VC High 3 bits |

MIPI\_TX\_EXT6 (0x536)

| BIT         | 7                | 6 | 5 | 4                | 3 | 2 | 1 | 0 |
|-------------|------------------|---|---|------------------|---|---|---|---|
| Field       | MAP_SRC_6_H[2:0] |   |   | MAP_DST_6_H[2:0] |   |   | – | – |
| Reset       | 0b000            |   |   | 0b000            |   |   | – | – |
| Access Type | Write, Read      |   |   | Write, Read      |   |   | – | – |

| BITFIELD    | BITS | DESCRIPTION                                                   | DECODE                                             |
|-------------|------|---------------------------------------------------------------|----------------------------------------------------|
| MAP_SRC_6_H | 7:5  | Mapping register source VC high 3 bits for extended mode      | 0bXXX: Mapping register source VC high 3 bits      |
| MAP_DST_6_H | 4:2  | Mapping register destination VC high 3 bits for extended mode | 0bXXX: Mapping register destination VC high 3 bits |

MIPI\_TX\_EXT7 (0x537)

| BIT         | 7                | 6 | 5 | 4                | 3 | 2 | 1 | 0 |
|-------------|------------------|---|---|------------------|---|---|---|---|
| Field       | MAP_SRC_7_H[2:0] |   |   | MAP_DST_7_H[2:0] |   |   | – | – |
| Reset       | 0b000            |   |   | 0b000            |   |   | – | – |
| Access Type | Write, Read      |   |   | Write, Read      |   |   | – | – |

| BITFIELD    | BITS | DESCRIPTION                                                   | DECODE                                             |
|-------------|------|---------------------------------------------------------------|----------------------------------------------------|
| MAP_SRC_7_H | 7:5  | Mapping register source VC high 3 bits for extended mode      | 0bXXX: Mapping register source VC high 3 bits      |
| MAP_DST_7_H | 4:2  | Mapping register destination VC high 3 bits for extended mode | 0bXXX: Mapping register destination VC high 3 bits |

MIPI\_TX\_EXT8 (0x538)

| BIT         | 7                | 6 | 5 | 4                | 3 | 2 | 1 | 0 |
|-------------|------------------|---|---|------------------|---|---|---|---|
| Field       | MAP_SRC_8_H[2:0] |   |   | MAP_DST_8_H[2:0] |   |   | – | – |
| Reset       | 0b000            |   |   | 0b000            |   |   | – | – |
| Access Type | Write, Read      |   |   | Write, Read      |   |   | – | – |

| BITFIELD    | BITS | DESCRIPTION                                              | DECODE                                        |
|-------------|------|----------------------------------------------------------|-----------------------------------------------|
| MAP_SRC_8_H | 7:5  | Mapping register source VC high 3 bits for extended mode | 0bXXX: Mapping register source VC high 3 bits |

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| BITFIELD    | BITS | DESCRIPTION                                                   | DECODE                                             |
|-------------|------|---------------------------------------------------------------|----------------------------------------------------|
| MAP_DST_8_H | 4:2  | Mapping register destination VC High 3 bits for extended mode | 0bXXX: Mapping register destination VC High 3 bits |

**MIPI\_TX\_EXT9 (0x539)**

| BIT         | 7                | 6 | 5 | 4                | 3 | 2 | 1 | 0 |
|-------------|------------------|---|---|------------------|---|---|---|---|
| Field       | MAP_SRC_9_H[2:0] |   |   | MAP_DST_9_H[2:0] |   |   | – | – |
| Reset       | 0b000            |   |   | 0b000            |   |   | – | – |
| Access Type | Write, Read      |   |   | Write, Read      |   |   | – | – |

| BITFIELD    | BITS | DESCRIPTION                                                   | DECODE                                             |
|-------------|------|---------------------------------------------------------------|----------------------------------------------------|
| MAP_SRC_9_H | 7:5  | Mapping register source VC high 3 bits for extended mode      | 0bXXX: Mapping register source VC High 3 bits      |
| MAP_DST_9_H | 4:2  | Mapping register destination VC High 3 bits for extended mode | 0bXXX: Mapping register destination VC high 3 bits |

**MIPI\_TX\_EXT10 (0x53A)**

| BIT         | 7                 | 6 | 5 | 4                 | 3 | 2 | 1 | 0 |
|-------------|-------------------|---|---|-------------------|---|---|---|---|
| Field       | MAP_SRC_10_H[2:0] |   |   | MAP_DST_10_H[2:0] |   |   | – | – |
| Reset       | 0b000             |   |   | 0b000             |   |   | – | – |
| Access Type | Write, Read       |   |   | Write, Read       |   |   | – | – |

| BITFIELD     | BITS | DESCRIPTION                                                   | DECODE                                             |
|--------------|------|---------------------------------------------------------------|----------------------------------------------------|
| MAP_SRC_10_H | 7:5  | Mapping register source VC high 3 bits for extended mode      | 0bXXX: Mapping register source VC high 3 bits      |
| MAP_DST_10_H | 4:2  | Mapping register destination VC high 3 bits for extended mode | 0bXXX: Mapping register destination VC high 3 bits |

**MIPI\_TX\_EXT11 (0x53B)**

| BIT         | 7                 | 6 | 5 | 4                 | 3 | 2 | 1 | 0 |
|-------------|-------------------|---|---|-------------------|---|---|---|---|
| Field       | MAP_SRC_11_H[2:0] |   |   | MAP_DST_11_H[2:0] |   |   | – | – |
| Reset       | 0b000             |   |   | 0b000             |   |   | – | – |
| Access Type | Write, Read       |   |   | Write, Read       |   |   | – | – |

| BITFIELD     | BITS | DESCRIPTION                                                   | DECODE                                             |
|--------------|------|---------------------------------------------------------------|----------------------------------------------------|
| MAP_SRC_11_H | 7:5  | Mapping register source VC High 3 bits for extended mode      | 0bXXX: Mapping register source VC high 3 bits      |
| MAP_DST_11_H | 4:2  | Mapping register destination VC high 3 bits for extended mode | 0bXXX: Mapping register destination VC High 3 bits |

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MIPI\_TX\_EXT12 (0x53C)

| BIT         | 7                 | 6 | 5 | 4                 | 3 | 2 | 1 | 0 |
|-------------|-------------------|---|---|-------------------|---|---|---|---|
| Field       | MAP_SRC_12_H[2:0] |   |   | MAP_DST_12_H[2:0] |   |   | – | – |
| Reset       | 0b000             |   |   | 0b000             |   |   | – | – |
| Access Type | Write, Read       |   |   | Write, Read       |   |   | – | – |

| BITFIELD     | BITS | DESCRIPTION                                                   | DECODE                                             |
|--------------|------|---------------------------------------------------------------|----------------------------------------------------|
| MAP_SRC_12_H | 7:5  | Mapping register source VC high 3 bits for extended mode      | 0bXXX: Mapping register source VC high 3 bits      |
| MAP_DST_12_H | 4:2  | Mapping register destination VC high 3 bits for extended mode | 0bXXX: Mapping register destination VC high 3 bits |

MIPI\_TX\_EXT13 (0x53D)

| BIT         | 7                 | 6 | 5 | 4                 | 3 | 2 | 1 | 0 |
|-------------|-------------------|---|---|-------------------|---|---|---|---|
| Field       | MAP_SRC_13_H[2:0] |   |   | MAP_DST_13_H[2:0] |   |   | – | – |
| Reset       | 0b000             |   |   | 0b000             |   |   | – | – |
| Access Type | Write, Read       |   |   | Write, Read       |   |   | – | – |

| BITFIELD     | BITS | DESCRIPTION                                                   | DECODE                                             |
|--------------|------|---------------------------------------------------------------|----------------------------------------------------|
| MAP_SRC_13_H | 7:5  | Mapping register source VC high 3 bits for extended mode      | 0bXXX: Mapping register source VC high 3 bits      |
| MAP_DST_13_H | 4:2  | Mapping register destination VC high 3 bits for extended mode | 0bXXX: Mapping register destination VC high 3 bits |

MIPI\_TX\_EXT14 (0x53E)

| BIT         | 7                 | 6 | 5 | 4                 | 3 | 2 | 1 | 0 |
|-------------|-------------------|---|---|-------------------|---|---|---|---|
| Field       | MAP_SRC_14_H[2:0] |   |   | MAP_DST_14_H[2:0] |   |   | – | – |
| Reset       | 0b000             |   |   | 0b000             |   |   | – | – |
| Access Type | Write, Read       |   |   | Write, Read       |   |   | – | – |

| BITFIELD     | BITS | DESCRIPTION                                                   | DECODE                                             |
|--------------|------|---------------------------------------------------------------|----------------------------------------------------|
| MAP_SRC_14_H | 7:5  | Mapping register source VC high 3 bits for extended mode      | 0bXXX: Mapping register source VC high 3 bits      |
| MAP_DST_14_H | 4:2  | Mapping register destination VC high 3 bits for extended mode | 0bXXX: Mapping register destination VC high 3 bits |

MIPI\_TX\_EXT15 (0x53F)

| BIT         | 7                 | 6 | 5 | 4                 | 3 | 2 | 1 | 0 |
|-------------|-------------------|---|---|-------------------|---|---|---|---|
| Field       | MAP_SRC_15_H[2:0] |   |   | MAP_DST_15_H[2:0] |   |   | – | – |
| Reset       | 0b000             |   |   | 0b000             |   |   | – | – |
| Access Type | Write, Read       |   |   | Write, Read       |   |   | – | – |

| BITFIELD     | BITS | DESCRIPTION                                              | DECODE                                        |
|--------------|------|----------------------------------------------------------|-----------------------------------------------|
| MAP_SRC_15_H | 7:5  | Mapping register source VC high 3 bits for extended mode | 0bXXX: Mapping register source VC high 3 bits |

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| BITFIELD     | BITS | DESCRIPTION                                                   | DECODE                                             |
|--------------|------|---------------------------------------------------------------|----------------------------------------------------|
| MAP_DST_15_H | 4:2  | Mapping register destination VC high 3 bits for extended mode | 0bXXX: Mapping register destination VC high 3 bits |

**UART\_PT\_0 (0x548)**

| BIT         | 7                  | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|--------------------|---|---|---|---|---|---|---|
| Field       | BITLEN_PT_1_L[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x96               |   |   |   |   |   |   |   |
| Access Type | Write, Read        |   |   |   |   |   |   |   |

| BITFIELD      | BITS | DESCRIPTION                                                                                                                                                                                  | DECODE                                                                   |
|---------------|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| BITLEN_PT_1_L | 7:0  | Low byte of custom UART bit length for pass-through UART Channel 1. Set this register to the UART bit length divided by 6.666ns (LSB 8 bits)<br>Set BITLEN_MAN_CFG_1 to 1 to use this value. | 0xXX: Low byte of custom UART bit length for pass-through UART Channel 1 |

**UART\_PT\_1 (0x549)**

| BIT         | 7 | 6 | 5                  | 4 | 3 | 2 | 1 | 0 |
|-------------|---|---|--------------------|---|---|---|---|---|
| Field       | – | – | BITLEN_PT_1_H[5:0] |   |   |   |   |   |
| Reset       | – | – | 0x0                |   |   |   |   |   |
| Access Type | – | – | Write, Read        |   |   |   |   |   |

| BITFIELD      | BITS | DESCRIPTION                                                                                                                                                                                   | DECODE                                                                    |
|---------------|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|
| BITLEN_PT_1_H | 5:0  | High byte of custom UART bit length for pass-through UART Channel 1. Set this register to the UART bit length divided by 6.666ns (MSB 6 bits)<br>Set BITLEN_MAN_CFG_1 to 1 to use this value. | 0xXX: High byte of custom UART bit length for pass-through UART Channel 1 |

**UART\_PT\_2 (0x54A)**

| BIT         | 7                  | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|--------------------|---|---|---|---|---|---|---|
| Field       | BITLEN_PT_2_L[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x96               |   |   |   |   |   |   |   |
| Access Type | Write, Read        |   |   |   |   |   |   |   |

| BITFIELD      | BITS | DESCRIPTION                                                                                                                                                                                  | DECODE                                                                   |
|---------------|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| BITLEN_PT_2_L | 7:0  | Low byte of custom UART bit length for pass-through UART Channel 2. Set this register to the UART bit length divided by 6.666ns (LSB 8 bits)<br>Set BITLEN_MAN_CFG_2 to 1 to use this value. | 0xXX: Low byte of custom UART bit length for pass-through UART Channel 2 |

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UART\_PT\_3 (0x54B)

| BIT         | 7 | 6 | 5                  | 4 | 3 | 2 | 1 | 0 |
|-------------|---|---|--------------------|---|---|---|---|---|
| Field       | – | – | BITLEN_PT_2_H[5:0] |   |   |   |   |   |
| Reset       | – | – | 0x0                |   |   |   |   |   |
| Access Type | – | – | Write, Read        |   |   |   |   |   |

| BITFIELD      | BITS | DESCRIPTION                                                                                                                                                                                   | DECODE                                                                    |
|---------------|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|
| BITLEN_PT_2_H | 5:0  | High byte of custom UART bit length for pass-through UART Channel 1. Set this register to the UART bit length divided by 6.666ns (MSB 6 bits)<br>Set BITLEN_MAN_CFG_1 to 2 to use this value. | 0xXX: High byte of custom UART bit length for pass-through UART Channel 2 |

I2C\_PT\_4 (0x550)

| BIT         | 7            | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|--------------|---|---|---|---|---|---|---|
| Field       | SRC_A_1[6:0] |   |   |   |   |   |   | – |
| Reset       | 0b00000000   |   |   |   |   |   |   | – |
| Access Type | Write, Read  |   |   |   |   |   |   | – |

| BITFIELD | BITS | DESCRIPTION                                                                                                                                                                    | DECODE                                             |
|----------|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|
| SRC_A_1  | 7:1  | I <sup>2</sup> C address translator source A<br>When I <sup>2</sup> C device address matches SRC_A_1, internal I <sup>2</sup> C master replaces the device address by DST_A_1. | 0xXX: I <sup>2</sup> C address translator source A |

I2C\_PT\_5 (0x551)

| BIT         | 7            | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|--------------|---|---|---|---|---|---|---|
| Field       | DST_A_1[6:0] |   |   |   |   |   |   | – |
| Reset       | 0b00000000   |   |   |   |   |   |   | – |
| Access Type | Write, Read  |   |   |   |   |   |   | – |

| BITFIELD | BITS | DESCRIPTION                                                                          | DECODE                                                  |
|----------|------|--------------------------------------------------------------------------------------|---------------------------------------------------------|
| DST_A_1  | 7:1  | I <sup>2</sup> C address translator destination A<br>See the description of SRC_A_1. | 0xXX: I <sup>2</sup> C address translator destination A |

I2C\_PT\_6 (0x552)

| BIT         | 7            | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|--------------|---|---|---|---|---|---|---|
| Field       | SRC_B_1[6:0] |   |   |   |   |   |   | – |
| Reset       | 0b00000000   |   |   |   |   |   |   | – |
| Access Type | Write, Read  |   |   |   |   |   |   | – |

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| BITFIELD | BITS | DESCRIPTION                                                                                                                                                                    | DECODE                                             |
|----------|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|
| SRC_B_1  | 7:1  | I <sup>2</sup> C address translator source B<br>When I <sup>2</sup> C device address matches SRC_B_1, internal I <sup>2</sup> C master replaces the device address by DST_B_1. | 0xXX: I <sup>2</sup> C address translator source B |

I2C\_PT\_7 (0x553)

| BIT         | 7            | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|--------------|---|---|---|---|---|---|---|
| Field       | DST_B_1[6:0] |   |   |   |   |   |   | – |
| Reset       | 0b0000000    |   |   |   |   |   |   | – |
| Access Type | Write, Read  |   |   |   |   |   |   | – |

| BITFIELD | BITS | DESCRIPTION                                                                          | DECODE                                                  |
|----------|------|--------------------------------------------------------------------------------------|---------------------------------------------------------|
| DST_B_1  | 7:1  | I <sup>2</sup> C address translator destination B<br>See the description of SRC_B_1. | 0xXX: I <sup>2</sup> C address translator destination B |

I2C\_PT\_8 (0x554)

| BIT         | 7            | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|--------------|---|---|---|---|---|---|---|
| Field       | SRC_A_2[6:0] |   |   |   |   |   |   | – |
| Reset       | 0b0000000    |   |   |   |   |   |   | – |
| Access Type | Write, Read  |   |   |   |   |   |   | – |

| BITFIELD | BITS | DESCRIPTION                                                                                                                                                                    | DECODE                                             |
|----------|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|
| SRC_A_2  | 7:1  | I <sup>2</sup> C address translator source A<br>When I <sup>2</sup> C device address matches SRC_A_2, internal I <sup>2</sup> C master replaces the device address by DST_A_2. | 0xXX: I <sup>2</sup> C address translator source A |

I2C\_PT\_9 (0x555)

| BIT         | 7            | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|--------------|---|---|---|---|---|---|---|
| Field       | DST_A_2[6:0] |   |   |   |   |   |   | – |
| Reset       | 0b0000000    |   |   |   |   |   |   | – |
| Access Type | Write, Read  |   |   |   |   |   |   | – |

| BITFIELD | BITS | DESCRIPTION                                                                          | DECODE                                                  |
|----------|------|--------------------------------------------------------------------------------------|---------------------------------------------------------|
| DST_A_2  | 7:1  | I <sup>2</sup> C address translator destination A<br>See the description of SRC_A_2. | 0xXX: I <sup>2</sup> C address translator destination A |

I2C\_PT\_11 (0x557)

| BIT         | 7            | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|--------------|---|---|---|---|---|---|---|
| Field       | DST_B_2[6:0] |   |   |   |   |   |   | – |
| Reset       | 0b0000000    |   |   |   |   |   |   | – |
| Access Type | Write, Read  |   |   |   |   |   |   | – |

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| BITFIELD | BITS | DESCRIPTION                                                                          | DECODE                                                  |
|----------|------|--------------------------------------------------------------------------------------|---------------------------------------------------------|
| DST_B_2  | 7:1  | I <sup>2</sup> C address translator destination B<br>See the description of SRC_B_2. | 0xXX: I <sup>2</sup> C address translator destination B |

**BW\_MEAS\_0 (0x559)**

| BIT         | 7               | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|-----------------|---|---|---|---|---|---|---|
| Field       | BW_USE_THR[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00            |   |   |   |   |   |   |   |
| Access Type | Write, Read     |   |   |   |   |   |   |   |

| BITFIELD   | BITS | DESCRIPTION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|------------|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BW_USE_THR | 7:0  | Bandwidth usage threshold<br>When this is non-zero, bandwidth usage (utilization) of the GMSL2 output is measured and reported at BW_INT bit and optionally at ERRB pin. Every 256 txclk cycles (every 0.85µs when link is running at 6G), the number of cycles used for packet transmission is counted and it is compared with the 8-bit value of this register.<br>If BW_USE_LT = 0, BW_INT is asserted when usage is greater than the threshold.<br>If BW_USE_LT = 1, BW_INT is asserted when usage is less than the threshold. |

**BW\_MEAS\_1 (0x55A)**

| BIT         | 7 | 6 | 5 | 4 | 3           | 2         | 1           | 0           |
|-------------|---|---|---|---|-------------|-----------|-------------|-------------|
| Field       | – | – | – | – | BW_USE_OEN  | BW_INT    | BW_USE_LINK | BW_USE_LT   |
| Reset       | – | – | – | – | 0b0         | 0b0       | 0b0         | 0b0         |
| Access Type | – | – | – | – | Write, Read | Read Only | Write, Read | Write, Read |

| BITFIELD    | BITS | DESCRIPTION                                                                                                                                                                                                                                                                   | DECODE                                                                                                                         |
|-------------|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| BW_USE_OEN  | 3    | Set to 1 to output live bandwidth usage comparison at ERRB pin                                                                                                                                                                                                                | 0b0: Do not output live bandwidth usage comparison<br>0b1: Output live bandwidth usage comparison                              |
| BW_INT      | 2    | Live bandwidth usage comparator output.<br>Active only when BW_USE_LT = 1                                                                                                                                                                                                     | 0b0: Bandwidth usage is less than threshold<br>0b1: Bandwidth usage exceeds threshold                                          |
| BW_USE_LINK | 1    | Selects which link is monitored for bandwidth usage.<br><br>In Single or Reverse Split Link configurations (LINK_MODE=1,2, or 3) the below register decode values apply.<br><br>In Dual-Link mode, the total of both links is monitored, so the decode values are irrelevant. | 0b0: Link A bandwidth usage monitored<br>0b1: Link B bandwidth usage monitored                                                 |
| BW_USE_LT   | 0    | Determines whether BW_INT is asserted when bandwidth exceeds threshold. See BW_USE_THR description.                                                                                                                                                                           | 0b0: BW_INT not asserted when bandwidth usage exceeds threshold<br>0b1: BW_INT asserted when bandwidth usage exceeds threshold |



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**CNT4 (0x55C)**

| BIT         | 7                     | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|-----------------------|---|---|---|---|---|---|---|
| Field       | VID_PXL_CRC_ERR0[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00                  |   |   |   |   |   |   |   |
| Access Type | Read Clears All       |   |   |   |   |   |   |   |

| BITFIELD         | BITS | DESCRIPTION                                                                                                            | DECODE                                                                  |
|------------------|------|------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|
| VID_PXL_CRC_ERR0 | 7:0  | Total number of video pixel CRC errors detected at video streams. Reset after reading or with the rising edge of LOCK. | 0xXX: Total number of video pixel CRC errors detected in video stream X |

**CNT5 (0x55D)**

| BIT         | 7                     | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|-----------------------|---|---|---|---|---|---|---|
| Field       | VID_PXL_CRC_ERR1[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00                  |   |   |   |   |   |   |   |
| Access Type | Read Clears All       |   |   |   |   |   |   |   |

| BITFIELD         | BITS | DESCRIPTION                                                                                                            | DECODE                                                                  |
|------------------|------|------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|
| VID_PXL_CRC_ERR1 | 7:0  | Total number of video pixel CRC errors detected at video streams. Reset after reading or with the rising edge of LOCK. | 0xXX: Total number of video pixel CRC errors detected in video stream Y |

**CNT6 (0x55E)**

| BIT         | 7                     | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|-----------------------|---|---|---|---|---|---|---|
| Field       | VID_PXL_CRC_ERR2[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00                  |   |   |   |   |   |   |   |
| Access Type | Read Clears All       |   |   |   |   |   |   |   |

| BITFIELD         | BITS | DESCRIPTION                                                                                                            | DECODE                                                                  |
|------------------|------|------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|
| VID_PXL_CRC_ERR2 | 7:0  | Total number of video pixel CRC errors detected at video streams. Reset after reading or with the rising edge of LOCK. | 0xXX: Total number of video pixel CRC errors detected in video stream Z |

**CNT7 (0x55F)**

| BIT         | 7                     | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|-----------------------|---|---|---|---|---|---|---|
| Field       | VID_PXL_CRC_ERR3[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00                  |   |   |   |   |   |   |   |
| Access Type | Read Clears All       |   |   |   |   |   |   |   |

| BITFIELD         | BITS | DESCRIPTION                                                                                                            | DECODE                                                                  |
|------------------|------|------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|
| VID_PXL_CRC_ERR3 | 7:0  | Total number of video pixel CRC errors detected at video streams. Reset after reading or with the rising edge of LOCK. | 0xXX: Total number of video pixel CRC errors detected in video stream U |

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GMSL1\_2 (0xB02)

| BIT         | 7 | 6 | 5 | 4 | 3 | 2 | 1           | 0 |
|-------------|---|---|---|---|---|---|-------------|---|
| Field       | – | – | – | – | – | – | SRNG[1:0]   |   |
| Reset       | – | – | – | – | – | – | 0b11        |   |
| Access Type | – | – | – | – | – | – | Write, Read |   |

| BITFIELD | BITS | DESCRIPTION            | DECODE                                                                              |
|----------|------|------------------------|-------------------------------------------------------------------------------------|
| SRNG     | 1:0  | Serial data rate range | 0x0: 0.5 to 1 Gbps<br>0x1: 1 to 1.5 Gbps<br>0x2: 1.5 to 4.5 Gbps<br>0x3: Autodetect |

GMSL1\_4 (0xB04)

| BIT         | 7 | 6           | 5           | 4 | 3           | 2 | 1           | 0           |
|-------------|---|-------------|-------------|---|-------------|---|-------------|-------------|
| Field       | – | OUTENB      | PRBSEN      | – | CC_PORT_SEL | – | REVCCEN     | FWDCEN      |
| Reset       | – | 0b0         | 0b0         | – | 0b0         | – | 0b1         | 0b1         |
| Access Type | – | Write, Read | Write, Read | – | Write, Read | – | Write, Read | Write, Read |

| BITFIELD    | BITS | DESCRIPTION                                                        | DECODE                                                                                              |
|-------------|------|--------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| OUTENB      | 6    | Disables outputs                                                   | 0b0: Outputs enabled<br>0b1: Outputs disabled                                                       |
| PRBSEN      | 5    | PRBS test enable (In HIBW mode set PRBS_TYPE = 0)                  | 0b0: Set device normal operation<br>0b1: Enable PRBS test                                           |
| CC_PORT_SEL | 3    | Selects which I <sup>2</sup> C/UART port is connected to this link | 0b0: Port 0 (SDA0_RX0, SCL0_TX0)<br>0b1: Port 1 (SDA1_RX1, SCL1_TX1)                                |
| REVCCEN     | 1    | Enables reverse control channel from deserializer                  | 0b0: Disable reverse control channel receiver<br>0b1: Enable reverse control channel receiver       |
| FWDCEN      | 0    | Enables forward control channel to deserializer                    | 0b0: Disable forward control channel transmitter<br>0b1: Enable forward control channel transmitter |

GMSL1\_5 (0xB05)

| BIT         | 7           | 6           | 5           | 4           | 3           | 2 | 1 | 0 |
|-------------|-------------|-------------|-------------|-------------|-------------|---|---|---|
| Field       | I2CMETHOD   | NO_REM_MST  | HVTR_MODE   | EN_EQ       | EQTUNE[3:0] |   |   |   |
| Reset       | 0b0         | 0x0         | 0b1         | 0b1         | 0x9         |   |   |   |
| Access Type | Write, Read | Write, Read | Write, Read | Write, Read | Write, Read |   |   |   |

| BITFIELD   | BITS | DESCRIPTION                                                                                                                                                                                   | DECODE                                                                                                                                                        |
|------------|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| I2CMETHOD  | 7    | I <sup>2</sup> C Method - Skips register address when converting UART to I <sup>2</sup> C                                                                                                     | 0b0: Send the register address during UART-to-I <sup>2</sup> C conversion<br>0b1: Do not send the register address during UART-to-I <sup>2</sup> C conversion |
| NO_REM_MST | 6    | When set to 1, indicates that there is no I <sup>2</sup> C master on remote side so this (local) chip should ignore any I <sup>2</sup> C packet initiation (start condition) from remote side | 0x0: I <sup>2</sup> C master may or may not be present on remote side<br>0x1: No I <sup>2</sup> C master on remote side                                       |

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| BITFIELD   | BITS | DESCRIPTION                                                                 | DECODE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|------------|------|-----------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| HVTR_MOD E | 5    | HV tracking allows continuous HSYNC format                                  | 0b0: Use partial periodic HV tracking<br>0b1: Use partial and full periodic HV tracking                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| EN_EQ      | 4    | Enables equalizer for manual and adaptive modes                             | 0b0: Disable equalization<br>0b1: Enable equalization                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| EQTUNE     | 3:0  | Equalizer boost level at 750 MHz (effective when adaptive EQ is turned off) | 0b0000: 1.6 dB manual EQ setting<br>0b0001: 2.1 dB manual EQ setting<br>0b0010: 2.8 dB manual EQ setting<br>0b0011: 3.5 dB manual EQ setting<br>0b0100: 4.3 dB manual EQ setting<br>0b0101: 5.2 dB manual EQ setting<br>0b0110: 6.3 dB manual EQ setting<br>0b0111: 7.3 dB manual EQ setting<br>0b1000: 8.5 dB manual EQ setting<br>0b1001: 9.7 dB manual EQ setting<br>0b1010: 11.0 dB manual EQ setting<br>0b1011: 12.2 dB manual EQ setting<br>0b1100: Reserved<br>0b1101: Reserved<br>0b1110: Reserved<br>0b1111: Reserved |

**GMSL1\_6 (0xB06)**

| BIT         | 7           | 6           | 5           | 4           | 3           | 2           | 1 | 0 |
|-------------|-------------|-------------|-------------|-------------|-------------|-------------|---|---|
| Field       | HIGHIMM     | MAX_RT_EN   | I2C_RT_EN   | GPI_COMP_EN | GPI_RT_EN   | HV_SRC[2:0] |   |   |
| Reset       | 0b0         | 0b1         | 0b1         | 0b0         | 0b1         | 0b111       |   |   |
| Access Type | Write, Read | Write, Read | Write, Read | Write, Read | Write, Read | Write, Read |   |   |

| BITFIELD    | BITS | DESCRIPTION                                                                             | DECODE                                                                                              |
|-------------|------|-----------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| HIGHIMM     | 7    | Reverse channel high immunity mode (initial value set by the HIM CFG value at power-up) | 0b0: Reverse channel high immunity mode disabled<br>0b1: Reverse channel high immunity mode enabled |
| MAX_RT_EN   | 6    | Maximum retransmission limit enable                                                     | 0b0: Disable maximum retransmission limit<br>0b1: Enable maximum retransmission limit               |
| I2C_RT_EN   | 5    | I <sup>2</sup> C retransmission enable                                                  | 0b0: Disable I <sup>2</sup> C retransmission<br>0b1: Enable I <sup>2</sup> C retransmission enable  |
| GPI_COMP_EN | 4    | GPI skew compensation enable                                                            | 0b0: Disable GPI skew compensation<br>0b1: Enable GPI skew compensation                             |
| GPI_RT_EN   | 3    | GPI retransmission enable                                                               | 0b0: Disable GPI retransmission<br>0b1: Enable GPI retransmission                                   |

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| BITFIELD | BITS | DESCRIPTION         | DECODE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|----------|------|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| HV_SRC   | 2:0  | HS_VS bit selection | <p>0b000: Use D18/D19 for HS/VS (use this setting when the serializer is a 3.125Gbps device or if HIBW mode is used; otherwise, this setting is for use with the MAX9273 when DBL = 0 or HVEN = 1)</p> <p>0b001: Use D14/D15 for HS/VS (for use with the MAX9271/ MAX96705 when DBL = 0 or HVEN = 1)</p> <p>0b010: Use D12/D13 for HS/VS (for use with the MAX96707 when DBL = 0 or HVEN = 1)</p> <p>0b011: Use D0/D1 for HS/VS (for use with the MAX9271/ MAX9273/MAX96705/MAX96707 when DBL = 1 and HVEN = 0)</p> <p>0b100: Reserved</p> <p>0b101: Reserved</p> <p>0b110: Automatically determine the source of HSYNC/VSYNC (for use with the MAX96707)</p> <p>0b111: Automatically determine the source of HSYNC/VSYNC (for use with the MAX96705)</p> |

GMSL1\_7 (0xB07)

| BIT         | 7           | 6           | 5           | 4 | 3           | 2           | 1 | 0           |
|-------------|-------------|-------------|-------------|---|-------------|-------------|---|-------------|
| Field       | DBL         | DRS         | BWS         | – | HIBW        | HVEN        | – | PXL_CRC     |
| Reset       | 0b0         | 0b0         | 0b0         | – | 0b0         | 0b0         | – | 0b0         |
| Access Type | Write, Read | Write, Read | Write, Read | – | Write, Read | Write, Read | – | Write, Read |

| BITFIELD | BITS | DESCRIPTION                                                                                                                                                                                                                                                                                                        | DECODE                                                                                                                                           |
|----------|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| DBL      | 7    | Double-output mode                                                                                                                                                                                                                                                                                                 | <p>0b0: Use single-rate output</p> <p>0b1: Use double-rate output (2x word rate at 1/2x width)</p>                                               |
| DRS      | 6    | Data rate select                                                                                                                                                                                                                                                                                                   | <p>0b0: Use normal data rate output</p> <p>0b1: Use 1/2 rate data output (for use with low data rates)</p>                                       |
| BWS      | 5    | <p>Bus width select:</p> <p>BWS=0, HIBW=0: 24 total bits, 2 bits are used for control, 22 bits are usable by video</p> <p>BWS=0, HIBW=1: 27 total bits, 3 bits are used for control, 24 bits are usable by video</p> <p>BWS=1, HIBW=0: 32 total bits, 2 bits are used for control, 30 bits are usable by video</p> | <p>0b0: Set bus width for 22-/24-bit bus, 24-/27-bit mode (depending on HIBW setting)</p> <p>0b1: Set bus width for 30-bit bus (32-bit mode)</p> |
| HIBW     | 3    | High-bandwidth mode                                                                                                                                                                                                                                                                                                | <p>0b0: Disable high-bandwidth mode</p> <p>0b1: Enable high-bandwidth mode (when BWS = 0)</p>                                                    |
| HVEN     | 2    | HS/VS encoding enable                                                                                                                                                                                                                                                                                              | <p>0b0: Disable HS/VS encoding</p> <p>0b1: Enable HS/VS encoding</p>                                                                             |
| PXL_CRC  | 0    | Pixel error detection type (this is controllable by pin when LCCEN = 0)                                                                                                                                                                                                                                            | <p>0b0: Use 1-bit parity (compatible with all devices)</p> <p>0b1: Use 6-bit CRC</p>                                                             |

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GMSL1\_8 (0xB08)

| BIT         | 7 | 6 | 5           | 4           | 3 | 2           | 1                  | 0 |
|-------------|---|---|-------------|-------------|---|-------------|--------------------|---|
| Field       | – | – | GPI_EN      | EN_FSYNC_TX | – | PKTCC_EN    | CC_CRC_LENGTH[1:0] |   |
| Reset       | – | – | 0b1         | 0b0         | – | 0b0         | 0b01               |   |
| Access Type | – | – | Write, Read | Write, Read | – | Write, Read | Write, Read        |   |

| BITFIELD      | BITS | DESCRIPTION                                          | DECODE                                                                                          |
|---------------|------|------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| GPI_EN        | 5    | Enables GPI-to-GPO signal transmission to serializer | 0b0: Disable GPI-to-GPO transmission<br>0b1: Enable GPI-to-GPO transmission                     |
| EN_FSYNC_TX   | 4    | Enables frame sync signal transmission               | 0b0: Disable frame sync signal transmission<br>0b1: Enable frame sync signal transmission       |
| PKTCC_EN      | 2    | Enables packet-based control-channel mode            | 0b0: Disable packet-based control-channel mode<br>0b1: Enable packet-based control-channel mode |
| CC_CRC_LENGTH | 1:0  | Control channel CRC length                           | 0b00: 1-bit CRC<br>0b01: 5-bit CRC<br>0b10: 8-bit CRC<br>0b11: Reserved                         |

GMSL1\_D (0xB0D)

| BIT         | 7           | 6 | 5 | 4 | 3 | 2              | 1 | 0 |
|-------------|-------------|---|---|---|---|----------------|---|---|
| Field       | I2C_LOC_ACK | – | – | – | – | HS_TRACK_FSYNC | – | – |
| Reset       | 0b0         | – | – | – | – | 0b0            | – | – |
| Access Type | Write, Read | – | – | – | – | Write, Read    | – | – |

| BITFIELD       | BITS | DESCRIPTION                                                                                           | DECODE                                                                                                                                      |
|----------------|------|-------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| I2C_LOC_ACK    | 7    | Enable I2C-to-I2C slave local acknowledge when forward channel is not available                       | 0b0: Disable local acknowledge when forward channel is not available<br>0b1: Enable local acknowledge when forward channel is not available |
| HS_TRACK_FSYNC | 2    | This bit selects whether to allow infinite-length vertical blanking or to reset HLOCKED with VLOCKED. | 0x0: Allow infinite length vertical blanking<br>0x1: Reset HLOCKED with VLOCKED                                                             |

GMSL1\_E (0xB0E)

| BIT         | 7            | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|--------------|---|---|---|---|---|---|---|
| Field       | DET_THR[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00         |   |   |   |   |   |   |   |
| Access Type | Write, Read  |   |   |   |   |   |   |   |

| BITFIELD | BITS | DESCRIPTION                   | DECODE                                              |
|----------|------|-------------------------------|-----------------------------------------------------|
| DET_THR  | 7:0  | Threshold for detected errors | 0xXX: Number of errors for detected error threshold |

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GMSL1\_F (0xB0F)

| BIT         | 7 | 6           | 5           | 4           | 3           | 2 | 1 | 0           |
|-------------|---|-------------|-------------|-------------|-------------|---|---|-------------|
| Field       | – | EN_DE_FILT  | EN_HS_FILT  | EN_VS_FILT  | DE_EN       | – | – | PRBS_TYPE   |
| Reset       | – | 0b0         | 0b0         | 0b0         | 0b1         | – | – | 0b1         |
| Access Type | – | Write, Read | Write, Read | Write, Read | Write, Read | – | – | Write, Read |

| BITFIELD   | BITS | DESCRIPTION                                       | DECODE                                                                                |
|------------|------|---------------------------------------------------|---------------------------------------------------------------------------------------|
| EN_DE_FILT | 6    | Enable glitch filtering on DE                     | 0b0: Disable DE glitch filtering<br>0b1: Enable DE glitch filtering                   |
| EN_HS_FILT | 5    | Enable glitch filtering on HS                     | 0b0: Disable HS glitch filtering<br>0b1: Enable HS glitch filtering                   |
| EN_VS_FILT | 4    | Enable glitch filtering on VS                     | 0b0: Disable VS glitch filtering<br>0b1: Enable VS glitch filtering                   |
| DE_EN      | 3    | Enable processing separate HS and DE signals      | 0b0: Disable processing HS and DE signals<br>0b1: Enable processing HS and DE signals |
| PRBS_TYPE  | 0    | PRBS type select (n HIBW mode, set PRBS_TYPE = 0) | 0b0: GMSL legacy style PRBS test<br>0b1: MAX9272 style PRBS test                      |

GMSL1\_10 (0xB10)

| BIT         | 7              | 6 | 5           | 4                 | 3 | 2 | 1           | 0 |
|-------------|----------------|---|-------------|-------------------|---|---|-------------|---|
| Field       | RCEG_TYPE[1:0] |   | RCEG_BOUND  | RCEG_ERR_NUM[3:0] |   |   | RCEG_EN     |   |
| Reset       | 0b00           |   | 0b0         | 0x1               |   |   | 0b0         |   |
| Access Type | Write, Read    |   | Write, Read | Write, Read       |   |   | Write, Read |   |

| BITFIELD     | BITS | DESCRIPTION                                                                       | DECODE                                                                                              |
|--------------|------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| RCEG_TYPE    | 7:6  | Reverse channel generated error type                                              | 0b00: Random<br>0b01: Short burst<br>0b10: Long burst<br>0b11: Long burst                           |
| RCEG_BOUND   | 5    | Reverse channel generated error boundary (Effective when RCEG_TYPE = 0x)          | 0b0: Errors are unbounded to symbols<br>0b1: Errors are bounded to symbols                          |
| RCEG_ERR_NUM | 4:1  | Number of RCEG errors generated with each request (Effective when RCEG_TYPE = 0x) | 0xX: Number of errors generated per request                                                         |
| RCEG_EN      | 0    | Enable reverse channel error generator                                            | 0b0: Disable reverse channel error generator<br>0b1: Enable reverse channel error generator enabled |

GMSL1\_11 (0xB11)

| BIT         | 7                  | 6 | 5 | 4 | 3                    | 2 | 1                    | 0 |
|-------------|--------------------|---|---|---|----------------------|---|----------------------|---|
| Field       | RCEG_ERR_RATE[3:0] |   |   |   | RCEG_LO_BST_PRB[1:0] |   | RCEG_LO_BST_LEN[1:0] |   |
| Reset       | 0xF                |   |   |   | 0b00                 |   | 0b00                 |   |
| Access Type | Write, Read        |   |   |   | Write, Read          |   | Write, Read          |   |

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| BITFIELD        | BITS | DESCRIPTION                                                                                                  | DECODE                                                                                                      |
|-----------------|------|--------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| RCEG_ERR_RATE   | 7:4  | Error generation rate in terms of bit time = $2^{-(RCEG\_ERR\_RATE+3)}$ .<br>(Effective when RCEG_TYPE = 0x) | 0x0: Rate is $2^{-3}$<br>0x1: Rate is $2^{-4}$<br>0x2: Rate is $2^{-5}$<br>.<br>.<br>0xF: Rate is $2^{-18}$ |
| RCEG_LO_BST_PRB | 3:2  | Long burst error probability. Effective when RCEG_TYPE = 10.                                                 | 0b00: 1/1024<br>0b01: 1/128<br>0b10: 1/32<br>0b11: 1/8                                                      |
| RCEG_LO_BST_LEN | 1:0  | Long burst error length in terms of bit time. Effective when RCEG_TYPE = 10.                                 | 0b00: Continuous<br>0b01: 128 (~150µs)<br>0b10: 8192 (~9.83ms)<br>0b11: 1048576 (~1.26s)                    |

**GMSL1\_12 (0xB12)**

| BIT         | 7 | 6             | 5                 | 4 | 3                 | 2 | 1             | 0               |
|-------------|---|---------------|-------------------|---|-------------------|---|---------------|-----------------|
| Field       | – | CC_CRC_ERR_EN | LINE_CRC_LOC[1:0] |   | LINE_CRC_EN_GMSL1 | – | MAX_RT_ERR_EN | RCEG_ERR_PER_EN |
| Reset       | – | 0b1           | 0b01              |   | 0b0               | – | 0b1           | 0b0             |
| Access Type | – | Write, Read   | Write, Read       |   | Write, Read       | – | Write, Read   | Write, Read     |

| BITFIELD          | BITS | DESCRIPTION                                                         | DECODE                                                                                                                  |
|-------------------|------|---------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| CC_CRC_ERR_EN     | 6    | Enable reporting of (CC_CRC_ERR_CNT > 0) on the ERRB pin            | 0b0: Disable reporting of errors on ERRB<br>0b1: Enable reporting of errors on ERRB                                     |
| LINE_CRC_LOC      | 5:4  | Video line CRC insertion location                                   | 0b00: [1..4]<br>0x01: [5..8]<br>0x10: [9..12]<br>0x11: [13..16]                                                         |
| LINE_CRC_EN_GMSL1 | 3    | Video line CRC enable                                               | 0b0: Disable video line CRC<br>0x1: Enable video line CRC                                                               |
| MAX_RT_ERR_EN     | 1    | Enable reflection of maximum retransmission error on the ERRORB pin | 0b0: Disable maximum retransmission error on the ERROR pin<br>0b1: Enable maximum retransmission error on the ERROR pin |
| RCEG_ERR_PER_EN   | 0    | Periodic error generation enable. Effective when RCEG_TYPE = 0x.    | 0b0: Disable periodic error generator<br>0b1: Enable periodic error generator                                           |

**GMSL1\_13 (0xB13)**

| BIT         | 7           | 6               | 5                  | 4                   | 3 | 2 | 1 | 0 |
|-------------|-------------|-----------------|--------------------|---------------------|---|---|---|---|
| Field       | EOM_EN_G1   | EOM_PER_MODE_G1 | EOM_MAN_TRG_REQ_G1 | EOM_MIN_THR_G1[4:0] |   |   |   |   |
| Reset       | 0b1         | 0b1             | 0b0                | 0b00000             |   |   |   |   |
| Access Type | Write, Read | Write, Read     | Write, Read        | Write, Read         |   |   |   |   |



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| BITFIELD           | BITS | DESCRIPTION                                                                                                                          | DECODE                                                                                               |
|--------------------|------|--------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| EOM_EN_G1          | 7    | Eye-opening monitor (EOM) enable                                                                                                     | 0b0: Disable EOM<br>0x1: Enable EOM                                                                  |
| EOM_PER_MODE_G1    | 6    | Eye-opening monitor (EOM) periodic mode select                                                                                       | 0b0: Set EOM to use non-periodic mode<br>0x1: Set EOM to use periodic mode                           |
| EOM_MAN_TRG_REQ_G1 | 5    | Eye-opening monitor (EOM) manual trigger request.<br>Valid on the rising edge of this bit when not in periodic mode.                 | 0x0: Do not trigger EOM.<br>0x1: Manually trigger the EOM                                            |
| EOM_MIN_THR_G1     | 4:0  | Eye-opening minimum threshold (in terms of percent) for flagging ERRORB pin.<br>(Percentages increment 3.125% per bitfield setting.) | 0b00000: Disabled<br>0b00001: 3.125%<br>0b00010: 6.25%<br>0b00011: 9.375%<br>.<br>.<br>0b11111: 100% |

**GMSL1\_14 (0xB14)**

| BIT         | 7           | 6            | 5               | 4                | 3 | 2 | 1 | 0 |
|-------------|-------------|--------------|-----------------|------------------|---|---|---|---|
| Field       | AEQ_EN      | AEQ_PER_MODE | AEQ_MAN_TRG_REQ | EOM_PER_THR[4:0] |   |   |   |   |
| Reset       | 0b1         | 0b0          | 0b0             | 0b00000          |   |   |   |   |
| Access Type | Write, Read | Write, Read  | Write, Read     | Write, Read      |   |   |   |   |

| BITFIELD        | BITS | DESCRIPTION                                                                                                          | DECODE                                                                                                                 |
|-----------------|------|----------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| AEQ_EN          | 7    | Adaptive equalization (AEQ) enable                                                                                   | 0b0: Disable AEQ<br>0x1: Enable AEQ                                                                                    |
| AEQ_PER_MODE    | 6    | Adaptive equalizer periodic mode select                                                                              | 0b0: Set AEQ to use non-periodic mode<br>0x1: Set AEQ to use periodic mode                                             |
| AEQ_MAN_TRG_REQ | 5    | Enables adaptive equalizer manual fine tune request. Valid on the rising edge of this bit when not in periodic mode. | 0x0: Do not trigger AEQ fine tuning<br>0x1: Manually trigger the AEQ fine tuning                                       |
| EOM_PER_THR     | 4:0  | Eye-opening threshold to trigger a fine tune operation                                                               | 0b00000: Eye-opening threshold is disabled<br>0b10000: 50% eye-opening triggers fine-tune operation<br>OTHER: Reserved |

**GMSL1\_15 (0xB15)**

| BIT         | 7            | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|--------------|---|---|---|---|---|---|---|
| Field       | DET_ERR[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00         |   |   |   |   |   |   |   |
| Access Type | Read Only    |   |   |   |   |   |   |   |

| BITFIELD | BITS | DESCRIPTION            | DECODE                          |
|----------|------|------------------------|---------------------------------|
| DET_ERR  | 7:0  | Detected error counter | 0xXX: Number of detected errors |



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**GMSL1\_16 (0xB16)**

| BIT         | 7             | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|---------------|---|---|---|---|---|---|---|
| Field       | PRBS_ERR[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00          |   |   |   |   |   |   |   |
| Access Type | Read Only     |   |   |   |   |   |   |   |

| BITFIELD | BITS | DESCRIPTION        | DECODE                               |
|----------|------|--------------------|--------------------------------------|
| PRBS_ERR | 7:0  | PRBS error counter | 0xXX: Number of detected PRBS errors |

**GMSL1\_17 (0xB17)**

| BIT         | 7                   | 6              | 5         | 4         | 3              | 2 | 1 | 0 |
|-------------|---------------------|----------------|-----------|-----------|----------------|---|---|---|
| Field       | I2C_TIMED_OUT_GMSL1 | MAX_RT_ERR_I2C | PRBS_OK   | GPI_IN    | MAX_RT_ERR_GPI | – | – | – |
| Reset       | 0b0                 | 0b0            | 0b0       | 0b0       | 0b0            | – | – | – |
| Access Type | Read Only           | Read Only      | Read Only | Read Only | Read Only      | – | – | – |

| BITFIELD            | BITS | DESCRIPTION                                                                                                                                             | DECODE                                                                                                                 |
|---------------------|------|---------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| I2C_TIMED_OUT_GMSL1 | 7    | I2C channel timed out flag                                                                                                                              | 0b0: I2C channel not timed out<br>0b1: I2C channel timed out                                                           |
| MAX_RT_ERR_I2C      | 6    | Maximum retransmission error flag. Cleared when read.                                                                                                   | 0b0: No control-channel retransmission error<br>0b1: Control-channel retransmission maximum limit reached              |
| PRBS_OK             | 5    | MAX9271/73 compatible PRBS test for link is completed normally. Check PRBS_ERR register for the PRBS success. For other SERDES read PRBS_ERR registers. | 0b0: No MAX9271/MAX9273-compatible PRBS test completed<br>0b1: MAX9271/MAX9273-compatible PRBS test completed normally |
| GPI_IN              | 4    | GPI pin level                                                                                                                                           | 0b0: GPI is input low<br>0b1: GPI is input high                                                                        |
| MAX_RT_ERR_GPI      | 3    | Maximum retransmission error flag. Cleared when read.                                                                                                   | 0b0: No control-channel retransmission error<br>0b1: Control-channel retransmission maximum limit reached              |

**GMSL1\_18 (0xB18)**

| BIT         | 7                | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|------------------|---|---|---|---|---|---|---|
| Field       | CC_RETR_CNT[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00             |   |   |   |   |   |   |   |
| Access Type | Read Only        |   |   |   |   |   |   |   |

| BITFIELD    | BITS | DESCRIPTION                     | DECODE                                    |
|-------------|------|---------------------------------|-------------------------------------------|
| CC_RETR_CNT | 7:0  | I2C packet retransmission count | 0xXX: Number of I2C packets retransmitted |

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**GMSL1\_19 (0xB19)**

| BIT           | 7                  | 6                                              | 5 | 4 | 3                                          | 2 | 1 | 0 |
|---------------|--------------------|------------------------------------------------|---|---|--------------------------------------------|---|---|---|
| Field         | CC_CRC_ERRCNT[7:0] |                                                |   |   |                                            |   |   |   |
| Reset         | 0x00               |                                                |   |   |                                            |   |   |   |
| Access Type   | Read Only          |                                                |   |   |                                            |   |   |   |
| BITFIELD      | BITS               | DESCRIPTION                                    |   |   | DECODE                                     |   |   |   |
| CC_CRC_ERRCNT | 7:0                | Packet-based control channel CRC error counter |   |   | 0xXX: Number of control channel CRC errors |   |   |   |

**GMSL1\_1A (0xB1A)**

| BIT          | 7                 | 6                                          | 5 | 4 | 3                                                | 2 | 1 | 0 |
|--------------|-------------------|--------------------------------------------|---|---|--------------------------------------------------|---|---|---|
| Field        | RCEG_ERR_CNT[7:0] |                                            |   |   |                                                  |   |   |   |
| Reset        | 0x00              |                                            |   |   |                                                  |   |   |   |
| Access Type  | Read Only         |                                            |   |   |                                                  |   |   |   |
| BITFIELD     | BITS              | DESCRIPTION                                |   |   | DECODE                                           |   |   |   |
| RCEG_ERR_CNT | 7:0               | Control channel number of generated errors |   |   | 0xXX: Number of control channel generated errors |   |   |   |

**GMSL1\_1B (0xB1B)**

| BIT          | 7    | 6                                                         | 5 | 4 | 3                                                               | 2            | 1 | 0 |
|--------------|------|-----------------------------------------------------------|---|---|-----------------------------------------------------------------|--------------|---|---|
| Field        | –    | –                                                         | – | – | –                                                               | LINE_CRC_ERR | – | – |
| Reset        | –    | –                                                         | – | – | –                                                               | 0b0          | – | – |
| Access Type  | –    | –                                                         | – | – | –                                                               | Read Only    | – | – |
| BITFIELD     | BITS | DESCRIPTION                                               |   |   | DECODE                                                          |              |   |   |
| LINE_CRC_ERR | 2    | CRC error bit. Latched on error - cleared to 0 when read. |   |   | 0b0: Video line CRC ok<br>0b1: Video line CRC mismatch detected |              |   |   |

**GMSL1\_1C (0xB1C)**

| BIT           | 7    | 6                                                             | 5                  | 4 | 3                                                                                                                                      | 2 | 1 | 0 |
|---------------|------|---------------------------------------------------------------|--------------------|---|----------------------------------------------------------------------------------------------------------------------------------------|---|---|---|
| Field         | –    | –                                                             | EOM_EYE_WIDTH[5:0] |   |                                                                                                                                        |   |   |   |
| Reset         | –    | –                                                             | 0b000000           |   |                                                                                                                                        |   |   |   |
| Access Type   | –    | –                                                             | Read Only          |   |                                                                                                                                        |   |   |   |
| BITFIELD      | BITS | DESCRIPTION                                                   |                    |   | DECODE                                                                                                                                 |   |   |   |
| EOM_EYE_WIDTH | 5:0  | Measured eye-opening. Opening width = EOM_EYE_WIDTH/63 x 100% |                    |   | 0b000000: Width is 0%<br>0b000001: Width is 1/63 x 100%<br>0b000010: Width is 2/63 x 100%<br>.<br>.<br>0b111111: Width is 63/63 x 100% |   |   |   |

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GMSL1\_1D (0xB1D)

| BIT            | 7 | 6 | 5 | 4                  | 3            | 2 | 1 | 0 |
|----------------|---|---|---|--------------------|--------------|---|---|---|
| Field          | – | – | – | UNDERBO<br>OST_DET | AEQ_BST[3:0] |   |   |   |
| Reset          | – | – | – | 0b0                | 0x0          |   |   |   |
| Access<br>Type | – | – | – | Read Only          | Read Only    |   |   |   |

| BITFIELD           | BITS | DESCRIPTION                                                                                              | DECODE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|--------------------|------|----------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| UNDERBOO<br>ST_DET | 4    | Under boost detected                                                                                     | 0b0: Normal Operation<br>0x1: Underboost (at maximum AEQ gain) detected                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| AEQ_BST            | 3:0  | Adaptive equalizer boost value. Selected adaptive equalizer value; settings correspond to gain at 750MHz | 0b0000: 1.6 dB manual EQ setting<br>0b0001: 2.1 dB manual EQ setting<br>0b0010: 2.8 dB manual EQ setting<br>0b0011: 3.5 dB manual EQ setting<br>0b0100: 4.3 dB manual EQ setting<br>0b0101: 5.2 dB manual EQ setting<br>0b0110: 6.3 dB manual EQ setting<br>0b0111: 7.3 dB manual EQ setting<br>0b1000: 8.5 dB manual EQ setting<br>0b1001: 9.7 dB manual EQ setting<br>0b1010: 11.0 dB manual EQ setting<br>0b1011: 12.2 dB manual EQ setting<br>0b1100: Reserved<br>0b1101: Reserved<br>0b1110: Reserved<br>0b1111: Reserved |

GMSL1\_20 (0xB20)

| BIT            | 7                | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|----------------|------------------|---|---|---|---|---|---|---|
| Field          | CRC_VALUE_0[7:0] |   |   |   |   |   |   |   |
| Reset          | 0x00             |   |   |   |   |   |   |   |
| Access<br>Type | Read Only        |   |   |   |   |   |   |   |

| BITFIELD        | BITS | DESCRIPTION                                  | DECODE                        |
|-----------------|------|----------------------------------------------|-------------------------------|
| CRC_VALUE<br>_0 | 7:0  | Bits [7:0] of CRC output for the latest line | 0xXX: CRC[7:0] of latest line |

GMSL1\_21 (0xB21)

| BIT            | 7                | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|----------------|------------------|---|---|---|---|---|---|---|
| Field          | CRC_VALUE_1[7:0] |   |   |   |   |   |   |   |
| Reset          | 0x00             |   |   |   |   |   |   |   |
| Access<br>Type | Read Only        |   |   |   |   |   |   |   |

| BITFIELD        | BITS | DESCRIPTION                                   | DECODE                         |
|-----------------|------|-----------------------------------------------|--------------------------------|
| CRC_VALUE<br>_1 | 7:0  | Bits [15:8] of CRC output for the latest line | 0xXX: CRC[15:8] of latest line |

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**GMSL1\_22 (0xB22)**

| BIT         | 7                | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|------------------|---|---|---|---|---|---|---|
| Field       | CRC_VALUE_2[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00             |   |   |   |   |   |   |   |
| Access Type | Read Only        |   |   |   |   |   |   |   |

| BITFIELD    | BITS | DESCRIPTION                                    | DECODE                          |
|-------------|------|------------------------------------------------|---------------------------------|
| CRC_VALUE_2 | 7:0  | Bits [23:16] of CRC output for the latest line | 0xXX: CRC[23:16] of latest line |

**GMSL1\_23 (0xB23)**

| BIT         | 7                | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|------------------|---|---|---|---|---|---|---|
| Field       | CRC_VALUE_3[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00             |   |   |   |   |   |   |   |
| Access Type | Read Only        |   |   |   |   |   |   |   |

| BITFIELD    | BITS | DESCRIPTION                                    | DECODE                          |
|-------------|------|------------------------------------------------|---------------------------------|
| CRC_VALUE_3 | 7:0  | Bits [31:24] of CRC output for the latest line | 0xXX: CRC[31:24] of latest line |

**GMSL1\_96 (0xB96)**

| BIT         | 7                        | 6 | 5 | 4 | 3 | 2           | 1             | 0            |
|-------------|--------------------------|---|---|---|---|-------------|---------------|--------------|
| Field       | CONV_GMSL1_DATATYPE[4:0] |   |   |   |   | DIS_HIBW_E  | CONV_GMSL1_EN | DBL_ALIGN_TO |
| Reset       | 0b00000                  |   |   |   |   | 0b0         | 0b0           | 0b1          |
| Access Type | Write, Read              |   |   |   |   | Write, Read | Write, Read   | Write, Read  |

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| BITFIELD            | BITS | DESCRIPTION                                                                   | DECODE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|---------------------|------|-------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CONV_GMSL1_DATATYPE | 7:3  | Convert from GMSL1 color format mapping to GMSL2 CSI transmitter color format | 0x0: RGB888 OLDI<br>0x1: RGB565<br>0x2: RGB666<br>0x3: YUV 422 8-bit Mux mode (use yuv_8_10_mux_mode)<br>0x4: YUV 422 10-bit Mux mode (use yuv_8_10_mux_mode)<br>0x5: RAW8 single<br>0x6: RAW10 single<br>0x7: RAW12 single<br>0x8: RAW14<br>0x9: User-defined Generic 24-bit<br>0xA: User-defined YUV422 12-bit<br>0xB: User-defined Generic 8-bit<br>0xC<br>0xD<br>0xE<br>0xF<br>0x10: RGB888 VESA<br>0x11<br>0x12<br>0x13: YUV 422 8-bit Normal mode<br>0x14: YUV 422 10-bit Normal mode<br>0x15: RAW8 double (use alt_mem_map8)<br>0x16: RAW10 double (use alt_mem_map10)<br>0x17: RAW12 double (use alt_mem_map12)<br>0x18<br>0x19<br>0x1A<br>0x1B<br>0x1C<br>0x1D<br>0x1E<br>0x1F |
| DIS_HIBW_E          | 2    | Disables disparity error detection in HIBW                                    | 0b0: Enable disparity error detection<br>0b1: Disable disparity error detection                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| CONV_GMSL1_EN       | 1    | Enables conversion from GMSL1 color format mapping to GMSL2 CSI transmitter   | 0b0: Disable conversion<br>0b1: Enable conversion                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| DBL_ALIGN_TO        | 0    | HBM DBL mode type                                                             | 0b0: DBL-Single mode with alignment<br>0b1: DBL-DBL mode with no alignment (d)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

**GMSL1\_CA (0xBCA)**

| BIT         | 7 | 6 | 5 | 4 | 3 | 2         | 1         | 0         |
|-------------|---|---|---|---|---|-----------|-----------|-----------|
| Field       | — | — | — | — | — | PHASELOCK | WBLOCK_G1 | DATAOK    |
| Reset       | — | — | — | — | — | 0b0       | 0b0       | 0b0       |
| Access Type | — | — | — | — | — | Read Only | Read Only | Read Only |

| BITFIELD  | BITS | DESCRIPTION | DECODE                                 |
|-----------|------|-------------|----------------------------------------|
| PHASELOCK | 2    | CDR locked  | 0b0: CDR not locked<br>0b1: CDR locked |

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| BITFIELD      | BITS | DESCRIPTION                        | DECODE                                                                                |
|---------------|------|------------------------------------|---------------------------------------------------------------------------------------|
| WBLOCK_G<br>1 | 1    | Word boundary detected             | 0b0: No word boundary detected<br>0b1: Word boundary detected                         |
| DATAOK        | 0    | Data activity detected on the link | 0b0: No data activity detected on the link<br>0b1: Data activity detected on the link |

**GMSL1\_CB (0xBCB)**

| BIT            | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0             |
|----------------|---|---|---|---|---|---|---|---------------|
| Field          | – | – | – | – | – | – | – | LOCKED_G<br>1 |
| Reset          | – | – | – | – | – | – | – | 0b0           |
| Access<br>Type | – | – | – | – | – | – | – | Read Only     |

| BITFIELD  | BITS | DESCRIPTION | DECODE                                   |
|-----------|------|-------------|------------------------------------------|
| LOCKED_G1 | 0    | Link locked | 0b0: Link not locked<br>0b1: Link locked |

**GMSL1\_D1 (0xBD1)**

| BIT            | 7                 | 6 | 5 | 4 | 3                | 2 | 1 | 0 |
|----------------|-------------------|---|---|---|------------------|---|---|---|
| Field          | CNTL_OUT_ORD[2:0] |   |   |   | CNTL_OUT_EN[4:0] |   |   |   |
| Reset          | 0x0               |   |   |   | 0x00000          |   |   |   |
| Access<br>Type | Write, Read       |   |   |   | Write, Read      |   |   |   |

| BITFIELD         | BITS | DESCRIPTION                                       | DECODE                                                                                                                                                                                                                                                                                                                                                                    |
|------------------|------|---------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CNTL_OUT_<br>ORD | 7:5  | Internal CNTL_OUT order to pins CNTL4 to<br>CNTL0 | 0x0: 4, 3, 2, 1, 0<br>0x1: 3, 2, 1, 0, 4<br>0x2: 2, 1, 0, 4, 3<br>0x3: 1, 0, 4, 3, 2<br>0x4: 0, 1, 2, 3, 4<br>0x5: 1, 2, 3, 4, 0<br>0x6: 2, 3, 4, 0, 1<br>0x7: 3, 4, 0, 1, 2                                                                                                                                                                                              |
| CNTL_OUT_<br>EN  | 4:0  | CNTL output enable for CNTL 0, 1, 2, 3, 4         | 0bXXXX0: Disable control output 0<br>0bXXXX1: Enable control output 0<br>0bXXX0X: Disable control output 1<br>0bXXX1X: Enable control output 1<br>0bXX0XX: Disable control output 2<br>0bXX1XX: Enable control output 2<br>0bX0XXX: Disable control output 3<br>0bX1XXX: Enable control output 3<br>0b0XXXX: Disable control output 4<br>0b1XXXX: Enable control output 4 |

**GMSL1\_2 (0xC02)**

| BIT            | 7 | 6 | 5 | 4 | 3 | 2 | 1           | 0 |
|----------------|---|---|---|---|---|---|-------------|---|
| Field          | – | – | – | – | – | – | SRNG[1:0]   |   |
| Reset          | – | – | – | – | – | – | 0b11        |   |
| Access<br>Type | – | – | – | – | – | – | Write, Read |   |

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| BITFIELD | BITS | DESCRIPTION            | DECODE                                                                                  |
|----------|------|------------------------|-----------------------------------------------------------------------------------------|
| SRNG     | 1:0  | Serial data rate range | 0b00: 0.5 to 1 Gbps<br>0b01: 1 to 1.5 Gbps<br>0b10: 1.5 to 4.5 Gbps<br>0b11: Autodetect |

**GMSL1\_4 (0xC04)**

| BIT         | 7 | 6           | 5           | 4 | 3           | 2 | 1           | 0           |
|-------------|---|-------------|-------------|---|-------------|---|-------------|-------------|
| Field       | – | OUTENB      | PRBSEN      | – | CC_PORT_SEL | – | REVCCEN     | FWDCEN      |
| Reset       | – | 0b0         | 0b0         | – | 0b0         | – | 0b1         | 0b1         |
| Access Type | – | Write, Read | Write, Read | – | Write, Read | – | Write, Read | Write, Read |

| BITFIELD    | BITS | DESCRIPTION                                                        | DECODE                                                                                              |
|-------------|------|--------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| OUTENB      | 6    | Disables outputs                                                   | 0b0: Outputs enabled<br>0b1: Outputs disabled                                                       |
| PRBSEN      | 5    | PRBS test enable (In HIBW mode set PRBS_TYPE = 0)                  | 0b0: Set device normal operation<br>0b1: Enable PRBS test                                           |
| CC_PORT_SEL | 3    | Selects which I <sup>2</sup> C/UART port is connected to this link | 0b0: Port 0 (SDA0_RX0, SCL0_TX0)<br>0b1: Port 1 (SDA1_RX1, SCL1_TX1)                                |
| REVCCEN     | 1    | Enables reverse control channel from deserializer                  | 0b0: Disable reverse control channel receiver<br>0b1: Enable reverse control channel receiver       |
| FWDCEN      | 0    | Enables forward control channel to deserializer                    | 0b0: Disable forward control channel transmitter<br>0b1: Enable forward control channel transmitter |

**GMSL1\_5 (0xC05)**

| BIT         | 7           | 6           | 5           | 4           | 3           | 2 | 1 | 0 |
|-------------|-------------|-------------|-------------|-------------|-------------|---|---|---|
| Field       | I2CMETHOD   | NO_REM_MST  | HVTR_MODE   | EN_EQ       | EQTUNE[3:0] |   |   |   |
| Reset       | 0b0         | 0x0         | 0b1         | 0b1         | 0x9         |   |   |   |
| Access Type | Write, Read | Write, Read | Write, Read | Write, Read | Write, Read |   |   |   |

| BITFIELD   | BITS | DESCRIPTION                                                                                                                                                                                | DECODE                                                                                                                                                        |
|------------|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| I2CMETHOD  | 7    | I <sup>2</sup> C Method - Skip register address when converting UART to I <sup>2</sup> C                                                                                                   | 0b0: Send the register address during UART-to-I <sup>2</sup> C conversion<br>0b1: Do not send the register address during UART-to-I <sup>2</sup> C conversion |
| NO_REM_MST | 6    | Sets to 1 to indicate that there is no I <sup>2</sup> C master on remote side so this (local) chip should ignore any I <sup>2</sup> C packet initiation (start condition) from remote side | 0b0: I <sup>2</sup> C master may or may not be present on remote side<br>0b1: No I <sup>2</sup> C master on remote side                                       |
| HVTR_MODE  | 5    | HV tracking allows continuous HSYNC format                                                                                                                                                 | 0b0: Use partial periodic HV tracking<br>0b1: Use partial and full periodic HV tracking                                                                       |
| EN_EQ      | 4    | Enables equalizer for manual and adaptive modes                                                                                                                                            | 0b0: Disable equalization<br>0b1: Enable equalization                                                                                                         |

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| BITFIELD | BITS | DESCRIPTION                                                                    | DECODE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|----------|------|--------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| EQTUNE   | 3:0  | Equalizer boost level at 750 MHz<br>(effective when Adaptive EQ is turned off) | 0b0000: 1.6 dB manual EQ setting<br>0b0001: 2.1 dB manual EQ setting<br>0b0010: 2.8 dB manual EQ setting<br>0b0011: 3.5 dB manual EQ setting<br>0b0100: 4.3 dB manual EQ setting<br>0b0101: 5.2 dB manual EQ setting<br>0b0110: 6.3 dB manual EQ setting<br>0b0111: 7.3 dB manual EQ setting<br>0b1000: 8.5 dB manual EQ setting<br>0b1001: 9.7 dB manual EQ setting<br>0b1010: 11.0 dB manual EQ setting<br>0b1011: 12.2 dB manual EQ setting<br>0b1100: Reserved<br>0b1101: Reserved<br>0b1110: Reserved<br>0b1111: Reserved |

**GMSL1\_6 (0xC06)**

| BIT         | 7           | 6           | 5           | 4           | 3           | 2           | 1 | 0 |
|-------------|-------------|-------------|-------------|-------------|-------------|-------------|---|---|
| Field       | HIGHIMM     | MAX_RT_EN   | I2C_RT_EN   | GPI_COMP_EN | GPI_RT_EN   | HV_SRC[2:0] |   |   |
| Reset       | 0b0         | 0b1         | 0b1         | 0b0         | 0b1         | 0b111       |   |   |
| Access Type | Write, Read | Write, Read | Write, Read | Write, Read | Write, Read | Write, Read |   |   |

| BITFIELD    | BITS | DESCRIPTION                                                                                | DECODE                                                                                              |
|-------------|------|--------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| HIGHIMM     | 7    | Reverse channel high immunity mode<br>(initial value set by the HIM CFG value at power-up) | 0b0: Reverse channel high immunity mode disabled<br>0b1: Reverse channel high immunity mode enabled |
| MAX_RT_EN   | 6    | Maximum retransmission limit enable                                                        | 0b0: Disable maximum retransmission limit<br>0b1: Enable maximum retransmission limit               |
| I2C_RT_EN   | 5    | I <sup>2</sup> C retransmission enable                                                     | 0b0: Disable I <sup>2</sup> C retransmission<br>0b1: Enable I <sup>2</sup> C retransmission enable  |
| GPI_COMP_EN | 4    | GPI skew compensation enable                                                               | 0b0: Disable GPI skew compensation<br>0b1: Enable GPI skew compensation                             |
| GPI_RT_EN   | 3    | GPI retransmission enable                                                                  | 0b0: Disable GPI retransmission<br>0b1: Enable GPI retransmission                                   |



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| BITFIELD | BITS | DESCRIPTION         | DECODE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|----------|------|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| HV_SRC   | 2:0  | HS_VS bit selection | <p>0b000: Use D18/D19 for HS/VS (use this setting when the serializer is a 3.125Gbps device or if HIBW mode is used; otherwise, this setting is for use with the MAX9273 when DBL = 0 or HVEN = 1)</p> <p>0b001: Use D14/D15 for HS/VS (for use with the MAX9271/ MAX96705 when DBL = 0 or HVEN = 1)</p> <p>0b010: Use D12/D13 for HS/VS (for use with the MAX96707 when DBL = 0 or HVEN = 1)</p> <p>0b011: Use D0/D1 for HS/VS (for use with the MAX9271/ MAX9273/MAX96705/MAX96707 when DBL = 1 and HVEN = 0)</p> <p>0b100: Reserved</p> <p>0b101: Reserved</p> <p>0b110: Automatically determine the source of HSYNC/VSYNC (for use with the MAX96707)</p> <p>0b111: Automatically determine the source of HSYNC/VSYNC (for use with the MAX96705)</p> |

GMSL1\_7 (0xC07)

| BIT         | 7           | 6           | 5           | 4 | 3           | 2           | 1 | 0           |
|-------------|-------------|-------------|-------------|---|-------------|-------------|---|-------------|
| Field       | DBL         | DRS         | BWS         | – | HIBW        | HVEN        | – | PXL_CRC     |
| Reset       | 0b0         | 0b0         | 0b0         | – | 0b0         | 0b0         | – | 0b0         |
| Access Type | Write, Read | Write, Read | Write, Read | – | Write, Read | Write, Read | – | Write, Read |

| BITFIELD | BITS | DESCRIPTION                                                                                                                                                                                                                                                                                                       | DECODE                                                                                                                                           |
|----------|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| DBL      | 7    | Double-output mode                                                                                                                                                                                                                                                                                                | <p>0b0: Use single-rate output</p> <p>0b1: Use double-rate output (2x word rate at 1/2x width)</p>                                               |
| DRS      | 6    | Data rate select                                                                                                                                                                                                                                                                                                  | <p>0b0: Use normal data rate output</p> <p>0b1: Use 1/2 rate data output (for use with low data rates)</p>                                       |
| BWS      | 5    | <p>Bus width select</p> <p>BWS=0, HIBW=0: 24 total bits, 2 bits are used for control, 22 bits are usable by video</p> <p>BWS=0, HIBW=1: 27 total bits, 3 bits are used for control, 24 bits are usable by video</p> <p>BWS=1, HIBW=0: 32 total bits, 2 bits are used for control, 30 bits are usable by video</p> | <p>0b0: Set bus width for 22-/24-bit bus, 24-/27-bit mode (depending on HIBW setting)</p> <p>0b1: Set bus width for 30-bit bus (32-bit mode)</p> |
| HIBW     | 3    | Enables high-bandwidth mode                                                                                                                                                                                                                                                                                       | <p>0b0: Disable high-bandwidth mode</p> <p>0b1: Enable high-bandwidth mode (when BWS = 0)</p>                                                    |
| HVEN     | 2    | HS/VS encoding enable                                                                                                                                                                                                                                                                                             | <p>0b0: Disable HS/VS encoding</p> <p>0b1: Enable HS/VS encoding</p>                                                                             |
| PXL_CRC  | 0    | Pixel error detection type (this is controllable by pin when LCCEN = 0)                                                                                                                                                                                                                                           | <p>0b0: Use 1-bit parity (compatible with all devices)</p> <p>0b1: Use 6-bit CRC</p>                                                             |

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GMSL1\_8 (0xC08)

| BIT         | 7 | 6 | 5           | 4           | 3 | 2           | 1                  | 0 |
|-------------|---|---|-------------|-------------|---|-------------|--------------------|---|
| Field       | – | – | GPI_EN      | EN_FSYNC_TX | – | PKTCC_EN    | CC_CRC_LENGTH[1:0] |   |
| Reset       | – | – | 0b1         | 0b0         | – | 0b0         | 0b01               |   |
| Access Type | – | – | Write, Read | Write, Read | – | Write, Read | Write, Read        |   |

| BITFIELD      | BITS | DESCRIPTION                                          | DECODE                                                                                          |
|---------------|------|------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| GPI_EN        | 5    | Enables GPI-to-GPO signal transmission to serializer | 0b0: Disable GPI-to-GPO transmission<br>0b1: Enable GPI-to-GPO transmission                     |
| EN_FSYNC_TX   | 4    | Enables frame sync signal transmission               | 0b0: Disable frame sync signal transmission<br>0b1: Enable frame sync signal transmission       |
| PKTCC_EN      | 2    | Packet-based control channel mode enable             | 0b0: Disable packet-based control-channel mode<br>0b1: Enable packet-based control-channel mode |
| CC_CRC_LENGTH | 1:0  | Control channel CRC length                           | 0b00: 1-bit CRC<br>0b01: 5-bit CRC<br>0b10: 8-bit CRC<br>0b11: Reserved                         |

GMSL1\_D (0xC0D)

| BIT         | 7           | 6 | 5 | 4 | 3 | 2              | 1 | 0 |
|-------------|-------------|---|---|---|---|----------------|---|---|
| Field       | I2C_LOC_ACK | – | – | – | – | HS_TRACK_FSYNC | – | – |
| Reset       | 0b0         | – | – | – | – | 0b0            | – | – |
| Access Type | Write, Read | – | – | – | – | Write, Read    | – | – |

| BITFIELD       | BITS | DESCRIPTION                                                                                 | DECODE                                                                                                                                      |
|----------------|------|---------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| I2C_LOC_ACK    | 7    | Enables I2C-to-I2C slave local acknowledge when forward channel is not available            | 0b0: Disable local acknowledge when forward channel is not available<br>0b1: Enable local acknowledge when forward channel is not available |
| HS_TRACK_FSYNC | 2    | Selects whether to allow infinite-length vertical blanking or to lose HLOCKED with VLOCKED. | 0b0: Allow infinite length vertical blanking<br>0b1: Lose HLOCKED with VLOCKED                                                              |

GMSL1\_E (0xC0E)

| BIT         | 7            | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|--------------|---|---|---|---|---|---|---|
| Field       | DET_THR[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00         |   |   |   |   |   |   |   |
| Access Type | Write, Read  |   |   |   |   |   |   |   |

| BITFIELD | BITS | DESCRIPTION                   | DECODE                                              |
|----------|------|-------------------------------|-----------------------------------------------------|
| DET_THR  | 7:0  | Threshold for detected errors | 0xXX: Number of errors for detected error threshold |

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GMSL1\_F (0xC0F)

| BIT         | 7 | 6           | 5           | 4           | 3           | 2 | 1 | 0           |
|-------------|---|-------------|-------------|-------------|-------------|---|---|-------------|
| Field       | – | EN_DE_FILT  | EN_HS_FILT  | EN_VS_FILT  | DE_EN       | – | – | PRBS_TTYPE  |
| Reset       | – | 0b0         | 0b0         | 0b0         | 0b1         | – | – | 0b1         |
| Access Type | – | Write, Read | Write, Read | Write, Read | Write, Read | – | – | Write, Read |

| BITFIELD   | BITS | DESCRIPTION                                        | DECODE                                                                                |
|------------|------|----------------------------------------------------|---------------------------------------------------------------------------------------|
| EN_DE_FILT | 6    | Enables glitch filtering on DE                     | 0b0: Disable DE glitch filtering<br>0b1: Enable DE glitch filtering                   |
| EN_HS_FILT | 5    | Enables glitch filtering on HS                     | 0b0: Disable HS glitch filtering<br>0b1: Enable HS glitch filtering                   |
| EN_VS_FILT | 4    | Enables glitch filtering on VS                     | 0b0: Disable VS glitch filtering<br>0b1: Enable VS glitch filtering                   |
| DE_EN      | 3    | Enables processing separate HS and DE signals      | 0b0: Disable processing HS and DE signals<br>0b1: Enable processing HS and DE signals |
| PRBS_TYPE  | 0    | PRBS type select (in HIBW mode, set PRBS_TYPE = 0) | 0b0: GMSL legacy style PRBS test<br>0b1: MAX9272 style PRBS test                      |

GMSL1\_10 (0xC10)

| BIT         | 7              | 6 | 5           | 4                 | 3 | 2 | 1           | 0 |
|-------------|----------------|---|-------------|-------------------|---|---|-------------|---|
| Field       | RCEG_TYPE[1:0] |   | RCEG_BOUND  | RCEG_ERR_NUM[3:0] |   |   | RCEG_EN     |   |
| Reset       | 0b00           |   | 0b0         | 0x1               |   |   | 0b0         |   |
| Access Type | Write, Read    |   | Write, Read | Write, Read       |   |   | Write, Read |   |

| BITFIELD     | BITS | DESCRIPTION                                                                       | DECODE                                                                                              |
|--------------|------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| RCEG_TYPE    | 7:6  | Reverse channel generated error type                                              | 0b00: Random<br>0b01: Short burst<br>0b10: Long burst<br>0b11: Long burst                           |
| RCEG_BOUND   | 5    | Reverse channel generated error boundary (Effective when RCEG_TYPE = 0x)          | 0b0: Errors are unbounded to symbols<br>0b1: Errors are bounded to symbols                          |
| RCEG_ERR_NUM | 4:1  | Number of RCEG errors generated with each request (Effective when RCEG_TYPE = 0x) | 0xX: Number of errors generated per request                                                         |
| RCEG_EN      | 0    | Enables reverse channel error generator                                           | 0b0: Disable reverse channel error generator<br>0b1: Enable reverse channel error generator enabled |

GMSL1\_11 (0xC11)

| BIT         | 7                  | 6 | 5 | 4 | 3                    | 2 | 1                    | 0 |
|-------------|--------------------|---|---|---|----------------------|---|----------------------|---|
| Field       | RCEG_ERR_RATE[3:0] |   |   |   | RCEG_LO_BST_PRB[1:0] |   | RCEG_LO_BST_LEN[1:0] |   |
| Reset       | 0xF                |   |   |   | 0b00                 |   | 0b00                 |   |
| Access Type | Write, Read        |   |   |   | Write, Read          |   | Write, Read          |   |

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| BITFIELD        | BITS | DESCRIPTION                                                                                                  | DECODE                                                                                                      |
|-----------------|------|--------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| RCEG_ERR_RATE   | 7:4  | Error generation rate in terms of bit time = $2^{-(RCEG\_ERR\_RATE+3)}$ .<br>(Effective when RCEG_TYPE = 0x) | 0x0: Rate is $2^{-3}$<br>0x1: Rate is $2^{-4}$<br>0x2: Rate is $2^{-5}$<br>.<br>.<br>0xF: Rate is $2^{-18}$ |
| RCEG_LO_BST_PRB | 3:2  | Long burst error probability. Effective when RCEG_TYPE = 10.                                                 | 0b00: 1/1024<br>0b01: 1/128<br>0b10: 1/32<br>0b11: 1/8                                                      |
| RCEG_LO_BST_LEN | 1:0  | Long burst error length in terms of bit time. Effective when RCEG_TYPE = 10.                                 | 0b00: Continuous<br>0b01: 128 (~150µs)<br>0b10: 8192 (~9.83ms)<br>0b11: 1048576 (~1.26s)                    |

**GMSL1\_12 (0xC12)**

| BIT         | 7 | 6             | 5                 | 4 | 3                 | 2 | 1             | 0               |
|-------------|---|---------------|-------------------|---|-------------------|---|---------------|-----------------|
| Field       | – | CC_CRC_ERR_EN | LINE_CRC_LOC[1:0] |   | LINE_CRC_EN_GMSL1 | – | MAX_RT_ERR_EN | RCEG_ERR_PER_EN |
| Reset       | – | 0b1           | 0b01              |   | 0b0               | – | 0b1           | 0b0             |
| Access Type | – | Write, Read   | Write, Read       |   | Write, Read       | – | Write, Read   | Write, Read     |

| BITFIELD          | BITS | DESCRIPTION                                                          | DECODE                                                                                                                  |
|-------------------|------|----------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| CC_CRC_ERR_EN     | 6    | Enables reporting of (CC_CRC_ERR_CNT > 0) on the ERRB pin            | 0b0: Disable reporting of errors on ERRB<br>0b1: Enable reporting of errors on ERRB                                     |
| LINE_CRC_LOC      | 5:4  | Video line CRC insertion location                                    | 0b00: [1..4]<br>0x01: [5..8]<br>0x10: [9..12]<br>0x11: [13..16]                                                         |
| LINE_CRC_EN_GMSL1 | 3    | Video line CRC enable                                                | 0b0: Disable video line CRC<br>0x1: Enable video line CRC                                                               |
| MAX_RT_ERR_EN     | 1    | Enables reflection of maximum retransmission error on the ERRORB pin | 0b0: Disable maximum retransmission error on the ERROR pin<br>0b1: Enable maximum retransmission error on the ERROR pin |
| RCEG_ERR_PER_EN   | 0    | Periodic error generation enable. Effective when RCEG_TYPE = 0x.     | 0b0: Disable periodic error generator<br>0b1: Enable periodic error generator                                           |

**GMSL1\_13 (0xC13)**

| BIT         | 7           | 6               | 5                  | 4                   | 3 | 2 | 1 | 0 |
|-------------|-------------|-----------------|--------------------|---------------------|---|---|---|---|
| Field       | EOM_EN_G1   | EOM_PER_MODE_G1 | EOM_MAN_TRG_REQ_G1 | EOM_MIN_THR_G1[4:0] |   |   |   |   |
| Reset       | 0b1         | 0b1             | 0b0                | 0b00000             |   |   |   |   |
| Access Type | Write, Read | Write, Read     | Write, Read        | Write, Read         |   |   |   |   |

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| BITFIELD           | BITS | DESCRIPTION                                                                                                                          | DECODE                                                                                               |
|--------------------|------|--------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| EOM_EN_G1          | 7    | Enables eye-opening monitor (EOM)                                                                                                    | 0b0: Disable EOM<br>0x1: Enable EOM                                                                  |
| EOM_PER_MODE_G1    | 6    | Eye-opening monitor (EOM) periodic mode select                                                                                       | 0b0: Set EOM to use non-periodic mode<br>0x1: Set EOM to use periodic mode                           |
| EOM_MAN_TRG_REQ_G1 | 5    | Eye-opening monitor (EOM) manual trigger request.<br>Valid on the rising edge of this bit when not in periodic mode.                 | 0x0: Do not trigger EOM.<br>0x1: Manually trigger the EOM                                            |
| EOM_MIN_THR_G1     | 4:0  | Eye-opening minimum threshold (in terms of percent) for flagging ERRORB pin.<br>(Percentages increment 3.125% per bitfield setting.) | 0b00000: Disabled<br>0b00001: 3.125%<br>0b00010: 6.25%<br>0b00011: 9.375%<br>.<br>.<br>0b11111: 100% |

**GMSL1\_14 (0xC14)**

| BIT         | 7           | 6            | 5               | 4                | 3 | 2 | 1 | 0 |
|-------------|-------------|--------------|-----------------|------------------|---|---|---|---|
| Field       | AEQ_EN      | AEQ_PER_MODE | AEQ_MAN_TRG_REQ | EOM_PER_THR[4:0] |   |   |   |   |
| Reset       | 0b1         | 0b0          | 0b0             | 0b00000          |   |   |   |   |
| Access Type | Write, Read | Write, Read  | Write, Read     | Write, Read      |   |   |   |   |

| BITFIELD        | BITS | DESCRIPTION                                                                                                         | DECODE                                                                                                                 |
|-----------------|------|---------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| AEQ_EN          | 7    | Adaptive equalization enable                                                                                        | 0b0: Disable AEQ<br>0x1: Enable AEQ                                                                                    |
| AEQ_PER_MODE    | 6    | Adaptive equalizer periodic mode select                                                                             | 0b0: Set AEQ to use nonperiodic mode<br>0x1: Set AEQ to use periodic mode                                              |
| AEQ_MAN_TRG_REQ | 5    | Adaptive equalizer manual fine tune request enable. Valid on the rising edge of this bit when not in periodic mode. | 0x0: Do not trigger AEQ fine tuning<br>0x1: Manually trigger the AEQ fine tuning                                       |
| EOM_PER_THR     | 4:0  | Eye-opening threshold to trigger a fine tune operation                                                              | 0b00000: Eye-opening threshold is disabled<br>0b10000: 50% eye opening triggers fine-tune operation<br>OTHER: Reserved |

**GMSL1\_15 (0xC15)**

| BIT         | 7            | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|--------------|---|---|---|---|---|---|---|
| Field       | DET_ERR[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00         |   |   |   |   |   |   |   |
| Access Type | Read Only    |   |   |   |   |   |   |   |

| BITFIELD | BITS | DESCRIPTION            | DECODE                          |
|----------|------|------------------------|---------------------------------|
| DET_ERR  | 7:0  | Detected error counter | 0xXX: Number of detected errors |

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**GMSL1\_16 (0xC16)**

| BIT         | 7             | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|---------------|---|---|---|---|---|---|---|
| Field       | PRBS_ERR[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00          |   |   |   |   |   |   |   |
| Access Type | Read Only     |   |   |   |   |   |   |   |

| BITFIELD | BITS | DESCRIPTION        | DECODE                               |
|----------|------|--------------------|--------------------------------------|
| PRBS_ERR | 7:0  | PRBS error counter | 0xXX: Number of detected PRBS errors |

**GMSL1\_17 (0xC17)**

| BIT         | 7                   | 6              | 5         | 4         | 3              | 2 | 1 | 0 |
|-------------|---------------------|----------------|-----------|-----------|----------------|---|---|---|
| Field       | I2C_TIMED_OUT_GMSL1 | MAX_RT_ERR_I2C | PRBS_OK   | GPI_IN    | MAX_RT_ERR_GPI | – | – | – |
| Reset       | 0b0                 | 0b0            | 0b0       | 0b0       | 0b0            | – | – | – |
| Access Type | Read Only           | Read Only      | Read Only | Read Only | Read Only      | – | – | – |

| BITFIELD            | BITS | DESCRIPTION                                                                                                                                                  | DECODE                                                                                                                 |
|---------------------|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| I2C_TIMED_OUT_GMSL1 | 7    | I2C channel timed out flag                                                                                                                                   | 0b0: I2C channel not timed out<br>0b1: I2C channel timed out                                                           |
| MAX_RT_ERR_I2C      | 6    | Maximum retransmission error flag. Cleared when read.                                                                                                        | 0b0: No control channel retransmission error<br>0b1: Control channel retransmission maximum limit reached              |
| PRBS_OK             | 5    | MAX9271/73 compatible PRBS test to determine if link completed normally. Check PRBS_ERR register for PRBS success. For other SERDES read PRBS_ERR registers. | 0b0: No MAX9271/MAX9273-compatible PRBS test completed<br>0b1: MAX9271/MAX9273-compatible PRBS test completed normally |
| GPI_IN              | 4    | GPI pin level                                                                                                                                                | 0b0: GPI is input low<br>0b1: GPI is input high                                                                        |
| MAX_RT_ERR_GPI      | 3    | Maximum retransmission error flag. Cleared when read.                                                                                                        | 0b0: No control-channel retransmission error<br>0b1: Control-channel retransmission maximum limit reached              |

**GMSL1\_18 (0xC18)**

| BIT         | 7                | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|------------------|---|---|---|---|---|---|---|
| Field       | CC_RETR_CNT[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00             |   |   |   |   |   |   |   |
| Access Type | Read Only        |   |   |   |   |   |   |   |

| BITFIELD    | BITS | DESCRIPTION                 | DECODE                                    |
|-------------|------|-----------------------------|-------------------------------------------|
| CC_RETR_CNT | 7:0  | I2C packet retransmit count | 0xXX: Number of I2C packets retransmitted |

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**GMSL1\_19 (0xC19)**

| BIT           | 7                  | 6                                              | 5 | 4 | 3                                          | 2 | 1 | 0 |
|---------------|--------------------|------------------------------------------------|---|---|--------------------------------------------|---|---|---|
| Field         | CC_CRC_ERRCNT[7:0] |                                                |   |   |                                            |   |   |   |
| Reset         | 0x00               |                                                |   |   |                                            |   |   |   |
| Access Type   | Read Only          |                                                |   |   |                                            |   |   |   |
| BITFIELD      | BITS               | DESCRIPTION                                    |   |   | DECODE                                     |   |   |   |
| CC_CRC_ERRCNT | 7:0                | Packet-based control-channel CRC error counter |   |   | 0xXX: Number of control-channel CRC errors |   |   |   |

**GMSL1\_1A (0xC1A)**

| BIT          | 7                 | 6                                          | 5 | 4 | 3                                                | 2 | 1 | 0 |
|--------------|-------------------|--------------------------------------------|---|---|--------------------------------------------------|---|---|---|
| Field        | RCEG_ERR_CNT[7:0] |                                            |   |   |                                                  |   |   |   |
| Reset        | 0x00              |                                            |   |   |                                                  |   |   |   |
| Access Type  | Read Only         |                                            |   |   |                                                  |   |   |   |
| BITFIELD     | BITS              | DESCRIPTION                                |   |   | DECODE                                           |   |   |   |
| RCEG_ERR_CNT | 7:0               | Control channel number of generated errors |   |   | 0xXX: Number of control channel generated errors |   |   |   |

**GMSL1\_1B (0xC1B)**

| BIT          | 7    | 6                                                         | 5 | 4 | 3                                                               | 2            | 1 | 0 |
|--------------|------|-----------------------------------------------------------|---|---|-----------------------------------------------------------------|--------------|---|---|
| Field        | –    | –                                                         | – | – | –                                                               | LINE_CRC_ERR | – | – |
| Reset        | –    | –                                                         | – | – | –                                                               | 0b0          | – | – |
| Access Type  | –    | –                                                         | – | – | –                                                               | Read Only    | – | – |
| BITFIELD     | BITS | DESCRIPTION                                               |   |   | DECODE                                                          |              |   |   |
| LINE_CRC_ERR | 2    | CRC error bit. Latched on error - cleared to 0 when read. |   |   | 0b0: Video line CRC ok<br>0b1: Video line CRC mismatch detected |              |   |   |

**GMSL1\_1C (0xC1C)**

| BIT           | 7    | 6                                                             | 5                  | 4 | 3                                                                                                                                      | 2 | 1 | 0 |
|---------------|------|---------------------------------------------------------------|--------------------|---|----------------------------------------------------------------------------------------------------------------------------------------|---|---|---|
| Field         | –    | –                                                             | EOM_EYE_WIDTH[5:0] |   |                                                                                                                                        |   |   |   |
| Reset         | –    | –                                                             | 0b000000           |   |                                                                                                                                        |   |   |   |
| Access Type   | –    | –                                                             | Read Only          |   |                                                                                                                                        |   |   |   |
| BITFIELD      | BITS | DESCRIPTION                                                   |                    |   | DECODE                                                                                                                                 |   |   |   |
| EOM_EYE_WIDTH | 5:0  | Measured eye-opening. Opening width = EOM_EYE_WIDTH/63 x 100% |                    |   | 0b000000: Width is 0%<br>0b000001: Width is 1/63 x 100%<br>0b000010: Width is 2/63 x 100%<br>.<br>.<br>0b111111: Width is 63/63 x 100% |   |   |   |

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GMSL1\_1D (0xC1D)

| BIT            | 7 | 6 | 5 | 4                  | 3            | 2 | 1 | 0 |
|----------------|---|---|---|--------------------|--------------|---|---|---|
| Field          | – | – | – | UNDERBO<br>OST_DET | AEQ_BST[3:0] |   |   |   |
| Reset          | – | – | – | 0b0                | 0x0          |   |   |   |
| Access<br>Type | – | – | – | Read Only          | Read Only    |   |   |   |

| BITFIELD           | BITS | DESCRIPTION                                                                                              | DECODE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|--------------------|------|----------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| UNDERBOO<br>ST_DET | 4    | Under boost detected                                                                                     | 0b0: Normal Operation<br>0x1: Underboost (at maximum AEQ gain) detected                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| AEQ_BST            | 3:0  | Adaptive equalizer boost value. Selected adaptive equalizer value; settings correspond to gain at 1.5GHz | 0b0000: 0 dB manual EQ setting<br>0b0001: 0 dB manual EQ setting<br>0b0010: 1.0 dB manual EQ setting<br>0b0011: 2.6 dB manual EQ setting<br>0b0100: 4.0 dB manual EQ setting<br>0b0101: 5.8 dB manual EQ setting<br>0b0110: 7.2 dB manual EQ setting<br>0b0111: 8.6 dB manual EQ setting<br>0b1000: 10.4 dB manual EQ setting<br>0b1001: 11.7 dB manual EQ setting<br>0b1010: 13.5 dB manual EQ setting<br>0b1011: 14.5 dB manual EQ setting<br>0b1100: Reserved<br>0b1101: Reserved<br>0b1110: Reserved<br>0b1111: Reserved |

GMSL1\_20 (0xC20)

| BIT            | 7                | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|----------------|------------------|---|---|---|---|---|---|---|
| Field          | CRC_VALUE_0[7:0] |   |   |   |   |   |   |   |
| Reset          | 0x00             |   |   |   |   |   |   |   |
| Access<br>Type | Read Only        |   |   |   |   |   |   |   |

| BITFIELD        | BITS | DESCRIPTION                                  | DECODE                        |
|-----------------|------|----------------------------------------------|-------------------------------|
| CRC_VALUE<br>_0 | 7:0  | Bits [7:0] of CRC output for the latest line | 0xXX: CRC[7:0] of latest line |

GMSL1\_21 (0xC21)

| BIT            | 7                | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|----------------|------------------|---|---|---|---|---|---|---|
| Field          | CRC_VALUE_1[7:0] |   |   |   |   |   |   |   |
| Reset          | 0x00             |   |   |   |   |   |   |   |
| Access<br>Type | Read Only        |   |   |   |   |   |   |   |

| BITFIELD        | BITS | DESCRIPTION                                   | DECODE                         |
|-----------------|------|-----------------------------------------------|--------------------------------|
| CRC_VALUE<br>_1 | 7:0  | Bits [15:8] of CRC output for the latest line | 0xXX: CRC[15:8] of latest line |



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**GMSL1\_22 (0xC22)**

| BIT         | 7                | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|------------------|---|---|---|---|---|---|---|
| Field       | CRC_VALUE_2[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00             |   |   |   |   |   |   |   |
| Access Type | Read Only        |   |   |   |   |   |   |   |

| BITFIELD    | BITS | DESCRIPTION                                    | DECODE                          |
|-------------|------|------------------------------------------------|---------------------------------|
| CRC_VALUE_2 | 7:0  | Bits [23:16] of CRC output for the latest line | 0xXX: CRC[23:16] of latest line |

**GMSL1\_23 (0xC23)**

| BIT         | 7                | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|------------------|---|---|---|---|---|---|---|
| Field       | CRC_VALUE_3[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00             |   |   |   |   |   |   |   |
| Access Type | Read Only        |   |   |   |   |   |   |   |

| BITFIELD    | BITS | DESCRIPTION                                    | DECODE                          |
|-------------|------|------------------------------------------------|---------------------------------|
| CRC_VALUE_3 | 7:0  | Bits [31:24] of CRC output for the latest line | 0xXX: CRC[31:24] of latest line |

**GMSL1\_96 (0xC96)**

| BIT         | 7                        | 6 | 5 | 4 | 3 | 2           | 1             | 0            |
|-------------|--------------------------|---|---|---|---|-------------|---------------|--------------|
| Field       | CONV_GMSL1_DATATYPE[4:0] |   |   |   |   | DIS_HIBW_E  | CONV_GMSL1_EN | DBL_ALIGN_TO |
| Reset       | 0b00000                  |   |   |   |   | 0b0         | 0b0           | 0b1          |
| Access Type | Write, Read              |   |   |   |   | Write, Read | Write, Read   | Write, Read  |

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| BITFIELD            | BITS | DESCRIPTION                                                                   | DECODE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|---------------------|------|-------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CONV_GMSL1_DATATYPE | 7:3  | Convert from GMSL1 color format mapping to GMSL2 CSI transmitter color format | 0x0: RGB888 OLDI<br>0x1: RGB565<br>0x2: RGB666<br>0x3: YUV 422 8-bit Mux mode (use yuv_8_10_mux_mode)<br>0x4: YUV 422 10-bit Mux mode (use yuv_8_10_mux_mode)<br>0x5: RAW8 single<br>0x6: RAW10 single<br>0x7: RAW12 single<br>0x8: RAW14<br>0x9: User-defined Generic 24-bit<br>0xA: User-defined YUV422 12-bit<br>0xB: User-defined Generic 8-bit<br>0xC<br>0xD<br>0xE<br>0xF<br>0x10: RGB888 VESA<br>0x11<br>0x12<br>0x13: YUV 422 8-bit Normal mode<br>0x14: YUV 422 10-bit Normal mode<br>0x15: RAW8 double (use alt_mem_map8)<br>0x16: RAW10 double (use alt_mem_map10)<br>0x17: RAW12 double (use alt_mem_map12)<br>0x18<br>0x19<br>0x1A<br>0x1B<br>0x1C<br>0x1D<br>0x1E<br>0x1F |
| DIS_HIBW_E          | 2    | Disable disparity error detection in HIBW                                     | 0b0: Enable disparity error detection<br>0b1: Disable disparity error detection                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| CONV_GMSL1_EN       | 1    | Enable conversion from GMSL1 color format mapping to GMSL2 CSI transmitter    | 0b0: Disable conversion<br>0b1: Enable conversion                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| DBL_ALIGN_TO        | 0    | HBM DBL mode type                                                             | 0b0: DBL-Single mode with alignment<br>0b1: DBL-DBL mode with no alignment (d)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

**GMSL1\_CA (0xCCA)**

| BIT         | 7 | 6 | 5 | 4 | 3 | 2         | 1         | 0         |
|-------------|---|---|---|---|---|-----------|-----------|-----------|
| Field       | — | — | — | — | — | PHASELOCK | WBLOCK_G1 | DATAOK    |
| Reset       | — | — | — | — | — | 0b0       | 0b0       | 0b0       |
| Access Type | — | — | — | — | — | Read Only | Read Only | Read Only |

| BITFIELD  | BITS | DESCRIPTION | DECODE                                 |
|-----------|------|-------------|----------------------------------------|
| PHASELOCK | 2    | CDR locked  | 0b0: CDR not locked<br>0b1: CDR locked |

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Appendix A: Register Map and Tables

| BITFIELD      | BITS | DESCRIPTION                        | DECODE                                                                                |
|---------------|------|------------------------------------|---------------------------------------------------------------------------------------|
| WBLOCK_G<br>1 | 1    | Word boundary detected             | 0b0: No word boundary detected<br>0b1: Word boundary detected                         |
| DATAOK        | 0    | Data activity detected on the link | 0b0: No data activity detected on the link<br>0b1: Data activity detected on the link |

**GMSL1\_D1 (0xCD1)**

| BIT            | 7                 | 6 | 5 | 4                | 3 | 2 | 1 | 0 |
|----------------|-------------------|---|---|------------------|---|---|---|---|
| Field          | CNTL_OUT_ORD[2:0] |   |   | CNTL_OUT_EN[4:0] |   |   |   |   |
| Reset          | 0x0               |   |   | 0x00000          |   |   |   |   |
| Access<br>Type | Write, Read       |   |   | Write, Read      |   |   |   |   |

| BITFIELD         | BITS | DESCRIPTION                                    | DECODE                                                                                                                                                                                                                                                                                                                                                                    |
|------------------|------|------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CNTL_OUT_<br>ORD | 7:5  | Internal CNTL_OUT order to pins CNTL4 to CNTL0 | 0x0: 4, 3, 2, 1, 0<br>0x1: 3, 2, 1, 0, 4<br>0x2: 2, 1, 0, 4, 3<br>0x3: 1, 0, 4, 3, 2<br>0x4: 0, 1, 2, 3, 4<br>0x5: 1, 2, 3, 4, 0<br>0x6: 2, 3, 4, 0, 1<br>0x7: 3, 4, 0, 1, 2                                                                                                                                                                                              |
| CNTL_OUT_<br>EN  | 4:0  | CNTL output enable for CNTL 0, 1, 2, 3, 4      | 0bXXXX0: Disable control output 0<br>0bXXXX1: Enable control output 0<br>0bXXX0X: Disable control output 1<br>0bXXX1X: Enable control output 1<br>0bXX0XX: Disable control output 2<br>0bXX1XX: Enable control output 2<br>0bX0XXX: Disable control output 3<br>0bX1XXX: Enable control output 3<br>0b0XXXX: Disable control output 4<br>0b1XXXX: Enable control output 4 |

**GMSL1\_EN (0xF00)**

| BIT            | 7 | 6 | 5 | 4 | 3 | 2 | 1           | 0           |
|----------------|---|---|---|---|---|---|-------------|-------------|
| Field          | — | — | — | — | — | — | LINK_EN_B   | LINK_EN_A   |
| Reset          | — | — | — | — | — | — | 0x1         | 0x1         |
| Access<br>Type | — | — | — | — | — | — | Write, Read | Write, Read |

| BITFIELD  | BITS | DESCRIPTION    | DECODE                                    |
|-----------|------|----------------|-------------------------------------------|
| LINK_EN_B | 1    | Enables link B | 0b0: Disable Link B<br>0b1: Enable Link B |
| LINK_EN_A | 0    | Enables link A | 0b0: Disable Link A<br>0b1: Enable Link A |

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**GMSL1\_UART (0xF01)**

| BIT            | 7 | 6 | 5 | 4 | 3 | 2 | 1                         | 0                         |
|----------------|---|---|---|---|---|---|---------------------------|---------------------------|
| Field          | – | – | – | – | – | – | GMSL1_UA<br>RT_ARB_T<br>O | GMSL1_UA<br>RT_ARB_E<br>N |
| Reset          | – | – | – | – | – | – | 0x0                       | 0x0                       |
| Access<br>Type | – | – | – | – | – | – | Write, Read               | Write, Read               |

| BITFIELD              | BITS | DESCRIPTION                                                        | DECODE                      |
|-----------------------|------|--------------------------------------------------------------------|-----------------------------|
| GMSL1_UAR<br>T_ARB_TO | 1    | GMSL1 UART arbitration timeout value                               | 0b0: 256µs<br>0b1: 4ms      |
| GMSL1_UAR<br>T_ARB_EN | 0    | UART arbitration enable on GMSL1 forward<br>control channel links. | 0b0: Disable<br>0b1: Enable |

**COMMON1 (0xF02)**

| BIT            | 7 | 6 | 5 | 4 | 3 | 2 | 1                          | 0                          |
|----------------|---|---|---|---|---|---|----------------------------|----------------------------|
| Field          | – | – | – | – | – | – | REM_ACK_<br>ACKED_G1<br>_B | REM_ACK_<br>ACKED_G1<br>_A |
| Reset          | – | – | – | – | – | – | 0b0                        | 0b0                        |
| Access<br>Type | – | – | – | – | – | – | Read Only                  | Read Only                  |

| BITFIELD               | BITS | DESCRIPTION             | DECODE                                                           |
|------------------------|------|-------------------------|------------------------------------------------------------------|
| REM_ACK_A<br>CKED_G1_B | 1    | Remote ACK acknowledged | 0b0: Remote ACK not acknowledged<br>0b1: Remote ACK acknowledged |
| REM_ACK_A<br>CKED_G1_A | 0    | Remote ACK acknowledged | 0b0: Remote ACK not acknowledged<br>0b1: Remote ACK acknowledged |

**GMSL1\_ERR\_OEN (0xF03)**

| BIT            | 7 | 6 | 5 | 4 | 3 | 2 | 1                 | 0                 |
|----------------|---|---|---|---|---|---|-------------------|-------------------|
| Field          | – | – | – | – | – | – | G1_B_ERR_<br>_OEN | G1_A_ERR_<br>_OEN |
| Reset          | – | – | – | – | – | – | 0b1               | 0b1               |
| Access<br>Type | – | – | – | – | – | – | Write, Read       | Write, Read       |

| BITFIELD         | BITS | DESCRIPTION                                                            | DECODE                                          |
|------------------|------|------------------------------------------------------------------------|-------------------------------------------------|
| G1_B_ERR_<br>OEN | 1    | Enable reporting of GMSL1 Link B errors<br>(G1_B_ERR_FLAG) at ERRB pin | 0b0: Disable reporting<br>0b1: Enable reporting |
| G1_A_ERR_<br>OEN | 0    | Enable reporting of GMSL1 Link A errors<br>(G1_A_ERR_FLAG) at ERRB pin | 0b0: Disable reporting<br>0b1: Enable reporting |

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GMSL1\_ERR\_FLAG (0xF04)

| BIT         | 7 | 6 | 5 | 4 | 3 | 2 | 1             | 0             |
|-------------|---|---|---|---|---|---|---------------|---------------|
| Field       | – | – | – | – | – | – | G1_B_ERR_FLAG | G1_A_ERR_FLAG |
| Reset       | – | – | – | – | – | – | 0b0           | 0b0           |
| Access Type | – | – | – | – | – | – | Read Only     | Read Only     |

| BITFIELD      | BITS | DESCRIPTION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | DECODE                                                   |
|---------------|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|
| G1_B_ERR_FLAG | 1    | GMSL1 Link B error flag. When PRBS test is enabled, this flag asserted if at least one PRBS error has been detected. When PRBS test is not enabled, the flag is asserted when any of these conditions is true:<br>1) Number of detected decoding errors is greater than the detected error threshold (DET_ERR > DET_THR)<br>2) Measured eye-opening is less than or equal to eye-opening threshold (EOM_EYE_WIDTH ≤ EOM_MIN_THR_G1)<br>3) Adaptive EQ has detected an under boost<br>4) Video Line CRC error is detected<br>5) Maximum retransmission count exceeded in PKTCC communication<br>6) CRC errors detected in PKTCC communication | 0b0: Error flag not asserted<br>0b1: Error flag asserted |
| G1_A_ERR_FLAG | 0    | GMSL1 Link A error flag. When PRBS test is enabled, this flag is asserted if at least one PRBS error has been detected. When PRBS test is not enabled, the flag is asserted when any of these conditions is true:<br>1) Number of detected decoding errors is greater than the detected error threshold (DET_ERR > DET_THR)<br>2) Measured eye-opening is less than or equal to eye-opening threshold (EOM_EYE_WIDTH ≤ EOM_MIN_THR_G1)<br>3) Adaptive EQ has detected under boost<br>4) Video Line CRC error is detected<br>5) Maximum retransmission count exceeded in PKTCC communication<br>6) CRC errors detected in PKTCC communication | 0b0: Error flag not asserted<br>0b1: Error flag asserted |

I2C\_0 (0xF05)

| BIT         | 7 | 6 | 5              | 4 | 3 | 2              | 1 | 0 |
|-------------|---|---|----------------|---|---|----------------|---|---|
| Field       | – | – | G1_SLV_SH[1:0] |   | – | G1_SLV_TO[2:0] |   |   |
| Reset       | – | – | 0b10           |   | – | 0b110          |   |   |
| Access Type | – | – | Write, Read    |   | – | Write, Read    |   |   |

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| BITFIELD  | BITS | DESCRIPTION                                                                                                                                                                                                                                                                                                                   | DECODE                                                                                                                                                                        |
|-----------|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| G1_SLV_SH | 5:4  | GMSL1 I <sup>2</sup> C-to-I <sup>2</sup> C slave setup and hold time setting (setup, hold). Configures the interval between SDA and SCL transitions when driven by the internal I <sup>2</sup> C slave. Set this according to the I <sup>2</sup> C speed mode.<br><br>Please see register groups CC_G2P* for GMSL2 registers. | 0b00: Set for I <sup>2</sup> C Fast-mode Plus speed<br>0b01: Set for I <sup>2</sup> C Fast-mode speed<br>0b10: Set for I <sup>2</sup> C standard-mode speed<br>0b11: Reserved |
| G1_SLV_TO | 2:0  | GMSL1 I <sup>2</sup> C-to-I <sup>2</sup> C slave timeout setting<br>Internal GMSL1 I <sup>2</sup> C slave times out after the configured duration if it does not receive any response while waiting for a packet from remote device.<br><br>Please see register groups CC_G2P* for GMSL2 registers. (Addr. 0x500 - 0x6B0)     | 0b000: 16μs<br>0b001: 1ms<br>0b010: 2ms<br>0b011: 4ms<br>0b100: 8ms<br>0b101: 16ms<br>0b110: 32ms<br>0b111: Disabled                                                          |

I2C\_1 (0xF06)

| BIT         | 7                   | 6              | 5 | 4 | 3 | 2              | 1 | 0 |
|-------------|---------------------|----------------|---|---|---|----------------|---|---|
| Field       | EN_I2C_LO<br>OPBACK | G1_MST_BT[2:0] |   |   | – | G1_MST_TO[2:0] |   |   |
| Reset       | 0x0                 | 0b101          |   |   | – | 0b110          |   |   |
| Access Type | Write, Read         | Write, Read    |   |   | – | Write, Read    |   |   |

| BITFIELD            | BITS | DESCRIPTION                                                                                                                                                                                                                                                                                                                                       | DECODE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|---------------------|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| EN_I2C_LO<br>OPBACK | 7    | Enable I <sup>2</sup> C loopback between links in G1 + G2 or G1 + G1 aggregation mode. When I <sup>2</sup> C loopback is enabled, the I <sup>2</sup> C transactions generated at the leaf-end of on one link are reflected to the leaf-end of the other link                                                                                      | 0x0: Disable loopback<br>0x1: Enable loopback                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| G1_MST_BT           | 6:4  | GMSL1 I <sup>2</sup> C-to-I <sup>2</sup> C master bit rate setting, configures the I <sup>2</sup> C bit rate used by the internal I <sup>2</sup> C master (in the device on remote side from the external I <sup>2</sup> C master) Set this according to the I <sup>2</sup> C speed mode.<br><br>See register groups CC_G2P* for GMSL2 registers. | 0b000: 9.92Kbps - Set for I <sup>2</sup> C Standard mode speed<br>0b001: 33.2Kbps - Set for I <sup>2</sup> C Standard mode speed<br>0b010: 99.2Kbps - Set for I <sup>2</sup> C Standard or Fast-mode speed<br>0b011: 123Kbps - Set for I <sup>2</sup> C Fast-mode speed<br>0b100: 203Kbps - Set for I <sup>2</sup> C Fast-mode speed<br>0b101: 397Kbps - Set for I <sup>2</sup> C Fast or Fast-mode Plus speed<br>0b110: 625Kbps - Set for I <sup>2</sup> C Fast-mode Plus speed<br>0b111: 980Kbps - Set for I <sup>2</sup> C Fast-mode Plus speed |
| G1_MST_TO           | 2:0  | GMSL1 I <sup>2</sup> C-to-I <sup>2</sup> C master timeout setting<br>Internal GMSL1 I <sup>2</sup> C master times out after the configured duration if it does not receive any response while waiting for a packet from the remote device.<br><br>See register groups CC_G2P* for GMSL2 registers. (Addr. 0x500 - 0x6B0)                          | 0b000: 16μs<br>0b001: 1ms<br>0b010: 2ms<br>0b011: 4ms<br>0b100: 8ms<br>0b101: 16ms<br>0b110: 32ms<br>0b111: Disabled                                                                                                                                                                                                                                                                                                                                                                                                                               |

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I2C\_2 (0xF07)

| BIT         | 7             | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|---------------|---|---|---|---|---|---|---|
| Field       | G1_SRC_A[6:0] |   |   |   |   |   |   | – |
| Reset       | 0b0000000     |   |   |   |   |   |   | – |
| Access Type | Write, Read   |   |   |   |   |   |   | – |

| BITFIELD | BITS | DESCRIPTION                                                                                                                                                                                                                                                                               | DECODE                                     |
|----------|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|
| G1_SRC_A | 7:1  | <p>GMSL1 I<sup>2</sup>C address translator source A<br/>When I<sup>2</sup>C device address matches I<sup>2</sup>C SRC_A, internal I<sup>2</sup>C master (on remote side) replaces the device address by I<sup>2</sup>C DST_A.</p> <p>See register groups CC_G2P* for GMSL2 registers.</p> | 0bXXXXXXX: Value of I <sup>2</sup> C SRC_A |

I2C\_3 (0xF08)

| BIT         | 7             | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|---------------|---|---|---|---|---|---|---|
| Field       | G1_DST_A[6:0] |   |   |   |   |   |   | – |
| Reset       | 0b0000000     |   |   |   |   |   |   | – |
| Access Type | Write, Read   |   |   |   |   |   |   | – |

| BITFIELD | BITS | DESCRIPTION                                                                                                                                                           | DECODE                                     |
|----------|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|
| G1_DST_A | 7:1  | <p>GMSL1 I<sup>2</sup>C address translator destination A<br/>See the description of I<sup>2</sup>C SRC_A.</p> <p>See register groups CC_G2P* for GMSL2 registers.</p> | 0bXXXXXXX: Value of I <sup>2</sup> C DST_A |

I2C\_4 (0xF09)

| BIT         | 7             | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|---------------|---|---|---|---|---|---|---|
| Field       | G1_SRC_B[6:0] |   |   |   |   |   |   | – |
| Reset       | 0b0000000     |   |   |   |   |   |   | – |
| Access Type | Write, Read   |   |   |   |   |   |   | – |

| BITFIELD | BITS | DESCRIPTION                                                                                                                                                                                                                                                                               | DECODE                                     |
|----------|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|
| G1_SRC_B | 7:1  | <p>GMSL1 I<sup>2</sup>C address translator source B<br/>When I<sup>2</sup>C device address matches I<sup>2</sup>C SRC_B, internal I<sup>2</sup>C master (on remote side) replaces the device address by I<sup>2</sup>C DST_B.</p> <p>See register groups CC_G2P* for GMSL2 registers.</p> | 0bXXXXXXX: Value of I <sup>2</sup> C SRC_B |

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I2C\_5 (0xF0A)

| BIT         | 7             | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|---------------|---|---|---|---|---|---|---|
| Field       | G1_DST_B[6:0] |   |   |   |   |   |   | – |
| Reset       | 0b0000000     |   |   |   |   |   |   | – |
| Access Type | Write, Read   |   |   |   |   |   |   | – |

| BITFIELD | BITS | DESCRIPTION                                                                                                                                                       | DECODE                                     |
|----------|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|
| G1_DST_B | 7:1  | GMSL1 I <sup>2</sup> C address translator destination B<br>See the description of I <sup>2</sup> C SRC_B.<br><br>See register groups CC_G2P* for GMSL2 registers. | 0bXXXXXXX: Value of I <sup>2</sup> C DST_B |

SPI\_CC\_WR (0x1300)

Internal write location for SPI to Control Channel bytes

SPI\_CC\_RD (0x1380)

Internal read location for SPI to Control Channel bytes

RLMS3 (0x1403, 0x1503)

| BIT         | 7           | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|-------------|---|---|---|---|---|---|---|
| Field       | AdaptEn     | – | – | – | – | – | – | – |
| Reset       | 0x0         | – | – | – | – | – | – | – |
| Access Type | Write, Read | – | – | – | – | – | – | – |

| BITFIELD | BITS | DESCRIPTION          | DECODE                                                                            |
|----------|------|----------------------|-----------------------------------------------------------------------------------|
| AdaptEn  | 7    | Adapt process enable | 0b0: Manual adaptation process disabled<br>0b1: Manual adaptation process enabled |

RLMS4 (0x1404, 0x1504)

| BIT         | 7                   | 6 | 5 | 4 | 3                | 2 | 1            | 0           |
|-------------|---------------------|---|---|---|------------------|---|--------------|-------------|
| Field       | EOM_CHK_AMOUNT[3:0] |   |   |   | EOM_CHK_THR[1:0] |   | EOM_PER_MODE | EOM_EN      |
| Reset       | 0x4                 |   |   |   | 0x2              |   | 0x1          | 0x1         |
| Access Type | Write, Read         |   |   |   | Write, Read      |   | Write, Read  | Write, Read |

| BITFIELD       | BITS | DESCRIPTION                                                                                                                                                       | DECODE                                                                                            |
|----------------|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| EOM_CHK_AMOUNT | 7:4  | A factor (N) used to select the order of number of observations in each eye monitor window. N is used in the equation:<br>Observations = $6.29 \times 10^{(N+2)}$ | 0xX: N factor                                                                                     |
| EOM_CHK_THR    | 3:2  | Eye-opening monitor number of error bits to allow in a measurement window                                                                                         | 0b00: Allow no errors<br>0b01: Allow 1 error<br>0b10: Allow 2 errors<br>0b11: Allow 3 errors      |
| EOM_PER_MODE   | 1    | Eye-opening monitor periodic mode enable                                                                                                                          | 0b0: Eye-opening monitor periodic mode disabled<br>0b1: Eye-opening monitor periodic mode enabled |



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| BITFIELD | BITS | DESCRIPTION                | DECODE                                                                |
|----------|------|----------------------------|-----------------------------------------------------------------------|
| EOM_EN   | 0    | Eye-opening monitor enable | 0b0: Eye-opening monitor disabled<br>0b1: Eye-opening monitor enabled |

**RLMS5 (0x1405, 0x1505)**

| BIT         | 7               | 6                | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|-----------------|------------------|---|---|---|---|---|---|
| Field       | EOM_MAN_TRG_REQ | EOM_MIN_THR[6:0] |   |   |   |   |   |   |
| Reset       | 0x0             | 0x10             |   |   |   |   |   |   |
| Access Type | Write Only      | Write, Read      |   |   |   |   |   |   |

| BITFIELD        | BITS | DESCRIPTION                                                                                                                                     | DECODE                                                            |
|-----------------|------|-------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|
| EOM_MAN_TRG_REQ | 7    | Eye-opening monitor manual trigger. For use when periodic mode is disabled.                                                                     | 0b0: No action<br>0b1: Eye-opening monitor manual trigger request |
| EOM_MIN_THR     | 6:0  | The eye-opening monitor minimum threshold as defined by the equation: % eye-opening = EOM_MIN_THR/64. If the value is zero the EOM is disabled. | 0bXXXXXXX: Eye-opening monitor minimum threshold factor           |

**RLMS6 (0x1406, 0x1506)**

| BIT         | 7           | 6                | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|-------------|------------------|---|---|---|---|---|---|
| Field       | EOM_PV_MODE | EOM_RST_THR[6:0] |   |   |   |   |   |   |
| Reset       | 0x1         | 0x00             |   |   |   |   |   |   |
| Access Type | Write, Read | Write, Read      |   |   |   |   |   |   |

| BITFIELD    | BITS | DESCRIPTION                                                                                                                                                     | DECODE                                                     |
|-------------|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|
| EOM_PV_MODE | 7    | Eye-opening is measured vertically or horizontally                                                                                                              | 0b0: Vertical opening mode<br>0b1: Horizontal opening mode |
| EOM_RST_THR | 6:0  | The eye-opening monitor refresh threshold as defined by the equation: % eye opening = EOM_MIN_THR/64. If the value is zero the eye-opening monitor is disabled. | 0bXXXXXXX: Eye-opening monitor refresh threshold factor    |

**RLMS7 (0x1407, 0x1507)**

| BIT         | 7         | 6         | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|-----------|-----------|---|---|---|---|---|---|
| Field       | EOM_DONE  | EOM[6:0]  |   |   |   |   |   |   |
| Reset       | 0x0       | 0x00      |   |   |   |   |   |   |
| Access Type | Read Only | Read Only |   |   |   |   |   |   |

| BITFIELD | BITS | DESCRIPTION                                    | DECODE                                                                     |
|----------|------|------------------------------------------------|----------------------------------------------------------------------------|
| EOM_DONE | 7    | Eye-opening monitor measurement done           | 0b0: Eye-opening monitor not complete<br>0b1: Eye-opening monitor complete |
| EOM      | 6:0  | Last completed eye-opening monitor observation | 0bXXXXXXX: Eye-monitor opening measurement result                          |

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**RLMS34 (0x1434, 0x1534)**

| BIT           | 7                  | 6                                      | 5 | 4 | 3                                                       | 2 | 1 | 0 |
|---------------|--------------------|----------------------------------------|---|---|---------------------------------------------------------|---|---|---|
| Field         | EyeMonPerCntL[7:0] |                                        |   |   |                                                         |   |   |   |
| Reset         | 0x00               |                                        |   |   |                                                         |   |   |   |
| Access Type   | Write, Read        |                                        |   |   |                                                         |   |   |   |
| BITFIELD      | BITS               | DESCRIPTION                            |   |   | DECODE                                                  |   |   |   |
| EyeMonPerCntL | 7:0                | Eye monitor period count (RxClk20) LSB |   |   | 0xXX: Eye monitor period count (Least Significant Byte) |   |   |   |

**RLMS35 (0x1435, 0x1535)**

| BIT           | 7                  | 6                                      | 5 | 4 | 3                                                      | 2 | 1 | 0 |
|---------------|--------------------|----------------------------------------|---|---|--------------------------------------------------------|---|---|---|
| Field         | EyeMonPerCntH[7:0] |                                        |   |   |                                                        |   |   |   |
| Reset         | 0x00               |                                        |   |   |                                                        |   |   |   |
| Access Type   | Write, Read        |                                        |   |   |                                                        |   |   |   |
| BITFIELD      | BITS               | DESCRIPTION                            |   |   | DECODE                                                 |   |   |   |
| EyeMonPerCntH | 7:0                | Eye monitor period count (RxClk20) MSB |   |   | 0xXX: Eye monitor period count (Most Significant Byte) |   |   |   |

**RLMS37 (0x1437, 0x1537)**

| BIT          | 7    | 6                                                                                                             | 5 | 4          | 3                                                                                          | 2           | 1           | 0           |
|--------------|------|---------------------------------------------------------------------------------------------------------------|---|------------|--------------------------------------------------------------------------------------------|-------------|-------------|-------------|
| Field        | –    | –                                                                                                             | – | EyeMonDone | EyeMonCntClr                                                                               | EyeMonStart | EyeMonPh    | EyeMonDPol  |
| Reset        | –    | –                                                                                                             | – | 0x0        | 0x0                                                                                        | 0x0         | 0x0         | 0x0         |
| Access Type  | –    | –                                                                                                             | – | Read Only  | Write Only                                                                                 | Write Only  | Write, Read | Write, Read |
| BITFIELD     | BITS | DESCRIPTION                                                                                                   |   |            | DECODE                                                                                     |             |             |             |
| EyeMonDone   | 4    | Eye monitor period complete (read-only, reset on start)                                                       |   |            | 0b0: Eye monitor data collection not complete<br>0b1: Eye monitor data collection complete |             |             |             |
| EyeMonCntClr | 3    | Eye monitor error/valid count clear (one-shot). Read-back is EyeMonClrPL from long pulse generation.          |   |            | 0b0: NA<br>0b1: Clear eye monitor data collection counters                                 |             |             |             |
| EyeMonStart  | 2    | Eye monitor start (one-shot). Read-back is EyeMonStClrPL from long pulse generation for both start and clear. |   |            | 0b0: NA<br>0b1: Start eye monitor data collection                                          |             |             |             |
| EyeMonPh     | 1    | Eye monitor phase                                                                                             |   |            | 0b0: Eye monitor search early phase<br>0b1: Eye monitor search late phase                  |             |             |             |
| EyeMonDPol   | 0    | Eye monitor data polarity                                                                                     |   |            | 0b0: Eye monitor search for 1's<br>0b1: Eye monitor search for 0's                         |             |             |             |

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**RLMS39 (0x1439, 0x1539)**

| BIT         | 7                  | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|--------------------|---|---|---|---|---|---|---|
| Field       | EyeMonErrCntH[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00               |   |   |   |   |   |   |   |
| Access Type | Read Only          |   |   |   |   |   |   |   |

| BITFIELD      | BITS | DESCRIPTION                         | DECODE                                                |
|---------------|------|-------------------------------------|-------------------------------------------------------|
| EyeMonErrCntH | 7:0  | Eye monitor error count (read-only) | 0xXX: Eye monitor error count (Most Significant Byte) |

**RLMS3A (0x143A, 0x153A)**

| BIT         | 7                  | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|--------------------|---|---|---|---|---|---|---|
| Field       | EyeMonValCntL[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00               |   |   |   |   |   |   |   |
| Access Type | Read Only          |   |   |   |   |   |   |   |

| BITFIELD      | BITS | DESCRIPTION                               | DECODE                                                 |
|---------------|------|-------------------------------------------|--------------------------------------------------------|
| EyeMonValCntL | 7:0  | Eye monitor valid (hit) count (read-only) | 0xXX: Eye monitor valid count (Least Significant Byte) |

**RLMS3B (0x143B, 0x153B)**

| BIT         | 7                  | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|--------------------|---|---|---|---|---|---|---|
| Field       | EyeMonValCntH[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00               |   |   |   |   |   |   |   |
| Access Type | Read Only          |   |   |   |   |   |   |   |

| BITFIELD      | BITS | DESCRIPTION                               | DECODE                                                |
|---------------|------|-------------------------------------------|-------------------------------------------------------|
| EyeMonValCntH | 7:0  | Eye monitor valid (hit) count (read-only) | 0xXX: Eye monitor valid count (Most Significant Byte) |

**RLMS3D (0x143D, 0x153D)**

| BIT         | 7            | 6 | 5 | 4 | 3 | 2 | 1 | 0               |
|-------------|--------------|---|---|---|---|---|---|-----------------|
| Field       | ErrChPh[6:0] |   |   |   |   |   |   | ErrChPhToggleEn |
| Reset       | 0x00         |   |   |   |   |   |   | 0x1             |
| Access Type | Read Only    |   |   |   |   |   |   | Write, Read     |

| BITFIELD        | BITS | DESCRIPTION                       | DECODE                                                                              |
|-----------------|------|-----------------------------------|-------------------------------------------------------------------------------------|
| ErrChPh         | 7:1  | Error channel phase (read-only)   | 0bXXXXXXX: 5.6 degrees per step                                                     |
| ErrChPhToggleEn | 0    | Error channel phase toggle enable | 0b0: Use primary phase only<br>0b1: Auto toggle phase between primary and secondary |

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**RLMS3E (0x143E, 0x153E)**

| BIT         | 7            | 6               | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|--------------|-----------------|---|---|---|---|---|---|
| Field       | ErrChPhSecTA | ErrChPhSec[6:0] |   |   |   |   |   |   |
| Reset       | 0x1          | 0x33            |   |   |   |   |   |   |
| Access Type | Write, Read  | Write, Read     |   |   |   |   |   |   |

| BITFIELD     | BITS | DESCRIPTION                                 | DECODE                                                    |
|--------------|------|---------------------------------------------|-----------------------------------------------------------|
| ErrChPhSecTA | 7    | Error channel phase secondary timing adjust | 0b0: Timing adjust disabled<br>0b1: Timing adjust enabled |
| ErrChPhSec   | 6:0  | Error channel phase secondary (odd)         | 0bxxxxxx: Error channel phase secondary (odd)             |

**RLMS3F (0x143F, 0x153F)**

| BIT         | 7            | 6               | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|--------------|-----------------|---|---|---|---|---|---|
| Field       | ErrChPhPriTA | ErrChPhPri[6:0] |   |   |   |   |   |   |
| Reset       | 0x0          | 0x72            |   |   |   |   |   |   |
| Access Type | Write, Read  | Write, Read     |   |   |   |   |   |   |

| BITFIELD     | BITS | DESCRIPTION                               | DECODE                                                    |
|--------------|------|-------------------------------------------|-----------------------------------------------------------|
| ErrChPhPriTA | 7    | Error channel phase primary timing adjust | 0b0: Timing adjust disabled<br>0b1: Timing adjust enabled |
| ErrChPhPri   | 6:0  | Error channel phase primary (even)        | 0bxxxxxx: Error channel phase primary (even)              |

**RLMS58 (0x1458, 0x1558)**

| BIT         | 7 | 6              | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|---|----------------|---|---|---|---|---|---|
| Field       | – | ErrChVTh1[6:0] |   |   |   |   |   |   |
| Reset       | – | 0x18           |   |   |   |   |   |   |
| Access Type | – | Write, Read    |   |   |   |   |   |   |

| BITFIELD  | BITS | DESCRIPTION                             | DECODE                                                                                                                                |
|-----------|------|-----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| ErrChVTh1 | 6:0  | Error channel target amplitude for ones | 0bxxxxxx: Format: {sign, Magnitude[5:0]}<br>where<br>sign: 1 = negative, 0 = positive<br>Magnitude:: binary amplitude 4.7mV per count |

**RLMS59 (0x1459, 0x1559)**

| BIT         | 7 | 6              | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|---|----------------|---|---|---|---|---|---|
| Field       | – | ErrChVTh0[6:0] |   |   |   |   |   |   |
| Reset       | – | 0x58           |   |   |   |   |   |   |
| Access Type | – | Write, Read    |   |   |   |   |   |   |

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| BITFIELD  | BITS | DESCRIPTION                              | DECODE                                                                                                                                |
|-----------|------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| ErrChVTh0 | 6:0  | Error channel target amplitude for zeros | 0bxxxxxx: Format: {sign, Magnitude[5:0]}<br>where<br>sign: 1 = negative, 0 = positive<br>Magnitude:: binary amplitude 4.7mV per count |

**RLMS5B (0x145B, 0x155B)**

| BIT         | 7 | 6             | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|---|---------------|---|---|---|---|---|---|
| Field       | – | ErrChVTh[6:0] |   |   |   |   |   |   |
| Reset       | – | 0x00          |   |   |   |   |   |   |
| Access Type | – | Read Only     |   |   |   |   |   |   |

| BITFIELD | BITS | DESCRIPTION                         |
|----------|------|-------------------------------------|
| ErrChVTh | 6:0  | Error channel threshold (read-only) |

**RLMS64 (0x1464, 0x1564)**

| BIT         | 7 | 6 | 5 | 4 | 3 | 2 | 1              | 0 |
|-------------|---|---|---|---|---|---|----------------|---|
| Field       | – | – | – | – | – | – | TxSSCMode[1:0] |   |
| Reset       | – | – | – | – | – | – | 0x0            |   |
| Access Type | – | – | – | – | – | – | Write, Read    |   |

| BITFIELD  | BITS | DESCRIPTION             | DECODE                                                                                             |
|-----------|------|-------------------------|----------------------------------------------------------------------------------------------------|
| TxSSCMode | 1:0  | Tx spread spectrum mode | 0b00: No spread spectrum (manual phase)<br>0b01: Center spread<br>0b10: Reserved<br>0b11: Reserved |

**RLMS70 (0x1470, 0x1570)**

| BIT         | 7 | 6                 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|---|-------------------|---|---|---|---|---|---|
| Field       | – | TxSSCFrqCtrl[6:0] |   |   |   |   |   |   |
| Reset       | – | 0x01              |   |   |   |   |   |   |
| Access Type | – | Write, Read       |   |   |   |   |   |   |

| BITFIELD     | BITS | DESCRIPTION                          | DECODE                                                  |
|--------------|------|--------------------------------------|---------------------------------------------------------|
| TxSSCFrqCtrl | 6:0  | Tx spread spectrum frequency control | 0bXXXXXXXX: Tx spread spectrum center frequency control |

**RLMS71 (0x1471, 0x1571)**

| BIT         | 7 | 6                  | 5 | 4 | 3 | 2 | 1 | 0           |
|-------------|---|--------------------|---|---|---|---|---|-------------|
| Field       | – | TxSSCCenSprSt[5:0] |   |   |   |   |   | TxSSCEn     |
| Reset       | – | 0x01               |   |   |   |   |   | 0x0         |
| Access Type | – | Write, Read        |   |   |   |   |   | Write, Read |

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| BITFIELD          | BITS | DESCRIPTION                                      | DECODE                                                              |
|-------------------|------|--------------------------------------------------|---------------------------------------------------------------------|
| TxSSCCenS<br>prSt | 6:1  | Tx spread spectrum center spread startup control | 0bXXXXXX: Tx spread spectrum center spread startup control          |
| TxSSCEn           | 0    | Tx spread spectrum enable                        | 0b0: Tx spread spectrum disabled<br>0b1: Tx spread spectrum enabled |

**RLMS72 (0x1472, 0x1572)**

| BIT         | 7                | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|------------------|---|---|---|---|---|---|---|
| Field       | TxSSCPreScL[7:0] |   |   |   |   |   |   |   |
| Reset       | 0xCF             |   |   |   |   |   |   |   |
| Access Type | Write, Read      |   |   |   |   |   |   |   |

| BITFIELD         | BITS | DESCRIPTION                                      | DECODE                                                 |
|------------------|------|--------------------------------------------------|--------------------------------------------------------|
| TxSSCPreSc<br>IL | 7:0  | Tx spread spectrum frequency pre-scaler low byte | 0xXX: Tx spread spectrum frequency pre-scaler low byte |

**RLMS73 (0x1473, 0x1573)**

| BIT         | 7 | 6 | 5 | 4 | 3 | 2                | 1 | 0 |
|-------------|---|---|---|---|---|------------------|---|---|
| Field       | – | – | – | – | – | TxSSCPreScH[2:0] |   |   |
| Reset       | – | – | – | – | – | 0x0              |   |   |
| Access Type | – | – | – | – | – | Write, Read      |   |   |

| BITFIELD         | BITS | DESCRIPTION                                       | DECODE                                                   |
|------------------|------|---------------------------------------------------|----------------------------------------------------------|
| TxSSCPreSc<br>IH | 2:0  | Tx spread spectrum frequency pre-scaler high bits | 0bXXX: Tx spread spectrum frequency pre-scaler high bits |

**RLMS74 (0x1474, 0x1574)**

| BIT         | 7             | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|---------------|---|---|---|---|---|---|---|
| Field       | TxSSCPhL[7:0] |   |   |   |   |   |   |   |
| Reset       | 0x00          |   |   |   |   |   |   |   |
| Access Type | Write, Read   |   |   |   |   |   |   |   |

| BITFIELD | BITS | DESCRIPTION                                    | DECODE                                                         |
|----------|------|------------------------------------------------|----------------------------------------------------------------|
| TxSSCPhL | 7:0  | Tx spread spectrum interpolator phase low byte | 0xXX: Tx spread spectrum frequency interpolator phase low byte |

**RLMS75 (0x1475, 0x1575)**

| BIT         | 7 | 6             | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|---|---------------|---|---|---|---|---|---|
| Field       | – | TxSSCPhH[6:0] |   |   |   |   |   |   |
| Reset       | – | 0x00          |   |   |   |   |   |   |
| Access Type | – | Write, Read   |   |   |   |   |   |   |

| BITFIELD | BITS | DESCRIPTION                           | DECODE                                                                |
|----------|------|---------------------------------------|-----------------------------------------------------------------------|
| TxSSCPhH | 6:0  | Tx spread spectrum interpolator phase | 0bXXXXXXXX: Tx spread spectrum frequency interpolator phase high bits |

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**RLMS76 (0x1476, 0x1576)**

| BIT         | 7 | 6 | 5 | 4 | 3 | 2 | 1                | 0 |
|-------------|---|---|---|---|---|---|------------------|---|
| Field       | – | – | – | – | – | – | TxSSCPhQuad[1:0] |   |
| Reset       | – | – | – | – | – | – | 0x0              |   |
| Access Type | – | – | – | – | – | – | Write, Read      |   |

| BITFIELD    | BITS | DESCRIPTION                                    | DECODE                                               |
|-------------|------|------------------------------------------------|------------------------------------------------------|
| TxSSCPhQuad | 1:0  | Tx spread spectrum interpolator phase quadrant | 0bXX: Tx spread spectrum interpolator phase quadrant |

**RLMS95 (0x1495, 0x1595)**

| BIT         | 7           | 6 | 5             | 4 | 3 | 2 | 1 | 0 |
|-------------|-------------|---|---------------|---|---|---|---|---|
| Field       | TxAmpManEn  | – | TxAmpMan[5:0] |   |   |   |   |   |
| Reset       | 0x0         | – | 0x29          |   |   |   |   |   |
| Access Type | Write, Read | – | Write, Read   |   |   |   |   |   |

| BITFIELD   | BITS | DESCRIPTION                  | DECODE                                                                            |
|------------|------|------------------------------|-----------------------------------------------------------------------------------|
| TxAmpManEn | 7    | Tx amplitude manual override | 0b0: Do not manually override Tx amplitude<br>0b1: Manually override Tx amplitude |
| TxAmpMan   | 5:0  | Tx amplitude                 | 0bXXXXXX: Binary amplitude 10mV per count                                         |

**RLMSA4 (0x14A4, 0x15A4)**

| BIT         | 7                 | 6 | 5 | 4            | 3 | 2 | 1 | 0 |
|-------------|-------------------|---|---|--------------|---|---|---|---|
| Field       | AEQ_PER_MULT[1:0] |   |   | AEQ_PER[5:0] |   |   |   |   |
| Reset       | 0x2               |   |   | 0x3D         |   |   |   |   |
| Access Type | Write, Read       |   |   | Write, Read  |   |   |   |   |

| BITFIELD     | BITS | DESCRIPTION                                                                                                                          | DECODE                                                          |
|--------------|------|--------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|
| AEQ_PER_MULT | 7:6  | Adaptive EQ period multiplier                                                                                                        | 0b00: 1ms<br>0b01: 4ms<br>0b10: 16ms<br>0b11: 64ms              |
| AEQ_PER      | 5:0  | Adaptive EQ period.<br>Periodic adaptation is disabled when value is 0.<br>Adaptive EQ period is (AEQ_PER value times AEQ_PER_MULT). | 0bXXXXXX: Only zeros have a special meaning.<br>See Description |

**RLMSAC (0x14AC, 0x15AC)**

| BIT         | 7                | 6                   | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|------------------|---------------------|---|---|---|---|---|---|
| Field       | ErrChPhSecTAFR3G | ErrChPhSecFR3G[6:0] |   |   |   |   |   |   |
| Reset       | 0x1              | 0x4D                |   |   |   |   |   |   |
| Access Type | Write, Read      | Write, Read         |   |   |   |   |   |   |

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| BITFIELD             | BITS | DESCRIPTION                                                                                                                           | DECODE                                                    |
|----------------------|------|---------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| ErrChPhSec<br>TAFR3G | 7    | Error channel phase secondary timing adjust                                                                                           | 0b0: Timing adjust disabled<br>0b1: Timing adjust enabled |
| ErrChPhSec<br>FR3G   | 6:0  | A 7-bit phase command used for eye monitoring and used during adaptation by the error channel. (Fast Receive, Secondary Phase, 3Gbps) | 0bXXXXXXX: 7'h4D                                          |

**RLMSAD (0x14AD, 0x15AD)**

| BIT         | 7                    | 6                   | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|----------------------|---------------------|---|---|---|---|---|---|
| Field       | ErrChPhPri<br>TAFR3G | ErrChPhPriFR3G[6:0] |   |   |   |   |   |   |
| Reset       | 0x0                  | 0x0D                |   |   |   |   |   |   |
| Access Type | Write, Read          | Write, Read         |   |   |   |   |   |   |

| BITFIELD             | BITS | DESCRIPTION                                                                                                                         | DECODE                                                    |
|----------------------|------|-------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| ErrChPhPriT<br>AFR3G | 7    | Error channel phase primary timing adjust                                                                                           | 0b0: Timing adjust disabled<br>0b1: Timing adjust enabled |
| ErrChPhPriF<br>R3G   | 6:0  | A 7-bit phase command used for eye monitoring and used during adaptation by the error channel. (Fast Receive, Primary Phase, 3Gbps) | 0bXXXXXXX: 7'h0D                                          |

**RLMSAE (0x14AE, 0x15AE)**

| BIT         | 7                     | 6                    | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|-----------------------|----------------------|---|---|---|---|---|---|
| Field       | ErrChPhSe<br>cTAFR1G5 | ErrChPhSecFR1G5[6:0] |   |   |   |   |   |   |
| Reset       | 0x1                   | 0x4E                 |   |   |   |   |   |   |
| Access Type | Write, Read           | Write, Read          |   |   |   |   |   |   |

| BITFIELD              | BITS | DESCRIPTION                                                                                                                             | DECODE                                                    |
|-----------------------|------|-----------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| ErrChPhSec<br>TAFR1G5 | 7    | Error channel phase secondary timing adjust                                                                                             | 0b0: Timing adjust disabled<br>0b1: Timing adjust enabled |
| ErrChPhSec<br>FR1G5   | 6:0  | A 7-bit phase command used for eye monitoring and used during adaptation by the error channel. (Fast Receive, Secondary Phase, 3.5Gbps) | 0bXXXXXXX: 7'h4E                                          |

**RLMSAF (0x14AF, 0x15AF)**

| BIT         | 7                     | 6                    | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|-----------------------|----------------------|---|---|---|---|---|---|
| Field       | ErrChPhPri<br>TAFR1G5 | ErrChPhPriFR1G5[6:0] |   |   |   |   |   |   |
| Reset       | 0x0                   | 0x0F                 |   |   |   |   |   |   |
| Access Type | Write, Read           | Write, Read          |   |   |   |   |   |   |



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| BITFIELD              | BITS | DESCRIPTION                                                                                                                           | DECODE                                                    |
|-----------------------|------|---------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| ErrChPhPriT<br>AFR1G5 | 7    | Error channel phase primary timing adjust                                                                                             | 0b0: Timing adjust disabled<br>0b1: Timing adjust enabled |
| ErrChPhPriF<br>R1G5   | 6:0  | A 7-bit phase command used for eye monitoring and used during adaptation by the error channel. (Fast Receive, Primary Phase, 3.5Gbps) | 0bXXXXXXX: 7'h0F                                          |

**RLMSB0 (0x14B0, 0x15B0)**

| BIT         | 7                     | 6                    | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|-----------------------|----------------------|---|---|---|---|---|---|
| Field       | ErrChPhSe<br>cTASR1G5 | ErrChPhSecSR1G5[6:0] |   |   |   |   |   |   |
| Reset       | 0x1                   | 0x34                 |   |   |   |   |   |   |
| Access Type | Write, Read           | Write, Read          |   |   |   |   |   |   |

| BITFIELD              | BITS | DESCRIPTION                                                                                                                             | DECODE                                                    |
|-----------------------|------|-----------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| ErrChPhSec<br>TASR1G5 | 7    | Error channel phase secondary timing adjust                                                                                             | 0b0: Timing adjust disabled<br>0b1: Timing adjust enabled |
| ErrChPhSec<br>SR1G5   | 6:0  | A 7-bit phase command used for eye monitoring and used during adaptation by the error channel. (Slow Receive, Secondary Phase, 3.5Gbps) | 0bXXXXXXX: 7'h34                                          |

**RLMSB1 (0x14B1, 0x15B1)**

| BIT         | 7                     | 6                    | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|-----------------------|----------------------|---|---|---|---|---|---|
| Field       | ErrChPhPri<br>TASR1G5 | ErrChPhPriSR1G5[6:0] |   |   |   |   |   |   |
| Reset       | 0x0                   | 0x73                 |   |   |   |   |   |   |
| Access Type | Write, Read           | Write, Read          |   |   |   |   |   |   |

| BITFIELD              | BITS | DESCRIPTION                                                                                                                           | DECODE                                                    |
|-----------------------|------|---------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| ErrChPhPriT<br>ASR1G5 | 7    | Error channel phase primary timing adjust                                                                                             | 0b0: Timing adjust disabled<br>0b1: Timing adjust enabled |
| ErrChPhPriS<br>R1G5   | 6:0  | A 7-bit phase command used for eye monitoring and used during adaptation by the error channel. (Slow Receive, Primary Phase, 3.5Gbps) | 0bXXXXXXX: 7'h73                                          |

**RLMSB2 (0x14B2, 0x15B2)**

| BIT         | 7                     | 6                    | 5 | 4 | 3 | 2 | 1 | 0 |
|-------------|-----------------------|----------------------|---|---|---|---|---|---|
| Field       | ErrChPhSe<br>cTASRG75 | ErrChPhSecSRG75[6:0] |   |   |   |   |   |   |
| Reset       | 0x1                   | 0x3E                 |   |   |   |   |   |   |
| Access Type | Write, Read           | Write, Read          |   |   |   |   |   |   |

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| BITFIELD              | BITS | DESCRIPTION                                                                                                                             | DECODE                                                    |
|-----------------------|------|-----------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| ErrChPhSec<br>TASRG75 | 7    | Error channel phase secondary timing adjust                                                                                             | 0b0: Timing adjust disabled<br>0b1: Timing adjust enabled |
| ErrChPhSec<br>SRG75   | 6:0  | A 7-bit phase command used for eye monitoring and used during adaptation by the error channel. (Slow Receive, Secondary Phase, 750Mbps) | 0bXXXXXXX: 7'h3E                                          |

**RLMSB3 (0x14B3, 0x15B3)**

| BIT            | 7                     | 6                    | 5 | 4 | 3 | 2 | 1 | 0 |
|----------------|-----------------------|----------------------|---|---|---|---|---|---|
| Field          | ErrChPhPri<br>TASRG75 | ErrChPhPriSRG75[6:0] |   |   |   |   |   |   |
| Reset          | 0x0                   | 0x7D                 |   |   |   |   |   |   |
| Access<br>Type | Write, Read           | Write, Read          |   |   |   |   |   |   |

| BITFIELD              | BITS | DESCRIPTION                                                                                                                           | DECODE                                                    |
|-----------------------|------|---------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| ErrChPhPriT<br>ASRG75 | 7    | Error channel phase primary timing adjust                                                                                             | 0b0: Timing adjust disabled<br>0b1: Timing adjust enabled |
| ErrChPhPriS<br>RG75   | 6:0  | A 7-bit phase command used for eye monitoring and used during adaptation by the error channel. (Slow Receive, Primary Phase, 750Mbps) | 0bXXXXXXX: 7'h7D                                          |

**RLMSB4 (0x14B4, 0x15B4)**

| BIT            | 7                          | 6                     | 5 | 4 | 3 | 2 | 1 | 0 |
|----------------|----------------------------|-----------------------|---|---|---|---|---|---|
| Field          | ErrChPhSe<br>cTASRG37<br>5 | ErrChPhSecSRG375[6:0] |   |   |   |   |   |   |
| Reset          | 0x1                        | 0x3C                  |   |   |   |   |   |   |
| Access<br>Type | Write, Read                | Write, Read           |   |   |   |   |   |   |

| BITFIELD               | BITS | DESCRIPTION                                                                                                                             | DECODE                                                    |
|------------------------|------|-----------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| ErrChPhSec<br>TASRG375 | 7    | Error channel phase secondary timing adjust                                                                                             | 0b0: Timing adjust disabled<br>0b1: Timing adjust enabled |
| ErrChPhSec<br>SRG375   | 6:0  | A 7-bit phase command used for eye monitoring and used during adaptation by the error channel. (Slow Receive, Secondary Phase, 375Mbps) | 0bXXXXXXX: 7'h3C                                          |

**RLMSB5 (0x14B5, 0x15B5)**

| BIT            | 7                      | 6                     | 5 | 4 | 3 | 2 | 1 | 0 |
|----------------|------------------------|-----------------------|---|---|---|---|---|---|
| Field          | ErrChPhPri<br>TASRG375 | ErrChPhPriSRG375[6:0] |   |   |   |   |   |   |
| Reset          | 0x0                    | 0x7B                  |   |   |   |   |   |   |
| Access<br>Type | Write, Read            | Write, Read           |   |   |   |   |   |   |

MAX9296B Dual 1 x 4 or 2  
x 2 CSI-2 GMSL2  
Deserializer

Appendix A: Register Map and Tables

| BITFIELD           | BITS | DESCRIPTION                                                                                                                           | DECODE                                                    |
|--------------------|------|---------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| ErrChPhPriTASRG375 | 7    | Error channel phase primary timing adjust                                                                                             | 0b0: Timing adjust disabled<br>0b1: Timing adjust enabled |
| ErrChPhPriSRG375   | 6:0  | A 7-bit phase command used for eye monitoring and used during adaptation by the error channel. (Slow Receive, Primary Phase, 375Mbps) | 0bXXXXXXX: 7'h7B                                          |

**RLMSB6 (0x14B6, 0x15B6)**

| BIT                 | 7                   | 6                                                                                                                                         | 5 | 4 | 3                                                         | 2 | 1 | 0 |
|---------------------|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------|---|---|-----------------------------------------------------------|---|---|---|
| Field               | ErrChPhSecTASRG1875 | ErrChPhSecSRG1875[6:0]                                                                                                                    |   |   |                                                           |   |   |   |
| Reset               | 0x1                 | 0x3B                                                                                                                                      |   |   |                                                           |   |   |   |
| Access Type         | Write, Read         | Write, Read                                                                                                                               |   |   |                                                           |   |   |   |
| BITFIELD            | BITS                | DESCRIPTION                                                                                                                               |   |   | DECODE                                                    |   |   |   |
| ErrChPhSecTASRG1875 | 7                   | Error channel phase secondary timing adjust                                                                                               |   |   | 0b0: Timing adjust disabled<br>0b1: Timing adjust enabled |   |   |   |
| ErrChPhSecSRG1875   | 6:0                 | A 7-bit phase command used for eye monitoring and used during adaptation by the error channel. (Slow Receive, Secondary Phase, 187.5Mbps) |   |   | 0bXXXXXXX: 7'h3B                                          |   |   |   |

**RLMSB7 (0x14B7, 0x15B7)**

| BIT                     | 7                           | 6                                                                                                                                       | 5 | 4 | 3                                                         | 2 | 1 | 0 |
|-------------------------|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|---|---|-----------------------------------------------------------|---|---|---|
| Field                   | ErrChPhPri<br>TASRG187<br>5 | ErrChPhPriSRG1875[6:0]                                                                                                                  |   |   |                                                           |   |   |   |
| Reset                   | 0x0                         | 0x7A                                                                                                                                    |   |   |                                                           |   |   |   |
| Access<br>Type          | Write, Read                 | Write, Read                                                                                                                             |   |   |                                                           |   |   |   |
| BITFIELD                | BITS                        | DESCRIPTION                                                                                                                             |   |   | DECODE                                                    |   |   |   |
| ErrChPhPriT<br>ASRG1875 | 7                           | Error channel phase primary timing adjust                                                                                               |   |   | 0b0: Timing adjust disabled<br>0b1: Timing adjust enabled |   |   |   |
| ErrChPhPriS<br>RG1875   | 6:0                         | A 7-bit phase command used for eye monitoring and used during adaptation by the error channel. (Slow Receive, Primary Phase, 187.5Mbps) |   |   | 0bXXXXXXX: 7'h7A                                          |   |   |   |

**RLMSC4 (0x14C4, 0x15C4)**

| BIT         | 7 | 6 | 5 | 4 | 3 | 2           | 1 | 0 |
|-------------|---|---|---|---|---|-------------|---|---|
| Field       | — | — | — | — | — | RevFast     | — | — |
| Reset       | — | — | — | — | — | 0b0         | — | — |
| Access Type | — | — | — | — | — | Write, Read | — | — |

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Appendix A: Register Map and Tables

| BITFIELD | BITS | DESCRIPTION                     | DECODE                                                                            |
|----------|------|---------------------------------|-----------------------------------------------------------------------------------|
| RevFast  | 2    | GMSL1 reverse channel fast mode | 0x0: Reverse channel fast mode disabled<br>0x1: Reverse channel fast mode enabled |

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## Appendix A: Register Map and Tables

For pricing, delivery, and ordering information, please visit Maxim Integrated's online storefront at <https://www.maximintegrated.com/en/storefront/storefront.html>.

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